

SIEMENS

MAGNETOM

MR

Functional Description

Aera

Skyra

Avanto^{fit}

Skyra^{fit}

Prisma

Prisma^{fit}

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Links

This document has a built-in hyperlink system to facilitate navigation. Text links are recognizable by their **BLUE** color. Graphical links are usually denoted with a triangle  placed in the upper left corner of any box within a graphic. Where this could not be realized an attempt has been made to indicate a link using some other means. Links, however, can be recognized when the cursor changes to a hand  when placed over a link.

Introduction

The Vision

The MAGNETOM Aera/Skyra are the first systems of the next MR platform generation, comprising HW and SW, which builds the basis of upcoming MR scanners. The new platform has to address the MR and workflow architecture and concepts of the next decade. The main focus has been prescribed as follows:

- Setting trends to excite customers and to make a real difference for their future business.
- Shaping MR as an outstanding part of the Siemens Healthcare solution business and workflow strategy.
- Platform future-proof in cost structure, with a concept for global development, manufacturing, and logistics.

Recent technological trends have been an increase of B_0 in gradient performance, in RF receive performance and parallel imaging techniques. B_0 strength and gradient performance have reached the limits for physiological exposure in clinical systems and are not expected to increase in the clinical market in foreseeable time. Experimental systems may still exceed these values.

Present or foreseeable trends in the MRI society are

- Cost reduction
- Workflow
- Higher level of automation
 - Integrated workflow optimization is in the focus because of the increasing cost pressure in medical business, and

the extending variety and complexity of clinical applications

- Application Range
- General trend to differentiation
 - In research systems, multi-channel RF transmission is pretty much in the focus albeit not fully exploited (limitation to low flip angles, long calculation times)
 - Increasing usage of hybrid solutions (MRI usage in operational theatre (BrainLab, IMRIS), together with PET, with radiation treatment, with HIFU, with stereotactical frames, etc.
- Patient Comfort
 - Wider bore sizes for increased patient comfort because the patients do now act increasingly in the role of customers.

Consequently, the primary focus of the new platform is the clinical routine market in terms of appearance and cost position. This market is understood to be more important than ever. However, solutions for the research market need to be provided as well. Demands of the research market shall be covered with product options where possible.

Technological trends which have an impact on the MRI business are

- the availability of new high speed network solutions
- the performance increase, denser packaging and cheaper availability of FPGAs which allows to use this technology more extensively and to build smaller and cheaper digital electronics.

- the increased computer performance, allowing for a higher degree of parallel services
- the availability of liquefying helium coolers for superconducting magnets (operating at 4 Kelvin)
- the foreseeable availability of new powerful optimizer software

It has been six years since the last product line was first introduced. With the advent of the Avanto and Espree systems came a slew of technological advancements over the product line before it, first and foremost the Tim technology. Tim has made a great impact in the field of MR diagnostic imaging not only through the coil and scan coverage made possible with Tim coils, but also through the many applicational tools and features that streamlined or automated the workflow process in setting up protocols and examinations. Of course many other features made the Avanto/Espree systems very attractive not only from a clinical standpoint, but from a patient friendliness and business standpoint as well. Such as the 0% LHe magnet system or reduction in service and maintenance costs through better reliability.

Consistent with the innovative and state-of-the-art technological engineering prowess of the MR business unit we are proud to introduce yet another line of MR scanner products incorporating both new and improved hardware and software technologies: the **MAGNETOM Aera** and **MAGNETOM Skyra**. In keeping with the extremely successful concept of the Avanto, Espree and Verio systems the Aera/Skyra line has taken the Tim technology to yet even greater heights with newer, better coils and a receive system having up to 128 independent RF channels can be used simultaneously in one single scan and in one single FOV, each generating an independent partial image. To realize this a new hardware platform has been designed using a very different approach to the function, placement, testing and integration of practically all scanner sub-systems. New concepts for measurement control, RF, patient handling, and cooling systems have not only lead to components with greater reliability and better performance but also more service friendly and cost

effective. This new systems approach also requires a new communications architecture, able to support early problem detection and reporting on a 24/7 basis!

System Configurations

The MAGNETOM Aera/Skyra system variants are:

- MAGNETOM Skyra: Tim [204x128] XQ Gradients
- MAGNETOM Skyra: Tim [204x64] XQ Gradients
- MAGNETOM Skyra: Tim [204x48] XQ Gradients
- MAGNETOM Skyra: Tim [204x24] XQ Gradient
- MAGNETOM Aera: Tim [204x64] XQ Gradients
- MAGNETOM Aera: Tim [204x48] XQ Gradients
- MAGNETOM Aera: Tim [204x48] XJ Gradients
- MAGNETOM Aera: Tim [204x24] XJ Gradients

The diagram below provides an overview of the Aera/Skyra hardware components. The blocks can be clicked to jump to that part of the Functional Description.

Restriction of Hazardous Substances

Most components manufactured for these systems conform to the RoHS directive on the restriction of the use of certain hazardous substances in electrical and electronic equipment. RoHS restricts the use of the following substances:

- 1. Lead (Pb)
- 2. Mercury (Hg)
- 3. Cadmium (Cd)
- 4. Hexavalent chromium (Cr6+)
- 5. Polybrominated biphenyls (PBB)
- 6. Polybrominated diphenyl ether (PBDE)

Components having an RoHS label are free from these materials or substances.

Figure 1 Aera/Skyra System Block Diagram

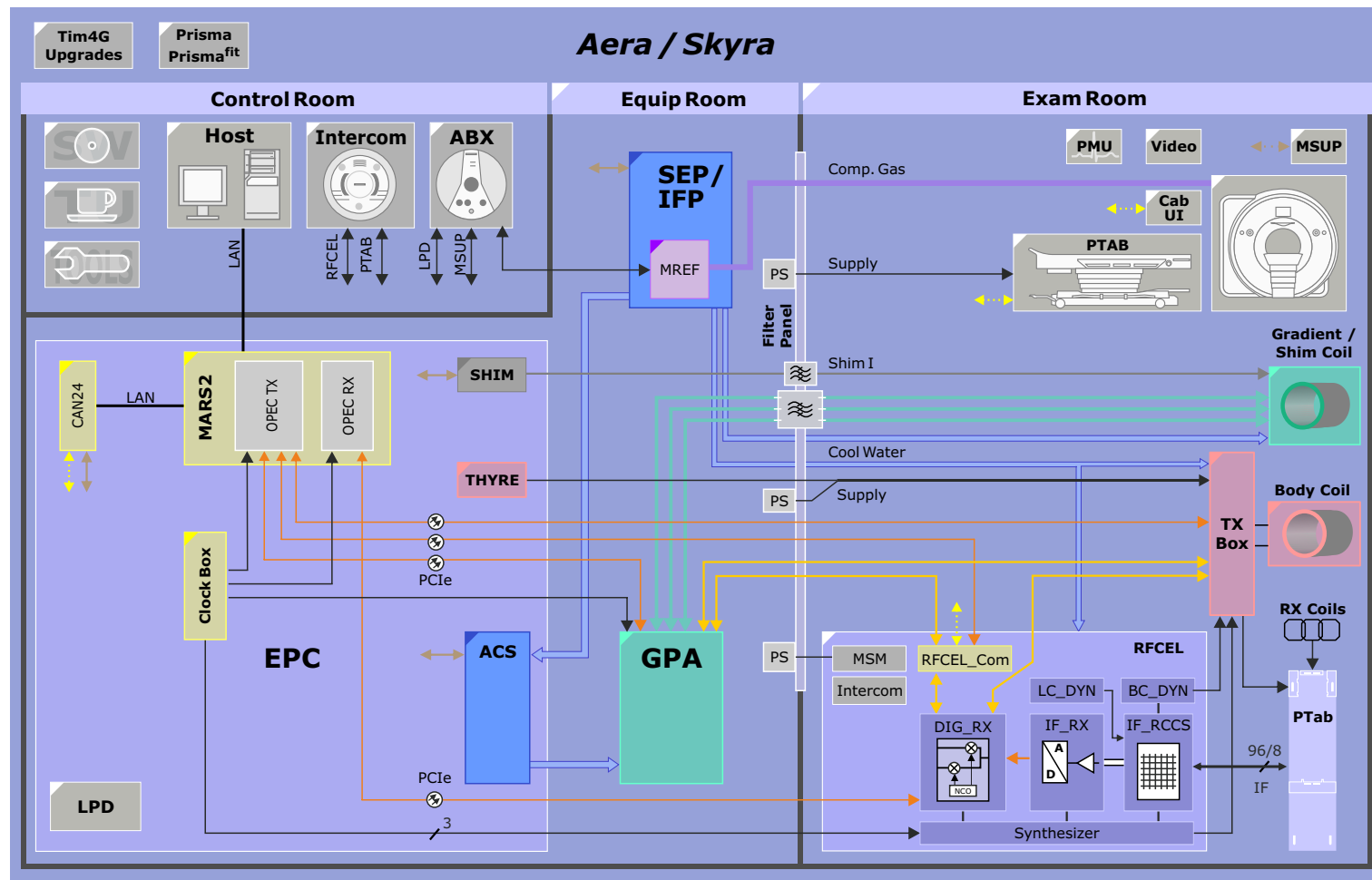
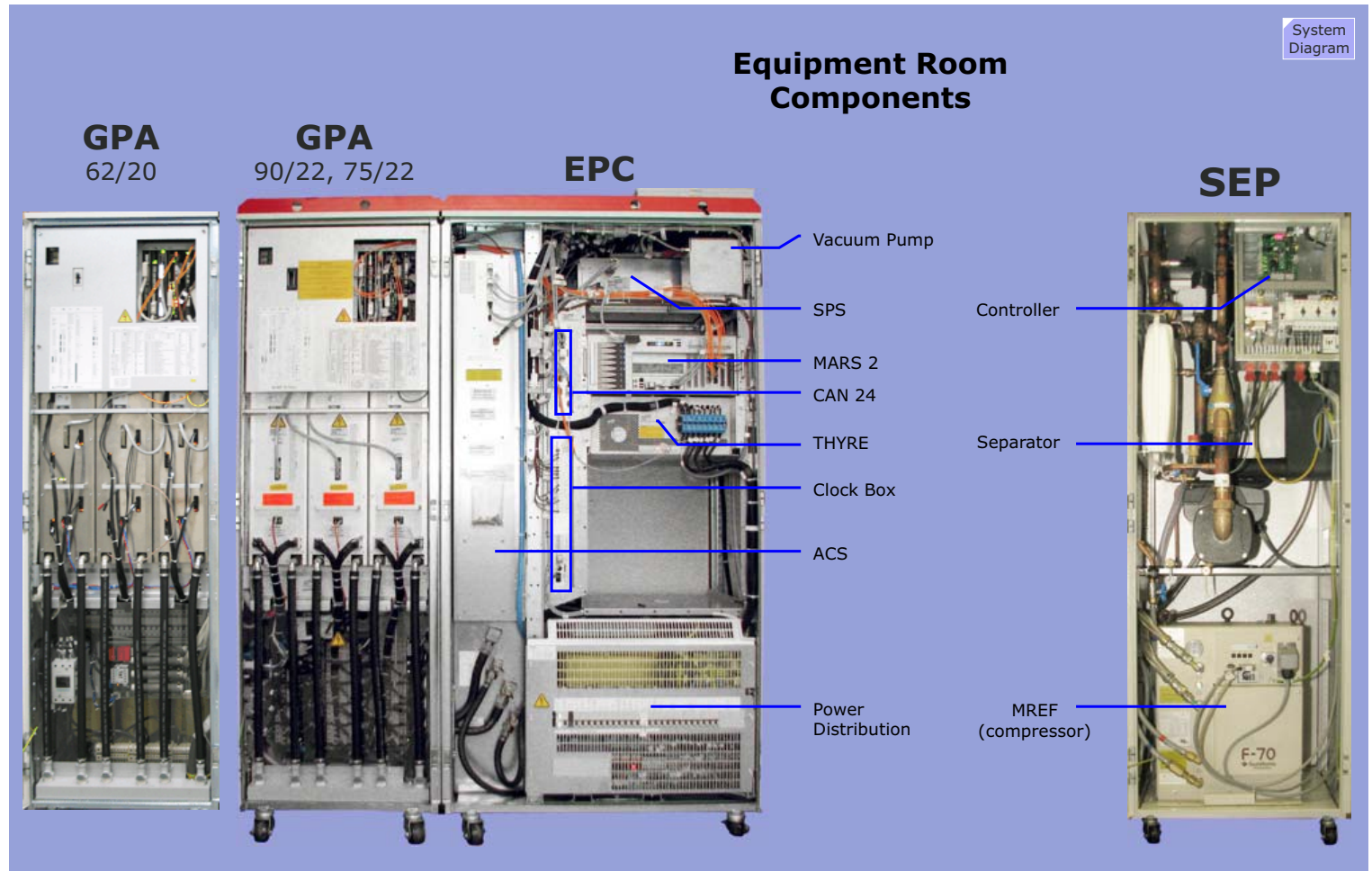


Figure 2 Equipment Room Components



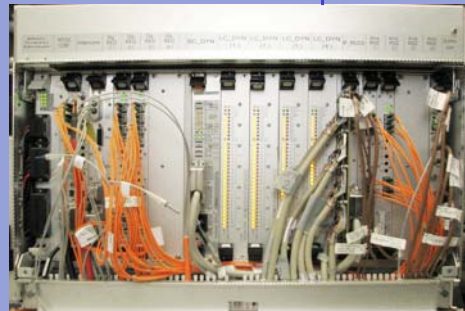
Exam Room Components

TX Box



A large, silver, rectangular metal enclosure with various ports and labels on the front panel.

RFCEL



A rack-mounted unit with a complex internal wiring structure, featuring numerous orange and white cables connected to various components.

Filter Panel (view from equip. room)



A view of the filter panel from the equipment room, showing a white metal enclosure with a yellow warning triangle. It includes a red box highlighting the internal wiring and a red line pointing to the DAVID component.

Labels: Grad Filter, RFCEL PS, TX Box PS, PTAB PS, DAVID

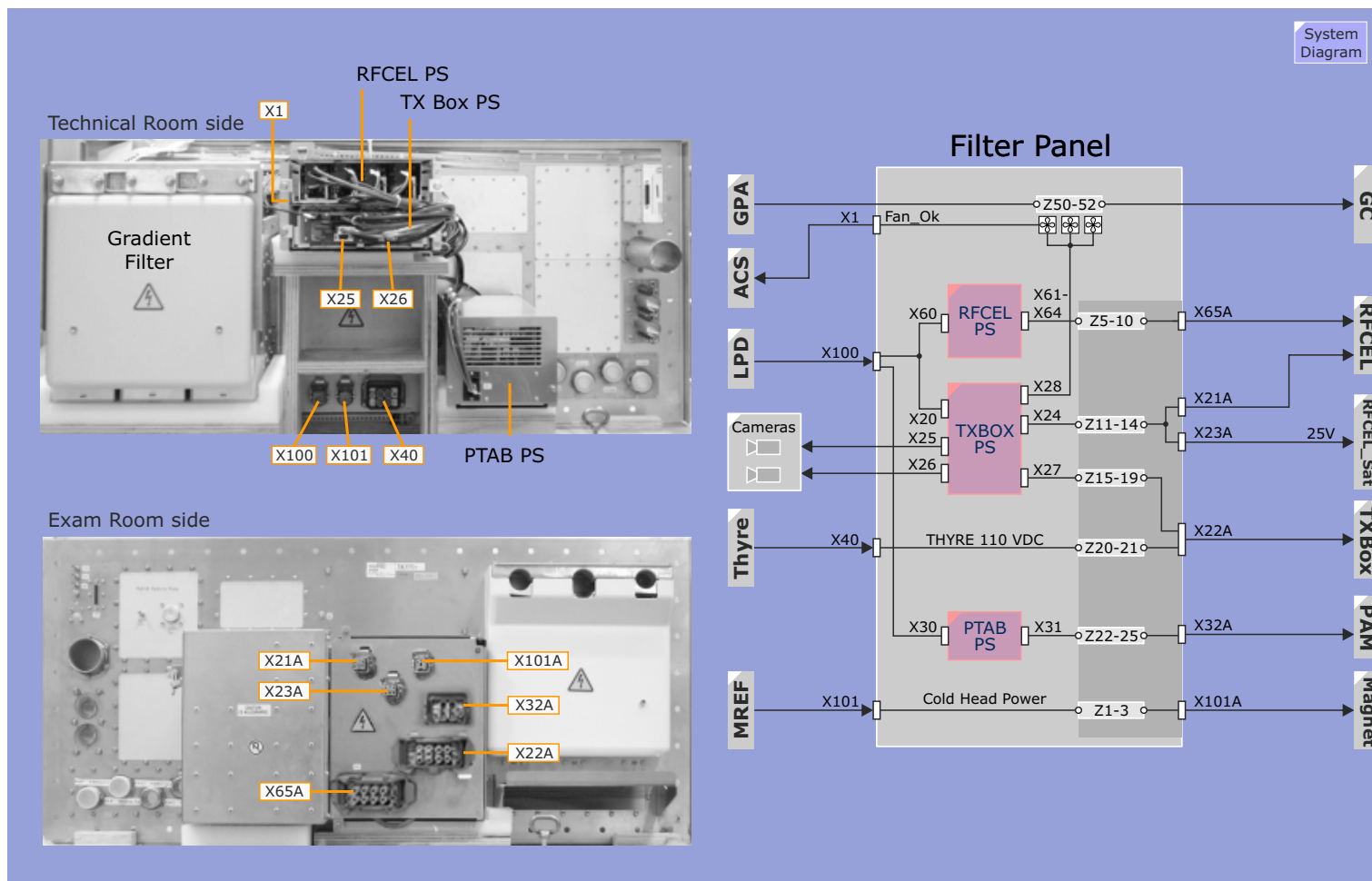
Filter Panel (view from exam room)



A view of the filter panel from the exam room, showing a white metal enclosure with a yellow warning triangle. It includes a red box highlighting the internal wiring and a red line pointing to the Shim Filter component.

Labels: Shim Filter, Power supply outputs, Grad Filter

Figure 4 Filter Panel, Power Supplies



List of Abbreviations:

System Documentation Manuals:

CB-Doc	Computer-based Documentation
DIA	Diagram Manual
FUN	Functional Description
INS	Installation Manual (2 volumes)
PG	Planning Guide
REP, ROP	Replacement of Parts
TSG	Trouble Shooting Guide

ABX	Alarm Box
ACS	Air Conditioning System
ANC	Active Noise Cancellation
ARV	Atmospheric Relief Valve (Magnet System)
AS	Active Shield
AVA	Actual Value Amplifier (Gradient System)
AVL	Auxiliary Vent Line
BC	Body Coil
BNC	Bayonet Connector named after Neil-Concelman
BPL	backplane
CAN	Controller Area Network
CAV	Current Actual Value (Gradient System)
CB	circuit breaker
CBB	Cabinet Building Block (components in EPC)
CCD	Charge-Coupled Device
CCR	Ceramic Carbon Resistor (Magnet System)
CCU	Current Converter Unit (Gradient System)
CH	Cold Head
COSLIO	CAN Open SLIO
CP	circular polarized
CPAN	Control Panel
cPCI	compact PCI
CTC	Cross Term Compensation
CUI	Cabin User Interface
CW	Control Word

DAFID	Data Filter - filter on filter panel
DDS	Direct Digital Synthesizer
DEV	Diameter Elliptical Volume (Magnet System)
Dot	Day optimizing throughput
DPD	Direct Position Drive (Patient Table)
DSV	Diameter Spherical Volume
DUCY	Duty Cycle (Gradient System)
DVI	Digital Visual Interface
ECC	Eddy Current Compensation
ECG	Electro-CardioGram
FP	equalizing filter parameter (IF_RX32 - REDUMU)
EIS	External Interference Shield
EPC	Electronic Power Cabinet
EPI	Echo Planar Imaging
EPR	Electronic Patient Record
ERDU	Emergency Run-Down Unit
ERNI	connector mfr., founded by Ernst Erni
ETH	ETHERnet
FCL	Fixed Current Lead
FDM	Frequency Domain Multiplexer
FGU	Field Generating Unit (Magnet, RF TX, Gradients)
FI	Filter (Gradient System)
fMRI	functional Magnetic Resonance Imaging
F/O	fibre optic
FOC	fiber optic cable
FoV	Field of View
FPGA	Field Programmable Gate Array
FRU	Field Replacable Unit
FSC	Full Scale
FWHM	Full Width at Half Maximum
GC	Gradient Coil
GCS	Gradient Cable Supervision
GIMLI	Generic Infrastructure Module for Link Interfaces
GPA	Gradient Power Amplifier
GPGPU	General Purpose GPU
GPU	Graphic Processor Unit
GraCon	Gradient Controller
GROIN	GRadiant Operation and INfrastructure Node

GSSU	Gradient Small Signal Unit	MWS	Mantelwellen Sperre (RF Trap)
HDD	Hard Disk Drive	N	connector type named after Paul Neill
ICE	Image Calculation Environment	N/C	normally closed
IDE	Integrated Device Equipment	NCO	Numerical Controlled Oscillator
IF	intermediate frequency	N/O	normally open
IFP	Interface & Filter Panel (Cooling System)	NOE	Nuclear Overhauser Effect
IGBT	Isolated Gate Bipolar Transistor	NTP	Network Time Protocol
iPAT	integrated Parallel Acquisition Technique	NUMARIS	Nuclear Medicine Acquisition and Reconstruction Imaging Software
IQ	Image Quality		
IRIS	Intelligent Runtime Imager System	ODU	mfgr of plugs (www.odu.de)
IRV	Internal Relief Valve (Magnet System)	OLM	Optical Length Meter (Patient Table)
LC	Local Coil	OPEC	Optical PCI Express switch Card
LNC	Low Noise Converter	OSC	Optical SCALE (Patient Table)
LO	Local Oscillator	OVC	Outer Vacuum Chamber
LP	linear polarized	PAM	Power Adaptor Module (Patient Table)
LPD	Line Power Distribution (same as PDS)	PAMO	Patient Monitor
LVD	Low Voltage Differential	PAT	Parallel Acquisition Technique
LVQD	Low Voltage Quench Detection	PCI	Peripheral Component Interconnect
LWL	Lichtwellenleiter (fiber optic cable)	PCIe	Peripheral Component Interconnect express
MARS 2	Measurement and Reconstruction System, Vers. 02	PDAU	Physiological Data Acquisition Unit
MBB	Magnet Building Block (components on magnet)	PDS	Power Distribution System (same as LPD)
MC	Measurement Control	PDD	Patient Data Display (Cabin User Interface)
MCIR	Measurement Control and Image Reconstruction	P/E	Pneumatic - Electric
MCM	Main Communication Module	PERU	Physiological ECG and Respiratory Unit
MDH	Measurement Data Header	PID	Proportional Integral Differential (or Derivative)
MNO	multi nuclei option (spectroscopy)	PLL	Phase Locked Loop
MOCO	MOtion COrrECTION	PMS	Patient Monitoring System
MOFI	deMODulator Filter	PMU	Physiological Measurement Unit
MPS	Magnet Power Supply (service tool)	PODO	Pump-out drop-off (magnet)
MPSU	Magnet Power Supply Unit (same as above)	POF	plastic optical fiber
MRAWP	MR Acquisition WorkPlace (new MRC)	PPU	Peripheral Pulse Unit
MRC	MR (Main) Console	PS	Power Supply, Power Stage
MREF	Refrigerator (i.e. Helium Compressor)	PSU	Power Supply Unit
MRIR	MR Image Reconstruction system (imager)	PSY	Power Stage Y (Patient Table)
MRSC	MR Satellite Console	psiA	psi Absolute (ref. to vacuum)
MRWP	MR Workplace (new MRSC)	psiG	psi Gauged (diff. to atmosphere)
MSM	Main Supply Module	PTAB	Patient TABLE
MSUP	Magnet Supervision	PTFE	PolyTetraFluoroEthylene

PWM	Pulse Width Modulation
QA	Quality Assurance
QLA	Quick LAtch - subminiature coax RF connector
RCCS	Receive Coil Channel Selector
RDL	Run Down Load (Magnet System)
RF_CEL	RF Cabin ELectronic
RF_CEL_COM	RF Cabin ELectronic COMMunications
RF_CEL_SAT	RF Cabin ELectronic SATellite
RFFIL	RF FILter
RFSWD	Radio Frequency Safety WatchDog
RTL	real time clock
RX	receive, receiver
SAS2.0	Serial Attached SCSI
SCA	Single Connection Adapter
SCT	Sequence Control Task
SEMO	SErvice MOde (Gradient System)
SEP	SEParator cabinet
SeSo	Service Software
SGA	switched gain amplifier
SLIO	Serial Linked Input/Output
SPI	Serial Peripheral Interface
SPS	Shim Power Supply
SR	Slew Rate
SSB	Single SideBand
STIMO	STIMulation MOnitor
STIMU	STIMUlation
SUVI	SUPerVISION (Gradient System)
SV	Solenoid (Bypass) Valve (Magnet System)
SWT	SWitch Terminal (Patient Table)
TALES	Transmit Antenna LLevel Sensor
TDM	Time Division Multiplexer
THYRE	A DC pwer supply for the TX Box
Tim	Total imaging matrix
Tim 4G	Total imaging matrix, 4th generation
TSG	Temperature Switch Gas (Magnet System)
TSW	Temperature Switch Water (Magnet System)
TTL	Transistor-Transistor Logic
TX	transmit

USB	Universal Serial Bus
VCG	Vector-CardioGram
VFB	Voltage FeedBack (Gradient System)
VGA	Video Graphics Array
VID	VIDeo (patient monitoring)
WXGA	Wide eXtended Graphics Array

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Software

Introduction

This description gives a general overview of the *syngoMR* software and its components. The main emphasis will be the service relevant aspects of the software. In most cases, repairing software problems will entail either fixing a bad configuration or re-installing software entirely.

Overview

Software has now become, by far, the single most expensive component of our MR systems. It has and is being developed by more than 300 programmers across four continents and is continually growing in functionality, size and number of unexpected features.

To gain our bearings, let's first define the software components running on the system. In general, the **software** components running in the system can be classified into one of three groups:

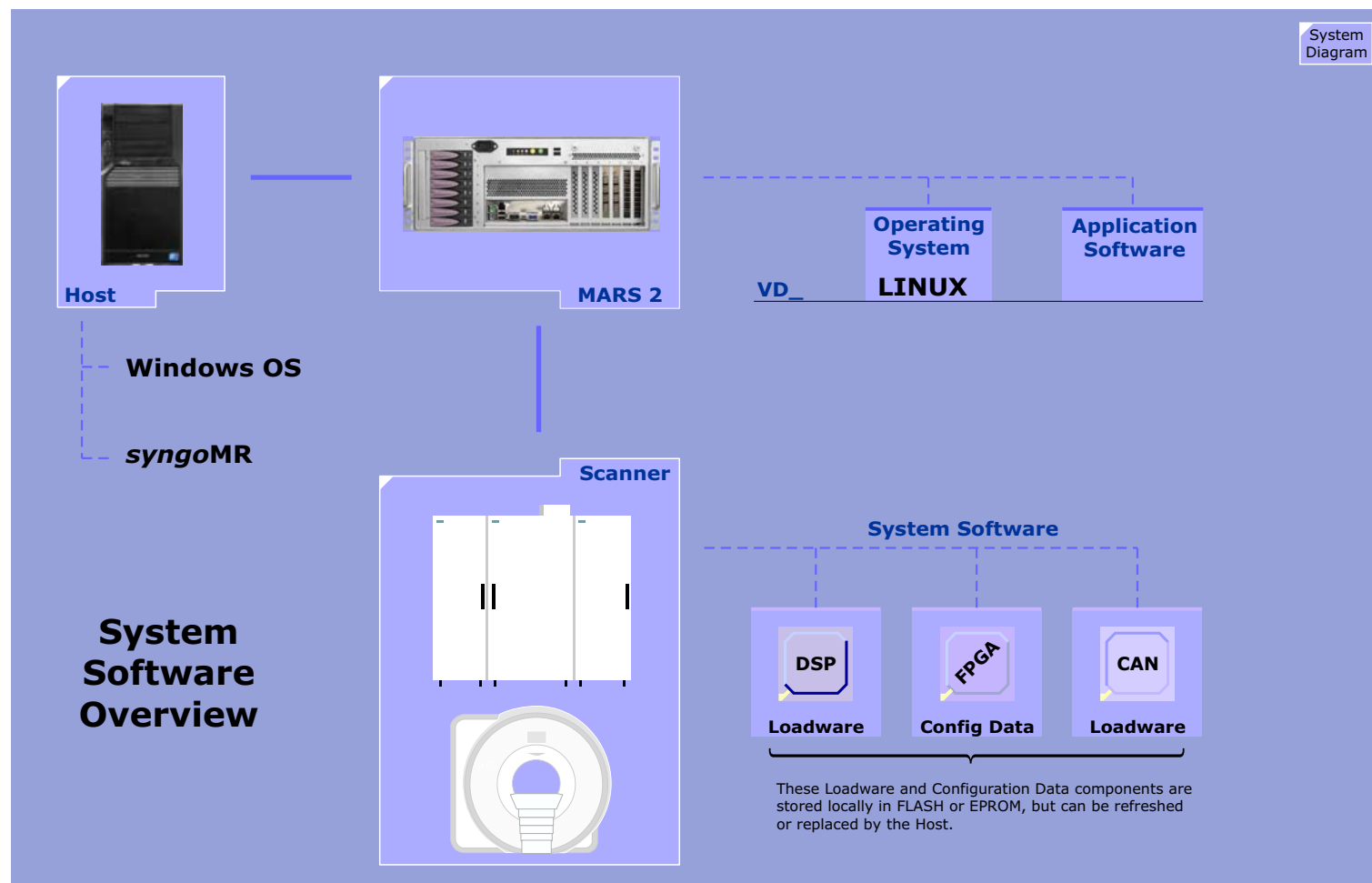
- Operating System (OS)
- Applications (*syngoMR*)
- Scanner Peripheral Control (DSP, FPGA loadware)

Of course, there can be no software without hardware. The **hardware** components running one or more of these software groups:

- syngo MR Acquisition Workplace (MRC) and syngo MR Workplace (MRSC) **Host**
- MARS (**Imager and Control**)
- **Scanner** hardware peripheral units such as various low-level hardware and peripheral controllers

The software components mentioned above which require interaction from and with the user or service engineer will be looked at more closely, that is primarily the software running on the Host. The MARS software is invisible to the user and generally does not require configuration or handling and therefore will not be covered, although some tips will be given which may help determine whether it is running properly. The scanner software is discussed in the MARS and Control section.

Figure 5 System Software Overview



MARS Software

The MARS software is pre-installed at the factory and automatically updated by the Host when a newer version is released. NO INSTALLATION NECESSARY!!! The operating system is a 64-bit Linux variant.

Host Software

The Host software includes a Microsoft OS and the *syngo* MR® software applications required for setting up scan protocols, image post-processing, image archiving functions and other organizational tasks such as providing the imager and hardware controllers with software during the scanner boot up procedure.

The software differences between syngo MR Acquisition Workplace (MRC) and syngo MR Workplace (MRSC) are determined by the **configuration** after a software installation.

syngo®

syngo is a software platform developed especially for medical systems and applications and provides the basis for modality specific control and applications as well as applications for a wide range of post-processing and other radiological workflow-related functions. *syngo* has a modular structure and can be divided into these major functional blocks :

- the *syngo* Platform
- the *syngo* Applications
- the *syngo* User Interface

syngo Platform

The *syngo* platform utilizes a Common Software Architecture (CSA) which provides fundamental functionalities such as basic image processing tools, a DICOM data model, an image database and networking capabilities.

A **Patient Browser** enables access to and allows one to navigate through the patient Database. Images in the database can be sent, exported and imported to and from remote network nodes or local mass storage media. The Patient Browser is **configurable** to adapt to individual workflow preferences.

syngo Applications

The common *syngo* applications are the same for all syngo implementations across all modalities and include the Viewing, Filming and 3D processing functionalities.

MR Specific *syngo* Applications

The application software developed for MR systems includes:

- Exam - image producing application
- 3D
- Argus
- BOLD
- Composing
- Mean Curve
- Perfusion MR
- Spectroscopy
- Nero 3D
- Tissue 4D
- Vessel View

syngo User Interface

syngo provides a user friendly graphical interface providing a common interface across all modalities, workstations, radiological and clinical information systems implementing the *syngo* software.

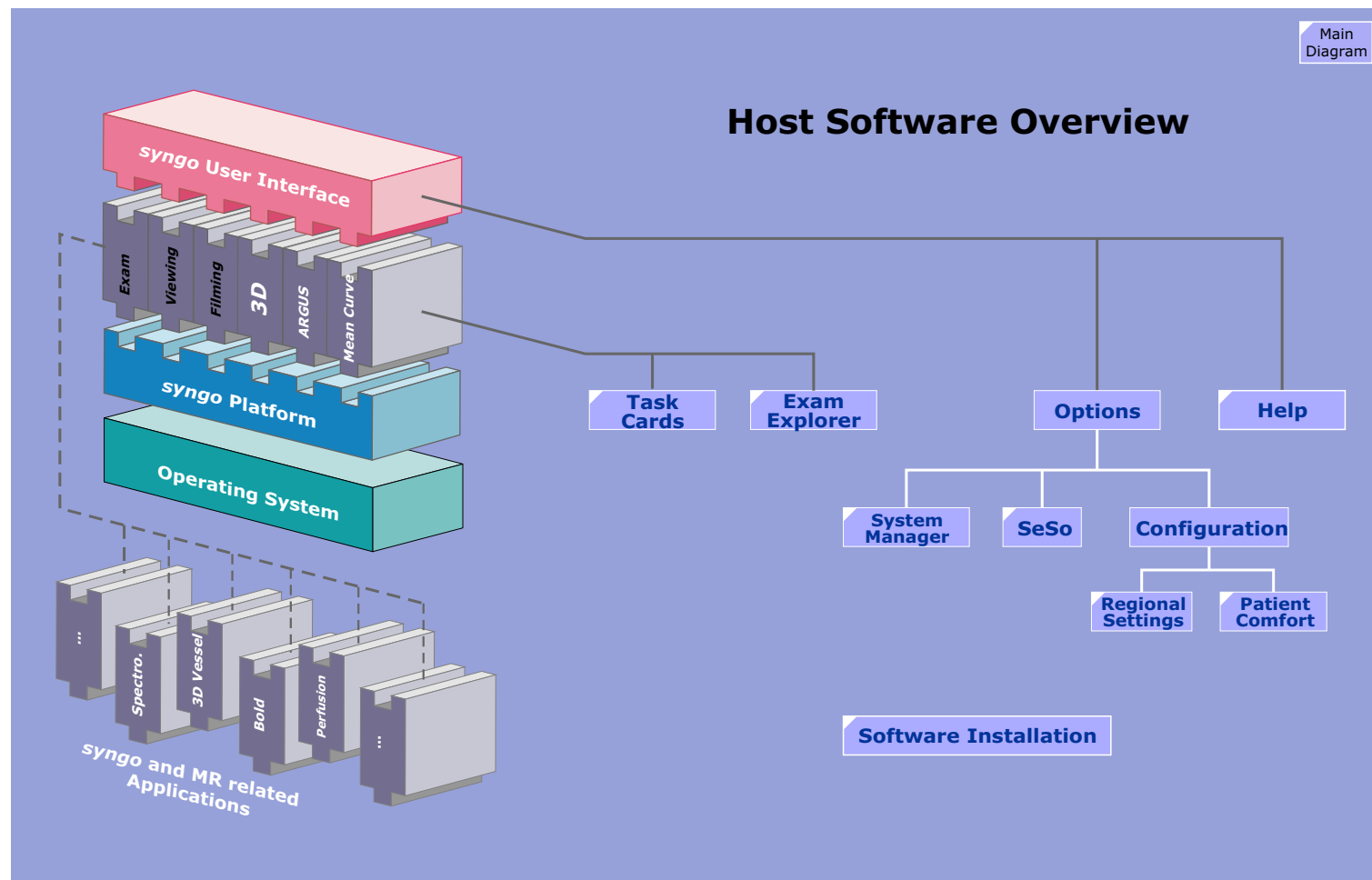
The *syngo* software together with the MR *specific application packages* is called *syngo* MR software

Much of the software just mentioned can be configured and adapted to the workflow requirements and personal preferences. This will be discussed throughout this section.

Scanner Software

The downloading of the scanner software is described in the [MARS and Control](#) section.

Figure 6 Host Software Overview



syngo MR

Task Cards

The *syngo* and *syngoMR* Applications are loaded onto Task Cards (see [Figure 7](#)) which are automatically started when the Host is booted or can be started manually.

The arrangement of the task cards reflects routine workflow in the hospital or practice and their layout supports the examination procedure.

Task Cards which are produced by the syngo software factory and available on all *syngo*-based modalities are :

- Viewing
- Filming
- 3D (option)

The **Filming** Task Card will only be loaded after a DICOM camera has been configured. If a DICOM camera is not physically available, it is possible to configure a dummy camera in order to get the Filming Task Card. This task card is an integral part of syngo and does not require additional licensing.

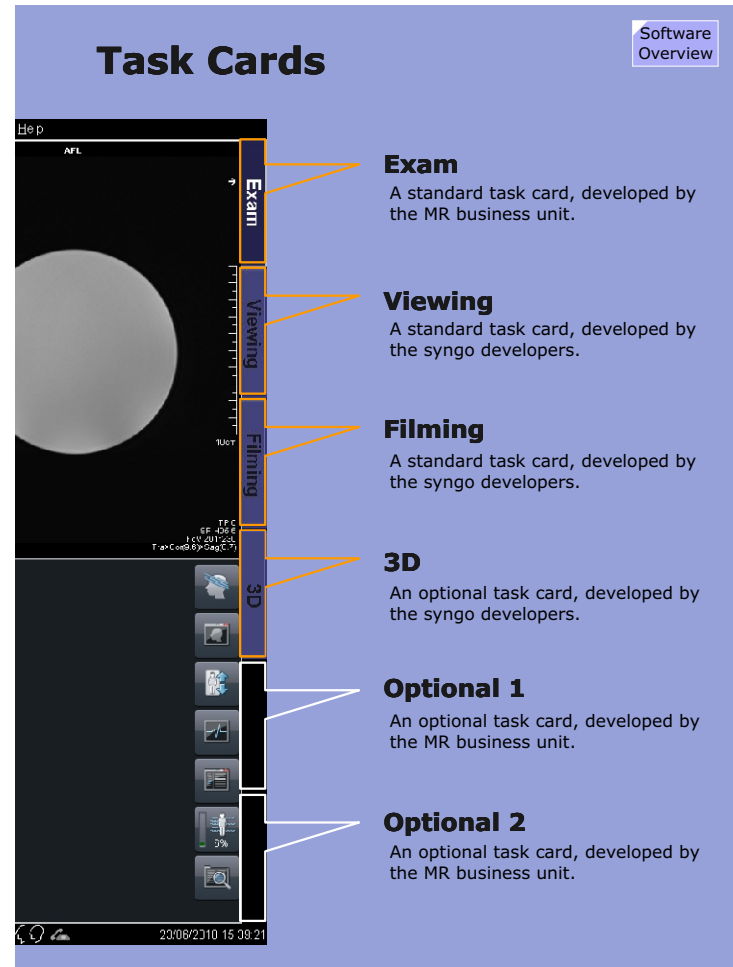
The **3D** Task Card is an optional package requiring additional licensing.

syngo MR specific Task Cards

- **Exam** - image producing application
 - **3D**
 - **Argus**
 - **BOLD**
 - **Composing**
 - **Mean Curve**
 - **Perfusion MR**
 - **Spectroscopy**
 - **Nero 3D**
 - **Tissue 4D**
 - **Vessel View**
- Task Cards in **orange** are optional

The **Exam** Task Card is used to define and run MR examinations and is only available on the MRC. All other MR specific task cards are optional and will require additional licenses to be enabled.

Figure 7 syngo MR Task Cards



Licensing

The 3D optional Task Card and optional software application packages require licenses before they can be used. Licenses must be ordered explicitly for the MRAWP (MRC) or MRWP (MRSC). A license issued for the MRAWP (MRC) will not run on the MRWP (MRSC) or vice versa.

Licenses can only be ordered from either the Licensing Center in Forchheim, Germany or from your Uptime Service Center or Regional Service Center in your country or time zone.

Task Card Menus

Figure 8 *The Menus (may deviate with software version)*

Patient Browser menu

Patient Applications Transfer Edit Queue Protocol View Image Tools Evaluation Scroll System Add-On Options Help

Viewing Task Card menu bar

Patient Applications Transfer Edit View Image Tools Evaluation Scroll System Options Help

Filming Task Card menu bar

Patient Applications Edit Film Image System Options Help

Although all the functions and commands found under these menus are described in minute detail in the Operators manual, there are several functions that are useful to know for Service engineers and will be discussed here :

- The Patient Browser
- The Transfer Menu
- The Exam Explorer
- The System Menu
- The Configuration Menu

Exam Explorer



The Exam Explorer is a program to view and manage the examination protocols using a graphical interface, a la Microsoft's Windows Explorer. With the Exam Explorer it is possible to copy or move protocols between **Regions**, **Exams** and **Programs** using the same drag and drop techniques known from the Microsoft Windows operating system. Right-clicking the protocol brings up a context menu providing commands to modify, rename, copy or delete protocols.

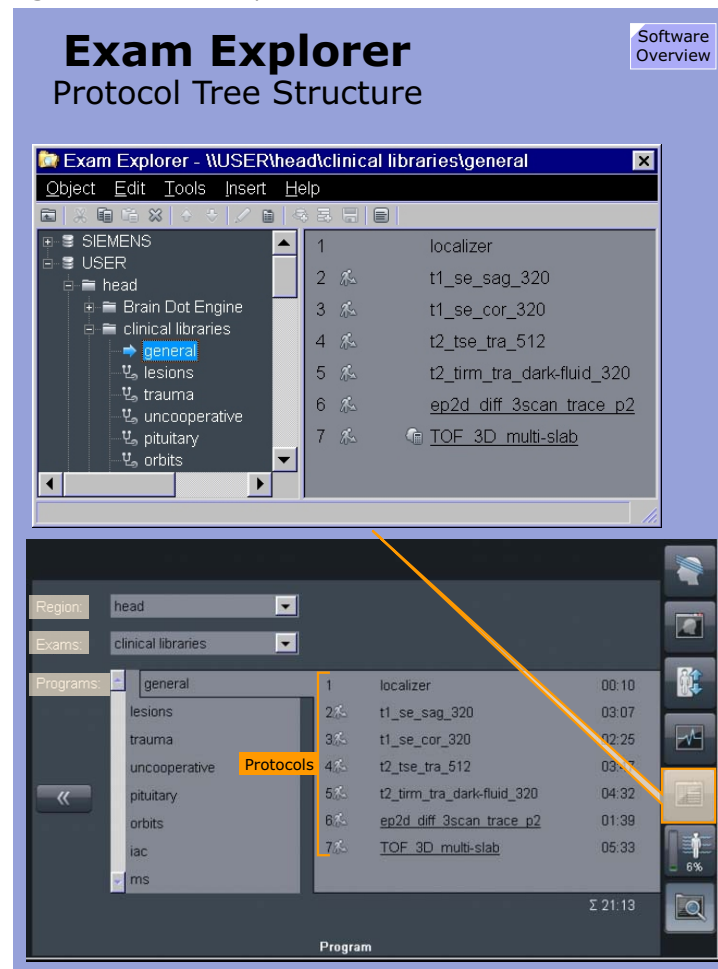
The Exam explorer can be opened from the Exam task card by either select **View > Exam Explorer** from the main menu or via the icon shown above.

The Protocol Trees

Protocols are found under two main groups: **SIEMENS** and **USER**. The protocols located under the SIEMENS tree are factory defaults and read only, they cannot be modified. The USER programs can be modified and are used by the customers for their examinations.

Protocols are organized within a strict hierarchy as shown in [Figure 9](#). The top level consists of **Regions**. Regions correspond to the anatomical region to be examined. Regions consist of **Exams**, which in turn contain **Programs** which are a grouping of **Protocols**.

Figure 9 Exam Explorer



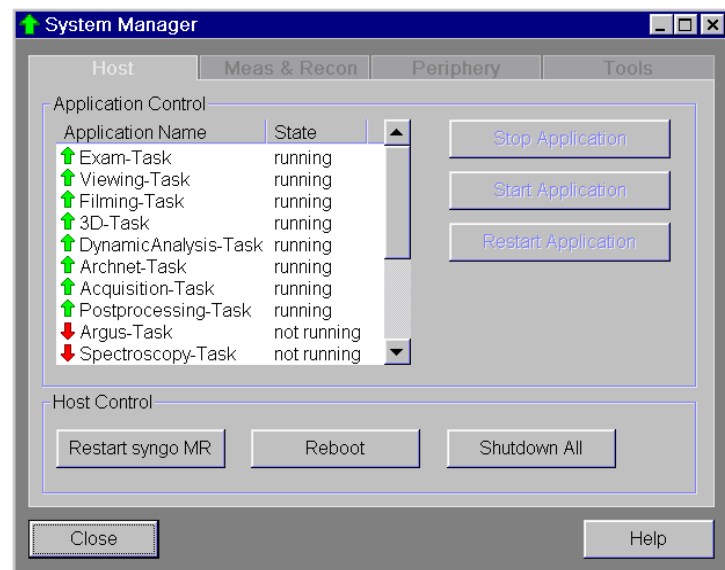
System Manager

The System Manager provides status information of the Host, Measurement and Reconstruction system and the Periphery units. It is invoked over the menu: **System > Control**.

Host Card

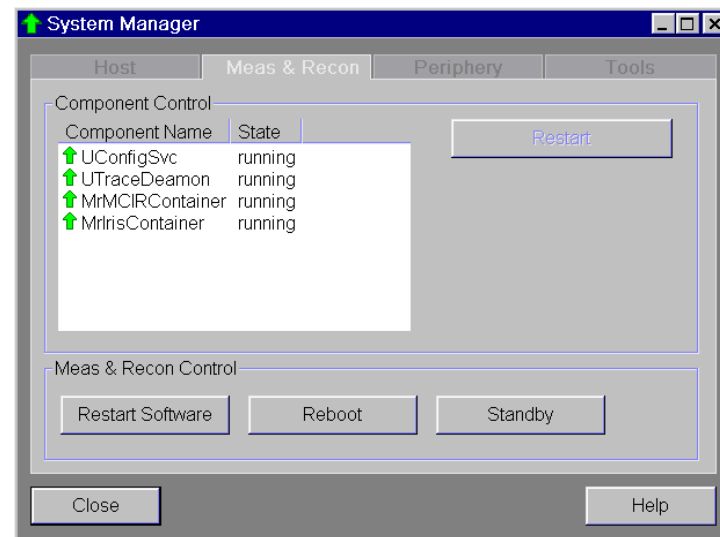
The System Manager / Host card displays the status of the loaded applications. The Exam-Task, Viewing-Task, Filming-Task and the 3D-Task are normally running (green arrow up). Optional tasks which are not loaded are marked with a red arrow down.

Applications which are hanging and not responding can be closed and restarted without having to shut down *syngo* MR or the operating system. It may work. But then again, it may not.



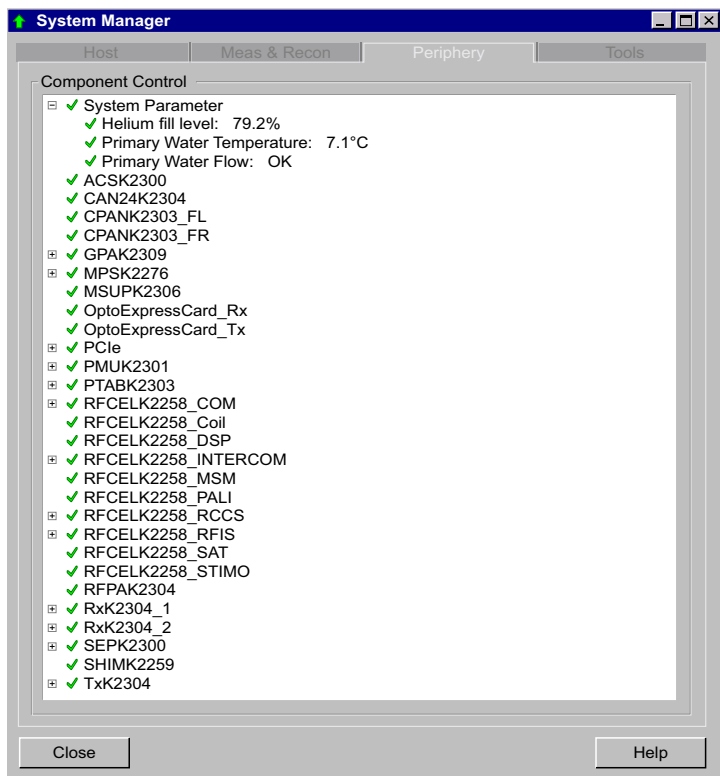
Measurement and Reconstruction System Card

The Measurement and Reconstruction System card displays the applications running on the MARS. From here you can reboot the MARS hardware or attempt to restart the single processes.



Periphery Card

The state information is displayed for all monitored hardware components (i.e., have CAN components or CPUs) within the MR scanner. These are listed under "Component Control". The current status of the components (OK, Not OK) is shown as green tick or red X. The green tick here indicates that all conditions required for an "OK" state have been met, i.e., no errors or faults occurred or is in an undefined state. The list is updated automatically.

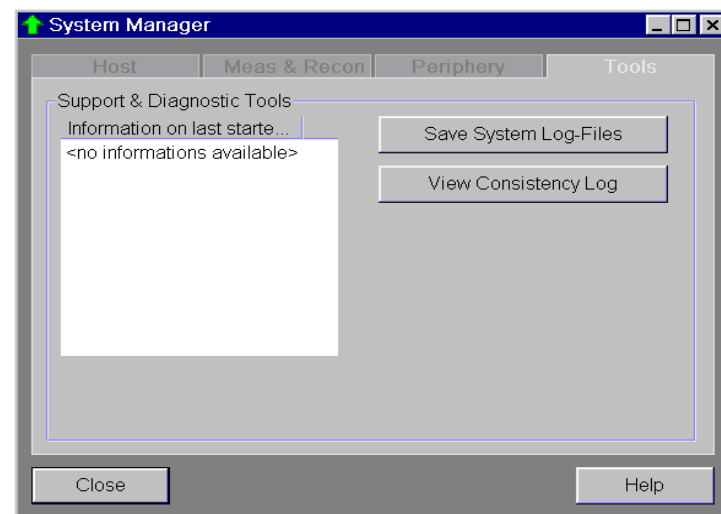


Tools Card

The Tools card is used to acquire diagnostic information or check important system files. If a software error occurs during scanning, it is recommended to create a MrSaveLog file immediately. It contains valuable diagnostic information valid at the time the error occurred. It is stored in the C:\MedCom\MriDiagnostic folder as a ZIP file.

The MR Consistency-Checker checks the system files for changes every time the system boots. The result is saved in a log file. You can view the content of this file in the Support & Diagnostic Tools window by clicking the View Consistency Log button.

CAUTION Never install additional, unreleased software on the system. Doing so could replace existing system files and corrupt the scanner software.



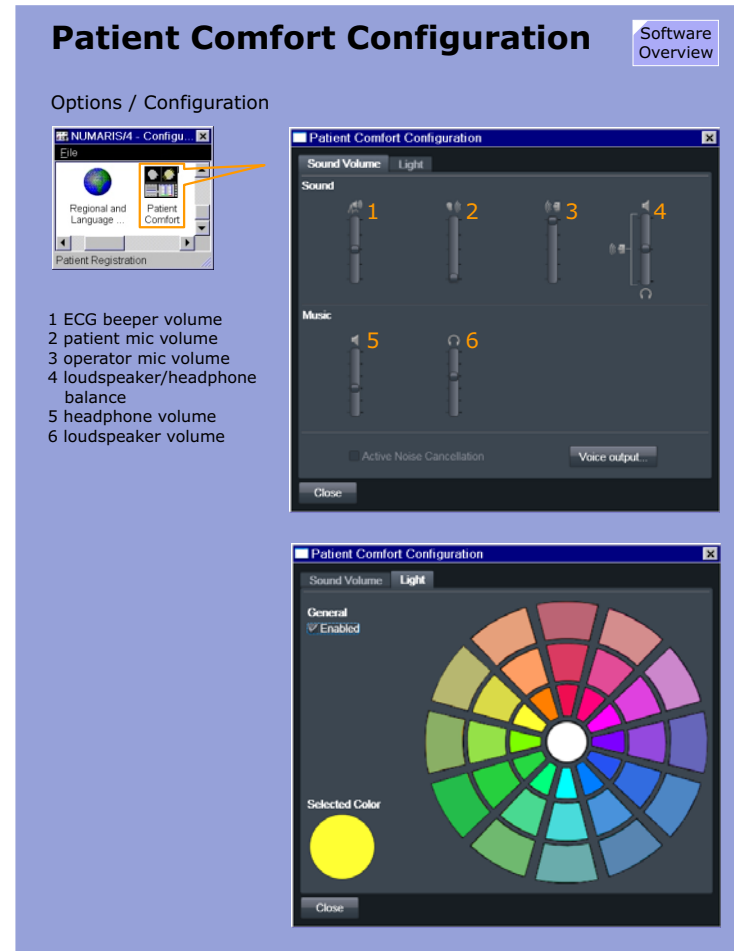
Patient Comfort

The user interface Sound Volume and Light can be configured under **Options > Configuration > Patient Comfort**.

The Sound Volume

Volume levels can be adjusted for

Figure 11 Patient Comfort Settings

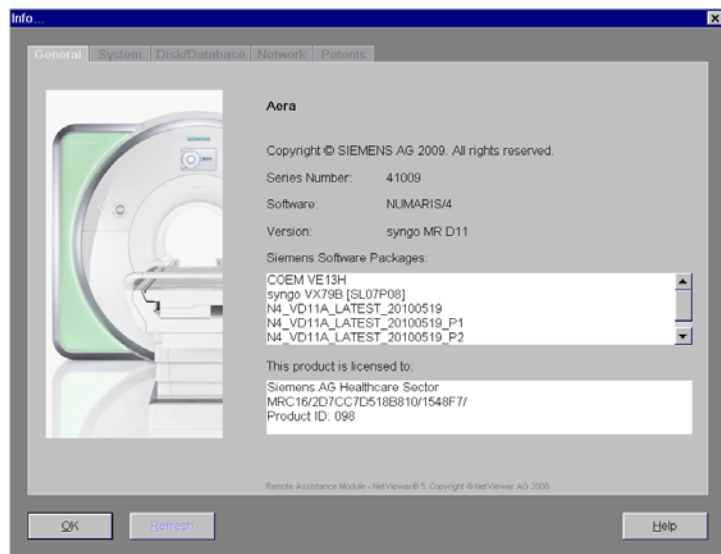


Help > Info

syngo MR also displays information about your scanner system. This includes information regarding memory capacity and availability. Select the menu **Help > Info** menu item.

General Card

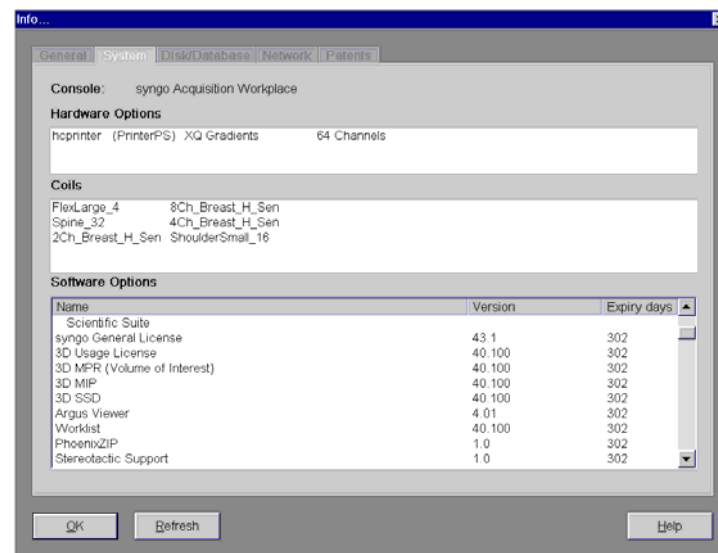
General system information is required, e.g. when a customer contacts Siemens Service regarding problems. Problems can be solved more quickly if we know the serial number of the system and the software version.



System Card

The System card provides further information about the scanner system. Here you will find information about hardware and software options installed as well as the available coils.

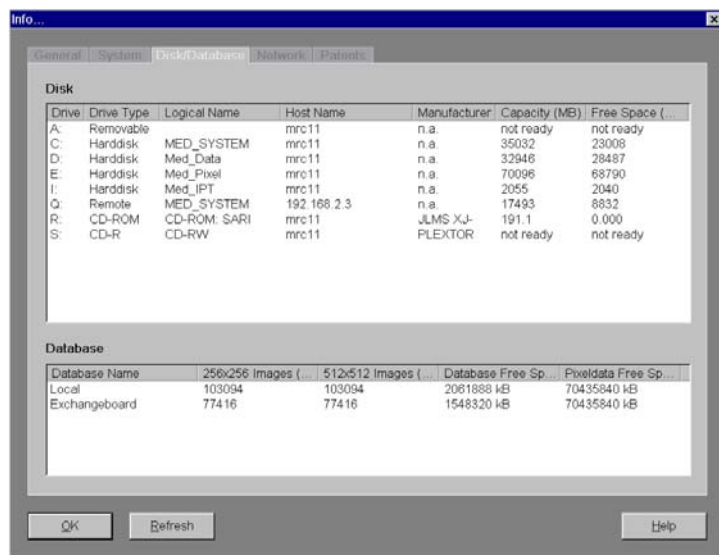
Also, the expiration date (if applicable) of software option licenses are shown.



Disk/Database Card

The **Disk** list displays all physical and logical (network) drives as well as their capacities and available space, expressed in MB and number of images.

All local databases are listed under Database. You are also shown how many images can still be stored in each matrix.

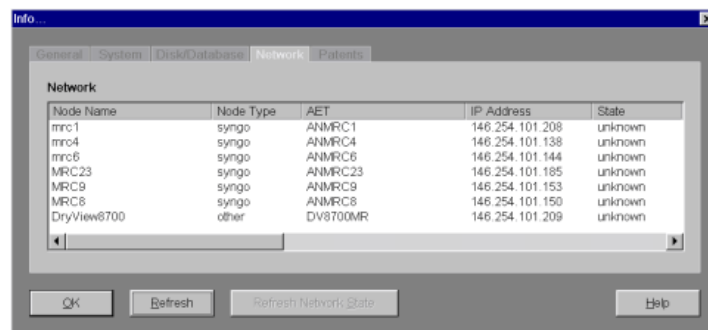


The following information is displayed:

- Name of the database
- Number of images 256 x 256 matrix images that can still be stored
- Number of images 512 x 512 matrix images that can still be stored
- Number of raw data sets that can still be stored

Network Card

If your console or satellite console is connected to a network, you may view network-specific information on the Network card such as the node names (computer names) and IP addresses.

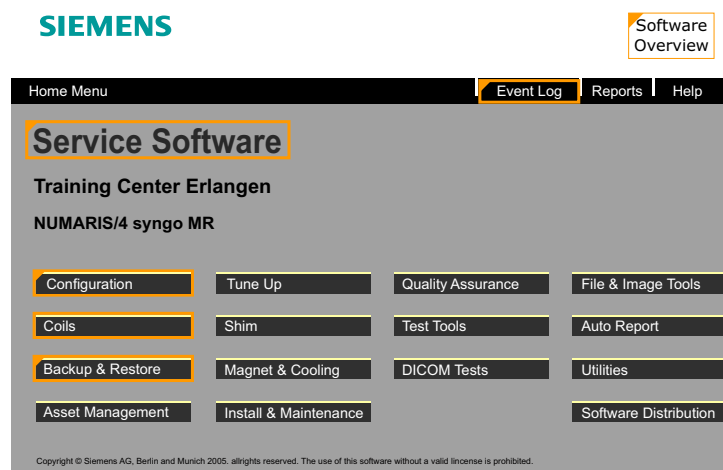


Service Software (SeSo)

Overview

After selecting Local Service in the menu Options Service and typing in the required password, the SeSo platform (shown below) will be displayed.

Figure 12 Service Software User Interface



SAM - Service Access Manager

SAM prevents or limits access to the various service functions via a password protection. [Figure 13](#) gives a list of the service levels and the available service functions for each level which has its own unique password. Services levels can be purchased from either the customer or even third arty service organizations according to the level of their expertise or competence. Service passwords are limited in time and must be renewed after a time period which is determined by Siemens.

Coils

This function was added in the VD13 software. The purpose of this new function is to enter information (serial number and part number) for the Body Coil and all Local Coils. As a result all SeSo masks involving the coils (TU, QA procedures) will be adapted to show only these coils. A capital idea!

The retrieval of the coil information for the Local Coils is automatic. Refer to the Help under the Coils mask for an explanation.

NOTE All TU and QA procedures that use the Body Coil will bring an error message if the Body Coil serial number has not been entered.

Local Access

SAM password protects the different service levels which allows the access of the available features to certain persons only.

Customer: The first level is accessible without password by the operator. Here only configuration parameters can be displayed.

In addition this level is used automatically if the operator selects the Event Log in the menu Options Service Eventlog.

SAM also distinguishes between local access and remote access.

There is the service password only for local access which allows to access the service software user interface without further limitations, since the CSE is at the system and has full control over the system anyway.

Remote Access

If wanting to log onto a system remotely there are three modes the local operator can select:

No Access - remote system access is denied and any pre-established connection will be immediately discontinued.

Limited Access - In this mode the remote connection is granted access privileges that allow only read / look type functions e.g. reading the Event Log, transfer images files etc.

Full Access - In this mode the operator relinquishes control and the remote connection is given full control over the system. The mouse and keyboard will be locked and the screen will be blocked. In this mode the remote session can perform service tests and system checks (QA). One limitation is no automated patient table movements will be allowed, for obvious safety reasons.

Figure 13 Service Levels

Service Level	Remote				Local
	No Access	Limited	Limited Permanent	Full	
Customer					QA (customer) Configuration (system ID)
In-house service	NO remote access possible	Functions the same as Limited Permanent, but limited in time.	Configuration (view) Backup (auto)	Configuration (edit) Restore (manual)	Backup & Restore (manual)
Basic			Service QA (view) Utilities: • Send Message • Process (list) • Component Mgr (list)	Service QA (scan) Tune Up Utilities: • Escape to OS • System (start/stop) • Process (start/stop) • Component Mgr (start/stop)	Tune Up
Advanced			Event Log Reports	Test Tools DICOM Test Tools	
Expert			File & Images Tools		
Siemens			Documentation Installation Protocol (view) Remote Application Support		Documentation Install Protocol (edit)

[illegible]

Event Log

In order to understand the different possibilities which can be selected in the Event Log, the following overview should be helpful. Note there are three sources of event messages:

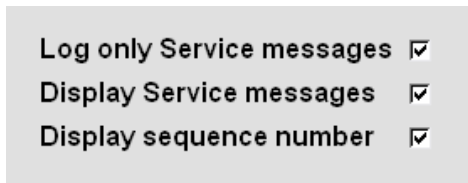
- the Windows OS
- Remote service (SAM)
- *syngo* MR

Regardless of the source, all messages are written into the same Event Log.

Pre-Filters

There is an additional pre-filter of the *syngo* MR messages that can be configured in the service software (see figure below). Enabling the "Log only Service messages" filter will block messages that are not service relevant (this should ALWAYS be enabled). The "Display service messages", when enabled, allows service messages to be entered (should also ALWAYS be enabled). The "display sequence number" will sort out the sequence of the error messages when more than one message is created at the same time. This is very useful since the first message is usually the one caused by the actual error, all following are often only the result of the first.

Figure 15 *syngo* MR Message Filters



These switches are set per default in the SeSo Configuration. If they are not set... well, we haven't noticed any difference.

Post Filters

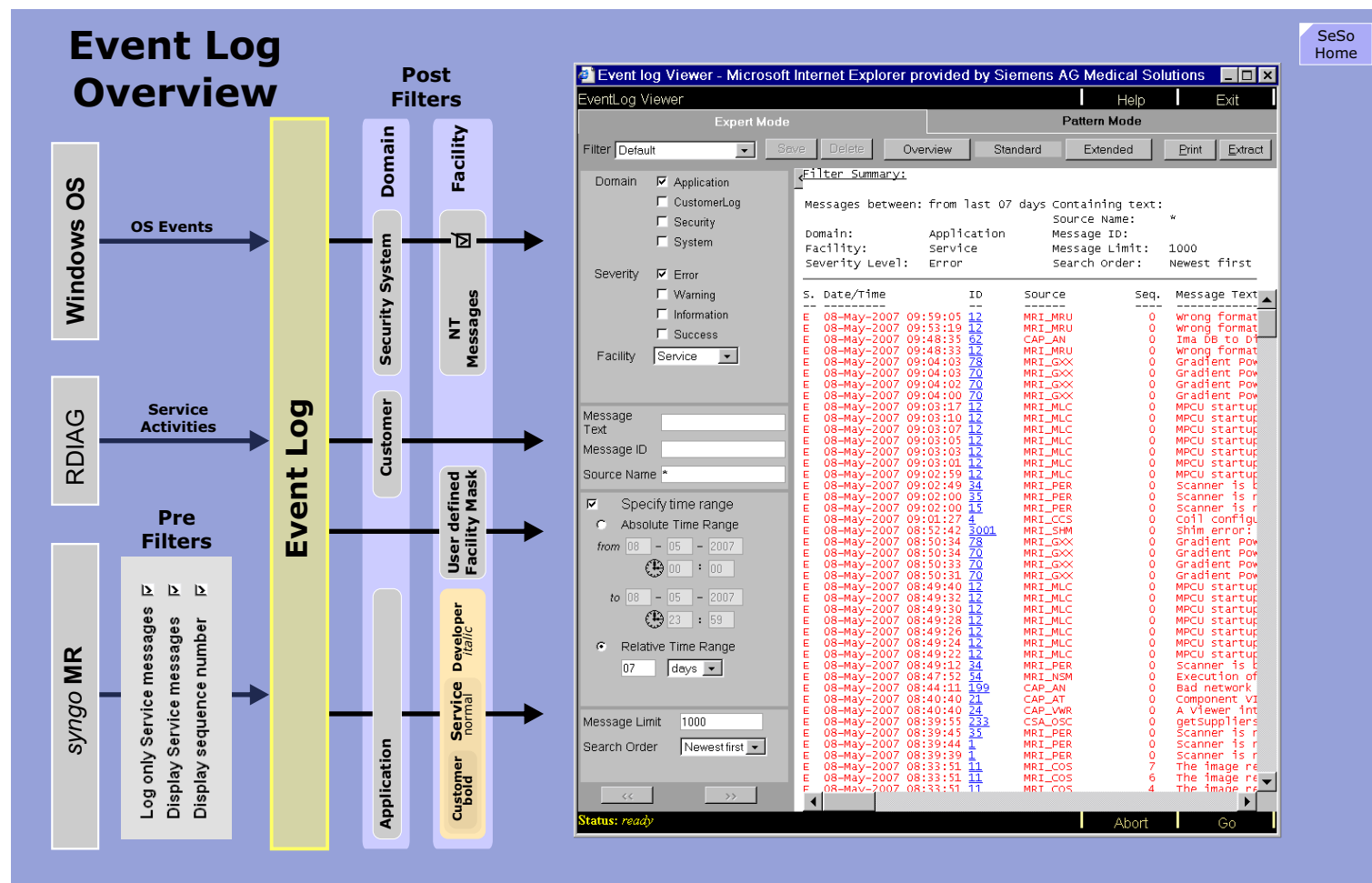
Domain

With the Domain filter only messages from the selected message source will be displayed.

Facility

For the display facility Service it depends on the service configuration parameter "Display Service Messages" whether service relevant Developer messages are displayed or not.

Figure 16 Overview of Event Log



SeSo Configuration

When entering the SeSo / Configuration page a List of system options is displayed with corresponding check boxes that can set if that option is to be configured (see [Figure 17](#)). The list of options is shown on the configuration mask on the left side and the list of the corresponding configuration pages on the right, with colored boxes showing the options to the corresponding configuration page(s).

Even if no options would be checked all the default configuration pages (pages in the list on without colored boxes in [Figure 17](#)) will still be available and must be configured.

The following configuration masks will be discussed on the following pages:

- Local Host - TCP/IP LAN (see [Figure 18](#))
- Measurement Settings (see [Figure 20](#))
- System Utilization (see [Figure 19](#))

Figure 17 SeSo Configuration Tool

Configuration

Main Configuration Page

System type

List of system options

Attached to Network	☒
Modem	☐
DICOM Print Devices	☒
DICOM Offline Devices	☒
DICOM Networking	☒
DICOM HIS/RIS	☒
Image Import/Export	☒
Paper Printer	☒
EPRI	☒
MR MBOX	☒

All configuration pages must be configured by default with the exception of the options listed on the initial configuration page, shown above. If the option is not checked, it will not appear in the configuration masks.

Local Host

- Site Info
- Console Type
- Console Options
- Country Code
- MAGNETOM LifeNet
- TCP/IP LAN
- Routing
- Users
- Monitor Type
- CAN24

Security

Settings

Service

Mail

FTP

Auto Transfers

Eventlog

Backup/Restore

DICOM

General

Character Set

Structured Reports

Offline Devices

Network nodes

Print Devices

HC Overview

LUT Files

HIS/RIS Nodes

Import/Export

Directories

EPRI

Server

External Devices

Paper Printers

PS LUT Files

System Mgmt

Master

Agent Controls

Applications

Pat. Registration

Viewer

CorRea

MPPS

Worklist Results

MR Exam DB

Measurement

System Type

Meas. Settings

Avail. Coils

System Utilization

MR Statistics

RIS/PACS

Chorus/PhoenixZip

Study Description

SeSo Home

Local Hosts - TCP/IP LAN

There are two networks implemented on the MR system:

- **internal:** connects the MRAWP (MRC) Host to the MARS.
- **External:** connects the MRAWP (MRC) Host to the MRWP (MRSC) Host and customer LAN

Internal Network

The network adapter used for the internal network has a default IP address range of 192.168.2.x. The IP addresses and net masks for the MARS is automatically created from the NIC2 address and loaded from the Host during the scanner boot:

- 192.168.2.1 - Host,
- 192.168.2.2 - MARS.

External Network

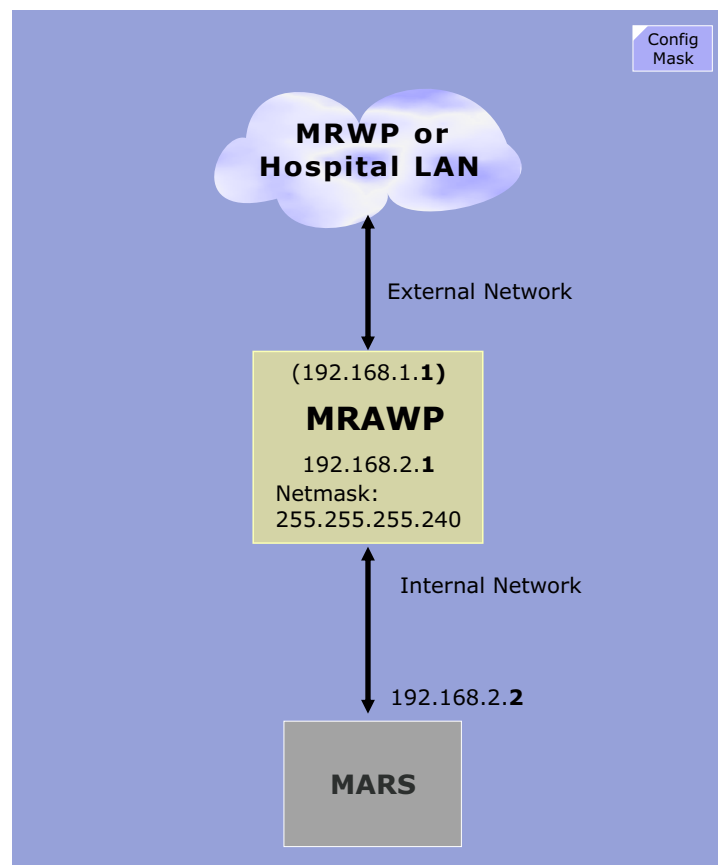
The network adapter used for the external LAN for connection the the customers LAN and to the MRWP (MRSC) if available. This LAN has NO default. Its IP address must be configured.

IP Conflicts

In some cases the customer's LAN will be using the same address as the default IP address for the internal LAN, in which case the internal LAN address must be changed to avoid conflicts.

NOTE Please refer to the Knowledge Base for a description on how to do this.

Figure 18 Host Network Topology



System Utilization

To be enabled with the purchase of this option. When activated the system will send a daily protocol of the number of patients scanned, which sequences were used and times for each along with the information selected in the mask below.

Figure 19 System Utilization

System Utilization Config Mask

Mandatory Segmentation Criteria

Region: Germany

Type of Institution: Imaging Center

Specialty: Teaching

Patient mix (In-patients vs Out-patients): more In-patients than Out-patients

System Type: High Field (1T or more)

System Utilization Activated ☒

Measurement Settings

A more appropriate title to this section would be "Hardware Configuration". The purpose of this mask is to select the installed hardware variants. Several components have variants depending usually on performance. The software needs to adapt itself to use the fullest abilities the hardware offers. And the customer paid for.

In the **first mask** the MAGNETOM model is selected. In the **second mask** (Figure 20) the variants are selected.

Figure 20 Measurement Settings

Measurement Settings for MAGNETOM Aera Config Mask

Note: You must have configured the system type before this dialog shows the correct measurement settings.

Number of Receiver Channels: 64

Gradient Power Amplifier: K2309_2250V_951A

Gradient Coil: GC 1.5XQ

Advanced SHIM Option: Installed

Stimulation Monitor: SAFE MODEL

SAR guide line: IEC

TX-Box: 1ch CP with LC

PMU: Installed

PMU Type: K2301

Patient Table Model: K2303

Patient Table Type: Mobile

Patient Table Range: 2019

Backside Scanner Control Unit: None

Cooling System: SEP 98SQ

Unit Type: Stationary

Comfort Kit: Installed

EPO: Installed

Inline Image Filter: enabled

Main Coil: Body

Backup & Restore

These functions allow scanner and site specific data, user protocols and service reports to be archived (backup) to CD media for restoring when the software needs to be reinstalled or if new software versions are installed.

Backup

There are several predefined backup packages that can be selected, each allowing the archiving of all or just a particular set of data. The only one used is the MasterBackup which includes everything that can be backed up.

After inserting a medical-grade CD into the S drive (upper most drive), choosing the drive and backup package the GO button starts the backup. The program displays the backup progress textually in the SeSo window. and when finished displays a READY message in the status line. The backup file copied to the CD has a naming convention <name of package>-<date>-<time>.ar

Example: Numaris4-10-02-2004-10-11-37.ar.

NOTE Use only medical grade backup media.

Restore

When necessary, a restore of a backup archive is made using the Restore command. The backup CD is inserted into drive R. When the CD is loaded a list of the backup file(s) can be selected in the Archives pull-down menu. Select the desired archive and press the GO button. As with the backup procedure the progress will be given in the SeSo window and a READY message on the status line when finished.

Backup Packages

There are a number of predefined packages in the backup which

allow archiving only certain items.

For information on what each package includes and which packages should be used refer to the service procedures and online help.

Master-Backup

A Master-Backup package includes everything. Use this.

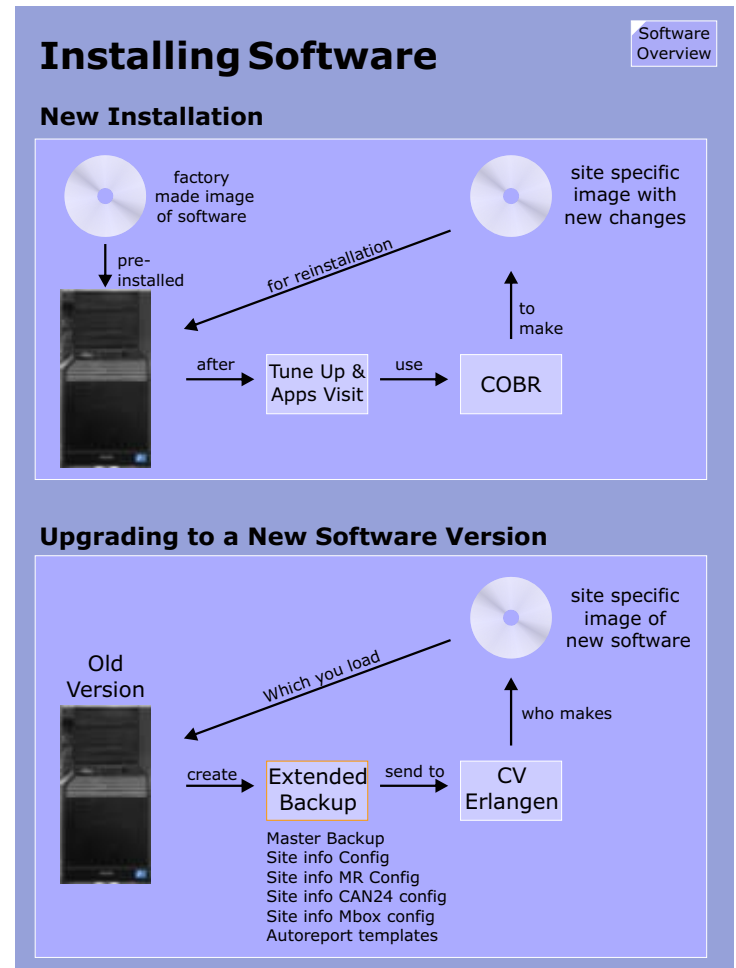
Software Installation

All systems (since early 2012) are only delivered with a COBR image DVD specific to the system.

The software installation is done only with the COBR backup and restore tools including a Ghost image.

NOTE Please refer to the software installation procedure in the Installation Software document for the corresponding software version.

Figure 21 VD11 Installation Scenario



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Introduction

The term Host refers to any computing element used to “host” the run-time software and applications of a system. The host for the MAGNETOM scanners is properly labelled **MRAWP** - MR Acquisition Workplace.

The Host computer is an off-the-shelf workstation platform. Details to the make and model are given in the designated section below. Due to the rapid developments in the PC industry the model, and eventually the manufacturer, will change with time.

The displays, however, are not off-the-shelf but are specially designed and manufactured for Healthcare for use on medical equipment.

MRAWP

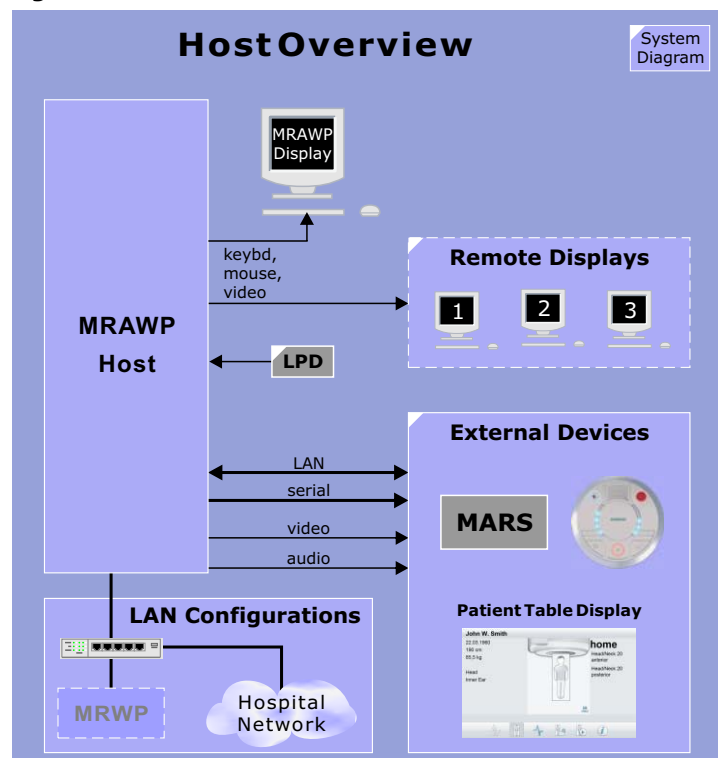
The major tasks of the MR Acquisition Workplace are:

- User Interface (*syngoMR*)
- image post-processing and image display functionalities
- archiving (mass storage, filming)

FRUs

See Spare parts Catalog!

Figure 22 MRAWP Host Overview



External Devices

The devices, that is, those instrumentality designed for a particular purpose, i.e., having their own special function independent of those functions being performed by the Host and thus found in their own housings externally to the Host and its affiliates, are referred to here as External Devices

The following connected entities falling under the category of External Devices:

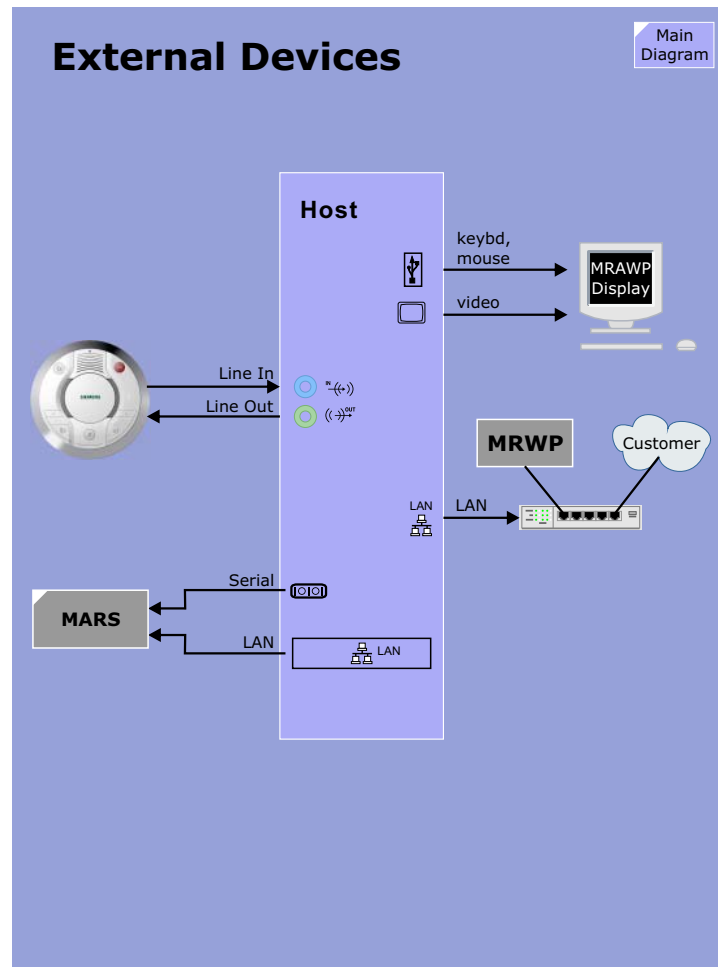
Device	Description
LAN (switch)	Internal - MARS 2 External - customer LAN
Dongle	The service software and all software and hardware options which require a license are keyed to a dongle with a unique ID. The Dongle is connected to any USB port at the rear side of the Host. The Dongle is tested by SeSo > TestTools > Host.
Intercom Unit	The headphone (line out) connection to the Intercom Unit provides music and voice command outputs to the patient. The microphone (line in) can be used to make recordings.
MARS	The main system controlling unit. Connected to the Host via a 1Gbit LAN and a serial port. The serial port is used only to log the boot process of the MARS. The resultant log file MrTTYRecorderBE.log , in ASCII text, is stored under C:\MedCom\log

Serial Connection to MARS 2

The MARS outputs the boot process steps over the serial interface to the Host. The Host only dumps this output into the **MrTTYRecorderBE.log** log file under C:\MedCom\log. It does not evaluate the log contents.

NOTE The serial cable MARS - MRAWP must be properly connected to both units. Otherwise, MARS boot problems will occur.

Figure 23 External Devices



Display Configurations

Figure 24 provides a current overview of additional display options.

Remote Monitors

In general, KVM-over-IP transmitters convert **K**eyboard, **V**ideo and **M**ouse signals into digital data which is then transportable over a LAN to a any remote location. The advantage is being able to use an existing LAN infrastructure to distribute KVM anywhere in the LAN boundaries without extra cabling.

For the remote monitor option only video is sent.

The converters have to be configured to adapt to the LAN IP range. If the KVM devices must traverse firewall boundaries, the firewall must be configured to open TCP ports the KVM devices may need.

Patient Data Display (PDD)

The PDD is driven directly by the Host. DVI-to-FO converters are used to route the video from the Host's graphic card to the PDD.

As option, a second PDD can be installed at the back side of the magnet. An additional DVI-splitter is supplied and must be connected to the Host's video 2 output of the graphic card.

In-Room Console

This option is no longer supported.

Figure 24 Display Configurations

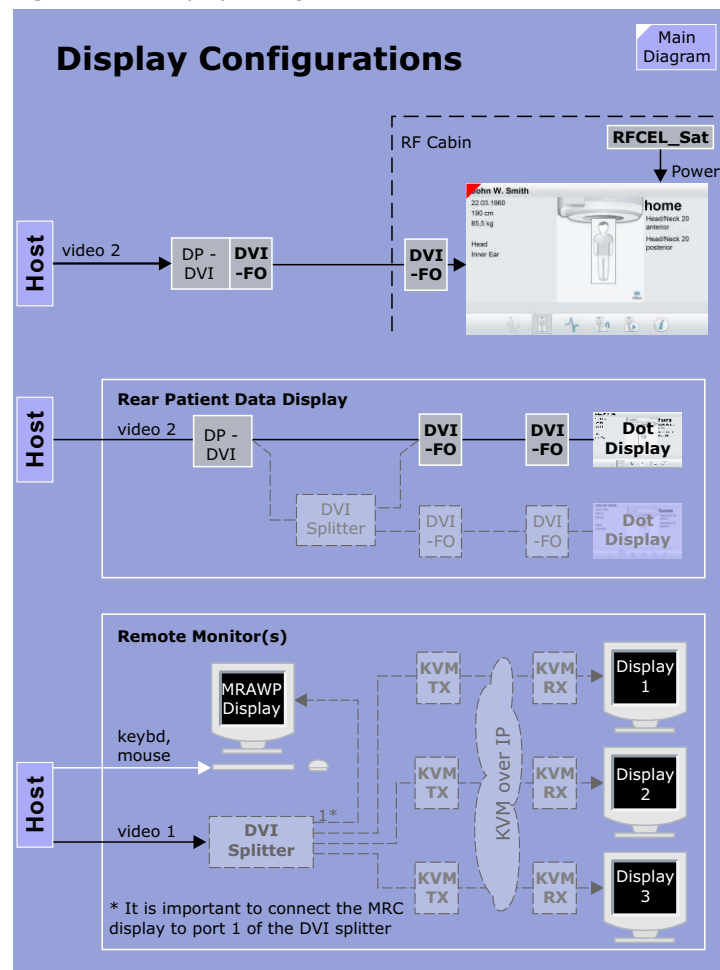
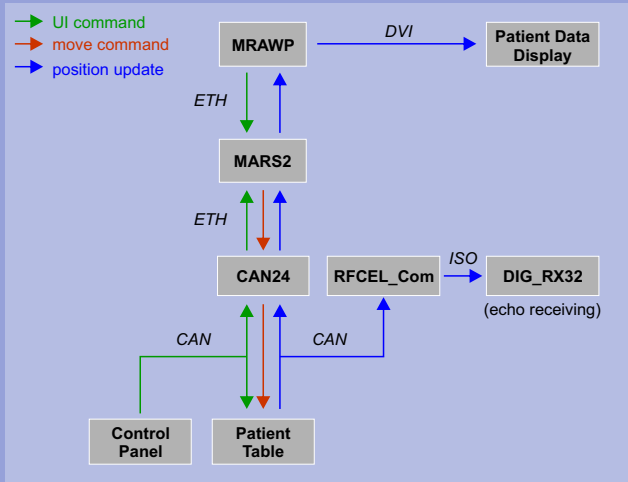
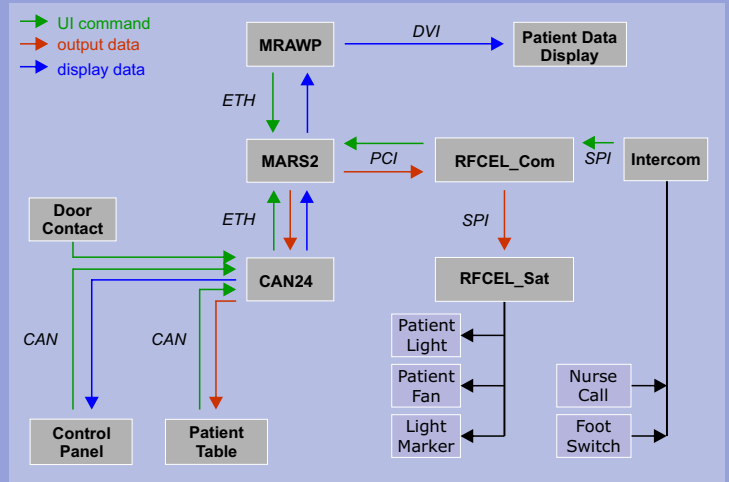


Figure 25 Information Flow Diagram

Table Movement and Position

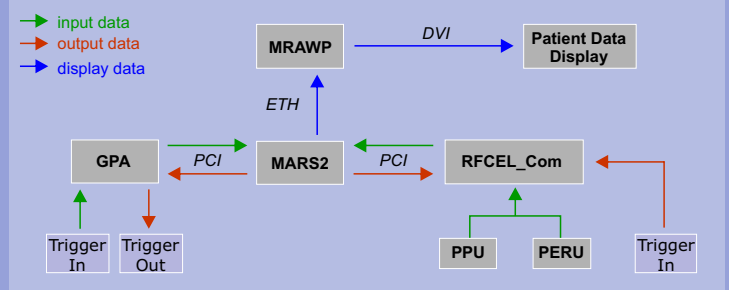


Cabin Interface



Information Flow Diagrams

Physiological Data Handling



Note: The diagram reflects the logic flow, NOT the physical layout.

MARS and Control

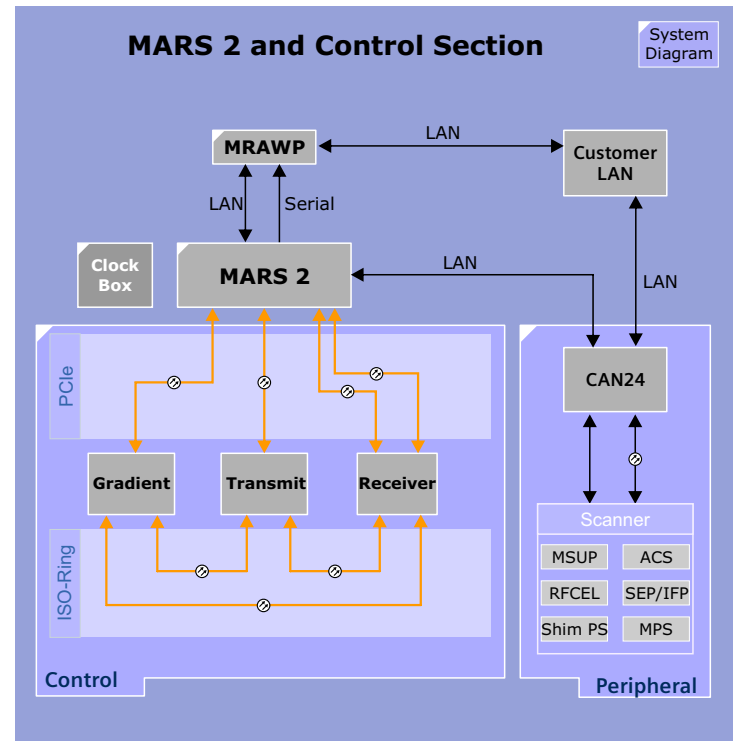
Introduction

The MARS 2 and Control section provides a description of the components responsible for or involved in the measurement control and communication functions. These functions were performed by components contained in a single electronics rack called the Measurement Control (MC) in previous systems. This is no longer the case in the daVinci architecture.

A comparison of components involved in the MC functions:

Function	OLD	NEW
Sequence control	MPCU in AMC (MMC)	MARS 2
Timing	PCI_TX, (MC4C40)	RFCEL_Com, ISO-Ring
Synchronization	Synthesizer (TX Module)	Clock Box
RF Pulse generation	PCI_TX, (MC4C40)	TX Box
SAR monitoring	PCI_Mon (MC4C40)	RFCEL_Com
Gradient pulse generation	PCI_TX (MC4C40)	GradCon in GPA
Stimulation monitoring	PCI_Mon, (MC1C40)	RFCEL_Com
Communications - periphery	PCI_CAN	CAN24
Communications - control	PCI_TX (MC4C40) - internal to the MC	all field generating and receiver units over PCIe and ISO-Ring networks
image reconstruction	MRIR	MARS 2

Figure 26 MARS and Control Overview Diagram



Measurement Control (MC)

Measurement control refers to those functions and tasks responsible for decoding the protocols (=sequences + parameters) sent down from the Host and generating the RF and gradient pulses as well as all static or dynamic timing and control signals for the RF, gradient and MR signal acquisition events as described by the sequence.

Prior Systems

The measurement control functions of earlier platforms were realized by components housed in a separate electronics rack unit called Measurement Control (MMC, AMC). DSPs and FPGAs as well as ASIC components on older systems were the basis of their designs.

daVinci System

The MC architecture of the daVinci platform has de-centralized the MC components and uses almost exclusively FPGAs as controlling elements. The sequence decoding is performed by the **MARS 2**. The MARS 2, however, is not only capable of running a single protocol but of managing a complete scan program.

The timing, signal generation and monitoring, however, are performed now directly in the field generating (RF and Gradient) sections described under Communications.

Synchronization

Exact timing and phase coherence of all applied field generating pulses and the acquisition of the MR signals is paramount to good image quality. The sequence control and signal generating components now being de-centralized a synchronization clock generator has become necessary. This is the responsibility of the **Clock Box**. The Clock Box employs an extremely stable crystal and clock generation circuitry for supplying frequency/phase stable clocks and sine waves at various frequencies to the scanner components requiring them.

Communications

The communications system in previous products used the CAN bus network to send commands to and receive error and status messages from the scanner peripheral components. The daVinci platform adds a second communications network called the **Control Communications** network consisting of a PCI Express (PCIe) and an ISO-Ring network.

Control Communications

- **PCI Express**

This bus system was developed to satisfy the high data throughput demands of today's graphics engines. This media is used by daVinci to transfer pulse and timing data from the MARS 2 to the field generating units and the huge volume of raw data from receiver components to the MARS 2.

- **ISO-Ring**

A proprietary communications media realized as a isochronous ring network. Isochronous meaning "equal in time", i.e., having the same length or occurring at the same time. This communications network is used for timing and synchronization data and real-time compensation data.

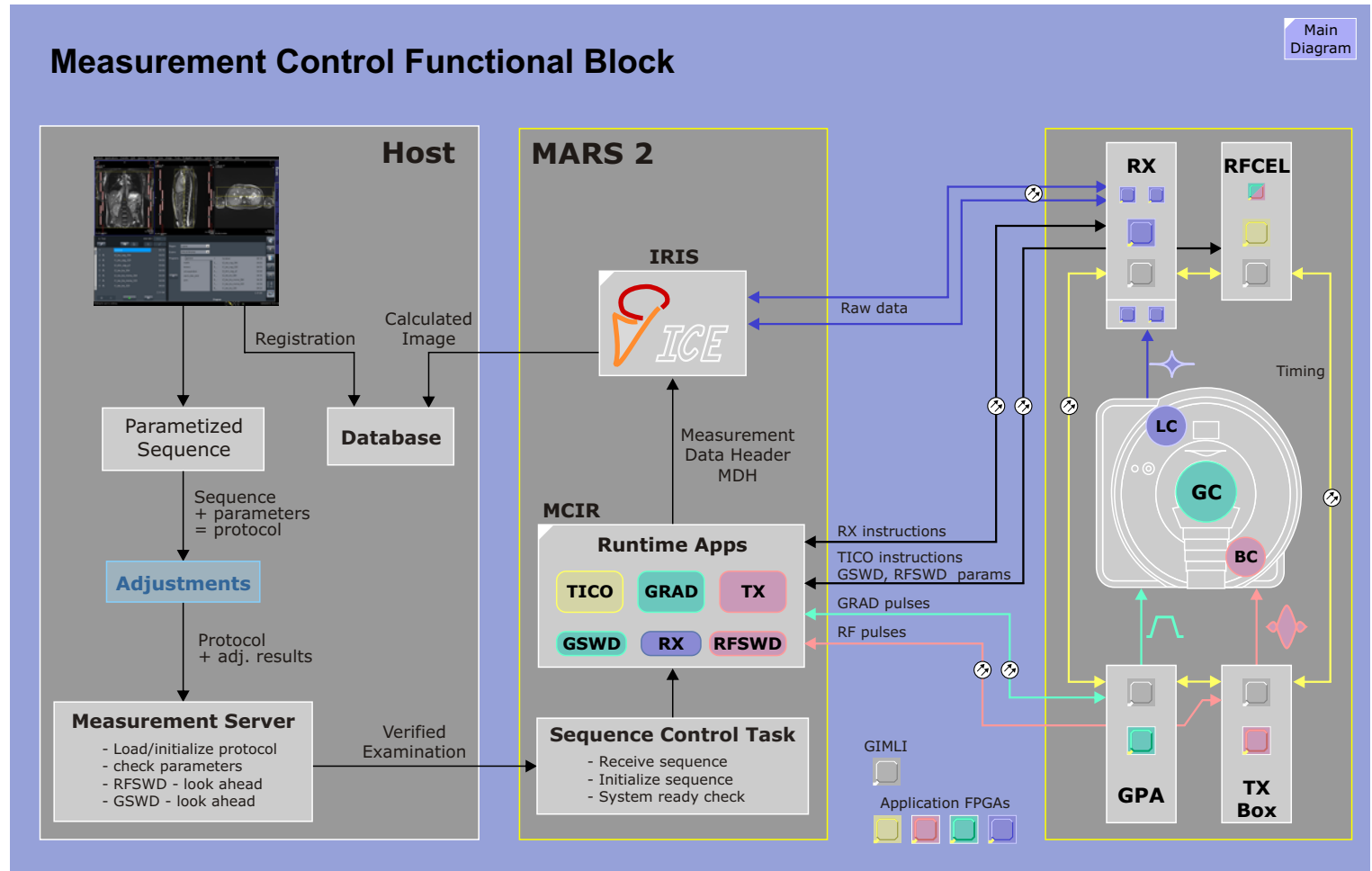
Peripheral Communications

This network is based on the same CAN protocol used in previous systems and for the same reasons. The controller is the **CAN24**, a self-contained CAN bridge able to control up to 3 CAN networks, only two of which are used in the current systems.

Image Reconstruction

The task of image reconstruction is also performed by the MARS 2. The acquired MR signals are processed directly by the receive components in the RFCEL and delivered to the MARS 2 over the PCIe network as raw data.

Figure 27 MARS 2 and Control Functional Overview



MARS 2

The MARS 2- **M**easurement (control) **A**nd (image) **R**econstruction **S**ystem (version) **2** - is as an industrial PC specifically tailored for scanner and sequence control and image reconstruction.

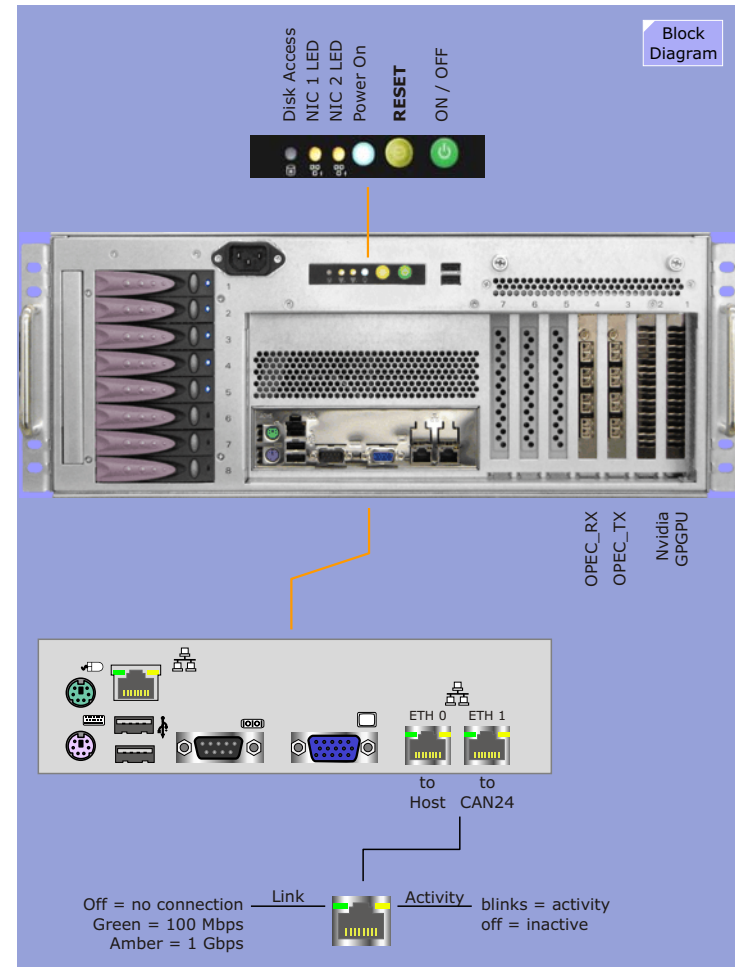
Hardware

The MARS is available in these hardware configurations:

	24 Channel	48 Channel	64 Channel	128 Channel
Weight	~ 30 kg			
CPU & Memory				
CPU	2*Xeon 64bit QuadCore 2.0GHz	2*Xeon 64bit QuadCore 2.4GHz	2*Xeon 64bit QuadCore 3.2GHz	2*Xeon 64bit QuadCore 3.46GHz
Memory	24 GB DDR3	48 GB DDR3	64 GB DDR3	128 GB DDR3
2.5" Hard Drives				
Raw Data	540 GB (2 drives)	187 GB (3 drives)	250 GB (4 drives)	500 GB (7 drives)
Front-side Connections				
ETH 0	1Gbit connection to Host			
ETH 1	1Gbit connection to CAN24			
RS232	boot messages to Host			
PCIe Slots				
1,2	-	-	Nvidia Tesla C1060	1x Nvidia Tesla C2075 (VD13)
3	TX OPEC			
4	- *	RX OPEC		
5,6	-	-	-	1x Nvidia Tesla C2075 (VD13)
7	-	-	-	RX OPEC

* The 24-channel configuration requires only 1 OPEC card.

Figure 28 MARS 2 Layout



Function

The MARS performs measurement control, image reconstruction and scanner reboot (software download) functions.

Measurement Control

Is able to handle a complete examination, not just single protocols
All system dependent data (configuration and TuneUp) is located here. Common architecture for interfaces: explicit, system independent. Provides a layer structure that allows flexible simulation.

Image Reconstruction

The image reconstruction task is to compute the final images from the digitized raw data supplied by the RFCEL. Up to seven hard drives are provided for temporary storage of the raw data. The drives are configured as a RAID 0 for speed.

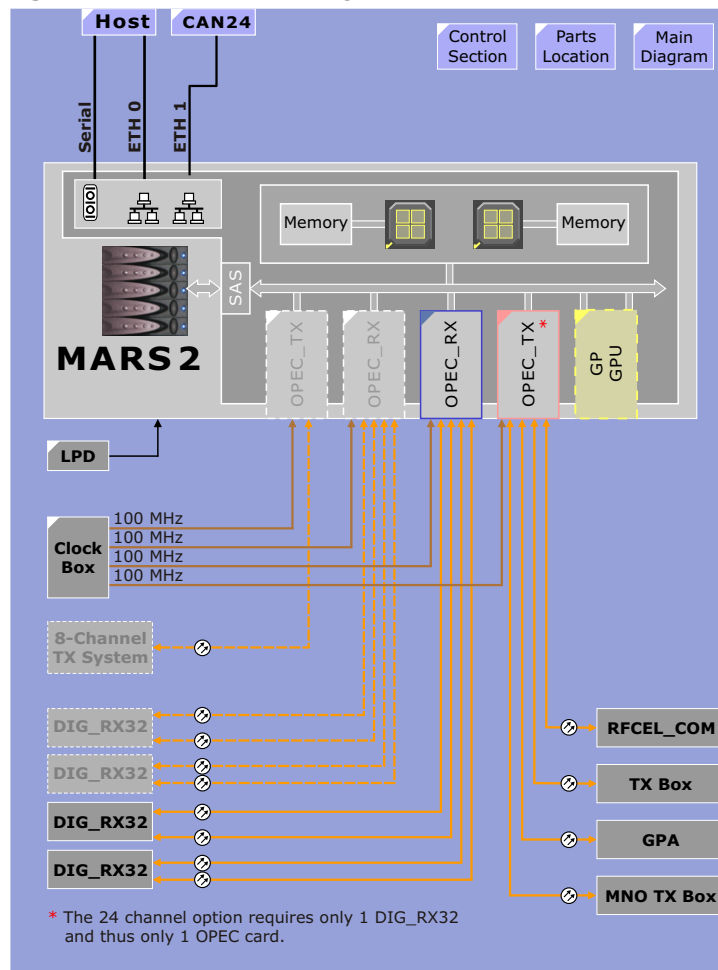
Optional post-processing application software also allows automated post-processing functions to be performed on either raw data or reconstructed images, as the case may be. This is known as Inline Processing. Some examples of Inline Processing features: PACE, Diffusion, 3D fMRI, Perfusion, StdDev, Subtraction, image compositing (combining several images into a single one).

RS232

The MARS outputs its boot progress to the Host over the serial interface. This data is logged and saved to the **MrTTYRecorderBE.log** file under C:\MedCom\log and may be helpful in identifying MARS boot problems.

NOTE The serial cable to the host must be connected when (re)booting, otherwise the MARS will not boot properly.

Figure 29 MARS 2 Block Diagram



OPEC

The Optical PCI Express Switch Card (OPEC) provides electric/fiber optic converters for up to 4 lanes. The switch buffers the internal PCIe bus from the end-points and provides flexibility in configuring the 4 fiber optic transceivers as 4 single lane ports, as 2 dual lane ports or 1 single four lane port.

Configuration Switch S2

A configuration switch, S2, is used to configure the OPEC card in either a TX or RX mode.

- The **TX mode** is used for connecting the GPA, TX Box and RFCEL_Com and for the 24 channel option the DIG_RX32.
- The **RX mode** is used for connecting the DIG_RX32 boards (in the RFCEL). For the 24 channel option this OPEC card is not installed.

100MHz Clock input

The PCIe switch also separates the PCI-Express clock domain inside the PC (fixed frequency or spread spectrum clocking) from the PCI Express clock domain used in the system (fixed frequency).

FO Transceiver Fault Monitoring

The FO transceivers implement a fault isolation feature performing real-time monitoring of:

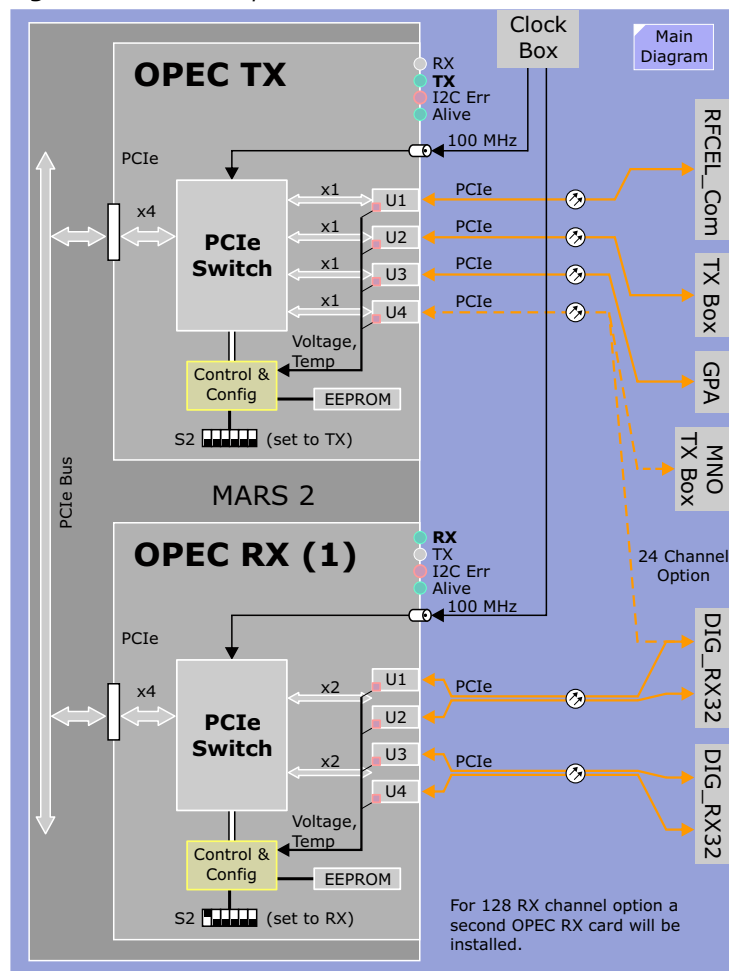
- transmitted and received optical power (light intensity)
- laser bias current
- temperature
- supply voltage

This information is read out from the transceivers via a serial interface.

LEDs

There are 16 LEDs on the board, but not visible from the outside.

Figure 30 OPEC - Optical PCIe Card



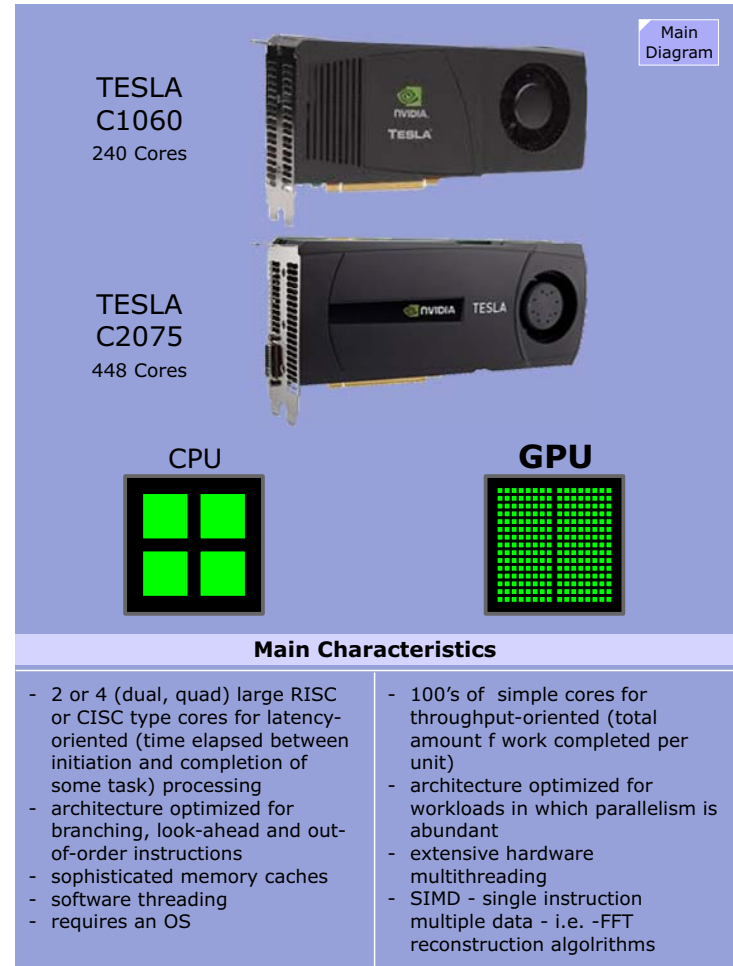
GPGPU

A General Purpose Graphics Processing Unit (GPGPU) is a device which uses a GPU architecture to perform computational operations for applications that require performing recursive (parallel) operations such as the FFT calculations for the image reconstruction and inline processing. In short: its Red Bull for computers.

The model for GPU computing is to use a CPU and GPU together in a heterogeneous computing model. The sequential part of the application runs on the CPU and the computationally-intensive part runs on the GPU. From the user's perspective, the application just runs faster because it is using the high-performance of the GPU to boost performance.

There are currently two versions of GPGPU from Nvidia dependant on software version and RX channel options. See the list on page 44.

Figure 31 NVIDIA Tesla GPGPU



Control Communications

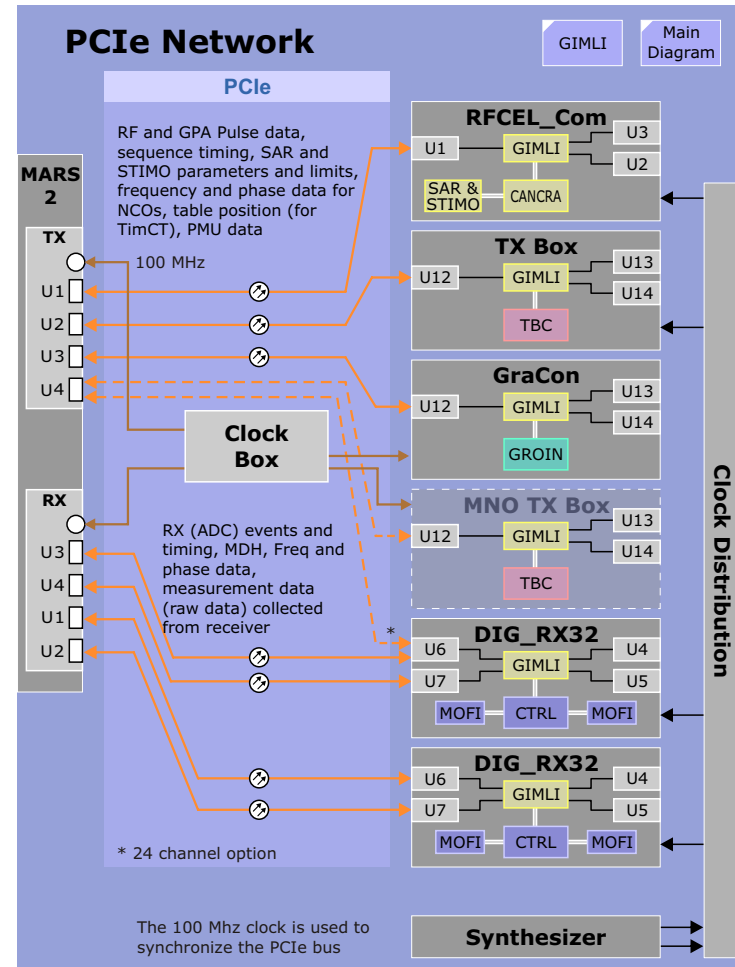
PCI Express

The **PCI Express (PCIe)** is known from the consumer PC industry as the data bus standard for PC peripheral cards. It was created primarily out of the need for greater data transport volume and speed between CPU and graphic accelerators (graphic cards). The standard has since evolved into a motherboard-level interconnect for uses in industrial applications. The version used here is a PCIe link for external peripherals as a point-to-point serial link based on fiber optic. The table below gives an overview of the data type transferred between the hardware participants in the control network.

Data transferred over PCIe bus	from	to
RF pulse data and sequence timing	MARS 2	TX, MNO
RF freq + phase data for TX and RX NCOs	MARS 2	TX, RX, MNO
Grad pulse data and sequence timing	MARS 2	GPA
RX event data and sequence timing	MARS 2	RX
static measurement data - configuration data for excitation path (TX, GC Site Data) or online patient safety (PALI, Stimo) parameters	MARS 2	TX, GPA, MNO
Raw data acquired during scan	RX	MARS 2
Patient table position	RFCEL	MARS 2 RX
PMU data (pulse, resp., ECG)	RFCEL/ MARS 2	MARS 2/ RFCEL

The **100 MHz** clock is used to synchronize this bus. This clock is generated on the **Clock Box** and distributed to the components in the technical room by the Clock Box and for the components in the exam room by the Synthesizer.

Figure 32 PCIe Network



ISO-Ring

The Iso-Ring is used for communicating broadcast data (from one source to many targets) such as B0 and other parameters affecting all units and real-time data that must be past onto ring members such as patient table position, PMU stamps, SAR and STIMO data and frequency-phase changes. It is also used to communicate severe problems in a component. This aborts the current sequence to guarantee patient safety (Heartbeat token).

Plausibility Check _ Safety Token (aka heartbeat)

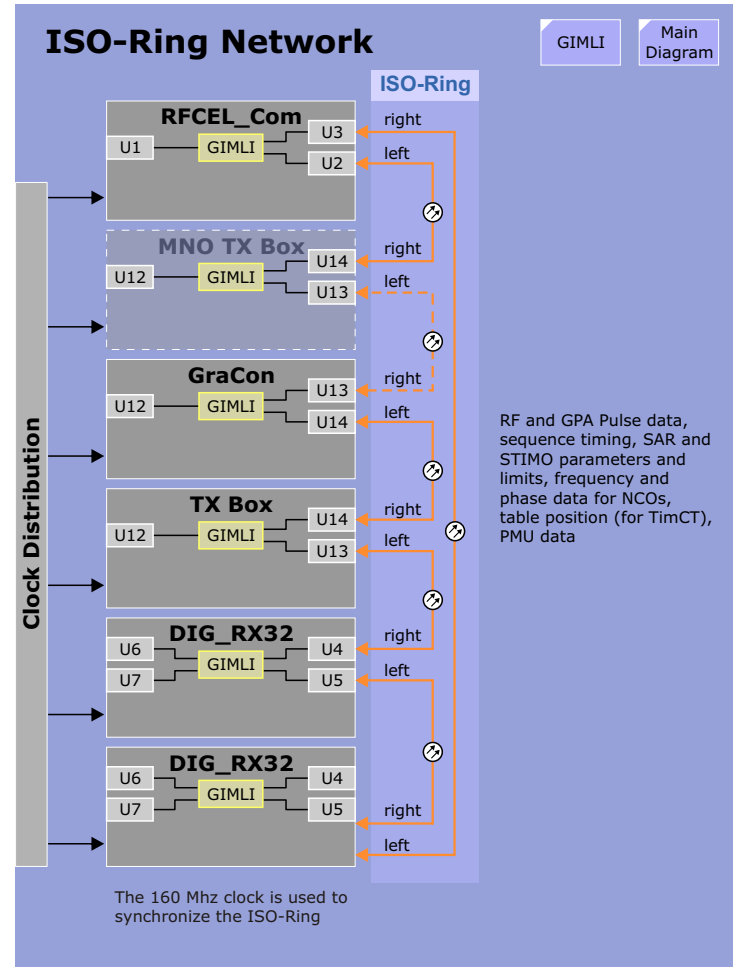
The communication network contains a method to signalize error conditions that require immediate entry of a fail-safe state. The method is to transmit a safety token across the ring every 10 us. Each component connected decodes the token and if there is no local error, it forwards this token to the next communication partner. If the token is dropped or ring communication ceases a communication error is flagged and the local component needs to enter the fail-safe state as soon as possible.

Any emergency stop condition is reported and forwarded using this Safety Token.

Data transferred over ISO-Ring bus	from	to
master sync clock	RFCEL	ALL
B0-ECC compensation	GPA	TX, RX
Gradient information created by the GPA SSU to be used e.g. by the PMU or patient intercom in aiding their respective filter algorithms	GPA	PMU, Intercom
Failsafe data to abort the measurement safely in case of error	ALL	ALL
Sequence Event Counter all 10 µs, all members of the ISO-Ring receive the current measurement time for synchronization	RFCEL	ALL
Patient table position	PTAB	RX, TX, MARS 2
PMU timestamps	RFCEL	RX

The **160 MHz** clock is used to synchronize this bus.

Figure 33 ISO-Ring Network



GIMLI

GIMLI FPGA provides a flexible Interface to an application specific FPGA that provides the components specific functionality. Both Upstream and Downstream DMA Channels are forwarded to this FPGA.

The GIMLI FPGA is loaded from a local flash memory that persistently stores the FPGA image. The FPGA image stored in the flash memory can be updated in system via the PCIe bus. It takes less than 3min to download a GIMLI FPGA image to the flash memory.

This local memory also provides at least one image for the application specific FPGA(s). It takes less than 3min to download an application specific FPGA image to the flash memory.

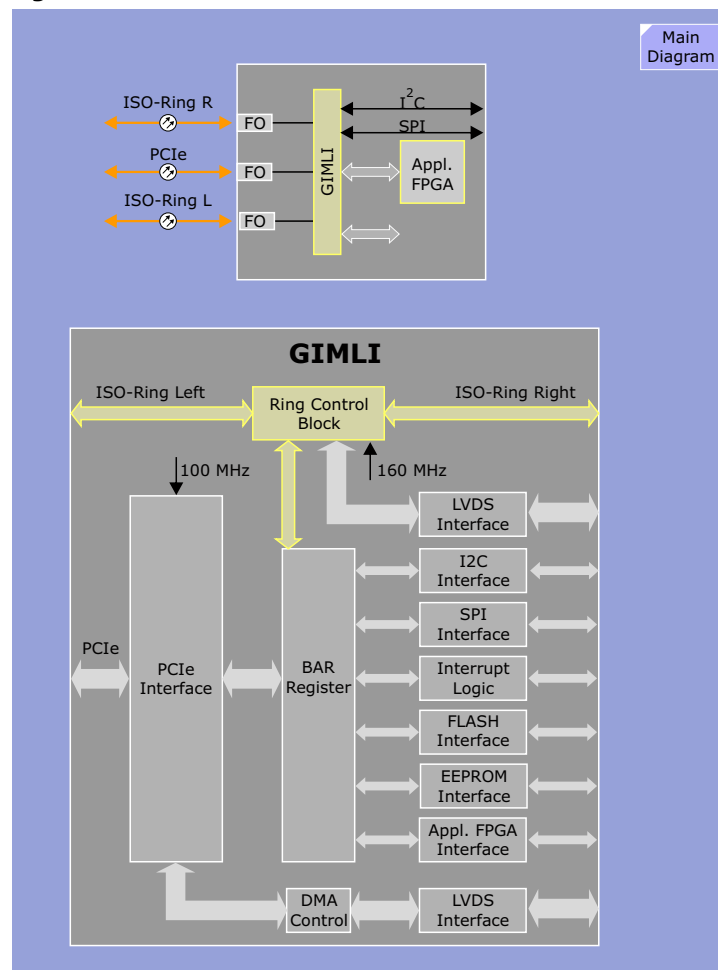
GIMLI FPGA provides programming support for the application specific FPGA. This can be controlled through the PCI-Express Target Interface.

GIMLI FPGA provides monitoring capabilities to all SFP Fibre Optics transceiver, as well as monitoring the FPGA temperatures. This is accessible through the PCI-Express Target Interface.

GIMLI (Generic Infrastructure Module for Link Interfaces) is an FPGA responsible for providing the communication links between the MARS 2 and the field generating units. The GIMLI and receiving components providing communication to:

- **PCI-Express:** 1 or 2 fiber optic lanes comprised of
 - a PCI-Express Target Interface.
 - two upstream and two downstream DMA Channels.
 - a messaged interrupt signal generator
- **Iso-Ring:** 2 fiber optic lanes in left and right directions

Figure 34 GIMLI Overview



Scanner Software

Loadware

All scanner components utilize one or more digital control components (i.e., CPU a DSP or FPGA) all of which require some form of software, loadware or configuration-ware files.

The task of retrieving and distributing these files belongs to the MARS 2. [Figure 36](#), page 53 gives a general overview of the BOOT process.

Please note that all scanner files reside on the Host. These can be updated with any new Software version.

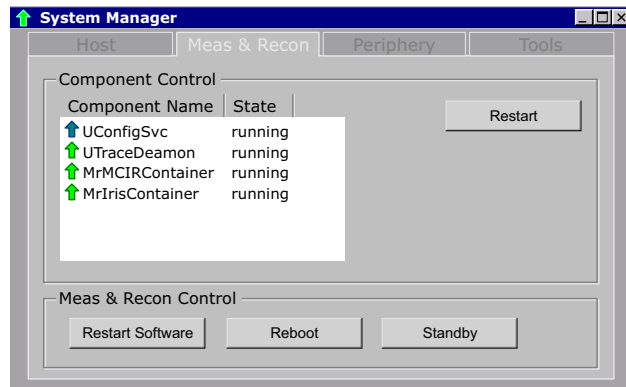
Restart/Reboot

- Restart Software

This function re-initializes the software modules. Intended for helping software engineers in their debugging.

- Reboot

This function resets the hardware and all software modules are loaded fresh. This function is required to reset errors in the scanner hardware.



Runtime Software

In reference to the functional block diagram [Figure 27](#), page 43 the two major functional tasks of the MARS is measurement control and image reconstruction and the respective programs can be seen in the Meas and Recon card. The current version of software will have these four software components.

- MCIR = Measurement Control and Image Reconstruction.
- IRIS = Intelligent Runtime Imager System

[Figure 35](#), page 52 provides an FPGA topology diagram of the Control Network intended to show the various dependencies between components. and [Figure 37](#), page 55 provides the CAN Bus Topology of the Peripheral Network.

Figure 35 FPGA Topology

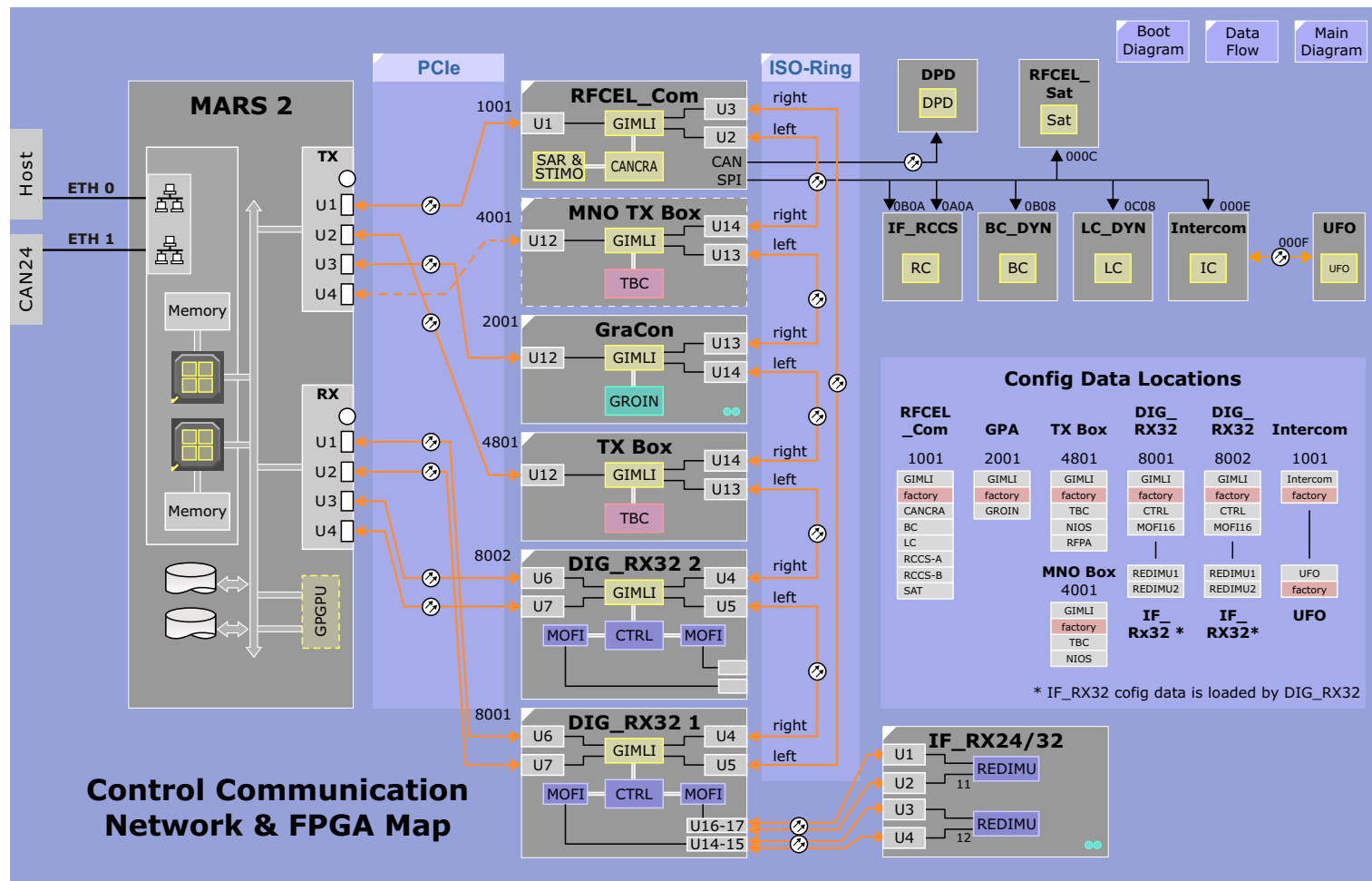
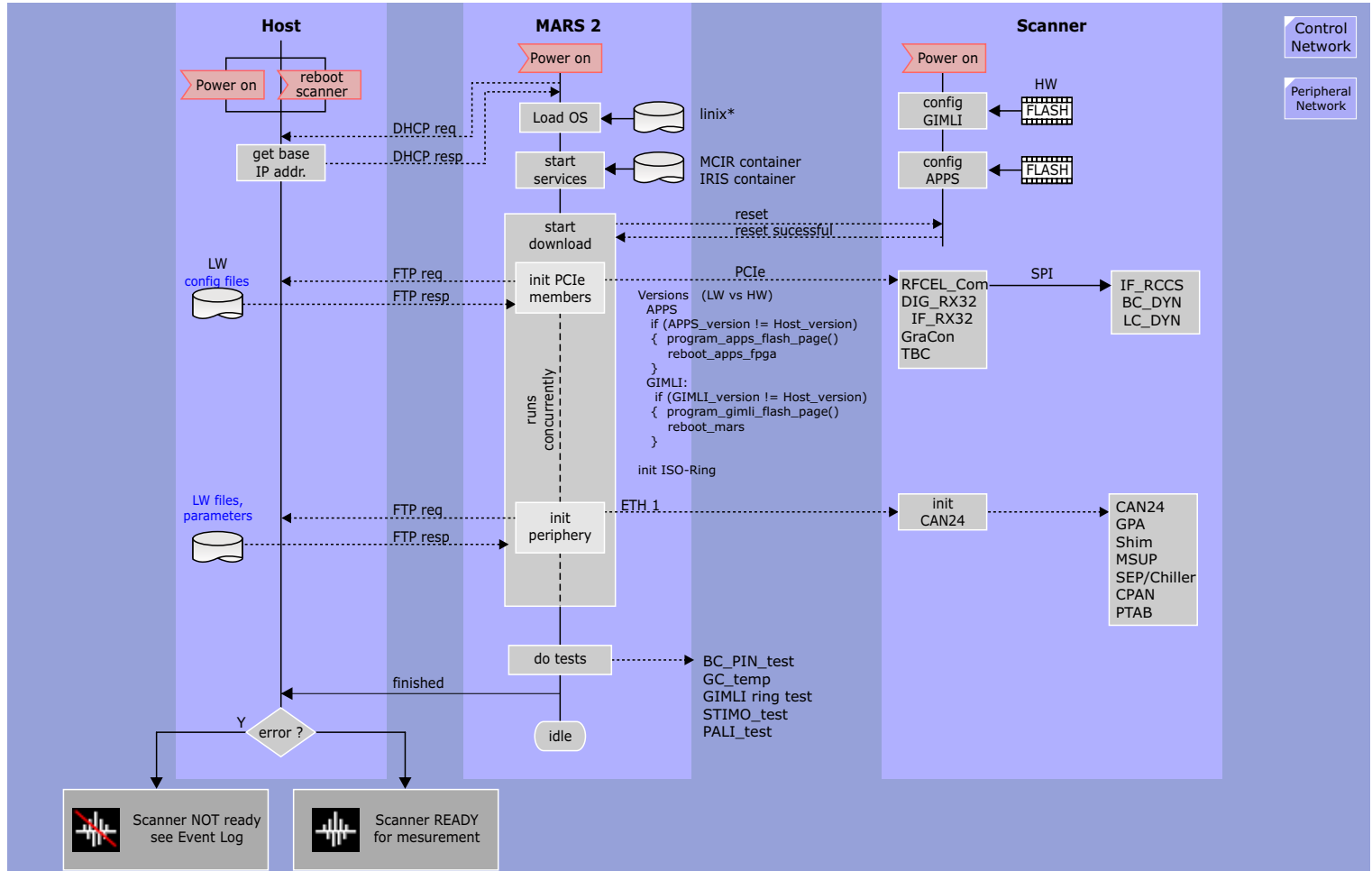


Figure 36 Scanner Boot Process Diagram



Peripheral Communications

This network is used to transfer:

- low-level hardware control commands from the MARS 2 to the scanner components
- status feedback from the scanner components to the MARS 2.

The network is realized using the CAN bus protocol with the CAN master located in the **CAN24**, an Ethernet-to-CAN bridge serving up to 3 CAN networks, only two of which are used in the current configuration. A second LAN interface has been provided to the CAN24 to allow reading out status of CAN components remotely via SRS.

CAN Open

The CAN Open protocol is an open, non-proprietary CAN protocol.

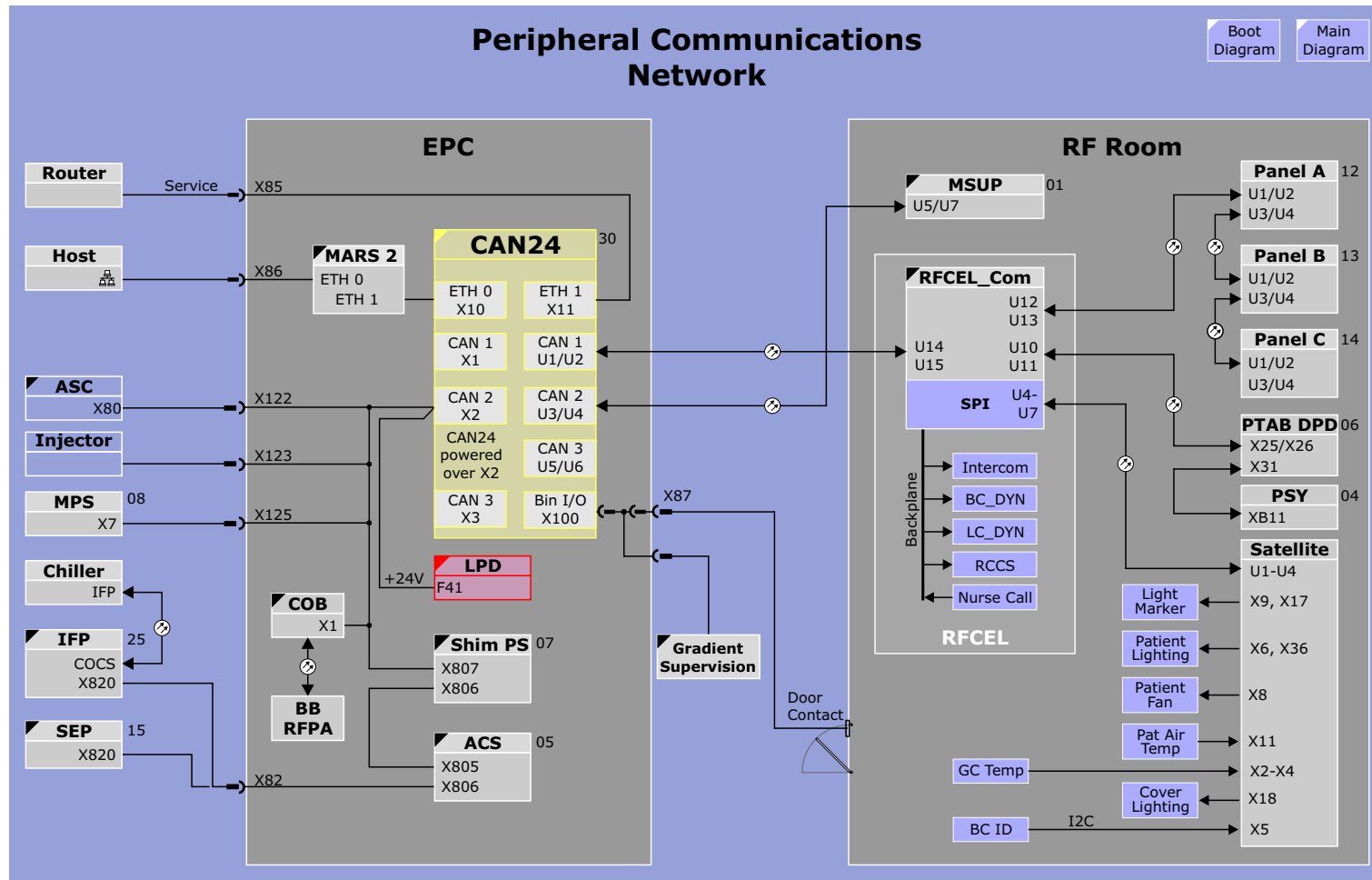
This protocol enjoys a wide acceptance and offers many advantages such as wide development tool support, auto configuration, plug and play capabilities, and reduction in programming overhead.

CAN Configuration

The CAN24 is a pseudo intelligent device that requires loadware. The version is checked on boot and updated if necessary.

Connected CAN components will also be checked for valid software versions and updated if necessary.

Figure 37 Peripheral Communications Network



CAN24

The CAN24 is an Ethernet-to-CAN gateway used to exchange CAN messages between MARS 2 and the peripheral CANOpen nodes. Normally, the CAN24 serves only as interface without doing any modifications or taking actions on certain messages.

When the system is switched off, however, it also serves as gateway between magnet and cooling components permanently running on CAN2 (MSUP, ACS, IFP, SEP) and the SRS server. The purpose is to automatically inform USC/RSC via SRS about important system faults that may require timely attendance (magnet quench, compressor faults, helium levels).

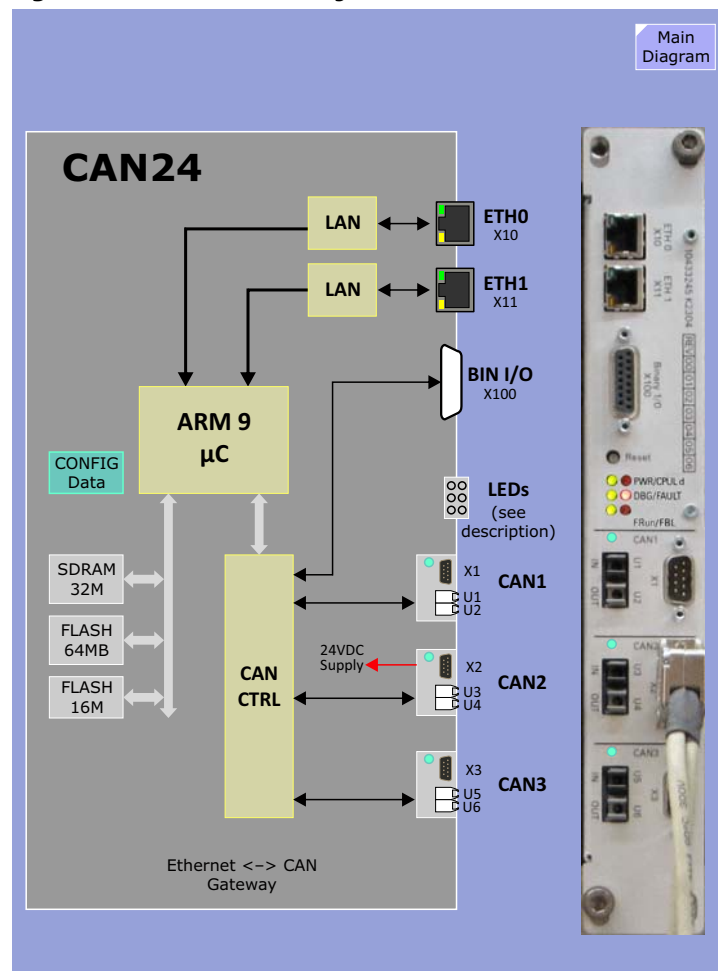
The Ethernet port to MARS 2 runs at 100 Mbps, using TCP/IP.

The Ethernet port to the SRS Router runs at 100 Mbps. The IP address for this port must be configured.

CAN24 is connected to service via CS-router, i.e. it gets its own IP-address (configurable via MARS 2 interface).

The CAN24 is rebooted when the scanner is started. To identify defective hardware a self test is to be implemented which is performed after PowerON, Reset or on request. The result is to be retrievable via status.

Figure 38 CAN24 Block Diagram



CAN logging

For service and test purposes a logging mechanism has been implemented, which records the CAN traffic for each CAN line separately. The log files can be read in via MARS 2 interface.

The log files have to contain:

- start date/time of logging
- stop date/time of logging
- time stamped (relative to start) CAN messages in raw format
- be readable via MARS 2 interface

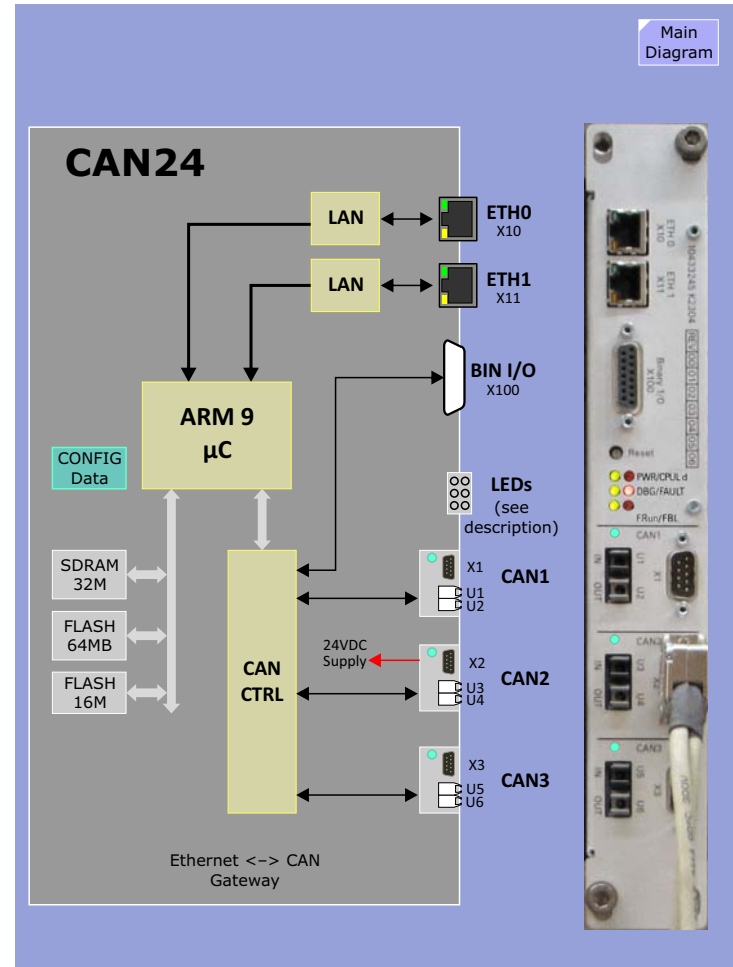
Connection to SRS Server

The CAN24 will have a separate SRS account, in addition to the scanner. The Link between to the CAN24 from the SRS server will not be possible as long as the scanner is powered on.

LEDs

LED	Description
PWR	24VDC supply present
CPU Ld	visualization of the ARM9 load (more ON = more STRESS)
DBG	Debug Mode (not used, but is normally on)
Fault	ON = connection to MARS lost
FRun	CAN controller is running (blinks @ 1Hz)
FBLd	ON = CAN controller buffer > 75% full OFF = CAN controller buffer < 50% full

Figure 39 CAN24 Block Diagram



Clock Distribution System

The de-centralized system control architecture imposes the necessity for a clock distribution. The diagram at right shows all clocks and their distribution to the various components. The Clock Box is the progenitor of all clocks (see description). The distribution of clocks to components in the technical room is over coax cables directly from the Clock Box. Two clocks are sent to the Synthesizer (in RFCEL) which is responsible for supplying components inside the exam room with clocks.

MARS 2 Power On Delay

Refer to [Figure 41](#).

The components connected to the PCIe network must be booted prior to starting the MARS 2. When the system has been turned on, a help contact of K2 (main power relay in LPD) signals the Clock Box and after **five seconds** later, the Clock Box activates the MARS 2 power relay K3. If the Clock Box was switched off (e.g. for service reasons) it waits an additional **3 minutes** to assure a stable clock, before switching on the MARS 2.

Figure 40 Clock Signal Distribution

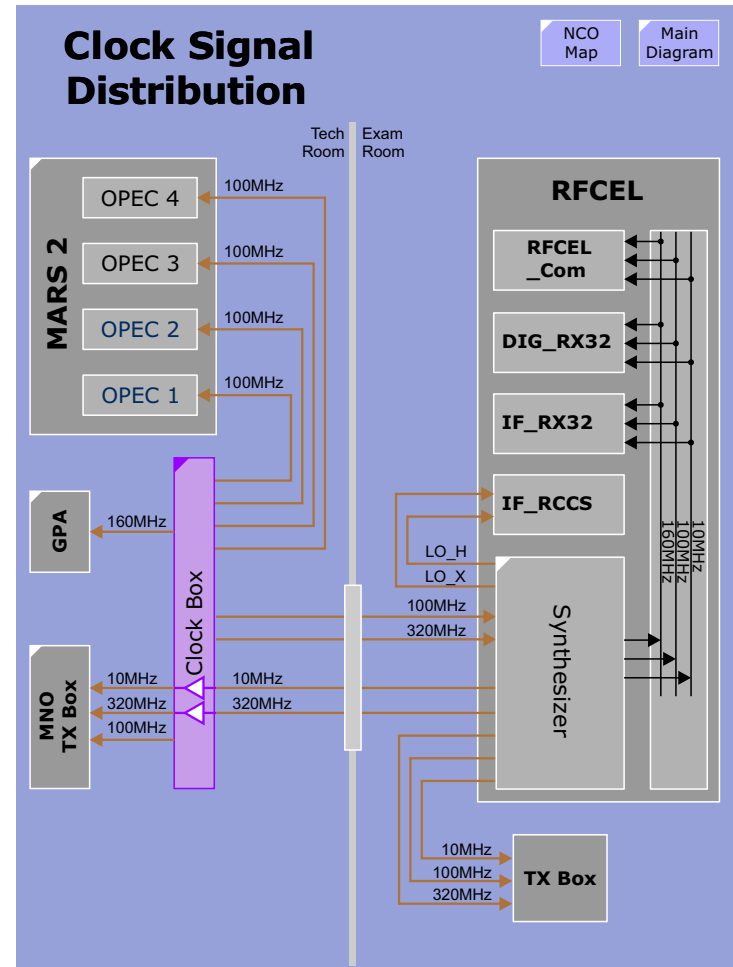
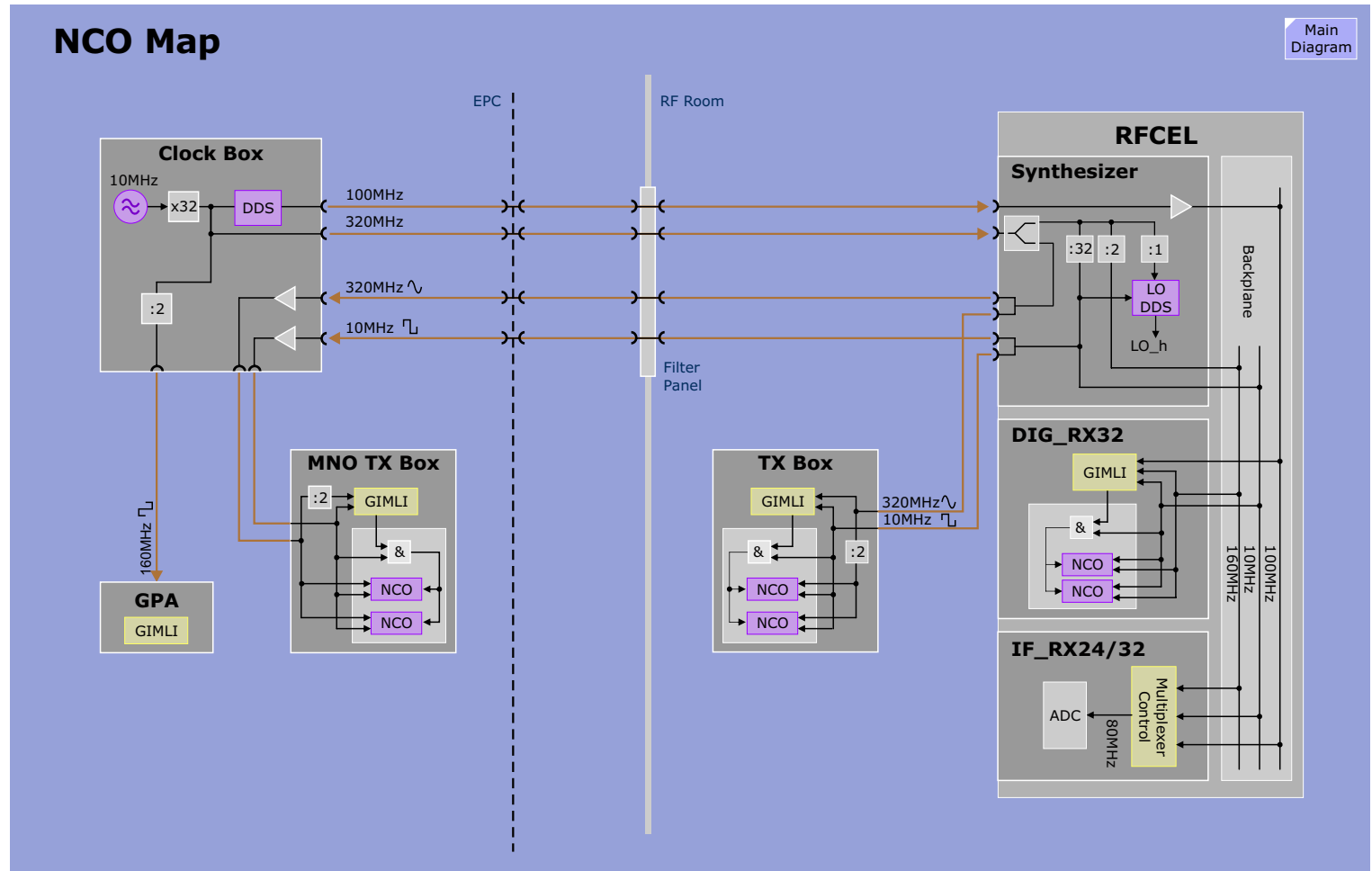


Figure 42 Map of all DDS and NCO Blocks



RF System

Introduction

The RF System components for the Aera/Skyra scanners can be grouped into these functional blocks:

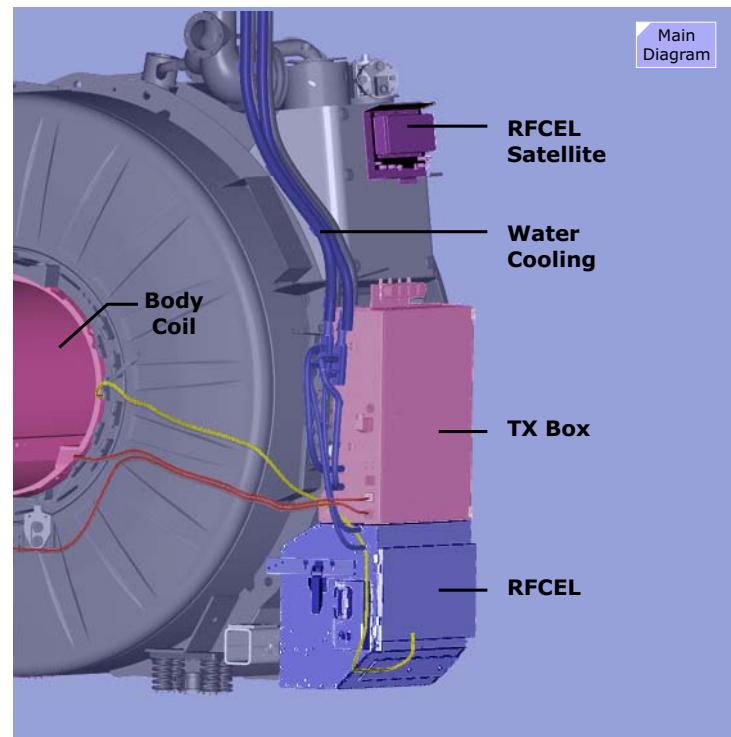
- Transmit Section
- Receive Section
- Coil Control (detuning, coil code)
- SAR Monitoring

The components belonging to these functional blocks and their locations are shown in the diagram at right.

- Transmit section
 - TX Box
 - Body Coil
- Receive section
 - Tim 4G Coils (not shown)
 - IF_RCCS, IF_RX24/32, Dig_RX32 - in RFCEL unit
- Coil Control
 - BC_Dyn, LC_Dyn - in RFCEL unit
- SAR Monitoring
 - RFCEL_Com, in RFCEL unit

These components do require and are dependant upon cool water for heat removal and stability.

Figure 43 Aera/Skyra RF System



Transmit and SAR Monitoring

The transmit section consists of those components which provide the functions of transmit signal generation, signal amplification, route and apply the RF transmit signals to the transmit coil(s) for proton excitation. These components must produce amplitude, frequency and phase stable RF pulses at power levels required to excite protons with flip angles up to 180° for patients weighing up to 200kg and capable of operating over a frequency range of 500 kHz around center frequencies for both 1.5 Tesla and 3 Tesla systems. The transmit section consists of:

- **THYRE** - DC power supply for TX Box
- **TXBOXPS** - low voltage power supply, located on Filter Panel.
- **TX Box** - contains all transmit components
- **Body Coil**

THYRE

A 110V DC power supply unit for the TX Box with enabling and internal monitoring. The TX Box power supply is located in the EPC cabinet.

TX Box

The TX Box fully integrates all components required to generate, amplify and route RF pulses to transmit coils and provide RF pulse samples for SAR monitoring. It is a no-maintenance FRU.

Body Coil

The BC is in essence the same as the Avanto BC with these exceptions:

- it has no pickup coils (no longer needed!)
- it includes an integrated FLASH memory containing ID information (SN, PN, etc.) for the Asset Management

SAR Monitoring

SAR monitoring is achieved by measuring the applied RF pulses to the patient during the scan in order to limit the overall RF deposition preventing excessive heating of patient tissue. The RF values are metered by directional couplers within the TX box. These values are sent to a DSP located on the RFCEL_Com in the RFCEL which calculates the SAR levels in real-time.

Receive and Coil Control

The receive section consists of the receive coils that pick up the MR signals off the patient and the receiver hardware components responsible for amplification, routing, acquisition and processing of the MR signals.

Tim 4G Coils

Tim's newly designed ultra-high density receive coils are combined with the highest channel configurations ever offered. And, with Tim's new patient adaptive technology, image quality and acquisition speed go to a whole new level.

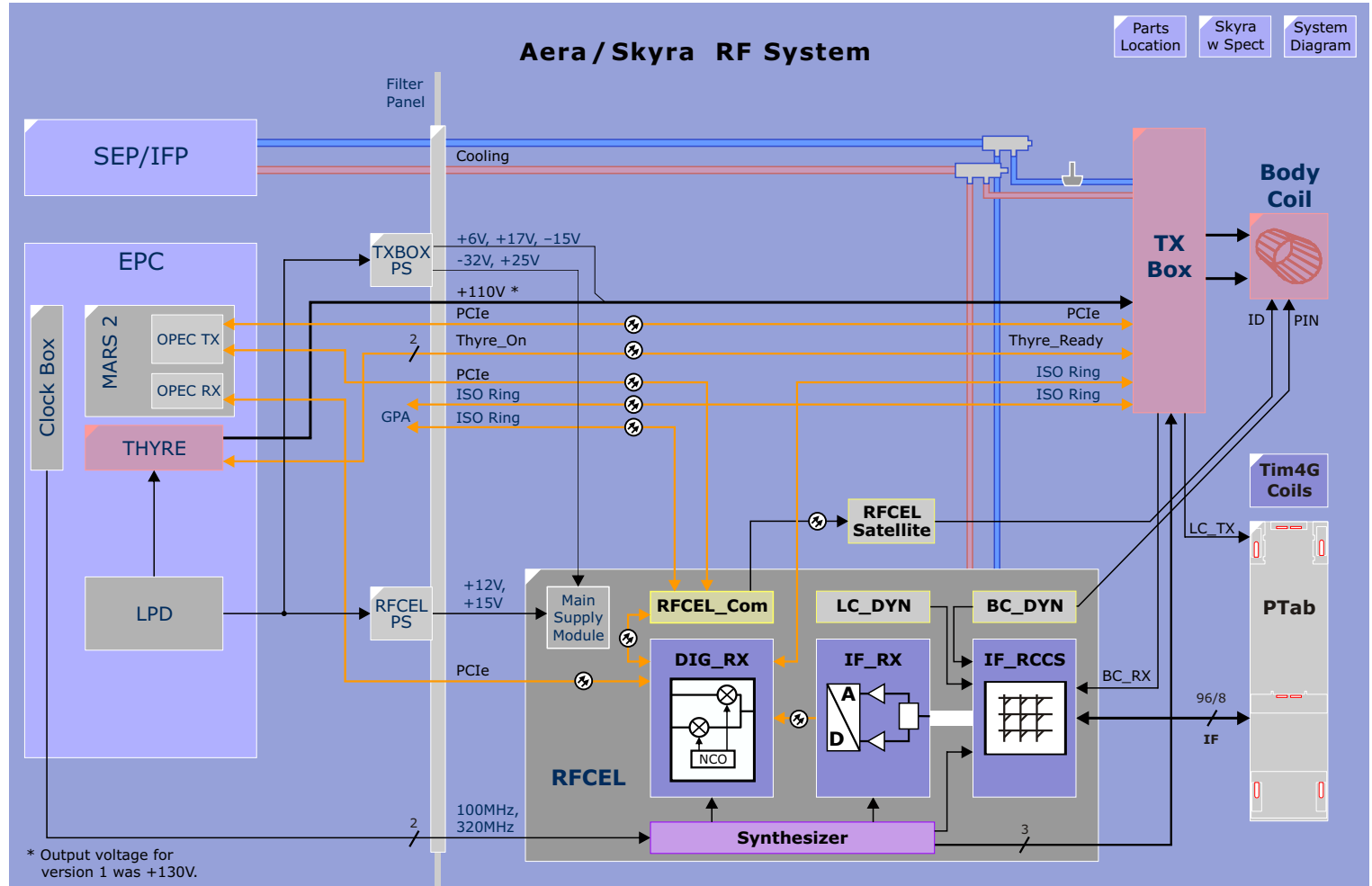
Now a maximum of 204 channels (coil elements) can be connected simultaneously and in one single scan and in one single FOV, 128 independent RF channels can be used simultaneously, each generating an independent partial image. Tim [204 x 128] enables more signal than ever before with all the channels needed to support ultra-high density coils.

DirectRF™ – Tim's new all digital-in / digital-out design integrates all RF transmit and receive components at the magnet, eliminating analog cables for true signal purity.

TimTX – Providing excellent B1 homogeneity and scalability to higher transmission channel configurations with the potential to unlock new applications.

New **DirectConnect™** coils eliminate the hassle of cables – speed exam setup, and improve SNR.

Figure 44 RF System Block Diagram



Several coils have been designed to plug directly into the coil plugs without cables for very user-friendly coil handling and easy coil setups. The Head and Spine Matrix coils and the Foot/Ankle coil plug into recessed **DirectConnect™** coil plugs. The large and small Shoulder coils plug directly into one of the **SlideConnect™** plugs. All other coils plug into the **SlideConnect™** connectors over cables.

All receive coils utilize an internal EEPROM for recognizing coil ID and storing coil-specific manufacturing information including serial number.

Receiver Hardware

Both the receiver and coil control hardware components have been placed into a single rack - the RFCEL - placed directly under the TX Box on the left side of the magnet.

The IF receive signals from the receive coils are input to the RFCEL. All receive-signal processing is performed within the RFCEL. Its output is digital raw data sent directly to the image reconstruction section over fiber-optic cables. This technology is given the name **DirectRF™**.

Coil Control

The components performing the coil control functions including coil recognition and coil detuning are also found in the RFCEL unit.

Coil Recognition

Coil recognition is realized with EEPROM located in the receiver coils. The EEPROM contains general manufacturing data such as coil name, part number and field strength as well as data specific to the coil including its serial number and the last system the coil was connected to.

Coil Detuning

Coil detuning is performed by switching PIN diodes strategically placed within the receive coils on and off for the receive and transmit phases. This function plays an important role in both

patient and coil hardware safety. The circuitry providing the voltage and current signals for switching the detune PIN diodes are found within the RFCEL.

Cooling

Both air and water cooling of the electronic circuitry and power devices. General air cooling is achieved with the use of heat exchangers and air circulation fans. Water cooled heat sinks are used where necessary to provide cooling of power devices.

Aera RF System Specifications:

- Operating frequency: 63.6 MHz
- RF amplifier peak RMS power 26.1 kW
- Transmit Amplitude Resolution: 16 bit
- Maximum number of simultaneous transmitter 1
- Pre-amplifier Bandwidth 1 MHz
- Receiver Signal Resolution: 32 bit
- Receiver Dynamic Range: 164 dB

Skyra RF System Specifications:

- Operating frequency: 123.2 MHz
- RF amplifier peak RMS power 37.5 kW
- Transmit Amplitude Resolution 16 bit
- Maximum number of simultaneous transmitter 1
- Pre-amplifier Bandwidth. 1 MHz
- Receiver Signal Resolution: 32 bit
- Receiver Dynamic Range: 164 dB

TX Box

All transmit section components including RF signal generator, power amplifier, transmit path routing, RF level metering, and associated control circuitry are integrated into the TBX. The components currently available:

1.5T Variations	Power	Weight	CP	Trueform
1 Channel	~26 kW	80 kg	YES	NO
3.0T Variations				
1.5 Channel	~37 kW	80 kg	YES	YES
2 Channel	~40 kW	85 kg	programmable	programmable

NOTE Early systems are equipped with a different TX Box (version 1). For details please refer to the Service Parts Catalog.

Main features of the TBX:

- narrowband, only for proton imaging 1,5T and 3T
- no maintenance or servicing requirements
- employs a power regulation loop to ensure stable and predictable RF power transmission
- FPGA based controller for signal processing, control of in box functions and communication with the host and periphery
- is an FRU, requires special replacement tool

TX Box Version 2

A second version was introduced using a lower supply voltage of +110 or +130 VDC, depending on version. There were some additional internal changes of various components to facilitate easier manufacturing. Because of the voltage differences between the TX Box versions they are not compatible and they each requires different THYRE power supplies.

Figure 45 TX Box Block Diagram

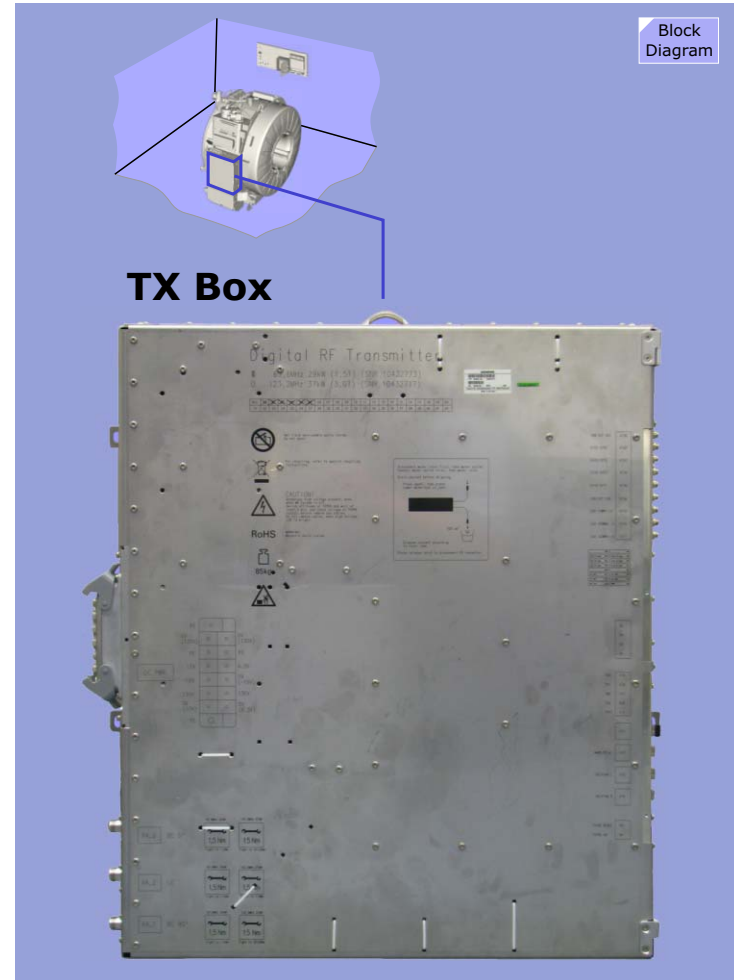
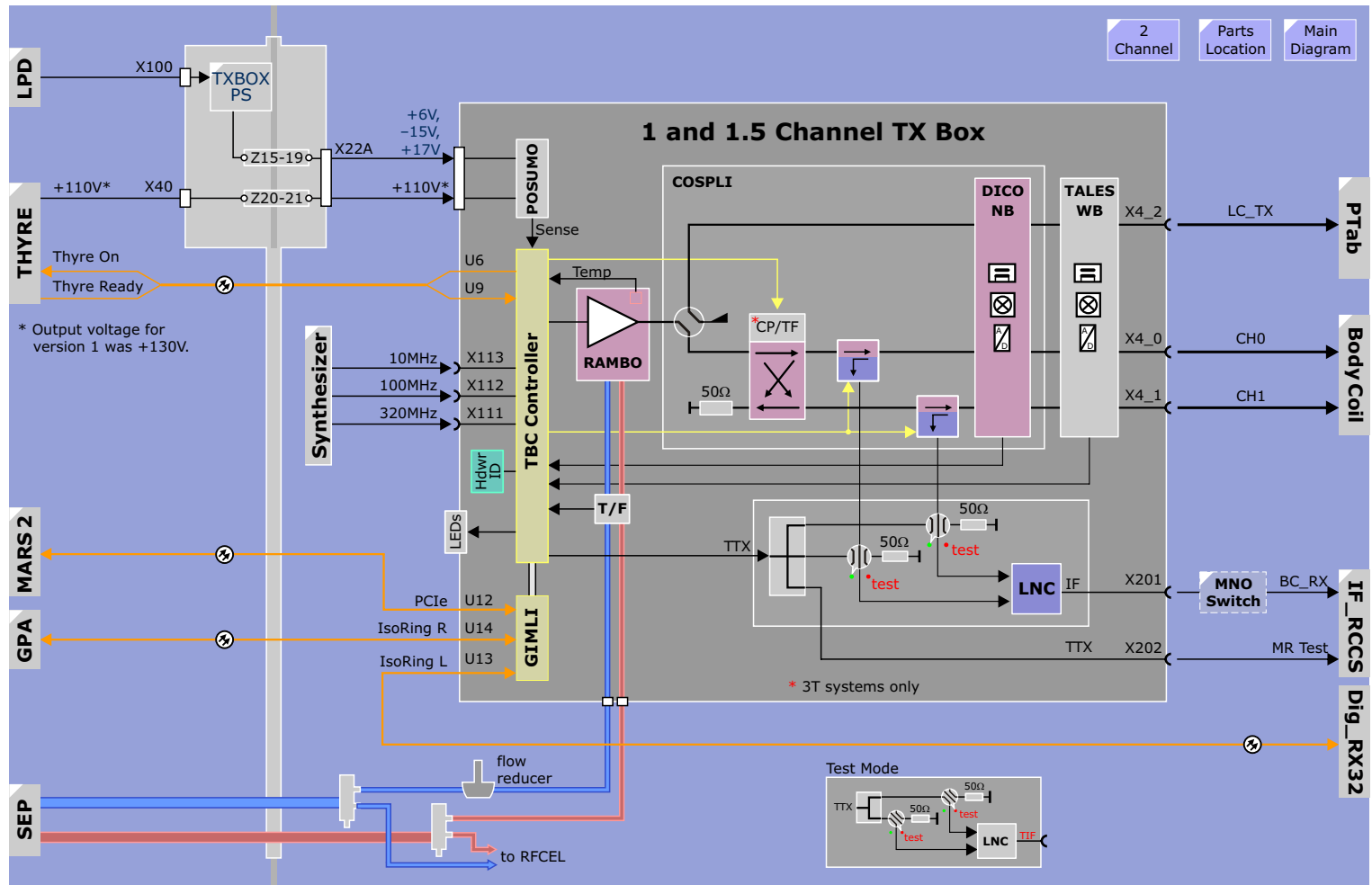


Figure 46 1 and 1.5 Channel TX Box Block Diagram



Inputs

Signal	Description
PCIe	configuration data, RF pulse commands, frequency and phase data, RX events
ISORing_R, L	Master sync clock, B0 shift, SAR data, sequence clock,
THYRE_Ready	2.5MHz = ok ON = over voltage OFF = under voltage
+110VDC (+130VDC for version 1)	supply voltage for the amplifier (POSUMO, RAMBO)
+6, -15, +17	supply voltages for the TX Box controller (TBC)
10, 100, 320 MHz Clocks	Synchronization clocks for RF (10,320MHz) and PCIe bus (100MHz)

Function

The TBX consists of these functional blocks:

- Controller TBC
- Amplifier (RAMBO, POSUMO)
- Power Metering (COSPLI, TALES)

TBC Controller

The TBC Controller performs these functions:

- Optical Interface to MARS 2 and ISO Ring
- Pulse generation and digital modulation
- Amplitude and phase regulation
- Bias control
- RF Power metering (for internal control and SAR)

Amplifier

- 3-stage amplifier, with blank
- RF Splitter and Combiner
- switchable 90°/140° hybrid (1.5 channel TX Box only)
- output routing switch
- Voltage monitor with capacitive coupler

Power Metering

- calibrated-for-life metering devices (no TALES replacements required)
- metering devices capable of complex impedance measurements allowing accurate metering of RF levels both at the box output and at the coil
- metered values used to measure and monitor coil power losses in real-time
- measurements for the metering devices used as the basis of the SAR monitoring

With the power metering devices it is now possible to realize a component protection to limit real physical values, e.g. the breakdown voltages of capacitors or the dissipated average power in the coil. They can measure complex voltages which are corrected in the TX Box and then sent to the SAR monitor (DSP on RFCEL_Com). These values are the real voltage values at the output of the TX Box and the basis for the SAR calculation, component protection and the feedback loop for the multi-channel transmit coils.

Power Supply Monitoring

In the event one of the supply voltages should fail, then all must be switched off (transistor protection).

Unexpected RF and Over-voltage Check

The accumulated signals (over 10 us) from the broadband TALES for BC0, BC1 and LC Channels are continuously monitored and checked against over-voltage and "unexpected RF".

There are two levels to compare against. A lower value is active in between pulses to guard against unexpected RF while not actively sending. Implicitly this value also guards against over-voltage conditions.

The higher value applies during RF_ON and a short period thereafter and performs an over-voltage protection.

LO Frequency Supervision

The TBC has a frequency counter, that is connected to the LO DAC outputs. Thus it is possible to count edges of the LO. The supervisory circuit counts edges and provides the sampled number of edges in a register. MARS 2 Computer can read this register and check for proper LO setting.

Cooling Water and Heatsink Monitoring

The coolant water temperature and flow rate as well as heatsink temperature are continuously monitored. If programmed limits are exceeded, errors will be issued.

Outputs

Signal	Description
CH0, CH1	There are two 90° or 140°phase-shifted outputs to the Body Coil, for CP or Trueform operation.
LC_TX	Transmit output to LC connector 1. Not all TX Box versions have this output.
BC_RX	The IF transformed receive signal of the Body Coil. Only used when receiving with the Body Coil
TTX	A test signal at the 1H frequency. Is currently not used.
Thyre_On	Enable signal to the THYRE power supply
SAR Data	The measured and digitized forward and reflected RF energies are sent to the RFCEL_Com over the ISO Ring for use in SAR monitoring.

NOTE Before working on the TX Box, EPC circuit breakers F51/F52 for THYRE must first be switched-off and allow at least 5 min. for the capacitor banks to fully discharge, as indicated by the **LV+HV** LED.

LEDs

TX Box 1

State	Name	LED	LED	Name	State
	--	V69	V68	--	
1.5T - V56,57 on by voltage present	High Voltage	V58	V59	High Voltage	3T - V56,57,60,55 on by voltage present
	High Voltage	V60	V55	High Voltage	
	High Voltage	V56	V57	High Voltage	
	--	V66	V67	--	
	PLL RX3 (not used)	V27	V26	PLL RX2 (MaRS PCIe)	green - ok red - error
green - ok red - error	PLL RX1 (ISO Ring L)	V25	V24	PLL RX0 (ISO Ring R)	green - ok red - error
green - ok red - error	PLL TX1 (160MHz GIMLI)	V23	V22	CORE (Heartbeat GIMLI)	blink - ok red - error

TX Box 2

State	Name	LED	LED	Name	State
green - ok red - error	Ring OK	V69	V68	Appl FPGA OK	green - ok red - error
not used	--	V52	V53	--	not used
on=voltage present	High Voltage (3T only)	V60	V51	High Voltage (3T only)	on=voltage present
	High Voltage (1.5T+3T)	V28	V8	High Voltage (1.5T+3T)	
green - ok red - error	App FPGA Pwr On	V66	V67	Power On Reset	green - ok red - error
not used	--	V27	V26	MARS PCIe	green - ok red - error
green - ok red - error	ISO Ring L	V25	V24	ISO Ring R	green - ok red - error
green - ok red - error	160 MHz GIMLI	V23	V22	Heartbeat GIMLI	blink - ok red - error
green - low volt present yellow - low + high volt		LV+HV		next to power connector	

2 Channel TX Box (TimTX TrueShape)

The 2-channel TX Box is a multi-channel transmitter option (for 3T systems only) consisting of two fully independent transmit channels, each being controlled separately.

Through interaction of the independent transmit channels selective excitation of a specific body area is possible.

Other advantages include:

- Higher image quality
- Less distortions, less blurring
- Higher B1 homogeneity due to zooming (ZOOMit)
- Better fat saturation
- Less motion and flow artifacts
- Increased spatial resolution in region of interest
- Higher speed
- Shorter echo trains
- Shorter scan time due to fewer phase-encoding steps

syngo® ZOOMit

syngo ZOOMit creates a zoom effect during MR imaging for faster scanning and improved image quality. Thanks to multi-channel technology with its fully flexible wave forms, you can select a small stripe instead of a large excitation plane. This will allow you to perform a targeted excitation. Data volume is reduced from the very start, as well as time effort for phase-encoding steps. The scan will be faster and image quality in the selected zoom area is considerably improved.

Inputs

The same as the 1 and 1.5 channel TX Box.

Outputs

Connections are the same as the 1 and 1.5 channel TX Box. Each channel however is independently controlled.

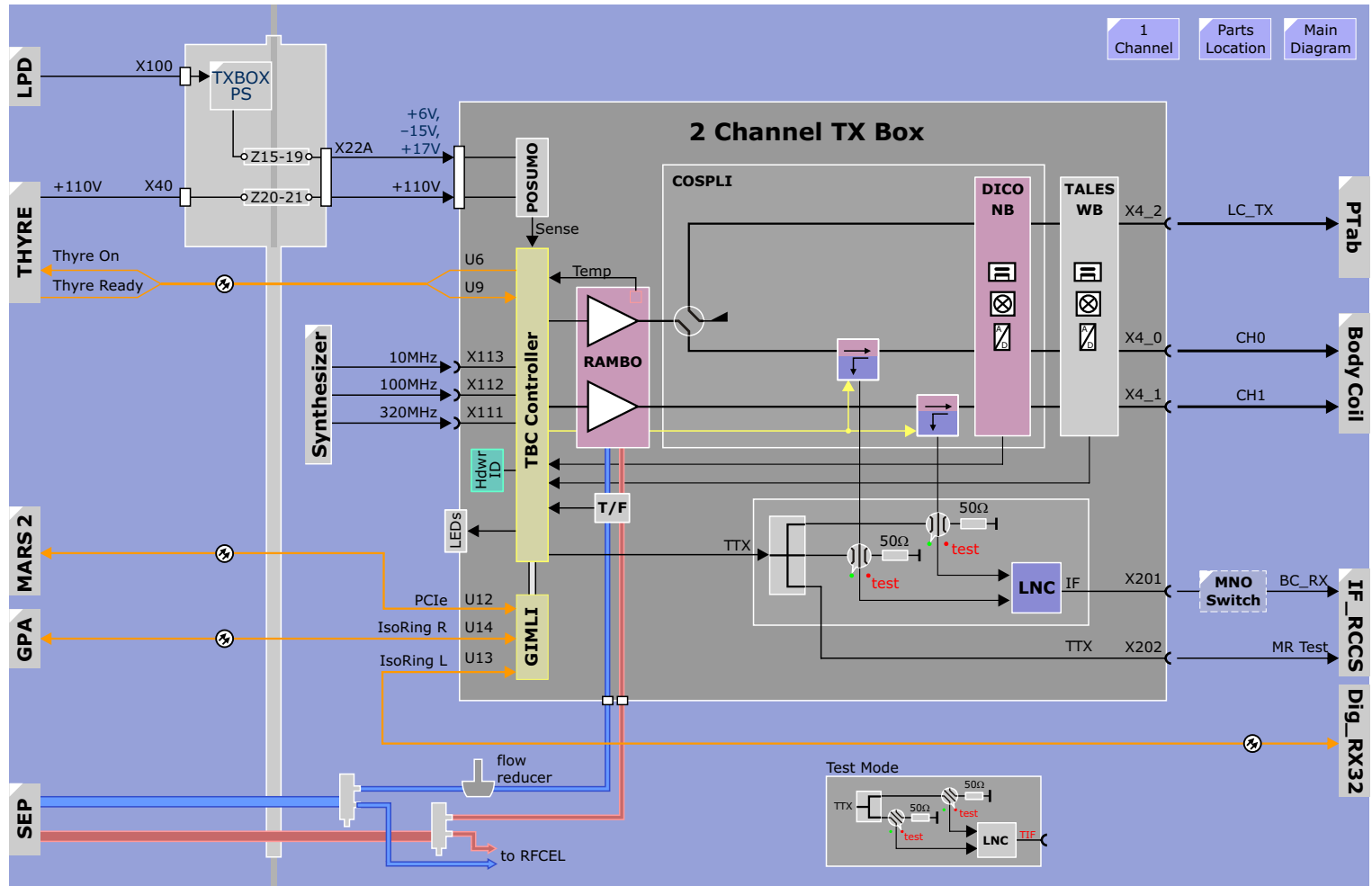
LEDs

The same as the 1 and 1.5 channel TX Box.

Body Coil

The 2-channel option also requires a modified Body Coil. This modified BC works using the same principle and its connections are the same.

Figure 47 2 Channel TX Box Block Diagram

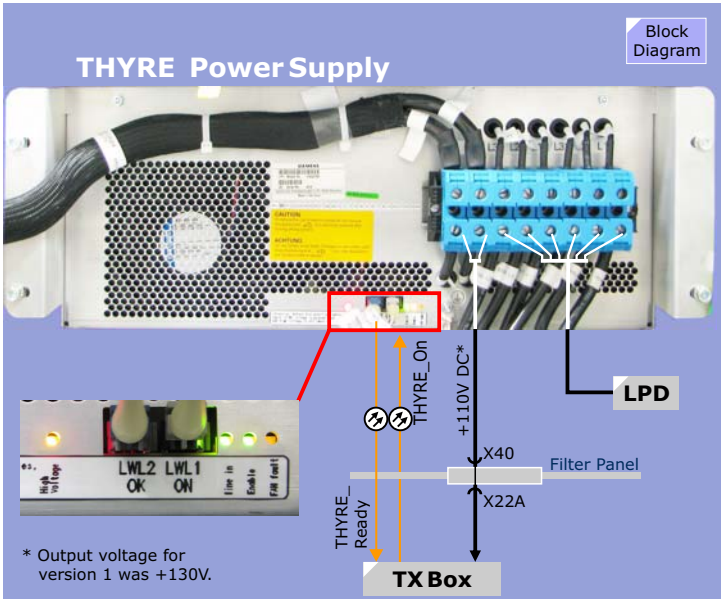


TX Box Power Supplies

THYRE

Is a DC power supply for the TX Box and is located in the EPC.

Figure 48 THYRE Power Supply



NOTE Refer to the Spare parts Catalog for exact information on the various versions of this component.

Function

The THYRE is a DC power supply for the TX Box. The output voltage is either 110V or 130V DC voltage depending on version.

Refer to the Spare Parts catalog for details. The voltage is monitored internally and errors are indicated with LEDs.

Inputs

Signal	Description
L1-L3, L1.1-L3.1	AC input from transformer. See "Procedures / THYRE / Check of Supply and Output Voltages THYRE" in TSG for details.
Thyre_On	When the signal toggles at 2,5MHz $\pm$ 1ppm the THYRE is enabled. A static HIGH or LOW level disables the THYRE.

Outputs

AC Input	Measures
K+, K-	Regulated 110V or 130V DC output
Thyre_Ready	This signal has the same definition as the control signal. If the signal toggles at 2,5MHz $\pm$ 1ppm the THYRE is ok. Static high level indicates overvoltage, static low level indicates an undervoltage condition.

LEDs

LED	Description
High voltage (yellow)	output voltage >24 V (cap bank charged!!!)
Line in (green)	AC supply voltages from LPD ok
Enable (green)	THYRE is enabled by TX Box
Fan fault (red)	THYRE cooling fan error

Low Voltage Power Supply

A logic power supply at the RF Filter Panel provides TX Box low voltages of +6, -15 and +17V DC. They are monitored by the TBX controller.

Body Coil

The Body Coil is a resonant transmit/receive antenna system. When used as a transmit coil it transduces current/voltage signals from the TX Box to an electro-magnetic field (designated B1 field), providing excitation of the proton nuclei in the MR-experiment.

Although primarily used as a transmit coil, the Body Coil can also be used as a receive coil, typically when making scout scans (topology images) used for slice positioning.

This multi functional use of the Body Coil requires a detune circuit to allow fast enable/disable of the coil when used only as a transmit coil (dynamic detuning) or to be disabled completely (static detuning) when not required, e.g. when using a transmit-capable Local Coil. For this PIN diodes are used.

1 Channel

Two phase shifted signals are applied to the resonator at 90° physical intervals. The resulting current flowing in the coil will be a sine wave. If the energy distribution between the two resonator halves is equal, a homogeneous B1 field will result. To improve the B1 homogeneity on 3T systems in the area of the abdomen or shoulders and to improve the SAR distribution in these areas the TrueForm™ mode is used. Trueform is an elliptical B1 field resulting from a phase shifting of the applied RF signals within the TX Box (which see). Selection between normal and TrueForm mode is determined by the examined Region in the *syngoMR* software.

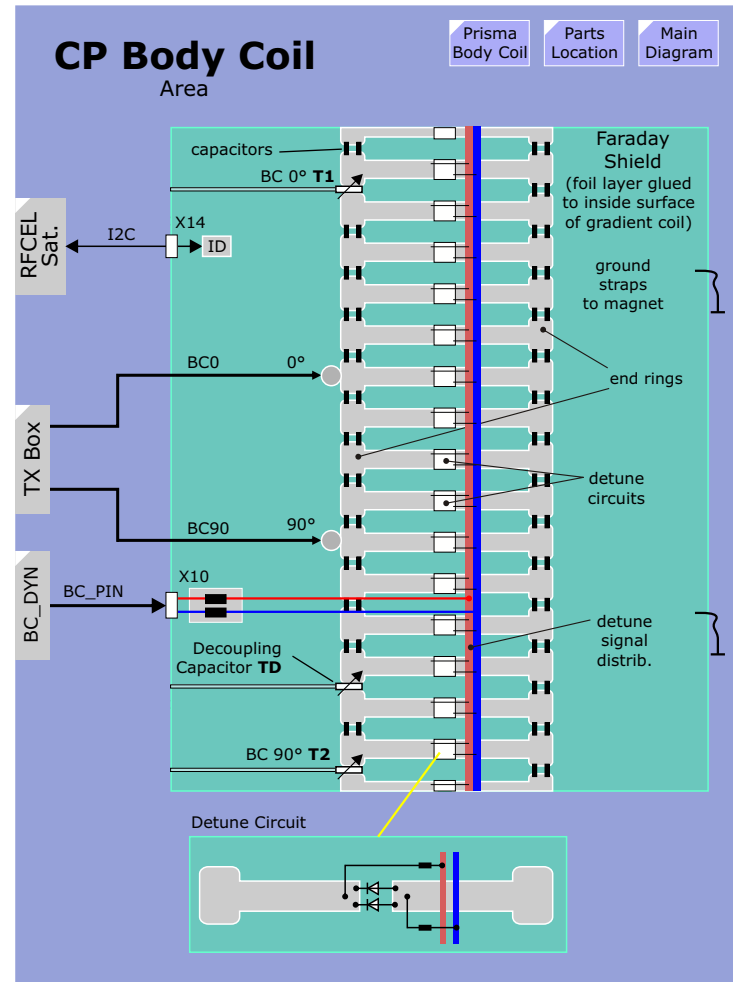
2 Channel

The 2 channel Body Coil has been optimized for 2-channel operation, yet it has the same connections and works on the same principle as the 1 channel coil.

Skyra Body Coil

The 123 MHz version has 32 elements, but works the same.

Figure 49 Body Coil Block Diagram



Decoupling (Transmission)

Being a CP coil, decoupling between resonator systems is an important issue. Strong coupling between the two resonator systems leads to high transmitter reference values and hence reduces the performance of the RF-transmit path. A procedure for checking and, if necessary, optimizing the decoupling is located in the Service Software platform under Tune Up / BC Tuning.

Detuning

When using local coils a detune circuit brings the Body Coil to off-resonance to prevent coupling with the Local Coils and thus absorbing any MR echo signal. For receive-only local coils the detuning is achieved dynamically (fast switching times). When using local coils capable of transmitting the Body Coil will be statically detuned. PIN diodes are the devices used to accomplish the detuning. A control voltage of -30 V is applied for dynamic detuning and a control voltage of approximately -200 V is used for static detuning. To tune the coil, the diodes are enabled with a positive voltage and a current of 5A. The PIN diode control signals are supplied by the BC_DYN board in the RFCEL.

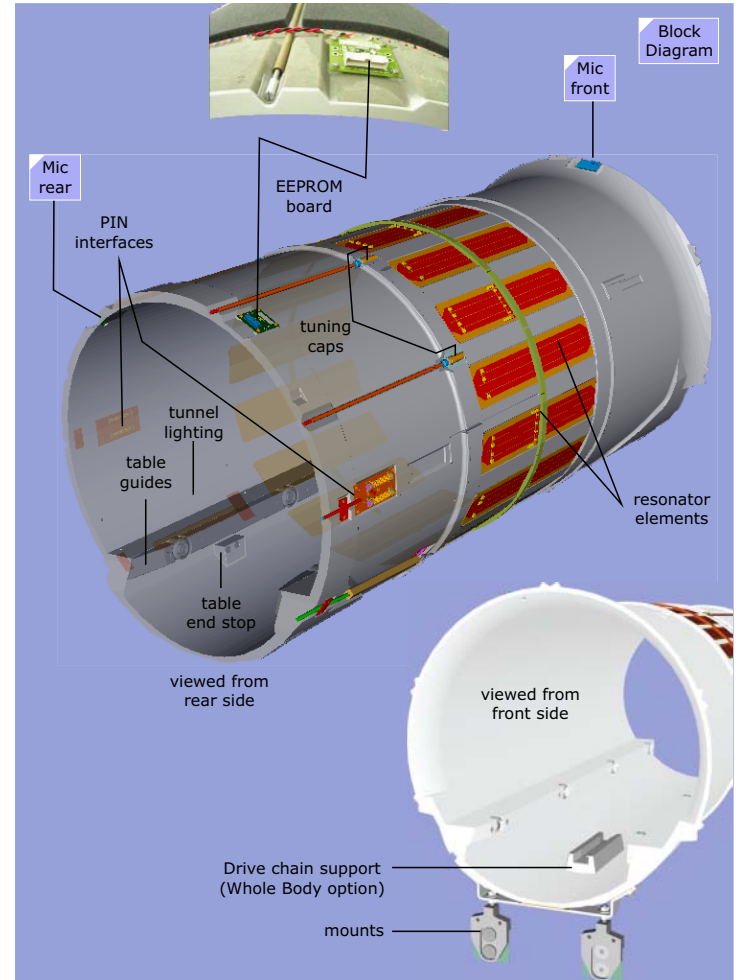
Faraday Shield

The RF shield (Faraday shield) is laminated on the inside of the Gradient Coil and connected over the magnet shell to ground. The RF shield reduces B1 field losses into the gradient coil as well as serves to prevent possible disturbances.

Specifications

Specification	Aera	Skyra
Design	16-element Birdcage	32-element Birdcage
Tube length/diameter	1.424 m / 70 cm	1,7 m / 70 cm
S/N	14	25
CPL	1500W	2kW/2.5kW(CP/TF)

Figure 50 Body Coil Parts Location



RFCEL - RF Cabin Electronics

Introduction

The RFCEL is a rack unit housing the components belonging to the receiver, coil control and SAR monitor as well as the intercom controller. All boards share a common backplane used to distribute both control signals and power supplies.

RFCEL Cooling

The RFCEL rack unit is actively cooled by a water-air heat exchanger and a fan blower assembly. This cooling also climates the TX Box, situated directly above the RFCEL unit. The RFCEL rack is hermetically sealed to avoid problems with dust, condense water from magnet, EMC problems etc.

RFCEL Contents

The RFCEL unit contains these boards:

- Main Supply Module - MSM
- **RFCEL_Com**
- Intercom
- DIG_RX32 (1, 2 or 4)
- BC_Dyn
- LC_Dyn (4x)
- IF_RCCS
- IF_RX32 (2 or 4 for 64 and 128 channels)
- IF_RX24 (1 or 2 for 24 and 48 channels)
- Synthesizer

The **RFCEL_Com** is the main controller for the receiver, coil control and SAR monitor functional blocks. It also acts as the ISO-Ring master, the controller for the PMU and several patient handling functions as well as cover lighting and BC identification.

The Intercom board is covered in the Patient Handling section.

Figure 51 RFCEL Rack Location

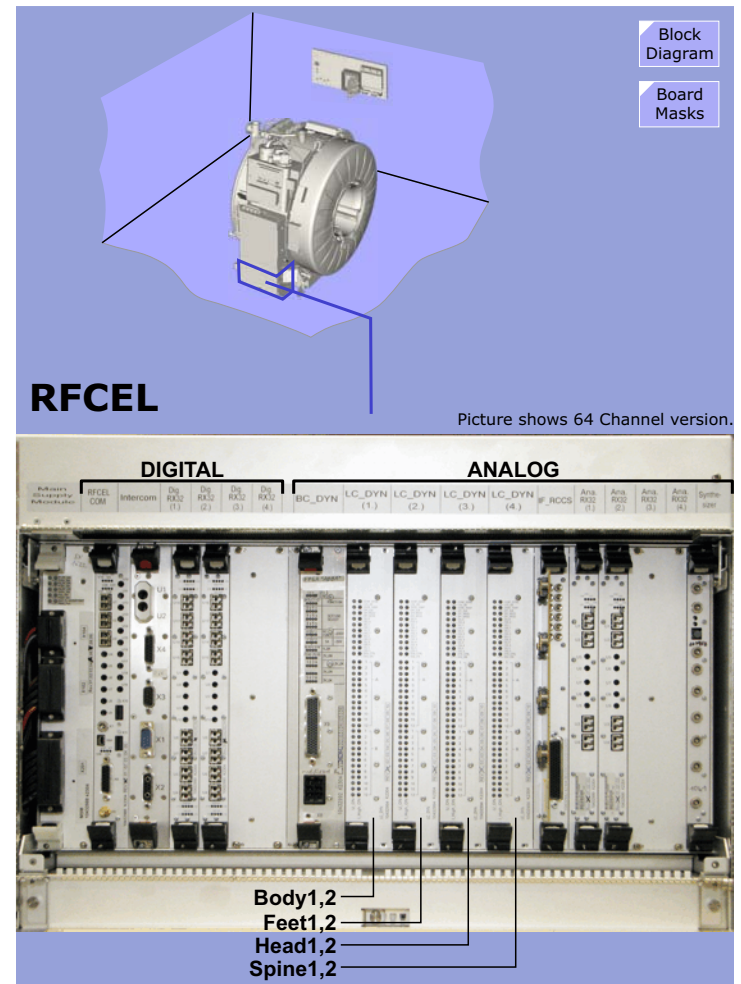
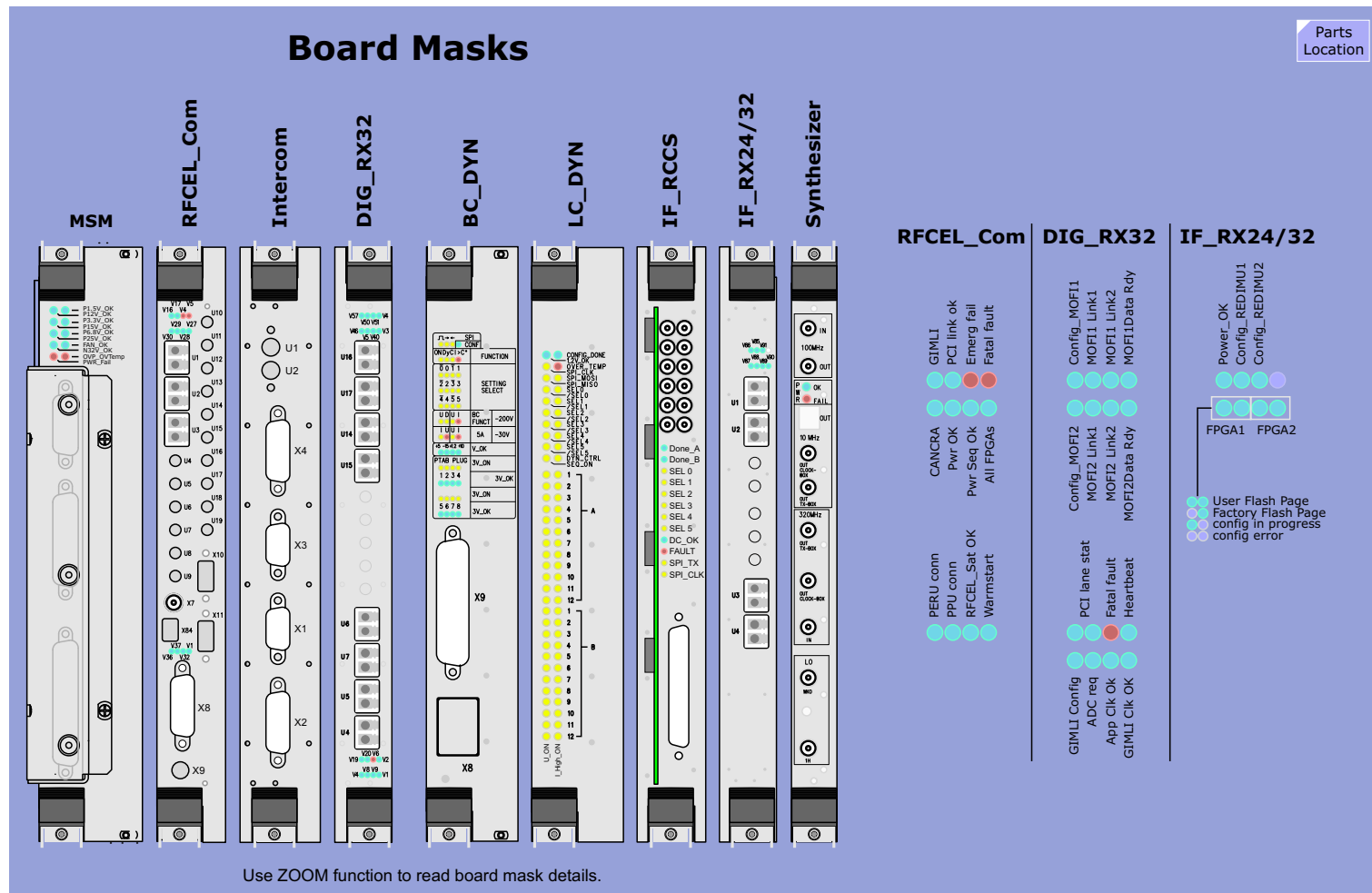


Figure 52 RFCEL Board Masks and LEDs



The daVinci Coil Interface

The received MR signal within the coil is mixed and transmitted on an intermediate frequency (IF). Two different intermediate frequencies are realized, allowing transmitting two MR signals independently over one cable using frequency domain multiplex. The coil connector is designed for 24 receive elements.

The two intermediate frequencies are fIF1 (8 MHz) and fIF2 (12 MHz). With fMR being the MR system frequency, two local oscillator (LO) frequencies of are fMR - fIF1 and fMR+fIF2 are fed to the coil through the IF_RCCS. The MR signal at frequency fMR is mixed to fIF1 for one coil elements and to fIF2 for an adjacent one.

The reason for this approach is to save cables and connector pins, and to design the receiving chain identically for different MR system frequencies, i.e. for 1.5T and 3T Systems. Provision is made to prevent the use of local coils with the incorrect frequency on the scanner, i.e. 1.5T coils cannot be used on the 3T scanner and vice versa.

Tim (Total imaging matrix) 4G Technology

Tim advances MR imaging – again – with 4G flexibility, accuracy, and speed. It's Siemens ultimate innovation technology that unlocks imaging power like never before. Tim's newly designed ultra high density receive coils are combined with the highest channel configurations ever offered. And, with Tim's new patient adaptive technology, image quality and acquisition speed go to a whole new level. Customers can think more exams per day. Every day.

Now up to 204 coil elements read in with up to 128 channels **[204x128]** – more signal than ever before with all the channels needed to support ultra-high density coils.

DirectRF – Tim's new all digital-in / digital-out design integrates all RF transmit and receive components at the magnet, eliminating analog cables for true signal purity.

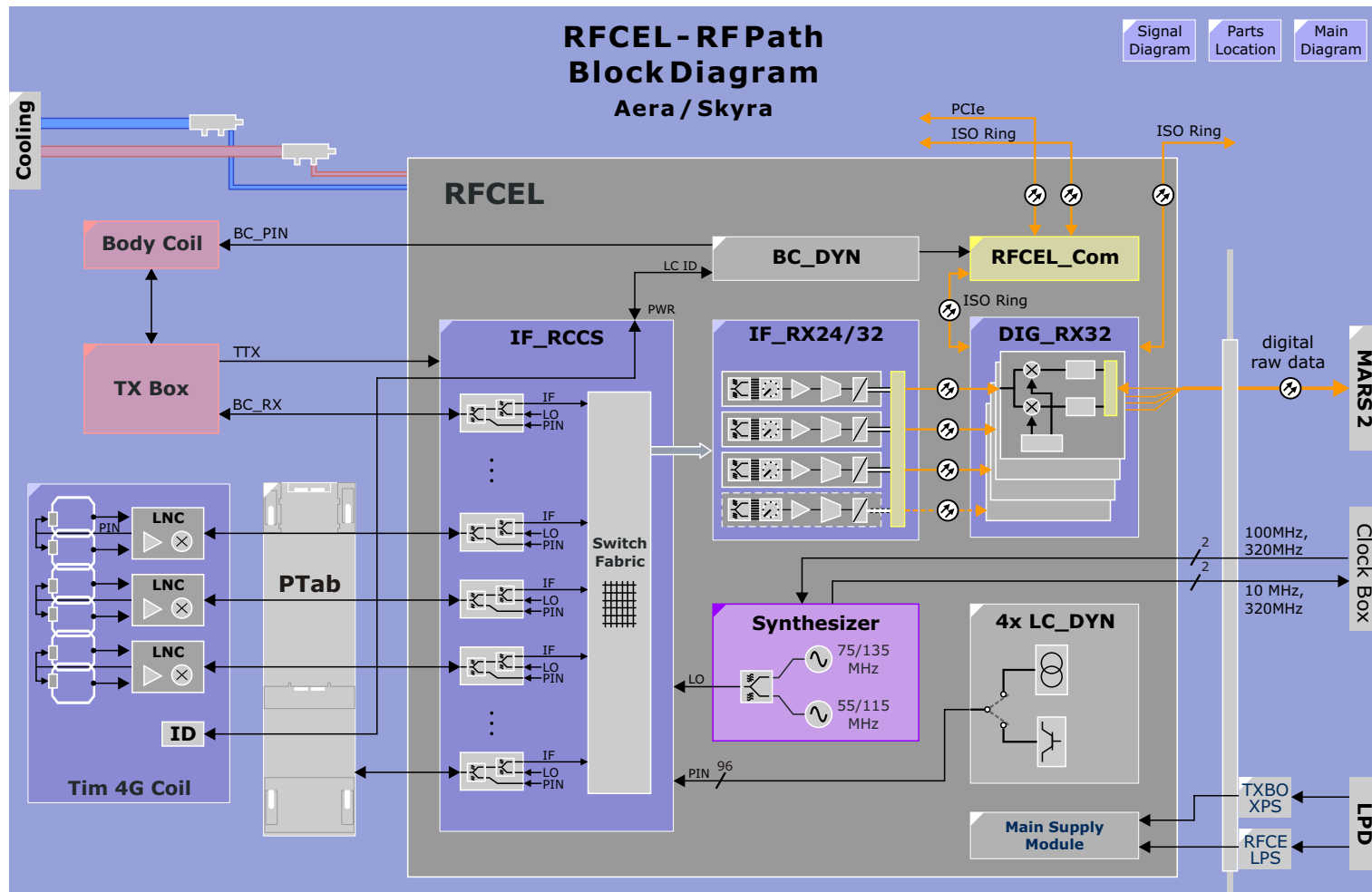
TimTX – Providing excellent B1 homogeneity and scalability to

higher transmission channel configurations with the potential to unlock new applications.

New **DirectConnect™** coils eliminate the hassle of cables – speed exam setup, and improve SNR.

To achieve the high receive coil and receive channel density an IF (intermediate frequency) technology is being used. The MR signals are mixed down to IF frequencies of 8 and 12 MHz inside the coil and bundled, with the use of diplexers, so that only one cable is required per coil element pair. This technique offers several advantages. Signals at IF frequencies incur lower R losses from cables and connectors which results in an improved SN. Also, cables and connectors designed for this frequency range are much cheaper than their RF counterparts. Signal bundling gives the obvious advantage of requiring fewer cables and thus allowing support for a higher number of coil elements. In addition to the IF-converted MR signals, the same signal cables are used to transfer the coil detune signals as well, providing an additional reduction in cabling.

Figure 53 RFCEL - RF Path Block Diagram



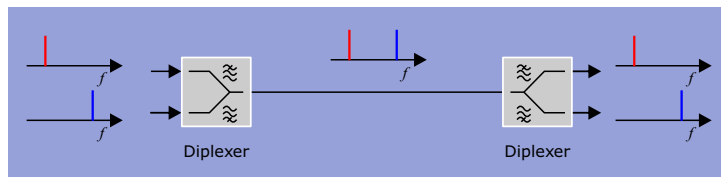
RX Path

Signal Processing

Figure 55, page 79 provides an overview of the overall signal flow and processing from the receive coil elements to the receiver components.

The MR signals picked up by the coil elements are input to a Low Noise Converter (**LNC**). Each LNC serves an adjacent coil element pair, so Tim4G coils will always have an even number of coil elements. The signals from the coil elements will be at the system frequency of 63 or 123 MHz (the 1H frequency). These will first be amplified with low-noise amplifiers. The amplified signals are applied to mixers whose LO signals are supplied by the Synthesizer. The two intermediate frequencies are defined by the MR frequency and the two individual LO frequencies. The LO frequencies are chosen so that the two intermediate frequencies are located symmetrically above and below the 10MHz system sampling frequency, i.e., 8,2MHz and 11,8MHz respectively (10MHz $\pm$ 1,8MHz). Filters in front of the mixers provide image rejection (removal of "mirror frequencies" developed in the mixing process) and remove noise outside the receive band to further improve signal quality. The combining of the two IF-signals and the decoupling of the LO and detune signals from the coax interface is done by means of cascaded diplex filters. Figure 54 shows how a diplexer is used to combine or separate an IF signal pair. The LNC outputs go out the coils through the local coils connectors and cabling on the Patient Table and input to the IF_RCCS.

Figure 54 FDM - Diplexers



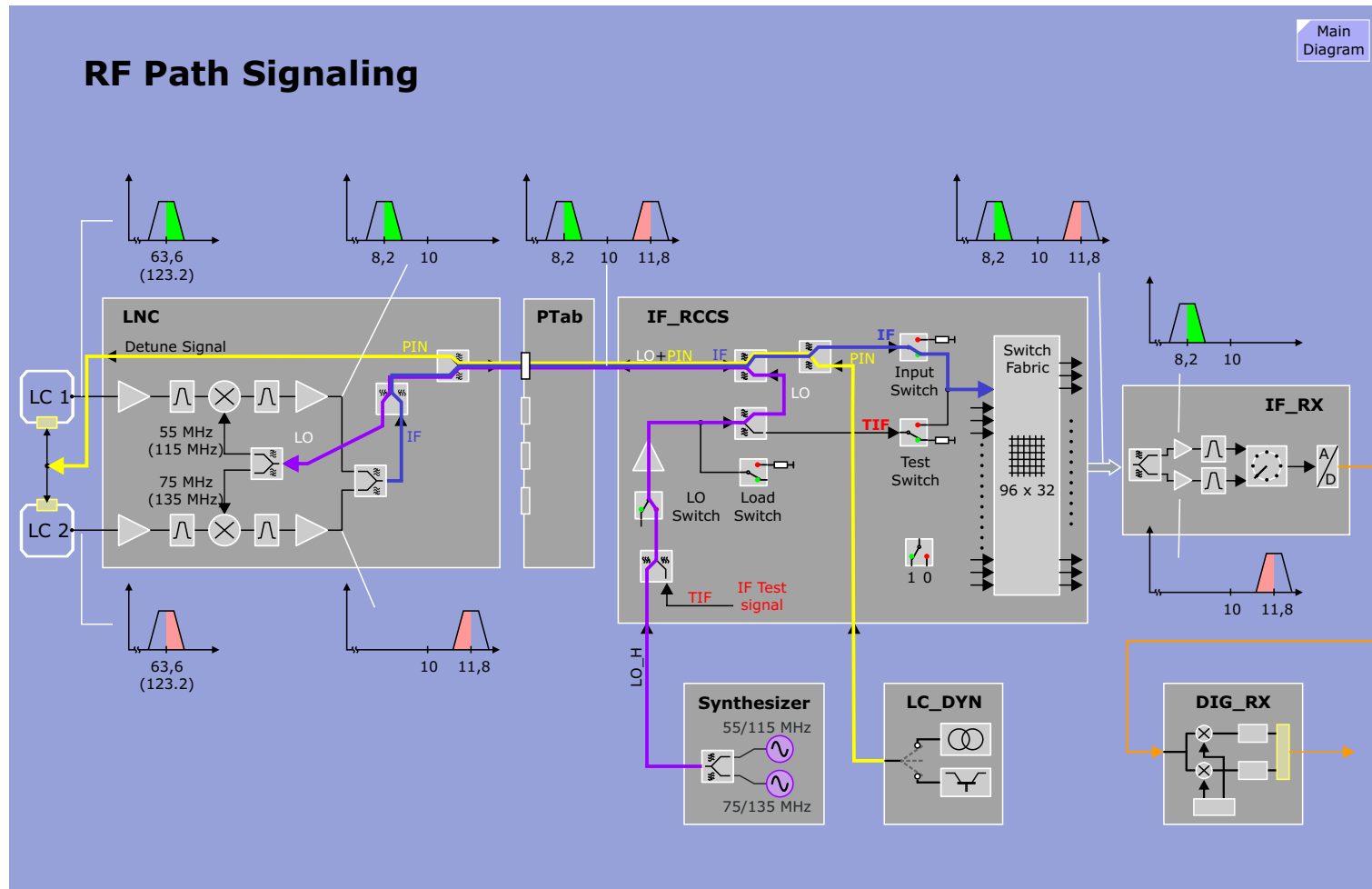
The **IF_RCCS** has the task of routing the IF inputs from the 8 coil connectors to the IF_RX24/32 boards using a Switch Fabric, an array of programmable switch nodes.

The IF_RCCS also provides the diplexers for combining/separating the LO mix frequencies and the coil detuning signals (PIN) to the coils from the IF signals coming in from the coils as well as switches used to insert 50 ohm terminators and/or redirect the IF signals for testing purposes. More on testing can be found in the Troubleshooting Guide (TSG) and in the description of the Service Plug located in the Service Tools section near the end of this document.

The IF signals will be routed through the Switch Fabric to one of the **IF_RX24/32** boards. The IF_RX24/32 has diplexers to separate the 8 and 12 MHz IF signal pairs into separate signals which are then digitized after being amplified and filtered once more.

The digitized IF signals will be sent to one last component (the **DIG_RX32**, not shown) where they will be processed and reduced to the baseband raw data through demodulation and decimation filtering. The raw data is then sent, over high-speed fiber optic cabling, to the image reconstruction processor in the **MARS 2**.

Figure 55 Receive Path Overview



IF_RCCS

Inputs

The IF_RCCS acts as a distribution terminal, or gateway, for the signals being routed to and from the Local Coils:

Signal	Description
PIN 1-96	PIN diode control signals
LO_H, LO_X	FDM IF LO signals for 1H (LO_H) and spectroscopy (LO_X) frequencies
LC_IF_In	FDM IF MR signals at 8 and 12 MHz
BC_RX	When imaging - receive signal from Body Coil When testing - the TIF signal
MR Test	a test signal at the MR frequency. Currently not used
3V, 10V	3V, 10V supplies for Local Coils
Board_Ctrl	see also RFCEL_Com description
SPI	serial interface to program and select setting tables
LC_Code	I2C bus to read out LC EEPROM containing coil information

Function

The IF_RCCS incorporates the following functions:

- IF-RX signal multiplexing
 - LO signal distribution
 - PIN diode control signal distribution
 - IF-Test signal routing
- IF-RX signal routing (Switch Fabric)

Board Control

There are several switches on the IF_RCCS board which must be set prior to each scan. These switches are controlled by FPGAs on the board which are programmed by the RFCEL_Com via an SPI interface. The maximum programming time should be no longer than 500ms. The function of these switches are described below.

- **Path Select Switches**
There are several path-select switches located in the LC Input Multiplexers and the Body Coil Input Multiplexers to be

set for spectroscopy and for testing. The Input Multiplexers are described in more detail on following pages.

- **Switch Fabric Switches**

The Switch Fabric consists of a programmable 96 x 32 array in which any of the 96 input receive lines can be routed to any of the up to 32 output lines which are connected to the first 2 of the IF_RX24 or IF_RX32 boards.

- For the 24 and 48 channel configurations the IF_RX24 boards will be used and the IF signals outputs will be routed in bundles of 12 (see diaram).
- For the 64 channel configuration the IF outputs are bundled in groups of 16 (see diaram).

- **Output Extension**

The additional Output Extension switch array can route only the input lines 65 through 96 to a second set of 32 output lines (**IF_Out 33-64**) which are hard-wired to the third and forth IF_RX32 boards. This section is only required for those coils having more than 64 coil elements. In which case a flexible routing of the input-to-output lines is not required.

- **Input Extension**

When using the Body Coil for receiving, the received signals from the Body Coil are switched to the first IF_RX24/32 board over the Input Extension array. The Input extension supplies up to 8 inputs. The number of actual required is dependant on the Body Coil type.

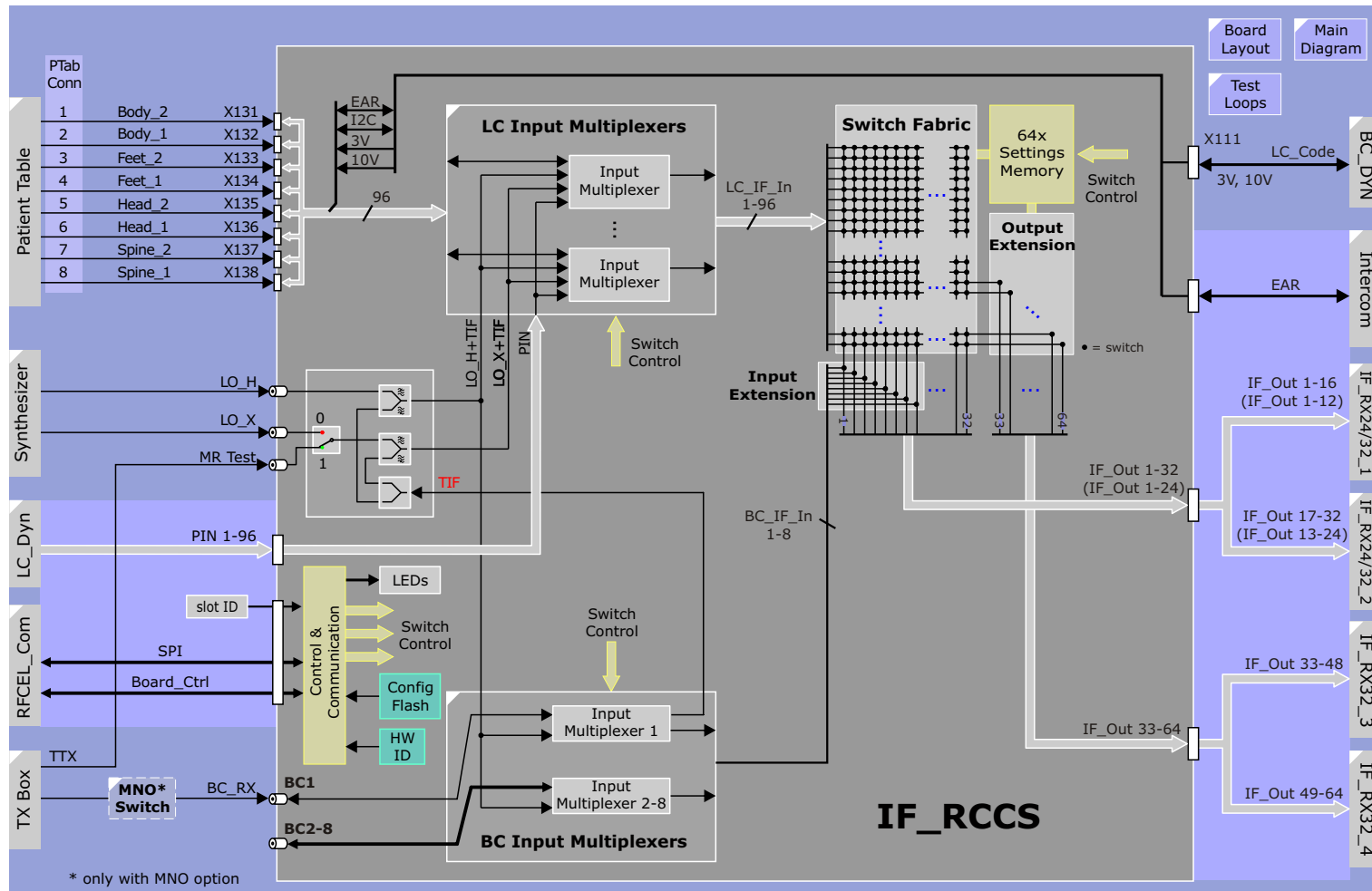
Switch Setting Monitoring

The FPGAs can detect these four errors:

- input setting programmed more than once
- invalid input switch setting
- output setting programmed more than once
- invalid output switch setting

These errors are reported back to the RFCEL_Com over the SPI interface.

Figure 56 IF_RCCS Block Diagram

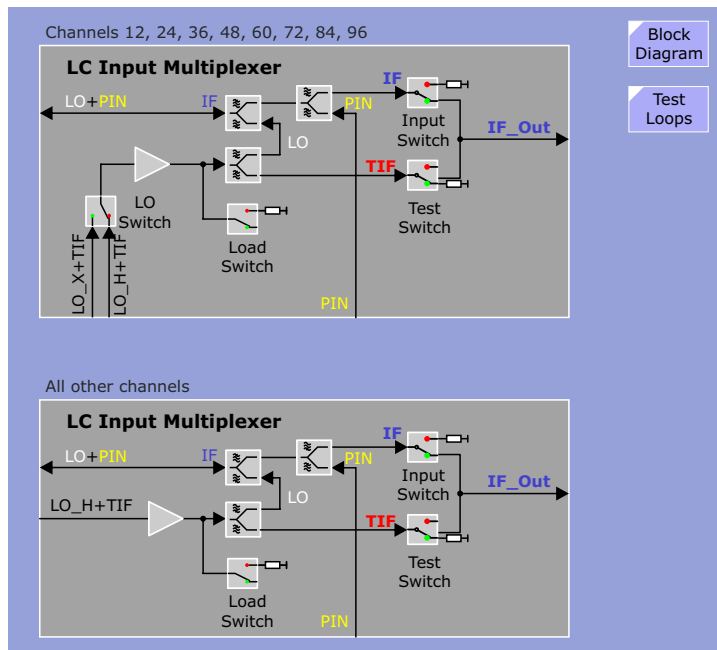


Local Coil Input Multiplexers

The diplexing of the IF LO signals and PIN diode control signal as well as the routing of the IF signals from the coils are performed in the LC Input Multiplexers on the IF_RCCS. Eight of these multiplexers have an additional LO switch which allows applying LO signals whose frequencies are for use in spectroscopy (LO_X).

The TIF signal is a Test IF signal that is used for testing purposes. This will be described in the TSG.

Figure 57 LC Input Multiplexers

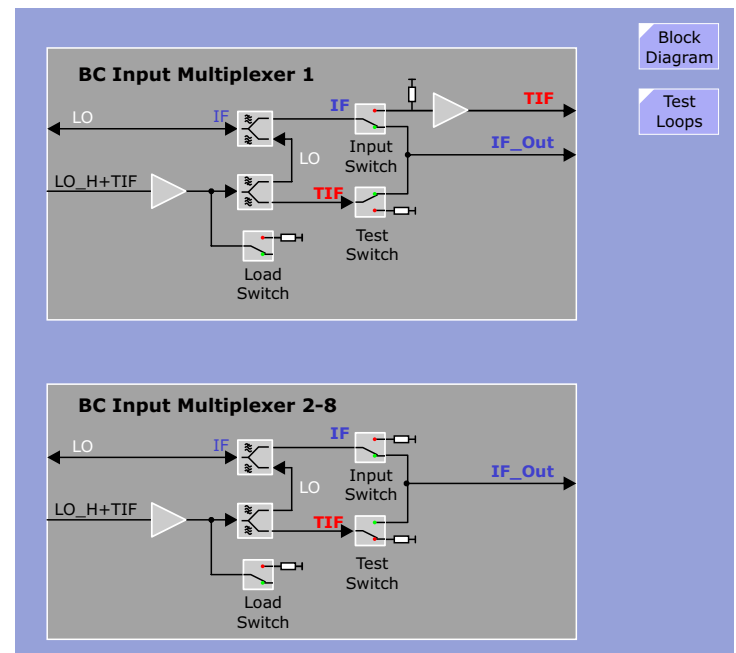


Body Coil Input Multiplexers

There are eight multiplexers for the Body Coil. How many are actually used is dependant on the Body Coil Type. The 1 and 1.5 channel Body Coils will only use BC Input Multiplexer 1, the others can be used for Body Coils having 2 or 8 channels.

BC Multiplexer 1 has an additional Input Switch which is used to route a Test IF signal (see main IF_RCCS block diagram) to all the multiplexers for testing purposes. This will be described in the TSG.

Figure 58 BC Input Multiplexers



Testing

The path-select switches mentioned above can be seen in the diagrams below. These switches are set by the Test software in SeSo.

OFF Mode

In the Off mode no signal path is provided from the external input to the corresponding input of the switch fabric. For each frequency range (IF and LO) a 50Ω termination is routed to the external input through diplexers. In the IF range the corresponding input of the switch fabric is also terminated with a 50Ω which is routed through one of the diplexers.

Internal Loop Test

In the Internal test loop mode an IF test signal can be routed to the input of the switch fabric. The external input is terminated with 50Ω as in the Off mode. The TIF test signal is only available if the input multiplexer in front of input 1 of the input extension is in the Off mode. Here the switches are set to route the TIF test signal to the Switch Fabric for testing the switch fabric node switches.

External Loop Test

In the External test loop mode the TIF test signal can be routed to the external input, i.e., to the local coil connectors or ERNI interconnections. The input to the switch fabric is still terminated with 50Ω as in the Off mode. The IF test signal is only available if the input multiplexer in front of channel 1 of the input extension is in the Off mode. Here the Service Plug is used to test for continuity (i.e., bad cables or connectors) and for testing the lower/higher LO signal amplitude. The TIF signal is acquired by an ADC in the Service Plug and evaluated by the test software.

Passive Mode

The Passive RX mode corresponds to the Imaging mode but no LO signal is available. This mode is provided for test purposes. Usually another input is set to the External test loop mode and is used to send an IF test signal to the external connectors. At the external

connector the service plug can be used to route this test signal to another IF_RCCS input which has to be set to Passive RX mode.

Outputs

Signal	Destination
PIN	Local Coils
FDM LO	Local Coils
3V, 10V	Local Coils
Local coil ID	RCFEL_Com via BC_DYN
EAR	headphone converter at the end of the Patient Table - this signal comes from the INtercom board.
IF_Out	FDM IF signals to IF_RX24/32

LEDs

LED	Description
Done_A	ON when FPGA row A properly configured
Done_B	ON when FPGA row B properly configured
SEL 0	these bit address the 64 Body Coil settings. ON indicates a logical 1.
SEL 1	
SEL 2	
SEL 3	
SEL 4	
SEL 5	
DC_OK	ON when all DC voltages are OK
Fault	ON when FPGAs not configured properly or a DC voltage is not OK
SPI_TX	blinks with SPI interface activity
SPI_CLK	blinks with SPI interface activity

Figure 59 LC Input Multiplexer Settings

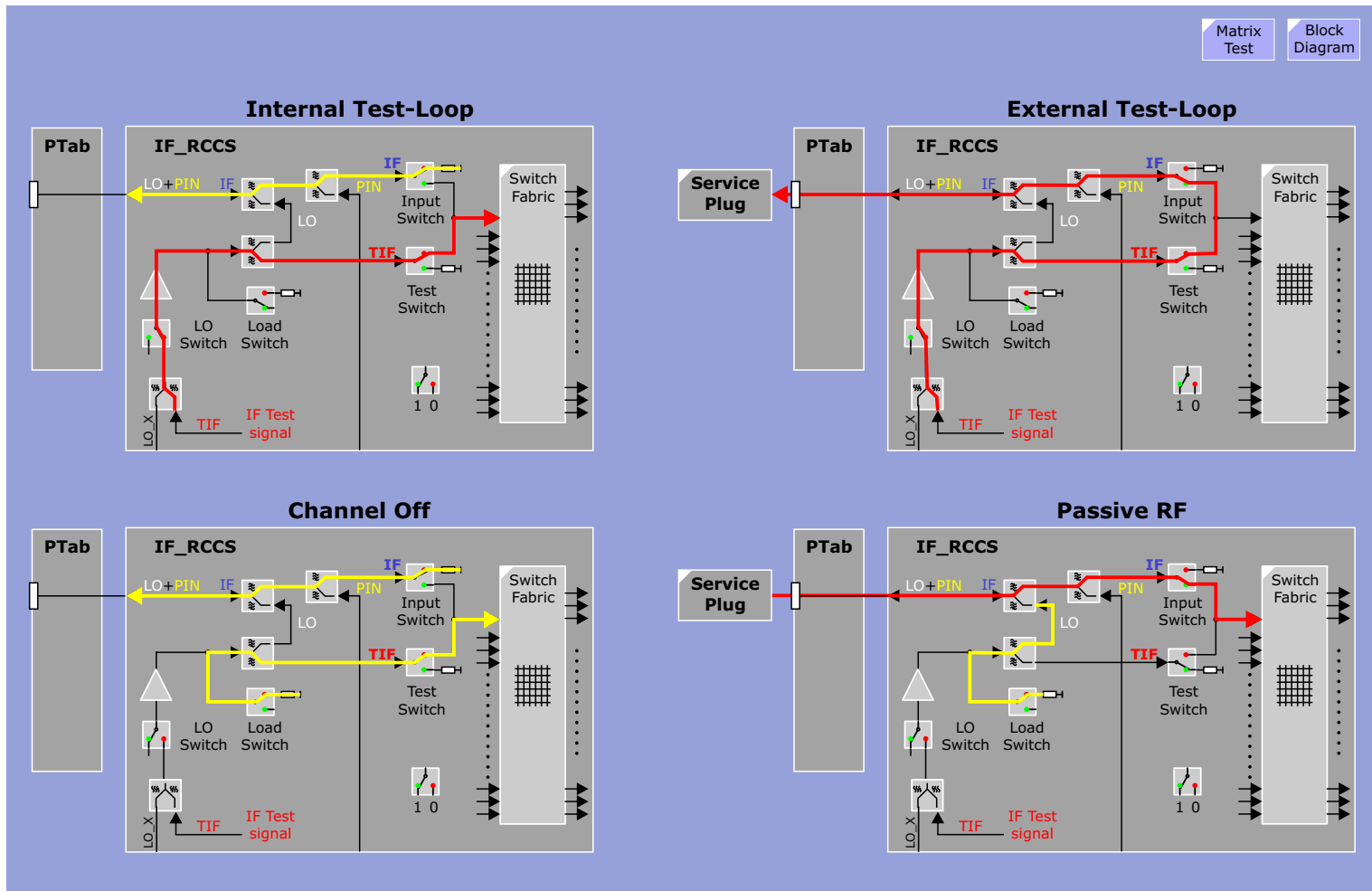
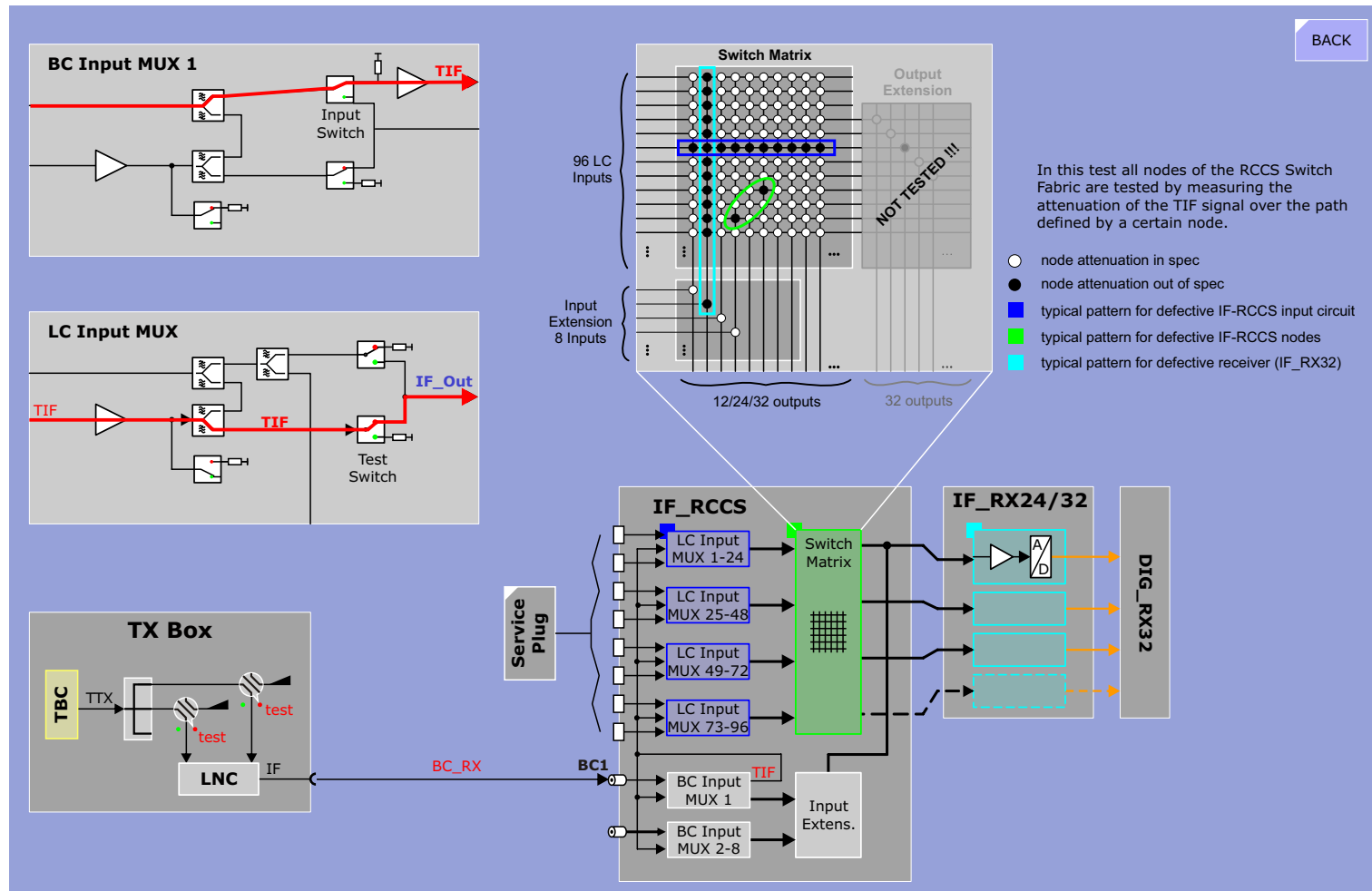


Figure 60 IF_RCCS / IF_RX32 Functional Test



IF_RX24/32

The IF_RX24/32 has the task to amplify and digitize the IF signals routed from the receive coils.

There are two versions of this board:

- IF_RX24 (24 channels) - used for the 24 and 48 channel configurations
- IF_RX32 (32 channels) - used for the 64 and 128 channel configurations

Inputs

Signal	Description
IF_Channel	FDM IF MR signals - frequency-division multiplexed (i.e., the combined 8 MHz and 12 MHz IF signals onto a signal line) from the local coils
Board_Ctrl	See also RFCEL_Com description
Clocks	10, 100 and 160 MHz clocks required by the FGPA and ADC
SGA_Ctrl	SGA high/low gain select signal from the DIG_RX32

Function

The incoming frequency domain multiplexed (FDM) IF-signals are de-multiplexed (split) into the two individual IF signals via diplexers. Each of the single IF signals is amplified, filtered and fed into a time division multiplexer (TDM). In total 8 IF signals (4 x 8,2 MHz + 4 x 11,8 MHz) are multiplexed in the time domain and routed to one single 16-bit ADC. The ADC runs on a fixed sampling rate (SR) of 80MS/s. As a result, each IF signal is continuously sampled with a sampling rate of 10MS/s regardless of the number of channels in use. An analog compressor is used to reduce the amplitude dynamic of the ADC input signal. This signal manipulation is reversed in the digital domain by a lookup table mapping of the individual samples.

The amplifiers are switched gain amplifiers (SGA) and the gain can be reduced by -6dB to provide a high/low-gain mode switch.

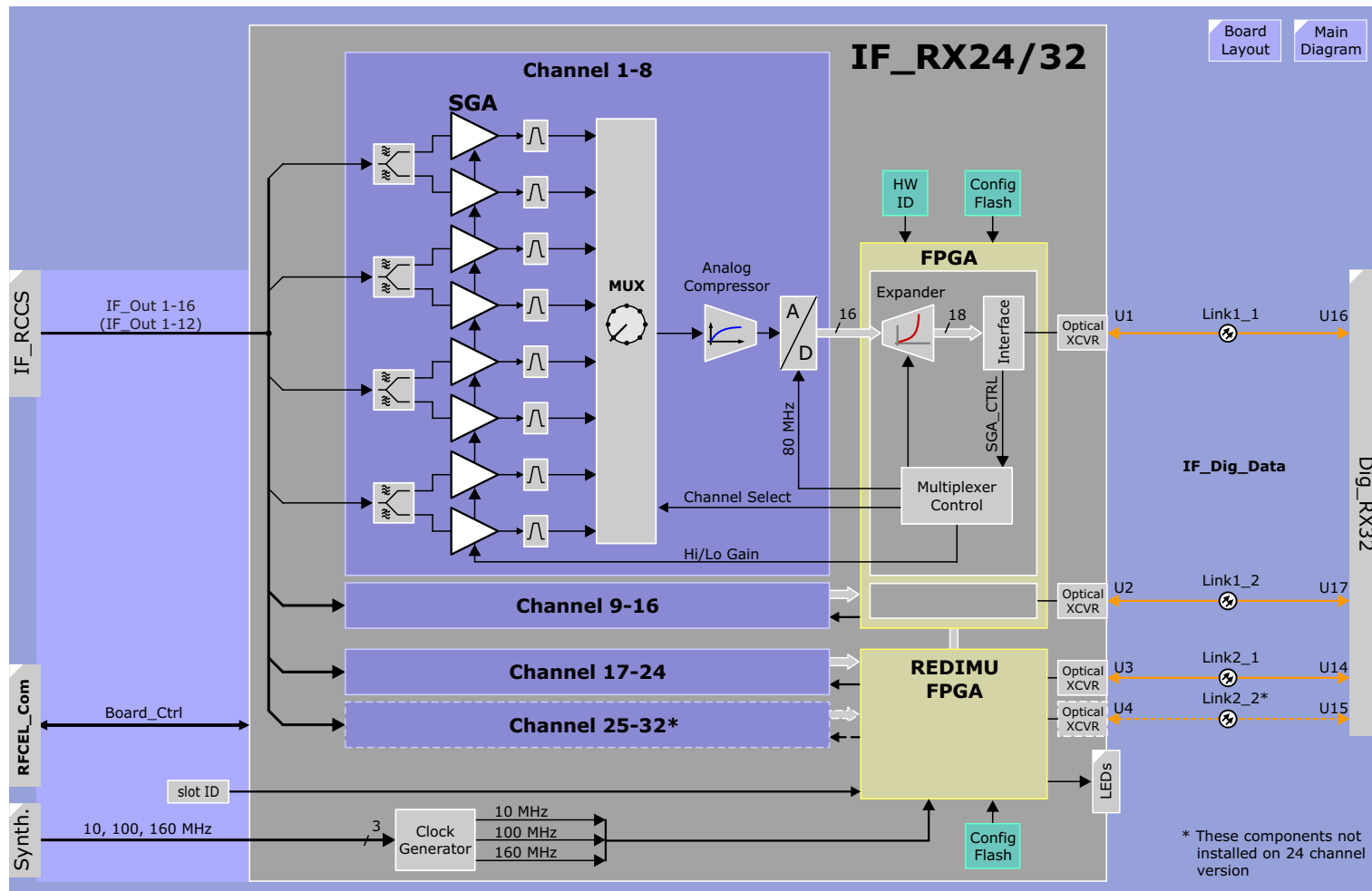
Outputs

The digitized IF signals are de-compressed to an 18-bit dynamic and sent to the DIG_RX32 board via high-speed optical fiber links.

LEDs

LED	Description
Row 1	
V86	Power OK
V85	Config done FPGA 1
V91	Config done FPGA 2
Row 2	
V87 / V88	ON/ON FPGA 1 configured from user page ON/OFF FPGA 1 configured from factory default page OFF/ON FPGA 1 configuring in progress OFF/OFF configuration failure (Config_done LED should also be OFF)
V89 / V90	ON/ON FPGA 2 configured from user page ON/OFF FPGA 2 configured from factory default page OFF/ON FPGA 2 configuring in progress OFF/OFF configuration failure (Config_done LED should also be OFF)

Figure 61 IF_RX32 Block Diagram



DIG_RX32

The DIG_RX32 board can process up to 32 digitized IF signals for digital demodulation and filtering.

Inputs

Signal	Description
IF_Dig_Data	ADC data from IF_RX24/32 board
Clocks	10MHz NCOs, demodulation 100MHz PCIe clock generation 160MHz Ring clock generation
Board_Ctrl	See also RFCEL_Com description
SGA_Ctrl	A command from the MARS 2 received over the PCIe bus

Function

The incoming digital IF signals are quadrature demodulated resulting in baseband data ($\Delta f = \text{PBW} \cdot \text{MA}$) which are low-pass filtered and decimated (sample rate reduction). The resultant data, called "measurement data", are then transferred to the MARS 2 computer via the two optical PCI Express lanes.

The main components of the board are:

- GIMLI FPGA - PCI express and ISO-Ring interface
- Control FPGA
- 2 MOFI16 FPGAs for demodulation and filtering
- 128 Mbyte DDR-RAM Data FIFO to overcome PCI bus latency
- 16 Mbyte QDR-RAM to group one echo for each channel
- Flash configuration to configure the four FPGAs

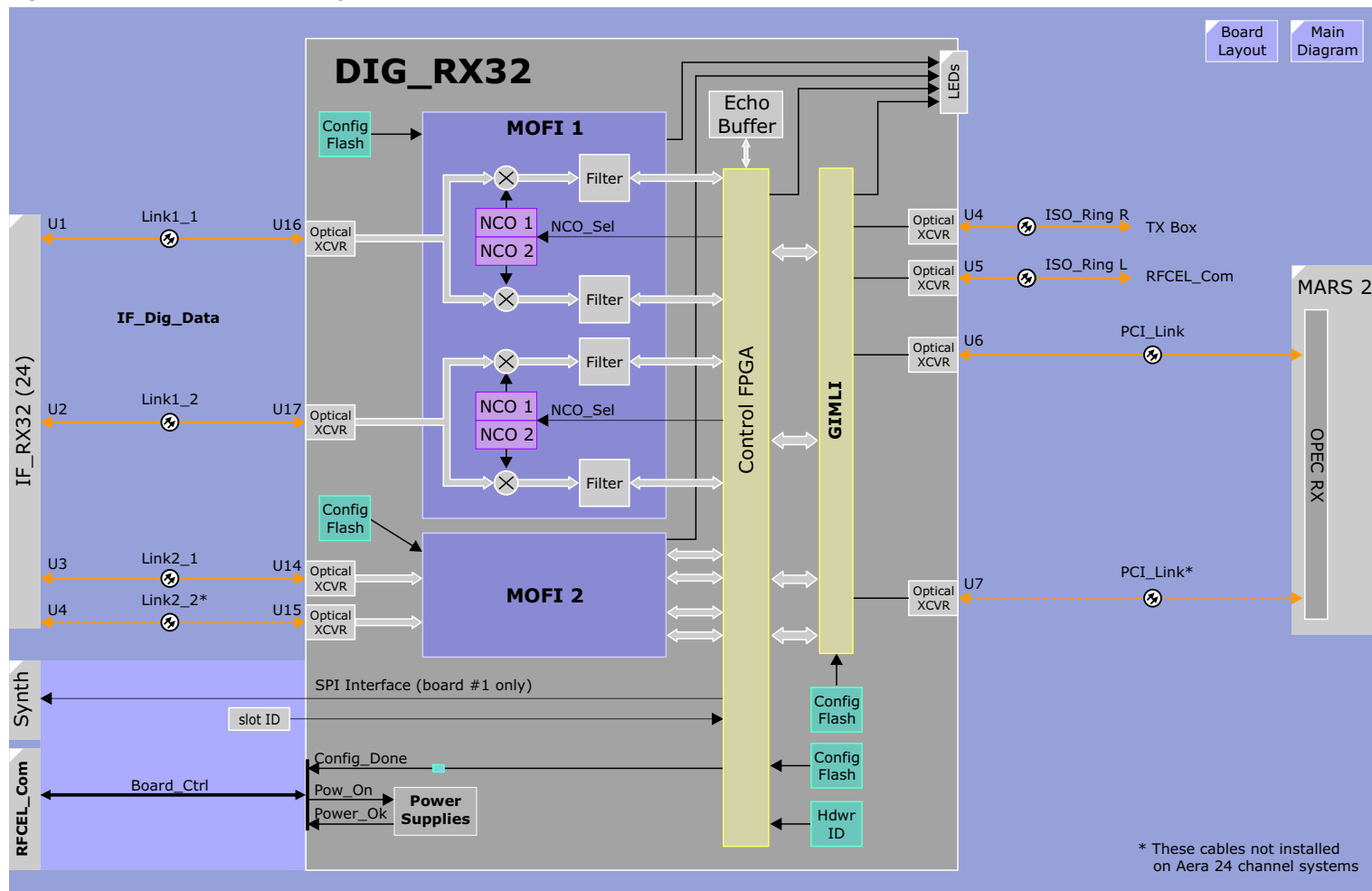
Outputs

Signal	Description
PCI Link	Digital measurement data to the MARS 2
SPI	SPI control interface to the Synthesizer
SGA_Ctrl	selection of high/low gain on IF_RX24/32

LEDs

LED	Name	Normal State	Description
Row 1 (left to right)			
V57	MOFI 1 config. done	ON	MOFI1 FPGA properly configured
V50	MOFI 1 Rx locked link1	ON	link to IF_RX24/32_1 is up
V51	MOFI 1 Rx locked link2	ON	link to IF_RX24/32_2 is up
V49	MOFI 1 filter data ready	OFF	Indicates that filter data of MOFI1 FPGA are available
Row 2			
V46	MOFI 2 config. done	ON	MOFI2 FPGA properly configured
V5	MOFI 2 Rx locked link1	ON	link to IF_RX32_3 is up
V40	MOFI 2 Rx locked link2	ON	link to IF_RX32_4 is up
V3	MOFI 2 filter data ready	OFF	Indicates MOFI2 filter data available
Row 3			
V19/ V20	PCIe lane status	OFF/ ON	OFF /ON 2 lanes up ON/OFF 1 lane up ON/ON Error
V6	Fatal Fault	OFF	STIMO or PALI has stopped the measurement
V26	GIMLI heartbeat	blink	GIMLI is operative
Row 4			
V4	Control FPGA config done	ON	Control FPGA properly configured
V8	ADC data request	OFF	Data request to the ADC of the IF_RX_ board is active
V9	Application Clock PLL locked	ON	Application clocks in the Control FPGA are properly available
V10	DMA Clock PLL locked	ON	Communication clock from the GIMLI FPGA is properly available

Figure 62 DIG_RX32 Block Diagram



Coil Control

The Coil Control functions include:

- static/dynamic coil tuning/detuning (coil selection)
- coil recognition

Coil Detuning

All transmit and receive coils require tune/detune signals to enable or disable the coil elements during the transmit and receive phases as needed:

- for receive only coils, to detune during transmission and tune during reception
- for transmit/receive capable local coils the control of the transmit and receive coil parts will be according to coil design

Coil detuning can be performed statically or dynamically, dependant on the coil selection in the protocol. All receive coils connected but not selected within the protocol will be statically detuned. Only receive coils selected in the protocol will be dynamically detuned and thus active during the receive cycle.

The **LC_DYN** supplies the tune/detune signals for the receive and transmit-capable local coils. The **BC_DYN** supplies the detune signals for the Body Coil.

To reduce heating effects the detune signal has 2 states:

- 100mA for active imaging
- 10mA for standby state

Coil Recognition

All local coils, transmit or receive, utilize EEPROM chips to store pertinent coil information including coil ID, coil name, serial number, manufacturer information, and serial number of the system the coil was last used on. The contents of the ID EEPROM is read out over a I2C bus task of identifying which local coils have been plugged in.

The **BC_DYN** provides the coil change circuitry and will interrupt

the RFCEL_COM if any of the 8 coil plugs have changed state. The RFCEL_Com can read the EEPROM of the coil via an I2C bus which runs through the BC_DYN and IF_RCCS.

This change will be reported by the RFCEL_COM to the measurement SW.

BC_Dyn

The BC_Dyn is part of the Coil Control section and provides several functions not relating to Body Coil control.

Inputs

Board control signals from the RFCEL_Com. [See here.](#)

Function

This board provides these functions:

- Local Coil Supply (3V and 10V)
- Power Supervision
- Heatsink temperature supervision
- Local Coil Plug Check (change detection)
- Local Coil Recognition (I²C bus)
- Body Coil Detune (PIN diode control)

Local Coil Supply

A 3V and a 10V supply provide power for the Local Coils. The 3V is generated from the 6.8V supply and the 10V from the 15V supply delivered to the board from the MSM over the backplane.

These voltages are monitored. Should a short circuit occur the supply will be disabled for protection.

Power Supervision

All voltages used on-board are hardware monitored. A failing voltage will cause an interrupt to the RFCEL_Com which will report it to host and abort the sequence.

Heat Sink Temp Monitoring

The temperature of the heat sinks used to cool power transistors on the board are monitored. If a temperature limit is exceeded an error message will be issued to the Host via the RFCEL_Com. The board will also be disabled and the sequence stopped if running.

Local Coil Plug Check

Each coil plug has two pins which are recessed, i.e., shorter than the others and so are last to make contact as a coil is plugged in or first to break contact when being pulled out. This feature is used to accomplish two tasks:

- to enable the power supply only after all connector pins have made contact or to disable the power supply before the plug pins have broken contact. This is done to prevent spikes on the 3V supply.
- to indicate a coil change so the RFCEL_Com can read out a coil ID or inform the Host that a coil has been removed.

Coil Recognition

In keeping with the new Asset Management strategy all Local Coils implement a EEPROM containing its coil code as well as the following manufacturing information:

- Coil Name
- Part number
- Serial number
- Revision level (for a given Version)
- Version level (PCB layout)
- Manufacturer ID
- Frequency / B0
- Serial # of system on which the coil has last used
- System Family

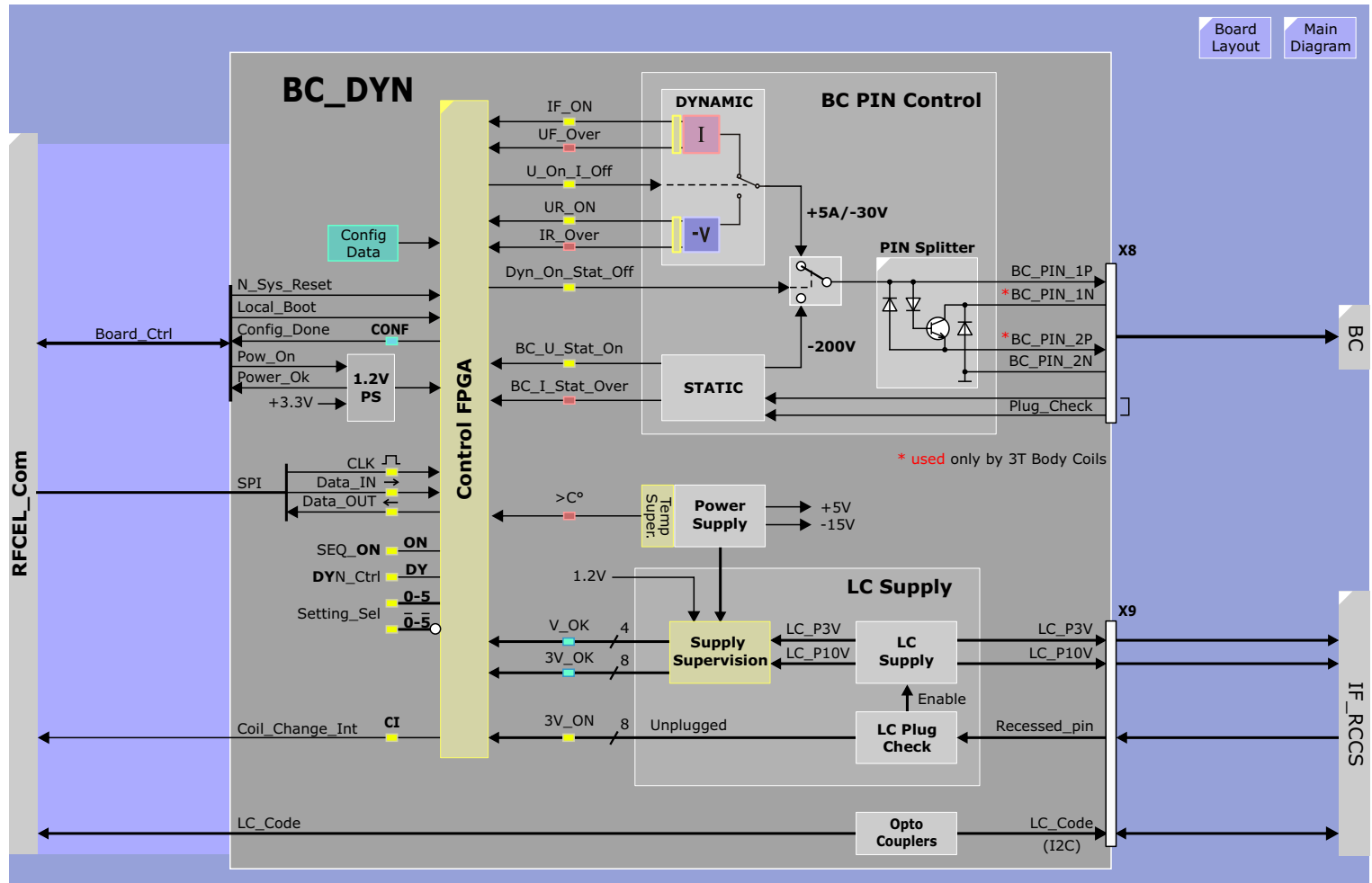
Coil code consists of two pieces of information:

- the code itself
- the EEPROM address (addresses used to indicate coil position for coils not fixed on table, e.g., Body18)

Plugging a coil (in or out) generates an interrupt. On interrupt all 8 addresses of the respective plug are read out and evaluated.

Should the interrupt occur during a scan, the scan will be aborted.

Figure 63 BC_DYN Block Diagram



Body Coil Detune

PIN diodes placed in the center of the Body Coil resonator elements are used to enable and disable the coil during the sequence during transmission and reception. Closing the diodes (current mode) tune the coil, opening the diodes (voltage mode) detunes the coil. When using a transmit-capable local coil, the Body Coil will be statically detuned by applying a very high voltage across the diodes.

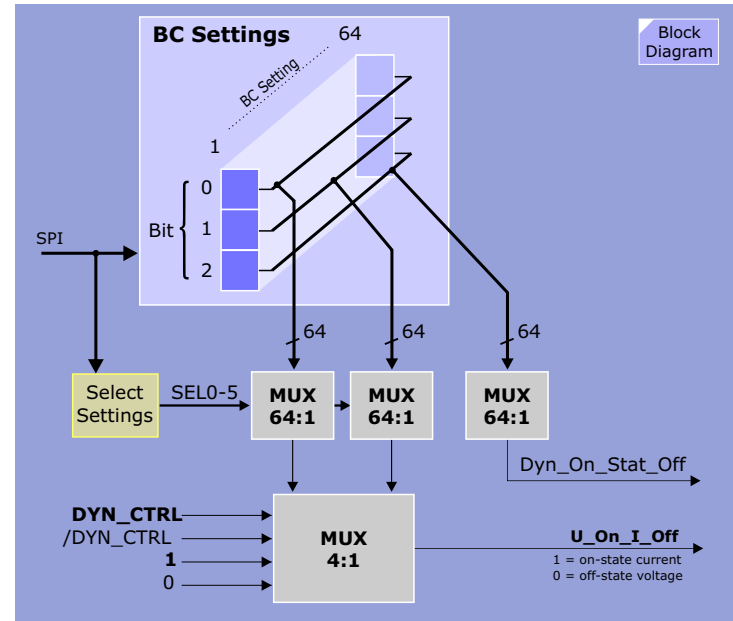
Tune State	Mode	Output to PINs
dynamically tuned	current	+5A $\pm$ 0,35 A (max voltage = 12V $\pm$ 2V)
dynamically detuned	voltage	-30 V $\pm$ 3V (max current = 60mA $\pm$ 35mA)
statically detuned	voltage	-200 V $\pm$ 20V (min. current = 1 mA)

Theoretically, the detune mode can be changed during the sequence with the use of the BC Settings which allow up to 64 different settings. The contents of the Bits 0-2 determine the output states of the **U_ON_I_OFF** and **DYN_ON_STAT_OFF** signals. The **SEL0-5** signals determine which of the 64 settings will be used to address the MUXs.

Body Coil plug Check

To prevent electrical shocks from the 200V a jumper on the PIN signal cable from the Body Coil acts as an interlock disabling the 200V should the cable be pulled.

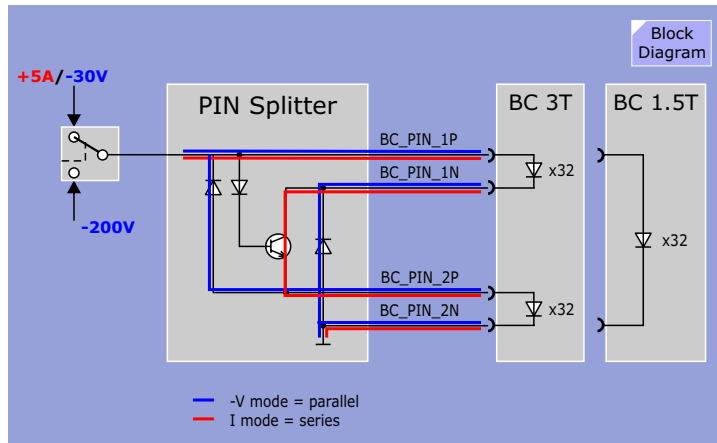
Figure 64 BC Settings



PIN Splitter

The Skyra Body Coils has double the number of PIN diodes (64) than Aera (32). If all 64 PIN diodes were in parallel the amount of current for enabling would exceed the 5A supply capability of the PIN diode control on the BC_DYN. To overcome this problem the PIN diodes in the Body Coil have been placed into two circuits of 32 diodes each. Thus when applying the 5A enable current the PIN Splitter supplies the 5A to these two circuits serially so that each circuit has 5A for each set of 32 diodes. When applying the disable voltage of -30V or -200V, the PIN Splitter supplies the voltage to the two circuits in parallel so that all 64 diodes get the same voltage. The polarity of the PIN diode signal switches the PIN Splitter between serial or parallel modes.

Figure 65 PIN Splitter



Outputs

- PIN control signals to Body Coil.
- Local Coil supply voltages

LEDs

LED	Description
	SPI clock active
	SPI data in
	SPI data out
CONF	indicates the FPGA was properly configured. This LED must be ON
ON	A sequence is running. Can be verified by a loud knocking.
Dy	Dynamic control signal - active
CI	coil interrupt - ON when a coil has been connected or disconnected.
>C°	heat sink temp too high
Setting Select 0-5	Settings memory addressing signals.
BC Funct U	U_ON_I_OFF - indicates whether -30V (ON) or 5A (OFF) is being switched to the output.
BC Funct D	DYN_ON_STAT_OFF - in dynamic mode when ON, in static mode when OFF
-200 V U	U_STAT_ON - -200 volts ok
-200 V I	I_STAT_OVER - too much current in reverse bias mode (possibly a short in cable or diode)
5A I	IF_ON - indicates 5A level is ok
5A U	UF_OVER - indicates no I flowing (voltage too high), possibly due to an open in cable or diode
-30 U	UR_On - indicates the -30V output voltage enabled and OK
-30 I	IR_OVER - current in reverse bias mode too high (possibly due to short in cable or diode).
V_OK	ON when +5, -15, +1.2, +10 voltages are OK
3V_ON	ON indicates a local coil is plugged in
3V_OK	ON when 3V voltage is OK

LC_Dyn

Inputs

Board control signals from the RFCEL_Com. [See here.](#)

Function

Local Coil Detuning

This Local Coils must be dynamically or statically detuned during the transmit phases of the sequence. PIN diodes are used as switches to enable or disable each coil element. The table below shows the levels for each:

Tune State	Mode	Output to PINs
tuned	voltage	-30 V $\pm 10\%$ (max short current = 70mA $\pm 20\%$)
active detuned	current	+108mA $\pm 10\%$ (max open voltage $\geq 11.5V$)
passive detuned	current	+10,8mA $\pm 10\%$ (unused coil elements during the RX phase)

Power Supply Monitoring

Several voltages are used by the board. these are monitored and in case of failure an error message will be issued to the Host via the RFCEL_Com.

Heat Sink Temp Monitoring

The temperature of the heat sinks used to cool power transistors on the board are monitored. If a temperature limit is exceeded an error message will be issued to the Host via the RFCEL_Com. The board will also be disabled and the sequence stopped if running.

Outputs

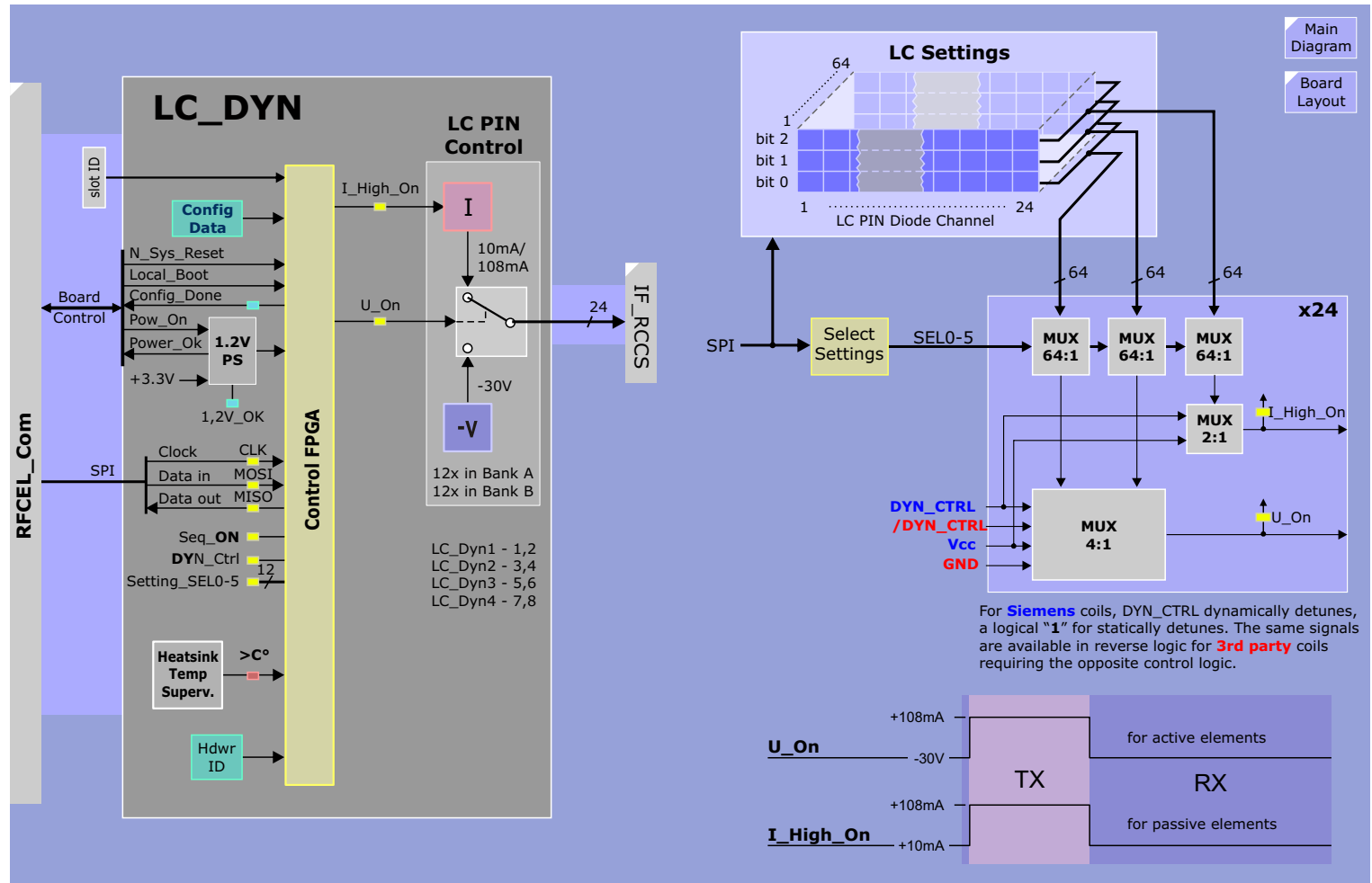
There are a total of 4 LC_DYN boards with each board assigned to a specific local coil connector:

Board	Feeds Connectors:
LC_DYN 1	Body 1, 2 - table connectors 1 and 2
LC_DYN 2	Feet 1, 2 - table connectors 3 and 4
LC_DYN 3	Head 1, 2 - table connectors 5 and 6
LC_DYN 4	Spine 1, 2 - table connectors 7 and 8

LEDs

LED	Description
1.2V_OK	ON when monitored 1.2V is OK
CONFIG_DONE	indicates the FPGA was properly configured. This LED must be ON
Over_Temp	heatsink temp exceeds limit
SPI_CLK	SPI clock active
SPI_MOSI	data in (master out slave in)
SPI_MISO	data out (master in slave out)
SEL 0-5 /SEL 0-5	SEL 0-5, addressing signals for selection of the settings table. The /SEL LEDs are for better visibility
SEQ_ON	A sequence is running. Can be verified by loud knocking sound...
Dyn_Ctrl	Dynamic control signal - active
U_ON	ON when -30V is being switched to output
I_High_ON	ON indicates 100mA being switched on to output only when U_ON is off.

Figure 66 LC_DYN Block Diagram



RFCEL_Com

The RFCEL_Com plays a central role in the control of all units within the RFCEL as well as Patient Handling components. As a ISO-Ring member, it performs the role of a master - the Lord of the Ring.

Inputs

PCIe and ISO-Ring bus - main inputs

Function

Functions performed by the RFCEL_Com:

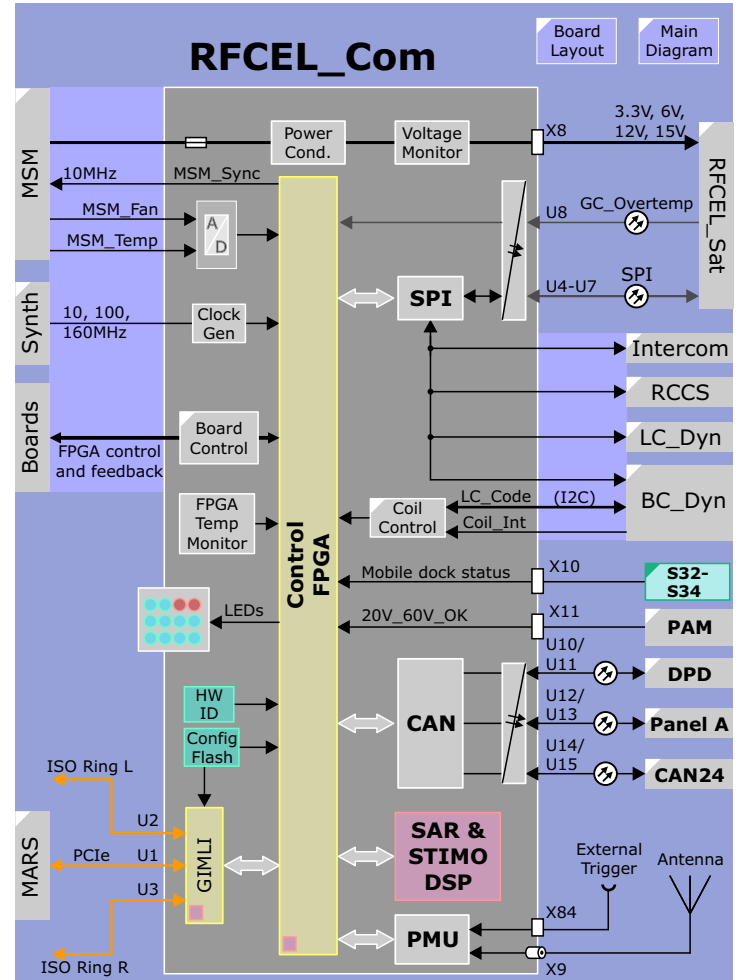
- ring master on GIMLI
- Sequence and PMU counters
- Control-word Instruction Queue for LC_DYN, BC_DYN and IF_RCCS
- Sequence Start, Stop and Halt mechanism
- Real time capture of patient table position
- f0 - interface
- DSP for PALI and STIMO
- PDAU Interface for wireless reception of Physiological Signals
- Coil Interface for read out of serial coil code and general access to the coil EEPROMs
- RFCEL System manager - board control for RFCEL boards

Communications Interface

A communication interface between the MARS 2 and to the following units via dedicated SPI links for control purposes:

- Patient Intercom Main module
- 4 * LC_DYN modules (settings table)
- BC_DYN module (settings table)
- IF_RCCS module
- RFCEL_SAT

Figure 67 RFCEL_Com Block Diagram



Sequence Engine

The sequence engine encompasses the sequence event counter which increments every 10 us. This counter is the central control unit of the scanner and all timing related activities are triggered dependent of the specific instruction header and the state of this counter. Therefore this counter and additional information is provided to all ring participants via the ISO-Ring.

The counter is started by the measurement SW with a start message. It stops on errors or a stop message or when a halt instruction is issued. It can be preloaded by the measurement SW with any value (typically 0) on the beginning of a sequence.

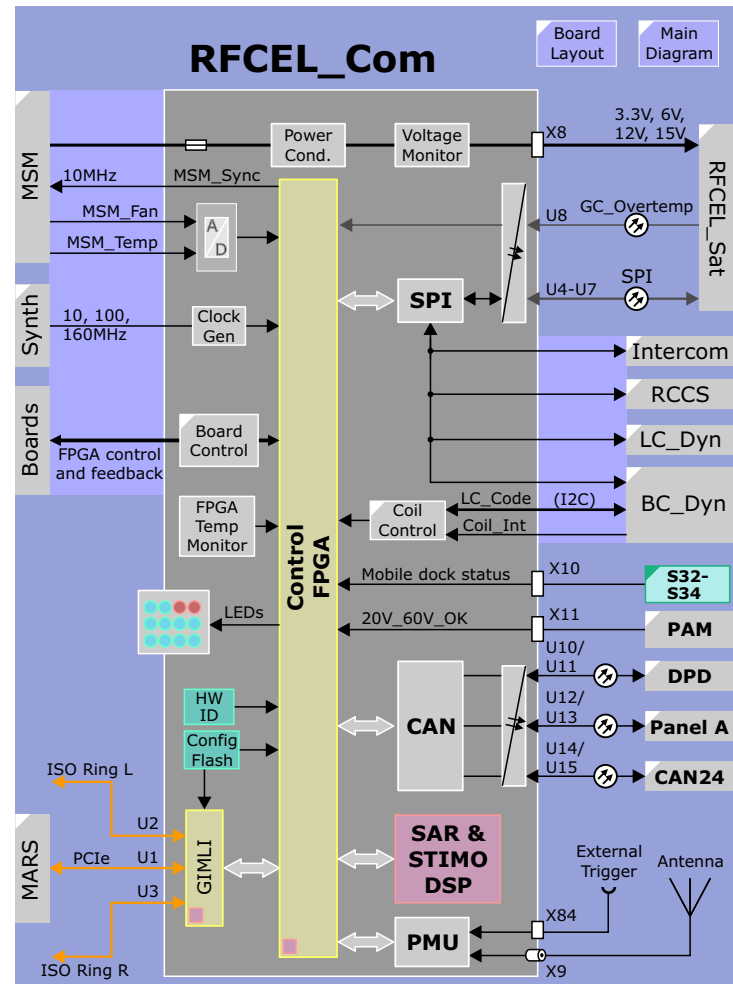
A special feature is the behavior that this counter will deduct 20 digits from its current state whenever it is started. This assures that agents who need a time budget before the original trigger time (e.g. gradient controllers) can start ahead up to 200us to compensate for their specific delays. The deducted phase will be hidden from the SW.

A further mechanism provides a call back function for the SW in the way that a wake up time can be programmed. If the actual sequence event counter is greater than or equal to the wake up time, an interrupt will be generated to the GIMLI. This provides a means for the SW to keep track of the real time situation of the scanner.

Two PMU related counters are also available. The counters increment every 2.5 ms and are transferred to the raw data measurement header via the ISO-Ring. Both counters can be initialized at any time to any value.

Another function in this context is the patient table position capture logic. This functional block snoops the CAN traffic between PTAB and Control Panels and CAN24/7 hub. Whenever a position update is generated by the PTAB, this will be captured and transferred to the raw data measurement header via the ISO-Ring.

Figure 68 RFCEL_Com Block Diagram



RFCEL System Manager - Board Control

The RFCEL_Com System Manager provides a means for a controlled power-up sequence of the RFCEL Rack to prevent power surges on the main supply module. FPGA Done and power OK signals of all boards are also monitored. Additionally it can reload FPGAs on the **BC_DYN**, **LC_DYN** and **IF_RCCS** by images stored in the GIMLI Flash. This feature is used to assure these boards always have the correct config data. After the first time the system is booted, the RFCEL_Com sets a warm start flag. When set, the RFCEL_Com knows it has the latest config data from the Host and boots the FPGAs of the boards directly, saving time.

- **POST**

A self test on power up is performed and if successful will illuminate V27.

- **Board Detection**

The boards which are available in the rack will be detected via their present pins and are flagged in the system status register.

- **FPGA Supervision**

The FPGA versions of the SPI-Slaves will be read out and flagged in the status array. If an update is required this will be performed.

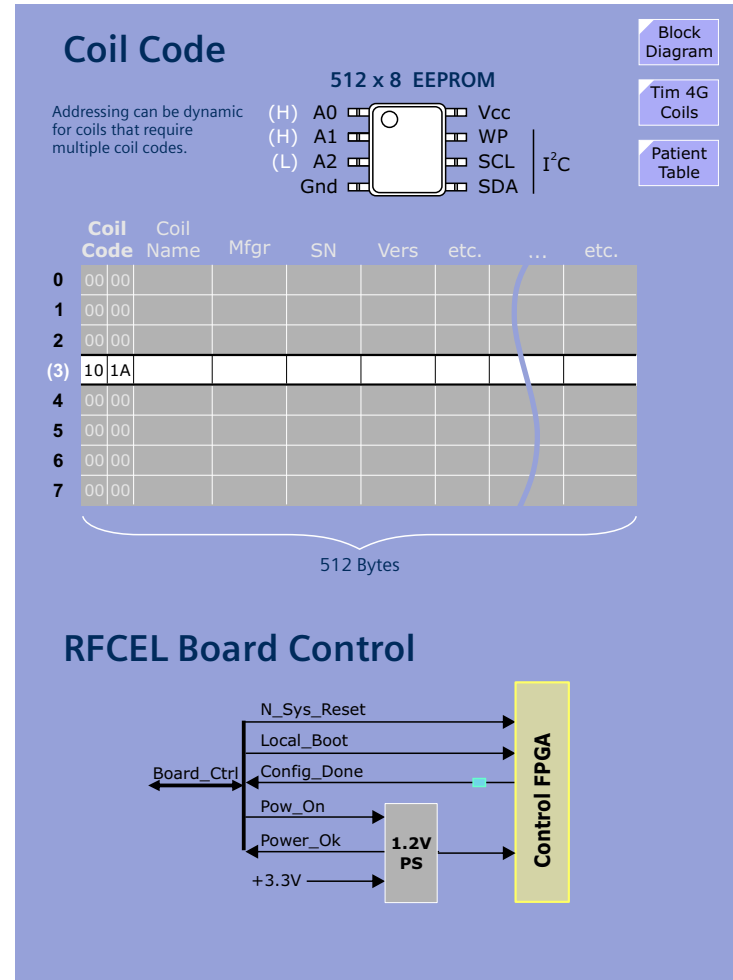
Coil Interface

The Coil Interface provides an access to the I2C EEPROMs residing in the local coils. When plugging or unplugging a coil the coil code will be retrieved at all 8 local coil plugs simultaneously. Afterwards an interrupt will be generated.

Parallel to reading out coil codes a coil consistency check is performed. If an interrupt from the BC_DYN (coil powered) or from the coil itself is issued it is assumed that at least one coil plug has a non-zero coil code present. Otherwise a coil consistency error is issued.

Besides this functionality the Coil Interface provides a means for reading out service relevant information at each plugged coil.

Figure 69 Coil Code and Board Control



DSP & Interface

The DSP provides an In Box and Out Box for communication purposes. Special interrupts are provided for communication and safety switch off features. Also TX narrow band and wide band data from up to 8 TX channels are sourced from the ISO-Ring. Set and current data from 4 gradient boxes are also supplied to the DSP.

The DSP can switch off the RF and gradient transmission via the ISO-Ring. In an emergency situation the TX power supplies and the GPA power stages can be deactivated also via the ISO-Ring.

PMU Interface

The PMU Interface provides an access to the wireless ECG- and Pulse transmitters. When enabled it transfers every 2.5 ms ECG data to the GIMLI.

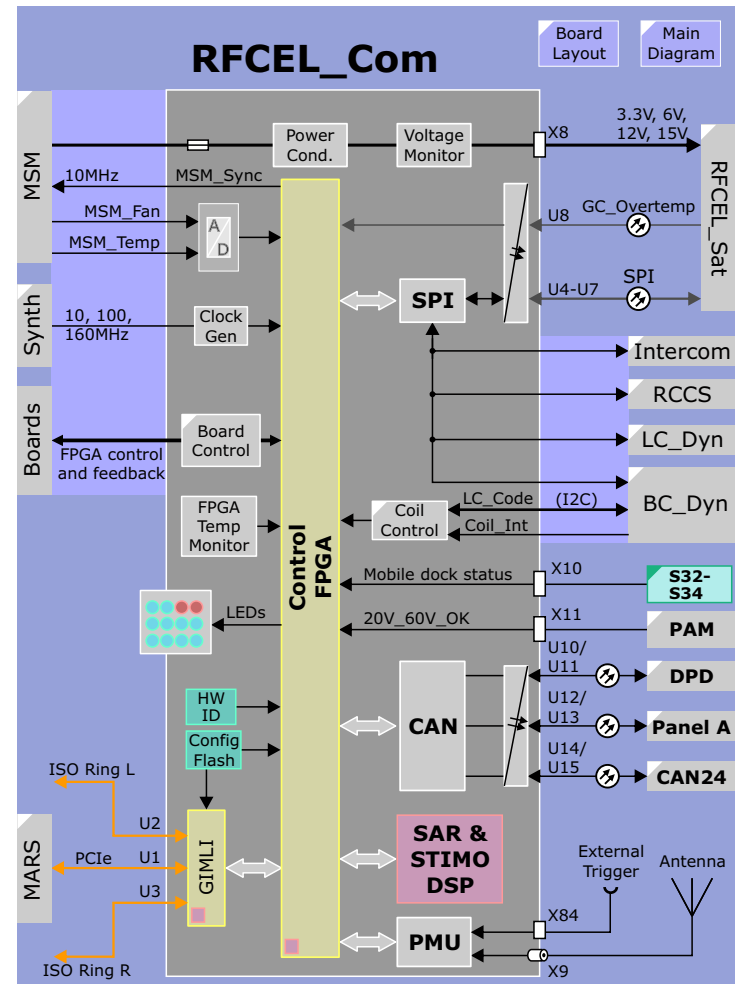
The PMU Interface is also responsible for providing a trigger-out signal to the GPA via the ISO-Ring. It also receives an external trigger in signals from the GPA via ISO-Ring and a special optically decoupled input on the RFCEL_COM front panel.

RFCEL Main Supply Supervision

The RFCEL_COM supervises power lines, fan operation and temperature on the main supply module according the following table:

Module	MSM Status
Power	1.5V Fault
Power	3.3V Fault
Power	6.5V Fault
Power	12V_1 Fault
Power	12V_2 Fault
Power	15V Fault
Power	25V Fault
Power	-32V Fault
FAN	FAN Fault
Temp	Overtemp

Figure 70 RFCEL_Com Block Diagram



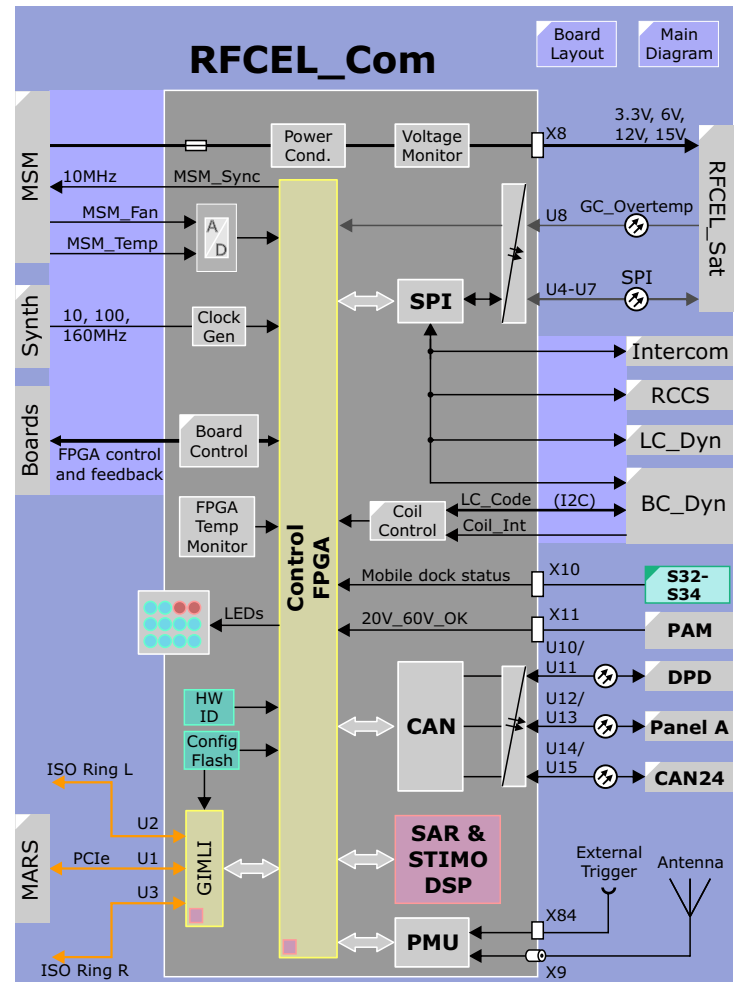
F0 Interface

The RFCEL_COM provides a interface which transfers the NCO portion of 1H and MNO center frequency to all RX and TX devices. This information is placed on ISO-Ring slots 47 and 48.

LEDs

LED	Color	Description
Row 1 (left to right)		
V16	GREEN	blinks when GIMLI FPGA in operation (heartbeat)
V17	GREEN	PCIe connection established
V4	RED	Emergency failure
V5	RED	Fatal fault
Row 2		
V30	GREEN	blinks when CANCELA FPGA in operation
V29	GREEN	ON when MSM power status is OK
V28	GREEN	ON when power-on sequence of RFCEL boards successful
V27	GREEN	ON when all FPGA IDs of RFCEL boards have been successfully retrieved
Row 3		
V36	GREEN	PERU connection established
V37	GREEN	PPU connection established
V32	GREEN	ON when RFCEL_Sat power is OK
V1	GREEN	ON when warmstart flag is set

Figure 71 RFCEL_Com Block Diagram



Main Supply Module

The MSM provides several supply voltages required by various boards within the RFCEL as well as the RFCEL Satellite, Control Panels and displays.

Inputs

- P12V, P15V - from RFCEL power supply on Filter Panel
- N32V, P25V - from TXBOX power supply on Filter Panel

Functions

The MSM module provides the following:

- generation of P1,5V, P3,3V and P6,8V
- Under-voltage supervision for all supply voltages
- Over-voltage, over-current protection for P1,5V, P3,3V, P6,8V
- Cooling air temperature detection
- RFCEL fan control

Voltage Generation

Low voltage supplies requiring high output currents (P1,5V, P3,3V, P6,8V) are generated from the P12V on-board using μ P controlled DC-DC converters.

Monitoring

- Undervoltage detection - supervision levels, see table. If a power fail is detected, the MR sequence has to be stopped immediately.
- The fan speed and therefore the air flow speed inside RFCEL is nearly constant and defined by a speed control input (output of MSM). A speed regulation as a function of the cooling temperature is not planned.
- fan rotation supervision - the fan outputs a supervision signal which is fed through the MSM to the RFCEL_Com.

- The cooling air temperature inside RFCEL is supervised. The sensor is located on the lower edge of the MSM board directly in the flow of the cooled air.
- The most critical position of the MSM is supervised by a **temperature** sensor. There are 2 detection limits:
 - **Limit 1:** If this limit is exceeded, an error message is sent to RFCEL_Com via the signal MSM_Overtemp_Error
 - **Limit 2:** If this limit is exceeded, the DC-DC modules will be switched off at once (component protection).
- The temperature limit values are defined by EPLAX and in dependence of the part temperature resistivity.

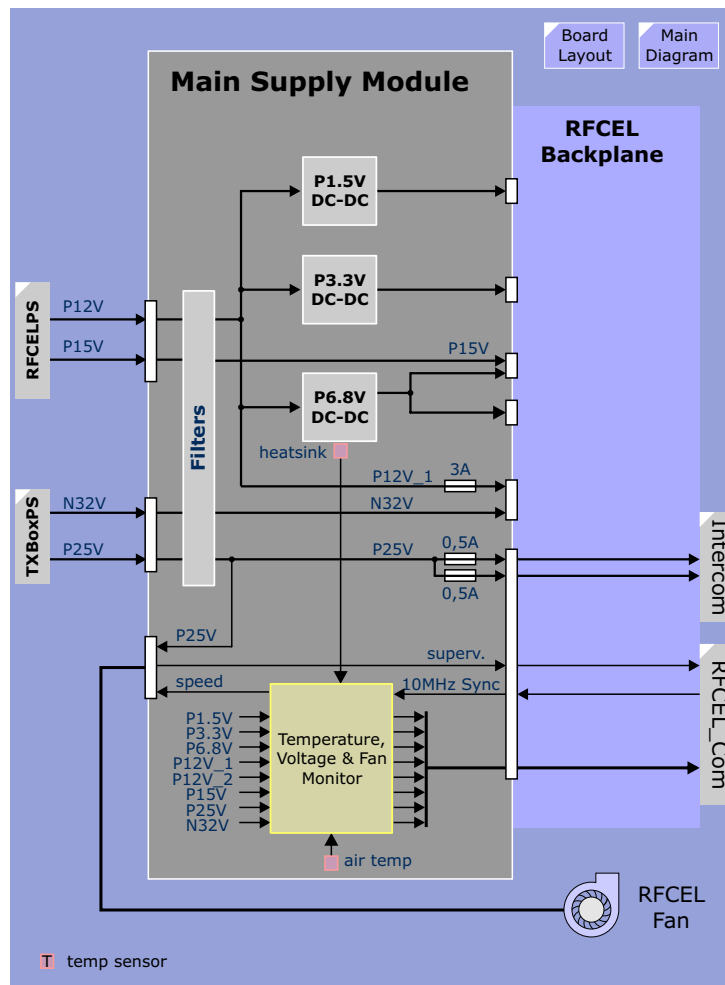
A separate μ P is responsible for DC-DC start up, over voltage supervision and shut down, temperature supervision etc. To reduce dependency on only one supply voltage, this circuit is supplied by P12V and P15V.

Supply Voltage	Undervoltage Limits
P1,5V	1,25V ... 1,35V
P3,3V	2,80V ... 3,00V
P 6,8V	5,60V ... 6,20V
P12V	10,00V ... 10,60V
P15V	12,00V ... 12,80V
P25V	21,50V ... 22,70V
N32V	26,00V ... 27,60V

Outputs

Regulated, filtered, clean, sauber, pure, no ripple DC voltages delivered to all RFCEL boards over the backplane.

Figure 72 Main Supply Module Block Diagram



LEDs

LED	Description
P1,5 V_OK P3,3 V_OK P6.8 V_OK	ON indicates voltage OK OFF indicates power failure blinks twice indicates over/temp blinks thrice indicates over/voltage   Note: A PWR_FAIL for these voltages and an OVTEMP error will occur a short time after a FAN error occurs (during higher load)
P12 V_OK P15 V_OK P25 V_OK N32 V_OK	ON indicates voltage OK OFF indicates power failure
FAN_OK	ON - fan is ok
OVP/ OVTEMP	ON - when an over temp detected BLINK - when over-voltage on P1.5V, P3.3V or P6.8V
PWR_FAIL	ON - when any power failure blinks twice indicates 10MHz clock not ok blinks thrice indicates no 10MHz clock   Note: Stops blinking 30s after the 1.5, 3.3 and 6.8V supplies are switched on.

LED Test

As a visual test both red LEDs will be switched on for 1 second, 1 second after the MSM start up procedure is completed.

Fuses

All fuses are PPTC (polymeric positive temperature coefficient) resettable fuses. Unlike conventional fuses, when excessive current flows through a PTC it heats up and its resistance goes high (open). When it has cooled, the fuse goes to normal a conduction state (low resistance) thus acting as an automatic reset fuse.

Synthesizer

Inputs

Signal	Description
320MHz	system reference clock
100MHz	GIMLI clock for the PCIe network synchronization
SPI	Control inputs for determining the frequency and phase of the DDS, delivered by the first DIG_RX32 board over the backplane via an SPI interface
Reset	

Function

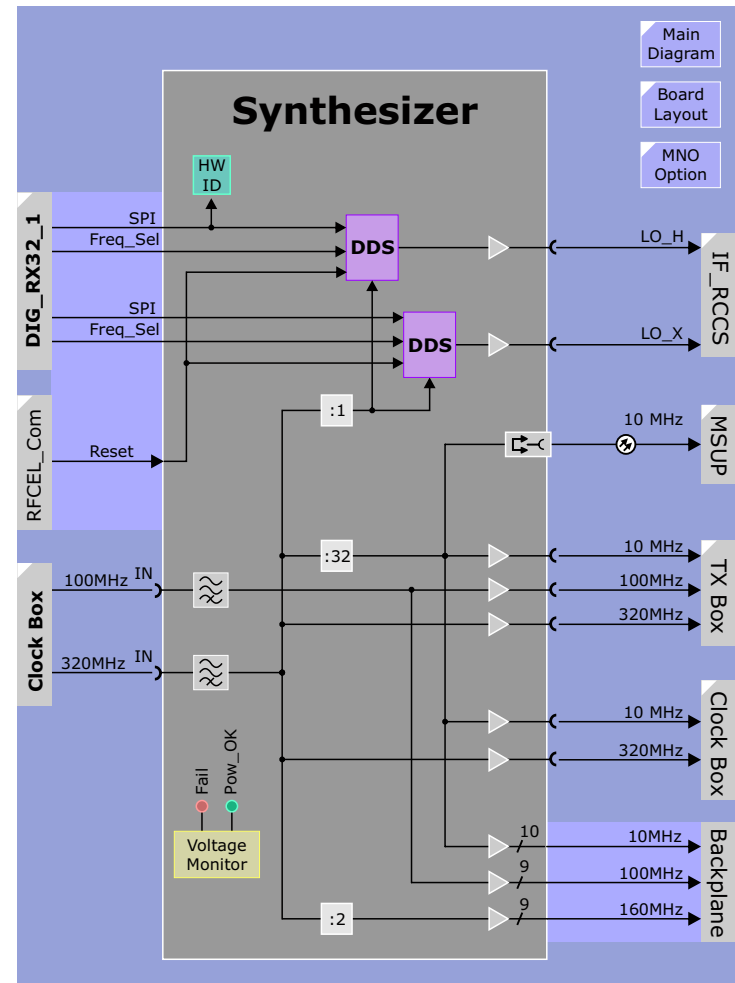
The synthesizer module generates and distributes various clock- and LO-signals.

The 10MHz and 160MHz output clock-signals are derived from the 320MHz reference input clock-signal by frequency dividing. The corresponding divider stages are synchronously reset after system power-up to ensure repeatable timing conditions.

The "two-tone" LO-signals are generated by modulating frequency-agile DDS signals with a fixed 10MHz signal by means of a digital up converter. The DDS/up-converter chip is programmed by means of the first DIG_RX32 module using a serial SPI bus.

The synthesizer LO-path contains frequency selective filters to remove spurious signals (e.g. the DDS alias signals). These filters are switchable to support 1,5T and 3T systems. The LO signal is supplied to the IF-RCCS via coaxial lines.

Figure 73 Synthesizer Block Diagram



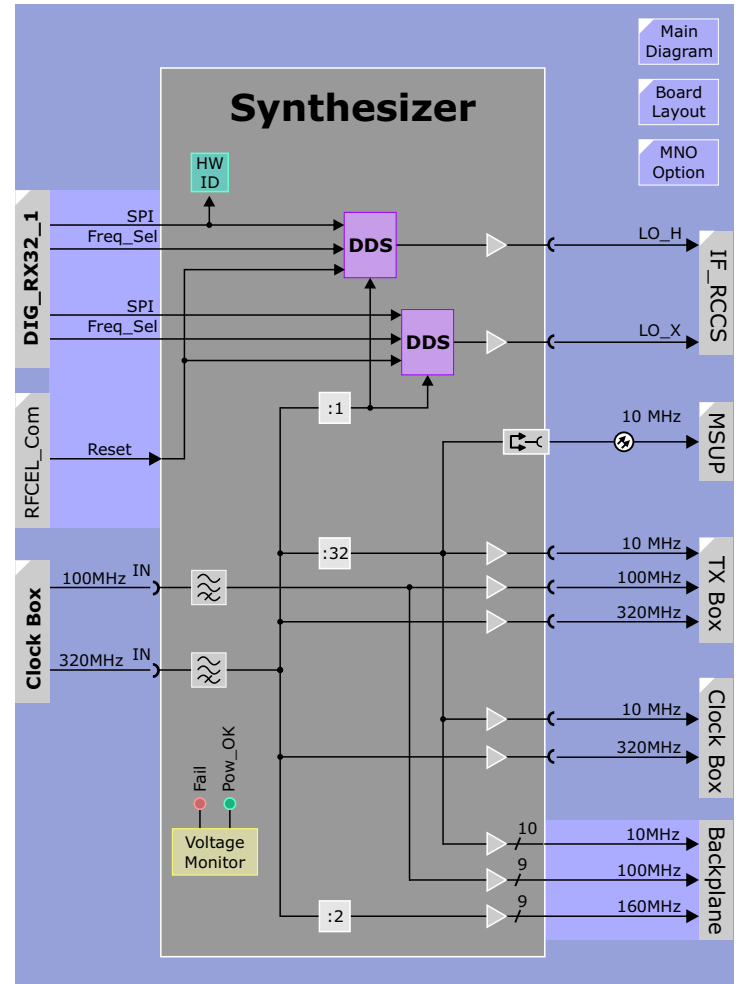
Outputs

Signal	Description																																								
LO_H	LO signals frequencies for proton (1H) imaging: - 55/75 MHz - 1.5T - 115/135 MHz - 3.0T																																								
LO_X	LO signal frequencies for supported spectroscopy nuclei @ 3T (in MHz): <table><tr><th>Nuclei</th><th>RF</th><th>LSB</th><th>USB</th></tr><tr><td>3He</td><td>93.85</td><td>102.5</td><td>82.5</td></tr><tr><td>7Li</td><td>47.88</td><td>40</td><td>60</td></tr><tr><td>13C</td><td>30.98</td><td>22.5</td><td>42.5</td></tr><tr><td>15N</td><td>12.48</td><td>20</td><td>--</td></tr><tr><td>17O</td><td>16.7</td><td>25</td><td>5</td></tr><tr><td>19F</td><td>115.9</td><td>107.5</td><td>127.5</td></tr><tr><td>23Na</td><td>32.59</td><td>25</td><td>45</td></tr><tr><td>31P</td><td>49.87</td><td>57.5</td><td>37.5</td></tr><tr><td>129Xe</td><td>34.08</td><td>42.5</td><td>22.5</td></tr></table>	Nuclei	RF	LSB	USB	3He	93.85	102.5	82.5	7Li	47.88	40	60	13C	30.98	22.5	42.5	15N	12.48	20	--	17O	16.7	25	5	19F	115.9	107.5	127.5	23Na	32.59	25	45	31P	49.87	57.5	37.5	129Xe	34.08	42.5	22.5
Nuclei	RF	LSB	USB																																						
3He	93.85	102.5	82.5																																						
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13C	30.98	22.5	42.5																																						
15N	12.48	20	--																																						
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23Na	32.59	25	45																																						
31P	49.87	57.5	37.5																																						
129Xe	34.08	42.5	22.5																																						
10 MHz	2x 50 ohm coax, frontplane, digital 1x optical fiber, frontplane, digital 10x backplane, LVDS, digital																																								
100 MHz	1x 50 ohm coax, frontplane, analog 9x backplane, LVDS, digital																																								
160 MHz	9x backplane, LVDS, digital. GIMLI clock for ISO-Ring network synchronization, used by RFCEL_Com and DIG_RX32 boards.s																																								
320 MHz	2x 50 ohm coax, frontplane, analog																																								

LEDs

LED	Description
Power OK	supplied voltages to the board are ok
Fail	voltage error

Figure 74 Synthesizer Block Diagram

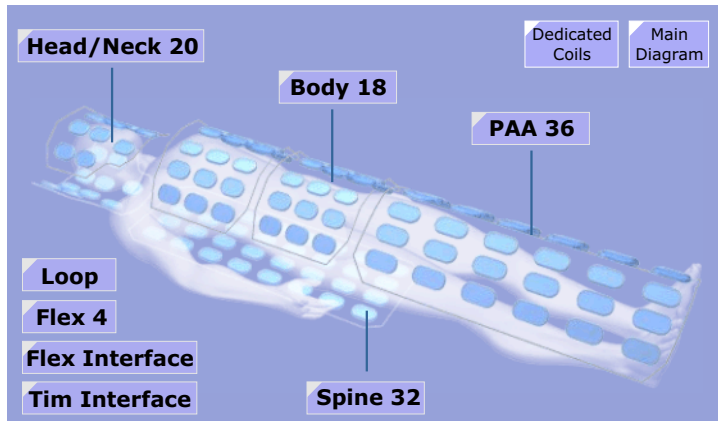


Tim 4G Coils

Tim 4G coils preserve the features having made Tim Matrix coils so successful, but have been enhanced ergonomically and mechanically through increased physical size, better scan coverage, elimination of connecting cables for **DirectConnect™** coils, and easy-use coil connectors for **SlideConnect™** coils.

From an electrical and interfacing point of view Tim 4G coils differ greatly from their predecessors in that Tim 4G coils utilize an IF technology which process the MR signals in-coil to an intermediate frequency of 8 and 12 MHz. This IF conversion yields major benefits:

- improved signal quality from reduction of frequency-induced resistive losses incurred in the cabling and connectors
- allows use of less costly cable and connector components
- ability to bundle 2 coil element signals over a single cable (**Dual-Density Signal Transfer** technology), allowing a higher level of component integration
- identical receive signal processing unit for all field strengths



1.5 T and 3 T Differences

There are three methods to distinguish 1.5T from 3T coils:

- Tim 4G coil connectors are mechanically keyed to prevent connection of 3T coils on 1.5T systems, or vice versa.
- the system is able to recognize a coil's frequency by its part number which is read out each time the coil is connected.
- the coils use color coding so the customer can instantly tell a coils operating frequency.



Coil Recognition

All Tim 4G coils employ EEPROMs which contain pertinent coil information including code, coil name, part number, SN, revision, version number, manufacturer, frequency, serial number of system where coil has been connected. The contents of the EEPROM is read out by the RFCEL_Com via an I2C bus each time the coil is connected. See the description "[Coil Code and Board Control](#)" on [page 99](#).

Head/Neck 20

The Head / Neck 20 coil belongs to the standard coil set which includes the Spine 32 coil.

Properties of this coil:

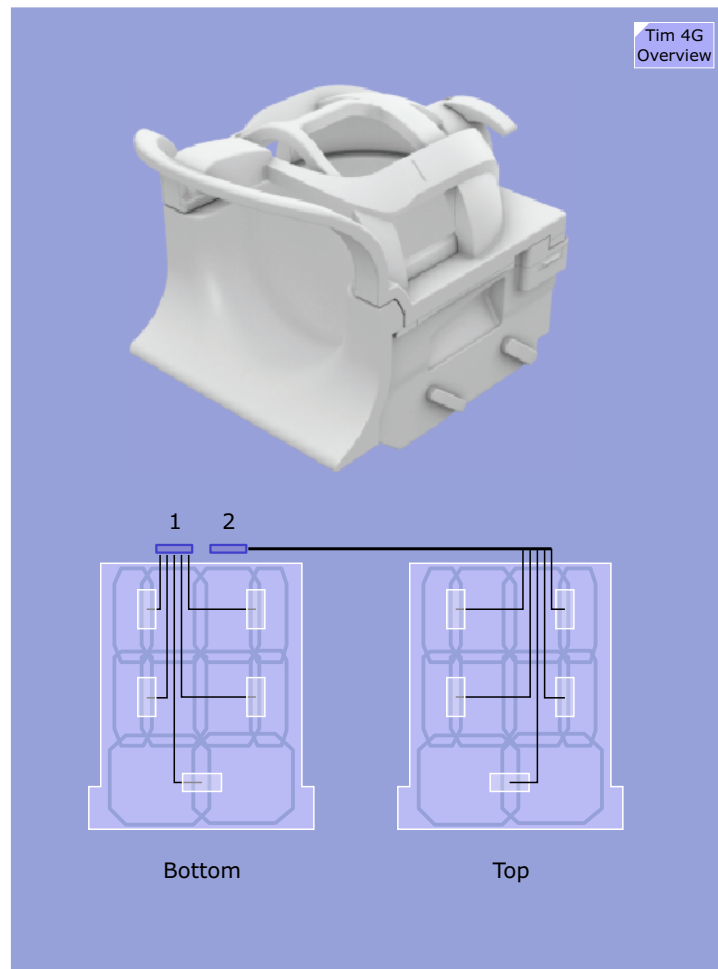
- 20 coil elements consisting of two rings of 8 elements for Head (4 on top and 4 on bottom) and one ring with 4 elements for Neck (2 top, 2 bottom).
- Cable-less coil with **DirectConnect™** technology
- Combined coil for head and neck examination for optimized workflow
- Upper coil part easy removed for highly claustrophobic patients, weighs only 1.7 kg
- Lower coil part may stay on the patient table for most of the examinations
- Smoothly integrated into the patient table with Spine 32
- Open patient-friendly design
- Cushioned head stabilizers (removable)
- No coil tuning
- iPAT-compatible in all directions
- Dual-Density Signal Transfer enables ultra-high density coil designs by integrating key RF components into the local coil
- Detachable double mirror

Applications include among others:

- Separate or combined Head / Neck examinations
- Head/Neck Angiography
- TMJ (temporomandibular joints)

	Application	Head and neck
Head/Neck 20 (DirectConnect)	Dimensions with mirror (L · W · H)	440 mm · 330 mm · 370 mm
	Weight	4.7 kg

Figure 75 Head/Neck 20



Spine 32

The Spine 32 is part of the standard system configuration.

Properties of this coil:

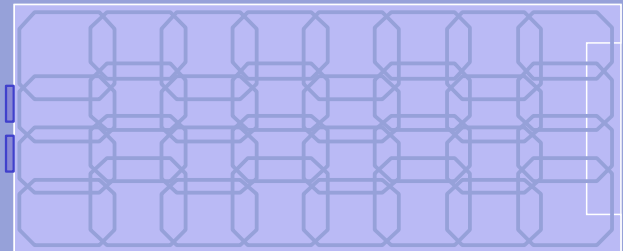
- 32-channel design with 32 integrated preamplifiers, 8 rows of 4 elements each
- Cable-less coil with DirectConnect™ technology
- Smoothly integrated into the patient table and streamlined with Head / Neck 20
- May remain on the patient table for nearly all exams
- No coil tuning
- iPAT-compatible in all directions
- Dual-Density Signal Transfer enables ultra-high density coil designs by integrating key RF components into the local coil

	Application area	Spine
Spine 32 (DirectConnect)	Dimensions	1200 mm · 489 mm · 63 mm (L · W · H)
	Weight	9.8 kg

Figure 76 Spine 32

Spine 32

Tim 4G
Overview



Body 18

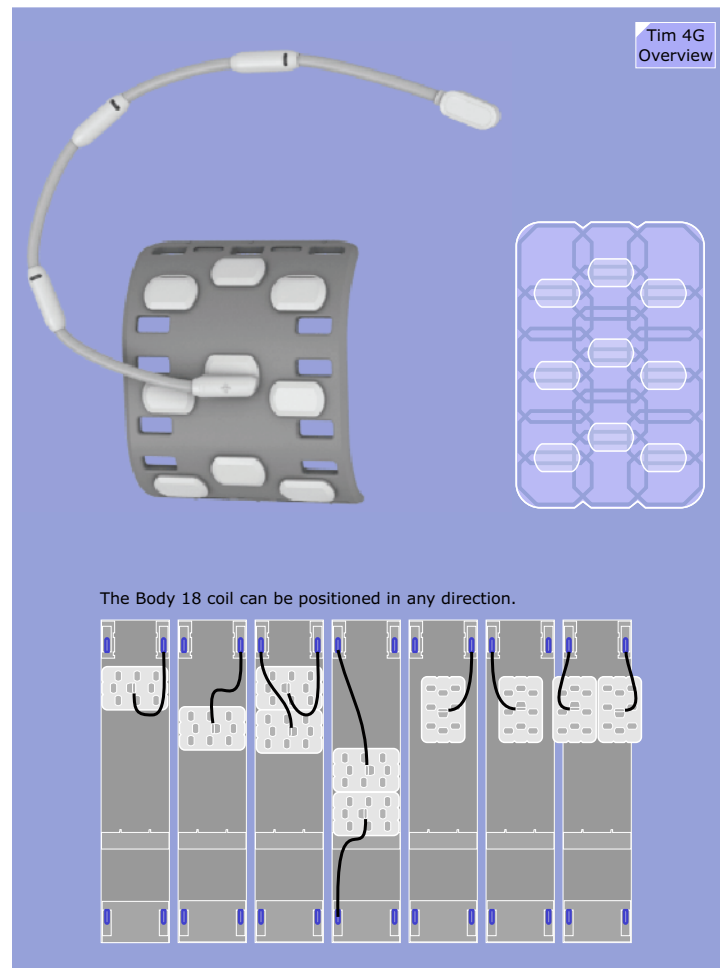
One Body 18 coil is part of the standard system configuration. Additional coils can be used in tandem for larger coverage.

Properties of this coil:

- 18-channel design with 18 integrated preamplifiers, with 3 rows of 6 elements each
- Operates in an integrated fashion with the Spine 32 (8 rows of 4 elements each)
- Can be combined with further Body 18 coils for larger coverage
- No coil tuning
- iPAT-compatible in all directions
- Dual-Density Signal Transfer enables ultra-high density coil designs by integrating key RF components into the local coil.
- SlideConnect™ technology for easy coil set up

Body 18 (SlideConnect)	Application area	Thorax, Heart, Abdomen, Pelvis, Hip
	Dimensions	385 mm · 590 mm (L · W)
	Weight	1.6 kg (on patient is 1,1 kg)

Figure 77 Body 18



Peripheral Angio 36

The PAA 36 is an option and must be purchased separately.

Properties of this coil:

- 36-channel design with 36 integrated preamplifiers, distributed over 6 planes with 6 elements each
- Operates in an integrated fashion with Body 18 coils and with the Spine 32 and for whole-body examinations also with the Head / Neck 20
- Can be utilized in head and feet first positions
- Both legs are independently covered with coil elements, maximizing the coil filling factor and the signal-to-noise ratio
- No tuning required
- iPAT-compatible in all directions
- Dual-Density Signal Transfer - two coil elements require only one receive line, minimizing cable requirements. Key RF components integrated into the coil
- SlideConnect™ technology for easy coil set up
- One cable only for easy handling

The orientation of the PAA will be derived from the connector used:

- connector Body1 or Body2 = Feet First;
- connector Feet1 or Feet2 = Head first).

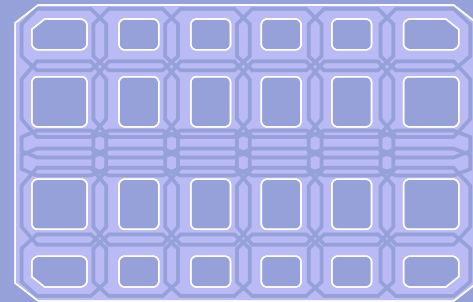
The PAA uses the same coil code for both orientations.

	Application	Leg arteries and veins
Peripheral Angio 36 (SlideConnect)	Dimensions with mirror (L · W · H)	440 mm · 330 mm · 370 mm
	Weight	4.7 kg

Figure 78 PAA 36

Peripheral Angio 36

Tim 4G
Overview



Flex 4

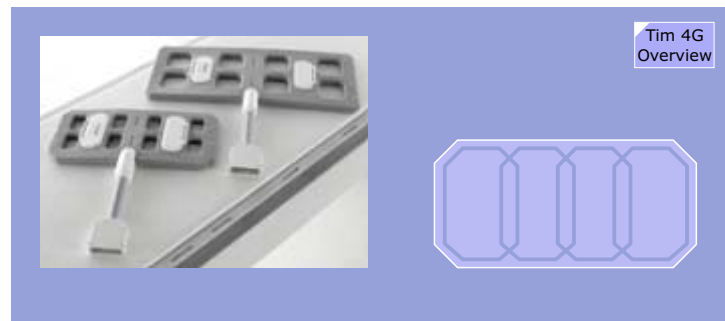
One Flex Large 4 and one Flex Small 4 are part of the standard system configuration.

Properties of this coil:

- connection via Flex Coil Interface 1.5 T
- allows flexible coil positioning
- all Flex coils use a common interface
- several Flex coil interfaces can be used simultaneously

	Flex Large 4	Flex Small 4
General	• Wrap-around coil made from soft and flexible material • 4-channel design • No coil tuning • iPAT-compatible in combination with other coils	
Applications	large regions such as medium to large shoulder, hip, and knee	small regions such as small to medium shoulder, wrist, elbow, and ankle
Weight	550 g	450 g
Diameter	516 mm × 224 mm	366 mm × 174 mm

Figure 79 Flex 4 Coils



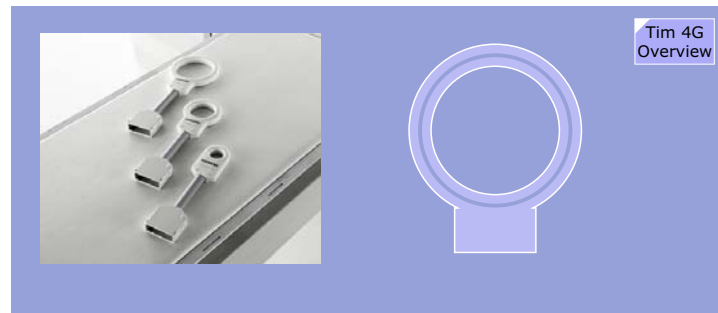
Loop Coils

The loop coils are optional and come in three terrific sizes: 4, 7 and 11 cm diameters.

- allows flexible coil positioning
- only one interface necessary for all Loop coils
- several Loop Coil Interfaces can be used simultaneously
- connection via Flex Coil Interface 1.5 T

	Loop, large	Loop, medium	Loop, small
General	• No coil tuning • iPAT-compatible in combination with other coils		
Applications	upper or lower extremities (e.g. shoulder, axilla)	inner ear, structure of wrist and fingers, pediatrics examinations	small structures near the surface (joints of fingers and toes, wrist, skin, temporo mandibular joints)
Weight	225 g	200 g	175 g
Diameter	110 mm	70 mm	40 mm

Figure 80 Loop Coils



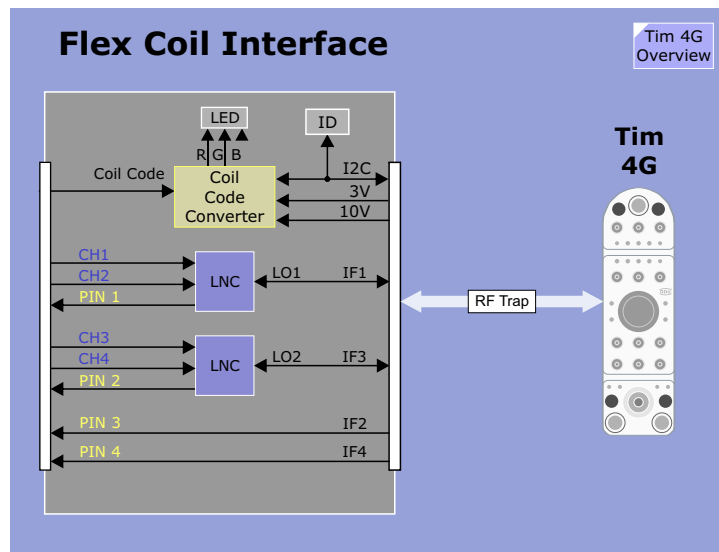
Flex Coil Interface

The Flex Coil Interface for use with both flex and loop coils provides these features:

- pre-amplification (LNC)
- RF-IF conversion (LNC)
- Coil Code conversion
- PIN signalling

The interface supports a maximum of 4 elements.

Figure 81 Tim Coil Interface Block Diagram



Tim Coil Interface

The Tim Coil Interface allows Tim Matrix coils to be used on the Aera/Skyra platform. Each Tim Interface allows connection of one 8-channel coil plug of a Tim Matrix coil. Tim Matrix coils with multiple plugs will require an equal number of additional interfaces. Tim Interfaces plug only into SlideConnectors allowing a maximum of 4 Tim Interfaces.

RF-IF Conversion

Tim Matrix coils output their received MR signal at their respective RF frequencies. The daVinci coils, however, convert the RF to an intermediate frequency (IF) of 8 and 12 MHz directly in the coil. The LNCs provides this RF-IF conversion.

Coil Code Conversion

The **Coil Code Converter** converts the resistive code of the Tim Matrix coils to a digital code and communicated to the system via an I2C bus. The same I2C bus is used to read out the hardware id EEPROM of the interface.

10V Coil Supply

The interface provides a **10V supply** over the receive lines as required by all Tim Matrix coils for their built-in pre-amplifiers.

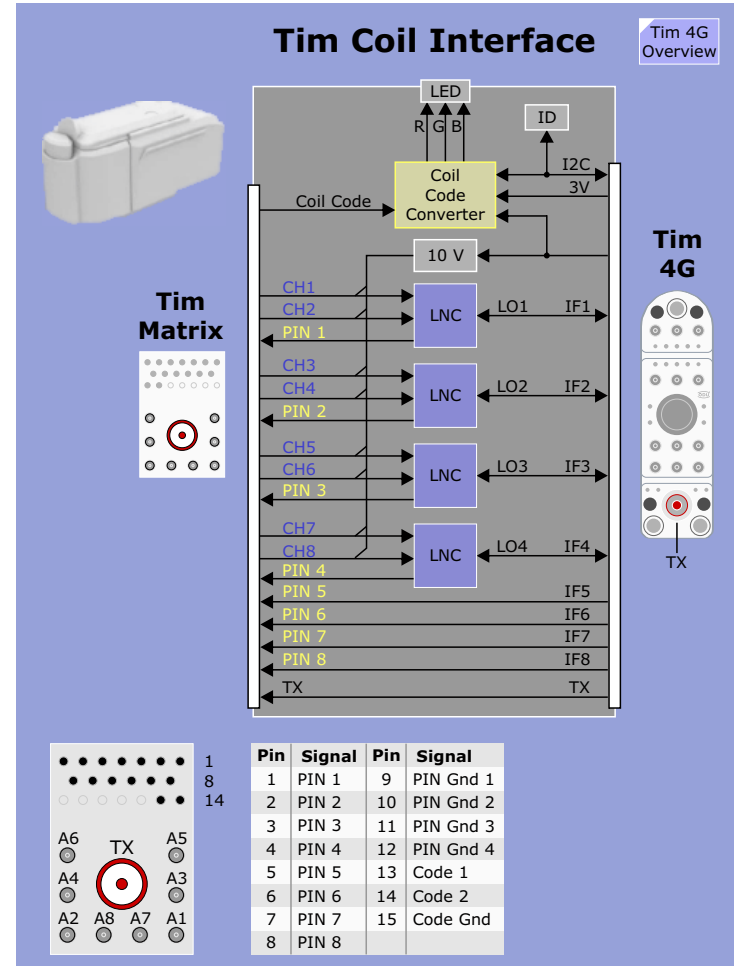
TX Pin

The Tim Interface can also transfer the TX signal (both X-nuclei and 1H) if connected to B1 or B2 respectively.

Hardware ID

The interface possesses a hardware ID EEPROM for asset management.

Figure 82 Tim Coil Interface Block Diagram



Dedicated Coils

A number of new dedicated Tim 4G coils are available:

Dedicated Coils		
Hand/Wrist 16 (SlideConnect)	Application	High-res hand images
	Dimensions with mirror (L · W · H)	332 mm · 215mm · 115mm
	Weight	2.8 kg
Foot/Ankle 16 (DirectConnect)	Application area	Spine
	Dimensions (L · W · H)	410 mm · 390mm · 310mm
	Weight	3.2 kg
Shoulder 16 Large (SlideConnect)	Application area	Best visualization of small anatomical structures (e.g. labrum)
	Opening	220 mm
	Weight	2.3 kg
Shoulder 16 Small (SlideConnect)	Application area	
	Opening	165 mm
	Weight	2.2 kg
CP Extremity Tim Interface	Application area	Multi purpose
	Dimensions (L · W · H)	405 mm · 270 mm · 290 mm
	Weight	6.5 kg
TX/RX 15 Channel Knee 2 Tim Interfaces	Application area	Multi purpose
	Dimensions (L · W · H)	256 mm · 360mm · 310 mm
	Weight	6.6 kg
Accessories	Flex Coil Interface	
	Tim Coil Interface	

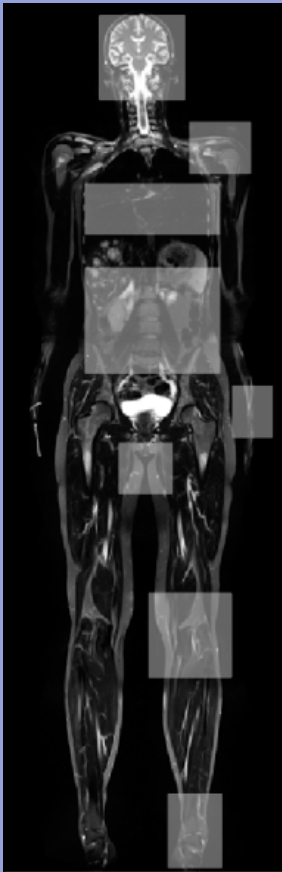
LEDs

LED	Description
R	Rev 00-02: - when plugged in: 3.3V too low (>2.6V) - after initialization: FLASH being programmed Rev >03: 10V error
G	a coil plug event (in or out)
B	Rev 00-02: - I2C bus activity Rev >03: - when plugged in: Processor running - after initialization: I2C activity (also when FLASH being

Figure 83 *Dedicated Coils*

Dedicated Coils

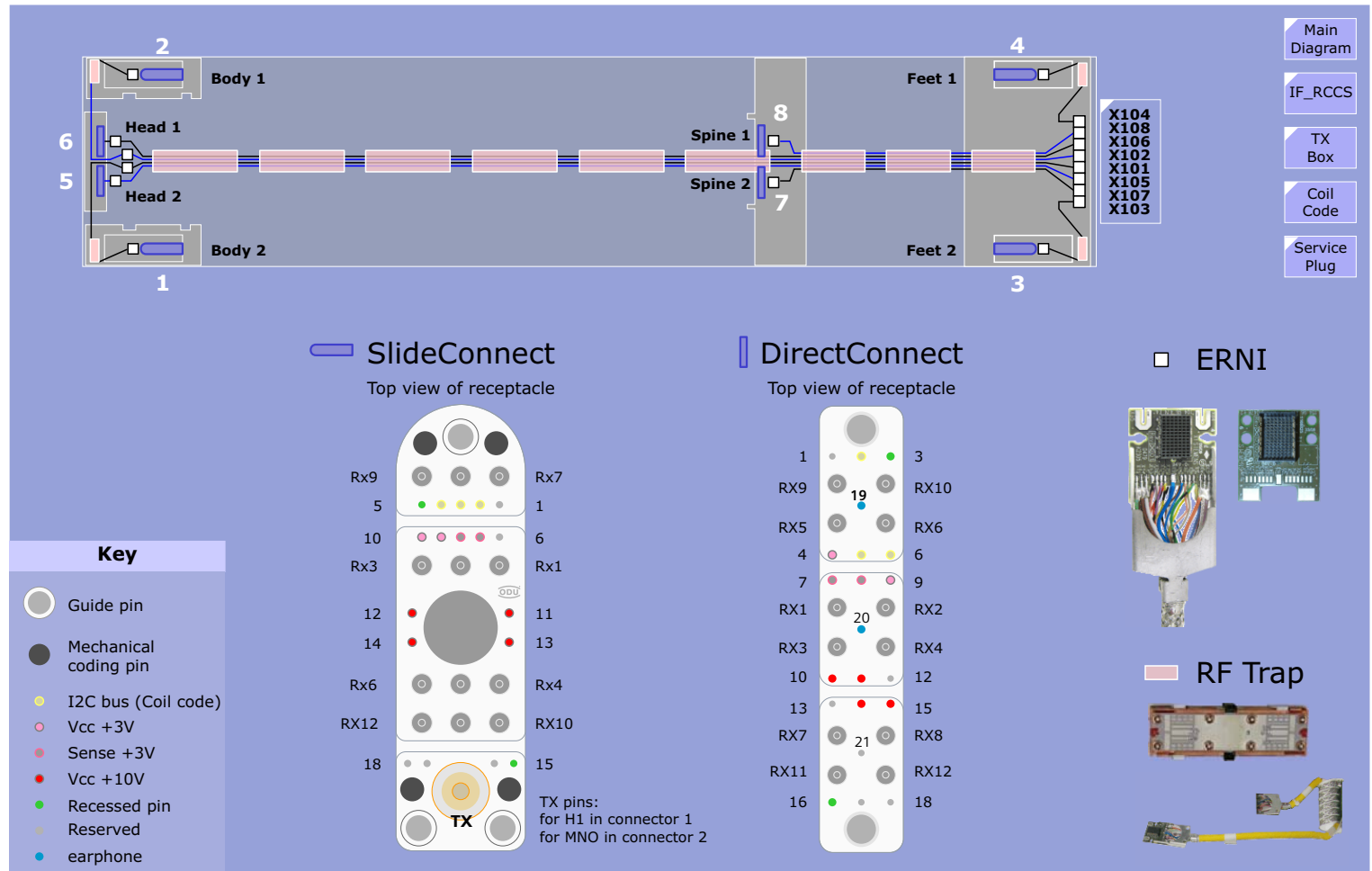




	Aera 1.5	Skyra 3.0
Head	Tx/Rx Head QED	Tx/Rx QED
Shoulder	16 ch SSMR	---- " ----
Breast	2/4/8 ch Sentinelle 4 ch BI Noras 16 ch AI Rapid	---- " ----
Multi Purpose	4 ch Flex sm & lg 4, 7, 11 cm Loop	---- " ----
Hand/Wrist	16 ch SSMR	---- " ----
Endorectal	Medrad	---- " ----
Knee	Tx/Rx Siemens 15 ch Tx/Rx QED	Tx/Rx QED ---- " ----
Foot/Ankle	16 ch SSMR	---- " ----

Main
Diagram

Figure 84 Coil Connectors and PTab Cable Inter-connections



Multi Nuclei Spectroscopy Option (MNO)

The MNO option (only for Skyra) provides the hardware components that allow performing MR experiments on these additional nuclei:

Nucleus	MHz	Nucleus	MHz	Nucleus	MHz
19.F	115,90	7.Li	47,88	13.C	30,98
3.He	93,85	129.Xe	34,08	17.O	16,70
31.P	49,87	23.Na	32,59		

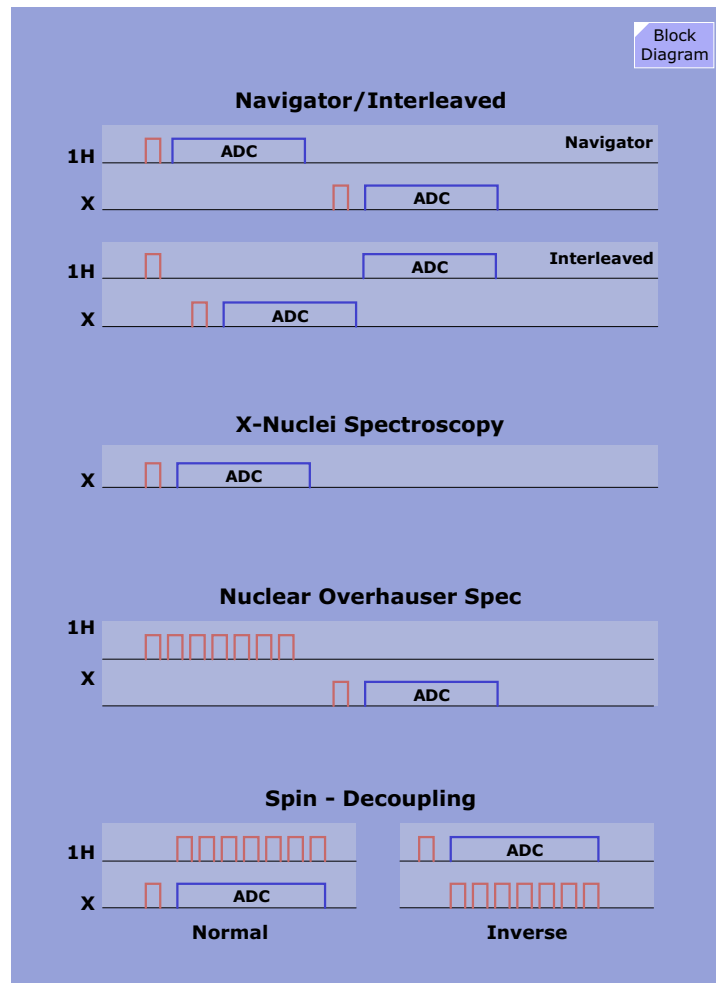
The diagram at right lists the spectroscopy applications supported.

It must be noted, that the Navigator and Interleaved procedures show reception of both water and non-water signals. This can only be achieved with double-resonant MNO coils. It is not possible to use MNO and non-MNO coils simultaneously.

The MNO option consists of these components:

- **BB RFPA** - the broad-band transmit amplifier for transmitting at MNO nuclei.
- **MNO TX Box** - generates the RF pulses at the MNO frequencies, supplies 2 control signals for the BB RFPA, provides filtering, RF metering for SAR monitoring and protect of internal components.
- **COB** - CAN Open bridge provides electrical to FO conversion of the CAN bus for the BB RFPA.
- **125 MHz Filter** - placed in the TX path between TX Box and Local Coil, required to prevent MNO transmitted pulses from being coupled into the TX Box and falsifying SAR.
- **MNO Coil Interface** - converts the received MNO signals into the 8/12 MHz IF of daVinci's receiver.
- **MNO Switch** - placed between the TX Box RX output and RCCS to prevent interference with the TX Box.

Figure 85 Supported Spectroscopy Measurements



MNO TX Box

This box is an FRU and is maintenance free.

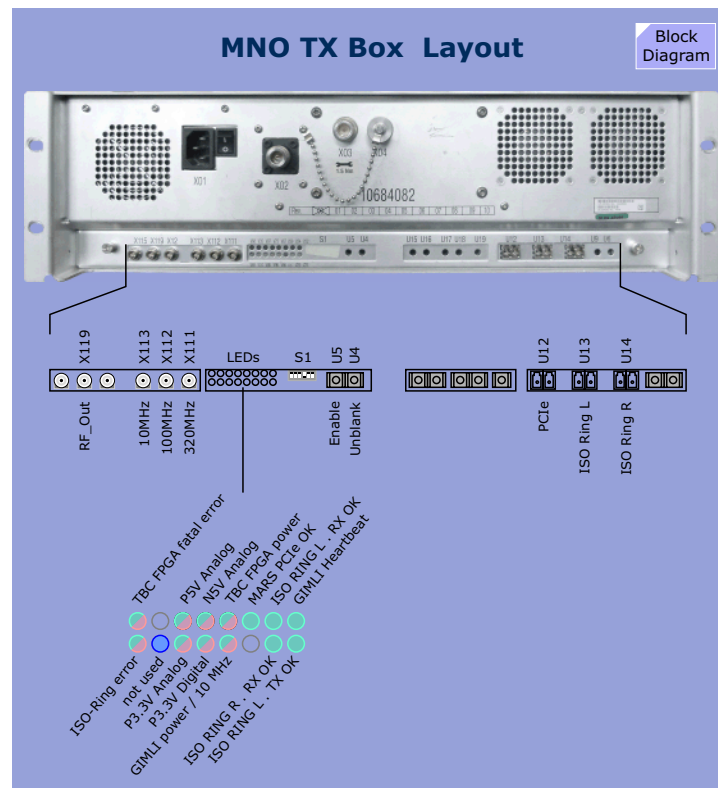
Function

The unit provides the RF pulse **signal generation** at the X-nuclei frequencies, and **meters** the amplified MNO RF pulses amplified by the external Broad Band RF amplifier (BB RFPA) via the DICO and TALES. It also supplies the Enable and Unblank control signals required by the BB RFPA. Additional **filters** are provided to filter out harmonics produced at MNO frequencies which could effect the metering devices and falsifying SAR and monitoring. A **125 MHz filter** is always present with an additional switchable **50 MHz filter** required for all MNO frequencies except 19F and 3He frequencies. The internal **50 ohm load** is used for the Tune-up and test procedures.

LEDs

LED		Description	LED		
Bottom Row L-R			Top Row L-R		
V69	G/R	ISO-Ring error	V68	G/R	TBC FPGA fatal error
V34	B/G	not used, but normally lights blue	V35	Y	not used
V98	G/R	P3.3V Analog	V97	G/R	P5V Analog
V96	G/R	P3.3V Digital	V75	G/R	N5V Analog
V66	G/R	GIMLI power / 10 MHz	V67	G/R	TBC FPGA power
			V26	G	MARS PCIe OK
V25	G	ISO RING R . RX OK	V24	G	ISO RING L . RX OK
V23	G	ISO RING L . TX OK	V22	G	GIMLI Heartbeat

Figure 87 MNO TX Box Layout



BB RFPA

Overview

The BB RFPA is a 55kg 19" rack-mounted water-cooled broad-band solid-state amplifier covering a frequency range of 17 to 123 MHz with a maximum output power level of 8 kW at 123 MHz. The unit is part of the MNO option and is used solely for MNO applications.

All connections to the unit have been placed on the back side for easy access. The unit is an FRU and maintenance-free. The unit is non-linear and must be calibrated. Calibration is performed by the RF Characteristic step of the system tune-up procedures. An RF characteristic has to be measured for each of the spectral (nuclei) frequencies. Testing of the RFPA is performed through the service platform.

Function

Power Up

The amplifier is supplied 3 Φ 400V AC power via circuit breaker F54 of the line power distribution.

WARNING Using F54 to switch the unit on or off can cause damage to the unit.

To power up the amplifier a CAN command is sent to set the internal enable signal. The external **Enable** signal must also be present (light ON). With both signals present, When the unit has finished booting it sends a **READY** command over the CAN bus when boot is finished. The **DC_RF** and **DC_Control** LEDs give a visual indication that the unit is ready.

Control and Monitoring

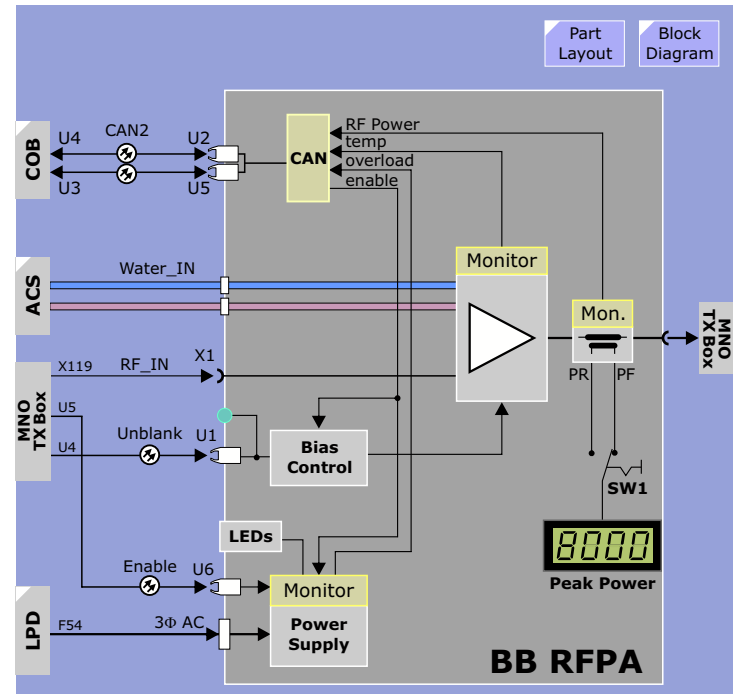
Internal monitoring provides detailed status information. The RFPA is largely autonomous as far as its control is concerned. A start up/ power up command is sent from the MNO TX Box and a READY

status is expected in return.

The RFPA has several integrated monitoring circuits keeping track of vital functions:

- operating voltages
- peak output power levels
- reflected power too high
- unblank duty-cycle
- power stage temperature.

Figure 88 BB RFPA



Inputs

Signal	Description
AC_IN	Three-phase AC voltage (400 V). Internal power supplies generate all required voltages.
CAN_Out (U2)	The CAN connections to the amplifier are non-conventional, so please follow the wiring diagrams.
CAN_In (U5)	The normal on/off switching of the amplifier as well as supplying status information is performed via CAN.
RF_IN	Main input from the MNO Box. The nominal input level for the full output power level (8 kW for 123 MHz) is 0 dBm.
UNBLANK	To reduce noise during reception, the amplifier is blanked during the receive cycle. FO light on = amplifier activated.
ENABLE	FO light out = amplifier shuts down.

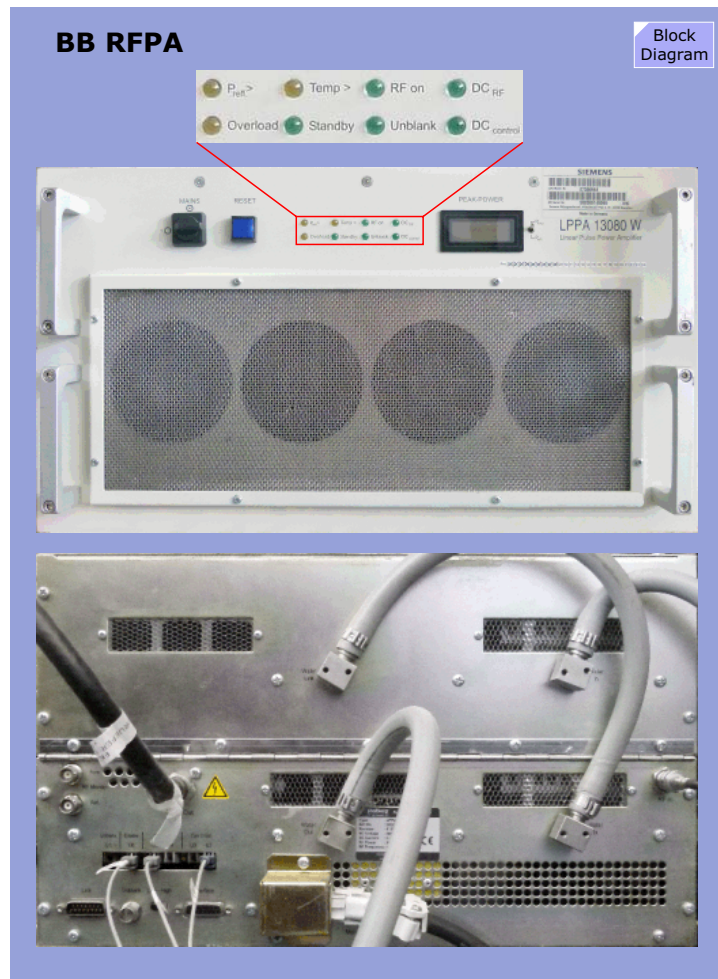
LEDs

LED	Description
PRefI>	ON = peak power limit exceeded (10% over max power)
Overload	ON = excessive DC current
Temp>	ON = temperature exceeded 50°C
Standby	ON = DC supply is disabled
RFON	ON = RF present at output (> 1% of the max power)
Unblank	ON = amplifier being unblanked
DCRF	ON = DC supply voltage for the RF stages is present
DCControl	ON = DC supply to controller is present
CAN TX/RX	ON = bus activity, not necessarily to/from the unit
RESET	ON = overload, the amplifier has been shut down

Switches, Buttons, LCD Panel

Device	Description
SW1	Selects active HIGH or active LOW unblank pulse. Is factory set to "low" and is mechanically fixed. Do not change!
Pfor / Prefl	selector switch for displaying forward or reflected RF power on the LCD display.
RESET	Can be used to manually reset the amplifier, i.e., after an overload. The lamp in the RESET button detects an overload status which has shut down the RF power
LCD	For display of forward or reflected RF power. The accuracy is not precise enough to be used for tests, especially at other frequencies (multi-nucleus option)

Figure 89 BB RFPA Layout



MNO Coil Interface

This interface is based on the Tim Coil Adapter and works in much the same manner. The Tim Coil Adapter must transform the MR signals received by the Tim coils into IF. The RF-IF conversion performed by the Tim Adapters is fixed at the system frequency of 64 MHz (Aera) or 123 MHz (Skyra). The MNO Coil Interface works at the 8 frequencies supported by the MNO option (see list below). The output of the MNO Interface is the specified 8 and 12 MHz IF of the daVinci platform receive system.

Table of conversion frequencies for the supported MNO nuclei:

Nuclei	F MR	LO1	LO2
19F	115,90	42.5	150/170
3He	93,85	67.5	170/150
31P	49,87	112.5	170/150
7Li	47,88	110.0	150/170
129Xe	34,08	127.5	170/150
23Na	32,59	125.0	150/170
13C	30,98	127.5	150/170
17O	16,70	145.0	170/150

The MNO Interface can only be plugged into the Slide Connect connectors, that is connectors 1 through 4 on the Patient Table. It is not possible to use MNO and 1H coils simultaneously.

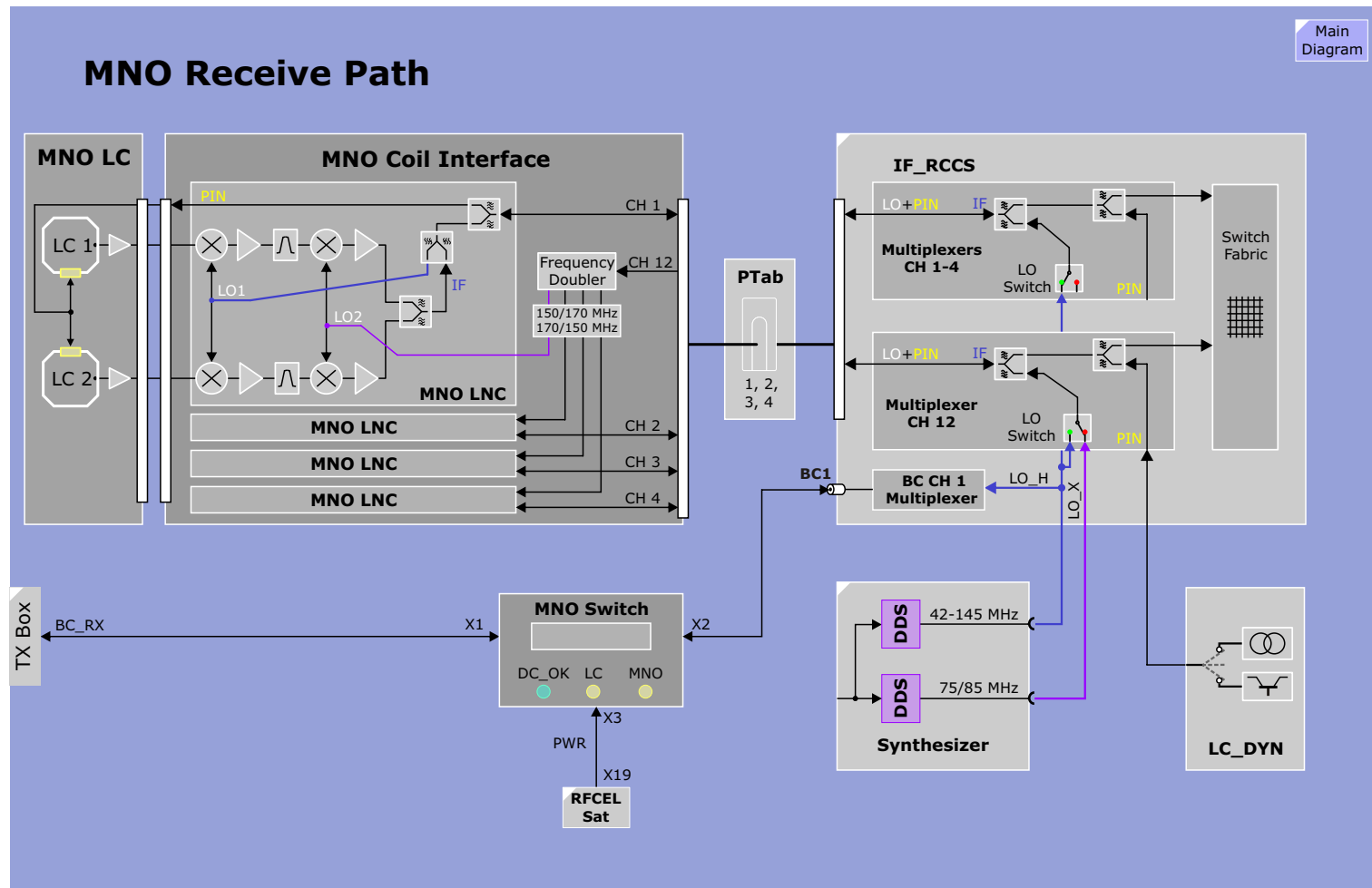
MNO Switch

This component blocks the LO_H signal going to the TX Box when MNO is in operation in order to prevent the LNC in the TX_Box from interfering with the MNO experiment (noise). The switching between normal and switched mode is performed based on the level of the LO_H signal. When in MNO operation the LO_H signal amplitude decreases significantly. This is the cue for the switch to block the signal altogether. When the LO_H signal amplitude is normal, the switch is deactivated and the receive path from the TX Box works as usual.

LEDs

LED	Description		
DC_OK	ON = DC supply to unit is OK.		
LC, MNO	LC	MNO	Mode
	OFF	OFF	BC imaging
	ON	OFF	LC imaging, proton frequency
	ON	ON	LC imaging, MNO frequency
	OFF	ON	undefined (not used)

Figure 90 MNO Receive Path



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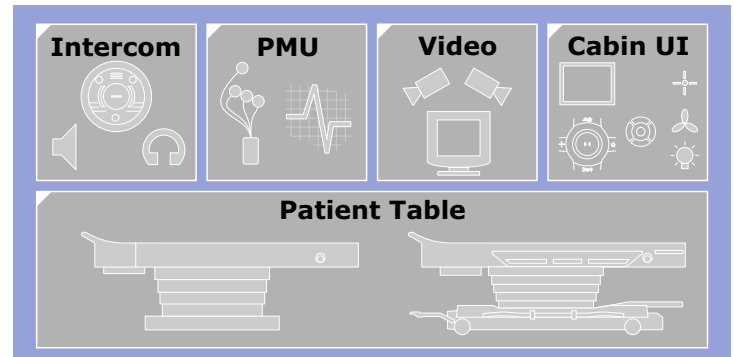
Patient Handling

Introduction

The Patient Handling for MAGNETOM Aera / Skyra is a new development providing many features intended to enhance operator workflow and maximize flexibility to the user.

The Patient Handling consists of the se components:

- **Patient Table (PTAB)**
- **Cabin User Interface (CUI)** - control panels, Dot Display
- **Intercom and Patient Video**
- **Physiological Measurement Unit (PMU)**



Patient Table

While a table's primary purpose is for quick, accurate and comfortable positioning of the patient within the scan volume it was a goal to design a table that is to be user friendly, durable, reliable and capable of handling the weights of 21 century sized individuals. And it should also be service friendly.

Towards achieving these goals the Tim Tables have incorporated drive technologies which include a **hydraulic drive** for vertical lift and a horizontal drive using a zipper-like **linked chain drive** mechanism which are hoped to prove to be long-lasting and dependable as well as requiring less service attention in the form of maintenance or repairs.

Further improvements can be found in the coil interfacing. Two coil connectors are being introduced:

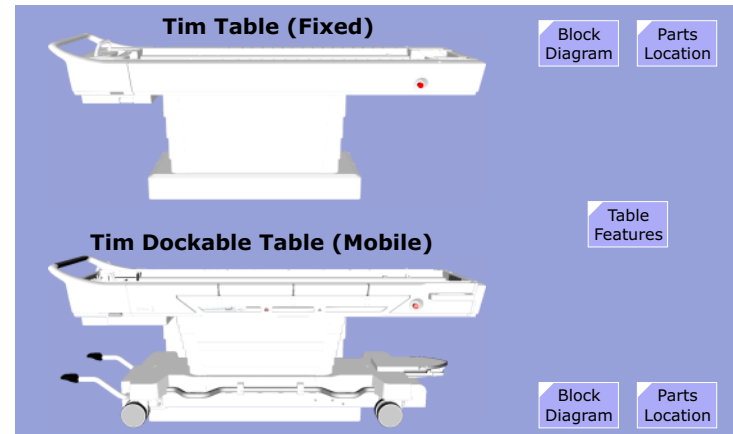
DirectConnect™ — connectors which are embedded in the table to allow a cable-less connection of fixed coils

SlideConnect™ — connectors for non-fixed coils

The Tim Tables are being offered in two variations:

- **Tim Table** (Fixed Table) - in the standard delivery
- **Tim Dockable Table** (Mobile Table) - an option

In meeting with the serviceability aspect, through attentive design all electronic devices and sensors and many mechanical parts are shared between the two tables. The mobile table, however, due to the many additional function and complexities, requires several additional or modified components as well. There are also frequency related differences in the coil connectors and RF traps in the table tops between the 1.5T and 3T systems, The coil connectors have been mechanically keyed to prevent using 1.5T coils on the 3T system and vice versa.

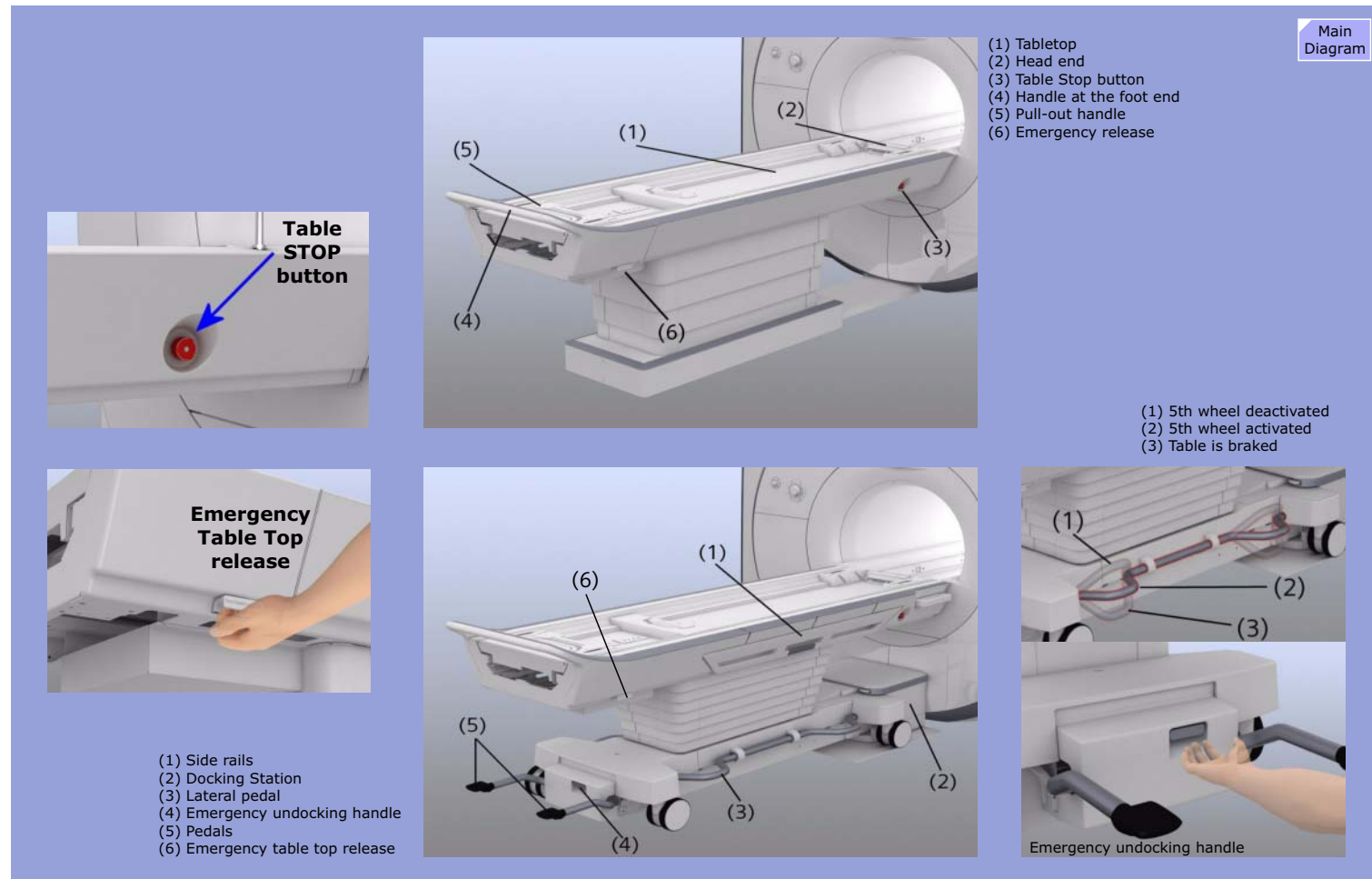


Specifications

Parameter	Value
Table length	2470 mm
Table Top length	2350 mm
Table width	760 mm
Table Top width	541 mm
Minimum height	520 mm - fixed table 560 mm docked/ 610 mm undocked - mobile table
Max. Load	250kg (200kg for Avanto fit)
Scan range	1400 mm (standard) 2050 mm (Whole Body option)
Max. horizontal speed	200 mm/s
Positioning accuracy	< +/- 0.5 mm

NOTE The scan range is defined by the syngo configuration and installed SW license.

Figure 91 Table Features



Tim Table (Fixed)

Overview

The Tim Table consists of the following components:

Mechanical

- Table Base
- Vertical Lift
- Support Frame
- Table Top

Electrical

- **PSY** – Power Stage Y D4110 - drive circuits for vertical motor
- **Y-Drive D4311** – motor driven hydraulic pump assembly for vertical lift
- **DPD** – Direct Position Drive D4320 - horizontal drive circuits and motor
- **PAM** – Power Adaptor Module D4120 - located behind the front magnet cover bottom-side, contains voltage regulators, voltage monitoring and switch-off circuits

Sensory

- **S10, S20** – optical sensors used to detect horizontal and vertical home positions respectively.
- **S11** – a microswitch used to detect the activation of the emergency Table Top release
- **S70, S71** – push-lock switches to emergency table stop.

Installation

To assure the table's lateral stability it is bolted onto a stainless steel base plate which must be glued to the RF-cabin floor during installation. In the rare event the table must be removed (e.g., gradient coil replacement) it is easily dismantled.

Figure 92 Base Mount Plate

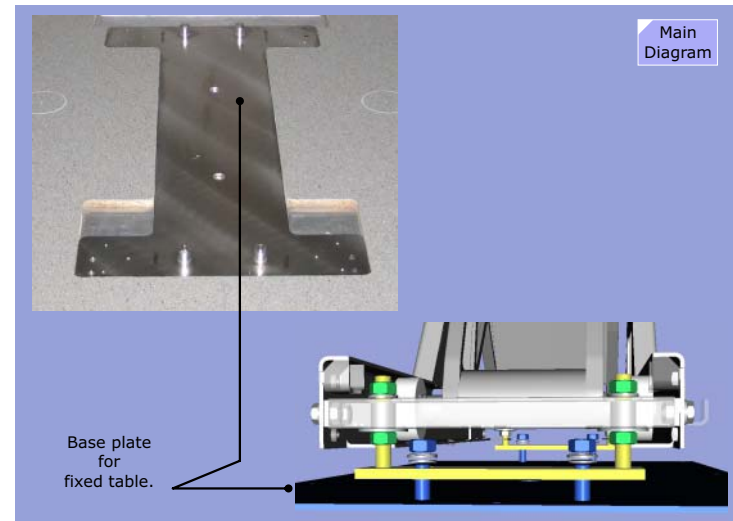


Figure 93 Tim Table Block Diagram

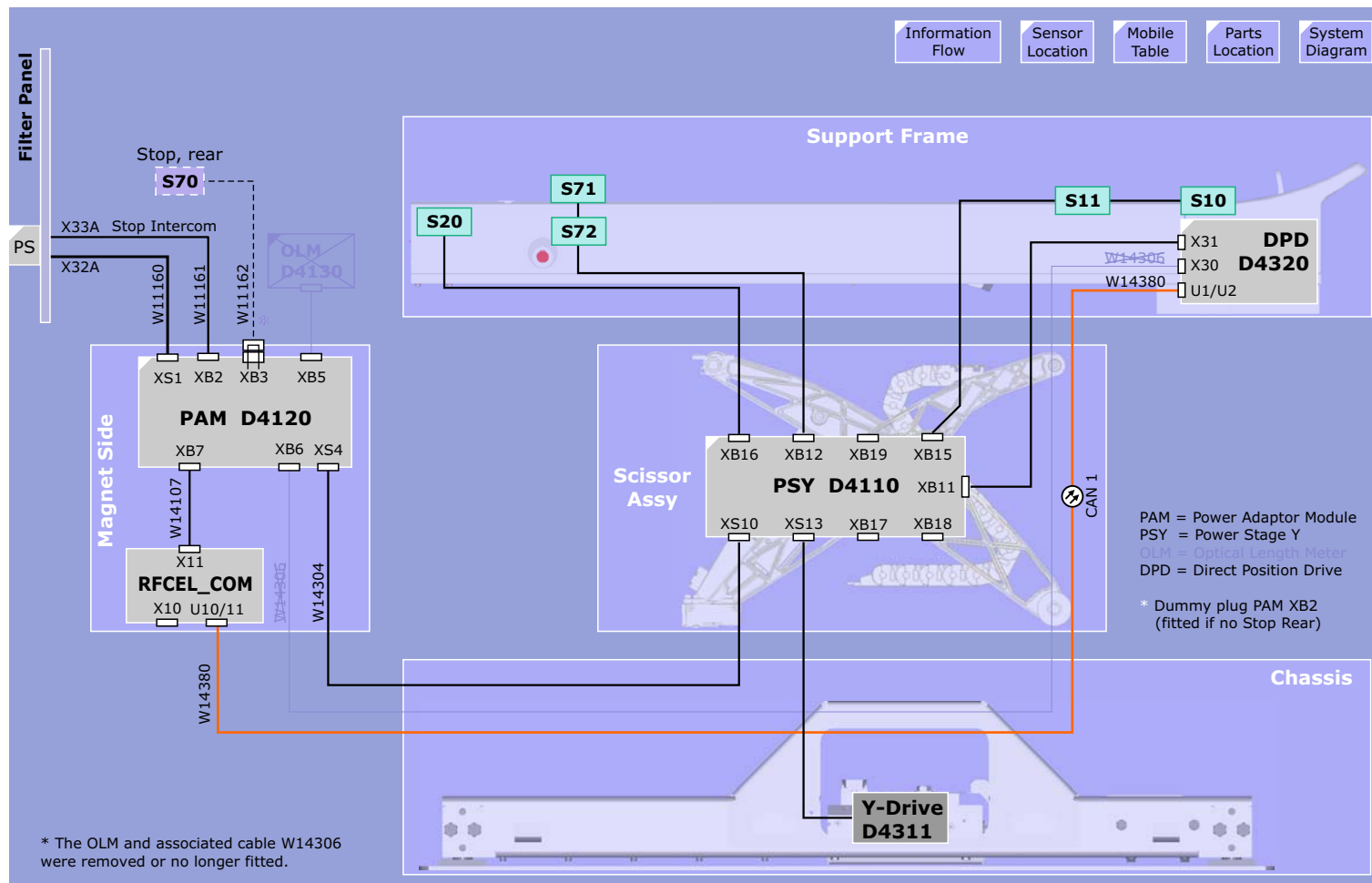
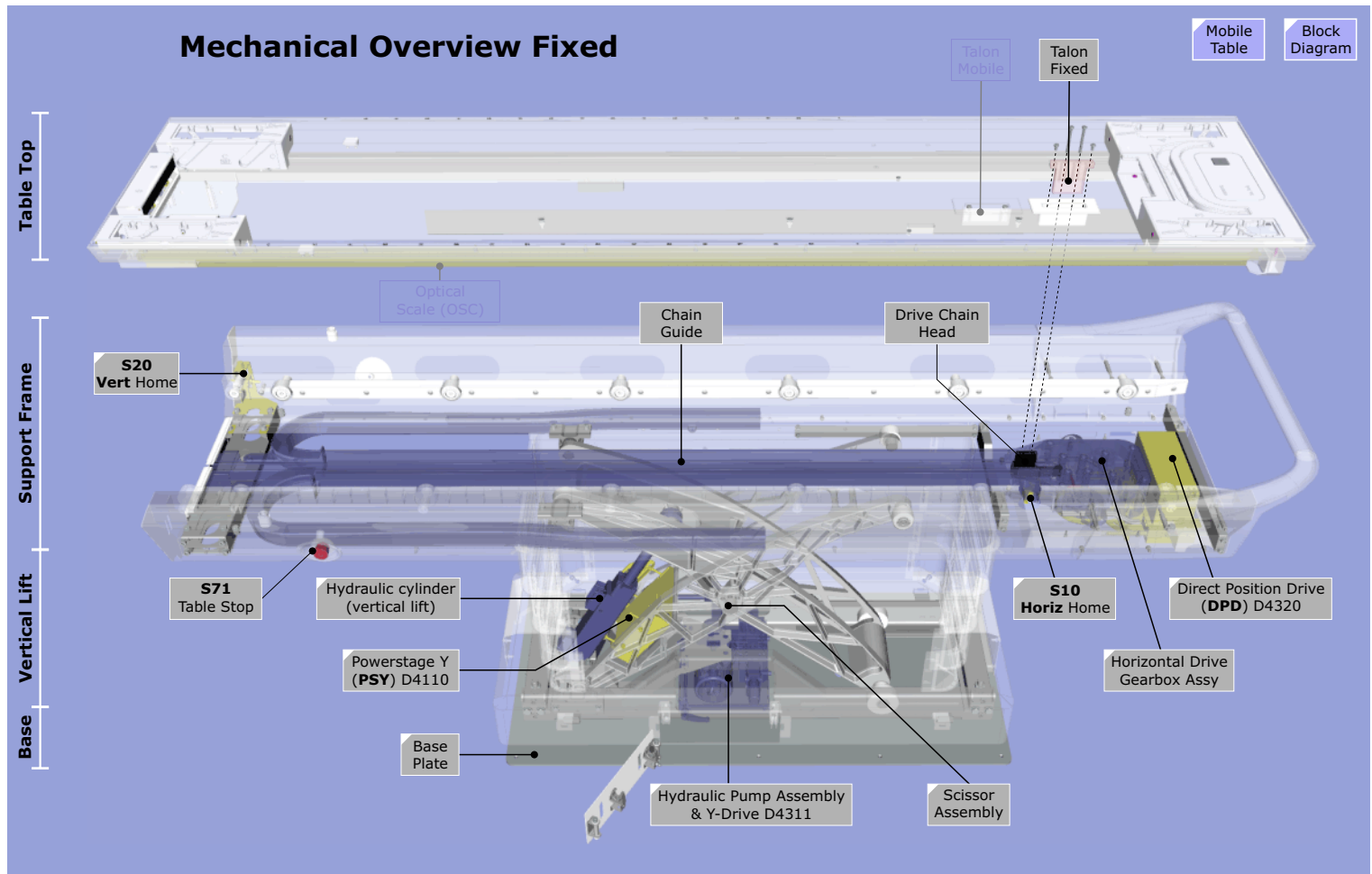


Figure 94 Tim Table Parts Location



Vertical Drive

The patient tables use a scissor mechanism powered by hydraulics to provide vertical lift for body weights respective of increasing weight trends seen accros the globe, especially in countries with a high density of McDonalds, Burger King and Kentucky Fried Chicken places. And Wendys and Taco Bells.

The vertical drive system consists of the following components:

- **Vertical Lift Assembly**
 - Scissor Assembly
 - Hydraulic Cylinder
 - Bearing Block
 - Rollers and Guide Plates
- **Hydraulic Pump Assembly**
 - Fixed: Y-drive D4311
 - Mobile: Y-drive D4312
- **S20 Vertical Home Sensor**
- **S60 Collision Detection** (mobile only)

Scissor Assembly

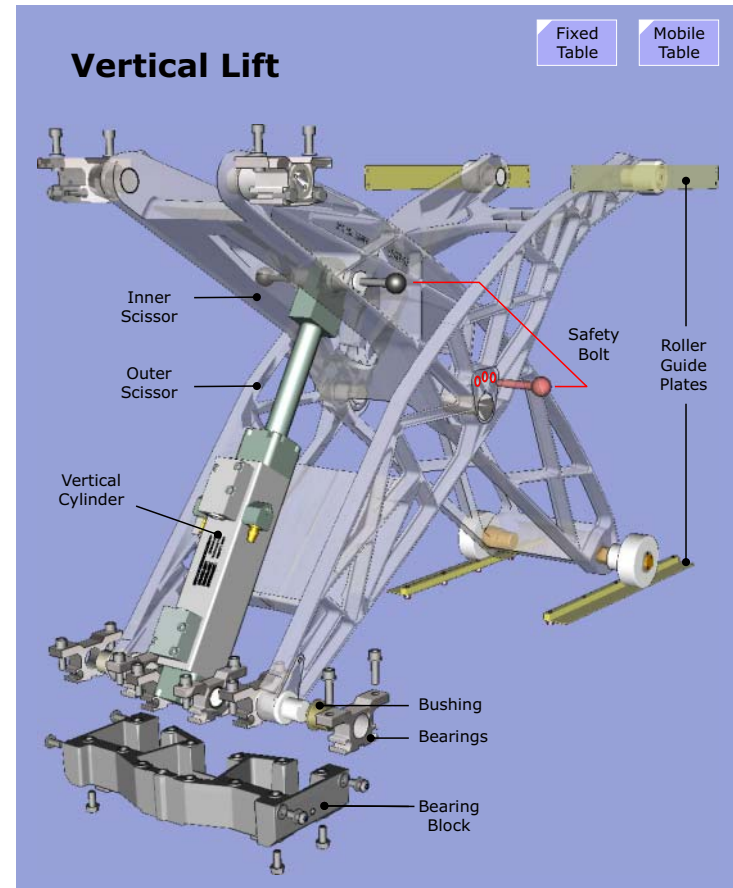
The lifting mechanism is a scissor assembly operated by a hydraulic cylinder. The assembly consists of an inner and an outer scissor hinged at their center via an axel. One end of the scissors are fixed to the base and to the Support Frame via bearings. On the opposite end are rollers providing smooth movement. Guides keep roller movement and traction even.

A **safety bolt** is provided to secure the scissors when performing repair or maintenance procedures.

Hydraulic Cylinder

The hydraulic cylinder generates the necessary forces to move the scissor assembly. It is connected via quick couplings to the Hydraulic pump assembly which builds up the necessary oil pressure for vertical lift.

Figure 95 Vertical Lift Mechanics



Hydraulic Pump Assembly

Hydraulic System

The vertical drive for a fixed table consists of a **motor-driven hydraulic pump**, several control valves and the **hydraulic fluid container**. Its oil level must be checked during Preventive Maintenance. Since also the hydraulic cylinder contains significant amounts of fluid, this level check must be performed at table down position.

Motorized Up

Refer to [Figure 97](#). For moving upwards, the small but highly efficient motor builds up the required oil pressure which drives the hydraulic cylinder (quick-coupling connection **A2**).

Motorized Down

For moving downwards, the motor is switched to reverse direction and builds up hydraulic pressure to open two control valves at the vertical cylinder (connection **S2**). This will release oil pressure from the hydraulic cylinder via connection **A2** and the table lowers down.

Manual Operation

Refer to [Figure 98](#). For manual operation, the Hydraulic pump assembly can be operated with a **service handle** and the **setting lever**.

NOTE A **service handle** for manual vertical lift is provided (under service access cover, at table base left-side).

Pump Output

For service, the hydraulic hoses from the Hydraulic pump assy to the vertical cylinder can be easily disconnected by means of self-locking **quick couplings** (connection **A2** and **S2**).

Figure 96 Hydraulic Pump Assembly

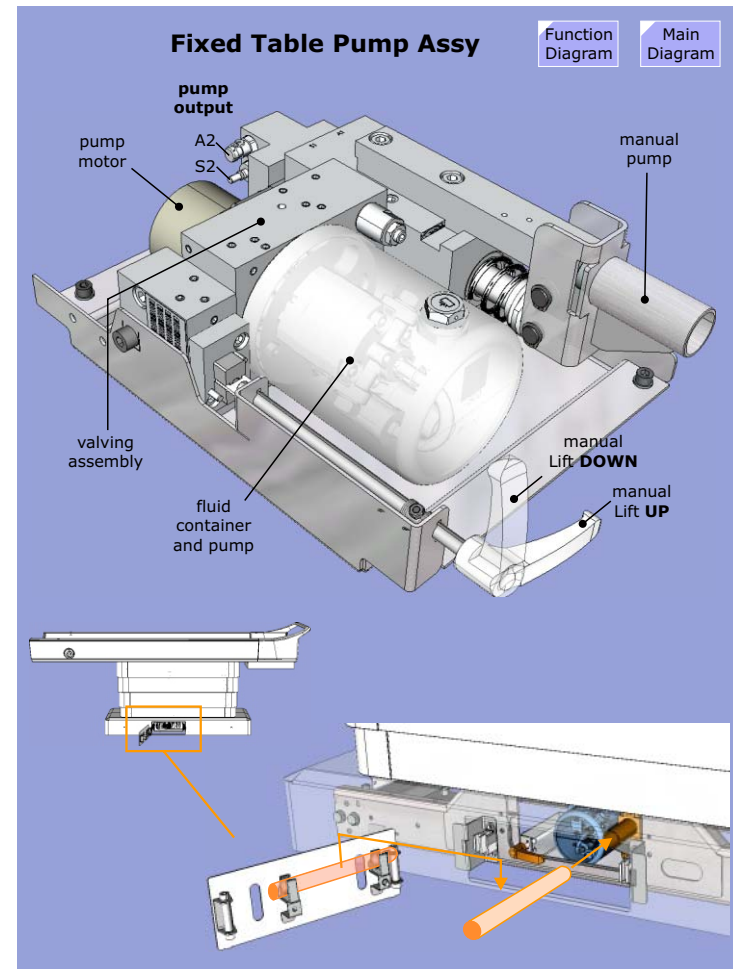


Figure 97 Hydraulic Overview Tim Table

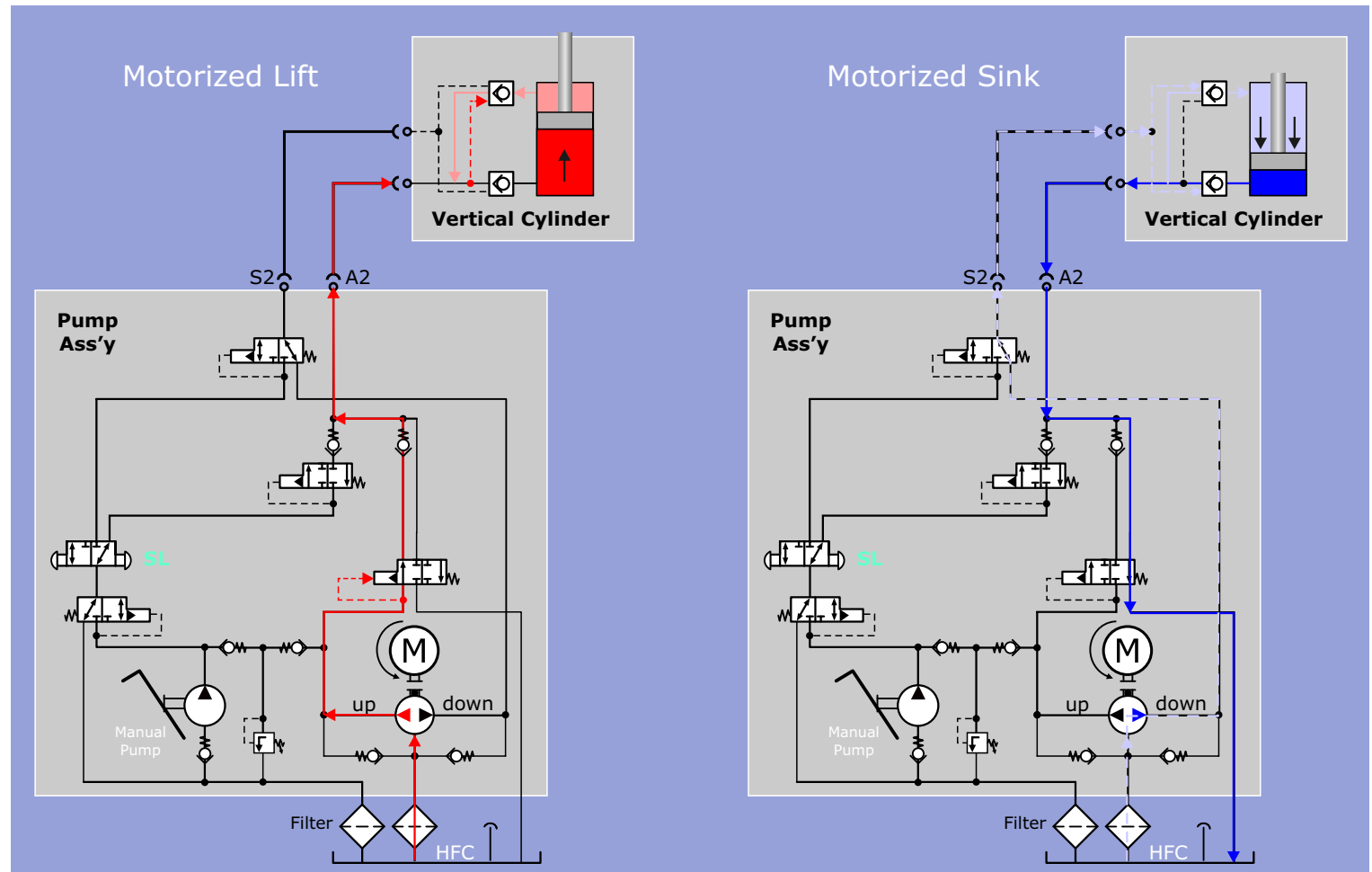
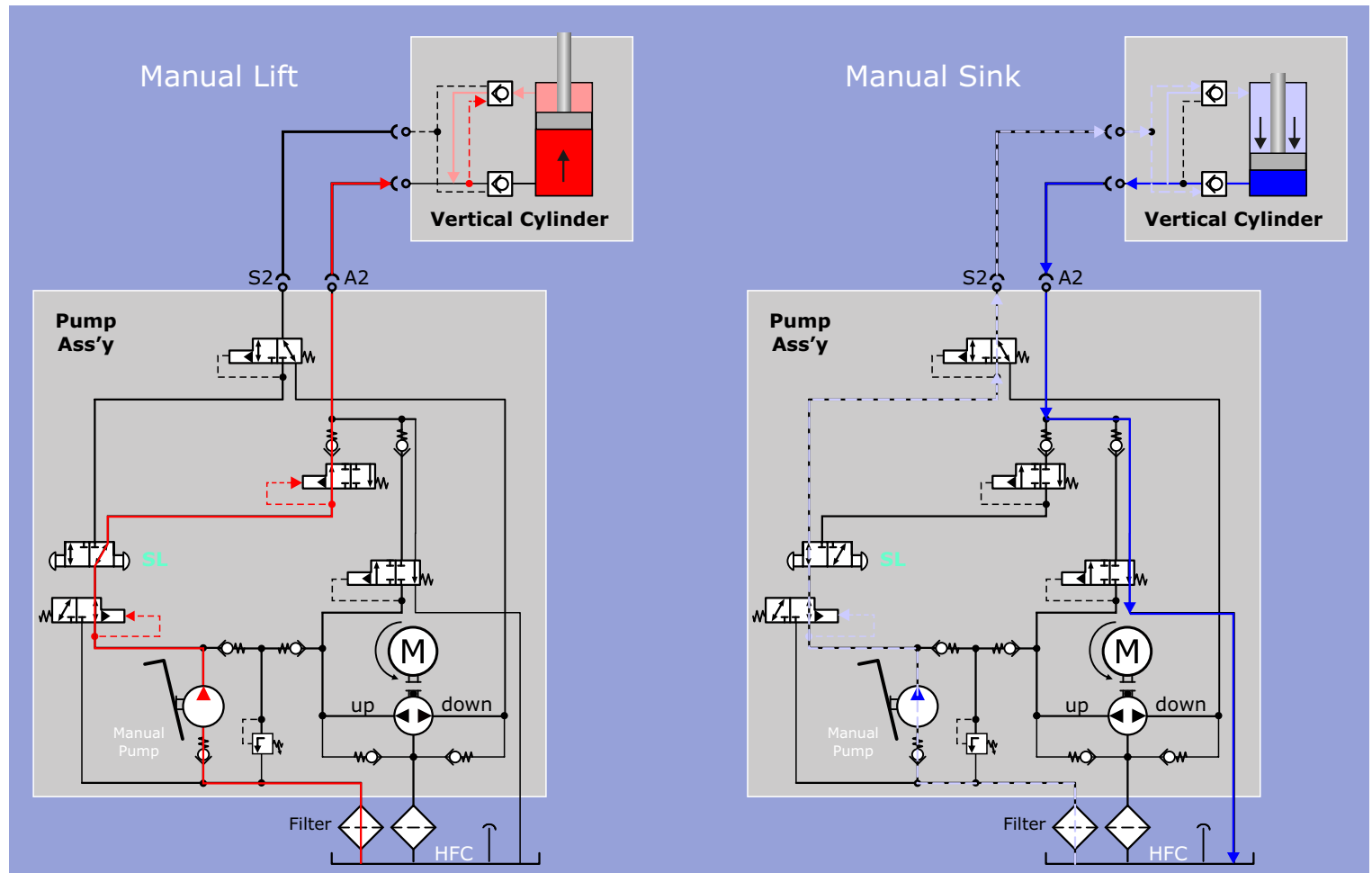


Figure 98 Hydraulic Overview Tim Table



S20 – Vertical HOME Sensor

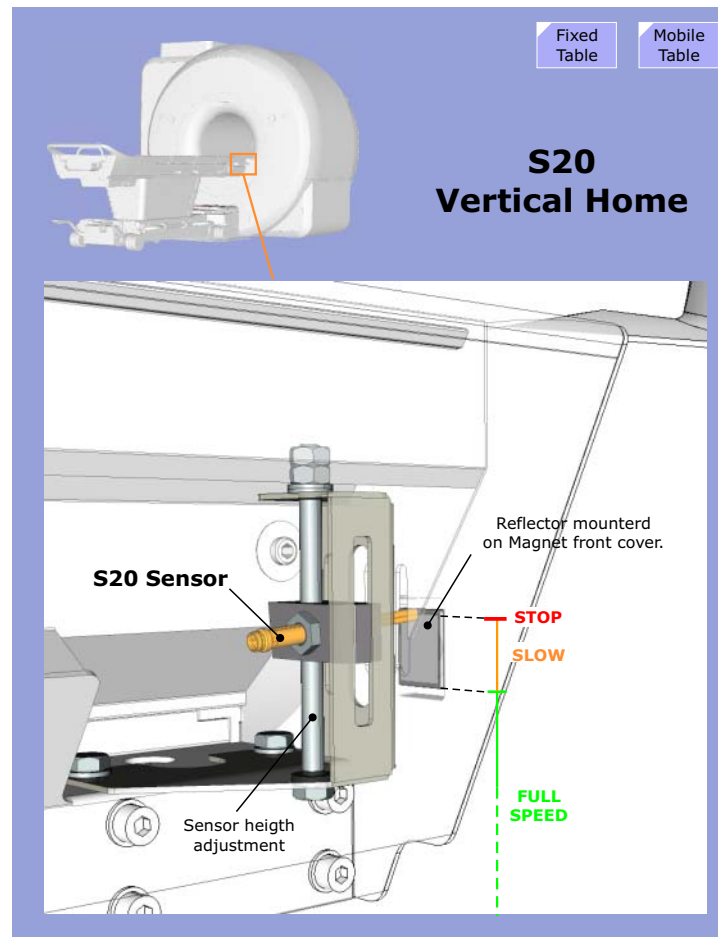
The precise upper position for transferring the Table Top into the bore is detected by the **optical sensor S20** at table front end, right-side. The position of sensor S20 is adjusted at the factory. The **reflector** can be adjusted to compensate differences in floor or magnet height. Should the amount of adjustment of the reflector not be enough, S20 can be adjusted as well. For this you have to remove the cover piece. Both procedures are described in the ROP.

Service Procedures

Tune Up - Table Height Adjustment

Each time after power-off (or docking), the table needs to search the precise S20 reflector position to establish the correct drive-in height for this specific magnet. This is achieved by an automated search procedure - moving up or down depending on the current vertical start position which is sensed by the PSY inclination sensor. The "Table Height" value defines the **lower vertical turn point** (below the reflector) where the system gives-up searching downwards and turns upwards to speed up the search process.

Figure 99 S20 - Vertical HOME Sensor



PSY – (Power Stage Y)

The PSY contains the **power stages for the vertical motor**. Its operating voltages are controlled by the Direct Position Drive (DPD). When the DPD has detected the Table Top closer than 50 cm to home position, the PSY is turned on. After home position is reached (optical switch S10), vertical movement is allowed.

Another important task of the PSY is to **evaluate the status of connected table switches and sensors** and to forward this information to the controller in the DPD. From there, MARS 2 or host can be read the information via CAN1 bus and display the corresponding patterns at the PDD or MRAWP.

Different to older systems, there are no mechanic switches for vertical position detection. In fact, just the coarse **vertical position** is sensed by a built-in absolute **inclination-sensor** inside the PSY. The precise drive-in height for transfer into the bore is determined by S20 optical sensor, see description below.

LED	color	Description
H10	green	+20 V_INT supply PSY ok (from PAM)

Fuses

Name	location	Rating
F10	PSY	4 A

Figure 100 PSY - Power Stage Y

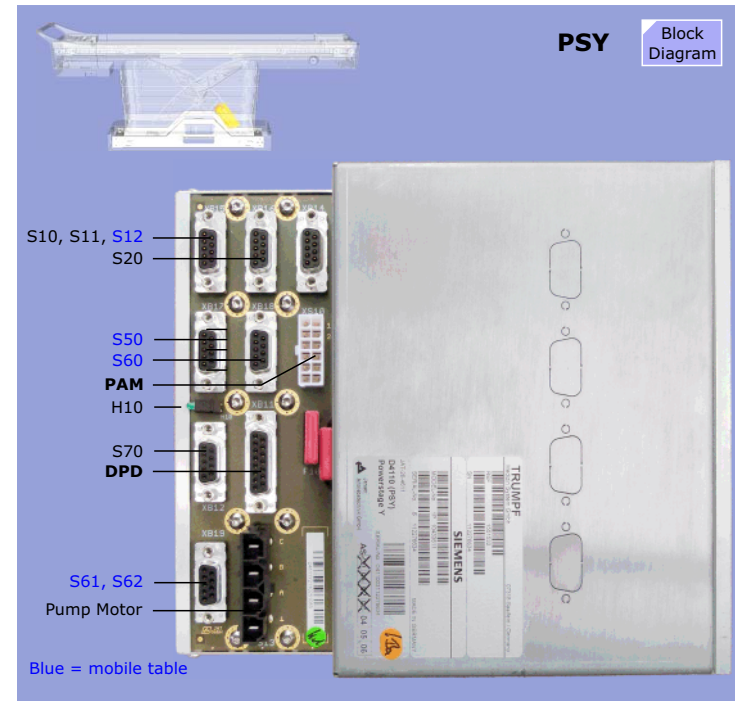
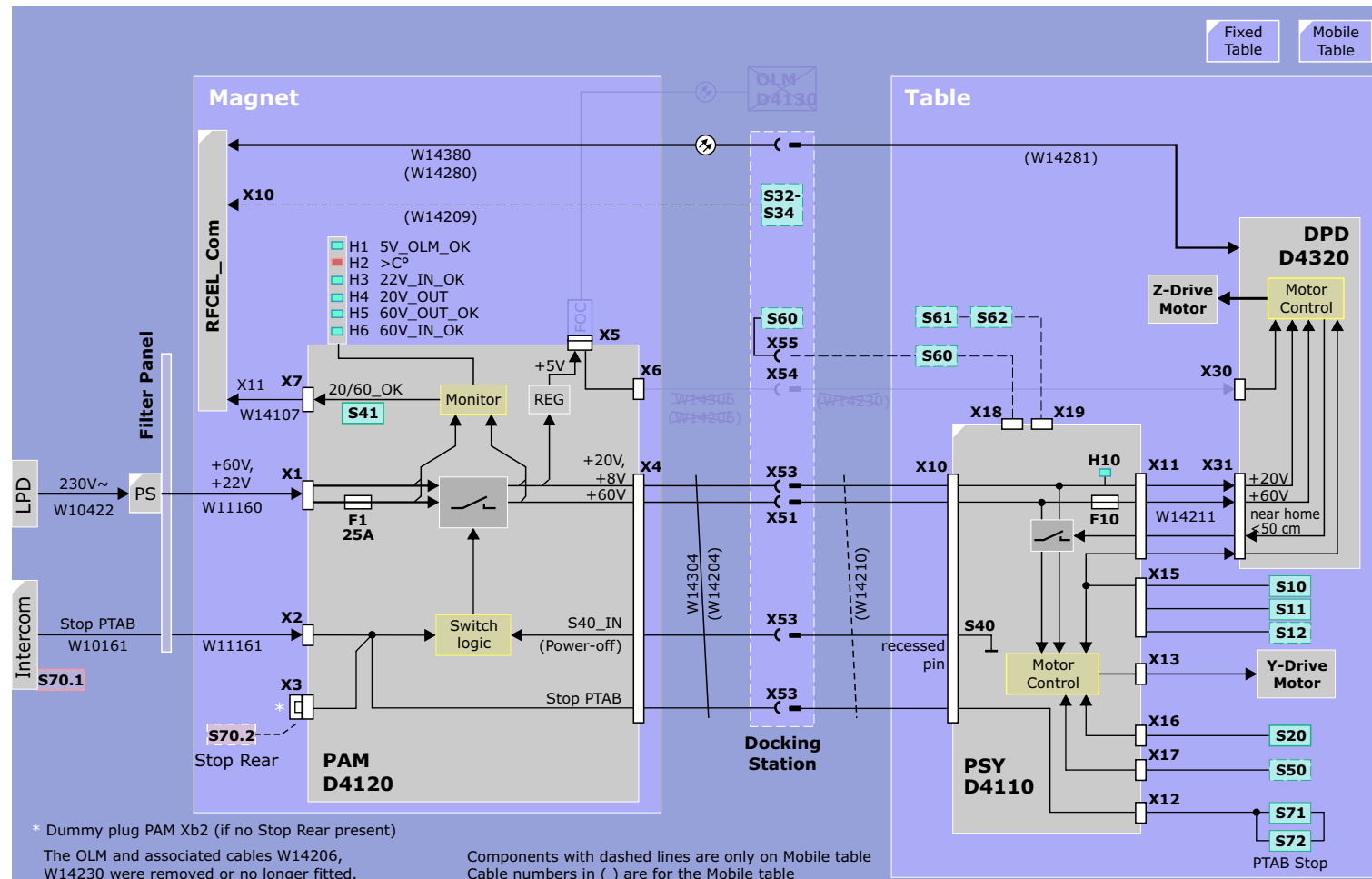


Figure 101 Electrical Components Block Diagram



PAM – Power Adaptor Module

Overview

The Patient Table electronics is supplied by the PTAB power supply, located on the filter plate. Its +22 and +60 voltage outputs are fed to the Power Adaptor Module (PAM) mounted on the right side of the RFCEL unit. The PAM contains regulators, power switch-off and monitoring-circuits.

PTAB voltage use:

- +22 V - for PTAB control circuits
- +60 V - for DPD and PSY motor circuits

Function

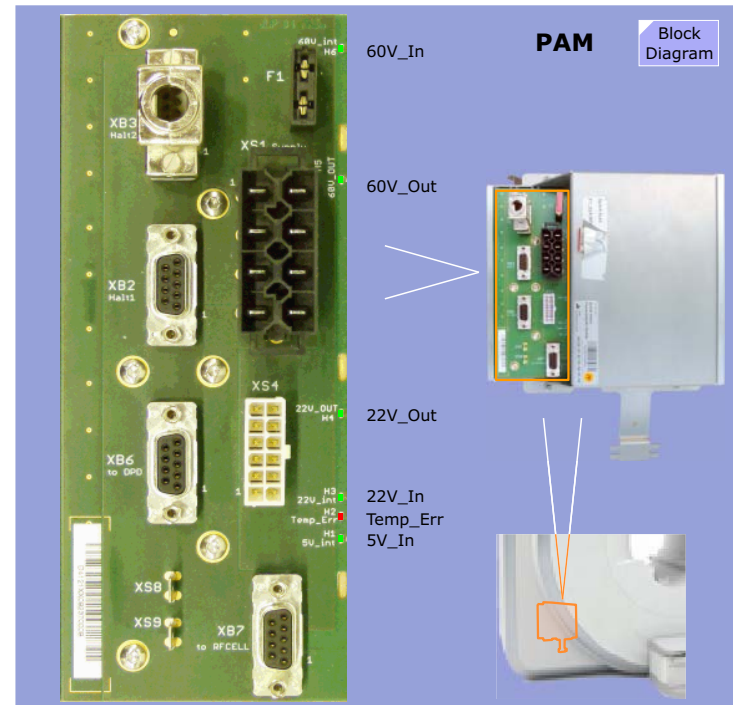
The PAM provides the following functions:

- Fuse for PTAB components (F1, 25 A)
- Switch on sequence for PTAB supply voltages
- Power on/off switching by dock/undock to avoid burned contacts (recessed pin on X53)
- Emergency Stop - disabling of drive systems in emergency
- voltage status reporting
- Supply voltage and interface

LEDs

LED	color	Description
H1	green	+5 V supply OLM ok
H2	red	PAM PCB over temperature
H3	green	+22 V_INT supply ok
H4	green	+20 V_OUT supply ok (for PSY and DPD logic)
H5	green	+60 V_OUT supply ok (motor supply voltage)
H6	green	+60 V_INT supply ok (from PTAB PS on filter panel)

Figure 102 Power Adaptor Module - PAM D4120



Jumpers

Name	Description
XB2	Table Stop circuit (removed for optional Rear Control Unit).

Test Points

+60 V	XS1 pin 1 (+) to pin 4 (gnd)
+22 V	XS1 pin 7 (+) to pin 5 (gnd)

Horizontal Drive

Overview

Horizontal movement is realized by a push-pull transport chain (works like a zipper) which is driven by a motor and gearbox.

The Horizontal Drive consists of the following major components:

- **Horizontal Drive Gear Box Assembly**
- **Horizontal "Direct Position Drive" (DPD) D4320**
- Optical Length Meter (OLM) D4130 [obsolete]
- Optical Scale (OSC) D4140 [obsolete]

Horizontal Drive Gear Box Assembly

A plastic gear box, located at the table foot-end. It is driven by the DPD over a toothed belt and contains a "zipper-like" **push-pull transport chain** for moving the Table Top. The chain is rigid when composed and flexible when its two belts are separated. Separated belts are captured by a belt storage at table head-end.

In case of an emergency, the Table Top can be decoupled from the Gear Box by an **Emergency Release Handle**.

DPD – Direct Position Drive

Overview

The DPD contains:

- Horizontal motor
- Table controller
- Horizontal drive electronics
- Incremental relative position encoder

The DPD is active during measurements, therefore the box is **electromagnetically shielded** to avoid RF-interference.

Table Controller

The DPD contains the table controller. Its intelligence is limited since most of the table functions are initiated and monitored by the MARS 2. An optical CAN1 interface is used for transferring table commands and status and error messages. The DPD is also responsible for power switching of the vertical drive ([page 137](#)).

The table controller contains a non-volatile memory to store important table data like:

- Table material number (Siemens part. no.)
- Table serial number
- Table utilization data (drive and docking cycles)

Scan Range Configuration

The allowed horizontal scan range is defined by the MR-system. A **standard system has 140 cm**. To expand the range to **up to 205 cm**, the optional **Tim Whole Body Suite license** must be installed, together with the suitable configuration setting in syngo. In cases where the clearance between magnet and back wall is not sufficient, the configured range can be restricted. Additional **mechanic end stops** which are installed inside the magnet funnel or at the optional rear support ensure the allowed range is never exceeded.

Table ID recognition

The software can recognize the 4 possible types of connected tables to assure a safe function:

- Tim Table 1.5 T
- Tim Table 3 T
- Tim Dockable Table 1.5 T
- Tim Dockable Table 3 T

The corresponding table material number (Siemens part no.) and Serial number is factory programmed into the DPD controller and can be read out automatically for the "Asset Management".

Figure 103 Horizontal Drive System - Fixed Table

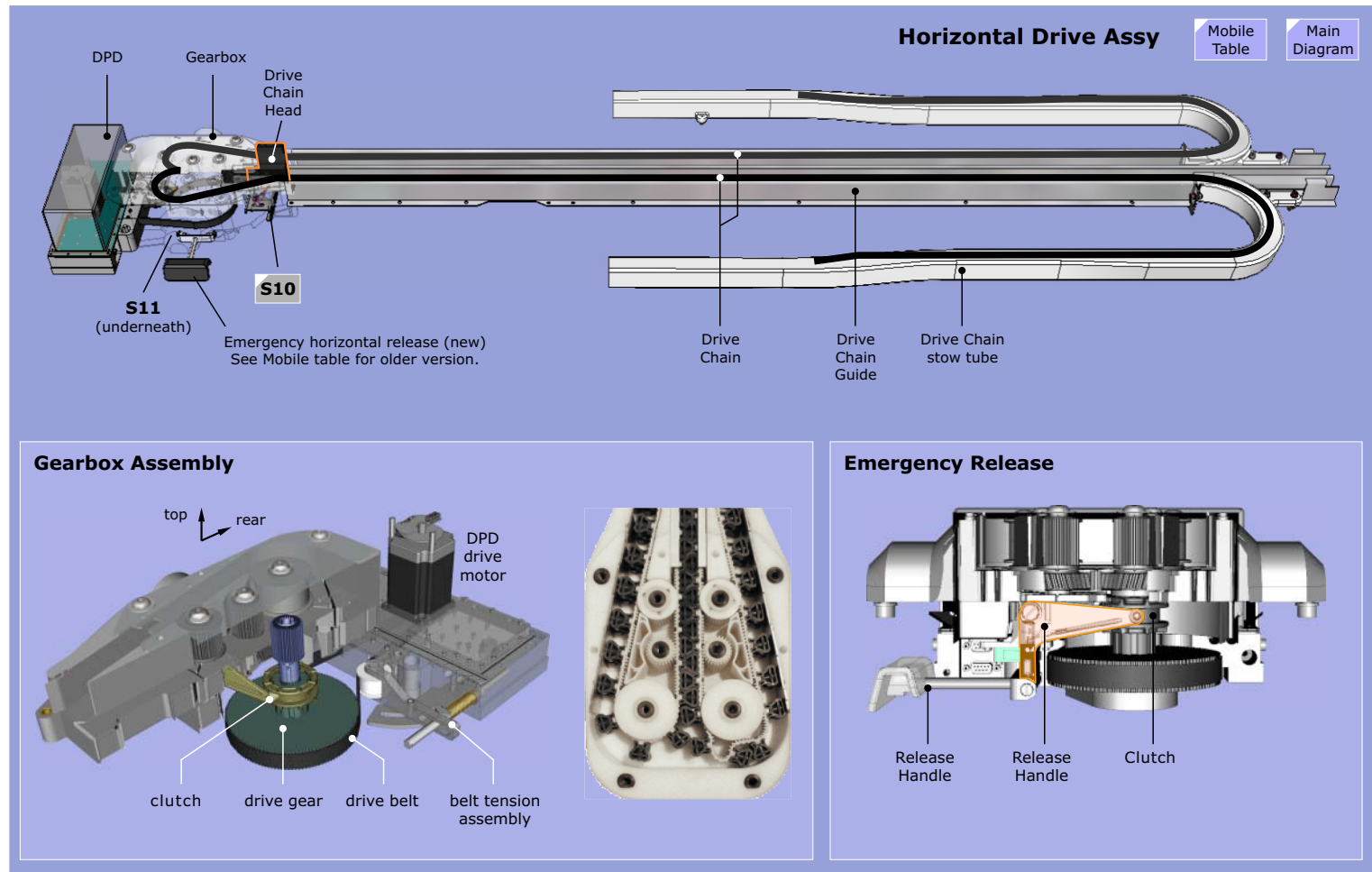
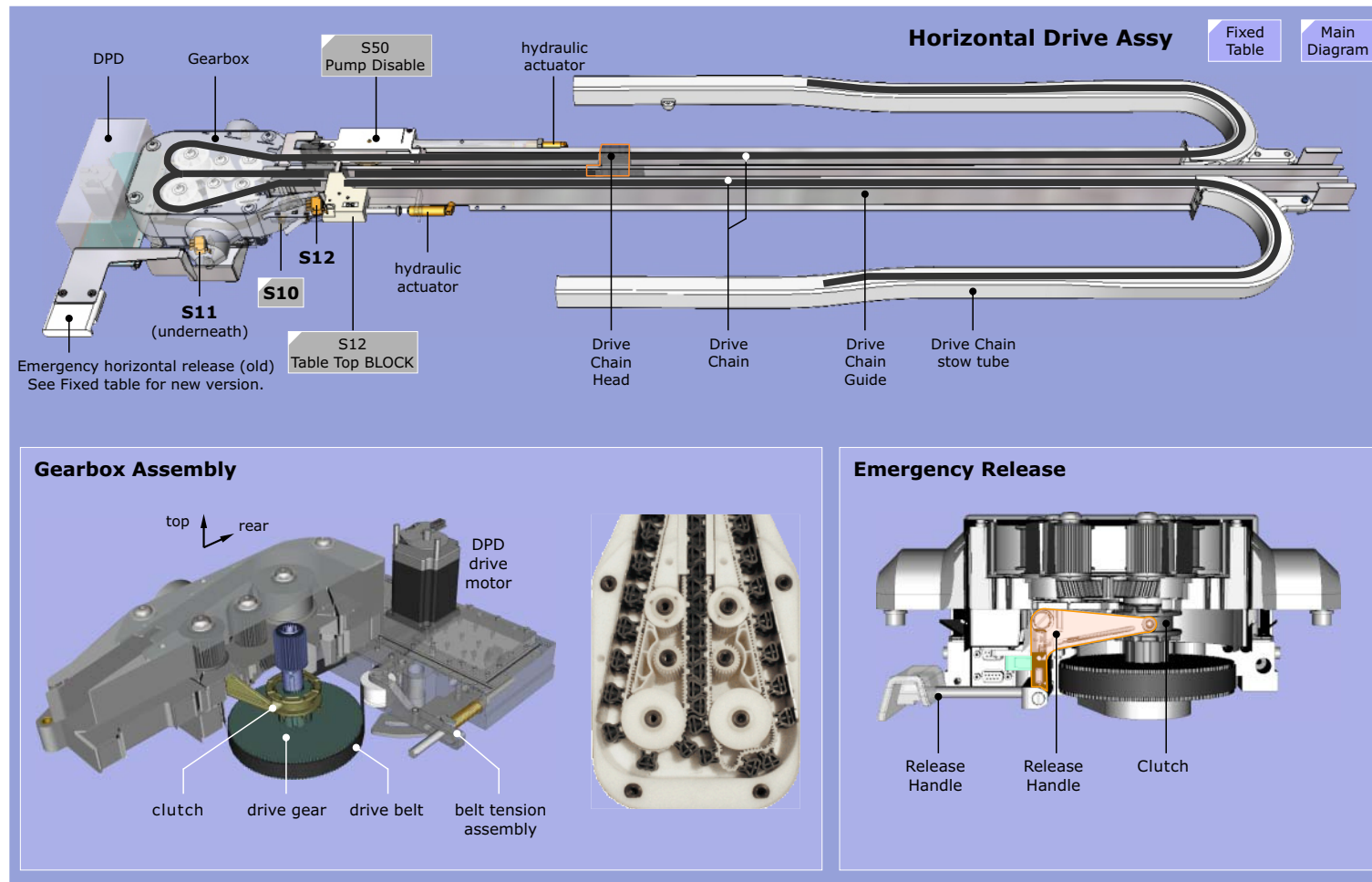


Figure 104 Horizontal Drive System - Mobile Table



Each time when the table is docked or powered up, the host performs a **plausibility check** by comparing the recognized table material number with the required system configuration data for field strength and table type. If a mismatch occurs, e.g. a 1.5 T table is docked at a MAGNETOM Skyra or a unknown serial number is detected, an error message is generated and the MARS 2 does not enable table travel.

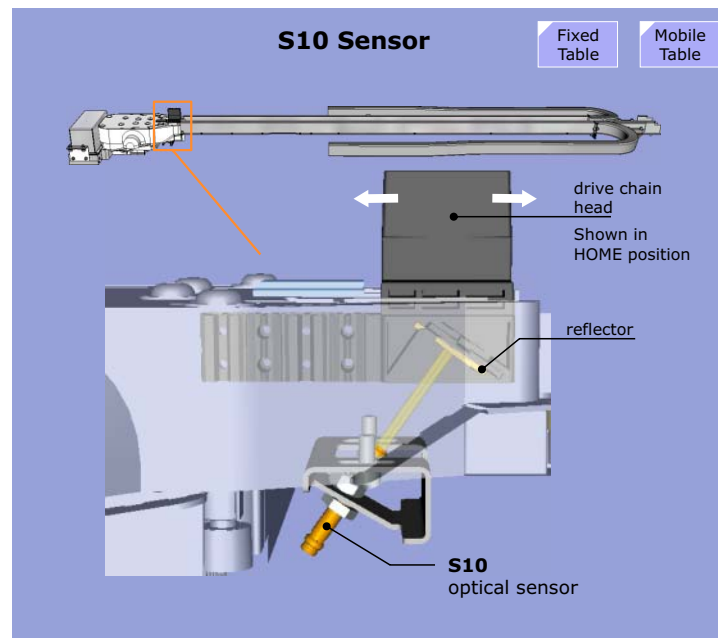
Service Procedures

Table ID programming

After replacing a DPD or to initialize the function of a unknown table, a **re-configuration of the DPD and system software is necessary**, otherwise the table is not functional. A standalone script ("mrverifyptab") is used to program the table material- and serial-number.

NOTE For the detailed DPD programming procedure please refer to the Replacement of Parts description.

Figure 105 Horizontal HOME Detector S10



OLM – (Optical Length Meter)

NOTE The OLM has been disabled for all systems. The following description is retained only for legacy reason.

Refer to SKB entries SKB0067279 and SKB0090953.

The OLM is an optical position sensor to detect horizontal Table Top position with high accuracy.

Function

The initial table reference position is detected by optical switch **S10** when moving the Table Top into its **HOME** position. Relative to this position, an **incremental encoder** inside the DPD keeps track of the horizontal position. When the **OSC** (underneath the Table Top) has reached the **OLM** it is used to obtain precise horizontal position information.

The OLM LEDs will be switched off when the Table comes into horizontal HOME position (S10).

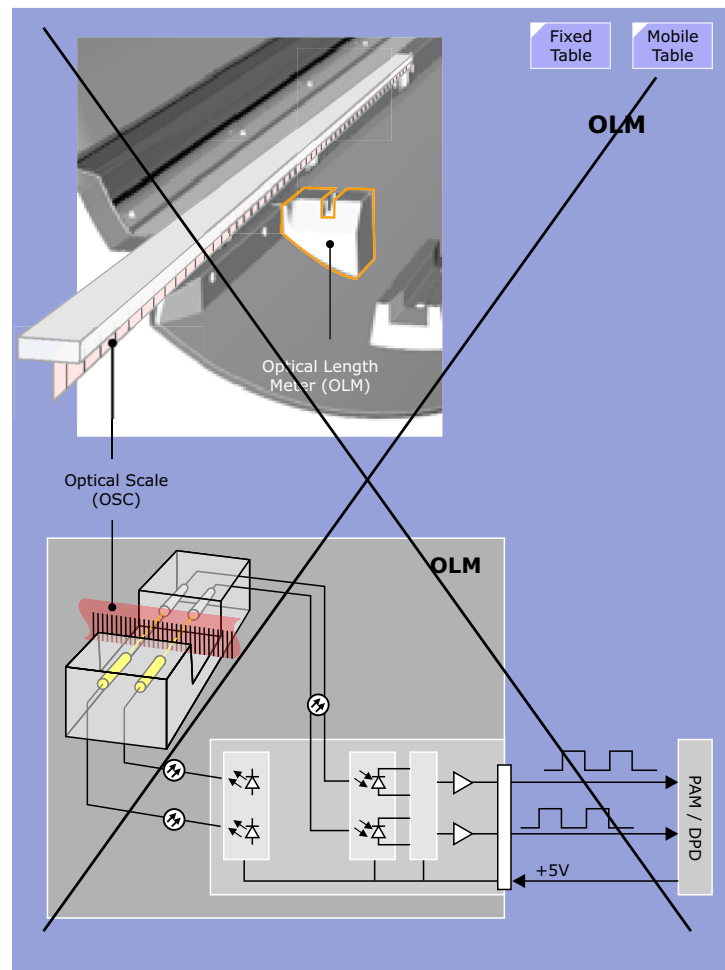
Auto Calibration

A test of the optical attenuation of the OLM is performed each time the table goes into horizontal HOME position (S10). If the light intensity falls below a limit a calibration of the OLM is initiated whereby the current is increased until the light intensity is within specifications.

OSC – (Optical Scale)

A plastic distance measurement scale for the OLM, mounted underneath the Table Top. The OSC is a FRU and can be easily replaced if damaged.

Figure 106 Optical Length Meter



Tim Dockable Table (Mobile)

Overview

The Tim Dockable Table is a consequent realization of a mobile table solution. It is believed to provide all must-have features such as patient comfort, maneuverability, agility, flexibility, ease of use, reliability and of course Tim-ready, i.e., built-in coil connectors.

Double-wheeled castor rollers provide a smooth travel and the implementation of a 5th wheel greatly improves maneuverability, especially around corners.

Smart mechanical ingenuity has met the challenge to provide a solution for a mobile table requiring just 2 pedals for both vertical lift and docking functions.

The **Docking Station** is an assembly fixed to the floor via a bonded base plate and provides an ingenious mechanical alignment mechanism assuring a successful docking between table and docking station regardless of approach angle. A connector assembly integrated within the dock assembly of the Docking Station and the Patient Table accommodates the various electrical, optical and pneumatic lines to and from the table and has been designed to assure good electrical connections as well as preventing possible damage to the connectors despite the weight of the table and patient.

Assemblies

The Tim Docking Table can be broken down into the following assemblies:

Electrical

- PSY, DPD, OSC, PAM

These components are identical for both tables.

Mechanical

- Table Base
 - Rollers and Brake System
 - 5th Wheel
 - Docking Unit
 - Hydraulic Pump Assembly
- Vertical Lift
 - Scissor Assembly
 - PSY
 - Hydraulic Cylinder
- Support Frame
 - Horizontal Drive
 - Vertical and Horizontal HOME sensory
- Table Top
 - Receive Coil Connectors

Sensory

- S10, S20, S11, S70, S71 – identical to Tim Table.
- **S12** – Horizontal Block
- **S32-34** – Dock
- **S50** – Pedal Block
- **S60** – Collision Detection
- **S61, S62** – Side Rail park position

Figure 107 Tim Dockable Table Electrical Diagram

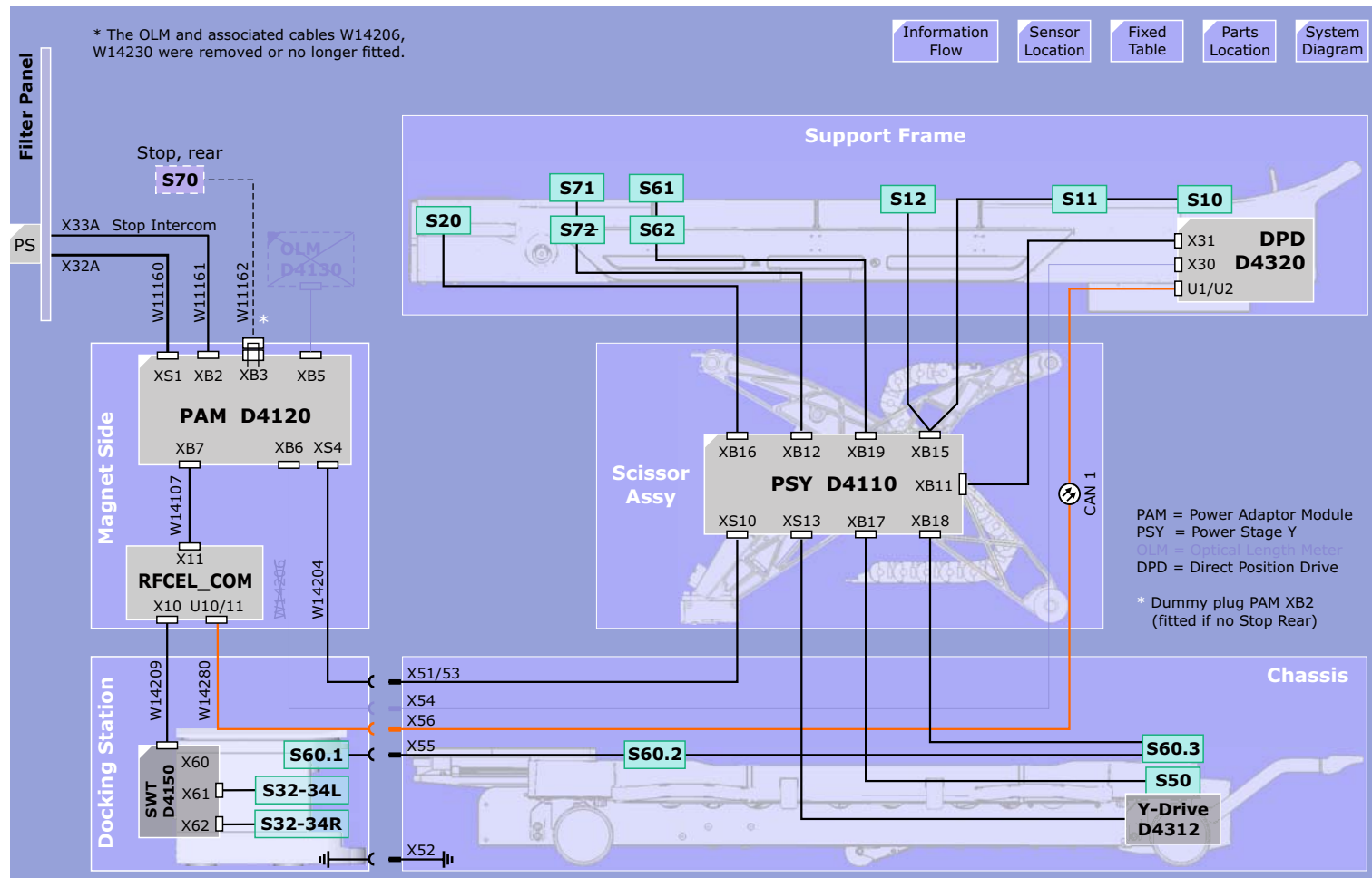
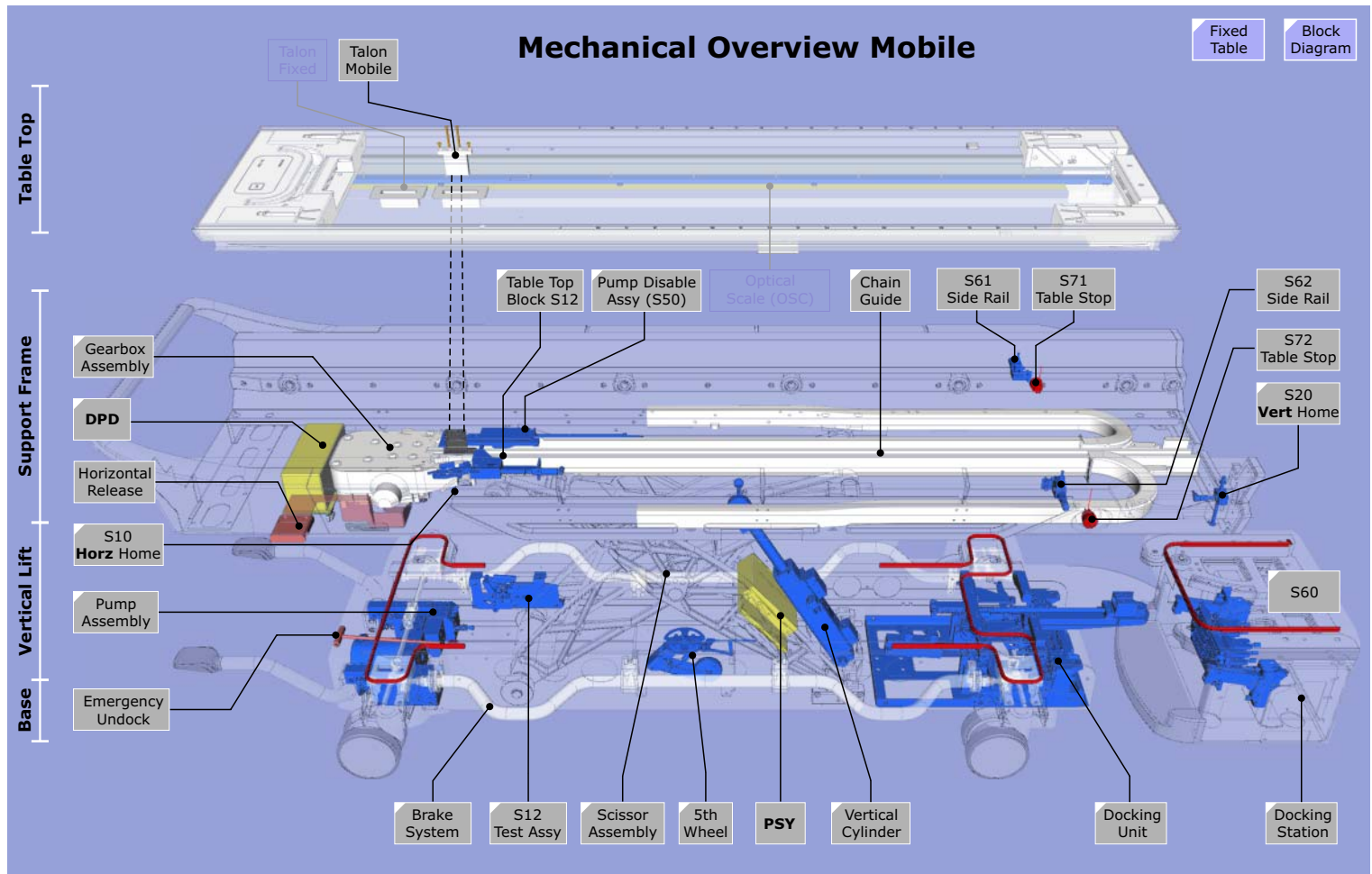


Figure 108 Tim Dockable Table Parts Location



Wheel and Brake System

The mobile patient table employs a set of twin-wheel castors with a central locking mechanism to provide good mobility and smooth tracking. The central locking mechanisms allow for quick locking of all wheels using a single brake engagement pedal located along the foot sides of the table. A centrally placed 5th wheel assembly has been added to improve maneuverability and is engaged or disengaged with the brake pedals.

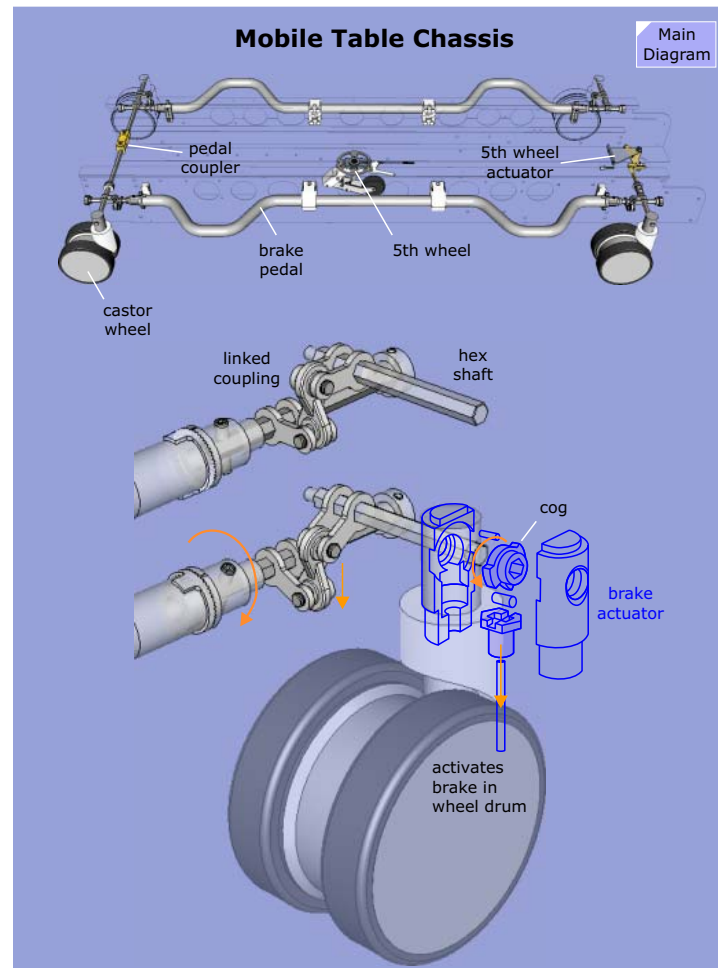
Brake Pedals

Both pedals are linked via a coupler at the back side of the table's chassis. Moving either pedal is mechanically coupled to the other brake pedal.

Brakes

To engage the built-in brake mechanism (highlighted blue line) the brake pedal is moved to the lower position. A cog in the brake actuator is rotated via a linkage coupling and hex shaft which in turn presses down a shaft engaging the brake in the wheel drum (not shown).

Figure 109 Wheel and Brake Assembly



5th Wheel Assembly

Engaged

The 5th wheel is engaged (locked in position) when the brake pedal is in the middle position. If previously disengaged the table needs only to be moved forwards (or backwards) to raster the lock the rotating lock.

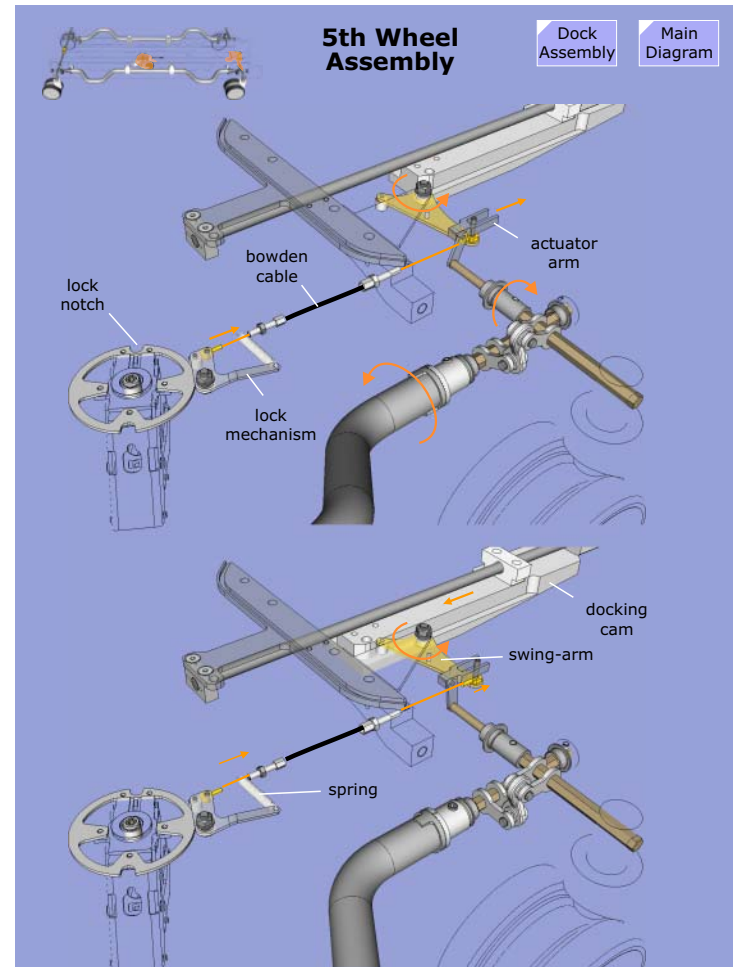
Disengaged

There are two actions that will disengage the 5th wheel:

- **Brake Pedal:** when the brake pedal is moved to the upper-most position an actuator arm is pushed forward pulling the Bowden cable attached to the spring-loaded locking mechanism on the 5th wheel assembly.
- **Docking Cam:** the 5th wheel is automatically disengaged when docking. This is necessary when the table is coming into the Docking Station at an angle and the table will be moving sideways as it straightens out during the dock process. When docking the swing-arm is pushed back by the cam pulling the Bowden cable

The spring assures the lock mechanism is re-engaged when the brake pedal is returned to the middle position.

Figure 110 5th Wheel Assembly



Vertical Lift

Overview

The vertical drive system consists of the following components:

- **Vertical Lift Assembly**
- **Hydraulic Pump Assembly**
- **S20 Vertical Home Sensor**
- **S60 Collision Detection**

Hydraulic Pump Assembly

Hydraulic System

The vertical drive for a mobile table fulfills the same basic functions like its counterpart for fixed tables.

In addition it contains two foot pedals for manual hydraulic operation of the vertical lift or the table docking cylinder.

Pump Outputs

The Pump has two outputs:

- **D1, D2** - go to the **docking cylinder** located in the Docking Unit at the front of the table.
- **S1, A1** - go to the **vertical lift hydraulic cylinder** located in the Scissor Assembly

Both sets of hydraulic lines connecting the pump outputs to the hydraulic cylinders use quick disconnects allowing for easy replacement.

Foot Pedals

The two foot pedals allow manual pumping and have context dependant functions:

- Right Pedal - table lift (when undocked) and dock
- Left Pedal - table down (when undocked) and undock

Pedal Function

The pedal function is context-dependant i.e., vertical lift when the table is undocked and docking function when the table is at the Docking Station. The switching is controlled by the hydraulic valves **V3** and **V4** which are operated by the Docking Cam ([which see](#)).

Pump Disable

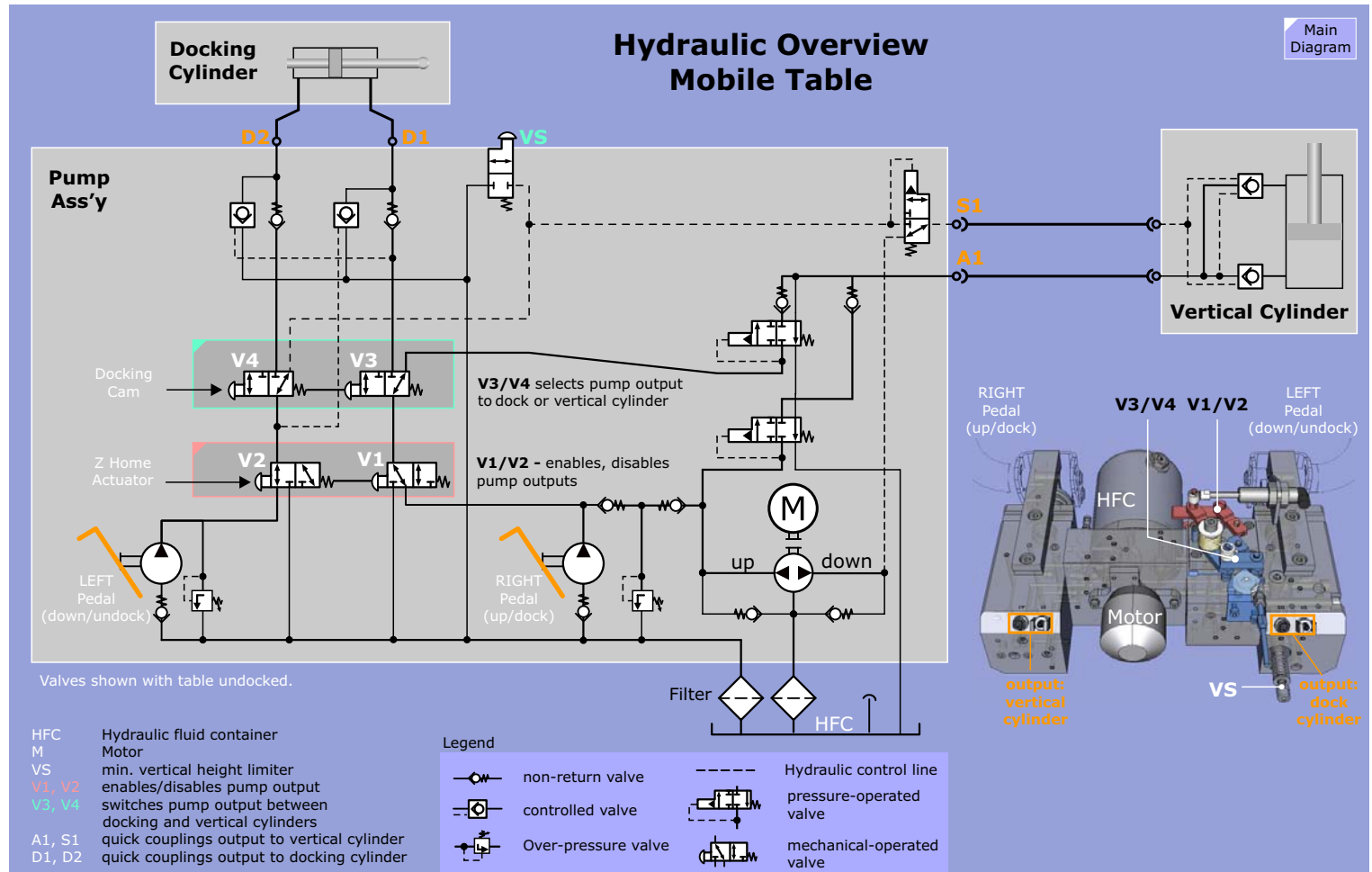
When the patient is moved into the bore (i.e., the Table Top leaves HOME position) the hydraulic pump must be disabled to prevent undocking the table. This is performed by **S50** and valves **V1/V2**.

When the Table Top leaves home position the hydraulic actuator of **S50** causes valve pair **V1/V2** to a position which causes the pump on the pump assembly to switch to the "pump disable" position and the pump is disabled. ([which see](#)).

Minimum Height Limiter

When the table is undocked, the electrical vertical collision detection circuit is unavailable. Therefore, the minimum table height above ground must be increased from **560 mm** to **610 mm** in order to prevent collisions (e.g. clamping hazard when a foot is positioned on top of the table chassis cover during down-movement). This function is controlled by the **hydraulic valve VS** which is actuated by a scissor wheel and shuts-off the vertical down function as soon as the table has reached the allowed minimum height.

Figure 111 Hydraulic Overview - Mobile Table



Pump Disable - V1/V2

The hydraulic pump must be disabled when the table top is no longer in HOME position, i.e. has been moved into the magnet. Valves V1 and V2 are responsible for this function.

Refer also to [Figure 111](#).

Pump Disable

As the table leaves the home position the drive chain head moves forward and releases a spring-loaded rack and pinion gear and allows the hydraulic actuator A to retract by releasing the hydraulic pressure on actuator B which is depressed by a retraction spring (see diagram at right). As this happens V1/V2 are pushed to the left disabling the pump output (pedals disabled).

Microswitch S50 ACTIVATES to indicate the status.

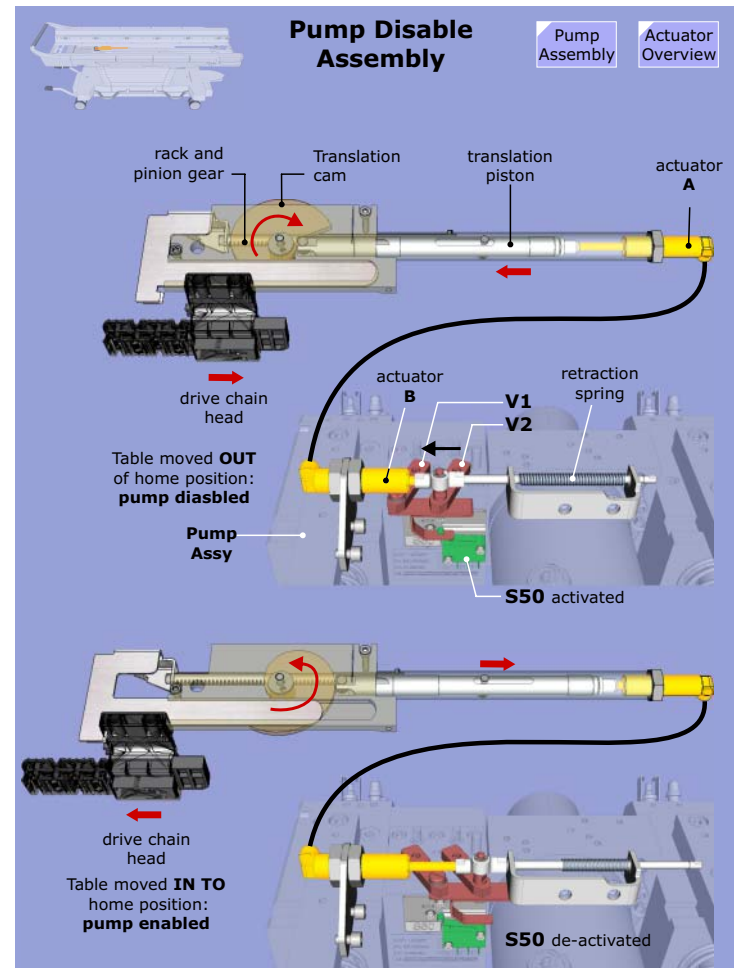
Overpressure valves in the pump prevent damage to the pump if the pedals are pushed down with force.

Pump Enable

As the table is driven into the HOME position the Pedal Lock mechanism CLOSSES actuator A building up pressure on actuator B which pushes V1/V2 to the right. The pump is now enabled again.

Microswitch S50 DE_ACTIVATES to indicate the status.

Figure 112 Pump Disable



Pump Assembly Switching - V3/V4

Valves V3 and V4 are responsible for switching the output of the pump assembly between the vertical hydraulic cylinder (for raising/lowering the table) and the docking hydraulic cylinder (for docking/undocking).

Valves V3 and V4 are coupled together and actuated by the Docking cam located in the Docking Unit found at the front part of the table.

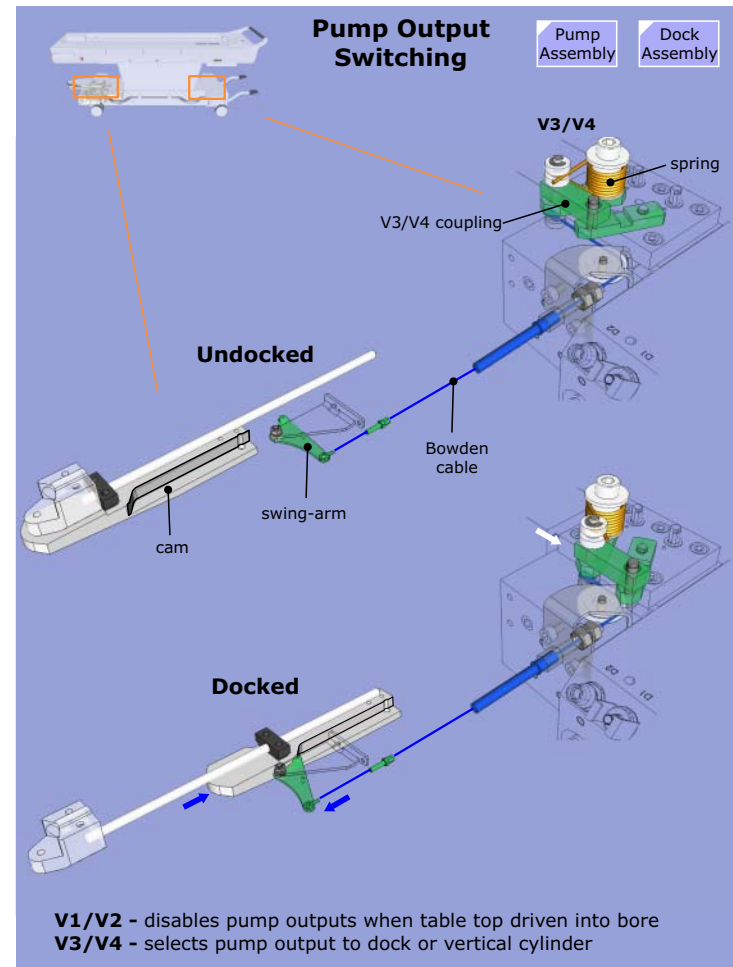
Docking

When the table is being docked the cam forces the swing-arm to pivot forward (blue arrows) pulling on a bowden cable connected to valve assembly V3/V4 moving it to the position which switches the pump output from the vertical cylinder to the dock cylinder (as seen in the diagram).

Undocking

As the table is undocked and pulled away the spring forces V3/V4 back to its original state and the pump output is now to the vertical lift hydraulic cylinder.

Figure 113 Switching Pump Output with V3/V4



S12 Test Assembly

Due to the safety importance of the Table Top Lock function (S12) a circuit has been implemented to test its function.

This test can only be exercised when the table is docked.

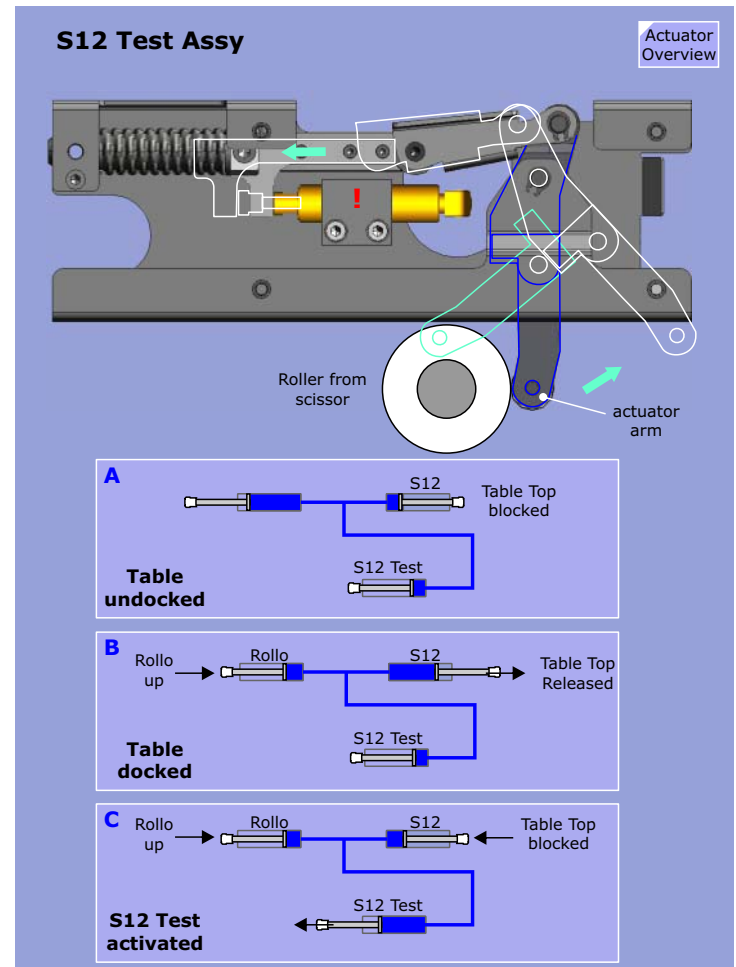
Function

When the table is lowered vertically the roller of the scissor assembly pushes the actuator arm back (white highlight). The arm in turn OPENS the actuator and pressure is released in the hydraulic circuit (bottom diagram). This releases the table block and switch S12 is activated verifying that it is functioning properly.

As the roller passes the arm completely it snaps back to its original position and the actuator is CLOSED increasing the pressure in the hydraulic circuit and the Table Block assembly S12.

The actuator arm is hinged so that when the table is raised it has no effect on the actuator (green highlight).

Figure 114 S12 Test Assembly



Vertical Collision Detection

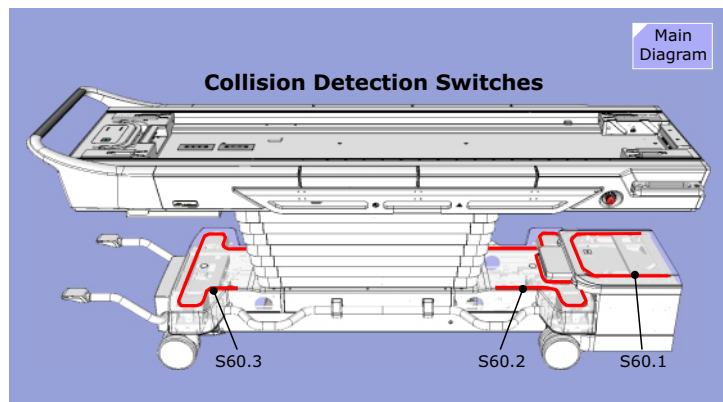
The Tim Dockable Table implements an electrical vertical collision protection system to provide crush protection to patient and hardware. It is realized using three pressure sensitive switches (hoses) connected in parallel:

- **S60-L1 Docking Station** – under top cover of the Docking Station
- **S60-L2 Table** – mounted top of the table chassis head-end
- **S60-L3 Table** – mounted top of the table chassis foot-end

The switches are normally open and will close when the hose is pinched. In order to be able to detect broken lines, a parallel terminator resistor (approx. 1.2 K Ω) is included in the circuit.

NOTE This circuit only works when the table is docked (connected). When undocked it is not functional.

Figure 115 S60 Collision Detection



Docking

Docking Mechanism

The Tim Dockable Tables attach themselves to a Docking Station located at the magnet's front-side mounted on a metal base plate which must be bonded to the floor during installation. All **electrical, fibre optic and air-hose connections** to the Patient Table are fed through this interface.

The Tim Dockable table is equipped with a **nozzle** which fits into the **cone** of the Docking Station to facilitate docking from an angle.

When approaching the magnet, the nozzle enters the cone and aligns the table and docking port. The **protective shutters** of Docking Station and Patient Table will be both opened to allow coupling of the docking connectors.

Figure 116 Dock Guidance

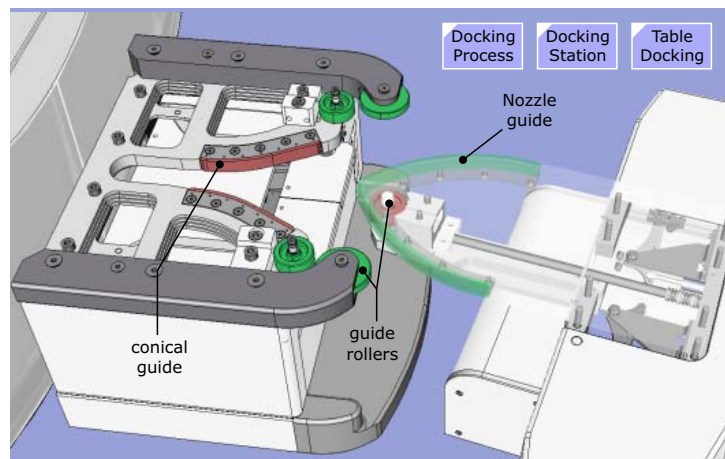
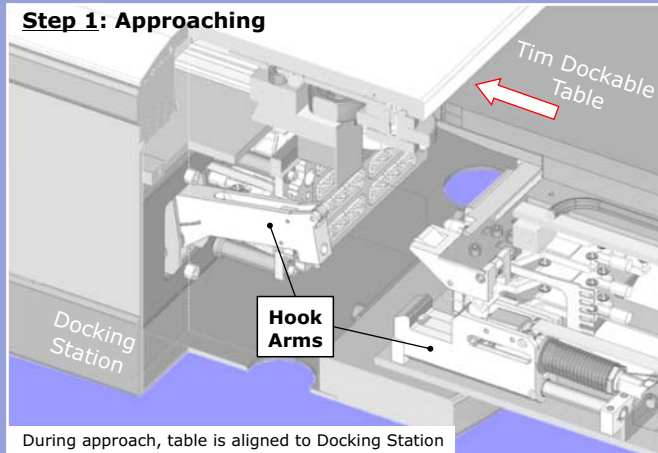
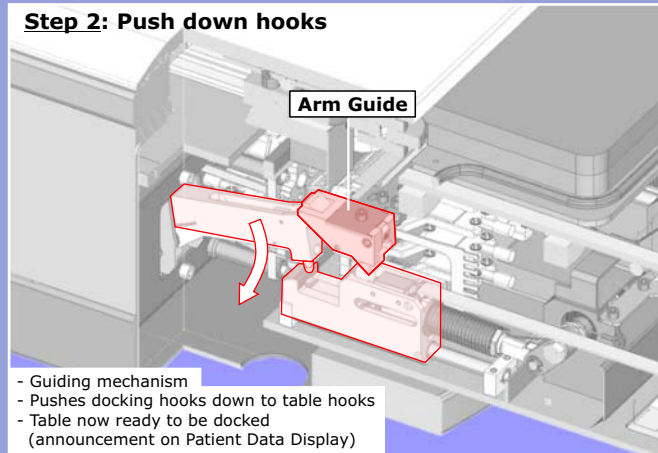


Figure 117 Docking Process

Step 1: Approaching

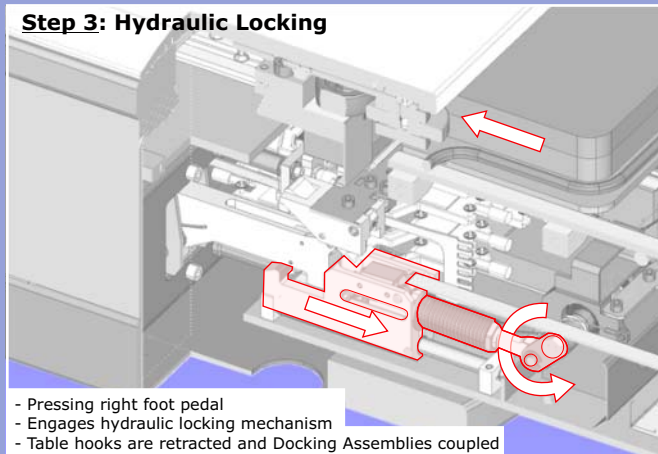


Step 2: Push down hooks

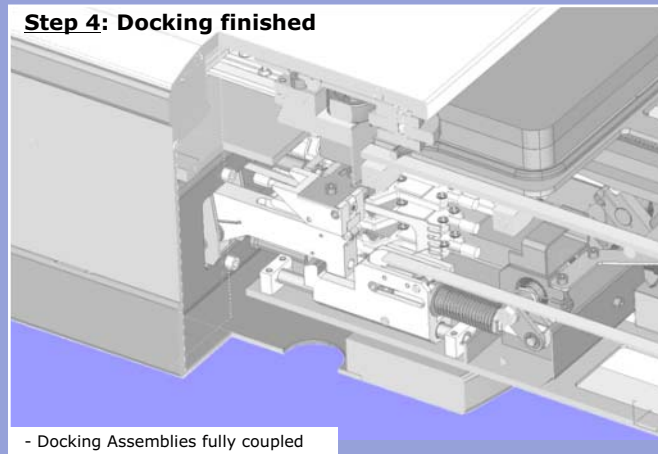


BACK

Step 3: Hydraulic Locking



Step 4: Docking finished

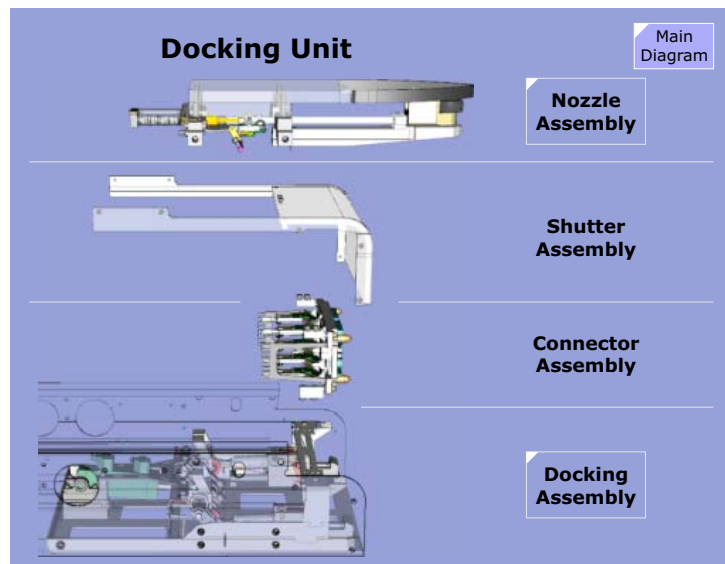


Docking Unit

The Docking Unit can be broken down into these four functional groups:

- **Nozzle Assembly** – consisting of an alignment guide comprised of a rounded nozzle element with rollers and a set of three actuators to initiate various docking-specific functions.
- **Connector Assembly** – consisting of all IF, control signal and voltage connections.
- **Docking Assembly** – responsible for securing the table to the Docking Station.
- **Shutter Assembly** – a retractable cover serving as a protection to the connector assembly when table is undocked.

Figure 118 Docking Unit



Connector Assembly

The figure at below provides a layout of the connectors on both connector assemblies.

The connector assemblies are attached using flexible mountings. Large guide pins on the table connector assembly fit into corresponding guide holes on the docking station connector assembly to assure a flush connection.

Connectors are replaceable parts.

Figure 119 Connector Assemblies

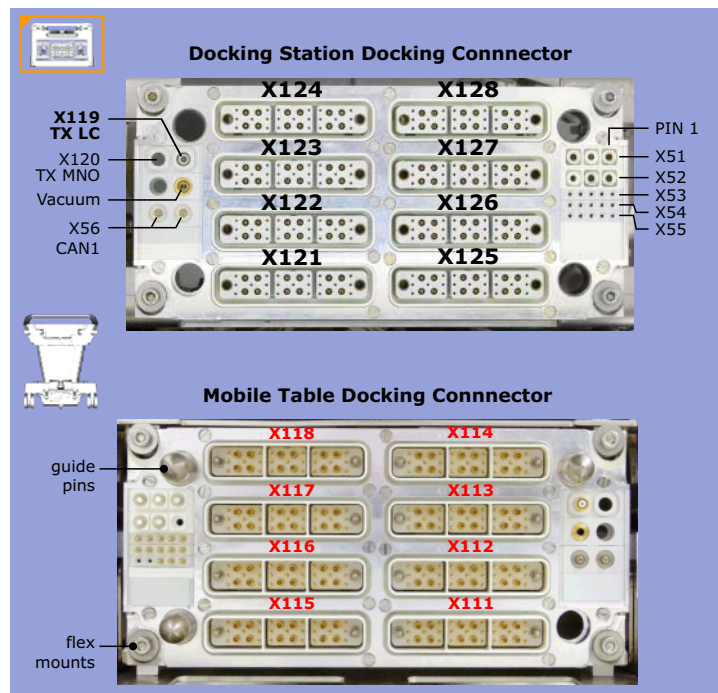


Table Docking Assembly

Several mechanical functions are to be found in the Docking Assembly, particularly the Nozzle Assembly and Dock Mechanism.

Nozzle Assembly

There are three functions which need to be performed when either docking or undocking the table.

When **docking** it is necessary to:

- release the table top which is locked when the table is not docked (S12)
- disengage the 5th wheel to ease the docking procedure, especially when docking at an angle
- switch over the function of the two foot pedals at the rear of the table from up/down to dock/undock (V3/V4)

When **Undocking** - The opposite is true.

Each of these functions are explained in detail elsewhere.

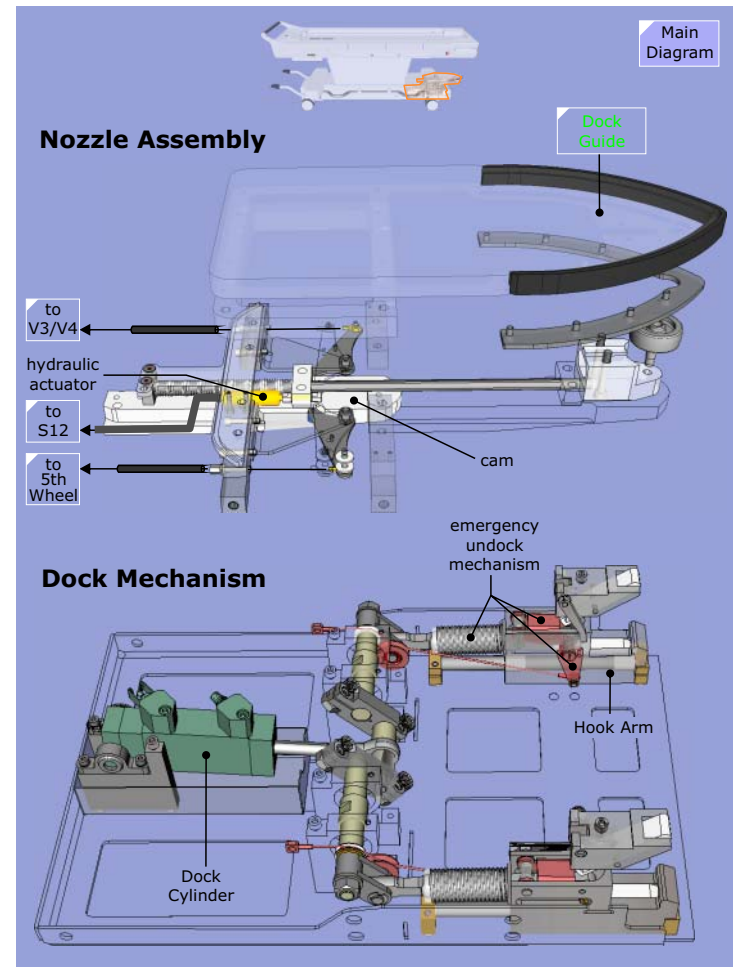
Dock Mechanism

The coupling of the table to the Docking Unit is achieved with a two-way hydraulic Dock cylinder connected to Hook arms. Retracting the Hook Arms pulls the table in the Docking Station to dock, pushing the Hook Arms out releases the table from the Docking Station.

Emergency Undock for Mobile Table

For emergency situations a Hook Arm release mechanism decouples the Hook Arms from its shafts are pushed forward with strong springs to release the table in a quick way. The emergency release mechanism is actuated with the Emergency Undock handle found on the bottom at the back of the table.

Figure 120 Table Docking Assembly

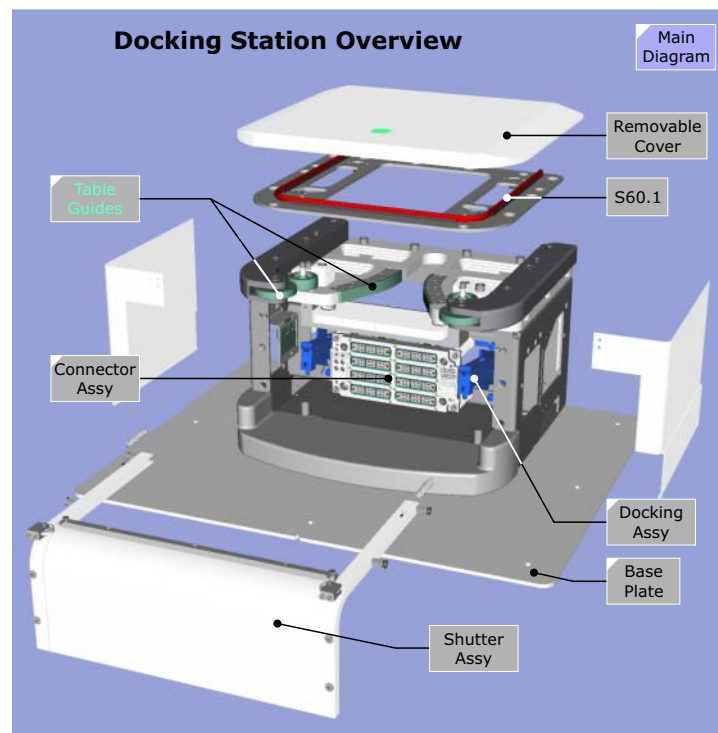


Docking Station

The Docking Station can be broken down into these assemblies or functions:

- **Cover Assembly** - the top cover is attached using snap-locks for easy removal. Integrated into the cover is an electrical pressure sensitive switch sensor (S60.1) which is used to detect blockages and prevent crushes.
- **Table Guides** - a conical guide to assure the table straightens up when the table is brought in at an angle. Important is, the table approach to the connector assembly is straight. Addition mechanisms on the connector assembly also aid in assuring the two connector assemblies make a flush contact. The pins of the connectors are very small and the table with patient weighs in at less 300kg. Not a good prognosis for the connectors if the guides don't perform their tasks.
- **Docking Assembly** - is responsible for a solid dock of the table to the station. It comprises a set of hook arms that latch onto similar hook assemblies found in the table's docking assembly. The docking hydraulic cylinder is responsible for pulling and fastening the two. Microswitches are located on the Docking Assembly to give feedback of the operation.
- **Connector Assembly** - provides all electrical connections to the system. Additional microswitches are placed on this assembly to indicate the two connectors have made good contact.
- **Shutter Assembly** - provides coverage of the unsightly innards of the Docking Station when the table isn't docked. It also protects the connectors from unwanted damage.
- **Covers** - the side covers and base moulding pieces are designed for easy removal and replacement.

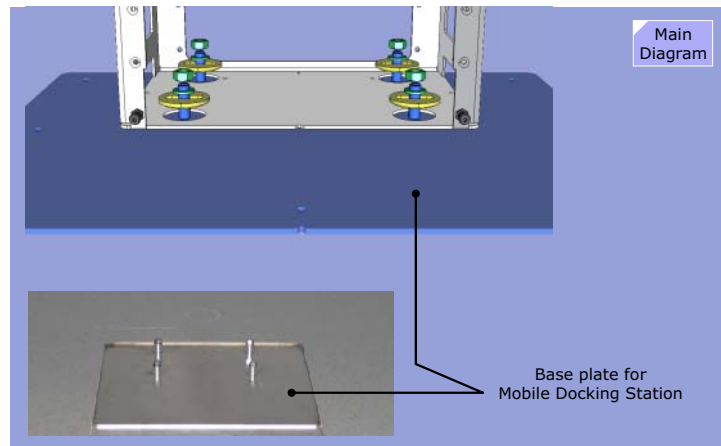
Figure 121 Dock Station



Installation

To assure the docking stability (the table weighs ~220kg, with patient a bit more, and when driven at full speed a bit more) the Docking Station is planted to the floor via a stainless steel base plate which is glued to the RF-cabin floor.

Figure 122 Base Mount Plate



Docking Assembly

The **Docking Assembly** is responsible for a solid dock of the table to the station. It comprises a set of hook arms that latch onto similar hook assemblies found in the table's docking assembly. The docking hydraulic cylinder is responsible for pulling and fastening the two.

Microswitches are located on the Docking Assembly to give feedback of the operation.

Figure 123 Dock Assembly

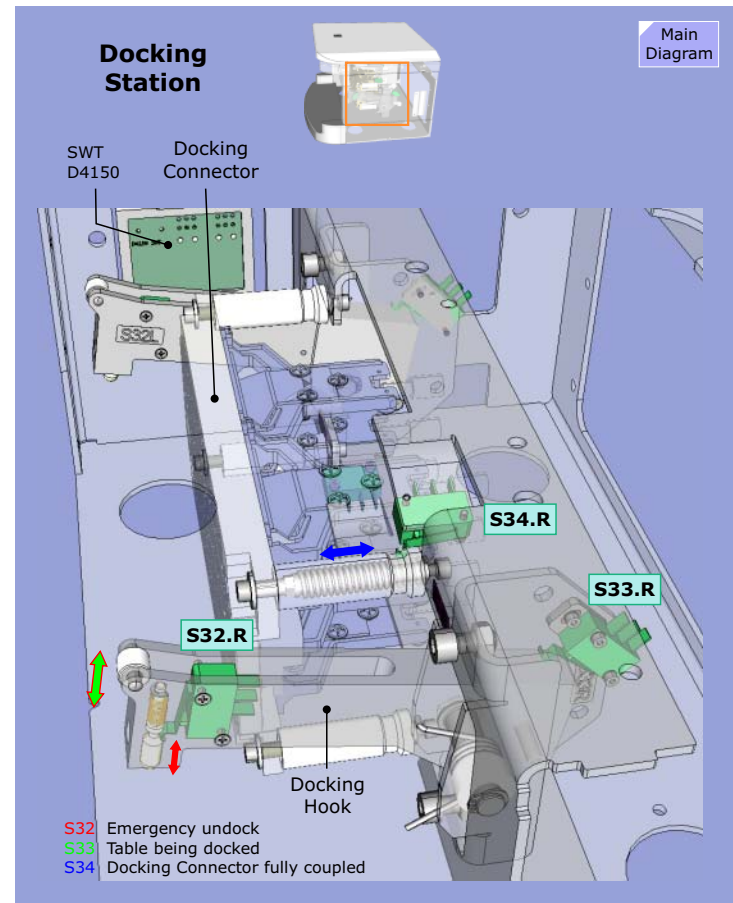
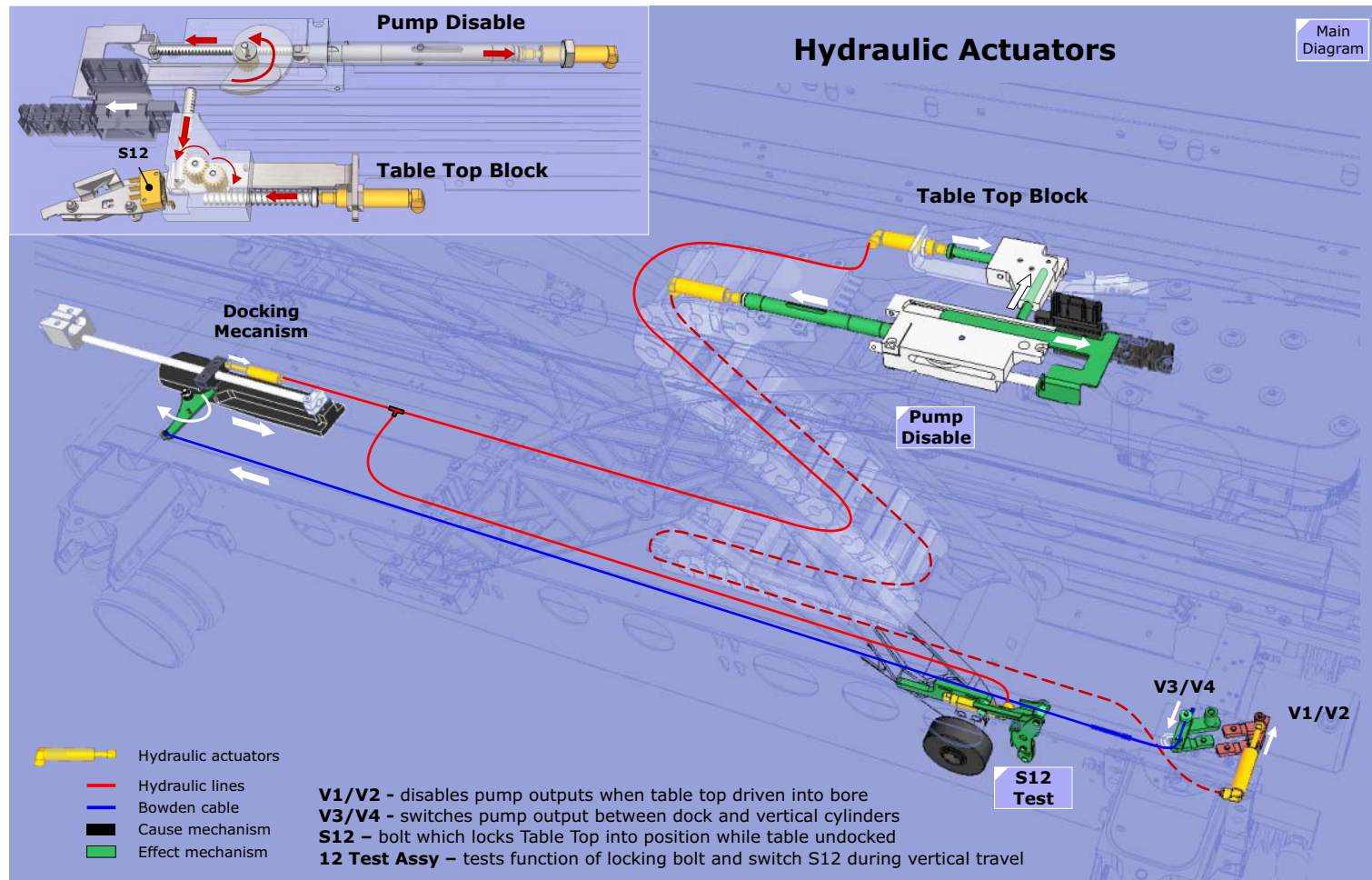


Figure 124 Tim Dockable Table - Mechanical Safety Interlocks



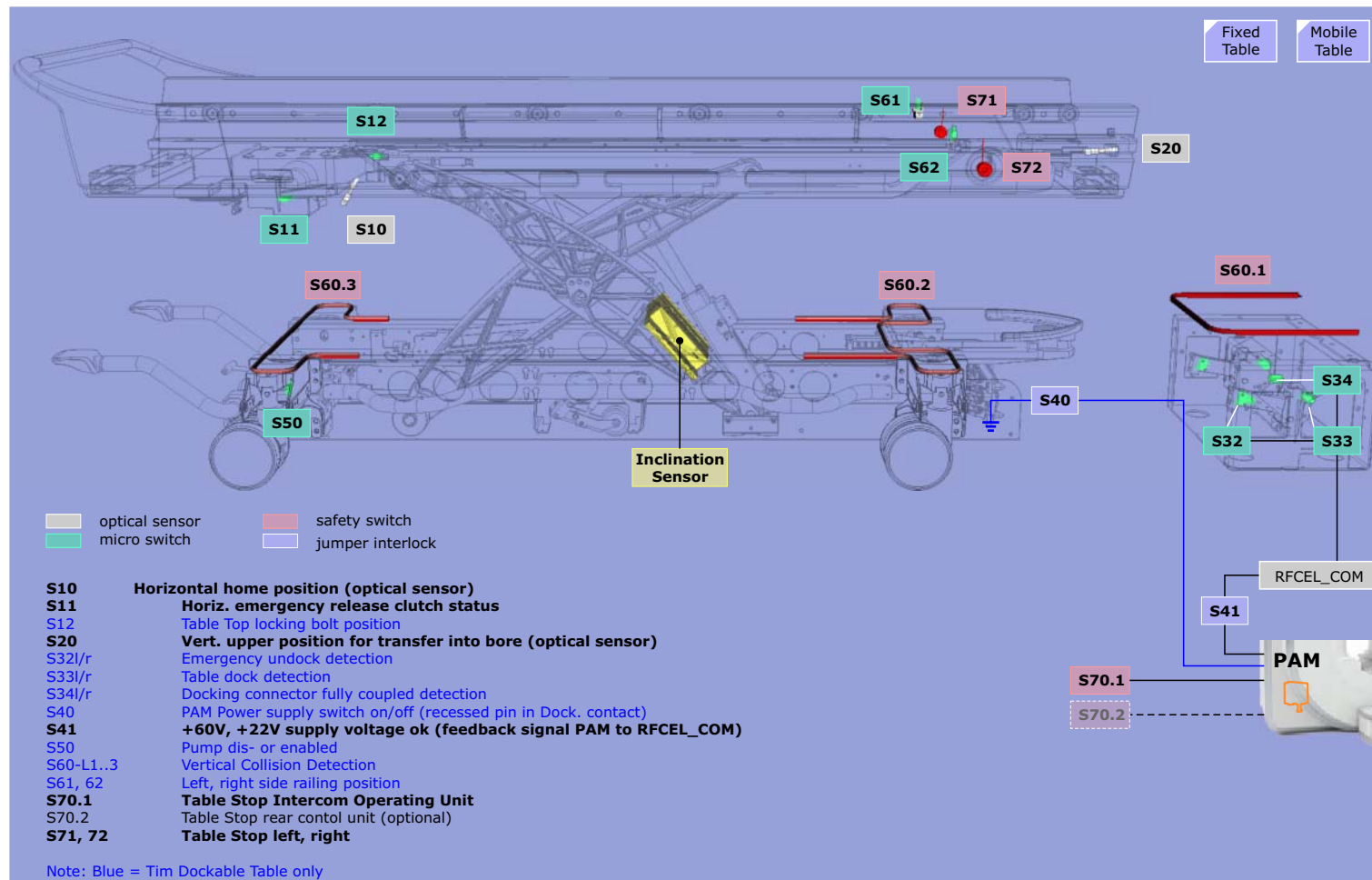
Switches

Switch	Location	Description	Status
S10	Gearbox	Optical Sensor, Home reference Z-position	activated in home position, vertical movement enabled
S11	Gearbox	Emergency clutch for horizontal drive	actuated when motor decoupled from transport chain (emergency clutch)
S12	Gearbox	Table plate locking bolt, secures plate against unwanted horizontal movement	actuated when locking bolt active (table in vertical mode just before lowest position or undocked)
S20	Support Frame, right side, head-end	Optical sensor, detects vertical transfer position into magnet	activated when vertical transfer position into bore is reached (and 30 mm above)
S32	Docking hook	Detection of emergency undock	released (= open) when the table is emergency undocked
S33	Docking hook	PTAB in docking range, docking possible	actuated (= closed) when docking hooks are pushed to downward position
S34	Docking Connector	Docking connector fully coupled (sufficient contact pressure)	released (= open) when fully locked
S40	Docking Connector, X53	Recessed ground pin in Docking Connector (X53, pin-1). Detects PTAB docked to magnet	engaged (= low level) when Docking Connector fully coupled. (enables PTAB supply voltages 22 V and 60 V at PAM)
S41	PAM	Potential free voltage feedback PAM, XB7, pin-1+2 to RFCEL_COM.	signal active (= low level) when 22 V and 60 V table supply from PAM is present.
S50	at Hydraulic pump assy mobile, foot-end	Monitoring switch for hydraulic valve position, i.e. pedal function is disabled when Table Top is not in home position Operated via hydraulic actuator from Table Top	actuated (= open) when Table Top is not in home position.

Switch	Location	Description	Status
S60	top of PTAB chassis and Docking Station	3 switch rails for vertical collision detection (obstacle under table)	actuated (= closed) when collision (a 1.2 K Ω terminator resistor connected in parallel for cable break detection)
S61	Support Frame, left-side	position of left side railing (horizontal movement enabled when hinged down)	actuated when left side railing is fully hinged down
S62	Support Frame, right-side	position of right side railing (horizontal movement enabled when hinged down)	actuated when right side railing is fully hinged down
S70.1	Intercom Operating Unit	PTAB Emergency Stop	actuated (= open) when table stop is pressed
S70.2	Magnet cover rear	PTAB Emergency Stop if optional rear panel is fitted. (connected in series to S71 and S72. If no S70.2 is fitted, a dummy jumper plug must be installed at PAM connector XB2)	actuated (= open) when rear table stop is pressed
S71	Support Frame, left-side	PTAB Emergency Stop 1	actuated (= open) when table stop is pressed
S72	Support Frame, right-side	PTAB Emergency Stop 2	actuated (= open) when table stop is pressed

Yellow fields in the table refer to switches found only on the Mobile Table.

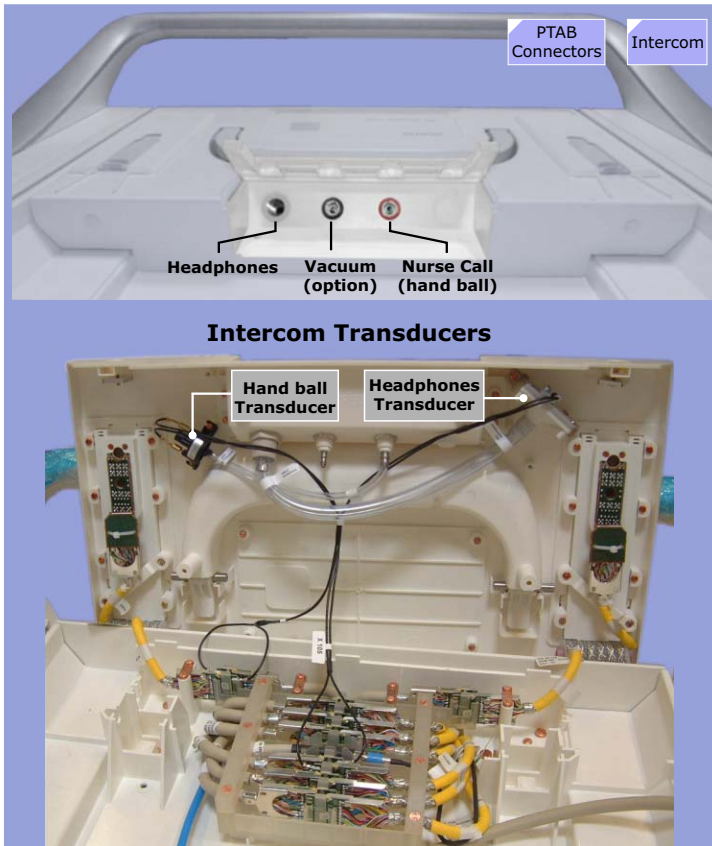
Figure 125 Patient Table Switches and Sensors Location



Coil and Auxiliary Connections

- Support of Tim (4G) coils
- Support of comfort connections – Head phone, Nurse call, Comfort kit (vacuum cushions)

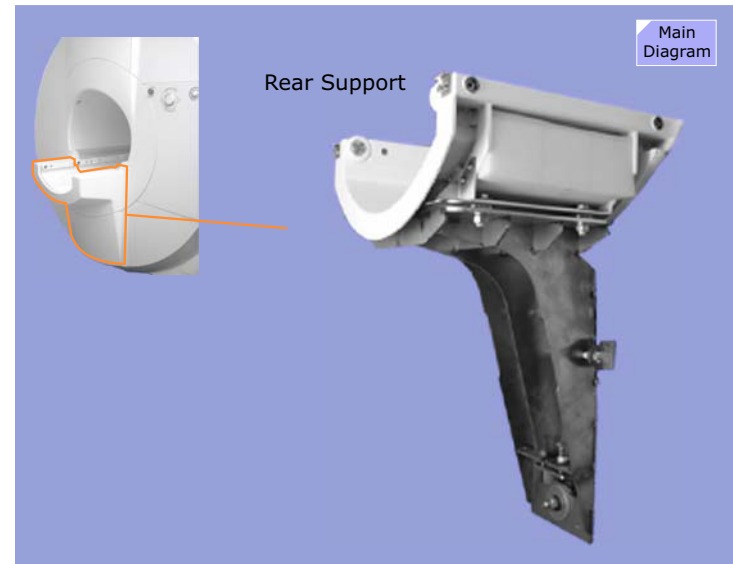
Figure 126 Coil and Auxiliary Connections



Whole Body option

For MAGNETOM Aera / Skyra the customer can purchase the Tim Whole Body Suite license which enables - together with the setting in syngo configuration - the **full scan range of up to 205 cm**. Whole Body also requires installation of the additional **Rear Support** at magnet back-end to support the Table Top.

Figure 127 Whole Body Option - Rear Support



Cabin User Interface (CUI)

Overview

The CUI consists of the following components:

- **Control Panels** - 2 at the front and a 3rd optional rear panel
- **Patient Data Displays (PDD)** aka "Dot Displays" - one at the front and optional rear PDD
- **Light Localizers** - one at the front and an optional rear laser light marker
- **Bore Lights** - 2 LED strips inside the magnet bore
- **Patient Fan** and **Air Temperature sensor** (2 sensors)

RFCEL_Sat

The RFCEL_Satellite provides several hardware functionalities for control of:

- Patient Light control (programmable current source, 5 levels)
- Patient Fan control (programmable, 5 levels)
- Light marker control (2x switchable supplies)
- Patient air temp monitoring (in °C)
- Gradient Coil temp monitoring (12 channels)
- Patient Data Display power switching (2x)
- Control Panel power distribution (with PTC fuses and LEDs)
- read out Body Coil ID (I²C bus)
- MNO Switch power supply (no fuse or LED)

Function

The RFCEL_Sat is controlled by the RFCEL_Com via an SPI interface(U1-U4).

Temperature Monitoring

- All **Gradient coil** NTC sensors can be checked against four programmable reference values. The alarm level is currently set for 85°C. If exceeded the scan is stopped. The error will automatically be reset when the coil temperature falls below 65°C (overtemp_recovery). Each sensor can be individually configured to react to one or more of the references or to none (i.e., masked out).
- The **patient air** sensors monitoring has a 5-50°C range with a 0.1° sensitivity.

Power Switching

Several components are supplied with power by the RFCEL_Sat some of which can be switched on and off. The light marker power is switched on only when localizing the patient. The Display Panels are switched off when measurements are begun.

LEDs

See [Figure 129](#).

Figure 128 Cabin User Interface Block Diagram

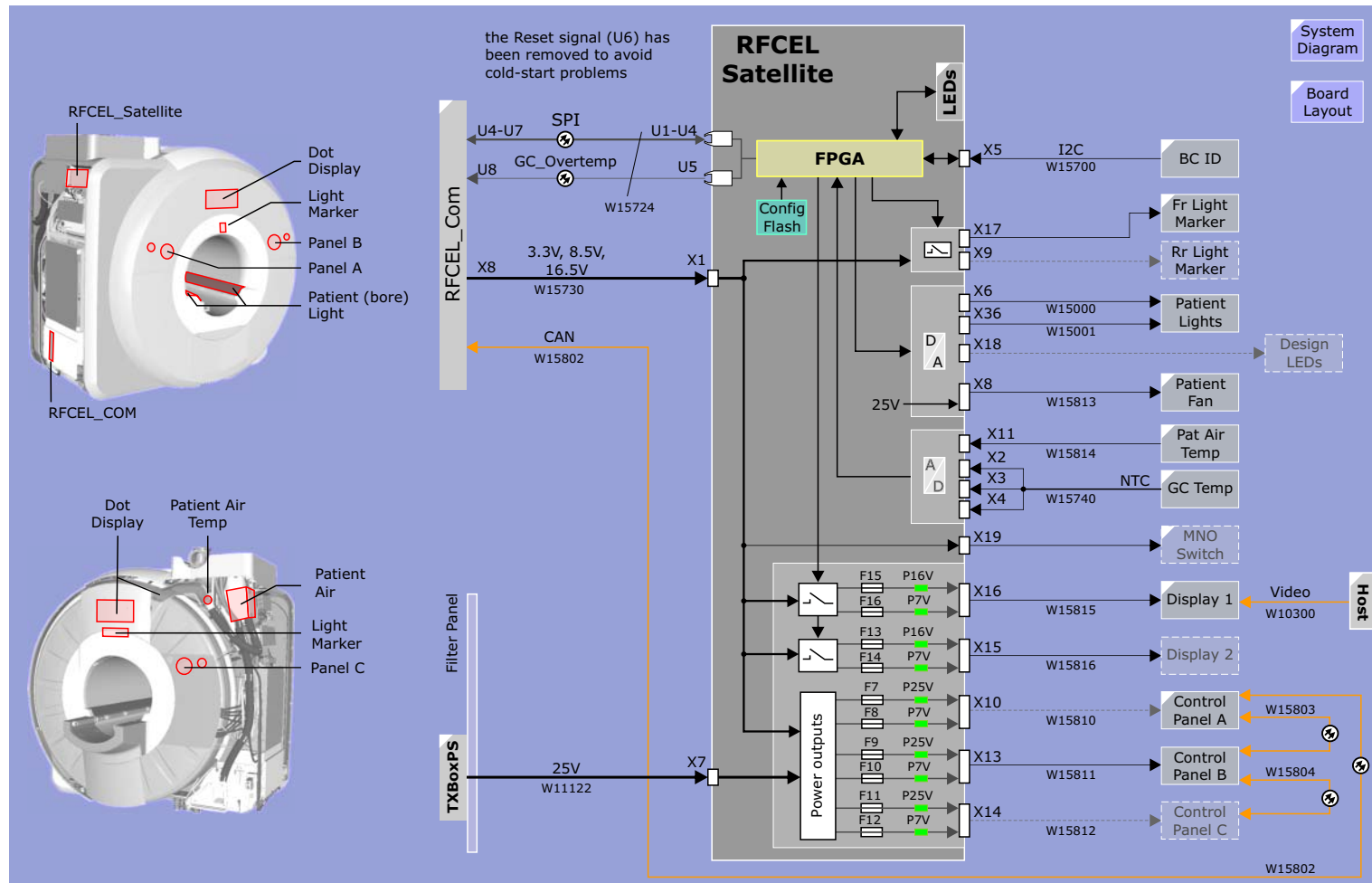
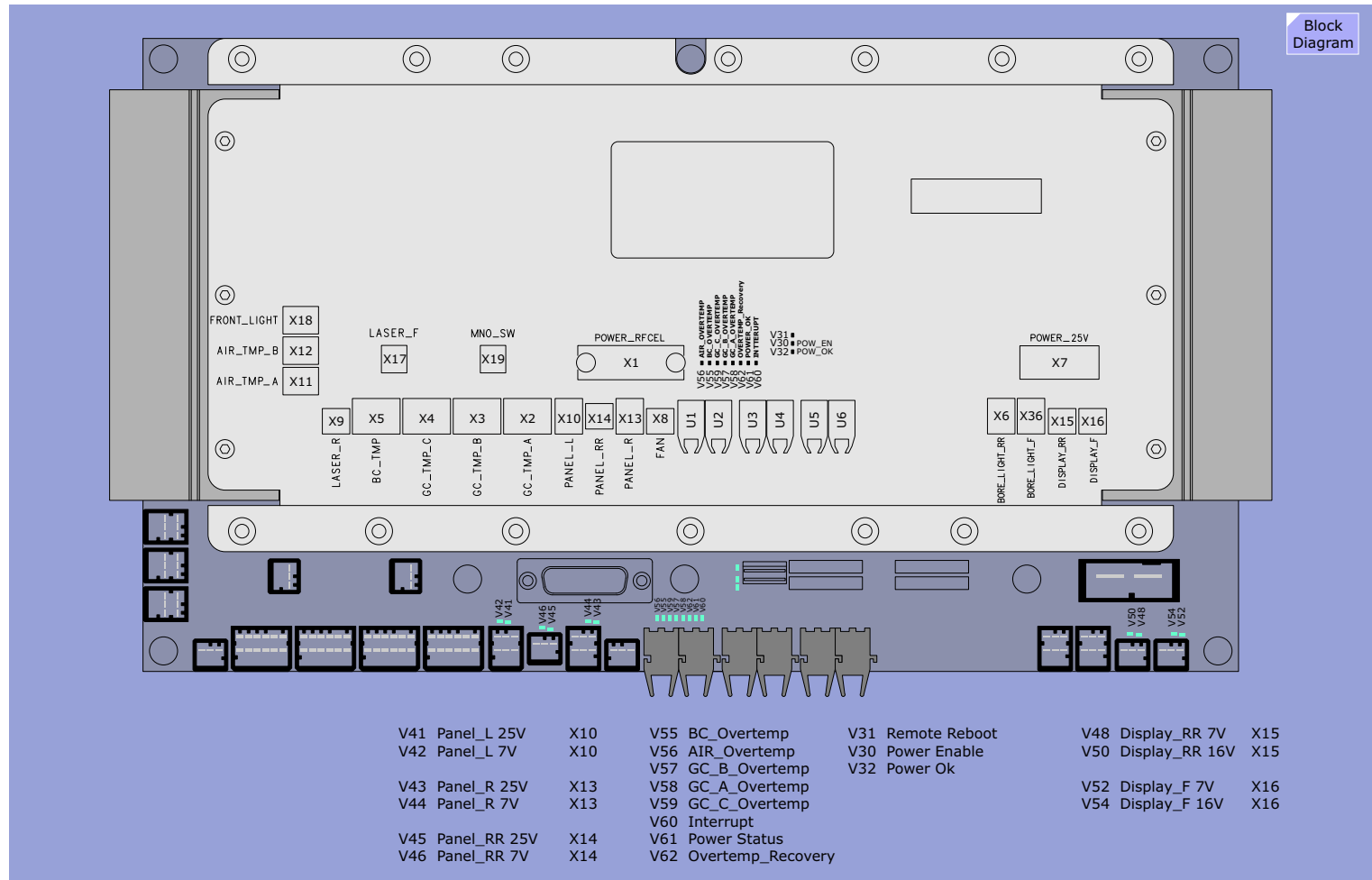


Figure 129 RFCEL_Sat Layout

Control Panel

The Control Panel has 2 combo button/jog wheel devices, the larger jog wheel is used for PTAB movement and the smaller jog wheel used to select menu functions. A controller is responsible for button/jog wheel sampling and debouncing as well as monitoring functions.

Function

Panel Controllers

Each controller communicates over a fiber optic CAN bus interface. Commands are sent only when a button or jog wheel is used. LED's are activated via commands from the scanner controller. The controllers are shielded and **remain powered on** during scans.

Panel Configuration

Up to 3 Control Panels can be connected to the system:

	CAN Name (in System>Control)	Panel ID (in EventLog)
Panel Front Left	CPANK2303_FL	18
Panel Front Right	CPANK2303_FR	19
Panel Rear (opt.)	CPANK2303_RR	20

The optional **rear panel** is identical to the front right panel. The second **PTAB-Stop Button** is NOT part of this Control Panel. Control Panels know their mounting position (left, right or rear) by coded wiring of their power cables. They require loadware which is supplied by the system during boot, if necessary.

Control Panel Self Test

When switched on, each Control Panel performs a self test to identify the following major errors:

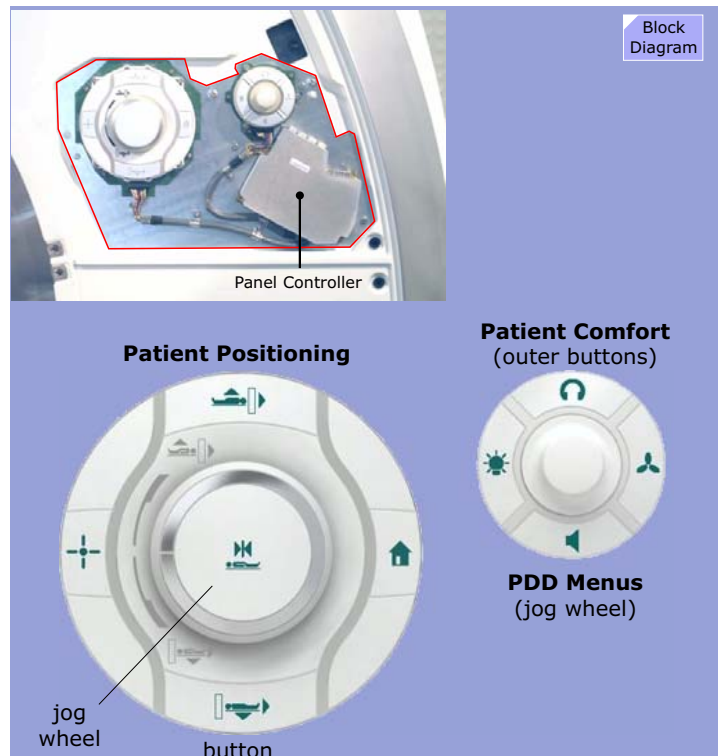
- Stuck buttons - the affected button flashes
- Large jog wheel blocked - all LEDs flash, wheel is deactivated
- No communication (CAN) - all LEDs on continuously

If self test is successful, all LEDs will be switched off temporarily. From now on the system control (MARS 2) determines the LED status.

Reset of Table Stop

For reset of table stops, press the **Up/In** and **Out/Down-Buttons** SIMULTANEOUSLY.

Figure 130 Control Panels



Patient Data Display (PDD)

- 13.3 inch 16:10 ratio, 1280 x 800 pixel, 262,144 colors
- video generated on the Host and transmitted via optical fiber
- switched off during measurements
- front PDD - display panel and controller are separated
- rear PDD (option) - same part number but display panel and controller board are "sandwiched" together

Function

PDD Contents

The PDD is used to display:

- Patient, coil, and PTAB information
- Physiological data
- the Comfort functions menu (selection made with small knob on control panel)

NOTE

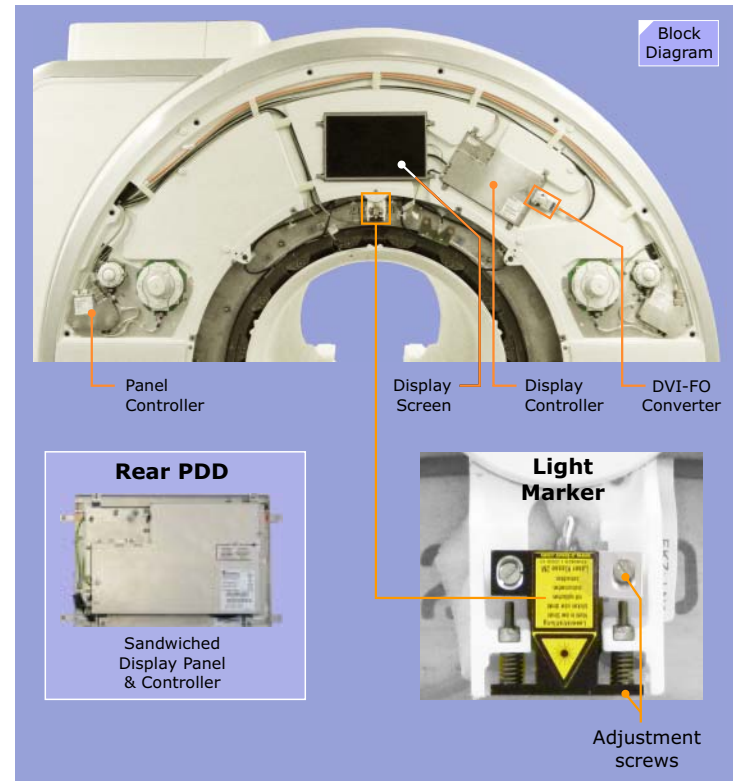
- When "Welcome to your MAGNETOM" is being displayed indicates no video signal present.
- Black screen indicates video signal present, but blanked (Windows screen-saver activated).
- The Screen-saver mode on the Host must be disabled (display settings) otherwise the PDD will be blanked together with the Host's monitor.

DVI-FO Converters

The video signal from the Host's graphic card is converted to FO via optical converters. The power supply for the DVI-FO transmitter on the Host side is provided by a USB port via an adapter-cable. The DVI receiver at the PDD is supplied with voltage by one pin of the PDD connector.

WARNING Pin 14 of the PDDs DVI connector carries +5 V supply voltage. Do not use regular electrical DVI-cables to connect the PDD to a graphics card! Otherwise, the graphics card can be damaged.

Figure 131 CUI Layout



Laser Light Localizers

The laser light localizer facilitates correct patient positioning. The system can be equipped with up to two light localizers:

- **Front Laser Light Localizer** - mounted behind top of the magnet front funnel
- **Back Laser Light Localizer** - option, mounted behind the back funnel.

The light markers are connect to the RFCEL_SAT board at X17 (front laser) and X9 (rear laser).

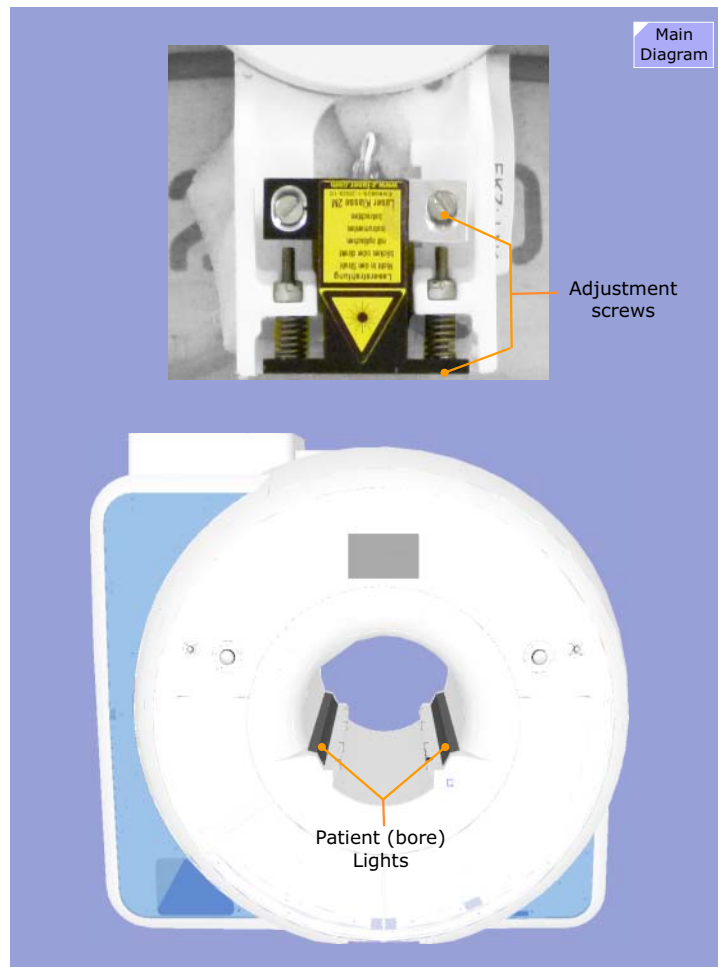
NOTE An optional rear Laser Light Localizer must be setup in the syngo configuration and adjusted under TuneUp>Light Marker.

For examinations implementing the Auto Align function a manual alignment of the laser light localizer to the coil is not necessary. If all relevant data (e.g. patient orientation, body height, examination) are registered, **auto-positioning** is indicated on the Patient Data Display at the magnet.

Bore Lights

MAGNETOM Aera and Skyra systems are equipped with tunnel lighting, realized with two LED strips left and right side of the table sideways. The power is supplied by the RFCEL_SAT with two separate cables (see [RFCEL_Sat diagram](#)). The LED brightness has 5 settings.

Figure 132 CUI Layout



Patient Fan, Air Temperature Sensor

Patient Fan

For increased patient comfort a variable-speed cooling fan, having five fan speeds, can be adjusted at the Control Panels or via syngo MR. The fan operating voltage is supplied by the RFCEL_SAT via cable W15831.

At magnet service end (lower right corner) you will find the fans' air inlet which is covered by a **replaceable filter mat**.

Air Temperature Sensor

The Air Temperature Sensor is located inside the patient air outlet of the rear cover.

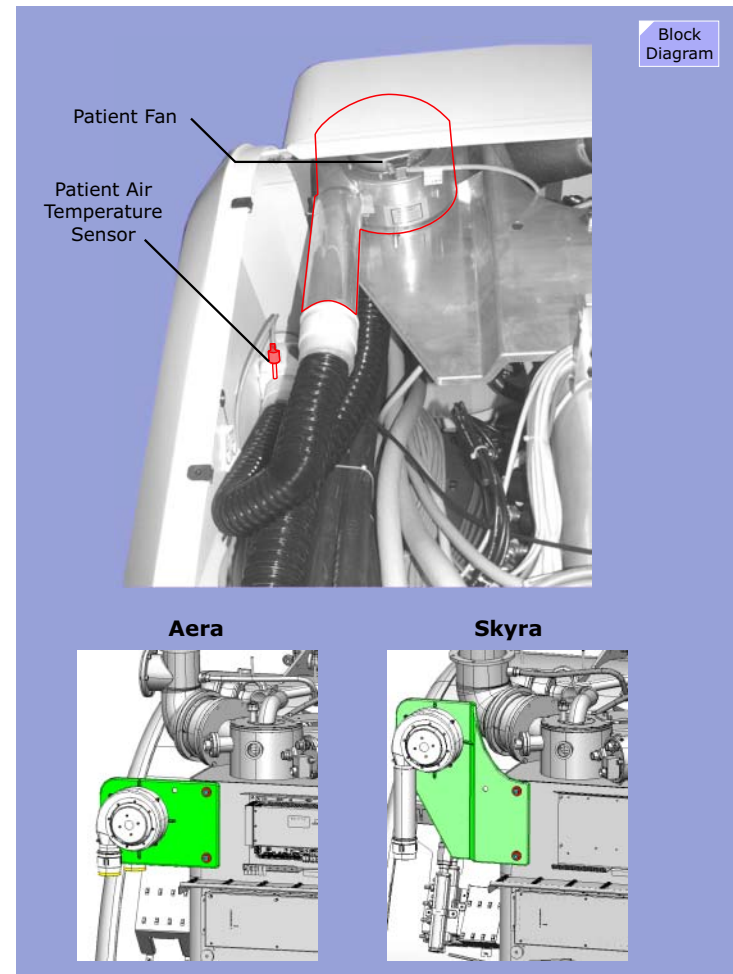
It consists of **two independent NTCs** which are connected via cable W15814 to the RFCEL_SAT. In case a sensor is broken or their reported temperature values differ more than 2° C, an error message is generated and no measurement is allowed.

Air Temperature Monitoring

Prior to starting a measurement, the RF-Safety Watchdog (RFSWD) software calculates the SAR look ahead parameters. Since the maximum allowed RF exposition to the patient also depends on the ambient air temperature, it has to be measured.

NOTE Above 24° C ambient air temperature, the RFSWD software starts to reduce the SAR limits accordingly.

Figure 133 Patient Fan, Air Temp Sensor



Intercom

Overview

The intercom allows personnel and patients to communicate during the examination. In addition, important operations like patient table stop are managed from the intercom.

Optionally, music or automatic voice outputs (e.g. breathing commands) can be played back in the examination room via the loudspeaker or headphones.

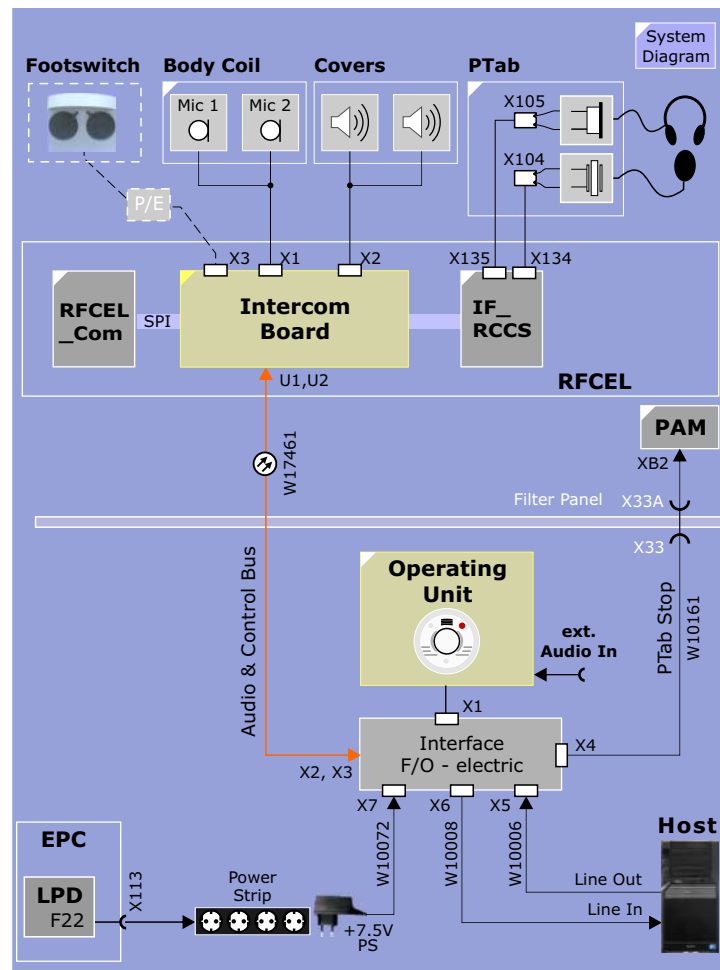
The Intercom consists of the following FRUs:

- **Operating Unit** – at the console, includes a F/O - electric interface, usually mounted underneath the console
- **Intercom Board** – at RFCEL
- **2 Loudspeakers** – at the magnet front cover left-side and magnet cover rear-side
- **2 Microphone Boards** – at Body Coil front and rear
- **Headphones**
- **Sound Transducer for headphones** – piezoelectric, at Table Top foot end (see right figure)
- **Squeeze Ball** – nurse call
- **P/E Transducer for squeeze ball** – pneumatic / electric transducer at Table Top foot end (see right figure)

Figure 134 Intercom Speakers



Figure 135 Intercom Block Diagram



Operating Unit

The operating unit is connected to the system via an electrical - fiber optic interface box, usually mounted underneath the console.

A Fibre-optic bus is used for transferring audio- and control-data to and from the Intercom Board in the RFCEL. The Table Stop signal is hard-wired via the interface box to the Power Adapter Module (PAM of PTAB).

Operating Elements

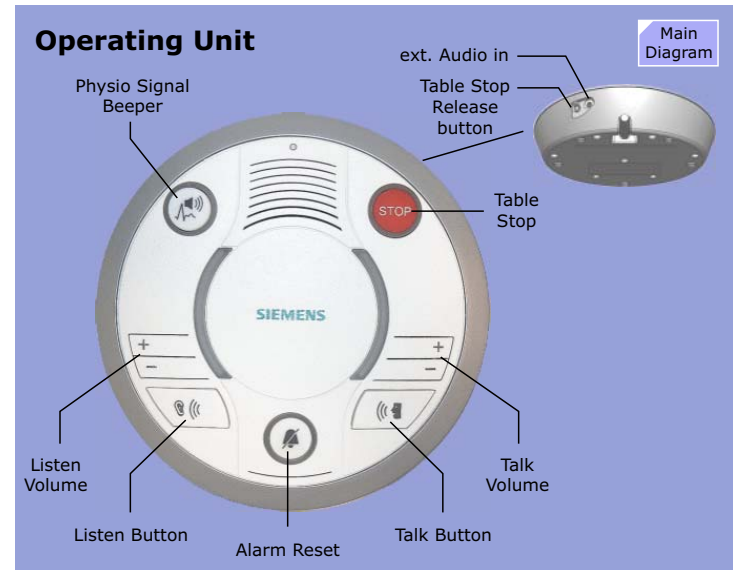
The following operating elements are available:

- Talk button - operator to patient
- Listen button - patient to operator
- Volume adjustment - talk and listen
- Alarm reset - for the patient alert (squeeze ball)
- Physio signal - switches on/off the physiological signal beep
- Table stop - stops table movement and measurement
- Table stop release - button at the Operating Units' rear-side
- External audio input - 3.5 mm audio socket at the Operating Units' rear side to connect an audio device (e.g. MP3 player)

NOTE To reset a Table Stop initiated at the Operating Unit, the Table Stop Release button at its rear-side must be pressed.

NOTE The volume presets are adjustable in syngo MR: Options > Configuration > Patient Comfort Config.

Figure 136 Intercom Operating Unit



Intercom Board

Overview

The major amplifier circuits but also the intelligence of the intercom is located at the Intercom Board in the RFCEL. The following interfaces are present here:

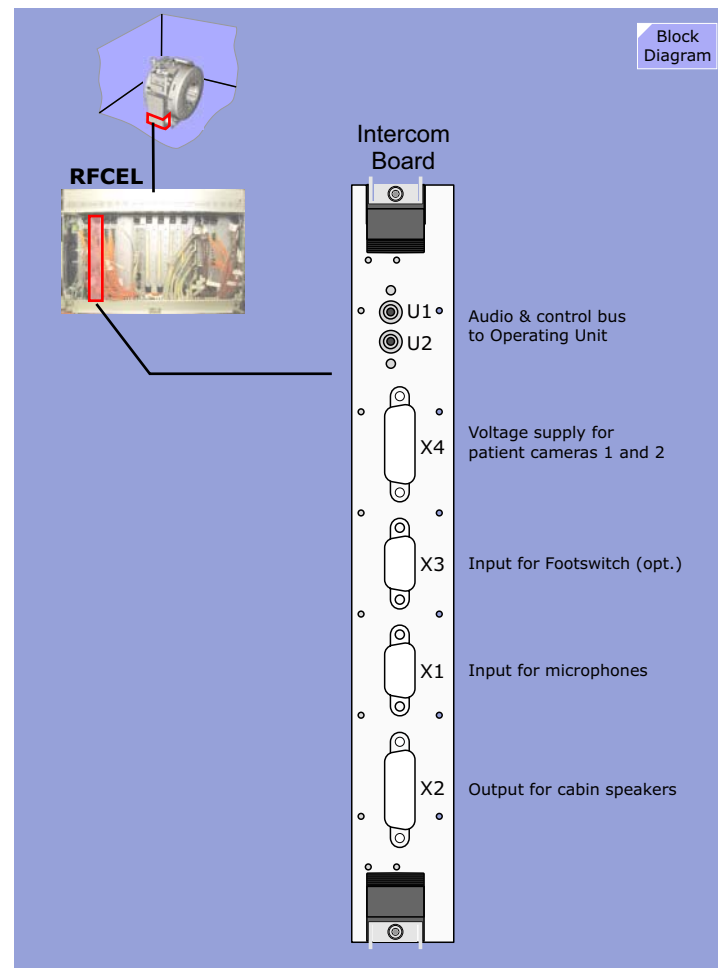
- Two inputs for the microphones - front connector X1
- One output for the two cabin speakers - front connector X2
- One input for the P/E converter of the Footswitch (option) - front connector X3
- Voltage supply patient cameras #1, #2 - front connector X4
- One output for the sound transducer (headphones) - backplane connection to the IF_RCCS and further to the PTab
- An optical bus to the operating unit for audio and control data - FOC front-connectors U1 and U2.
- Access to SPI-bus data - via backplane (e.g. update SW, gradient data for Active Noise Cancellation, etc.)
- Voltage supplies - via backplane to the RFCEL power supply. (+1,5 V, +3,3 V, +6 V, +15 V, +25 V, -32 V)

Active Noise Cancellation (ANC)

Understanding patient speech requires a sufficiently high volume setting which results in larger disturbances by measurement noise too. For improvement, the Active Noise Cancellation function under Options > Configuration > Patient Comfort Configuration is available. A self-adaptive algorithm at the Intercom Board processes and improves the audio signal from the two microphones at the BC. The online gradient data which is picked up via SPI from the backplane is used to calculate a reduction of gradient noise in the audio signal from patient.

NOTE The Active Noise Cancellation is supported from SW-version VD11D onwards.

Figure 137 Intercom Board Front View



Patient Video (option)

Overview

For monitoring the patients well-being, a video system can be optionally installed. It consists of a CCD camera and a **flat screen color display**. An optional **video combiner** allows simultaneous display of **up to four cameras**.

Function

Video Camera

Up to two in-room cameras can be installed inside the RF-cabin for optimized patient observation from magnet front-end and rear-end. They are supplied with power by the Intercom_Board of the RFCEL. In addition, up to two further cameras can be mounted for observing the situation outside the RF-cabin (waiting room, etc.). Their voltage is supplied by the TX-Box power supply (Filter Panel).

The video signal of each camera is routed by one fibre-optic cable directly to the video combiner (if more than one camera is used) or straight to the Patient Display.

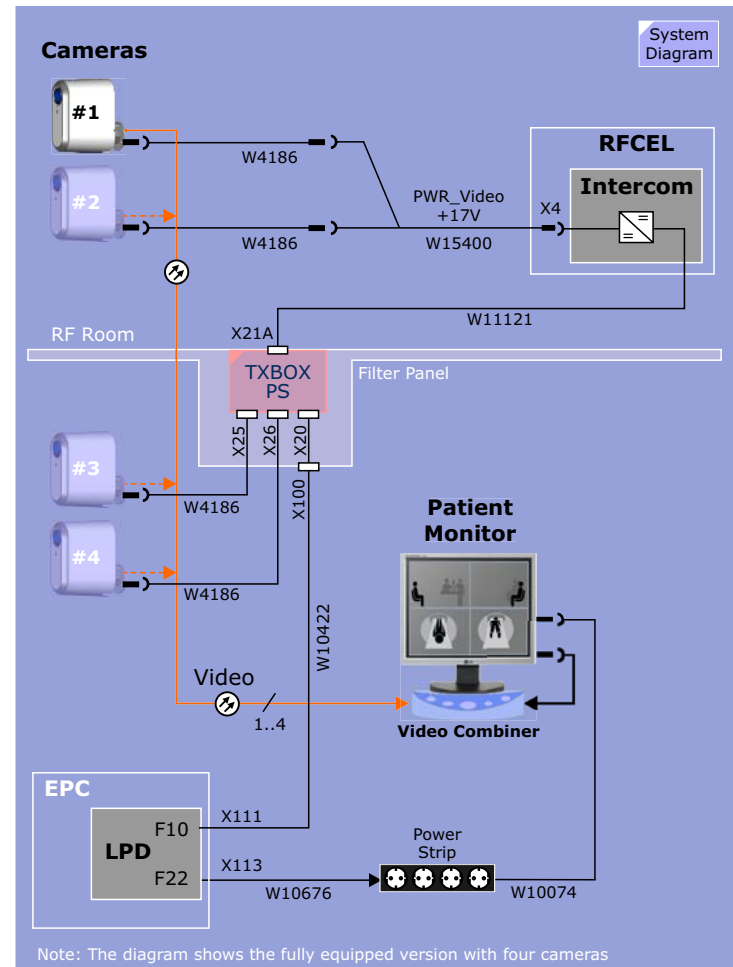
Patient Display

The video display is a color 800 × 600 pixel LCD display which may be positioned at the syngo Acquisition Workplace or at a convenient wall location. The necessary supply voltage is taken from the EPC cabinet (Fuse breaker F22) via the Console Line Distribution #2.

Video Combiner

Up to 3 additional cameras can be connected via an optional video combiner. The combiner can be set to view each camera separately or all at one (split view).

Figure 138 Patient Video Block Diagram



PMU

Overview

The Physiological Measurement Unit (PMU) provides physiological triggering for special cardiological and abdominal imaging techniques, resulting in improved image quality and increased diagnostic value of the images. For example, when running a heart study it is very important to know, whether specific images have been measured during the diastolic phase (heart muscle relaxed) or during the systolic phase (heart muscle contracted). The system control which executes the measurement (MARS 2), uses the physiological triggering signals for acquiring image data according to the desired physiological state.

Physiological signals available for triggering:

- Electrocardiogram (ECG)
- Respiratory (chest movement)
- Pulse (heart rate)
- Two external TTL trigger inputs (at Magnet Cover to RFCEL_COM and at Gradient, GSSU D68)
- Optical trigger output for fMRI (at Gradient, GSSU D68)

The PMU consists of the following components:

- **PERU** – a wireless Physiological ECG and Respiratory Unit
- **PPU** – a wireless Peripheral Pulse Unit sensor
- **Antenna** – to receive the PERU and PPU signals. Located at the top-right side of the magnet behind the front cover
- **RFCEL_COM** – the Physiological Data Acquisition Unit on the RFCEL_COM receives and processes the physiological signals wirelessly from the sensors

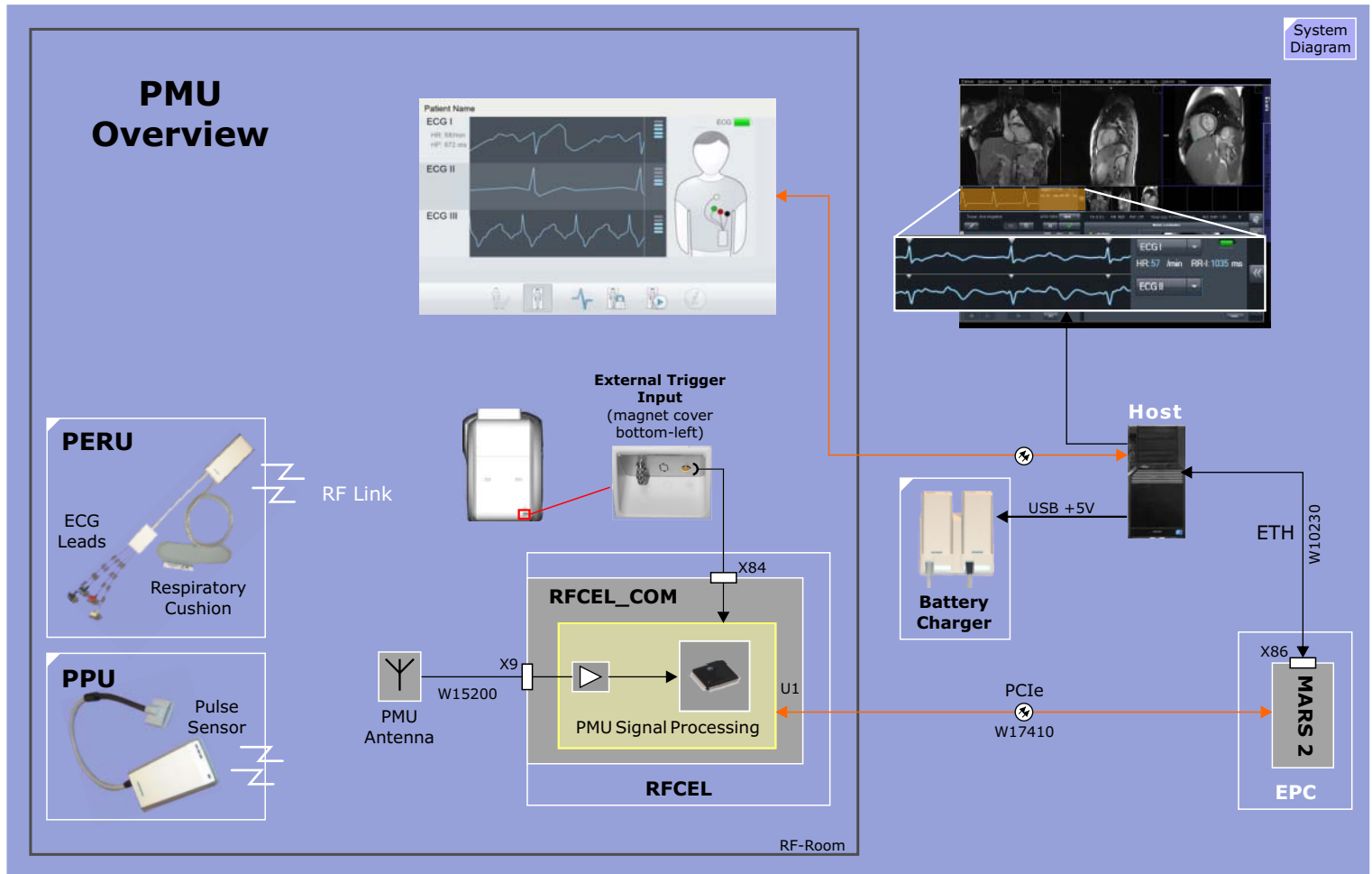
The measured data can be viewed at the Patient Data Display in the examination room and at the MRAWP (Physiological Display).

NOTE Only one PERU or PPU shall be located in the examination room simultaneously! Otherwise interference or unpredictable results may occur.

In order to achieve reliable physio triggering, a **learning phase** algorithm is running at the MARS 2 which **improves the ECG trigger quality**.

NOTE The ECG learning phase is only activated when the Table Top is located within 300 mm from Home position.

Figure 139 Physiological Measurement Unit (PMU) Block Diagram



PERU

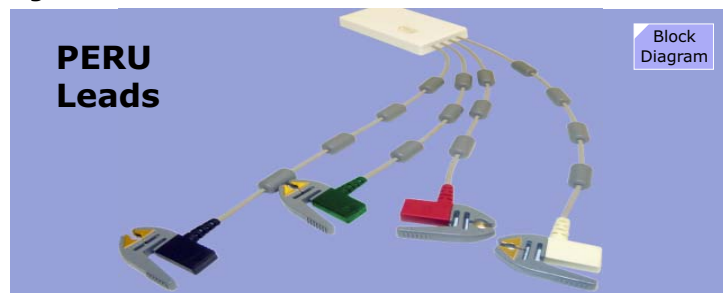
Function

The PERU (Physiological ECG and Respiratory Unit) is able to detect ECG and Respiratory signals (detected by a connected respiratory cushion) simultaneously. These signals are digitized and transmitted wirelessly using a proprietary 2.4 GHz transmission protocol. Maximum transmit power is 1 mW with a usable transmission radius of about 2 meters. Four ISM (Industrial Scientific and Medical Band) channels are used: 3 for ECG and 1 for respiratory. The unit has a magnet field sensor which detects the presence of the magnetic field (see LEDs).

NOTE Transmission is enabled only within a magnetic field.

The PERU has an integrated **Lithium-polymer battery** that can be **recharged in three hours** by a wall-mounted charging station. After being fully charged, the unit can **operate approximately 24 hours**.

Figure 140 PERU Leads



PPU

Function

The PPU (Peripheral Pulse Unit) detects the peripheral pulse of a patient by means of a fibre-optic sensor which is fixed with a finger adaptor (available in different sizes). The signal is converted into an electrical pulse, digitized and transmitted via 4 other dedicated RF channels in the 2.4 GHz ISM band (Industrial Scientific and Medical Band).

Magnetic field detection, RF transmit power and lithium battery parameters are identical to the PERU.

Figure 141 PPU Leads

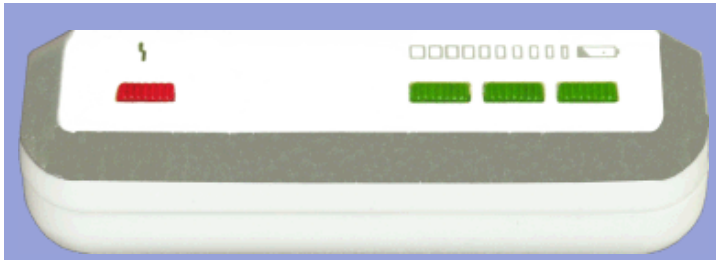


See "PERU, PPU LEDs" on page 180.

LEDs

LED	Status	Description
green	3 LEDs flash	Fully or nearly fully (2/3) charged accumulator
	2 LEDs flash	1/3 to 2/3 fully charged accumulator
	1 LED flashes	Nearly discharged accumulator, remaining operation duration is 1 hour
red	LED flashes regularly (as green LEDs)	Transmit function deactivated, unit is positioned outside the static magnetic field: NO electrode/application fault detected
	LED is off	Transmit function activated; unit is positioned on the patient table in the static magnetic field; NO electrode/application fault detected
	LED flashes rapidly	PERU: Electrode fault - one or more ECG electrodes are not applied correctly or fell off PPU: Application fault - Pulse sensor is not applied correctly at the finger

Figure 142 PERU, PPU LEDs



Service Routines

The PERU is equipped with a built in test-signal generator which is switched on if the PERU detects an electrode fault. Normally, this signal can not be selected by the RFCEL_COM and seen in the Physiological Display except when a **PMU test** in **SESO** was started.

In this case, the test signal can be displayed at the monitor, thus the whole signal path from PERU to RFCEL_COM, MARS 2 and

finally MRAWP can be tested.

NOTE In case the **PERU** detects an electrode fault and no PMU test is running, the RFCEL_COM zeros the ECG test signal.

NOTE For successfully testing the **PPU** in SeSo it must not be applied to a finger. Nevertheless, the finger adaptor must be fully attached to the unit.

Battery Charger

Function

A battery charger for PERU and PPU is delivered with the system. It can be wall-mounted and should be positioned close to the operator's console. The +5 VDC charging power is supplied either by a cable connection from two free USB connectors of the Host computer or an 230V/+5VDC external power supply, depending on the supplied hardware. Charging time for a discharged PERU and PPU battery is approx. three hours. The charging status is displayed by the green LED status bar at PERU and PPU.

If a PERU or PPU is stored for longer periods of time without being connected to the Battery Charger, it is recommended to connect the dummy plug (part of system delivery) to its power connector to avoid battery discharging.

Gradient System

Introduction

The gradient coils for both systems are actively shielded and incorporate the **TrueForm™** feature for optimal gradient field distributions within the human body. A list of the key specifications are given in a table below.

Gradient Amplifier Specifications

	GPA 62/20/F	GPA 90/22/F	GPA 75/22/F
System	Aera	Aera	Skyra
Option	XJ Gradients	XQ Gradients	XQ Gradients
Max. field	33 mT/m	45 mT/m	45 mT/m
Slew Rate	125	200	200
Max. Current	625 A	900 A	750 A
Max. Voltage	2000 V	2250 V	2250 V
Gradient Coil	GC 1.5 XJ	GC 1.5 XQ	GC 3.0
Shim Coils	not available	standard	standard

/F = Fastlink, i.e. PCIe/ISO-Ring optical interface

The **GPA 75/22** and **90/22** amplifiers are identical with the exception of the Power Stages.

The **GPA 62/20** amplifier is a new design. It shares most components with the 90/22/F cabinet, but has its Power Stage Power Supply rectifiers integrated into the Power Stages.

K2309 Amplifiers



XJ Gradient

Aera

62/20/F



XQ Gradients

Aera

90/22/F

Skyra

75/22/F

GPA 62/20/F Cabinet

Overview

The GPA 62.20.F amplifier is a switch-mode amplifier with cascade power stages and is the amplifier for the MAGNETOM Aera XJ Gradients. The Power Stages and Power Stage Supply are of a new design having the rectifiers of the Power Stage Supply integrated into the Power Stages.

Major Differences

Producing less power than the GPA 90(75)/22/F, the 62/20/F amplifier has down-sized these components:

- **Power Stages - lower performance**
- **Power Stage Supply - fewer parts, lower performance**
- mains breaker F1
- transformer
- secondary circuit breakers
- cooling water manifold

Performance Data

	GPA 62/20/F
System	Aera
Option	XJ Gradients
Maximum Gradient Field in mT/m	33
Slew Rate in T/m/s	125
Maximum Gradient Current	625 A
Power Stages	40 IGBTs
Maximum Voltage	2000 V
Gradient Coil	GC 1.5 XJ
Shim coils	not available

Figure 143 GPA 62/20/F Layout

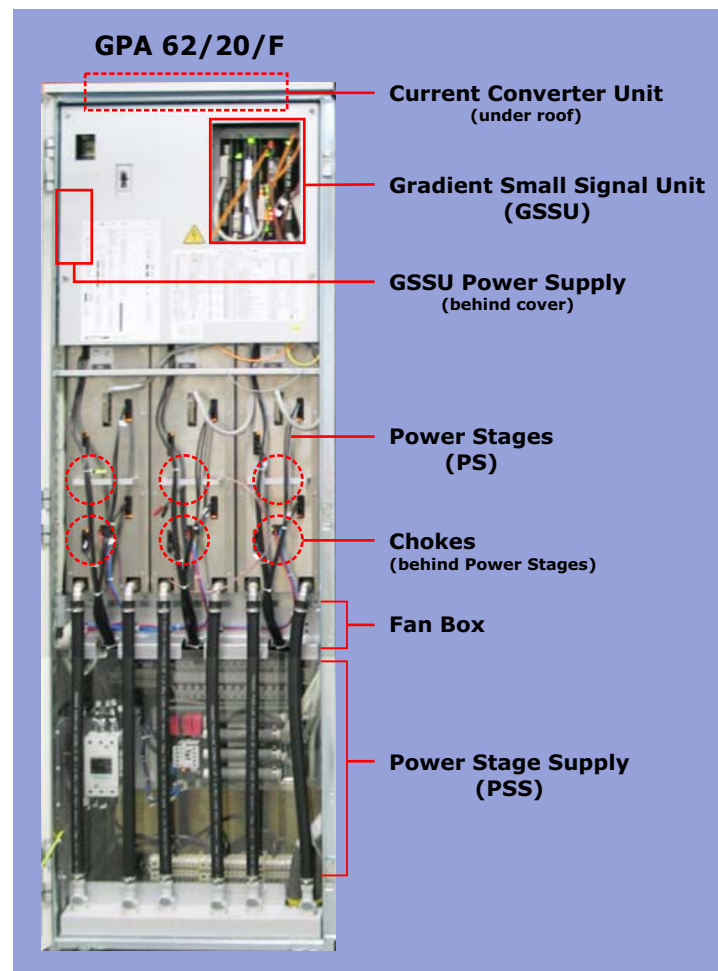
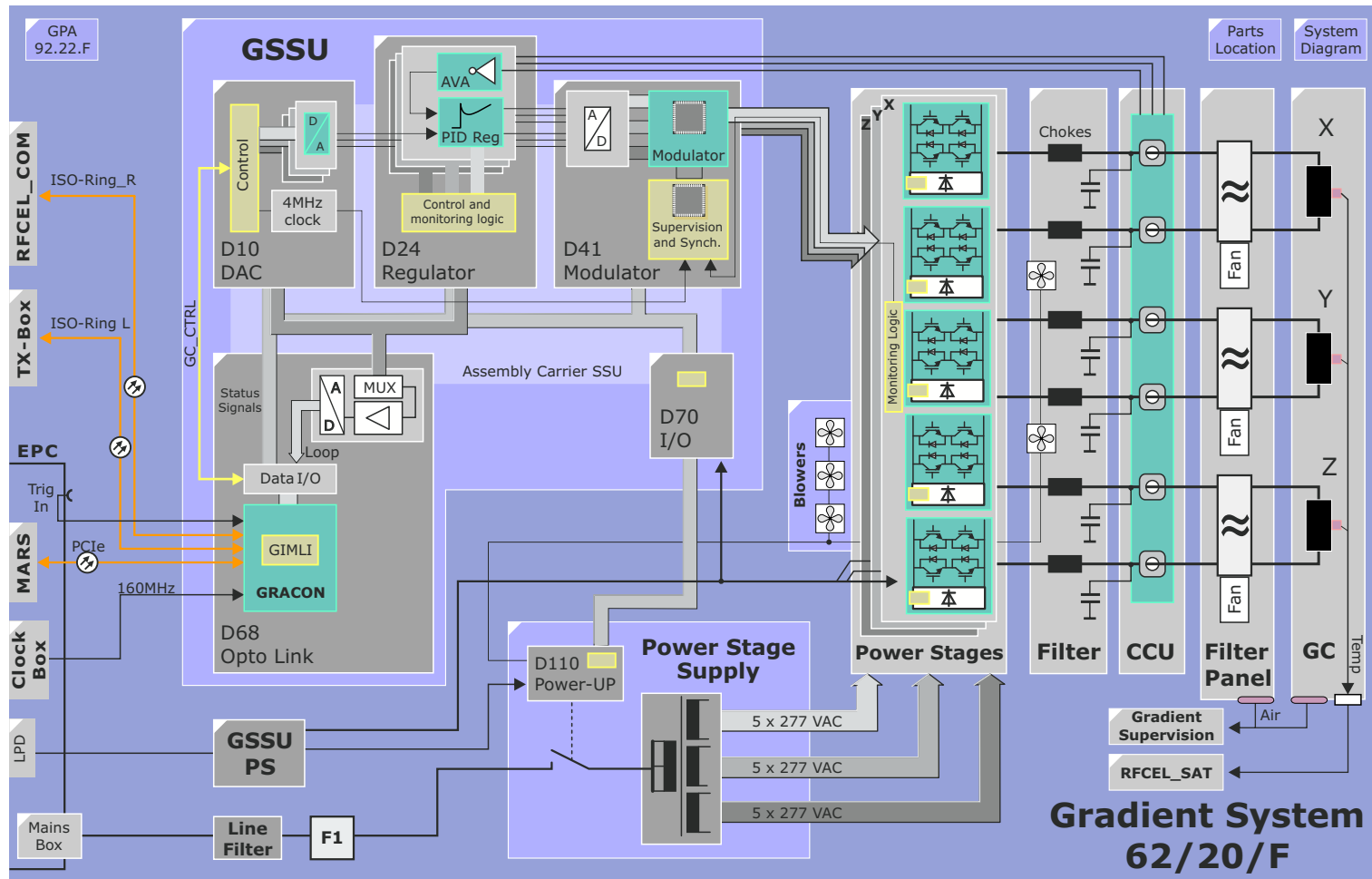


Figure 144 GPA 62/20/F Block Diagram



GPA 90(75)/22/F Cabinet

Overview

The GPA 90/22/F and 75/22/F are based on the K2298 / K2259 GPA of MAGNETOM Verio and MAGNETOM Avanto/Espree respectively. The following versions are offered:

	GPA 90/22/F	GPA 75/22/F
Option	XQ Gradients Aera	XQ Gradients Skyra
Maximum Gradient field in mT/m	45	45
Slew Rate in T/m/s	200	200
Maximum Gradient Current	900 A	750 A
Power Stages	60 IGBTs = Verio	50 IGBTs
Maximum Voltage	2250 V	2250 V
Gradient Coil	GC 1.5 XQ	GC 3.0
Shim coils	standard	

Differences to K2298 / K2259 GPA

- **Optical control interface** - The new system control architecture of MAGNETOM Aera / Skyra uses an optical PCIe and ISO-ring communication instead of CAN.
- **GSSU Power Supply** - no longer supplied by the GPA power section but directly from the EPC cabinet (fuse breaker F21)
- **Power Stages** - The GPA 90/22/F uses the Power Stages from Verio (60 IGBTs), the GPA 75/22/F is only used together with the Gradient Coil GC 3.0 which has a higher sensitivity. Therefore, a Power Stage with only 50 IGBTs is enough to achieve XQ Gradient performance.

Figure 145 GPA 90(75)/22/F Layout

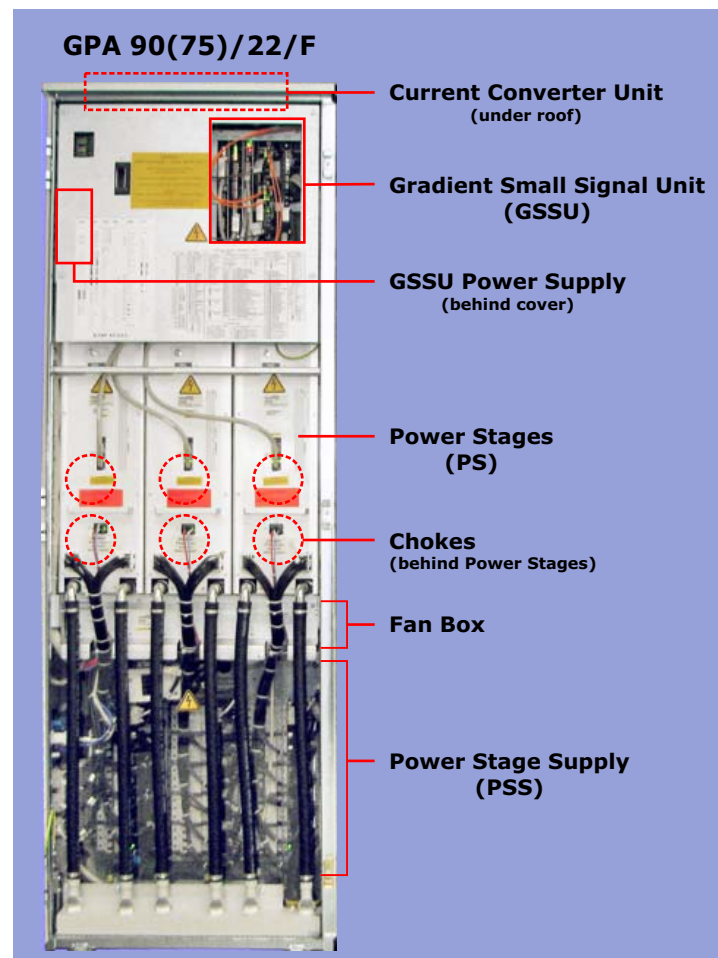
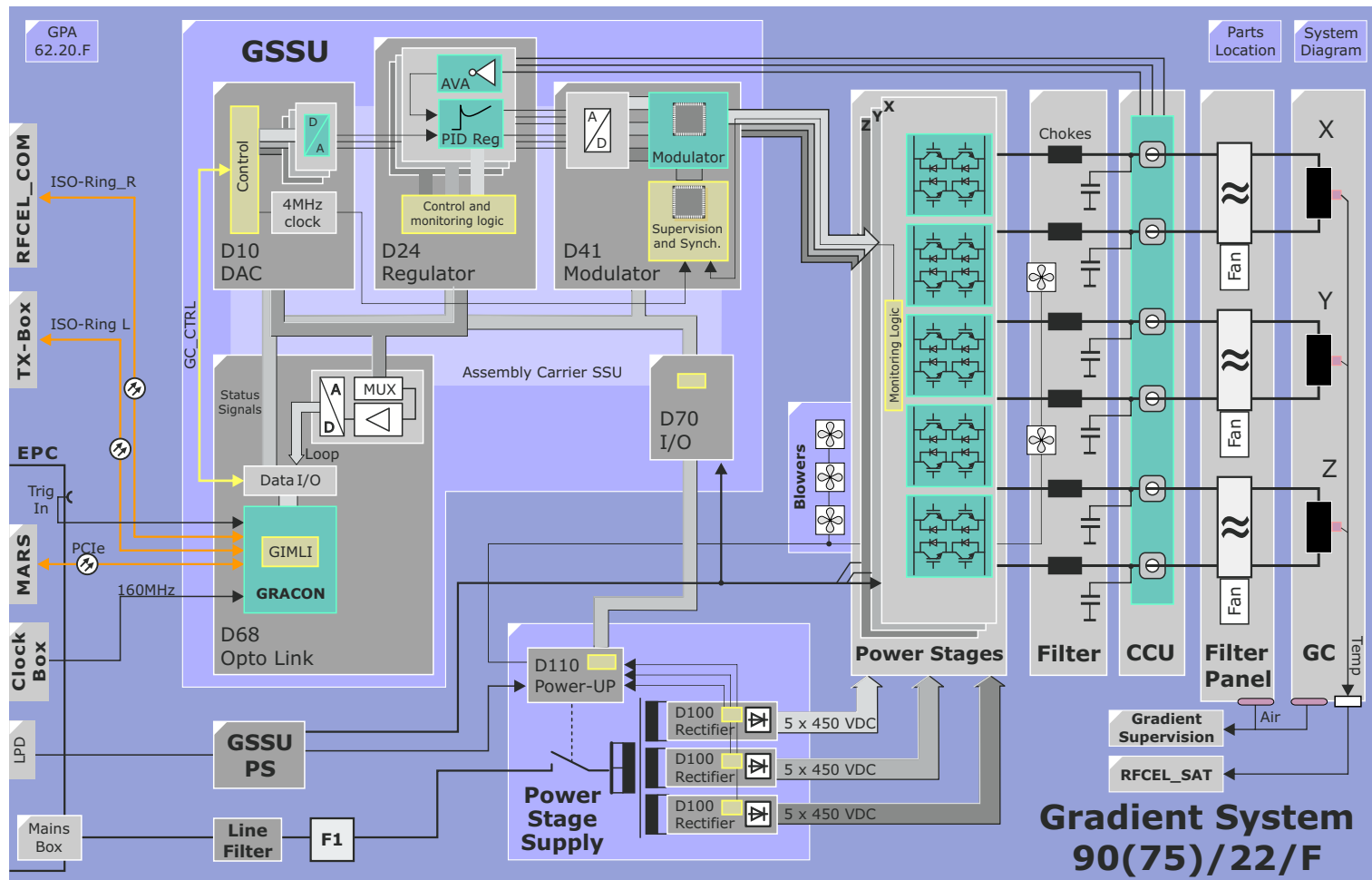


Figure 146 GPA 90(75)/22/F Block Diagram



Gradient Small Signal Unit (GSSU)

Overview

The Gradient Small Signal Unit (GSSU) is responsible for generating the pulse-width modulated signals, based on the digital data supplied by the MARS 2, for the switches in the Power Stage in order to drive a precise current in the Gradient Coils. Its additional task is the overall supervision of the GPA.

Additionally the GSSU provides a fibre-optic communications link and signal feedback to the MR System for service loop tests and Tune Up procedures.

The GSSU consists of the following PCBs or FRUs (Field Replaceable Units):

- Assembly Carrier with backplane and fan
- DAC D10
- Regulator D24
- Modulator D41
- Opto Link D68 (= D67 + D90)
- I/O Board D70

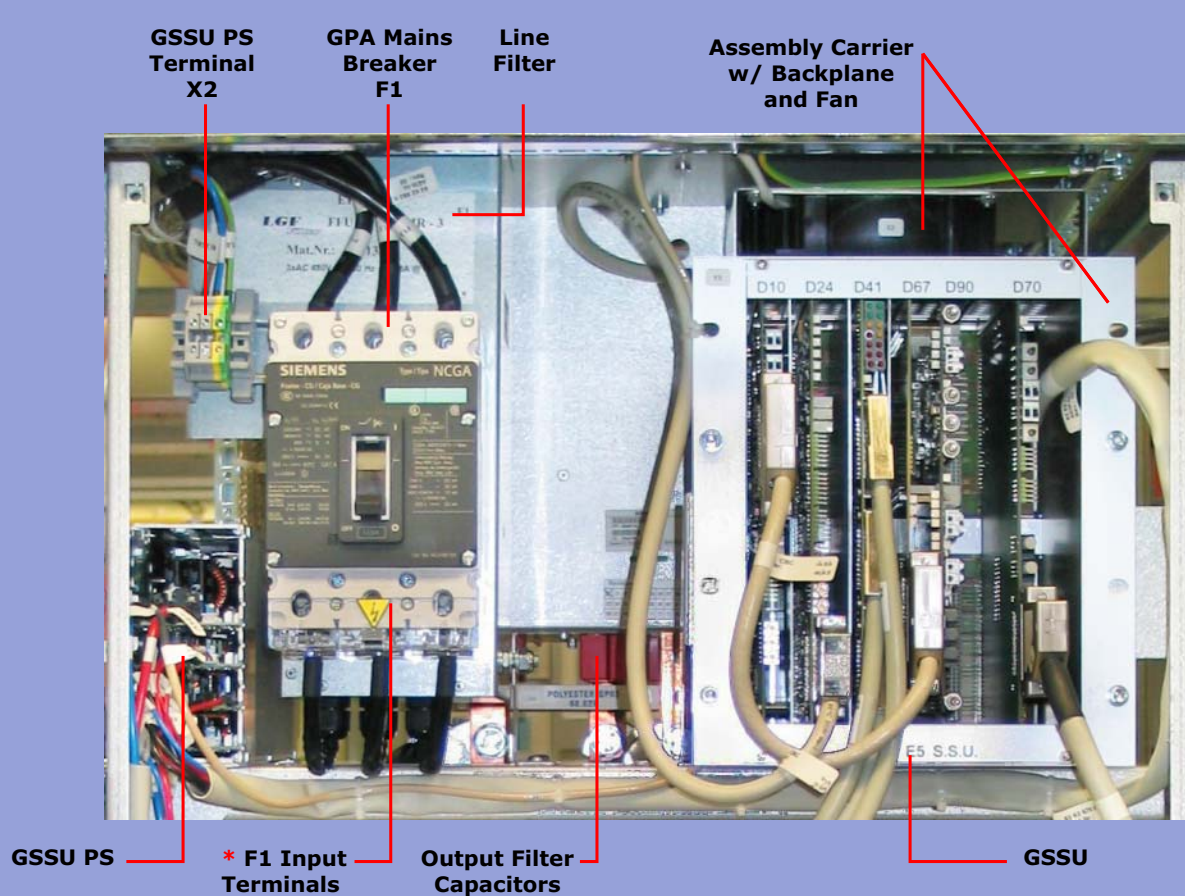
GSSU Assembly Carrier

Function

The GSSU Assembly Carrier consists of the mounting rack and Backplane and cooling fan.

A fan is built in on top of the Assembly Carrier.

Figure 147 GSSU Layout



GPA
62.20.F

GPA
90.22.F

* there is ALWAYS voltage present here, even when F1 is turned off!

D10 DAC

Overview

The DAC-board contains the D/A converters for X, Y, and Z axis. The digital pulse and compensation data is supplied by the MARS 2 over a PCIe fiber-optic link to the D68 Opto Link board which calculates the pulses and necessary compensations in real-time and forwards the gradient nominal data to the D10 DAC. Here, the D/A conversion takes place and the resulting analog nominal current values are sent to Regulator D24.

Function

Data Input Logic

The 20 bit amplitude data (**GC_CTRL**) for the three gradient axes is transferred multiplexed. The select signals **G_SEL** are used to clock in only new gradient data values to increase signal stability

Nominal Value

The main DAC output is fed over the backplane to the D24 Regulator board as the nominal gradient amplitude. A copy is also sent to the D68 Opto Link board for loop tests within the Service Software (DAC Functional Test).

The output voltage of the DACs (**DAC_X, Y, Z**) is ± 10 V at full scale. It corresponds to the maximum gradient current that is configured at the D24 Regulator board.

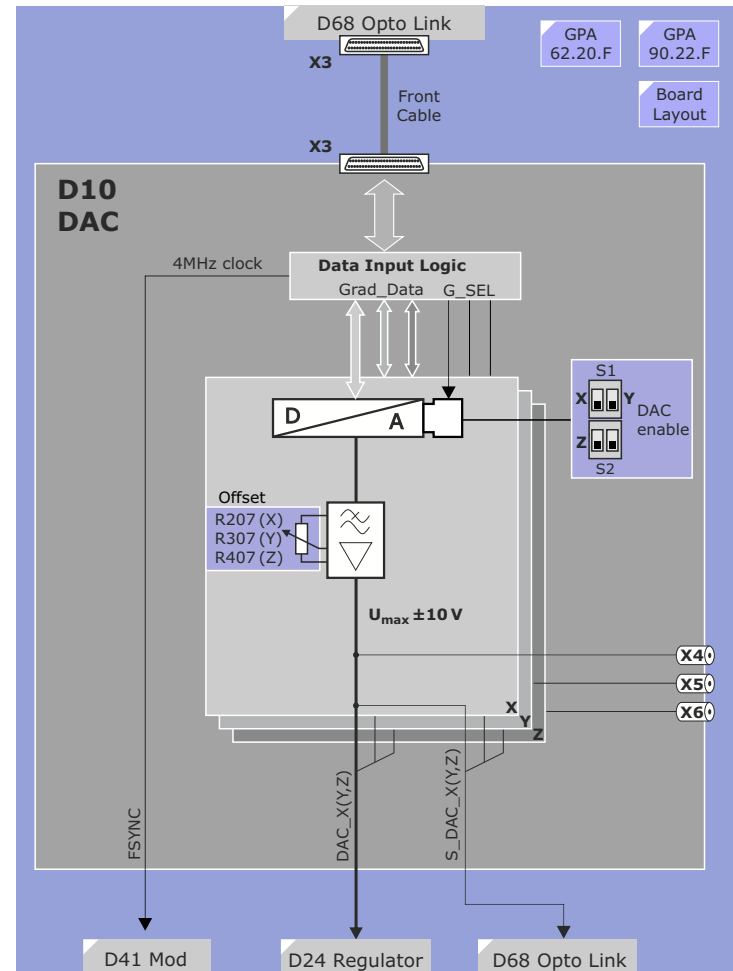
Synchronization

A 4 MHz clock signal is fed to D41 for pulse switch synchronization.

Loop-back Test

An input data loop-back test MARS 2-to-D10 can be run from the Service Software (DAC Functional Test). When a test is started, the 20 bit data of the X channel (**SU_Grad**) and the control lines are looped back to the MARS 2 to check the data path.

Figure 148 D10 DAC Block Diagram



Switches

Switches **S1** (left/right) and **S2** (left) can be used to zero the digital input of the DACs. This is necessary for factory offset adjustments using potentiometers R207(X), R307(Y) and R407(Z).

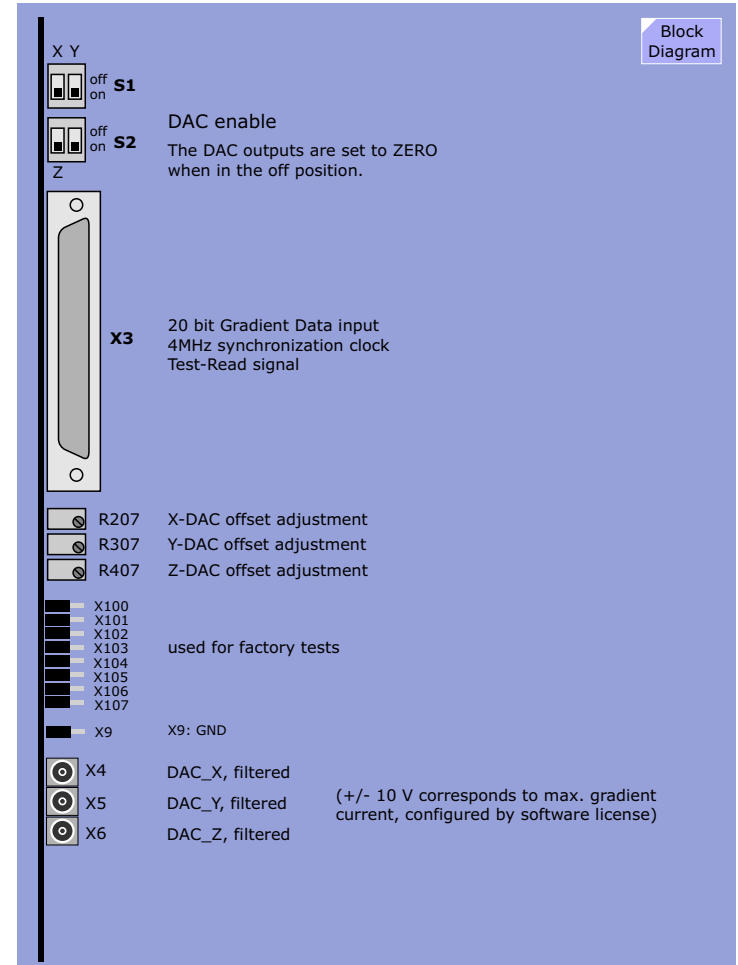
Switch	Description	Position
S1 left	DAC_X enable/disable	down/up
S1 right	DAC_Y enable/disable	down/up
S2 left	DAC_Z enable/disable	down/up
S2 right	not used	not used

Test Points

The DAC outputs, i.e. nominal values, (DAC_X, Y, Z) are available for monitoring via QLA connectors (X4, X5 and X6).

Test pins X100-X107 and X9 are used only for factory tests.

Figure 149 D10 Board Layout



D24 Regulator

Overview

The D24 includes the Regulator function for all three axes and performs the following functions:

- Actual Value Amplifier
- PID Regulator
- Regulator Parameter Control
- Monitoring and LEDs

Actual Value Amplifier

The AVA section is responsible for processing the current sensor outputs for use by the regulator and monitoring.

The DANFYSIK current sensors in the Current Converter Unit (CCU) deliver a signal proportional to the current driven through the Gradient Coil.

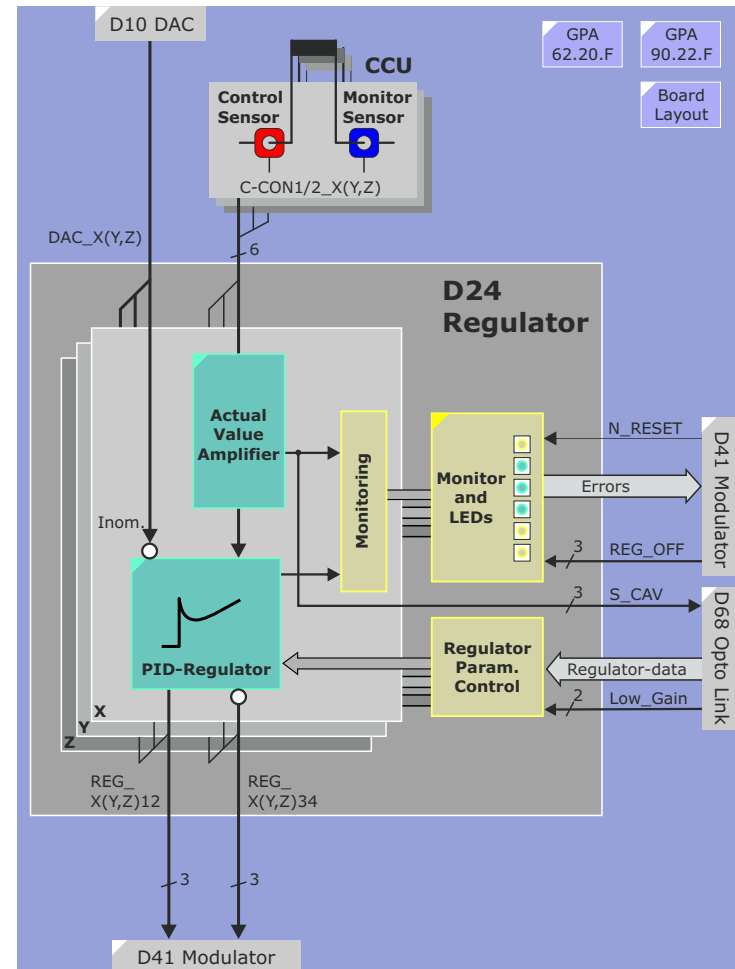
PID Regulator

The regulator has the task of controlling the voltage across the coil to establish the desired current waveform described by the nominal gradient pulse signal coming from the DAC and the actual current through the load. Since the load is highly inductive and the ramp times extremely short (μs) a PID regulator is used.

Monitoring

An amplifier that can produce up to 900A at voltages at 2000V requires monitoring. The monitoring on the D24 checks for and prevents conditions that could damage the hardware.

Figure 150 D24 Regulator Block Diagram



Function

Actual Value Amplifier (AVA)

The DANFYSIK current sensors in the Current Converter Unit (CCU) deliver a current (**C-CON1_X(Y,Z)**) that is proportional to the current driven through the Gradient Coil. That current is converted to a corresponding voltage level (1 V corresponds to 100 A).

The offset potentiometers R155(X), R189(Y) and R223(Z) have been factory adjusted for compensation of sensors and amplifier. Switches S3/4 on the D70 are used to ZERO the current for these adjustments.

The **C-CON2_X(Y,Z)** signal from the LEM current sensor is only used for monitoring purposes and is not amplified.

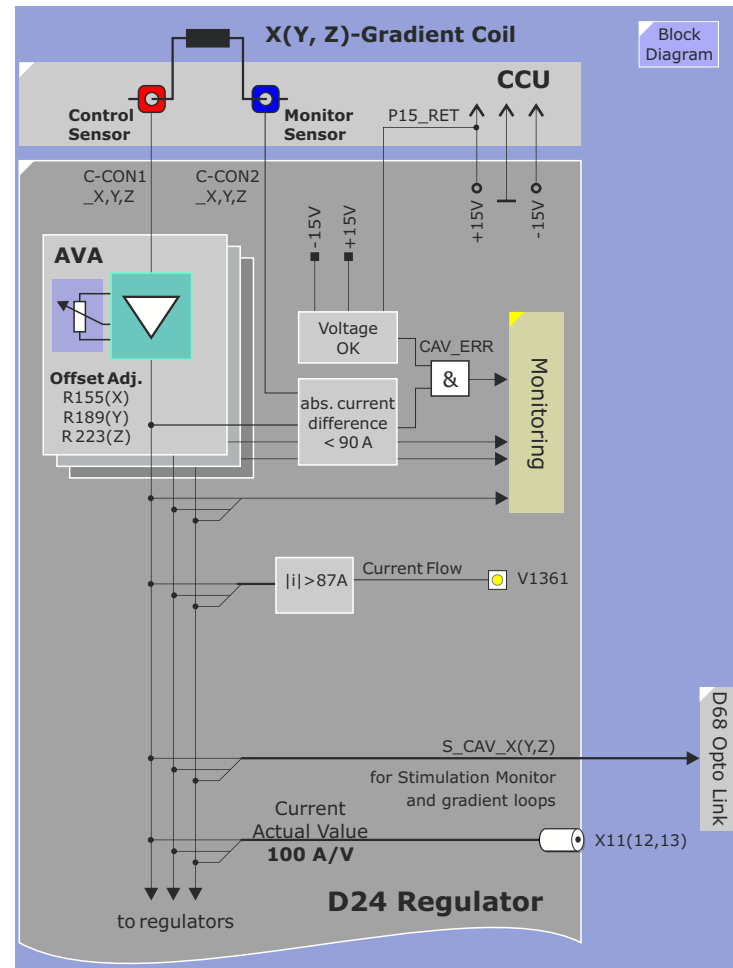
AVA Outputs

The AVA's main output signal is used for regulation. A copy (**S_CAV_X(Y, Z)**) is sent to the D68 Opto Link board where it is digitized and sent to the RFCEL_COM for peripheral nerve stimulation (STIMO) monitoring. In case the slew rate exceeds critical values, the RFCEL_COM sends a signal to the D68 which blocks the D41 Modulators and the gradient current is stopped. The same CAV signal is also used to evaluate the gradients during Tune Up and service test procedures (X, Y, Z-axis functional test).

In case an absolute current value of more than 87 A is detected in one of the axes, the yellow LED V1361 at the front side of the D24 lights up.

The current actual value (or I_{act}) is available at the front side QLA connectors X11(12,13).

Figure 151 D24 Actual Value Amplifier



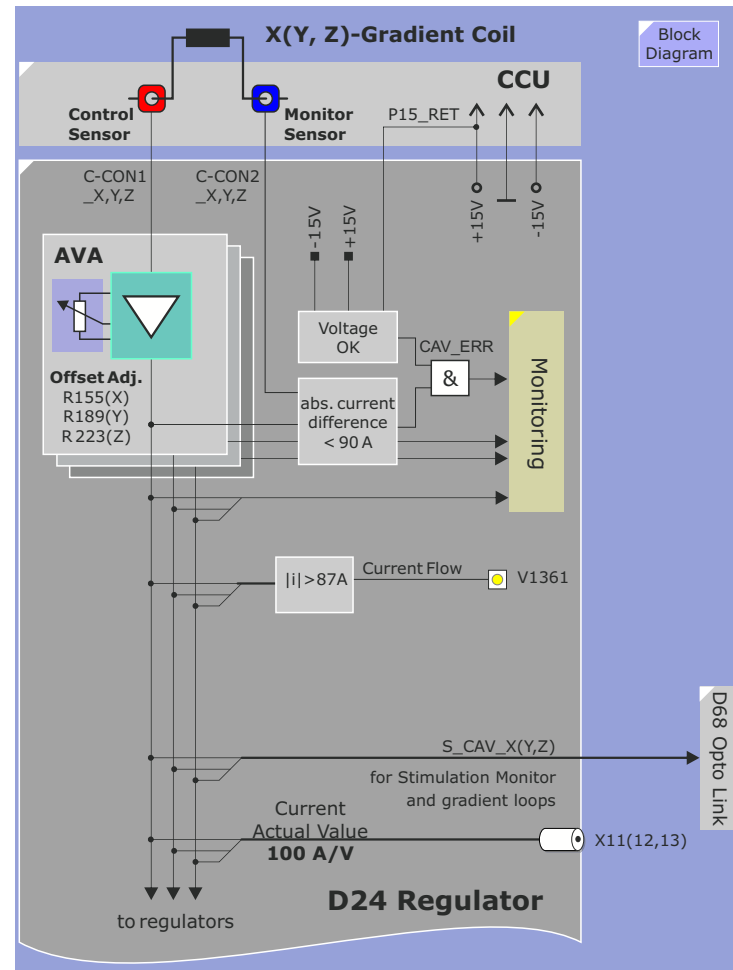
Current Sensor Monitoring

The DANFYSIK and LEM outputs are compared for differences in measured current of greater than 90 A. This redundancy is to assure that the current sensors are actually working. An undetected sensor malfunction would result in blowing up the Power Stages!

The DANFYSIK and LEM supply (± 15 V and ANA_GND) come from the D24 board and the +15 V is looped back as P15_RET-signal for monitoring. The ± 15 V supplies are also monitored.

If any of the above monitored signals indicates a fault, the respective current actual value error **CAV_ERR_X(Y, Z)** is sent to the D41 to disable the Modulator (for description of other errors and LEDs see D24 monitoring description).

Figure 152 D24 Actual Value Amplifier



PID-Regulator

The nominal value sent from the D10 DAC board is subtracted from the actual value coming from the AVA. This error signal is applied to the regulator circuit, based on a digitally controlled 7-bit PID-Regulator (P=proportional, I=integral, D=differential part).

The Regulator output is sent to the D41 Modulator board as inverted and non-inverted signals. The polarity of these signals determine the direction, their magnitude the amplitude of the output gradient current.

Regulator Parameter Control

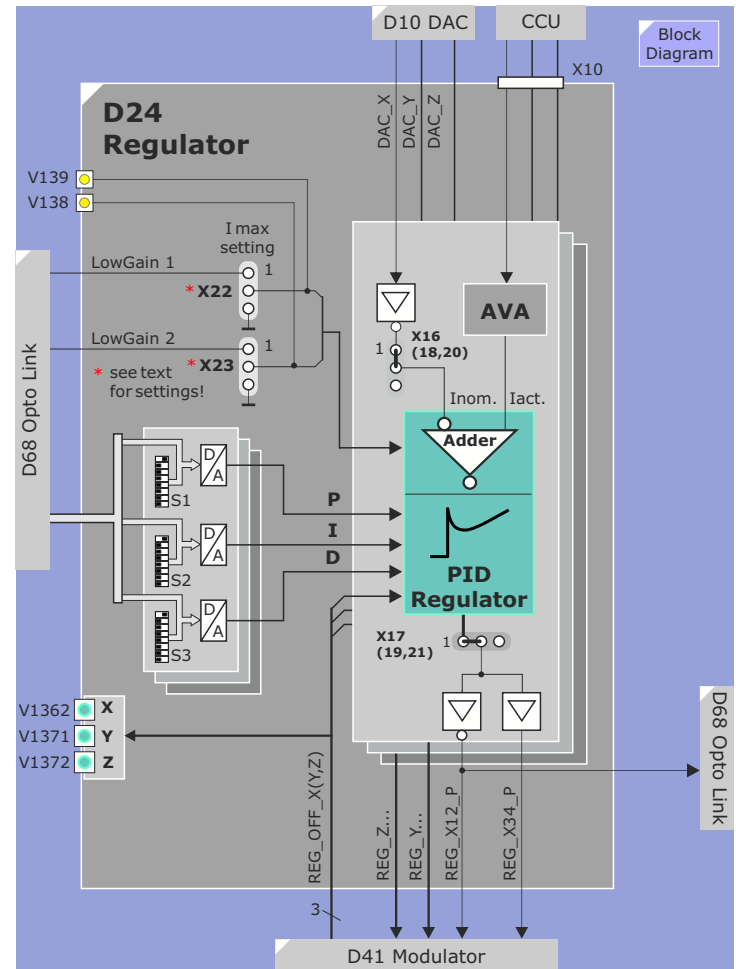
The initial setting of the Regulator after power on is determined by DIP-switches S1 to S9. The setting of switch 8 (off-position) on switch S3, S6 and S9 determines if this initial hardware setting can be overwritten by software via the D68. In that case, the downloaded site-specific Regulator parameters from the Tune Up are activated.

Switch	Function	#1	#2	#3	#4	#5	#6	#7	8
S1	X "P"	off	off	off	off	off	off	off	off
S2	X "I"	off	off	off	off	off	off	off	off
S3	X "D"	ON	ON	ON	ON	ON	ON	ON	off
S4	Y "P"	off	off	off	off	off	off	off	off
S5	Y "I"	off	off	off	off	off	off	off	off
S6	Y "D"	ON	ON	ON	ON	ON	ON	ON	off
S7	Z "P"	off	off	off	off	off	off	off	off
S8	Z "I"	off	off	off	off	off	off	off	off
S9	Z "D"	ON	ON	ON	ON	ON	ON	ON	off

Maximum Gradient Current Configuration

In order to adapt to the different available gradient performance levels but also to avoid damage to the Power Stages the maximum gradient current limits must be correctly configured with jumpers X22 and X23 on the D24 Regulator board (see [description of jumpers X22 and X23](#)).

Figure 153 D24 PID Regulator



D24 Monitoring

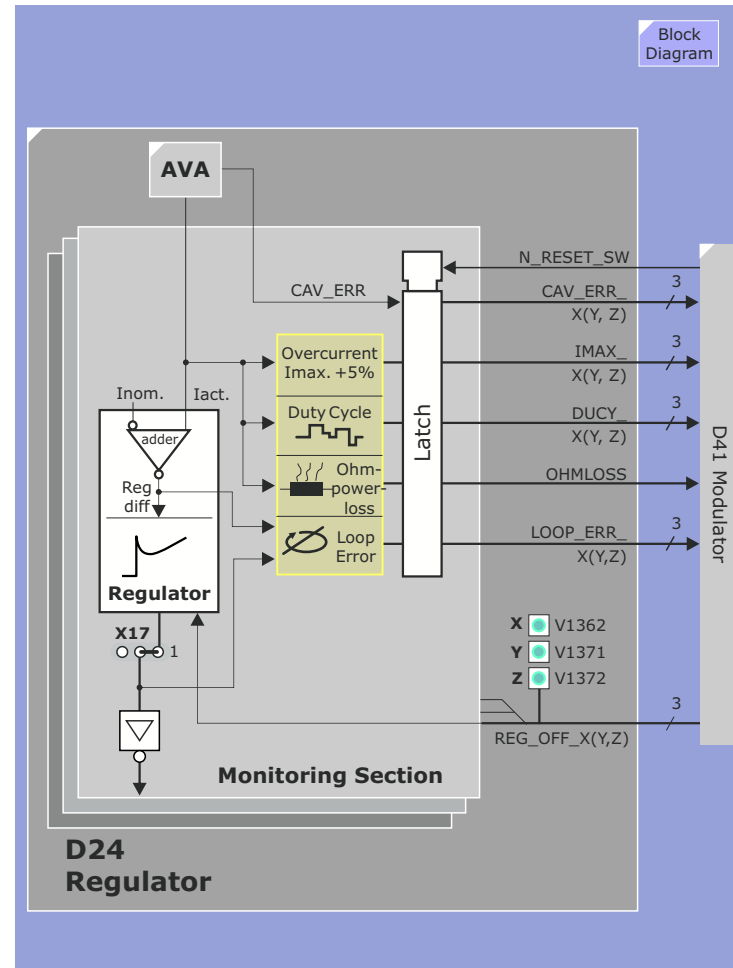
The following table shows the monitoring on the D24 board:

Error	Description
Over-current	Occurs when the gradient output current exceeds 5% of the maximum allowed output current
Duty Cycle	The CAV is integrated (time parameter in the μs range, i.e., fast acting to simulate the heat induced in the IGBT) and compared to a reference. The error occurs when the amplitude/time limit is exceeded.
Ohm power loss	The same as above, but the time of the integral is on the order of minutes to simulate the heat induced in the gradient cable and gradient filter.
Regulation (loop) error	Occurs when the difference between I_{nom} and I_{act} becomes too high (indicating a break in the regulation loop) or when the Regulator output signal is too high (indicating a Regulator malfunction)
CAV-error	- current sensor difference $>90\text{ A}$ 15 V return from current sensor; $\pm 15\text{ V}$ supply

Regulator Enable

The D41 Modulator uses the signal **REG_OFF_X(Y, Z)** to disable/enable the Regulator. The enable status is shown by three green LEDs on the front panel of the D24.

Figure 154 D24 Monitoring



LEDs

LED	Status	Description
V1361		current flow (one or more axes)
V1362		Regulator X enabled
V1371		Regulator Y enabled
V1372		Regulator Z enabled
V138/V139	V138	max. output current 491 A max. current 651 A (XJ Gradient, Aera) max. current 793 A (XQ Gradient, Skyra) max. current 951 A (XQ Gradient, Aera)
	V139	
		
		
		
		
		
		

Test Points

Refer also to sticker inside Gradient cabinet.

Potentiometers

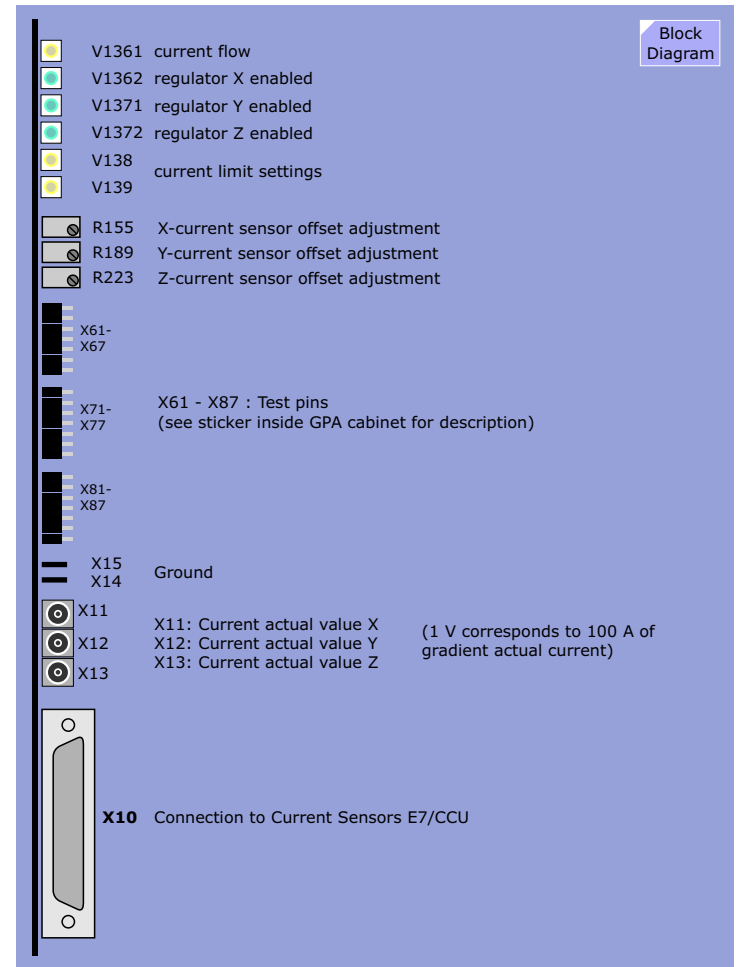
Potentiometer	Description
R155, R189, R223	X, Y, Z -current sensor offset adjustment

Jumpers

Jumper	Position	Description
X22 / X23		GPA max. current configuration (X22 = Bit 1, X23 = Bit 2)
	n.c. / n.c.	491 A (currently not used)
	2-3 / n.c.	651 A (XJ Gradient Aera)
	n.c. / 2-3	793 A (XQ Gradient Skyra)
	2-3 / 2-3	951 A (XQ Gradient Aera)

Jumpers X16-21 and X24 have to be set to position 1-2 for normal operation.

Figure 155 D24 Board Layout



D41 Modulator

Overview

The main duties of the Modulator are:

- **Modulation** - generation of pulse-width-modulated switching signals for the three axes.
- **Supervision** - monitoring of time-critical power stage related signals and latching of error signals from other boards.
- **Synchronization**
- **Power-UP control**

Function

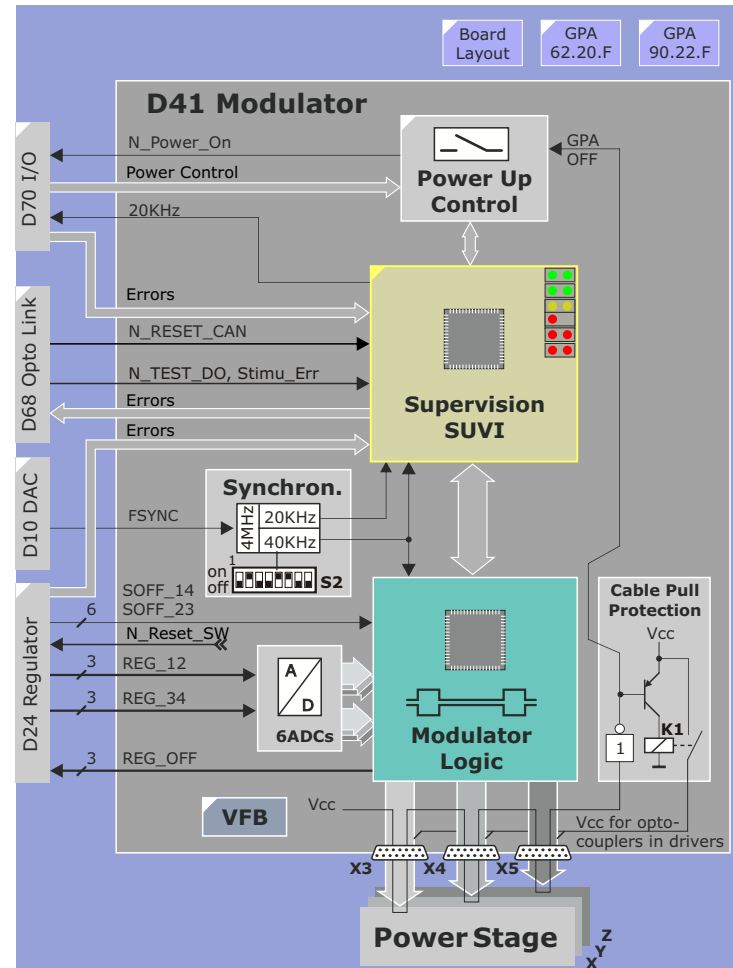
Pulse-Width Modulator

The D41 is based on a programmable logic (FPGAs). The D24 Regulator output signals **REG_X(Y, Z)_12(34)_P(N)** are analog-to-digital converted and fed to the Modulator chip where pulse width modulation is performed. The switching frequency of 40 kHz is derived from the 4MHz FSYNC signal using a programmable frequency divider which is configured by S2. The pulse width modulated switch signals for the IGBTs in the Power Stages are fed out over X3-X5 to the Power Stages using a standard SCSI-cable. The good high frequency characteristics and connector latches of SCSI standard make it perfect for this use. The connections of these cables are monitored with a wire interlock. Misconnections lead to a POWER OFF of the amplifier!

Slew Rate Adjustment

The maximum possible slew rate of the Gradient System is adjustable by setting limits for the pulse width modulator. The setting of **jumper X10 and X11** determines if software limits from the D68 or fixed values from the board itself are used (see jumper list).

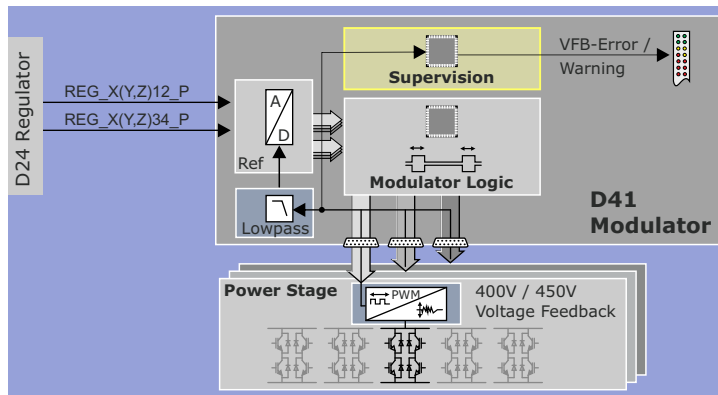
Figure 156 D41 Modulator Block Diagram



Power Stage Voltage Feedback (VFB)

The Power Stages and D41 Modulator are equipped with a feedback circuit that is used to compensate deviations in the line voltage to prevent gradient current instabilities. A sample of the 400/450V DC Power Stage supply voltage is made by the VFB circuit in the Power Stage and sent to the D41 where it is digitized and fed to the Modulator. The Modulator compensates for line fluctuations by modifying the pulse width modulated signals. If the power stage supply voltage is too far outside the limits or missing completely, an axis-dependent voltage feedback (VFB) error is generated.

Figure 157 D41 Modulator VFB



Voltage Monitoring

The D41 monitors the logical supply voltages (± 15 V and $+5$ V = VCC) from the GSSU Power Supply.

Modulator Logic Monitoring

- **Wrong controlling signal:** The proper timing and logic status of the PWM drive signals are continuously monitored (e.g. for cross-current (two switches of a half-bridge switched on at the same time) and dead-time delays). In

case of an error, the Modulator and Regulators are disabled.

- **Modulator defect:** The "clock fault", "wrong controlling signal" and some other D41-internal supervisions result in a "Modulator defect" error, indicated by the red LED (V123, left).
- **Cable pull protection:** A wire interlock (see Figure 156) is looped in series between pins 1 and 50 of each of the three cables to the Power Stages. If interrupted (cable not or improperly connected) the 5 V VCC supply to the optocouplers in the Power Stage's driver boards is removed and the PWM drive signals of all 3 stages is disabled and the "Cable pull protection" error is generated.
- **Power Stage cable defect:** During GPA Power-UP, the Modulator logic checks the functions of the output stages and the cable connection to the Power Stages by switching the output shortly (0.5 sec.) to "high impedance." The monitoring at the SUVI will measure "high level", but in case of a faulty Modulator output or a broken wire "low level" across the 20 k Ω pull-down resistor. A "Power Stage cable defect" error is generated as a result.
- **Oscillation:** Regulator loop oscillation with more than 10.5 kHz or has reached three times maximum level.
- **Clock fault:** The Modulator logic monitors the presence of the 164 MHz internal clock.

Figure 158 Power Stage Cable Test

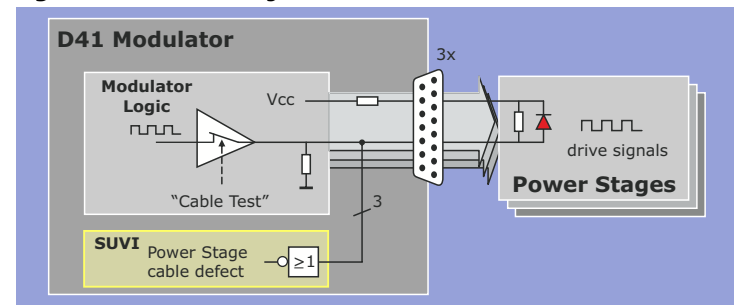
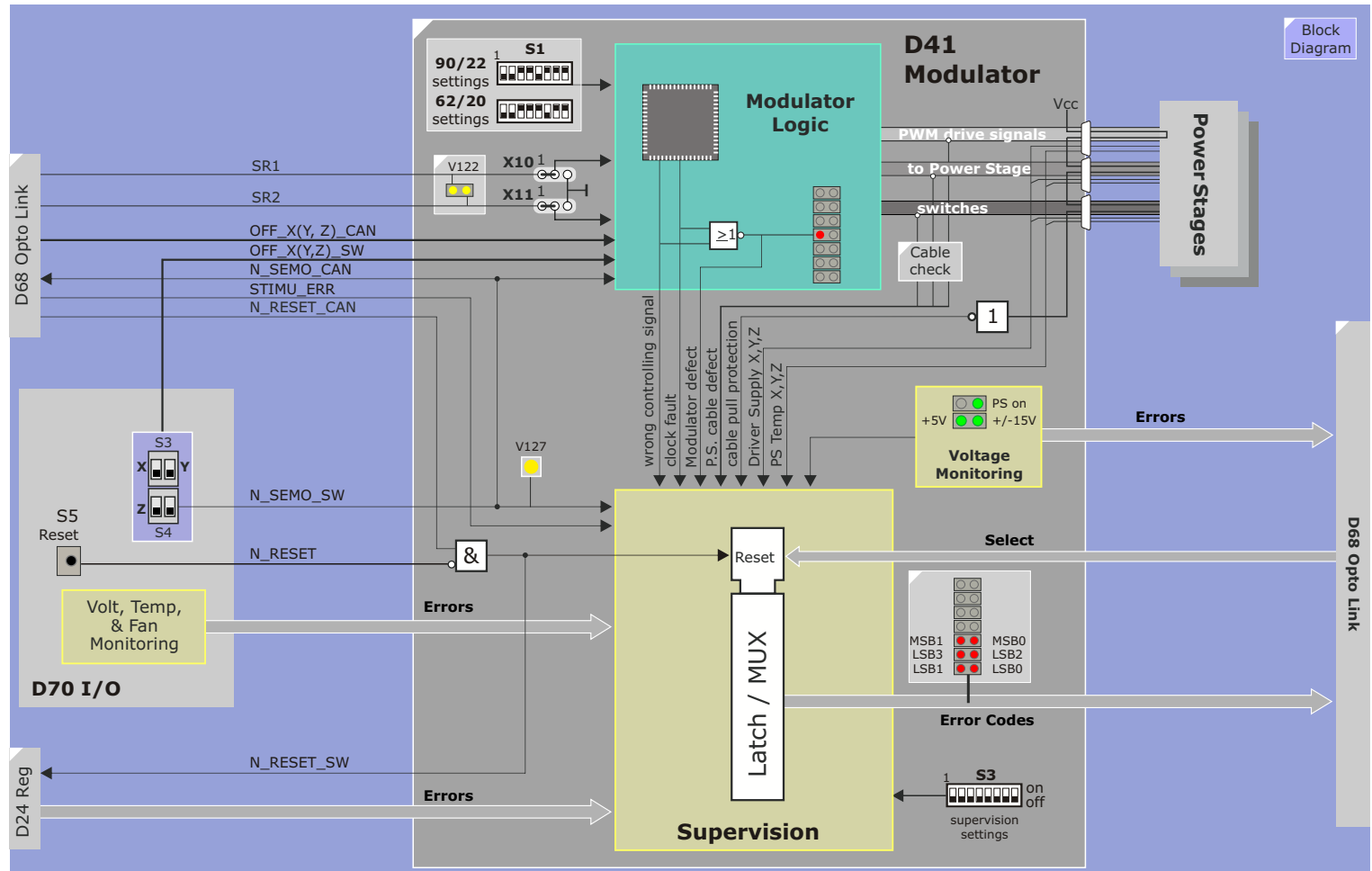


Figure 159 D41 Supervision, Display, Synchronization



Supervision Logic (SUVI)

The supervision logic (SUVI) reports the errors generated by the Modulator logic and handles the following additional errors:

- **Synchronization fault:** 20 kHz sync. not ok.
- **All Regulator errors:** reported by the D24 Regulator board.
- **GPA over-temperature, GPA voltage and fan monitoring:** reported by several monitor circuits via the D70 I/O board.
- **Stimulation errors:** reported by the Stimulation Monitor (in RFCEL_COM)

The SUVI error control logic can report 64 different error codes. These are addressed by the D68 board with Select lines. The error code is read out and displayed by 6 red LEDs at the front side of the board (see Figure 162). A warning or error message is generated and fed to the D68.

Errors reported to the D41 Modulator by other boards and errors occurring on the D41 are latched and sent to the D68 Opto Link board to be sent to the MARS 2 and ultimately to the Host to be logged in the event log and displayed to the customer.

Modulator Disable

The Modulators can be disabled manually with service switches **S3**, **S4** on the D70 (**OFF_X(Y,Z)_SW**) or by software via the **OFF_X(Y, Z)_CAN** signal from the D68. Both cases will disable the gradient current immediately. In addition, the Modulator can be disabled by software in case of an error.

Reset of Errors

All errors can be reset by the processor D68 (**N_RESET_CAN**) or manually by push button **S5** at the I/O-board D70.

Service Mode

To facilitate troubleshooting, the GPA can be switched to Service Mode with switch **S4.right** on the D70 in the upward position or by software via the **N_SEMO_CAN** signal from D68 Opto Link.

The following functions are affected by the Service Mode:

- **Modulator enable:** The Modulators are not disabled when the Power Stage supply voltage is switched off or is too low.
- **Synchronization fault:** The fault is handled as a "warning" only.
- **Closed loop error:** The fault is handled as a "warning" only.
- **Voltage feedback error:** The fault is handled as a "warning" only.
- **Oscillation error:** Regulator may oscillate for a slightly longer period of time until the fault is reported.

Switches

Switch	#1	#2	#3	#4	#5	#6	#7	8
S1 90(75)/22	off	off	ON	ON	off	ON	ON	ON
S1 62/20	off	off	ON	ON	ON	off	ON	ON
S2	off	ON	off	off	ON	ON	off	off
S3	off	off	off	off	off	off	off	off
S4 (D41 Rev.>00)	off	ON	ON	off	n.a.	n.a.	n.a.	n.a.

S1 Dead-time and Power Stage configuration

S2 Frequency settings

S3 Supervision settings

S4 Clock Phase

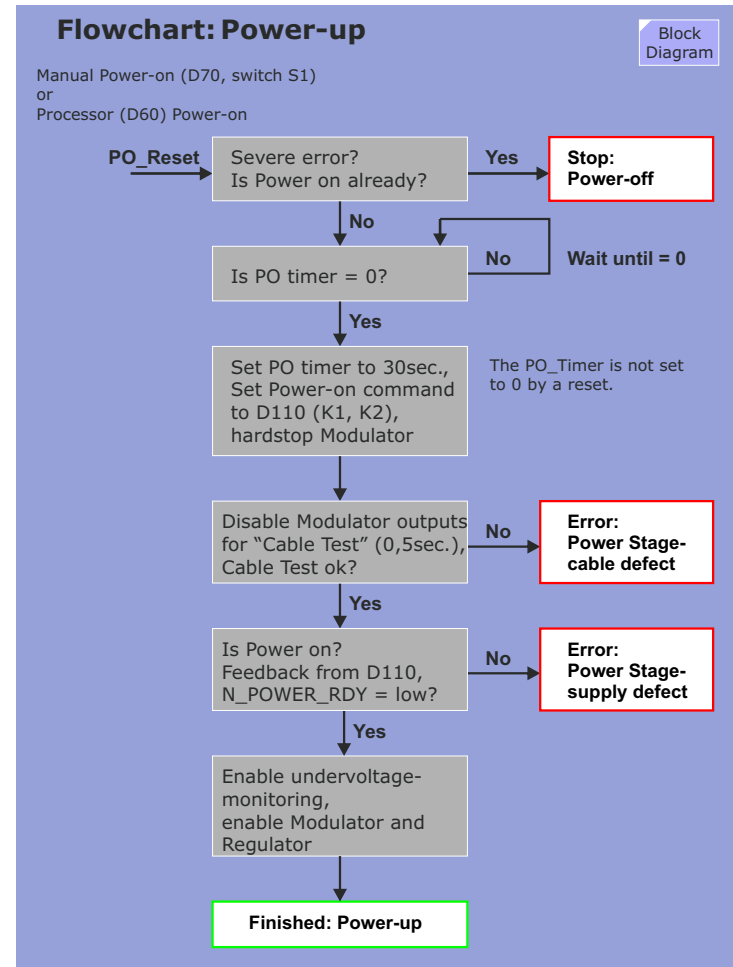
NOTE D41 boards from spare parts stock are delivered with default jumper setting which leads to the error "Modulator defect"! Correct the jumper setting before use!

GPA Power-UP

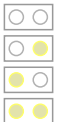
The D41 controls the Power Stage supply switching-on and -off (see also [D110 Power-UP](#) description). At the beginning of the Power-UP routine, all stored errors are reset, if a fault occurs later on, the D41 will power-off the supply and generate the corresponding error message.

NOTE Power-UP can take up to 30 seconds due to an internal timer at the D41 Modulator-board.

Figure 160 GPA Power Up Chart



LEDs

LED	Status	Description
V120 left	on	all Modulators on
V120 right	flashing	all Modulators on, but warning
V120 right	on	Power Stage power supply on
V121 left	on	Power status is changing
V121 right	on	
V122 left/ right		Note: values in brackets valid for GPA 62/20/F max. output voltage 562 V (500 V) max. voltage 1125 V (1000 V) max. voltage 1687 V (1500 V) max. voltage 2250 V (2000 V)
V123 left	on	Modulator defect
V123 right:	on	Warning
V124 V125 V126	see next page	Using a multiplexing method LEDS 124-125 are used to display 4 groups of 16 error codes each. LED 124 displays the group and LEDs 125, 126 display the 16 error codes using a binary pattern.
V127	on	Service Mode on

NOTE A minor problem is indicated as “warning” only (yellow LED V123/right lights up additionally).

Jumpers

Jumper	Position	Description
X9	1-2	Stimulation monitor is active
X10/ X11	1-2/ 1-2	Slew rate adjusted by D68
X13, X14	1-2	normal operation/ PCB test
X15	open	not used

Figure 161 D41 Board Layout

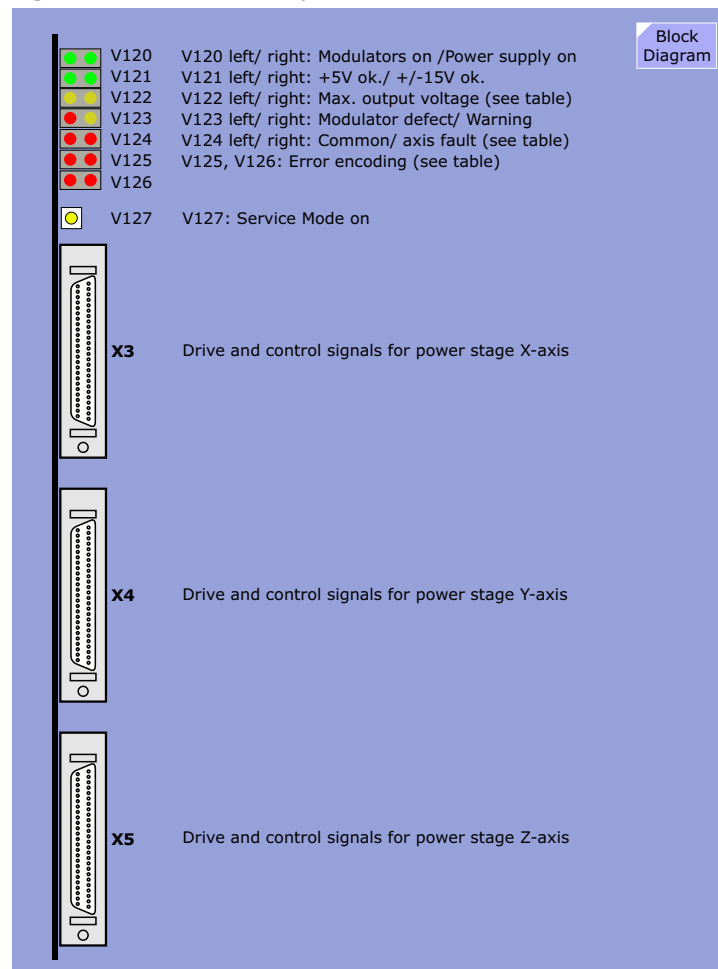


Figure 162 D41 LED Error Patterns

Block
Diagram

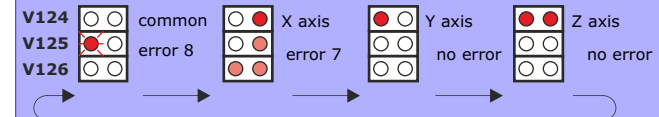
Error	LED V125/126	LED V124	Description
0		common	Regulator Oscillation Voltage feedback error (or Power Stage undervoltage for GPA 62/20/F)
1		common	GPA Modulator disabled by NUMARIS OFF from system
2		common	Soft stop from system switched off by D70 (S3, S4)
3		common	Fan-error Overcurrent, Imax. Limit
4		common	Control Signal to Power Stage defective, Modulator defective Duty cycle too high
5		common	Wire to Pow. Stage missing, cable pull protect. Ohm power loss too high
6		common	switched off by Stimulation Monitor current value (sensor) error
7		common	Replace D70 / set D70 X15 Pin A4-B4 Regulation (loop) error
8		common	Gradient Supervision error (VLM error) Modulator control signal freq. too high
9		common	Power Stage defect (Power on error; extreme power) Driver power supply defect (or Power Stage overvoltage for GPA 62/20/F)
10		common	GPA air temperature too high: >42°C warning - LED V123R on >47°C error Power Stage over-temperature (air or water)

Error	LED V125/126	LED V124	Description
11		common	Modulator off (3-phase PS-supply is off) Power Stage overvoltage
12		common	3-phase Power Stage supply defect Power Stage low voltage
13		common	Main Transformer over-temperature Rectifier over-temperature
14		common	Synchronization error Output inductor (choke) over-temp.
15		common	Configuration error D41 switch S1 Power Stage control cable defect

LED V124 :

	common		X axis		Y axis		Z axis
---	--------	---	--------	---	--------	---	--------

An example :



V124 indicates a common fault is being displayed and V125,126 show the error number 8 pattern. Also, this LED lights up very bright indicating this error occurred first.

Then in succession the X, Y and Z axis errors are displayed, whereby only the X axis is displaying an additional error, error number 7. Here the LEDs light up only at half brightness indicating it is a secondary error.

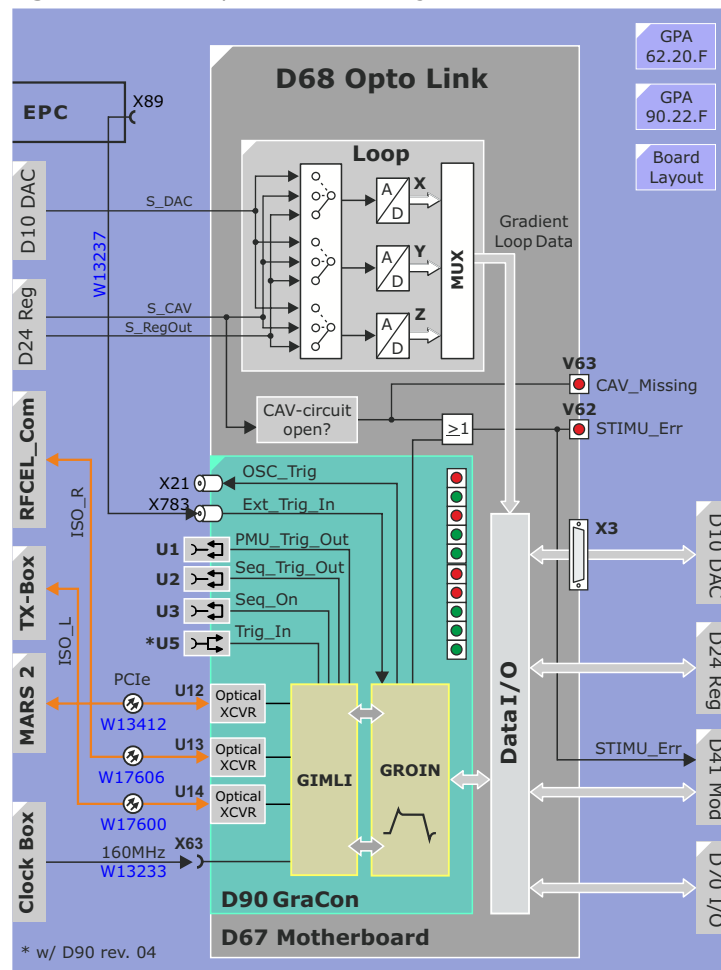
D68 (D67+D90) Opto Link

Overview

The D68 Opto Link is the GPA interface to the MR system. It consists of the D67 motherboard and the D90 GRACON board which plugs on the D67 as a piggy-back board. The D68 board assembly performs the following functions:

- **D67 Motherboard** - serves as data I/O for gradient nominal data and the multiplexed signals for stimulation monitoring, TuneUp and test loops.
- **D90 GraCon** piggy-back module consists of:
 - a **GIMLI** (Generic Infrastructure Module for Link Interfaces) for providing communication over a PCIe interface for communication and data transfer from/to MARS 2 and over the bi-directional optical ISO-ring interfaces for real-time data transfer and synchronization to other intelligent MR sub-components
 - an application specific FPGA **GROIN** (GRAdient Operation and Infrastructure Node) serves as a signal processor for the nominal gradient data and calculates the required compensations adjusted in Tune-Up.

Figure 163 D68 Opto Link Block Diagram



Function

System Interface

The GraCon communicates to the MR System by means of a **GIMLI** (Generic Infrastructure Module for Link Interfaces) which supplies two independent bus systems:

- **Optical PCIe bus** - FOC U12, comms to MARS 2
- **Optical ISO-ring Bus** - FOC U 13, ISO-ring R to RFCEL_COM and U14, ISO-ring L to TX-Box

The PCIe bus is used for receiving the digital gradient nominal data and configuration data but also for transmitting GPA status information or test signals back to the MARS 2.

The ISO-Ring bus is used to :

- receive sequence clock for synchronization
- send actual grad values to RFCEL_Com for STIMO monitoring
- send ECC-B₀ frequency shift values to the TX-Box

The GraCon has one QLA connection (X11) for the **160 MHz** clock signal supplied by the Clock Box:

Gradient Data Processing and Compensations

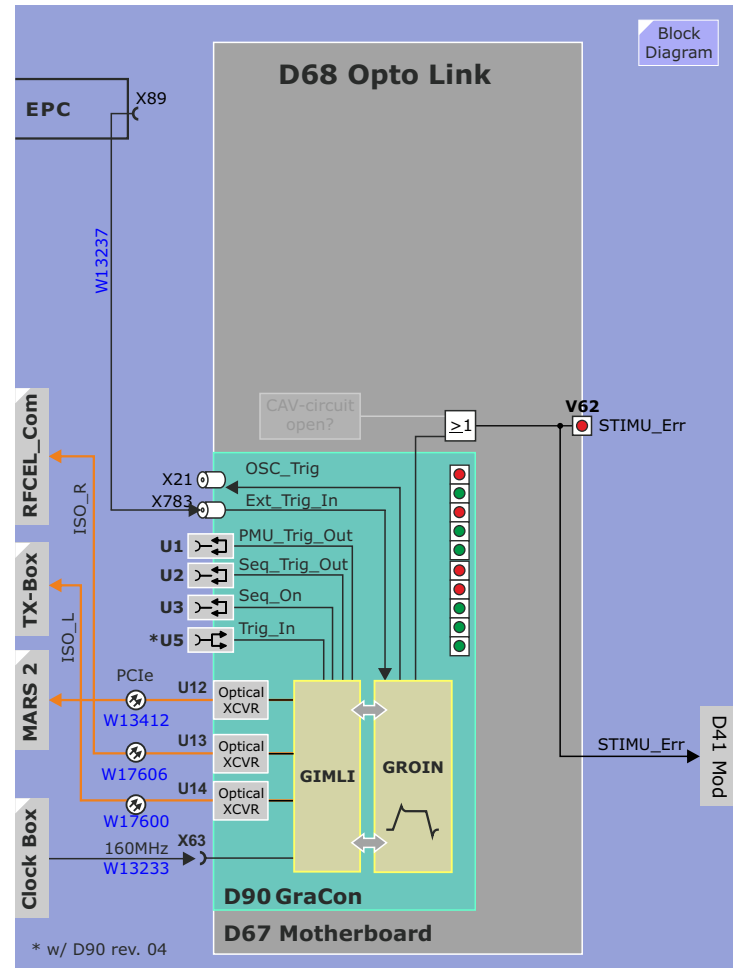
The GraCon receives the nominal gradient data from the MARS 2 over the PCIe bus. The **GROIN** FPGA calculates the required gradient nominal data in real-time and adds all necessary compensations previously determined in the Tune-Up (ECC/CTC, Gradient Delay). Finally, the resulting nominal gradient data is sent to the D10 DAC board via the front cable plug D68/X3 to D10/X3.

Trigger Signals

The D68 supplies some trigger features which can be accessed via GraCon front connectors:

- PMU Trigger Out - FOC U1
- Sequence Trigger Out - FOC U2
- Sequence On - FOC U3
- Oscilloscope Trigger - QLA connector X21
- External Trigger In - QLA connector X783

Figure 164 D68 Opto Link - System Interface



Signal Multiplexer (Loop)

The signal multiplexer can select 3 analog GSSU signals from each gradient axis for use in testing, Tune Up and QA measurements:

- **Current actual value** $S_CAV_X(Y, Z)$
- **Nominal value** $S_DAC_X(Y, Z)$
- **Regulator output** $S_REGOUT_X(Y, Z)$

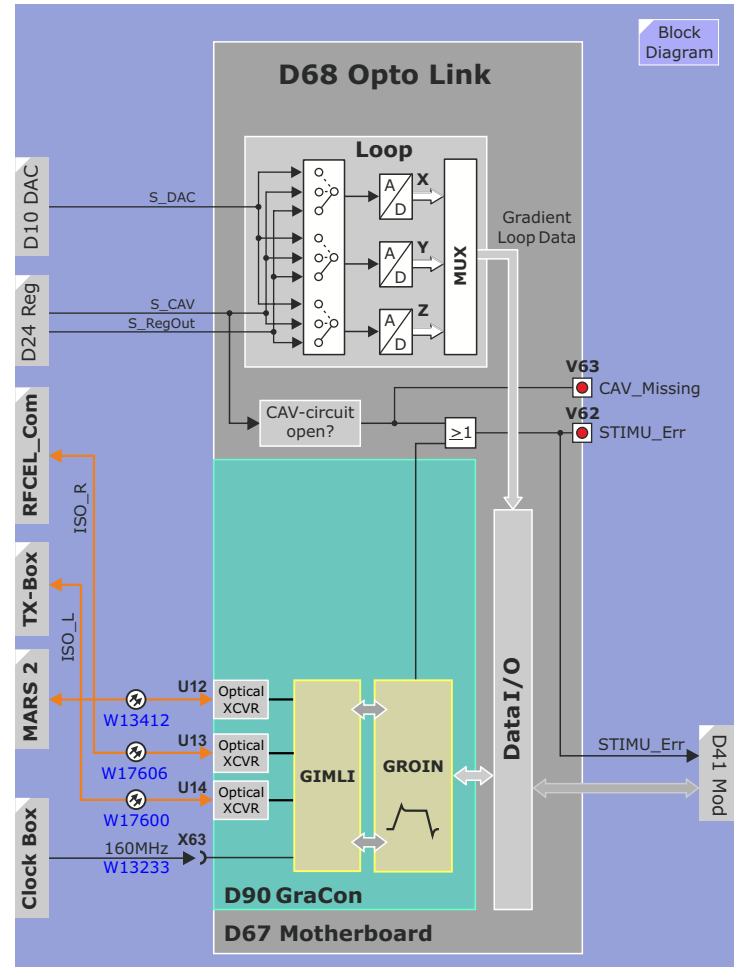
The signals are filtered and analog-to-digital converted and sent over the PCIe fiber-optic link to the MARS 2.

Stimulation Monitoring and Errors

The actual values (CAVs) are also used for stimulation monitoring. Since that function is safety relevant, a logic at the D68 monitors the integrity of this circuit (e. g. broken cable). In case of error, the "CAV-circuit open" error is sent over PCIe fiber-optic link to the MARS 2. LED V63 (actual value missing) and V62 (stimulation monitor) light up and the STIMU error blocks the modulator to stop the gradient current.

During imaging sequences, the 3 current actual values $S_CAV_X(Y, Z)$ are constantly fed as stimulation data over the ISO-ring fiber-optic link to the RFCEL_COM where the stimulation monitoring takes place. In case of exceeding critical stimulation thresholds, the RFCEL_COM sends the **STIMU_Err** signal to the D68 which will stop the modulator and LED V62 lights up.

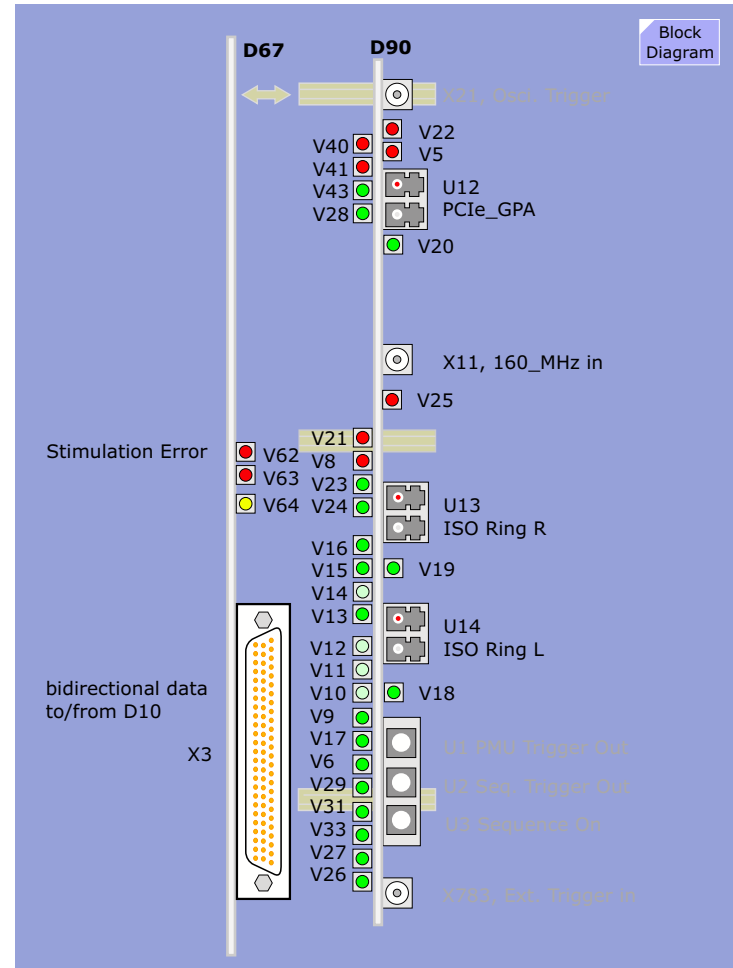
Figure 165 D68 Opto Link - Loop Back



LEDs

LED	color	Description
V5	red	Emergency Failure Alarm (e.g. PCB over-temp)
V6	green	TX PLL Locked Left Gimli
V8	red	nSTATUS Gimli
V9	green	1 PCIe Lane
V10	<green>	2 PCIe Lanes
V11	<green>	4 PCIe Lanes
V12	<green>	8 PCIe Lanes
V13	green	PCIe L0 State
V14	<green>	PCIe Reset
V15	green	PCIe PLL Locked
V16	green	Heartbeat Gimli
V17	green	TX PLL Locked Right Gimli
V18	green	RX Frequency Locked Left
V19	green	RX Frequency Locked Right
V21	red	nCONFIG Gimli
V22	red	Fatal Fault Alarm
V23	green	CONFIG_DONE Gimli
V24	green	INIT_DONE Gimli
V25	red	PowerOn Reset Alarm
V26/ V27	green	on/on = programmed from User page on/off = programmed from Factory page off/on = Programming in progress off/off = Programming failed
V28	green	INIT_DONE_Groin
V29	green	PLL Locked Groin
V31	green	DMA PLL Locked Groin
V33	green	Ring PLL Locked Groin
V40	red	nCONFIG Groin
V41	red	nSTATUS Groin
V43	green	CONFIG_DONE_Groin
V62	red	Stimulation error
V63	red	missing current actual value (from GraCon)
V64	yellow	Debugging (for developer only, normally off)

Figure 166 D68 Board Layout



Switches

Switch S1	Description	Position
GPA 90/22/F	S1.1 ... 4 = GPA part number (10513350)	S1.1 = off S1.2 = ON S1.3 = ON S1.4 = ON
	S1.5 ... 8 = Rev. Level (binary coded) S1.5 = 2^3 S1.6 = 2^2 S1.7 = 2^1 S1.8 = 2^0	Example for Rev. Level 5: S1.5 = off S1.6 = ON S1.7 = off S1.8 = ON
GPA 75/22/F	S1.1 ... 4 = GPA part number (10513313)	S1.1 = off S1.2 = ON S1.3 = ON S1.4 = off
	S1.5 ... 8 = Rev. Level (binary coded) S1.5 = 2^3 S1.6 = 2^2 S1.7 = 2^1 S1.8 = 2^0	Example for Rev. Level 5: S1.5 = off S1.6 = ON S1.7 = off S1.8 = ON
GPA 62/20/F	S1.1 ... 4 = GPA part number (10513280)	S1.1 = off S1.2 = ON S1.3 = off S1.4 = ON
	S1.5 ... 8 = Rev. Level (binary coded) S1.5 = 2^3 S1.6 = 2^2 S1.7 = 2^1 S1.8 = 2^0	Example for Rev. Level 5: S1.5 = off S1.6 = ON S1.7 = off S1.8 = ON

D70 I/O

Overview

The I/O board is an interface between the GSSU and the GSSU Power Supply and Power Stage section. Control and monitoring signals are fed via this board to the Modulator board D41 and the D68 Opto Link board. It also provides service function switches, test points and an air-temperature monitor.

Function

Power Switching

The D41 Modulator controls the power switching initiated by the D68 Opto Link or manually with push buttons **S1** and **S2** on the D70. See the [D110 Power-UP](#) description for details.

Voltage Supplies for GSSU

The GSSU Power Supply voltages are supplied to the GSSU boards through the D70 and distributed over the backplane. All voltages are buffered (capacitors) and the +5 V supply is protected by a 6.3 A fuse (F1) on-board.

GPA Air Temperature Monitoring

On-board NTCs are used for monitoring GPA cabinet air temperature. In case of over-temperature, the AIRTEMP_WARN (>42 °C) or AIRTEMP_ERR (>47 °C) signal informs the D41 Modulator.

Jumpers

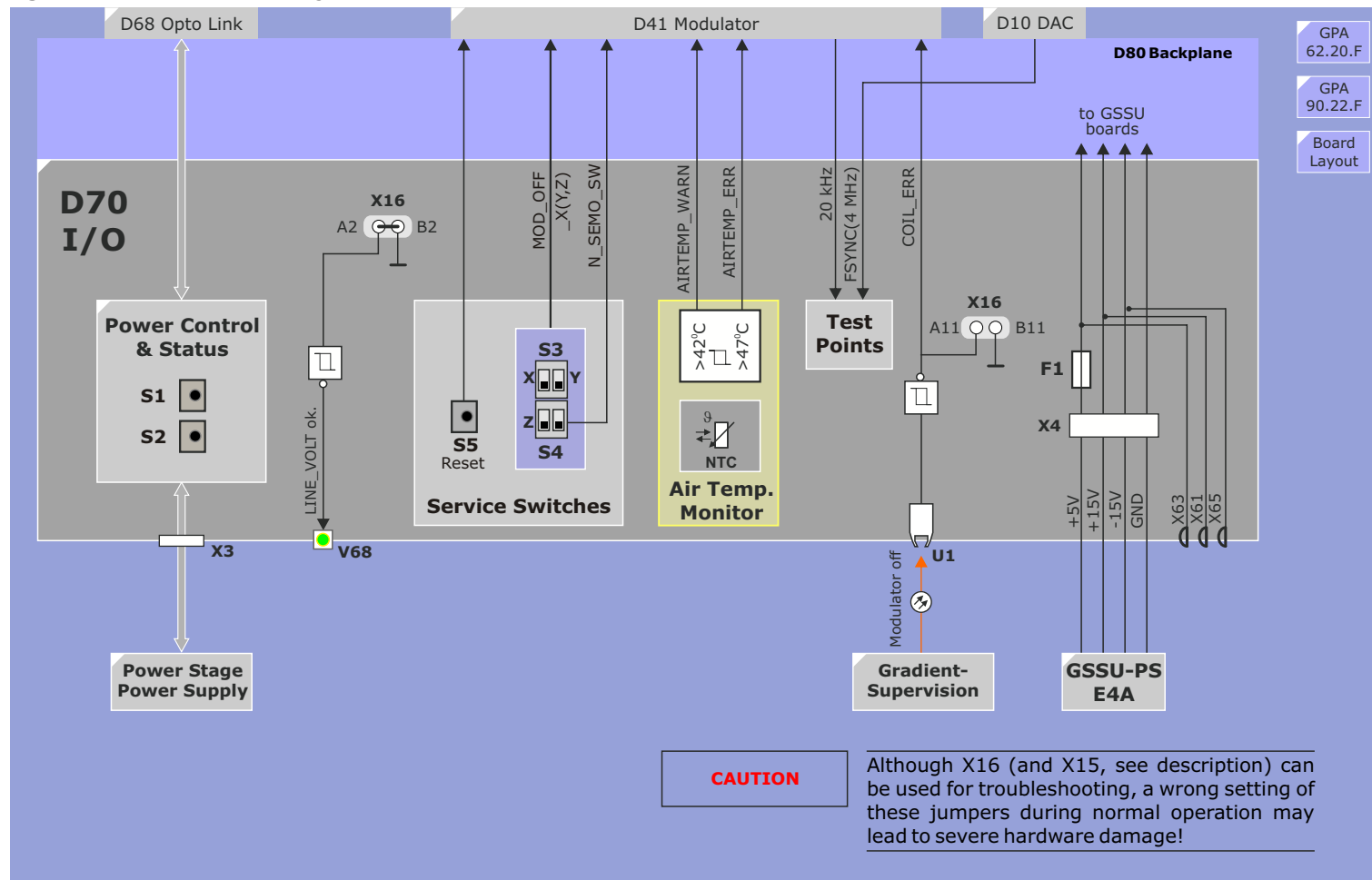
By means of inserting jumpers X15 and X16, each of the errors passing the D70 board can be bypassed.

WARNING X15 and X16 can be used for advanced troubleshooting. Setting these jumpers during normal operation may lead to severe hardware damage!

Jumper	Pins	Setting	Description
X15	A1-B1	open	Simulates N_POWER_READY
X15	A3-B3	open	Bypasses POWER_ON_ERR
X15	A4-B4	closed	Bypasses "spare error"
X15	A5-B5	open	Bypasses FAN_ERR
X15	A9-B9	open (closed)	Bypasses RC_TEMP_X error
X15	A10-B10	open (closed)	Bypasses RC_TEMP_Y error
X15	A11-B11	open (closed)	Bypasses RC_TEMP_Z error
X15	A12-B12	open	Bypasses FI_TEMP_X error
X15	A13-B13	open	Bypasses FI_TEMP_Y error
X15	A14-B14	open	Bypasses FI_TEMP_Z error
X16	A1-B1	open	Bypasses TRAFO_TEMP error
X16	A2-B2	closed	Simulates LINE_VOLT ok.
X16	A4-B4	open (closed)	Bypasses the UV_X error
X16	A5-B5	open (closed)	Bypasses the UV_Y error
X16	A6-B6	open (closed)	Bypasses the UV_Z error
X16	A7-B7	open (closed)	Bypasses the OVV_X error
X16	A8-B8	open (closed)	Bypasses the OVV_Y error
X16	A9-B9	open (closed)	Bypasses the OVV_Z error
X16	A11-B11	open	Bypasses the COIL_ERR (must be closed if no Gradient-Supervision installed)
X16	A12-B12	closed	Bypasses "Switch Coil" temp. monitoring
X16	A14-B14	open	Bypasses AIRTEMP_WARN/_ERR

Setting values in brackets () are for the GPA 62/20/F

Figure 167 D70 I/O Block Diagram



Switches

Switch	Description	Position
S1 (push button)	Power-on switch (mains power)	
S2 (push button)	Power-off switch (mains power)	
S3 left	Modulator X on/off	down/up
S3 right	Modulator Y on/off	down/up
S4 left	Modulator Z on/off	down/up
S4 right	Service Mode off/on	down/up
S5 (push button)	Reset of errors (at D41)	

LEDs

LED V68 (green) on the front side, indicating (AC secondary) LINE_VOLT ok is not used.

NOTE For GPA 62/20/F, the LED V68 is always on. The monitoring is bypassed with jumper X16 A2-B2.

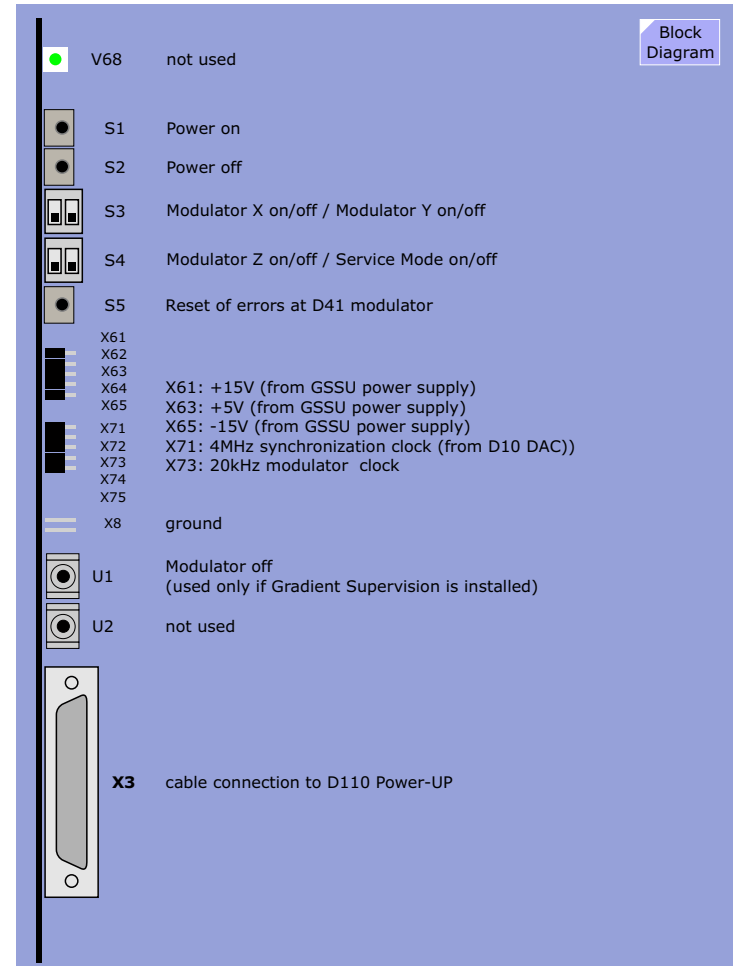
Fuses

There is a 6.3 A fast-blow fuse F1 on board for the +5 V voltage.

Test Points

+5 V (VCC), +/-15 V (from GSSU Power Supply) and two important timing signals: 4 MHz synchronization clock and the 20 kHz Modulator internal clock (see GSSU sticker inside GPA).

Figure 168 D70 I/O Board Layout



GPA 62/20/F

- Power Stage (PS)

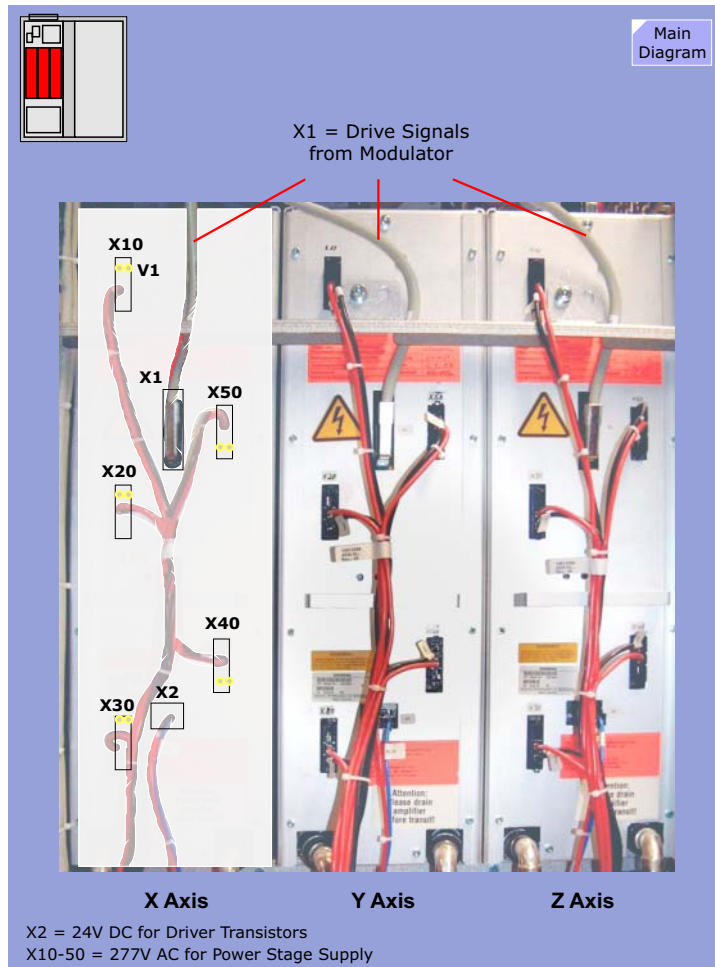
Overview

The Power Stage (PS) is a switch-mode current generator based on power IGBT-arrays. In order to allow a compact construction the power devices are mounted on water cooled heat sinks. Each Power Stage box consists of five so-called "H-bridges" that are connected in series (cascaded) to increase the output voltage. One "H-bridge" consists of four switches (IGBT modules) required to generate bipolar output current.

The following components can be found inside one Power Stage:

- Five water cooled H-bridges (cascades) with 4 IGBT switches each
- Five Rectifiers for the generation of the 400 VDC supplies
- Five 400 VDC capacitor blocks
- Integrated driver supply with galvanically insulated switch-mode power supply
- Integrated opto-transmitters and opto-receivers for galvanic insulation of drive- and monitoring-signals
- Auxiliary monitoring components

Figure 169 62/20/F Power Stage Layout



Function

Gradient Current Generation

The Power Stages consist of five H-bridges connected in series (cascaded). Each bridge is supplied with 400 VDC from the internal rectifiers for a maximum bipolar output of 2000 V @ 625 A. The individual IGBTs switches of the H-bridges are controlled by drivers which get the PWM signals - converted to fiber optic for galvanic isolation - from the D41 Modulator.

Voltage Supply

The PSS supplies 5 sets of 2-phase AC voltage inputs of 277V AC over connectors X10 - X50 which are rectified to provide the 400V DC supplies to the H-Bridges.

The presence of the five 400V DC voltages is indicated by five double-LEDs (yellow) close to the corresponding connector X10 - X50 front-side.

The rectifiers are equipped with over- (>470 VDC) and under-voltage (<80 VDC) monitors. Errors are fed to the D41 Modulator over connector X1 for reporting. For over-voltage the "Driver Power Supply Defect" is generated. An under-voltage leads to a "Voltage Feedback Error".

Voltage Feed Back (VFB)

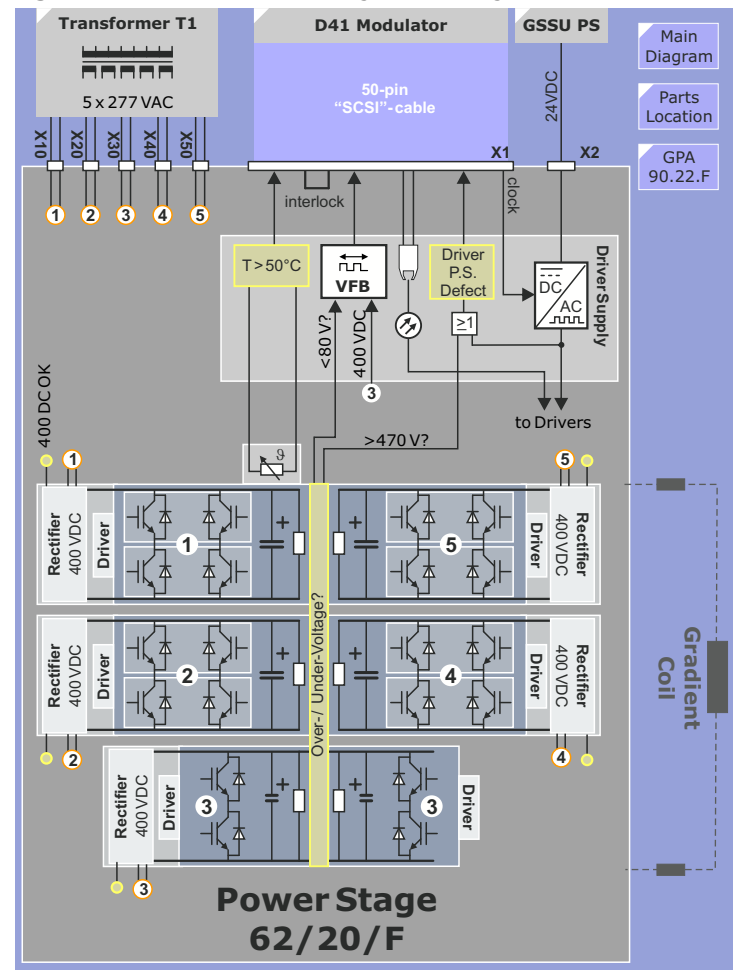
To reduce output current instabilities due to line voltage fluctuations a sample of the 400 VDC supply is processed and returned to the D41 Modulator which can compensate via the PWM.

Driver Voltage Supply and Monitoring

The Drivers get power from a 24 VDC supply from the GSSU Power Supply fed over connector X2 to a galvanically insulated Driver Supply.

If the Driver Supply voltage fails, the "Driver power supply defect" is generated to the D41 Modulator for reporting.

Figure 170 62/20/F Power Stage Block Diagram



The temperature of the water-cooled heat-sink for the IGBT switches is monitored via a thermal resistor. In case the temperature exceeds 50 °C, the D41 Modulator for reporting.

To avoid damage to the Power Stage transistors an interlock is implemented consisting of a jumper between the furthest two pins (pins 1 and 50) of the connector. If not properly connected the jumper connection is opened causing an interrupt and disables the GPA.

LED	Status	Description
V1 (5 yellow LEDs)	On	DC voltage present(>80 V)

[illegible]

GPA 90(75)/22/F - Power Stage (PS)

Overview

Two different types of Power Stages exist:

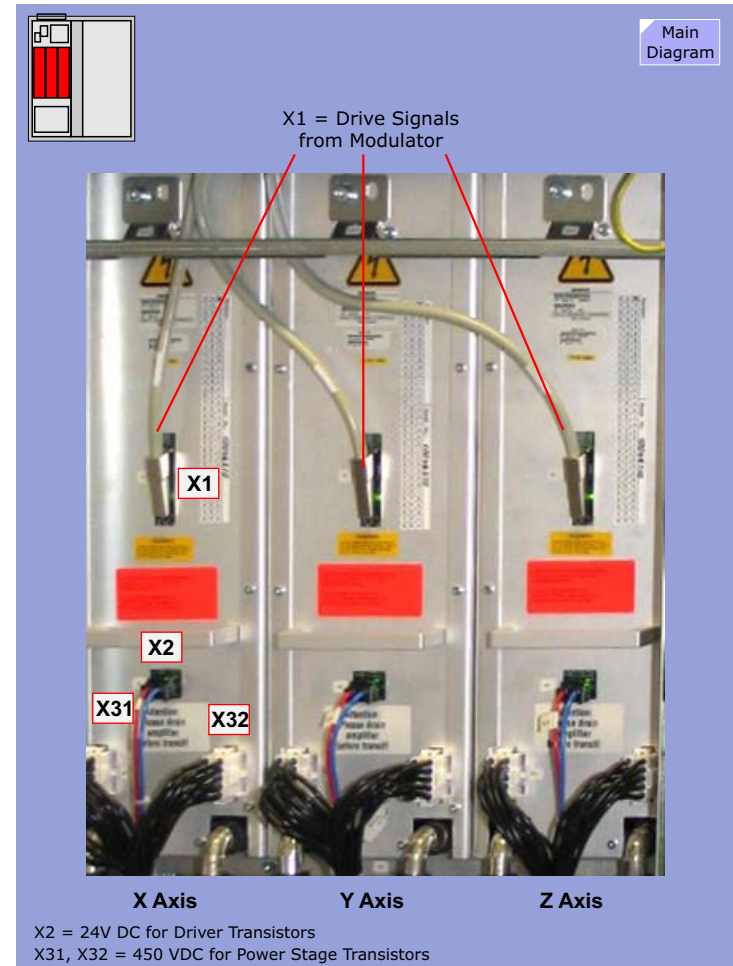
- **GPA 90/22/F** - 60 IGBTs, max. 900 A, 2250 V
- **GPA 75/22/F** - 50 IGBTs, max. 750 A, 2250 V

The following components can be found inside one Power Stage:

- Five water cooled H-bridges (cascades) with 4 IGBT switches each
- Five 450 VDC capacitor blocks
- Integrated driver supply with galvanically insulated switch-mode power supply
- Integrated opto-transmitters and opto-receivers for galvanic insulation of drive- and monitoring-signals
- Auxiliary monitoring components

NOTE Two different types of Power Stages are used for the GPA 90/22/F or GPA 75/22/F. They are not compatible!

Figure 172 90/22/F Power Stage Layout



Function

Gradient Current Generation

The Power Stages consist of five H-bridges connected in series, or cascade. Each bridge is supplied with 450 VDC from the D100 Rectifiers at connectors X31, X32. for a maximum bipolar output of 2250 V @ 900 A (750 A). The individual IGBTs switches of the H-bridges are controlled by the drivers which get the PWM signals - converted to fiber optic for galvanic isolation - from the D41 Modulator.

Voltage Feed Back (VFB)

To reduce output current instabilities due to line voltage fluctuations a sample of the 450 VDC supply is processed and returned to the D41 Modulator which can compensate via the PWM.

Driver Voltage Supply and Monitoring

The Drivers get power from a 24 VDC supply from the GSSU Power Supply fed over connector X2 to a galvanically insulated Driver Supply.

If the Driver Supply voltage fails, the "Driver power supply defect" is generated to the D41 Modulator for reporting.

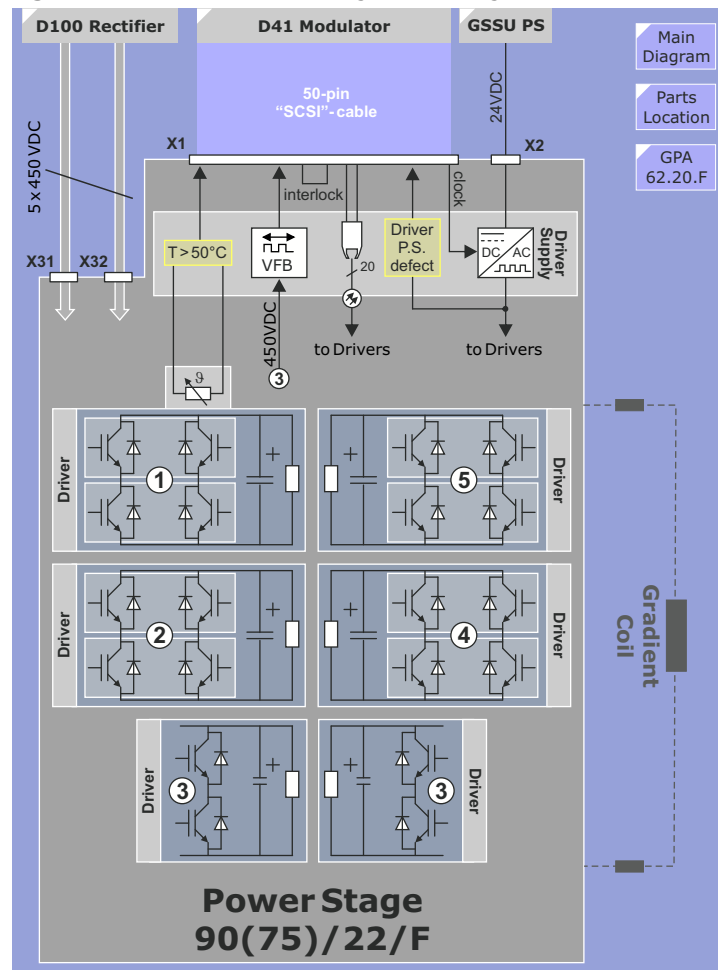
Temperature Monitoring

The temperature of the water-cooled heatsink for the IGBT switches is monitored via a thermal resistor. In case the temperature exceeds 50 °C, the D41 Modulator for reporting.

Cable Pull Protection

To avoid damage to the Power Stage transistors an interlock is implemented consisting of a jumper between the furthest two pins of the connector. If not properly connected the jumper connection is opened causing an interrupt and disables the GPA.

Figure 173 90/22/F Power Stage Block Diagram



GPA 62/20/F - Power Stage Supply (PSS)

Overview

Task

The Power Stage Supply (PSS) is responsible for providing the Power Stages with the supply voltages for each of its 5 cascades and to provide protection.

The PSS is a linear-type power supply consisting of:

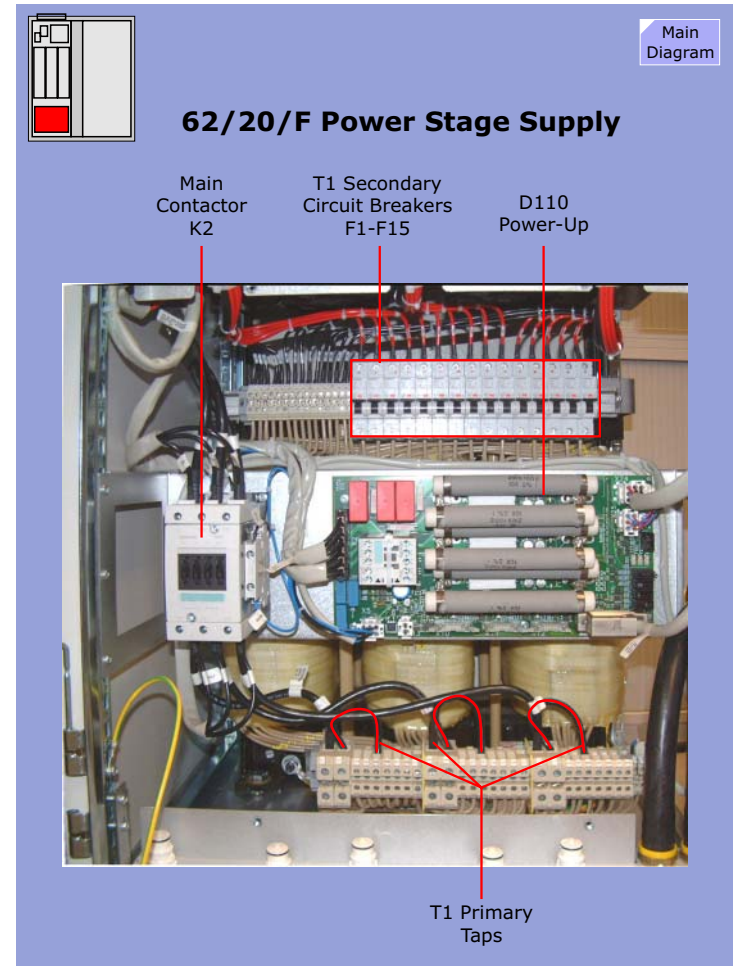
- transformer T1
- Power Up D110 board
- Contactor K2
- the rectifiers are located in the Power Stages

Protection

Due to the high voltage and current demands of the PS the PSS must also provide over-voltage and over-current protection.

Over-current protection is provided by way of **circuit breakers F1-F15** located at the output of the transformer. Over- and under-voltage monitoring is provided by the **D110 Power-Up** board and will signal the GPA controller to disable the **Main Contactor K2** to remove power in the event limits are exceeded.

Figure 174 62/20/F Power Stage Supply Layout



Mains Connections

The mains voltage (L1, L2, L3) is taken directly from the customer mains and fed over a line filter and a 80 A circuit breaker **F1** to transformer **T1**. Its primary tapings need to be adjusted to the local mains voltage. The presence of the mains voltage is indicated by LEDs **V301, V302, V303** (yellow) located on the D110 Power-UP board (description below).

To avoid extreme current peaks during the initial power-on sequence, the mains voltage is switched to transformer T1 via **K2** and **D110 Power-UP** board for a "soft start". Details are given in the description of the D110 Power-UP board.

The T1 secondary provides fifteen 277 VAC single phase outputs over 25 A circuit breakers **F1-F15** to the Power Stages. The rectifiers are built into each Power Stage and generate 5 x 400 VDC supply voltages for the cascades. Yellow LEDs, located near connector X10 - X50, are used to indicate the presence of the AC input voltages.

WARNING	The GPA is equipped with a dual power supply. Risk of electric shock! Switch off both circuit breakers F1 in the GPA and F21 in the EPC before working on the GPA. Remember, there is still power at the line filter and circuit breaker input terminals!
----------------	--

D110 Power-UP

The D110 is responsible for the following main functions:

- **Power switching** - main transformer T1 soft start
- **Overload detection** - for current limiting resistors
- **Temperature monitoring**

Function

GPA Power-on

The power switching is initiated by the **N_POW_ON_CAN** signal from the D68 or with push button **S1** on the D70.

The D41 first performs a test to check for a "Power Stage cable defect" (see D41 description). If successful, the **N_POWER_ON** signal is set and a "Transformer soft start" is performed.

Transformer Soft Start

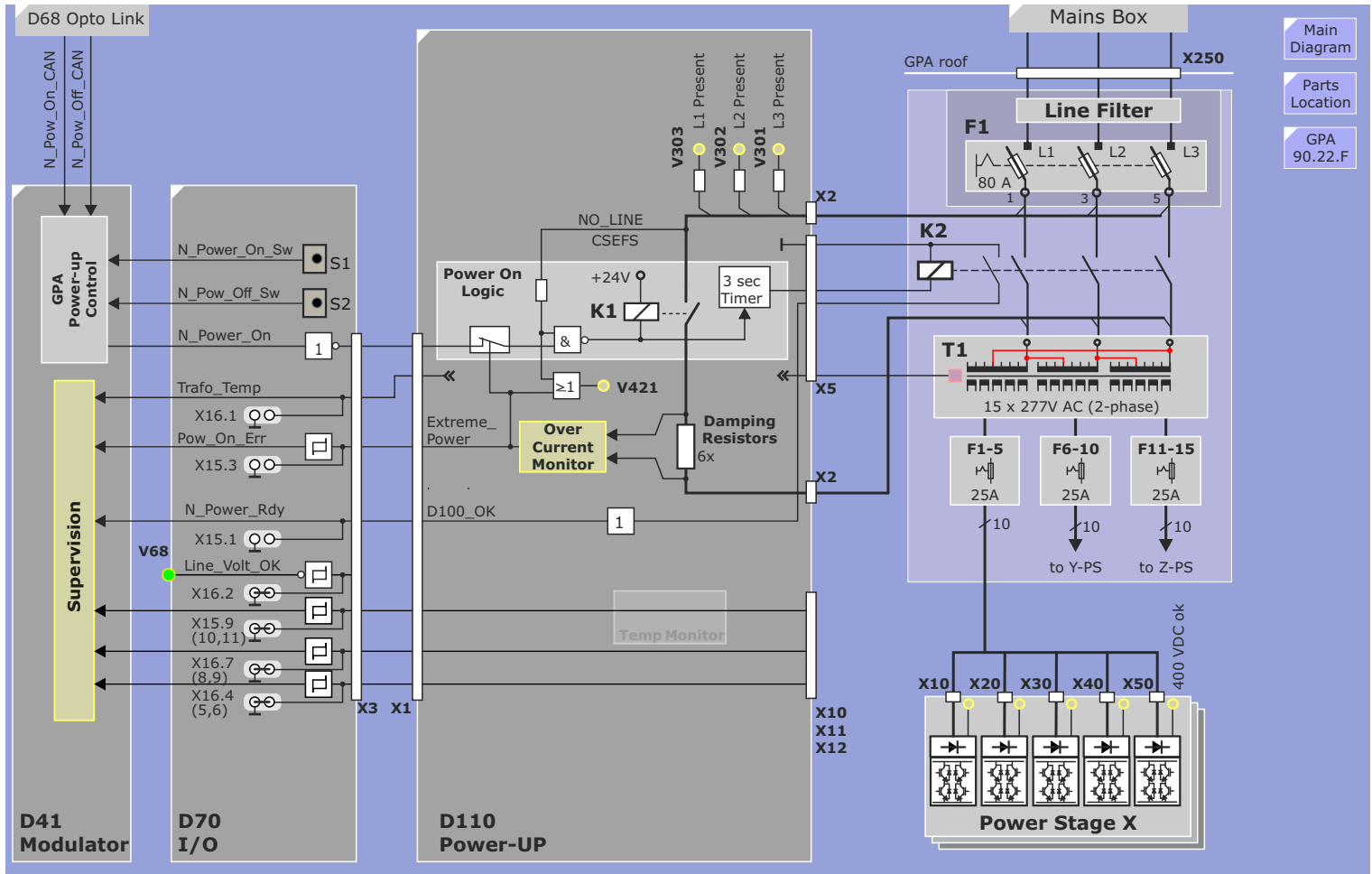
The Power-on logic energizes contactor **K1**, feeding the mains voltage to the transformer via damping resistors. After a delay of 3 seconds (and no errors present) contactor **K2** is enabled to establish a direct connection to the mains supply. K1 will be disabled as soon as K2 is engaged, to avoid potential overload conditions at the current limiting resistors.

An auxiliary contact on K2 is used to inform the modulator that the power-up was successful (**N_Power_Rdy**). All modulators and regulators are now enabled.

Overload Detection

If a line voltage is missing or a fault occurs, relays K1 and K2 will not be activated, or deactivated if already on. During power-up, the damping resistors are sensed by the over current monitor. In the event of an overload, the K1 control voltage is interrupted. The **Extreme_Power** error is generated and distributed to the D41 Modulator. The fault is also indicated by the yellow LED **V421**.

Figure 175 62/20/F Power Stage Supply Block Diagram



GPA Power-off

The GPA can be switched off by the D68 (**N_POW_OFF_CAN** signal). Push button **S2** on the D70 board can also be used (**N_POW_OFF_SW**). Both signals are fed to the D41 causing the modulators and Power Stage low voltage monitoring to be disabled and the **N_POWER_ON** signal is removed to disable relays K1 and K2.

Temperature Monitoring

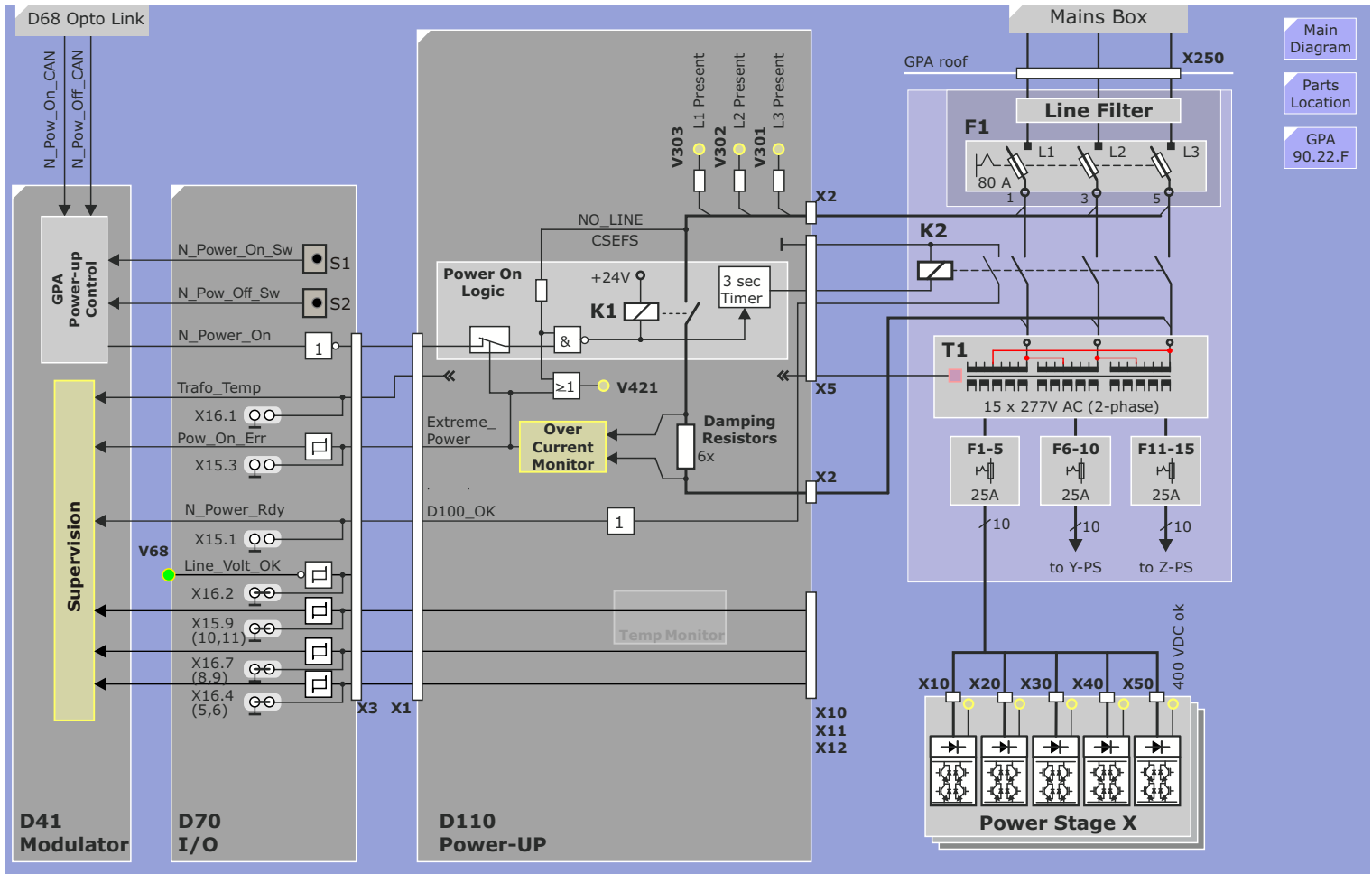
The D110 monitors the following temperatures:

- **Transformer T1** : is protected with an over-temperature switch that opens at 180° C (**TRAFO_TEMP**). The signal is sent to the D41 Modulator via the D70 I/O.
- **Chokes** : The output filter chokes of the Power Stages are equipped with temperature switches. In case the choke temperature is too high, they will open and the **FI_TEMP_X(Y, Z)** signal is fed to the D41 via the D70.

LEDs

LED	Status	Description
V303 (yellow)	on	line voltage present (phase L1)
V302 (yellow)	on	line voltage present (phase L2)
V301 (yellow)	on	line voltage present (phase L3)
V421 (yellow)	on	EXTREME_POWER (= charging resistor overload) or NO_LINE fault (= line voltage missing)

Figure 176 62/20/F Power Stage Supply Block Diagram



GPA 90(75)/22/F

- Power Stage Supply (PSS)

Overview

Task

The Power Stage Supply (PSS) is responsible for providing the Power Stages with the supply voltages for each of its 5 cascades and to provide protection.

The PSS is a linear-type power supply consisting of:

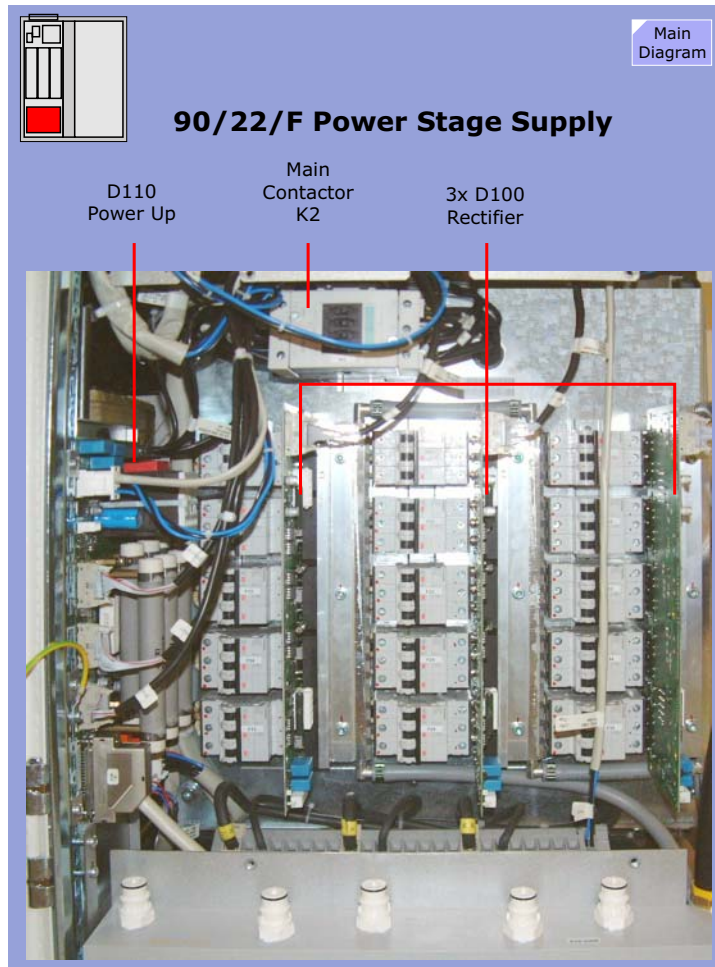
- transformer T1
- Power Up D110 board
- Contactor K2
- D100 Rectifier boards

Protection

Due to the high voltage and current demands of the PS the PSS must also provide over-voltage and over-current protection.

Over-current protection is provided by way of **circuit breakers F11-F35** located at the output of the transformer. Over- and under-voltage monitoring is provided by the **D110 Power-Up** board and will signal the GPA controller to disable the **Main Contactor K2** to remove power in the event limits are exceeded.

Figure 177 90/22/F Power Stage Supply Layout



Mains Connections

The mains voltage (L1, L2, L3) is taken directly from the customer mains and fed over a line filter and a 100 A circuit breaker **F1** to transformer **T1**. T1 primary tapplings need to be adjusted to the local mains voltage. The presence of the mains voltage is indicated by LEDs **V301**, **V302**, **V303** (yellow) on the D110 Power-UP board. (description below)

To avoid extreme current peaks during the initial power-on sequence, the mains voltage is switched to transformer T1 via **K2** and **D110 Power-UP** board for a "soft start". Details are given in the description of the D110 Power-UP board.

The T1 secondary provides fifteen 319 VAC 3-phase outputs over 32 A circuit breakers **F11-F35** to the three D100 Rectifier boards. After rectification, each D100 provides five 450 VDC outputs for the cascades in the Power Stages.

WARNING **The GPA is equipped with a dual power supply.**
Risk of electric shock!
Switch off both circuit breakers F1 in the GPA and F21 in the EPC before working on the GPA. Remember, there is still power at the line filter and circuit breaker input terminals!

D100 Rectifier

Function

Voltage Generation

Each of the D100 Rectifier boards is equipped with five 3-phase bridge rectifiers. The rectifiers have as input the 319 VAC secondary voltages of transformer T1. It outputs five independent 450 VDC supplies for each of the 5 H-bridges in the Power Stages.

Rectifier Voltage Monitoring

The incoming secondary AC voltages from transformer T1 are monitored. When all are in specification (319 VAC $\pm 10\%$), LED **V606** is lit but that status is not monitored at all.

The rectifier output DC voltages are monitored for over-(>515 VDC) and under voltage (<80 VDC). To indicate the presence of the DC voltage, five yellow LEDs (**V105**, **V205**, **V305**, **V405**, **V505**) light up as long as the voltage level is higher than 80 V. The corresponding error signals **O_VV_X(Y, Z)** and **N_VV_X(Y, Z)** are routed via the D70 to the D41 Modulator board.

Rectifier Temperature Sensor

The rectifiers are mounted on a water-cooled heatsink which is monitored using a PTC sensor. The sensor output is evaluated on the **D110 Power-UP** board. Should the heatsink temperature reach 60°C an error occurs.

LEDs

LED	State	Description
V606 (green)	ON	all secondary voltages are ok (319 V _{eff} +/- 10 %)
V105 - V505 (yellow)	ON	DC voltage present(>80 V)

D110 Power-UP

The D110 is responsible for the following main functions:

- **Power switching** - main transformer T1 soft start
- **Overload detection** - for current limiting resistors
- **Temperature monitoring**

Function

GPA Power-on

The power switching is initiated by the **N_POW_ON_CAN** signal from the D68 or with push button **S1** on the D70.

The D41 first performs a test to check for a "Power Stage cable defect" (see D41 description). If successful, the **N_POWER_ON** signal is set and a "soft start" is performed.

Transformer Soft Start

The Power-on logic energizes contactor **K1**, feeding the mains voltage to the transformer via damping resistors. After a delay of 3 seconds (and no errors present) contactor **K2** is enabled to establish a direct connection to the mains supply. K1 will be disabled as soon as K2 is engaged, to avoid potential overload conditions at the current limiting resistors.

An auxiliary contact on K2 is used to inform the modulator that the power-up was successful (**N_Power_Rdy**). All modulators and regulators are now enabled and the D41 enables its monitoring of the **N_VV_X(Y, Z)** signals.

Overload Detection

If a line voltage is missing or a fault occurs, relays K1 and K2 will not be activated, or deactivated if already on. During power-up, the damping resistors are sensed by the over current monitor. In the event of an overload, the K1 control voltage is interrupted. The "**Extreme_Power**" error is generated and distributed to the D41 Modulator. The fault is also indicated by the yellow LED **V421**.

GPA Power-off

The GPA can be switched off by the D68 (**N_POW_OFF_CAN** signal). Push button **S2** on the D70 board can also be used (**N_POW_OFF_SW**). Both signals are fed to the D41 causing the modulators and Power Stage low voltage monitoring to be disabled and the **N_POWER_ON** signal is removed to disable relays K1 and K2.

Temperature Monitoring

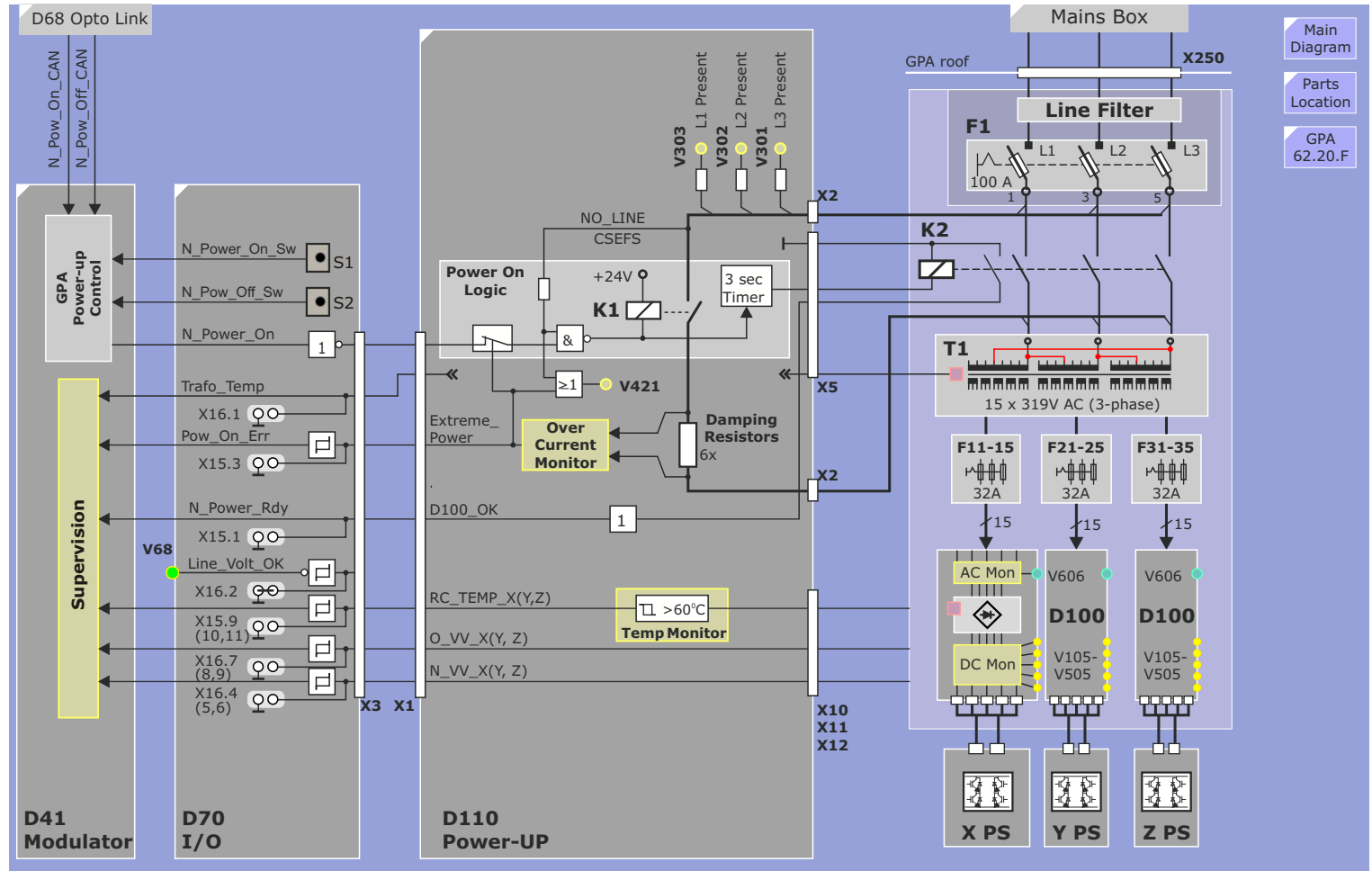
The D110 monitors the following temperatures:

- **Transformer T1** : is protected with an over-temperature switch that opens at 180° C (TRAFO_TEMP). The signal is sent to the D41 Modulator via the D70 I/O.
- **D100 Rectifier** : A PTC placed on each rectifier heat-sink monitors the temperature. When exceeding 60° C an error is reported to the D41 via D70.
- **Chokes** : The output filter chokes of the Power Stages are equipped with temperature switches. In case the choke temperature is too high, they will open and the FI_TEMP_X(Y, Z) signal is fed to the D41 via the D70.

LEDs

LED	Status	Description
V303 (yellow)	on	line voltage present (phase L1)
V302 (yellow)	on	line voltage present (phase L2)
V301 (yellow)	on	line voltage present (phase L3)
V421 (yellow)	on	EXTREME_POWER (= charging resistor overload) or NO_LINE fault (= line voltage missing)

Figure 178 90/22/F Power Stage Supply Block Diagram



GSSU Power Supply

Function

The GSSU Power Supply provides stabilized DC-voltages (+5 V and ± 15 V) for the GSSU boards, one 24 VDC for the drivers inside the Power Stage modules and 24 VDC for the relays and fans in the Power Stage Supply section.

Test Points

All stabilized DC voltages are adjustable with potentiometers on the respective power supply slot.

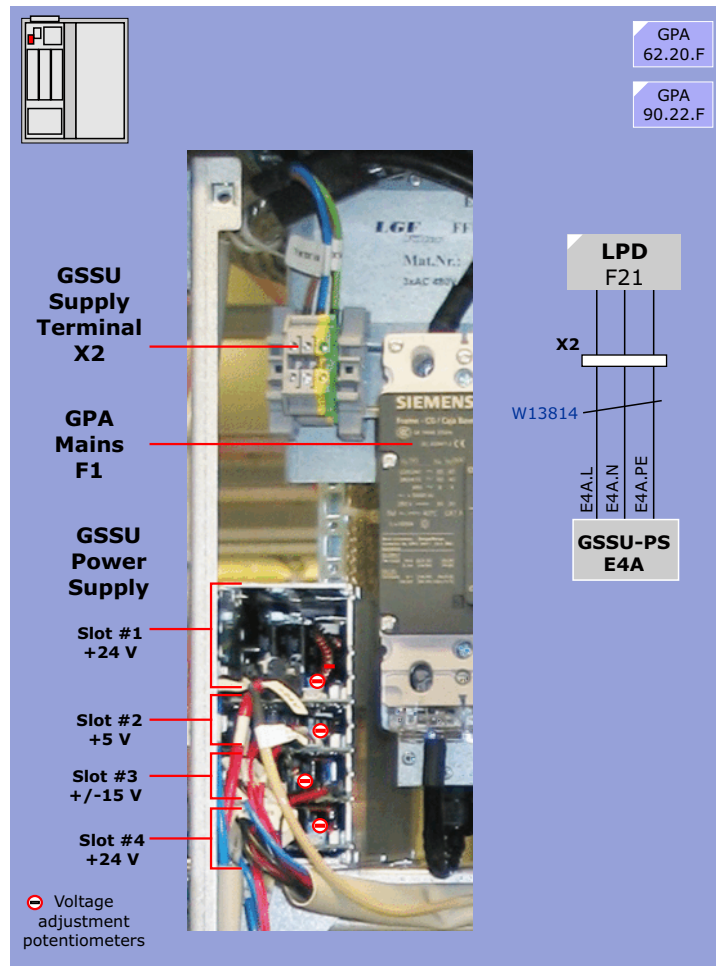
Slot #	Voltage
1	24.0 VDC $\pm$ 100 mV (driver supply) for Avanto fit: +17 VDC
2	5.075 VDC $\pm$ 25 mV (GSSU boards)
3	+/-15.05 VDC $\pm$ 25 mV (GSSU boards)
4	24.0 VDC $\pm$ 500 mV (Power Stage relays and fans)

NOTE Voltages are measured with disabled Modulators (S3 left/ right and S4 left at D70 set to upward position).

WARNING Risk of electrical shock! The GSSU Power Supply is connected to mains by circuit breaker F21 in the EPC cabinet! Switch off F21 EPC, before working on the GSSU Power Supply!

WARNING For Avanto fit and Skyra fit F21 in the EPC WILL NOT power off the GSSU. Instead use the F2 circuit braker found in the GPA, at same location as the Supply Terminal X2. Check CB-doc for additional details.

Figure 179 GSSU Power Supply



Current Converter Unit (CCU)

Overview

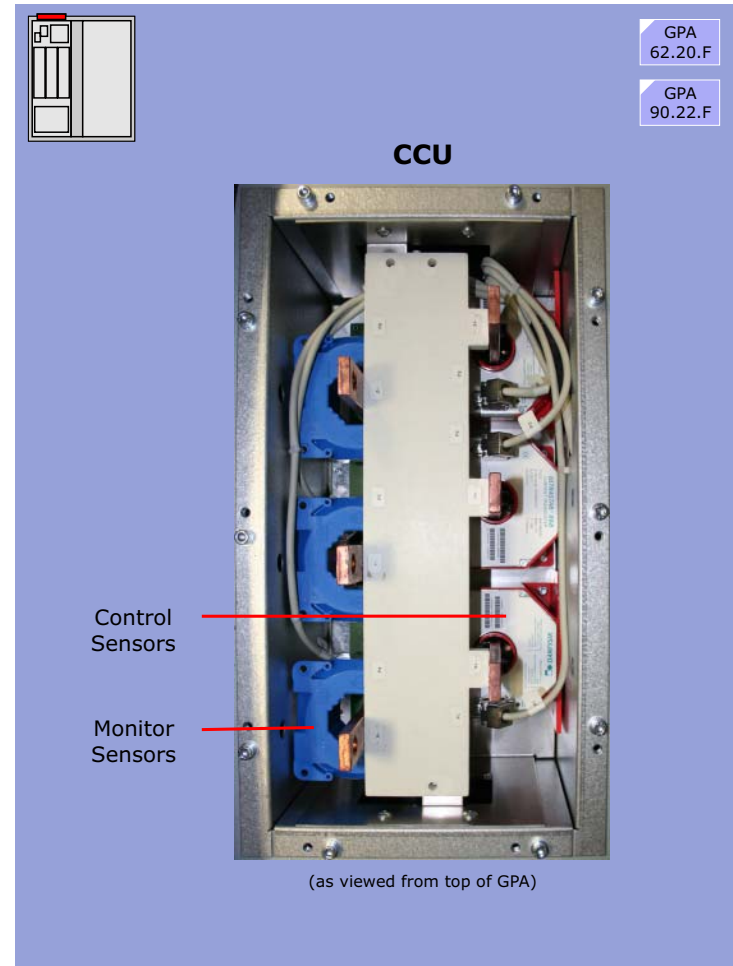
The Current Converter Unit (CCU) consists of six current sensors, three Control Sensors and three Monitor Sensors, a set for each of the three gradient coils. The **Control Sensors** are used to supply precise actual current values for the regulation. The gradient coils are floating so a second set of sensors, the **Monitor Sensors**, are required to verify no current is leaking to ground or to another gradient coil or a fault of a Control Sensor. This is done by comparing the outputs of each sensor pair.

If the current values of either sensor differs by a certain level the cabinet will be immediately disabled to prevent severe damage.

Function

See the [Actual Value Amplifier](#) section of the D24 Regulator description for a functional description of the current sensors.

Figure 180 Current Converter Unit (CCU)



Output Filter Chokes

Function

Six Output Filters, consisting of series inductive chokes and parallel capacitances, filter out switching ripple on the gradient current. The chokes are located in a rack (Choke-Set) behind the three Power Stages. The capacitors are found on the underside of the CCU.

For accessing the chokes, the Power Stages have to be removed.

Over Temperature Protection

Each choke is protected by a temperature switch that opens in case of over temperature. A corresponding **FI_TEMPX(Y, Z)** signal is generated and fed via D110 and D70 to the D41 Modulator-board.

FAN Box

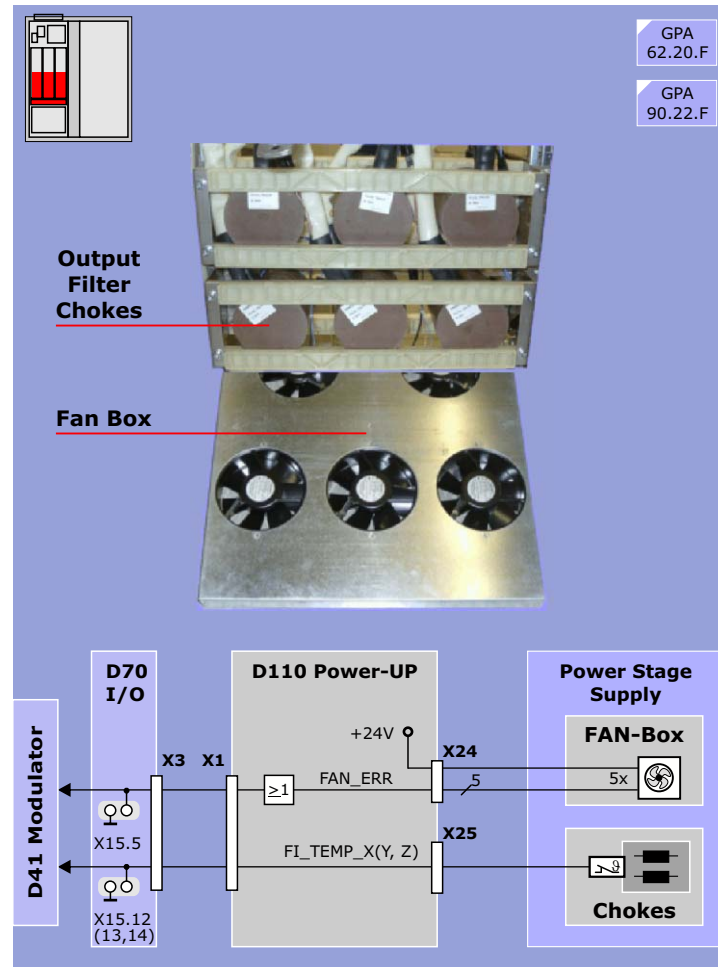
Function

For cooling the Power Stages and Output Filter-chokes, five 24 VDC fans are located in a box underneath.

Fan Monitoring

A feedback voltage from the five fans of the Power Stage and choke-section is used to check the proper function. If one of the signals fails, a **FAN_ERR** (low = ok) is reported via D70 to the D41 Modulator board.

Figure 181 Output Filter-Chokes and Fan Box



Gradient Coil (GC)

Overview

The Gradient Coil assembly consists of the active-shielded water-cooled Gradient Coil, five integrated high-order shim coils (optional) for active shimming and pockets for shim trays for placing shim-iron plates for the passive shim. All electrical connections are at the rear side of the magnet.

Coil Variants

For MAGNETOM Aera / Skyra three coil variants exist:

Coil	Description
Aera XJ GC 1.5 XJ	Connections: bolts with securing mechanisms E-shim Coils: NOT AVAILABLE
Aera XQ GC 1.5 XQ	Connections: heavy duty coaxial gradient connectors E-shim Coils: STANDARD
Skyra GC 3.0	Connections: heavy duty coaxial gradient connectors E-shim Coils: STANDARD

Function - Electrical

Coil Temperature Monitoring

12 NTCs are strategically placed at the gradient coil to monitor its temperature. Three of them are located close to the gradient coil power connectors to avoid major damage there.

All NTCs are connected to the **RFCEL_SAT** via cable W15740. A connector interlock (jumper in GC) makes sure the temperature monitoring connector is correctly plugged in. From RFCEL_SAT, the actual temperatures are reported to **RFCEL_COM** (cable W15724).

Coil over-temperature

In case of GC over-temperature (**> 85 °C**), the RFCEL_COM sets a

bit at the ISO-Ring which informs the GraCon to disable GPA. As a consequence, the GraCon informs the MARS 2 via the optical PCIe bus and the sequence is aborted.

NTC sensor test

All NTCs are tested **during scanner boot**. Evaluation logic determines whether a failed sensor will permit continued operation of the scanner.

A second step performs a **test of the switch-off circuit**. The temperature nominal limit is set to 10 °C and the temperature monitoring should disable GPA and scanner correctly. If the **test fails, an error message is generated** and the **scanner is blocked** from further measurements.

Function - Mechanical

Gradient Connectors

The GC 1.5 XQ and GC 3.0 gradient coils use high currents leading to substantial mechanical forces and heat. For this reason heavy-duty high-current coaxial connectors are used. They must be handled properly when installing and require attention during maintenance periods. A WBT has been especially prepared and is mandatory for CSEs to become indoctrinated on Gradient Coil connections. Also refer to the CB-DOC on the handling and care of gradient coil connections to avoid damage.

Coil Suspension

The gradient coil is suspended with three foam cushions at the bottom and four inflatable air cushions at the top. The cushions are inflated 0.5 bar for transport and deflated during the installation.

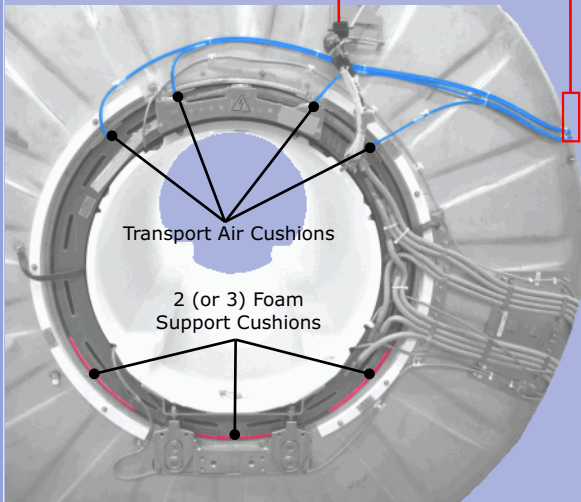
The first systems had manometers used to adjust the desired transport pressure. The manometers are no longer implemented on newer systems (mid 2013).

Figure 182 Gradient Coil and Suspension)

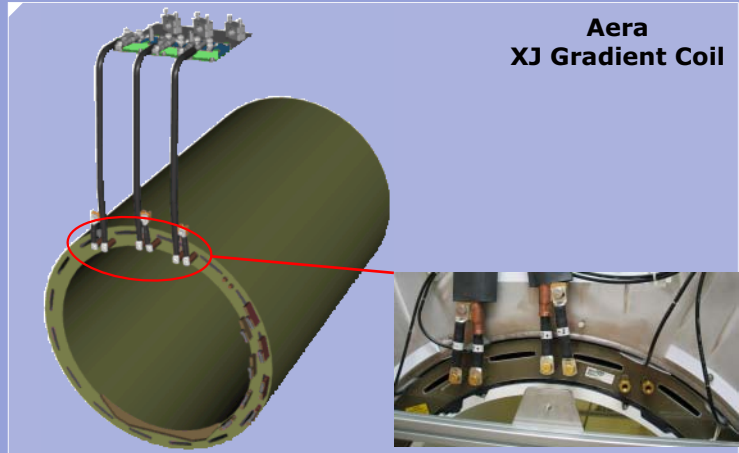
The manometers show the air pressure of the transport air cushions. Once the system is installed the air cushions are deflated and the manometers have no further function.



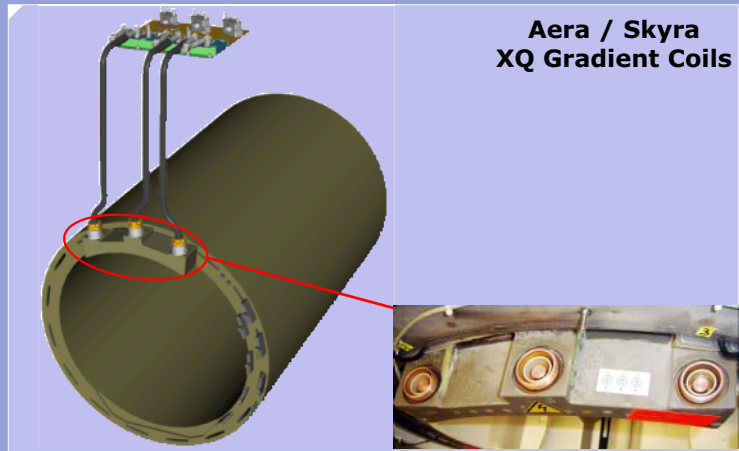
Shim Coil Connector



**Aera
XJ Gradient Coil**



**Aera / Skyra
XQ Gradient Coils**



Gradient Supervision

Overview

MAGNETOM Aera/Skyra systems are equipped with an externally mounted Gradient Supervision unit which prevents major damage to the MR-site in case the gradient connections should overheat in a fault situation (e.g. bad contacts). For that purpose a VESDA™ (Very Early Smoke Detection Apparatus) unit is used. The version manufactured for SIEMENS is called **SIEMENS_VLM**.

Function

Monitoring Principle

The SIEMENS_VLM incorporates a very sensitive, laser-based smoke detection unit which can detect overheated connections in a very early phase. A little pump continuously draws air through a hose system which connects to the relevant critical gradient connections which can overheat in a fault scenario.

The aspirated air samples are fed through a filter into the laser detection chamber inside the unit. Removal of dust and dirt from the air sample is quite important to keep the detector's optical surfaces free from contamination ensuring stable calibration and a long detector life as well as minimizing false alarms.

Inside the calibrated detection chamber the air sample is exposed to a laser light source. If the sample contains small particles (e.g. emitted by overheated gradient connections), light is scattered within the detection chamber and is immediately recognized by the highly sensitive optical receiver system.

Monitored Connections and Components

- EPC cabinet
- Filter panel gradient filters and connections
- Gradient connections at the magnet

Figure 183 Gradient Supervision SIEMENS_VLM

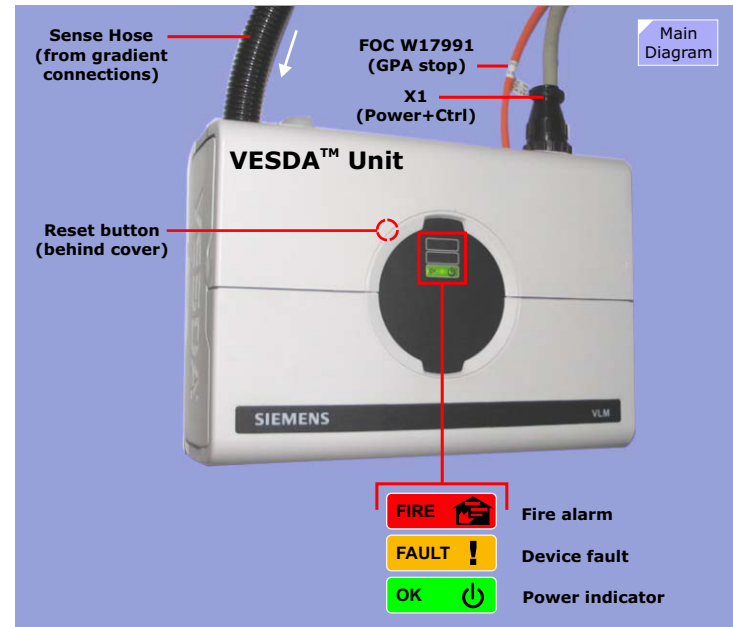
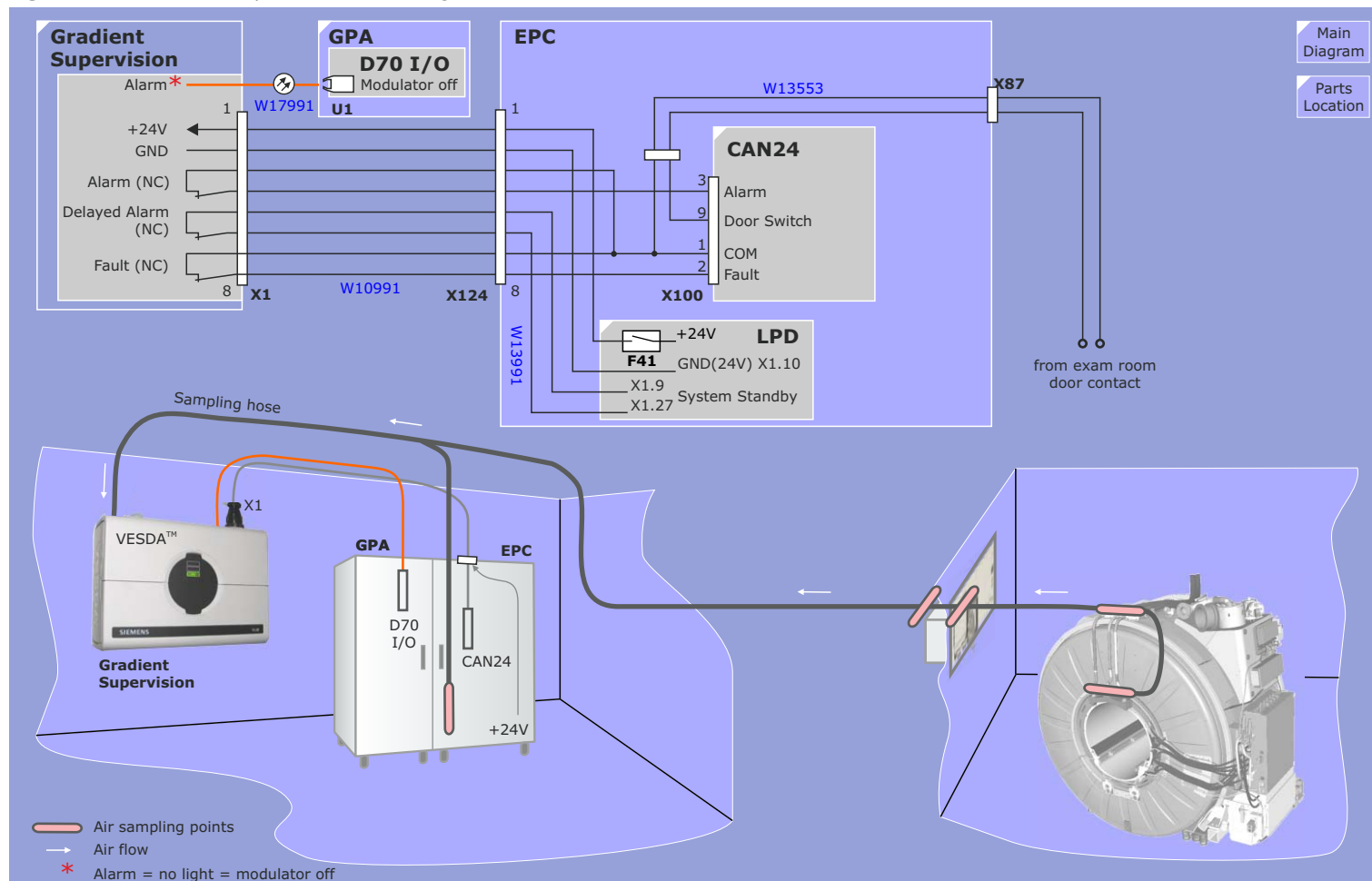


Figure 184 Gradient Supervision Block Diagram



Power Supply

The unit is connected via its plug X1 to +24 VDC supplied by the EPC cabinet (connector X124 at EPC roof). The supply voltage can be switched with circuit breaker F41 at the EPC.

Device Calibration

Warm-up

After an initial power-up, the yellow FAULT LED lights-up for 5 seconds and the SIEMENS_VLM runs through a 90 sec. warm-up cycle to function correctly.

Reference air-flow calibration

After initial installation of the SIEMENS_VLM its reference air flow nominal value needs to be calibrated one time.

NOTE For correct calibration, the sample hose system must be fully attached. Also, close RF-room door to set standard air pressure condition in the RF-cabin.

To start the calibration:

- The unit must be powered-up for more than 90 seconds (= completed warm-up)
- press the units' RESET button under the cover for 10 seconds until all 3 LEDs start flashing
- Release RESET button, all LEDs go off
- LEDs flash again, press RESET once more to confirm setting

The current air flow through the device (which depends on the individual system configuration and hose length) is now **set to 100%**.

Deviations larger than +/-20 % from the 100% nominal air flow (e.g. a blocked or disconnected sample hose) will result in a device fault.

Device Fault

Fault LED

In case of a malfunction the units **yellow FAULT LED** will be set.

In fault mode, the supervisions' sensitivity is potentially affected, but the operation of the MR-scanner is still possible. A constant light indicates a major fault, a flashing LED a minor problem.

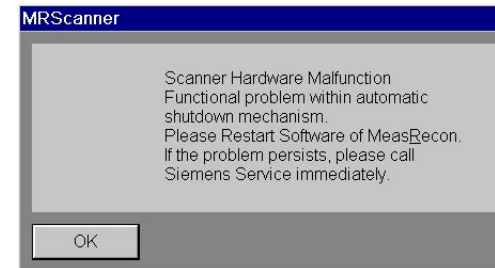
The following errors can cause a device fault:

- **Wrong air flow** - more than **+/-30 % deviation** (+/-20% for older systems, check SKB0078807) to the calibrated 100% reference air-flow value.
- **Excessive differential pressure** - between RF-cabin and equipment room. A pressure equalization tube must be fitted from the units exhaust back into the RF-cabin

Device Fault pop-up

The customer is informed with a pop-up window.

Figure 185 Device Fault pop-up



Device Fault Reset

A Device Fault will be **automatically reset** at the SIEMENS_VLM when the fault condition is not present anymore.

Fire Alarm

FIRE LED

In case smoke gases are detected within the monitored areas, the SIEMENS_VLM sets the **red FIRE LED**.

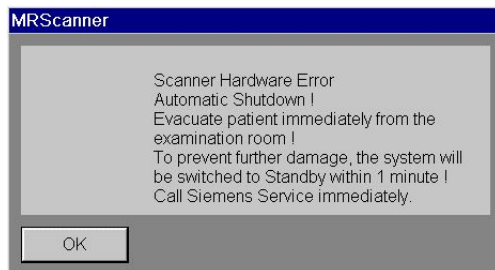
Figure 186 Fire Alarm LED



Alarm pop-up

The customer is informed about a fire alarm with a pop-up window.

Figure 187 Fire Alarm pop-up



Measurement abort

The MR-system is informed about the critical condition via the connection to the CAN24 (cable W10991). Consequently, a running measurement is aborted and the modulators of the GPA are blocked (via FOC W17991).

Power switch off

After **one minute** the MR-system is **switched to Standby** (via X1 to X124 EPC roof).

Fire Alarm Reset

A Fire Alarm condition is latched by the SIEMENS_VLM to inhibit further system power switch-on attempts.

A CSE can reset the fault by pressing the units' **RESET button**, to be found under the units' cover left to the indicator LEDs.

WARNING After a Fire Alarm event the system can not be released back to the customer for clinical use until a detailed check of the system is performed to identify its cause. Check CB-doc Troubleshooting guide -> Gradient Supervision for the necessary actions.

Jumpers

Jumper	Pins	Setting	Location	Description
Config	1-2	closed	under the units' cover, left of the FOC connector	Setting for Aera/Skyra: no alarm = FOC light on alarm = FOC light off

Switches

Switch	Location	Description
RESET	under the units' cover, left of the control LEDs	- Resets latched faults - Set-up button for device calibration

LEDs

LED	ON	FLASHING	Description
OK (green)	Power on	n.a.	Detector is powered up and in normal operation
FAULT (yellow)	Major Fault	Minor Fault	- Calibration of airflow required - Call service
FIRE (red)	Smoke detected	n.a.	- Alarm pop-up generated - GPA modulator will be disabled - MR-scanner will be switched off to standby 1 minute later

Service Procedures

The function of the Gradient Supervision must be checked every 12 month during maintenance.

NOTE For details please refer to the Safety-Related Tests Procedure in the CB-Doc.

Magnet System

Introduction

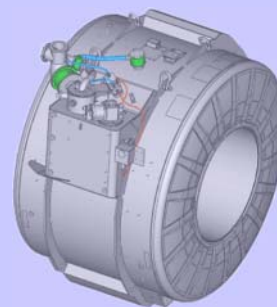
The magnets for the MAGNETOM Aera and MAGNETOM Skyra continue the successful trend of 70cm open-bore high-field systems while reducing even further the physical dimensions of previous magnets.

- **OR98 - 1.5 Tesla Aera magnet:** An ultra-short open-bore magnet only 137 cm long while maintaining a homogeneous volume of 50 x 50 x 45 cm (X,Y,Z). The covers add another 8 cm to the length, however their wide flared openings do not diminish the open design.
- **OR99 or OR103 - 3 Tesla Skyra magnet:** Similar in design to the OR98 but as a 3T magnet only marginally longer (163 cm) with the same elliptical homogenous volume of 50x50x45cm. The OR103 is a redesign of the OR99 magnet. It can be found on newer Skyra systems.

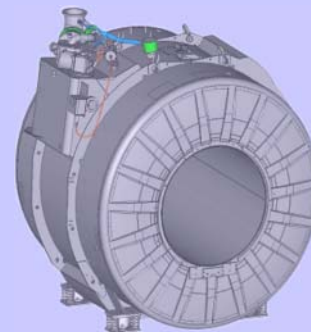
Both magnets are equipped with active shielding and EIS screen coils, FCLs, 4K cold heads, easy-access service ports. Field shimming to the environment is performed using passive iron shimming and the well known Magnet Power Supply 3600.

A combined side sock turret design includes both the service turret and the 4K Cold Head. The 4K recondensing Cold Head allows a "no-fill" requirement over the 10 year life-cycle period - making it a "zero helium" magnet under normal operating conditions.

A newly designed MSUP (same on all magnets) allows more interactive communication and improved diagnostic capabilities. The MSUP now includes the ERDU test load function as well as visual helium and temperature checks.



OR98
1.5 Tesla
Aera



OR99/103
3 Tesla
Skyra

The Magnet System for the MAGNETOM Aera / Skyra incorporates the newest features in magnet technology and include the following components:

- **OR98 (1.5 T) Magnet**

The operating field is 1.494 Tesla requiring a current of 511 amps (typical). The magnet generates the main B_0 magnetic field, producing a stray field of 0.5 mT at a distance of 4.0 m axially (through the bore) and 2.5 m in the radial planes.

The magnet is passively shimmed using a combination of Z2 iron ring-segments and iron shim plates. The Z2 ring-segments are fitted externally to each end of the magnet. The iron shimming is performed during the installation or, if necessary, after a quench.

The magnet homogeneity is calculated on a 50 x 50 x 45 cm DEV (Diameter Elliptical Volume X, Y, Z) on the iso-center within the bore of the magnet.

- **OR99/103 (3 T) Magnet**

The operating field is 2.894 Tesla at a typical operating current of 467 amps for the OR99 and 488 amps for the OR103. The magnet generates the main B_0 magnetic field, producing a stray field of 0.5 mT at a distance of 4.6 m axially (through the bore) and 2.6 m in the radial planes.

The magnet is passively shimmed using a combination of Z2 iron ring-segments and shim plates. The Z2 ring-segments are fitted externally to each end of the magnet. The iron shimming is performed during the installation or, if necessary, after a quench.

The magnet homogeneity is calculated on a 50 x 50 x 45 cm DEV (Diameter Elliptical Volume X, Y, Z).

- **Magnet Monitoring**

The MSUP is an intelligent unit which Monitors the magnet and refrigerator system functions and regulates the magnet pressure. Magnet pressure regulation is necessary with a Cold Head system capable of re-liquefying helium. This is the basis of the 0% helium loss system.

- **Alarm Box**

Provides magnet monitoring warning and alarm indicators for the system operator as well as for switching the system on and off, and a magnet STOP button for de-energizing the magnet in emergency situations.

- **Advanced High Order Shim**

Allows accurate shimming of local imaging volumes using a 5-channel second order (the largest) electrical shim system. The active shim is an option for 1.5 T and standard for 3 T systems. It consists of a high-precision, high stability shim power supply and the shim coils which are integrated into the gradient coil.

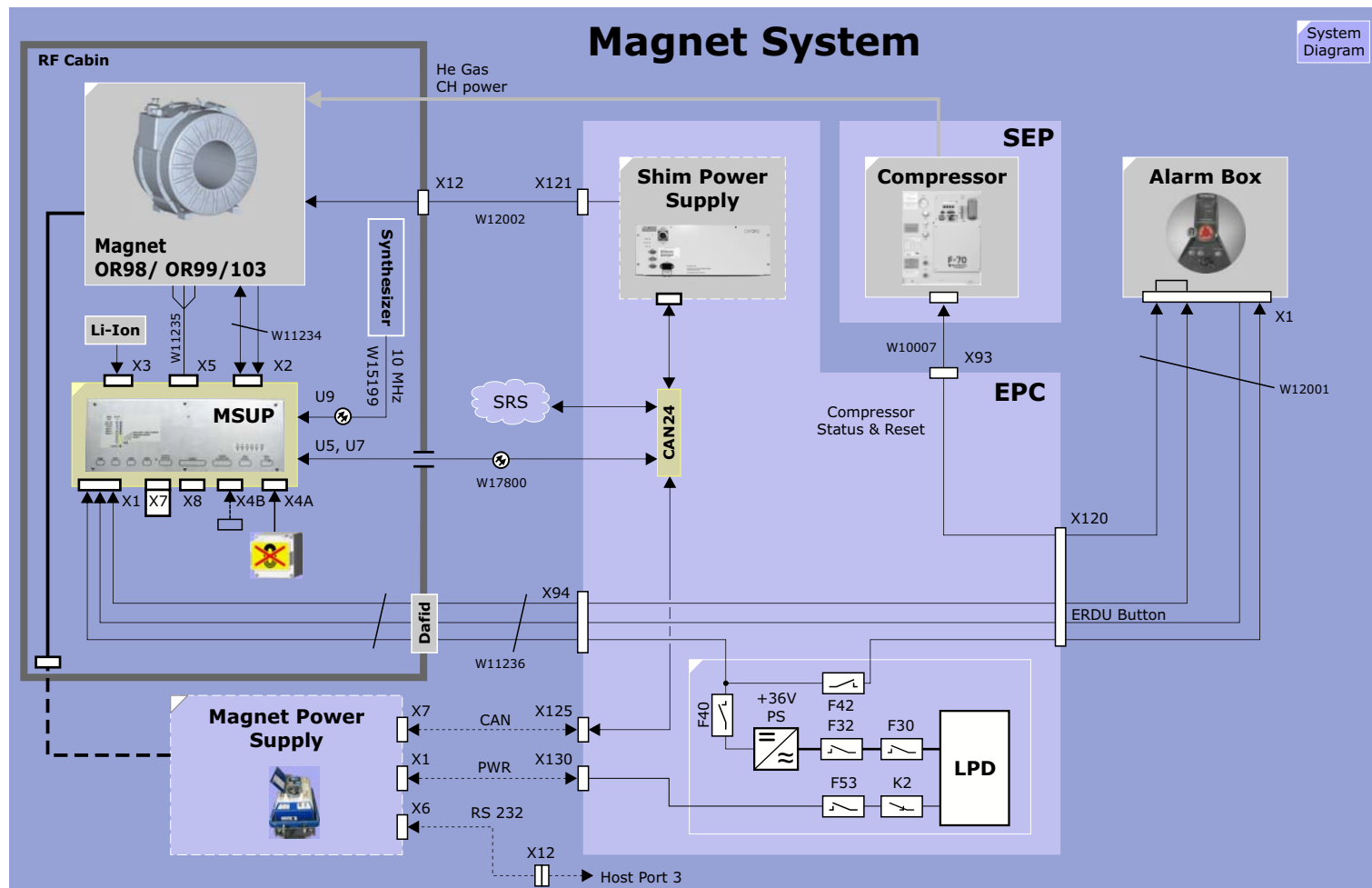
- **Magnet Power Supply (MPS 3600)**

A mobile service tool for energizing and de-energizing the magnet. It is not part of the system delivery.

- **Refrigerator System**

A SUMITOMO F70 He-Compressor and Cold Head unit provides cooling of the magnet's thermal shield and liquefaction of the helium gas.

Figure 188 Magnet System Block Diagram



OR98, OR99/103 Magnet

Overview

The OR98, OR99/103 magnets are actively shielded superconductive magnet assemblies sealed into an all welded stainless-steel vessel which is filled with liquid helium. This liquid helium vessel containing the magnet coils is surrounded by an actively cooled aluminium cyro-shield and suspended in a vacuumed outer steel vessel providing thermal isolation in order to reduce helium boil-off. Its second stage recondenses the boiled-off helium gas, hence a zero boil-off system!

The cryogenic and control circuitry access is provided by an integrated turret. A fixed current lead, made of stainless steel allows ramping of the magnet without opening the cryostat.

Specifications

Parameter	OR98	OR99/103*
Magnet Dimensions		
Height: (with vent elbow)	226 cm	221 cm
Installation height: (without vent elbow)	214.3 cm	214.3 cm
Width	205.4 cm	205.4 ± 0.5 cm
Length	137 cm	163 cm
Magnet inner bore diameter	90 cm	90 cm
Magnet Weights		
Without liquid helium	≤ 3040 kg	≤ 5770 kg (≤ 5350 kg)
With liquid helium (97 % level)	≤ 3200 kg	≤ 5920 kg (≤ 5500 kg)
Magnet Operating Parameters		
Nominal field strength	1.494 Tesla	2.894 Tesla
Operating current at nominal field	511 A	467 A (488 A)
0.5 mT fringe field distance	axial (z) 4.0 m radial (x, y) 2.5 m	axial (z) 4.6 m radial (x, y) 2.6 m
Max. field decay	< 0.1 ppm/h	< 0.1 ppm/h
Energizing time using MPS 3600 (pure ramp time, without checks and pressurization phase)	24 min.	90 min.
Homogeneity volume	50x50x45 cm DEV (Diameter Elliptical Volume X, Y, Z)	50x50x45 cm DEV (Diameter Elliptical Volume X, Y, Z)
Maximum shim iron mass	6.0 kg	6.0 kg
Magnet normal operating pressure (regulated by MSUP)	15.5 psia	15.5 psia
Magnet Helium Levels		

For current recommended helium levels check the SKB.

*Values for OR103 in brackets

Magnet Coil Components

The OR98 / OR99/103 magnets are actively shielded magnets consisting of a primary coil having 6 inner coils and 2 active shield coils to reduce the magnetic fringe field.

Switch Heater

A superconductive switch allows the magnet to be energized and de-energized.

During ramping the switch heaters are heated to become resistive allowing current to pass into the magnet coils. At the set current the heaters are turned off and the switch becomes superconducting allowing the magnet to remain at its operating current.

Quench Heaters

In case of emergency (accidents, fire, etc.), the magnetic field has to be de-energized immediately to avoid danger to personnel. Two quench heaters are placed close to the main coil. When activated, they immediately turn the superconducting magnet wire into its normal-conductive state. The coil current is discharged within 20 seconds (quench). The thermal energy released will boil off a significant amount of liquid helium.

Quench Protection Diodes

During a quench, up to 6 kV can be induced across the coil. To avoid damage, the switch protection diodes clamp the voltage to a maximum of approximately 12 V.

External Interference Screen (EIS)

The External Interference Screen (EIS) are extra coil windings placed on top primary coils. This coil picks up transient B_0 interference and shields the main coil from magnetic field fluctuation caused by ferrous objects that may move within the

locality, of the magnet field. (i.e. equipment).

Over time, the main magnetic field decays - caused by the rest resistance of the coil joints. Hence, a rising current is induced into the EIS coil, producing a counter-field that deteriorates the B_0 field homogeneity. Therefore, the EIS has to be reset (= EIS switch opened) once a day, controlled by a timer logic of the Magnet Supervision Unit (MSUP). To avoid image quality problems, the EIS is reset when the system is not utilized (typical set-up = 2:00 am).

Figure 189 Magnet Electrical Diagram

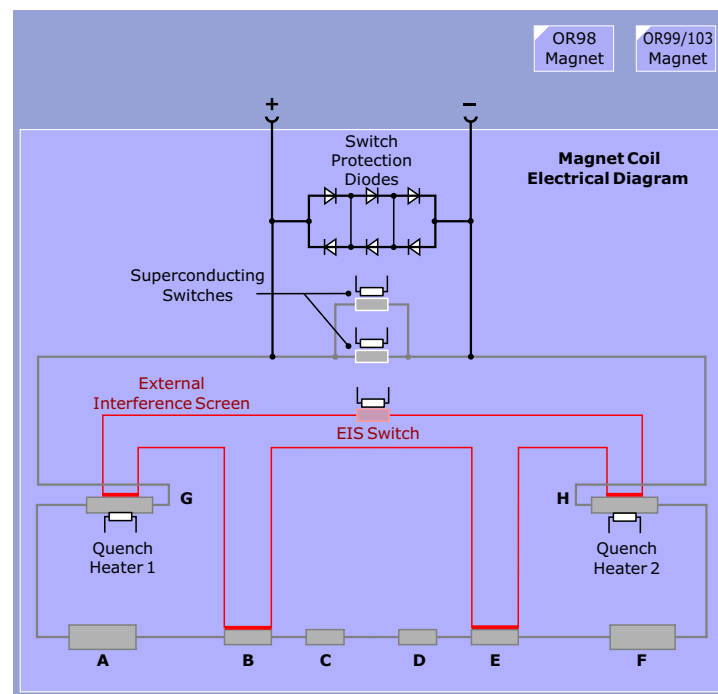


Figure 190 OR98 Magnet - Cross Sectional Diagram

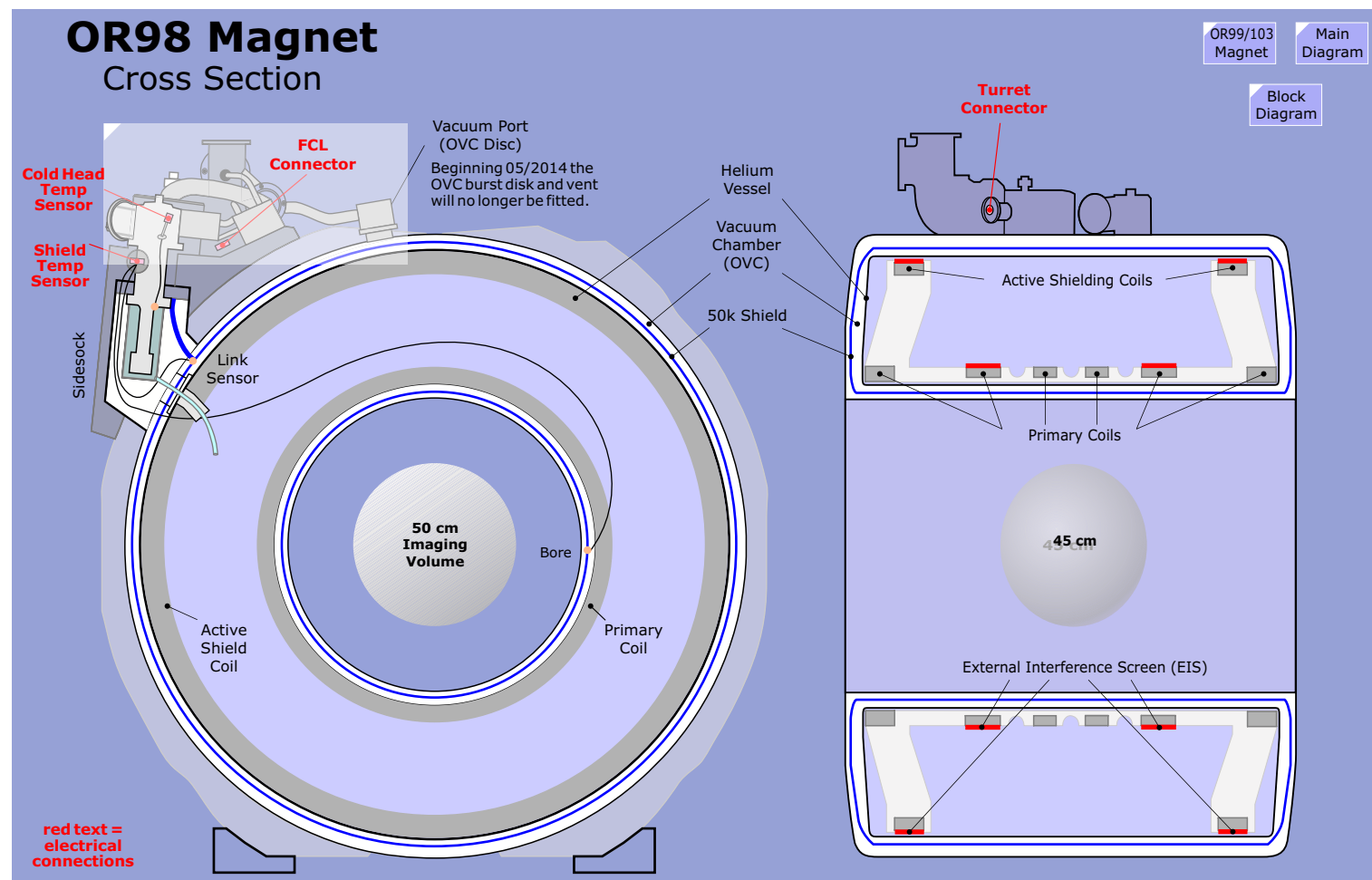
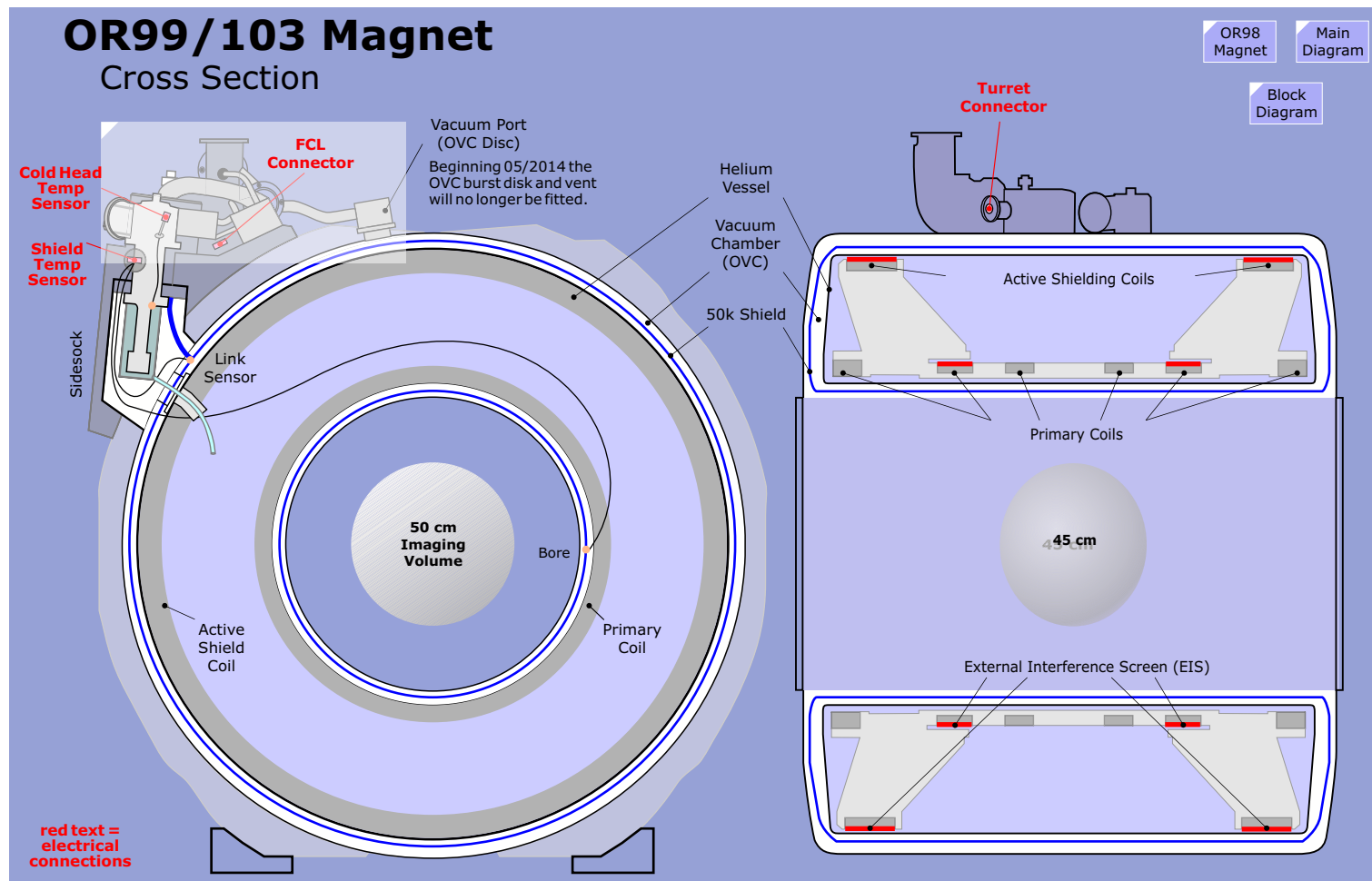


Figure 191 OR99/103 Magnet - Cross Sectional Diagram



Service Turret

General

The Service Turret is placed on the sidesock located on the left side of the magnet. It contains:

- Venting system - including valves and bursting disks
- Entry port for Helium Fill syphon
- Fixed Current Leads (FCL) for ramping
- Electrical Interface to the Magnet Supervision (MSUP)
- 4 K SUMITOMO Cold Head

Venting System

For location of venting system components, please refer to [figure 192](#).

Vent Valve

For the 1.5 T magnet OR98 a 16 psia (pounds per square inch absolute) vent valve is placed parallel to the Quench valve through which gas can vent when internal pressures exceed 16 psia. At the 3 T magnet OR99/103 a 17 psia valve is used to increase the magnets' buffer capacity for boiled-off helium gas.

System Bypass Valve

A hand-operated bypass valve, placed in parallel to the 16 psia valve, allows safe, controlled de-pressurization of the magnet helium vessel prior to helium fill and maintenance.

Cold Head Bypass Valve

This valve is only opened to allow boil-off gases to circulate through the Cold Head and turret assembly during transportation, or when the compressor will not be in operation for extended periods of time. This helps reduce the helium boil off.

NOTE The Cold Head bypass valve must be closed during normal operation of the Cold Head. Otherwise, no helium recondensing is possible.

Quench Valve

The quench valve is a spring-loaded plate, sealing on an O-ring face, which opens at 11 psig. The spring valve has in addition a graphite burst disk rated at 15 psig which will burst if the spring valve should not open.

NOTE After a quench, all burst disks and venting must again be leak checked. Please refer to the Troubleshooting Guide "Post Quench" procedure.

Auxiliary Vent Line and Bursting Disk

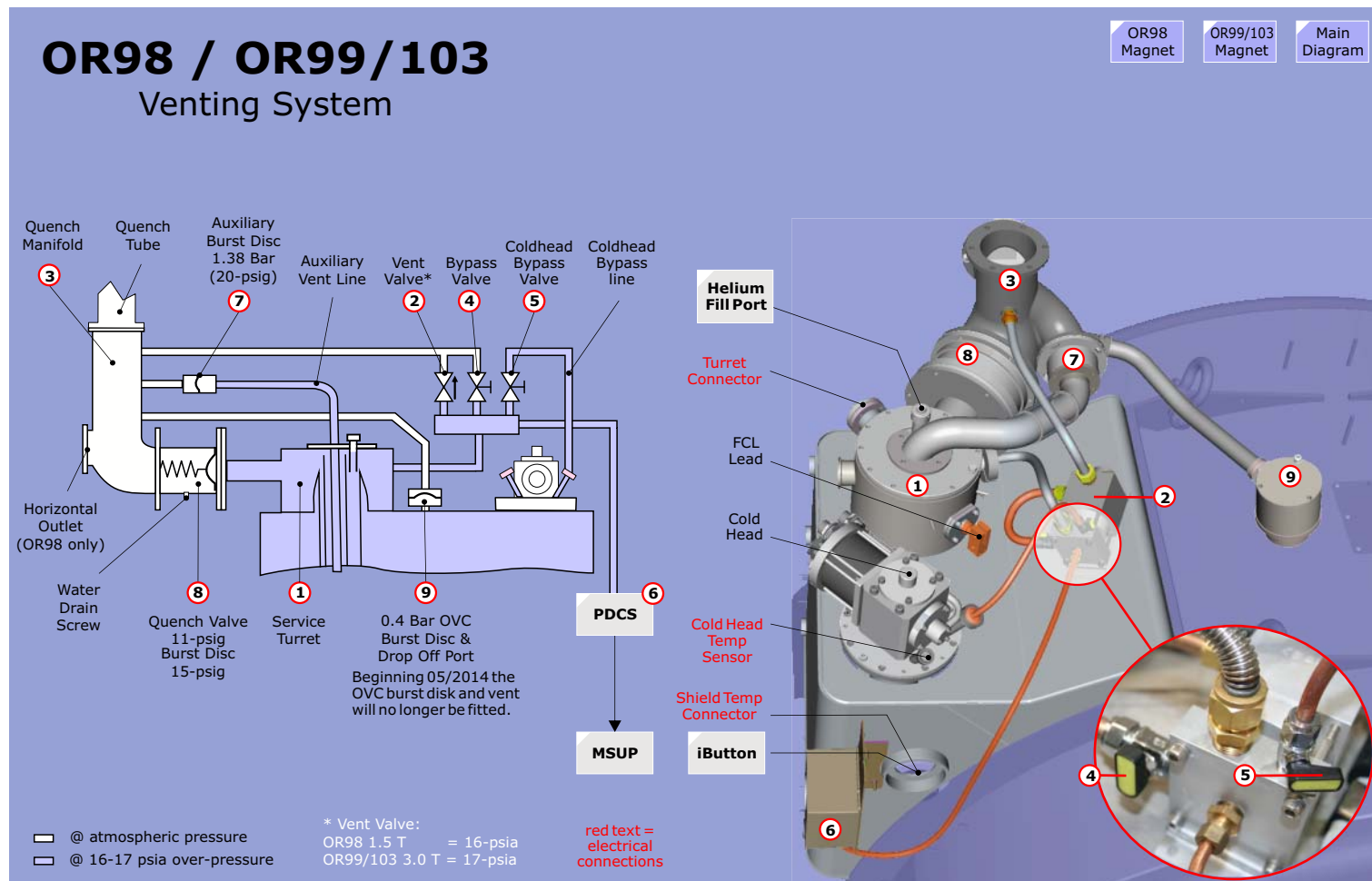
This vent line extends (internally) below the neck tube ensuring that there is an alternative vent path for the gas to escape, in the event that the magnet neck tube is restricted by the formation of ice.

A 1.38 bar (20 psig) stainless steel burst disk is fitted in line to the auxiliary vent. This is the secondary burst disk and under normal quench conditions will not break.

OVC Burst Disk and Vent Line

The OVC is fitted with a pressure relief port, protected by a 0.4 bar (5.8 psig) burst disk. In the event of a catastrophic failure when the inner helium vessel breaks and large quantities of helium will spill into the vacuum chamber causing over pressure. This vent is connected to the quench elbow, via the OVC vent line. Integrity of the disk is important to avoid helium gas on the seals of the PODO plate during filling, or a quench. This could result in degrading of the vacuum.

Figure 192 Venting System OR98 / OR99/103



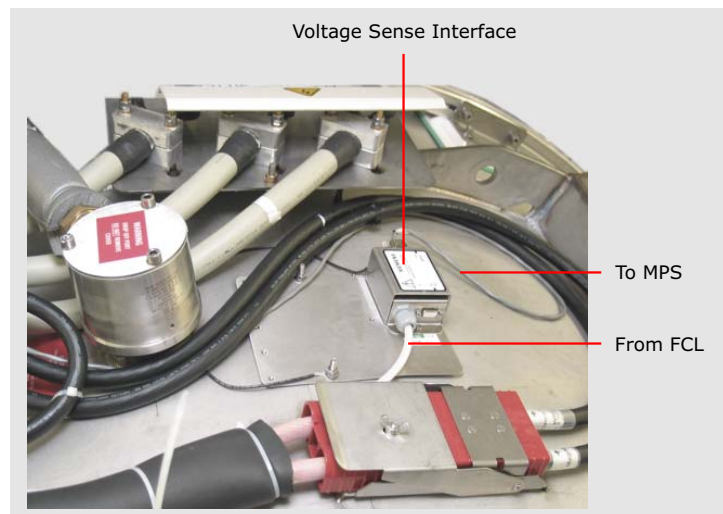
Fixed Current Lead

The magnet is ramped over the Fixed Current Lead (FCL) located on the Service Turret. There is 1 copper connector (positive) attached through the side of the turret through a gas tight ceramic feed-through. The negative connector is attached to the cryostat shell. The ramp cables are permanently connected and routed to the RF Filter Panel where the MPS can be connected.

Voltage Sense Interface

There is a little box close to the positive connector of the ramp cables. It contains a self-healing thermal fuse to protect the voltage-sense cables of the Fixed Current Lead against short circuits.

Figure 193 Voltage Sense Interface

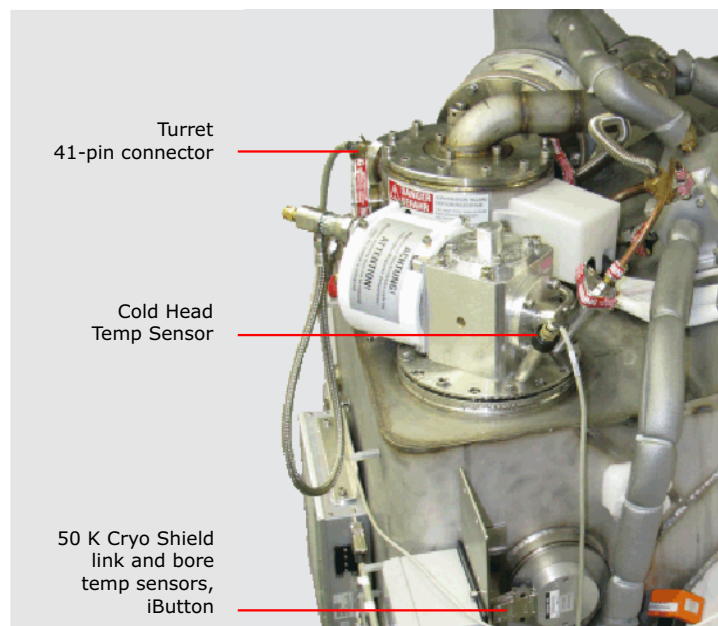


Electrical Interfaces to Magnet Supervision (MSUP)

The MSUP monitors and controls the following, via electrical interfaces:

- **Magnet** - For example helium level probes, switch heaters, etc. accessed by a **41-pin connector** at the service turret.
- **Cold Head** - 50 K stage temperature
- **Cryo Shield** - 50 K Shield link and bore temperature
- **Pressure Display & Control Switch** - an electric pressure transducer and analog dial gauge to monitor the magnet helium pressure. A low pressure switch to turn off the He-compressor in the event of low magnet pressure.

Figure 194 Electrical Connectors



Pressure Display & Control Switch (PDCS)

This Pressure Display & Control Switch integrates an electric pressure transducer, a pressure switch and a pressure gauge into a single unit.

Pressure Sensor

The pressure sensor outputs an electrical signal proportional to the pressure (in absolute psi = psia) and is used by the MSUP to regulate the magnet pressure.

Pressure Switch

All cryogenically cooled magnets are kept under a positive pressure to assure no air enters the magnet leading to ice blockages. Blockages could prohibit boil off gasses to escape leading to an explosion.

The switch is used as a fail-safe to turn off the Cold Head compressor in the event the magnet pressure falls to within 0.2 psi above atmosphere (0.2 psig). A large leak or deliberate depressurization of the magnet could be causes for loss of pressure.

Pressure Gauge

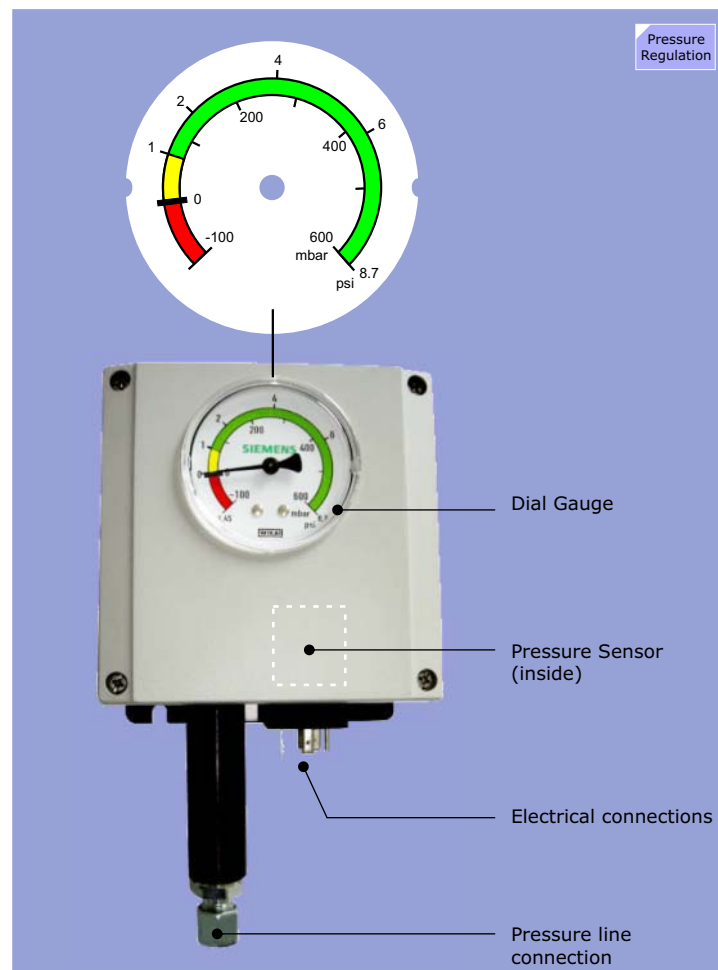
Additionally, the relative pressure is displayed on an analog dial gauge for service personnel.

CAUTION The pressure gauge indicates pressure at the venting system, NOT inside the magnet!

NOTE The pressure reading on the analog linear gauge may fluctuate with changes in atmospheric pressure!

The unit is protected by an over-pressure device. The pressure switch is calibrated at the factory and should not be tampered with. The pressure sensor does not require calibration.

Figure 195 Pressure Display & Control Switch



Magnet Supervision (MSUP)

Overview

The Magnet Monitoring supervises and controls magnets constantly during normal operation as well as during emergency ramp down. It consists of:

- Magnet Supervision (MSUP 2306)
- ERDU Battery

MSUP 2306

The MSUP 2306 is a **Field Replacable Unit (FRU)**. It contains 2 PCBs:

- MSUP PCB - mounted underside of the lid and is a spare part.
- ERDU PCB - mounted on the base. This single board is not replaceable, the complete MSUP unit must be ordered.

Location

The MSUP is located behind the magnet service side cover. The unit and is mounted on the magnet OVC. The ERDU battery is located to the right of the MSUP.

ERDU Battery

An external **24 V Lithium non-rechargeable battery** (with internal 5A fused output) is mounted on the magnet and provides a backup to the ERDU function.

Transit Mode

During magnet transportation, the standard 24 V Lithium Battery must stay disconnected to avoid discharge, thus leading to a reduction of its 10 year replacement interval.

For being able to read magnet data during transport (e.g. sample helium level), an auxiliary **+36 V Service power supply** can be connected to connector X1 at the MSUP.

Figure 196 MSUP 2306 Layout

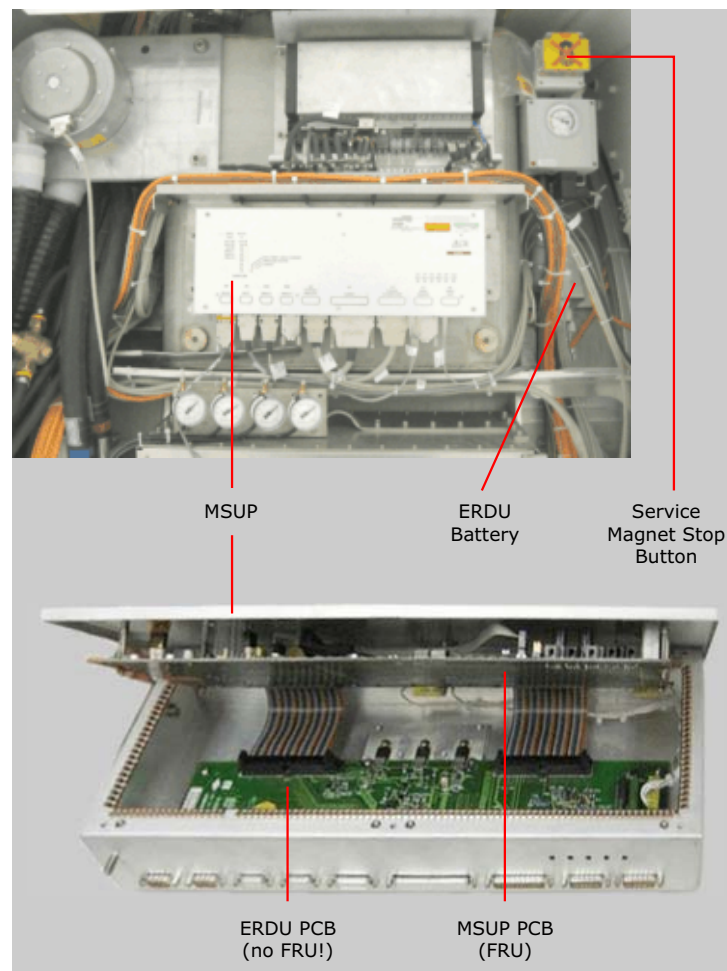
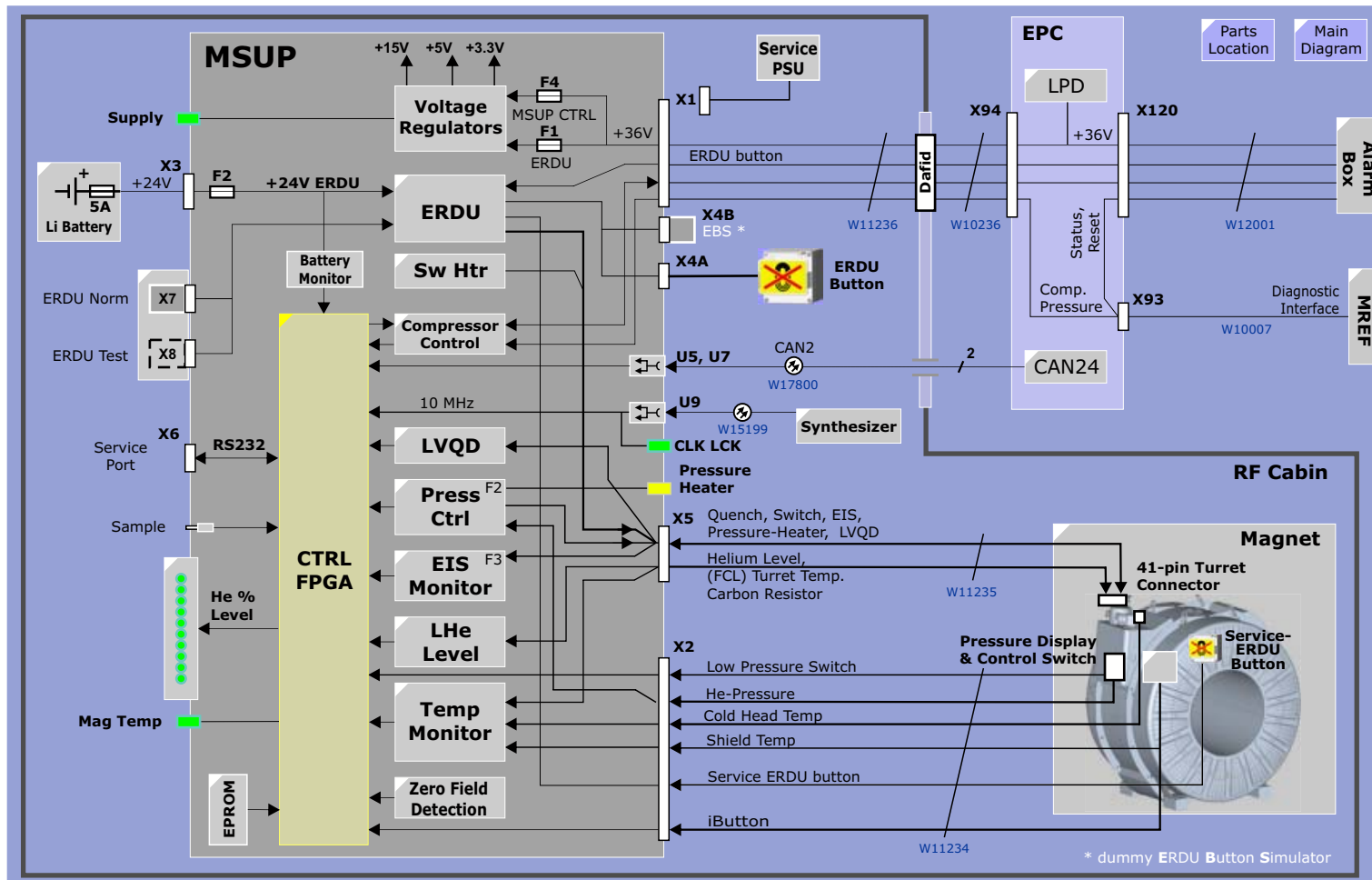


Figure 197 MSUP 2306 Block Diagram



Function

The MSUP supplies the following functions:

- Communication Interface for Status Messages and Errors
- Internal Self Test and Magnet Type Recognition
- Emergency Run Down Unit (ERDU)
- Battery Backup for ERDU
- Switch and Quench Heater Control and Monitoring
- External Interference Screen (EIS) Control
- Magnet Pressure Control and Monitoring
- Helium Level Monitoring
- Magnet Temperature Monitoring
- Zero Field Detection
- Magnet helium level indication on front panel with sample button.
- Compressor Control and Monitoring
- Low Voltage Quench Diagnostics (LVQD)
- Integrated ERDU test load.

Voltage Supply

The MSUP is supplied by the 36 VDC power supply of the LPD in the EPC, indicated by the green POWER ON LED at the MSUP lid. Additionally, the ERDU is backup by the 24 V ERDU battery (see description [page 246](#)).

CAN & CPU Module

CAN Bus

The **MSUP** communicates to the MARS 2 via a fiber optic CAN bus. Changes of the MSUP status and error messages are reported via CAN commands and MARS 2 to the host.

RS232 Serial Interface

For service purposes (e.g. log file analysis with **Magnet HSC**), a PC can be connected to **X6** of the MSUP and its program accessed

via a terminal emulator program (e.g., Putty). Communication parameters to be set are:

- 9600 Baud rate
- 8 bit, 1 start bit, no parity
- Flow Control = None
- File, properties, settings tab, emulation = VT100.

After pressing "Esc" and typing "R" at the Service PC, the communication link is enabled.

NOTE For a detailed description of communication parameters and commands please refer to the Troubleshooting Guide.

Synchronization clock 10 MHz

The CPU of the MSUP is continuously running, also during measurements. To avoid RF-interference to the MR-signal, the processor clock is synchronized to the MR system with a 10 MHz clock signal from the Synthesizer (FOC U9).

Internal Self Test and Magnet Type recognition

A built-in microprocessor controls the monitoring of all magnet sensors and internal circuits. All values are checked against standard parameters stored in its memory. The processor is equipped with a timer logic to trigger a sensor measurement at night. The actual time information is automatically downloaded from *syngo* MR (see also "[Supervisory Test Cycle](#)", page 250).

During power up, the processor initializes all MSUP components and performs a self test during which the presence of the +3.3 V, +5V, +15 V and +36 V supply voltages are checked. In addition, some reference voltages are checked to verify the proper function of the multiplexer and ADC circuit for the analog sensors (e.g. LHe-level measurement).

iButton

The iButton is a non-volatile memory device which stores a set of magnet specific configuration data. It is connected to the MSUP at X2 (CH/DIAG/PRESS/ID/VALVE).

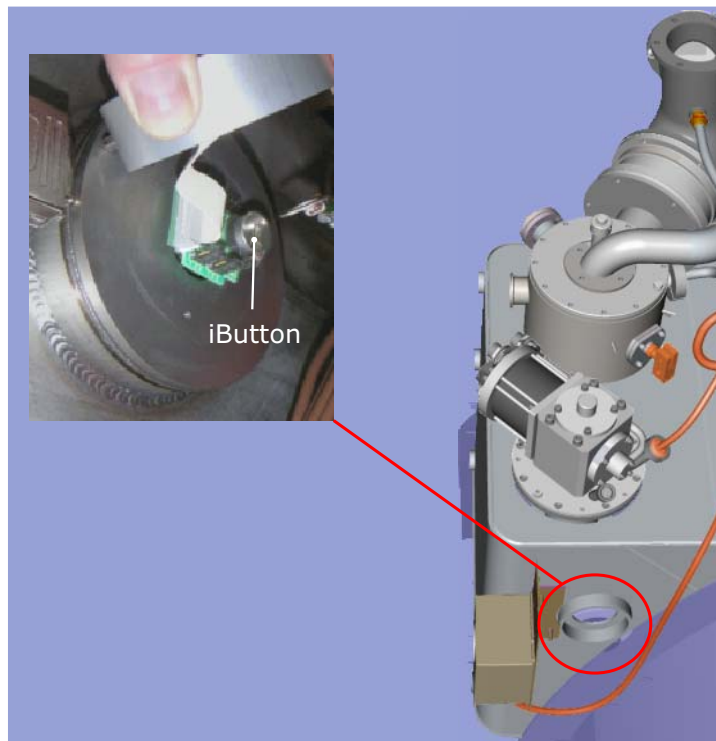
Magnet data is recovered from the iButton, via connector **X2**. The iButton is housed in the diagnostic port. The magnet type is determined by attempting to read the iButton located in the diagnostic port. If successful, all magnet parameters held in the iButton are read by the supervisory. That information is used to set up a connected MSUP 2306 accordingly.

The following most important Magnet data are stored in the iButton:

- Magnet type - OR98/OR99/OR103
- Magnet part- and serial-number
- Alarm Box part- and serial-number
- He probe calibration values for 0% and 100%
- Magnet pressure heater resistance
- Type of magnet resistors - carbon or ceramic
- Magnet resistor calibration values for He-, LN₂- and room-temp.
- Helium level warning threshold
- Supervisory test time
- Shield temperature warning- and alarm-thresholds
- FCL ramping max. temperature limit
- Last ramp current

NOTE The magnet parameter stored in the iButton are factory programmed and generally do not require any modification by CSEs on-site.

Figure 198 i Button Location



Supervisory Test Cycle

The **MSUP** can perform a Supervisory Test cycle to start a new measurement of:

- Helium level
- EIS reset

The Test cycle is completed within 30 seconds. It can be started by either:

- Pressing the "SAMPLE" button on the MSUP front panel
- CAN-bus commands
- Internal MSUP timer (sampling time can be set under *SeSo > Magnet&Cooling > Initialization*)

EEPROM

The **MSUP** EEPROM is used for three distinct purposes:

Storage of status entries

Status entries are stored with time and date stamp after each Supervisory Test. A record of the last 200 status entries is kept.

Storage of events

The following events are stored with time and date stamp and can be read out with the Service PC connected to the MSUP. (The last 30 readings are stored).

- Compressor OFF Time
- Compressor ON time
- ERDU ON time
- ERDU OFF Time
- ERDU Button pressed
- ERDU button released
- ERDU button missing
- Switch Heater ON time
- Switch Heater OFF time
- MSUP Power ON
- EIS Fault time

- EIS Fault cleared time
- Zero Field detected
- Battery Volts Warning
- High pressure warning
- Low pressure warning
- Compressor static pressure test start time (last 5 only)
- Compressor static pressure test stop time & test result (last 5 only)

Storage of user configured MSUP parameters

Some of the MSUP parameters can be configured by the user (e.g. calibration of both LHe-level meters in SeSo) and will be stored in the MSUP EEPROM and iButton:

- He probe calibration values for 0% and 100%
- He level warn point
- Magnet operating current
- Supervisory test time

Each of the parameters is communicated to SeSo via the CAN bus. Validity of the above data is checked permanently by checksum verification. If a deviation is found, the processor tries to recover the parameters from the iButton, or alternatively loads a default set of parameters for the OR98 / OR99/103 magnet.

Emergency Run Down Unit (ERDU)

ERDU Buttons

Every system has two ERDU buttons, one on the Alarm Box and one mounted in the magnet room, usually next to the door. A third button can be connected optionally. The button is to be pressed in an emergency situation involving the magnetic field. When activated, the **MSUP** applies power to the two quench heaters and the switch heater and the magnetic field will decay in less than 20 seconds.

ERDU Button Monitoring

A resistor/zener diode pair are connected in parallel to each ERDU button over which the MSUP drives a small constant current to generate a proportional voltage drop indicating the number of switches connected. In case a cable or switch is opened (button missing) or shorted (button activated!!!), the MSUP will generate a fault message and the yellow WARNING LED on the Alarm Box will light. An **ERDU button simulator plug** must be connected to the unused ERDU button connector to prevent an error due to a missing button.

When the ERDU has been activated by an ERDU button a time/date stamp of its activation and de-activation is recorded.

Quench Heater Monitoring

The quench and switch heater resistances are continually monitored by applying a small constant current over the heaters and measuring the voltage drop. An alarm is generated if a resistance value is found to be outside specification.

ERDU Battery

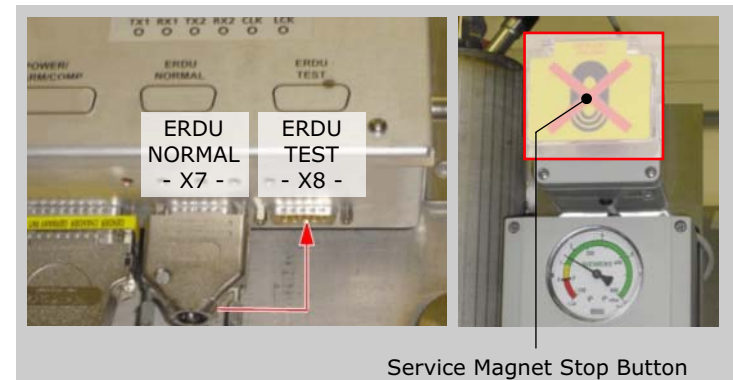
An external 24 V lithium battery backs up the ERDU function for 1 operation after power failure. The battery voltage can be displayed under *SeSo > Magnet&Cooling > Magnet Status*. The battery must be replaced after 10 years or if insufficient voltage is indicated.

ERDU Battery Test

The condition of the ERDU battery is continuously monitored. If the battery voltage is less than **+22.00 V**, a **warning** message is generated, the yellow WARNING LED on the Alarm Box lights and its buzzer beeps 3 times within 10 seconds.

If the battery voltage falls below a threshold of **+21.5 V**, an **error** message is generated and MR normal scanning is blocked. In addition, the yellow WARNING LED on the Alarm Box lights up and its buzzer beeps 3 times within 10 seconds.

Figure 199 ERDU Test Connector



ERDU Test Load

The MSUP has an integrated ERDU Test Load for testing the safety relevant functionality of the ERDU circuit. The **ERDU test** can be invoked by moving the **ERDU test connector** X7 (ERDU normal position) to X8 (ERDU test position) indicated by a continuous audible warn signal generated by the MSUP. In the test mode, the customer ERDU buttons are inactive and can be pressed for testing without quenching the magnet. The MSUP records the test results and report this data to the host. During the test mode, the **Service Magnet Stop Button** (located at magnet) is activated to be able to quench the magnet in case of hazards!



WARNING

When the ERDU test connector is disconnected from BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET. When the ERDU test connector is disconnected from MSUP (X7), the customer site Magnet Stop buttons WILL NOT quench the magnet. The MSUP beeper will sound.

Refer to the Safety-Related Test procedure for instructions on proper use of ERDU Test.

Switch Heater

The switch heater of the magnet is controlled by the **MSUP**. When the Magnet Power Supply is connected to the system, it can energize the switch heater via CAN bus commands to the MSUP as required. Like the quench heaters, the switch heater circuitry is permanently monitored.

Helium Level Monitoring

The magnet's helium level is measured with the help of two independent probes located within the helium vessel. Each probe is constructed out of a single strand of super-conducting niobium-tin wire. This wire is heated and its resistance measured, indicating the length of wire not immersed in liquid. To calculate the level in % the MSUP requires the wire resistances at 0 and 100% helium levels which must be entered into software (see paragraph on Calibration of LHe-level meters just below).

When the sampling is started, the **MSUP** performs a measurement of the liquid helium level in the magnet system. The reading is scaled between the calibration points to give a percentage value. Probe calibration data is stored in the MSUP, iButton, SeSo interface, and recorded in the SMT Magnet Acceptance Certificate.

Display of last sampled LHe-level

The LHe filling-level of the Magnet can be seen at:

- *syngo MR, System > Control > System Manager > Periphery*
- Service Software *SeSo > Magnet&Cooling > Magnet Status*
- He LVL% LED display on the MSUP front panel

LHe-level sampling

A new sampling of the LHe-level can be started:

- Automatically one time per day by internal MSUP timer (time setup in *SeSo > Magnet&Cooling > Initialization*)
- Manually by opening the *SeSo > Magnet&Cooling > Magnet Status* (additionally when updating)
- Manually with the **manual sample button** at the MSUP.

Calibration of LHe-level meters

The actual probe resistance values for 0 % and 100 % He-level for both probes are documented in the SMT Magnet Acceptance Certificate delivered with each magnet and need to be entered into software under *SeSo > Magnet&Cooling > Initialization*.

A **warning level** (possible range 40-90 %) can be adjusted in *SeSo > Magnet&Cooling > Alarm Settings*. If the level falls below this threshold, a warning message is generated and reported to *syngo MR*. In addition, the yellow Warning LED is lit and the buzzer beeps 2 times every 20 seconds.

The **alarm level** is fixed at 35%. If the level falls below that value, the yellow Warning LED is lit and the buzzer beeps 3 times every 10 seconds.

External Interference Screen (EIS) Control

To eliminate any accumulated current in the External Interference Screen (EIS) coil windings that might deteriorate the B_0 field homogeneity, the EIS coil has to be reset once each day. The **MSUP** automatically activates the EIS heater (15 seconds) at the time set under *SeSo > Magnet&Cooling > Magnet Setup*, usually set to 2:00 am to avoid image quality problems.

A manual EIS reset can be performed by retrieving the Magnet Status in SeSo. It is performed automatically during the shimming procedure. If the EIS heater is left on, the MSUP will switch it off automatically after 60 minutes (OR98) resp. 130 min. (OR99/103) to avoid excessive helium boil-off.

The presence of the EIS reset output current is monitored by the MSUP. In case of a fault (e.g. broken cable) a warning message is generated and the yellow WARNING LED on the Alarm Box lights.

A precautionary hardware circuit is included to prevent the situation of the EIS heater being permanently energized. An EIS heater 'crowbar' circuit ensures the output fuse to the heater is blown when a timer expires 210 minutes after recognizing the output having been continuously on.

Magnet Pressure Regulation

With a recondensing Cold Head system, a pressure regulation is required to prevent an under-pressure (= sub atmospheric) condition within the magnet which could cause air to ingress and freeze in the magnet vessel. The magnet pressure is regulated by the **MSUP** which gets the actual (absolute) pressure value from the Pressure Display & Control Switch unit. When the actual pressure value has fallen below the specified nominal value, the MSUP activates a pressure heater to boil off helium. By this means, a constant, regulated magnet pressure can be achieved.

The pressure regulation system can operate at four different modes:

- **Normal Mode**
The magnet pressure is regulated at 15.5psia
- **Hardware Control Mode**
A fault condition, the MSUP tries to regulate at 15.1psia
- **Manual Mode**
For service, to pressurize the magnet. The pressure heater is activated continuously
- **Safety Mode**
The magnet is almost fully depressurized, the He-compressor is turned-off by a safety switch to avoid under-pressure

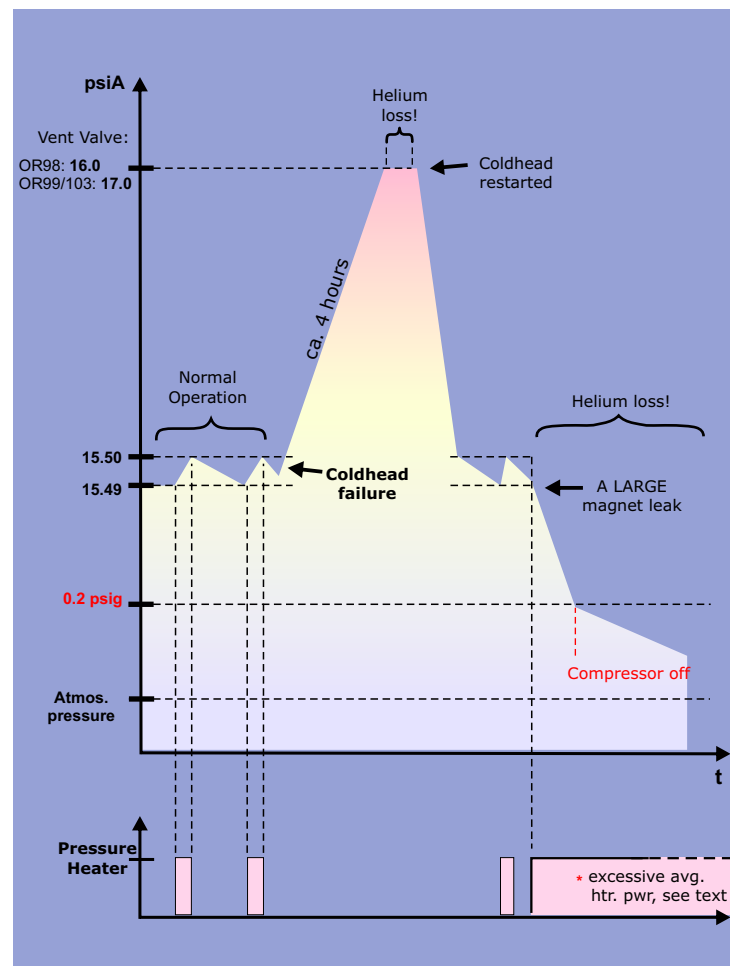
Normal Mode

During normal magnet operation the **MSUP** tries to regulate a constant magnet pressure around **15.5 psia**. When the pressure has fallen to 15.49 psia, the MSUP activates the pressure heater to boil off helium. When the pressure has risen to 15.50 psia, the heater is turned off.

Hardware Control Mode

In a service or fault condition (e.g. a leak), when the **MSUP** requires >2 W average heater power over a 4 hour period, the MSUP will switch into "hardware control mode". In this mode the MSUP reduces its regulation set-point to 15.1 psia and the wattage limit is ignored.

Figure 200 Magnet Pressure Regulation



The **MSUP** will stay in "hardware control mode" until the average heater power value or the whole MSUP is reset, which also resets the set-point back to 15.5 psia (see also following section "Reset of Pressure Heater Average Power Value"). To prevent burn-out the pressure regulation circuit is protected by autofuse F2 which activates automatically after 2.25 hours of continuous pressure heater operation.

Manual Mode

The pressure heater can be turned on manually (in SeSo) as a means to quickly build-up magnet pressure. When enabled manually, the duty cycle is 100% = approx. 30 W!!!. If forgotten, a timer switches off automatically after 50 minutes.

Safety Mode

If the magnet is almost fully depressurized (0.2 psig above atmospheric pressure), the He-compressor is turned-off by the pressure switch of the **Pressure Display & Control Switch (PDCS)** to avoid under-pressure.

Parameter	Specifications
max. Pressure Heater duty cycle for normal operation	20 %
Pressure Heater duty cycle in manual mode	100 %
max. Pressure Heater ON time	50 min.
System pressure high alarm	18.0 psia
System Pressure low alarm	15.0 psia

Pressure Heater Average Power Value

The **MSUP** constantly monitors the required pressure heater average power value (PHAP). This value, shown in SeSo under Magnet&Cooling > Magnet Status is displayed in Watts over a time period of 4 hours and provides diagnostic information about the performance of the pressure control system and Cold Head. During normal magnet operation, assuming the cold head is recondensing fine and there is no leak, a **typical PHAP of approx. 0.8 W/ 4 hours** can be seen. This value is slightly magnet- and cold-head

dependant and therefore recorded in the SMT Magnet Acceptance Document as **"Recondensing Margin"** value. Please note that this value can change during the day depending on the customers' system usage.

Unusual high or low PHAP values may indicate a magnet problem, e.g.:

- Magnet Leak (PHAP increased)
- Poor Cold Head performance (PHAP decreased)

Reset of Pressure Heater Average Power Value

After any service activity which may temporarily increase the PHAP value above normal (e.g. depressurizing, LHe-filling, etc...) the PHAP values must be manually reset to avoid that the **MSUP** falls into "hardware control mode".

The following steps need to be done in SeSo Magnet&Cooling > Magnet Status:

- Turn-on pressure heater manually until 15.6psia is reached
- Turn-off pressure heater and set back to "Automatic Mode"

NOTE When the MSUP power is turned-off, the Pressure Heater Average Power Values are also reset.

Magnet Temperature Monitoring

The following magnet temperatures can be checked by the MSUP:

- 50 K Cold Head temperature
- 50 K shield temperature (link, bore)
- Turret temperature (2x Fixed Current Lead = FCL)
- Internal magnet temperatures (4x Ceramic carbon resistors)

Temperature measurements are achieved by passing a constant current through the sensors/resistors and monitoring the voltage readings. Some of the readings are converted into temperature values.

During a manual sampling, the temperature of the magnet cryostat is displayed by the MAG TEMP / ERDU CURRENT LED at the MSUP:

MAG TEMP LED	Magnet Temperature
red	Room to Nitrogen
yellow	Nitrogen to Helium
green	Helium temperature normal

Zero-Field Detection

The MSUP contains a sensor to detect the absence of the main magnetic field (used for Quench detection function).

Low Voltage Quench Detection

In the event of a quench, the voltage change across each individual magnet coil is monitored and the first coil to reach the 'quench trigger level' is detected and recorded by the LVQD MSUP logfile. This data provides further diagnostics for SMT magnet service.

The coil voltages are picked up with potential taps on the wire joints across individual coils, and fed to a diagnostic PCB board on the magnet. Wires from the PCB board are fed up the magnet

turret loom to the 41-pin turret connector through to connector X5 at the MSUP.

After a quench has occurred the LVQD logfile can be saved to the host in SESO > Magnet & Cooling. Afterwards, the LVQD log at the MSUP can be reset.

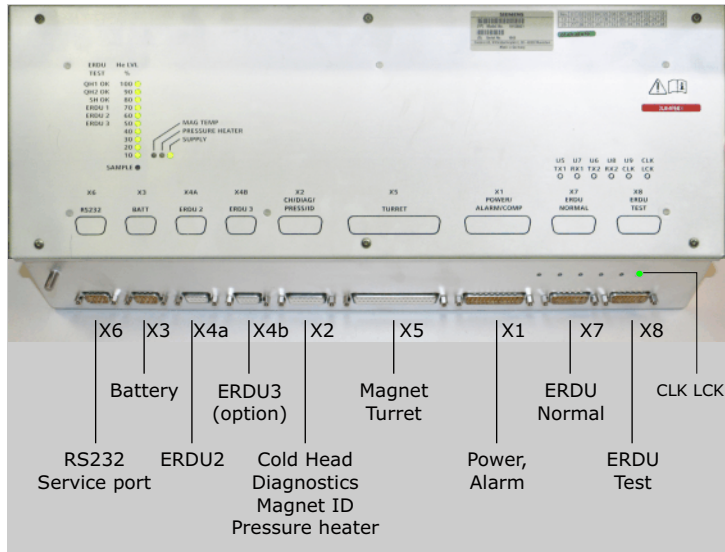
Specifications

Table 1 MSUP Parameter Specifications

Parameter	Spec. OR98 / OR99/103
Number of automatic LHe samples per day	1
He level alarm set	35 %
He level alarm clear	40 %
He level warn set	40 % (default values)
He level warn clear	42 % (default values)
Check on battery voltage warn level	22.00 V
Shield temp 1 alarm (50 K Link)	> 94 K
Shield temp 1 warn. (50 K Link)	> 77 K
Shield temp 2 alarm (50 K bore)	> 102 K
Shield temp 2 warn. (50 K bore)	> 84 K
Switch heater sensor alarm	80>Ohm>200
Quench heater 1 alarm	10>Ohm>30
Quench heater 2 alarm	10>Ohm>30
Maximum EIS ON time	60 / 130 min. (OR98 / OR99/103)
Maximum pressure heater ON time	50 min.
Pressure heater duty cycle for closed loop operation	20 %
System pressure, high control level	15.50 psia
System pressure, low control level	15.49 psia
System pressure, high alarm	18.0 psia
System pressure, low alarm	15.0 psia

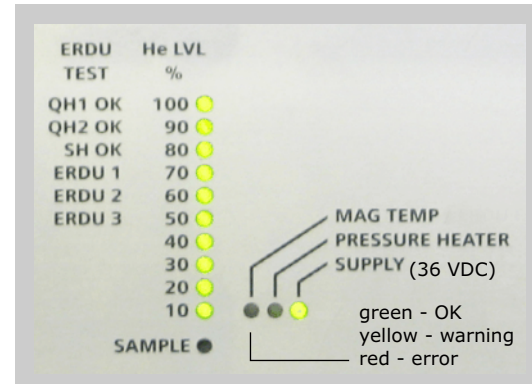
Connectors

Figure 201 MSUP Connectors



LEDs

Figure 202 MSUP LED display of helium level



LED	Color	Description
PRESSURE HEATER	yellow	Pressure Heater on
SUPPLY	green	+36 V MSUP supply voltage from EPC PSU present.
He LVL %	green/	Helium level/ERDU Test status
MAG TEMP	green/ yellow/ red	Helium temperature (normal) temp between nitrogen and Helium above nitrogen temperature
CLK LCK	green	10MHz Clock locked

Alarm Box

Overview

The Alarm Box (ABX) is an interface for:

- MR system power switching
- Magnet status indicator
- Remote control of He-compressor
- ERDU

Function

MR System Power Switching

The **System On** and **System Off** buttons are hard-wired to relays K10 and K11 in the Power Distributor responsible for these functions. See ["Power ON/OFF Relay Logic"](#) on page 292.

Figure 203 Alarm Box Layout

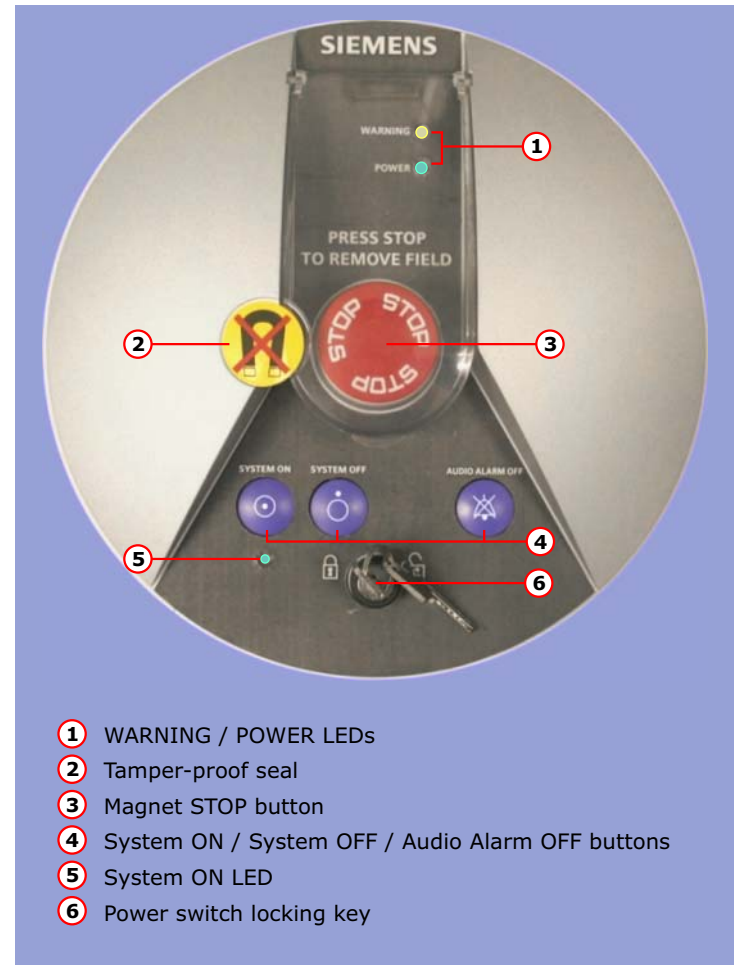
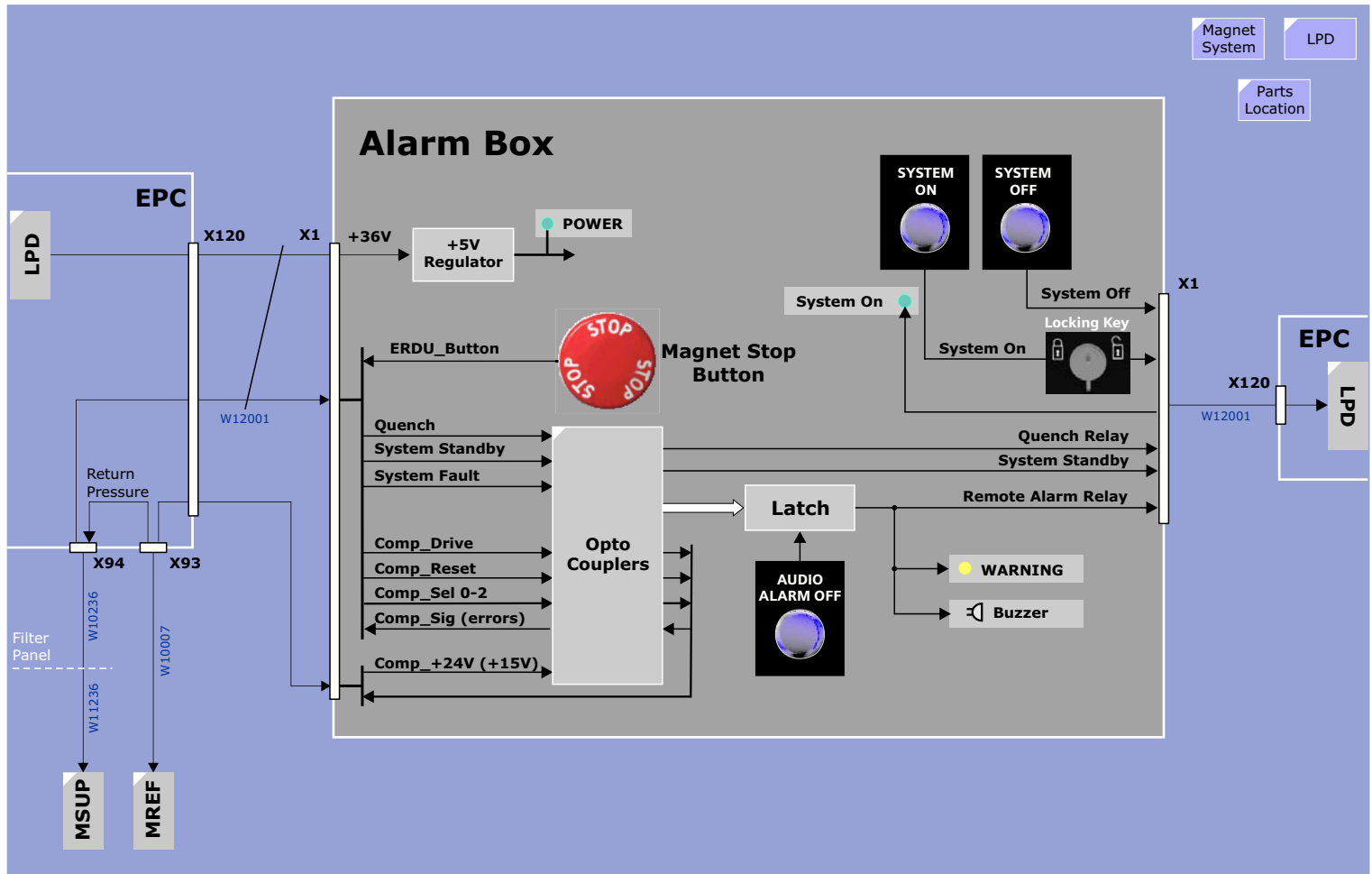


Figure 204 Alarm Box Block Diagram



Magnet Status Indicator

WARNING LED - Remote Alarm

Any of the following errors detected by the MSUP are indicated on the Alarm Box with the yellow **WARNING** LED:

- Mag Pressure Switch
- Compressor OFF
- HELIUM LEVEL
- Magnet Pressure
- ERDU Button
- EIS
- BATTERY LOW
- ERDU Heaters

Errors from the MSUP are opto-coupled and latched. The latch output sets the WARNING LED, an audible alarm buzzer and the Remote Alarm Relay signal to K17 in the LPD (see LPD diagram on [page 291](#)). The latch, and thus the WARNING LED, alarm buzzer and remote relay signal, can be reset by pressing the "AUDIO ALARM OFF" button.

Alarm Buzzer

The Alarm Box has 2 different style alarms:

- **Low priority alarm** - the alarm box **beeps twice every 20 seconds**, e.g. switch heater or quench heater monitor alarm, compressor off, helium warning or alarm, EIS incorrect drive alarm.
- **High priority alarm** - the alarm box **beeps three times every 10 seconds**, e.g. ERDU battery warning or alarm, ERDU button on or fault, ERDU circuit short, magnet high pressure (18 psia) or low pressure (15 psia).

The buzzer can be reset by pressing the "**AUDIO ALARM OFF**" button.

NOTE For certain self-resettable faults, e.g. compressor errors, the alarm buzzer stays on when the Warning LED is already reset. This is normal behavior. The buzzer can be reset with the "AUDIO ALARM OFF" button.

Compressor Monitoring

The **Alarm Box** electrically isolates the compressor signals to and from the Magnet Supervisory and informs the operator about compressor failure.

Compressor Control

In case of a compressor error, a **reset signal** is sent from the Alarm Box to restart the helium compressor - this is controlled by the MSUP. The F70 compressor has a 20-minute minimum restart interval in case the compressor has tripped off with over-temperature.

The MSUP also sends a compressor off error which sets the WARNING LED, the audible alarm and the Remote Alarm Relay signal.

NOTE For a detailed description of the Compressor Control, see page 266.

ERDU

The functional part of the **Emergency Rundown Unit** (ERDU) is performed by the MSUP, the Alarm Box only supplies one of the Magnet Stop buttons.

Red MAGNET STOP (ERDU) push-button

When the ERDU button is pressed the MSUP energizes the quench and switch heaters of the magnet. This results in a quench of the superconducting magnet coils and the magnetic field decays within 20 seconds.

To protect against accidental operation, the button cover is protected by a tamper-proof seal. The seal must be replaced after operation of the button.

NOTE The tamper seal is not the only proof of intentional quench. It is also logged in the MSUP log file MSUPHistory.log

Switches

Switch	Description
SYSTEM ON	All MR-System components are powered up
SYSTEM OFF	All MR-System components are switched off (except MSUP, MREF and EPC)
LOCKING KEY	To prevent from unauthorized power switching
STOP BUTTON	Emergency Rundown Unit (ERDU) for magnet
AUDIO ALARM OFF	Resets the relay contact and the buzzer

Connectors

Connector	Location	Description
X1	Alarm Box	Power switching signals, helium compressor control, MSUP control signals
X119	EPC roof	External customer alarm (pin 1 = N/O, pin 2 = N/C, pin 3 = common) Dedicated quench alarm (pin 4/5 = N/C)

LEDs

LED	Description
LINE POWER	Power is supplied to the alarm box
WARNING	- Liquid helium level below warning or alarm-thresholds - Quench / switch heater error or ERDU button fault - Magnet pressure too high / low - Compressor stopped (low pressure switch operated) - ERDU battery voltage warning or alarm - EIS drive error

Advanced High Order Shim

Overview

With help of this electric shim option, the local magnetic field inhomogeneities caused by the Patient can be improved.

Passive shimming with iron plates reduces low and high order inhomogeneities caused by magnet tolerances and ferromagnetic components in the environment. Patients (and phantoms) also cause local field inhomogeneities that have to be compensated for those applications sensitive to inferior shim, e.g. Spectroscopy, FatSat, EPI (Echo Planar Imaging), TGSE (Turbo Gradient Spin Echo).

The correction of linear inhomogeneities is done via offset currents applied to the gradient coils. Second order inhomogeneities are compensated by applying currents to 5 shim coils that are integrated into the gradient coil:

- A(2,0)
- A(2,1)
- B(2,1)
- A(2,2)
- B(2,2).

(see also Tune-up, Phantom Shim)

Customers who want to make use of these shim coils have to purchase the Advanced High Order Shim that consists of:

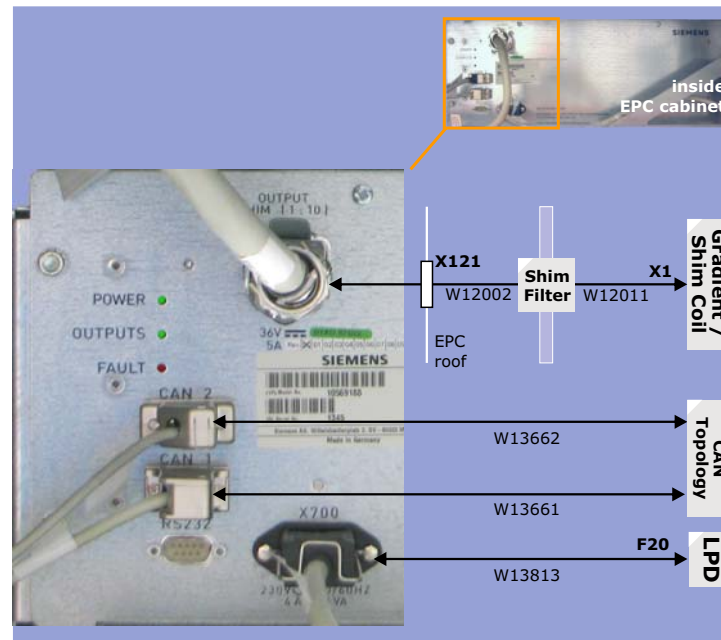
- Shim Power Supply Unit (SPS)
- Shim Filter Unit
- Shim cables

The SPS is mounted inside the EPC cabinet, the Shim Filter Unit is installed on the filter panel.

After installation of the hardware, the Advanced High Order Shim option has to be enabled in the *syngo* MR configuration.

Shim Power Supply (SPS)

Figure 205 SPS Block Diagram



General

The SPS is a self-contained unit mounted inside the EPC cabinet. It comprises one large printed circuit board containing most of the electronic parts and a small display board for front panel connectors and LEDs. The base of the unit contains two switched-mode power supplies and a cooling fan, together with a wiring loom and structural metal components.

Function

Five built-in switch-mode DC/DC converters are responsible for producing the shim currents for the 5 shim coils. The maximum current that can be supplied per channel is 5 A at a maximum output voltage of +/- 20 V. Current regulation is based on the principle of Pulse Width Modulation.

Shim currents are setup under *syngo* MR in the Tune-Up step "Phantom Shim". They are loaded automatically into the SPS via its CAN-bus connection. If required, these shim currents can be optimized temporarily for a patient measurement with help of the 3D-Shim. The SPS is able to report its status via CAN commands to *syngo* MR, i.e. error messages can be seen in the EventLog.

Specifications

Shim Power Supply Unit 2362	Specification
Weight:	13 kg (28.7 lb.)
Maximum magnetic field:	0-5 mT (50 Gauss)
Cooling:	air-cooled, internal fans
Line voltage:	198-253 VAC, 50/60 $\pm$ 1 Hz
Maximum current per channel:	$\pm$ 5 A DC, on all channels
Current drift due to self-heating:	< $\pm$ 5 mA change from initial current
Power dissipation:	< 250 W all outputs at 5 A, 20 V
Shim coil inductance:	<10 mH
Shim coil resistance A20: All other coils:	<1.5 Ohms <2.0 Ohms

LEDs

LED	Description
POWER, green	if lit, SPS is standby
OUTPUTS, green	Shim Outputs are switched on
FAULT, red	Error has occurred

Connectors

Channel#	Shim Coil	X121 pinout (EPC roof)	GC X1 Pinout
Channel 1 +	A20 +	1	14
Channel 1 -	A20 -	2	24
Channel 2 +	A21 +	3	1
Channel 2 -	A21 -	4	3
Channel 3 +	B21 +	5	10
Channel 3 -	B21 -	6	12
Channel 4 +	A22 +	7	26
Channel 4 -	A22 -	8	28
Channel 5 +	B22 +	9	35
Channel 5 -	B22 -	10	37

Shim Filter Unit

To supply the shim coils of the scanner inside the examination room, the output leads of the SPS are fed through a RF filter box, mounted at the filter plate. By that means, external interferences riding on the outside shim cables can not pass into the RF-cabin. The filters also protect the SPS from voltage transients induced by the gradient coil.

Specifications

Shim Filter Unit	Specification
Weight:	23 kg (50.7 lb.)
Maximum magnetic field:	0-10 mT (100 Gauss)
Cooling:	air-cooled, convection
Resistance of Filter Unit plus Cables:	< 1.2 Ohm, per channel
Power dissipation:	< 88 W all outputs at 5 A

Refrigerator System

Overview

The closed cycle cryogenic Refrigerator System consists of:

- **He Compressor** - water-cooled, provides compressed helium gas for the Cold Head.
- **Cold Head** - 4 K, recondensing of the boiled off helium and cooling of the 50 K magnet shield.
- **He Flexlines** - connection between He compressor and Cold Head.

Compressor + Cold Head are manufactured by SUMITOMO Heavy Industries Ltd., Japan.

He-Compressor F70

General

The compressor continuously draws low-pressure helium from the system return line. It compresses, cools and cleans the gas, then delivers it through the system gas supply line to the Cold Head.

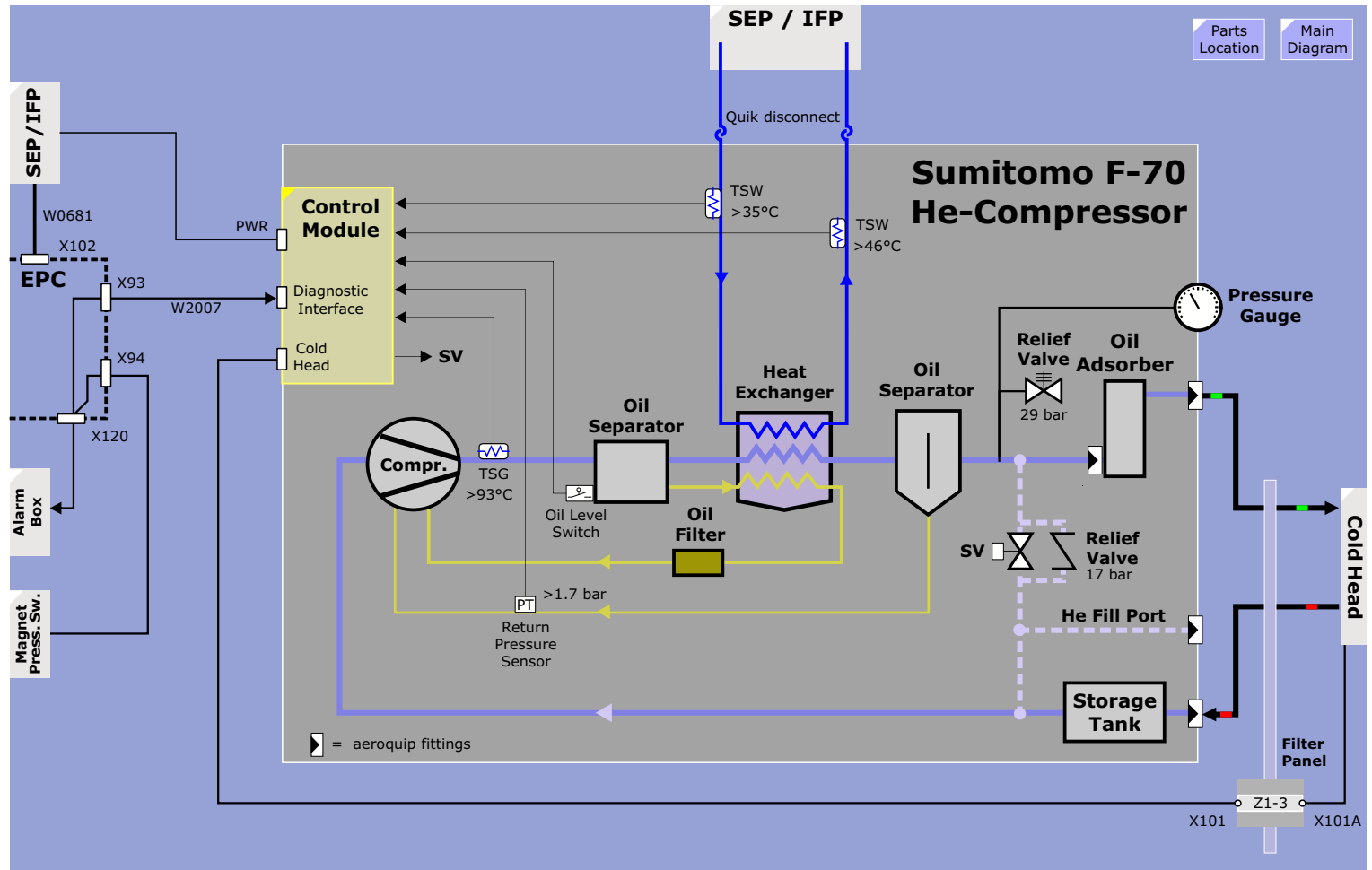
The He-Compressor comprises the modules:

- Compressor capsule
- Heat exchanger
- Gas purifiers
- Control module (electrical supply unit)

Figure 206 F70 Helium Compressor



Figure 207 F70 Compressor Block Diagram



Function

Compressor Capsule

The Cold Head needs to be supplied with highly pure He 5.0 grade gas (= purity >99.9990 %) in order to avoid internal contamination or freezing. When helium gas leaves the compressor capsule, the gas contains heat and compressor lubricant. Both must be removed.

Heat Exchanger

After being pressurized, the hot helium gas is passed through a helium/water-counter flow heat exchanger, located downstream of the compressor capsule. The gas temperature is lowered to about the temperature of the coolant.

A second oil/water-counter flow heat exchanger cools the oil of the compressor capsule.

Oil circulation is maintained by the difference in pressure between low and high-pressure helium. The temperature of the cooling water should be 5 °C to 25 °C at a flow rate of 6...9 l/min.

Gas Purifiers

Downstream from the heat exchanger, the helium is passed through an oil separator. The condensed oil is returned to the compressor capsule by means of capillary pipes.

Adsorber

The helium gas exiting the oil separator still contains small quantities of oil vapor. To prevent this oil from freezing inside the Cold Head, the adsorber removes any oil and moisture the gas is carrying. The adsorber is located behind the oil separators.

In accordance with the maintenance schedule, the adsorber must be replaced after every 30,000 hours of operation (~ 3 years).

Compressor Valves

Three important valves are built into the compressor:

- Atmospheric Relief Valve (**ARV**)
Prevents the compressor from operating at an unsafe pressure by venting to atmosphere.
- Solenoid Bypass Valve (**SV**)
Opens if compressor is switched off to equalize high and low pressure for Cold Head protection.
- Internal Relief Valve (**IRV**)
The internal relief valve opens to allow the compressor to be operated in the stand-alone mode or when the system gas lines are disconnected, to avoid overloading the motor.

Control Module

The compressor unit is equipped with several monitoring devices. Compressor status and errors are reported to the system via the Diagnostic Interface Connector located on the front panel.

The compressor ON/OFF status is displayed on the **Alarm Box** via the yellow Warning LED and is recorded in the *C:\MedCom\log\MSUCompressorStartHistory.log* and *MSUCompressorStopHistory.log* file.

Configuration Mode Switch

The configuration mode switch is located on the compressor front panel, and is used to configure the output from the DB25 remote control **Diagnostic Interface Connector**, depending on the magnet system type.

Configuration	Switch Position	Magnet System Type	ON/OFF
Mode 1	UP	10K Shield cooling	Enabled
Mode 2	DOWN	4K Re-condensing	Disabled

The **ON** and **OFF** buttons are **disabled** in configuration mode **2**, as the compressor is controlled by the magnet low pressure switch (PDCS), and MSUP via Alarm Box.



Setting the switch to MODE 1 will bypass the pressure control system resulting in sub-atmospheric pressure in the helium vessel - **risk of icing**. For OR98/OR99/103 magnets the configuration switch must be set to MODE 2.

External Control of He Compressor

- **Status and error reporting** to MSUP via Alarm box (Comp_Sig) - e.g. compressor running or off, helium under-pressure or over-temperature. In addition; the compressor helium pressure is returned to SeSo (Ret_Press)
- **Power-on** (Comp_Drive) signal from MSUP via Alarm box - this signal is set low if the MSUP is powered and there are no compressor control errors present.
- **Magnet low-pressure switch** - if magnet He pressure drops below "Atmospheric pressure +0.2 psig", the Magnet low-pressure switch will open. The MSUP shuts-down the compressor by disabling the Comp_Drive signal. The He reliquefaction process stops, thus preventing subatmospheric magnet pressure.
- **Reset signal** from MSUP via Alarm box (Comp_Reset) - to restart the compressor after a power or temperature failure.

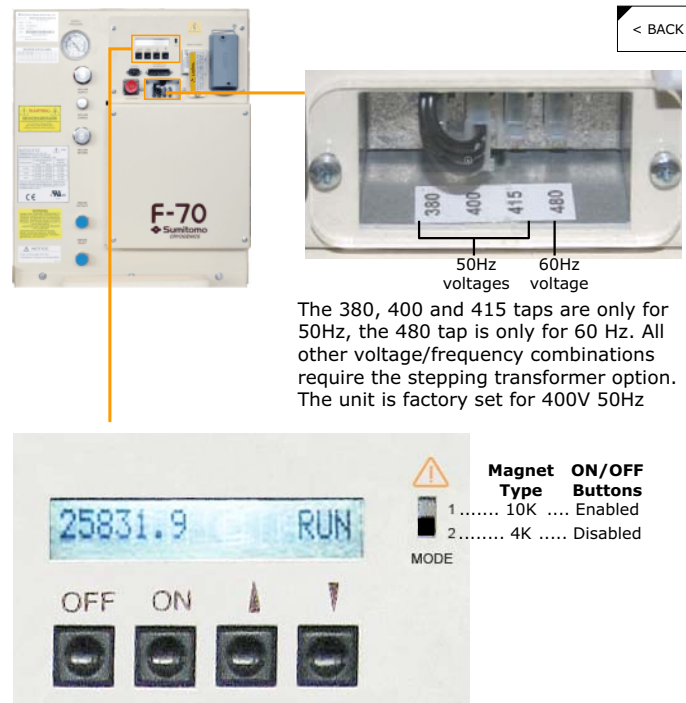
A female **DB-25 Diagnostic Interface Connector** is located on the front panel for external control, and to supply remote indication of the operating status of the compressor.

Mains Configuration

See Figure 208.

The Stepping Transformer option description is found "Installation / Options" in the CB-DOC.

Figure 208 F70 Helium Compressor



The first value shown on the display is the Run Time counter (total running hours since compressor last visited Sumitomo service center) and its state (RUN, STOP, OFF, ERROR).

Beginning with the firmware version 1.7 an **additional counter** was implemented which tracks the hours since the last adsorber change. Refer to the Preventative Maintenance instruction for details on viewing and resetting this counter.

Errors

Errors are displayed on the LCD System Display. The following compressor errors may occur:

- **Over-temperature** of Helium, cooling water and compressor capsule.
- Helium over or low pressure
- **Supply voltage errors** - Wrong voltage/frequency settings, reversed/missing phases.

NOTE If the compressor temperature exceeds limit it may take 20 to 60 minutes until the compressor can be reset.

Specifications

The compressor is shipped from the factory with a static pressure of 16.5 bar. The final pressure must be checked and should be adjusted to the required pressure at installation.

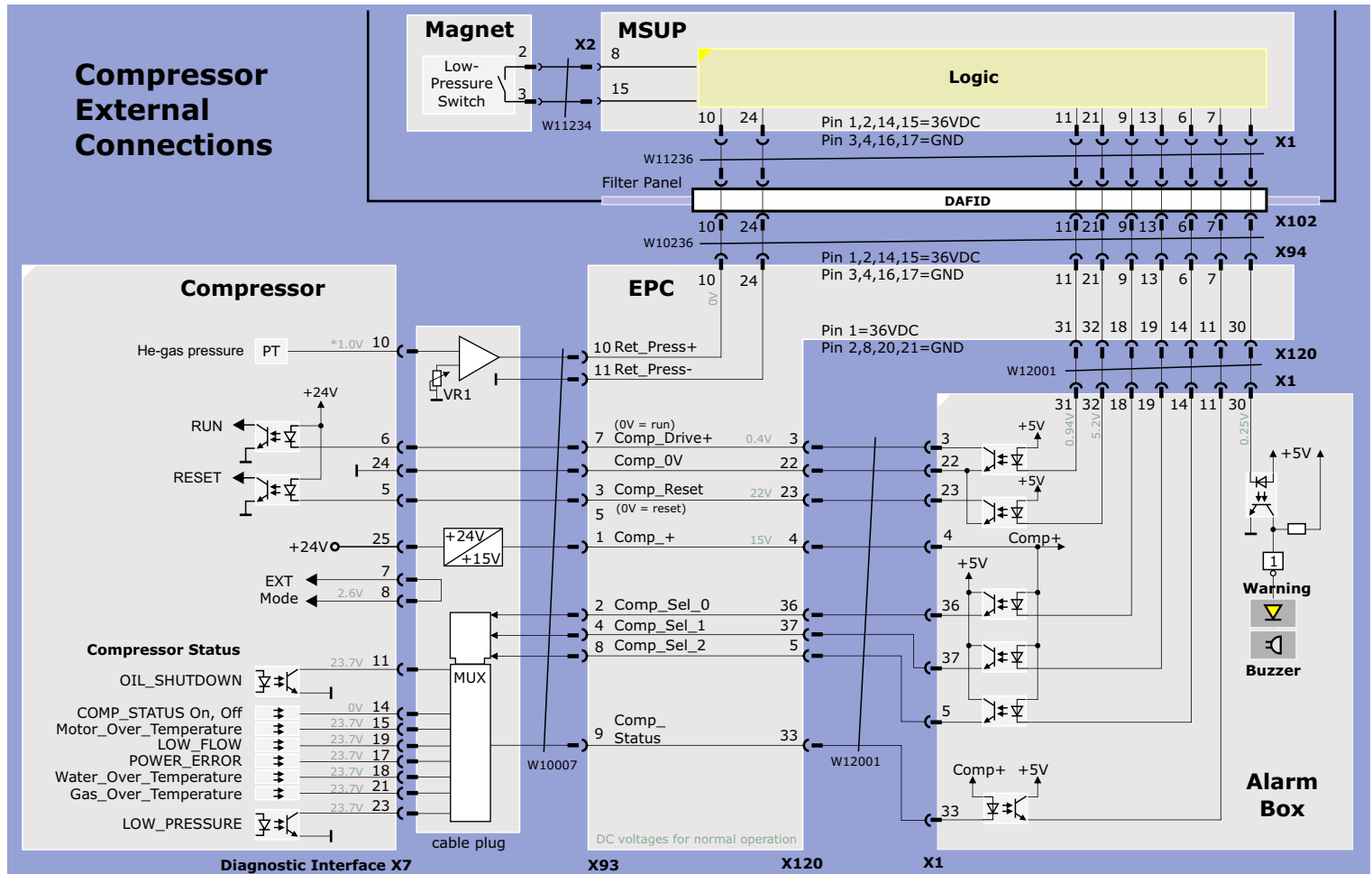
Equalization (static) pressure at 20⁰ C (68⁰ F) for 12 to 20 m long gas lines:

Specification	SUMITOMO F-70H
Water in and out fittings	2 x 1/2 inch MNPT
Circuit protection (fuses FU-1,2,3)	1 A, 600 V, class CC
Rated load current	16 A
Cooling water flow rate	4 K & 10 K Coldhead - 6 - 9 L/min. (1.6 to 2.4 gpm)
Cooling water inlet temperature	5 ⁰ C to 25 ⁰ C (41 ⁰ F to 77 ⁰ F)
Cooling water outlet temperature	45 ⁰ C (113 ⁰ F) maximum
Water filter	< 300 micron
Weight	100 kg (220 Lbs)

F70 Helium Gas Pressures

Cold Head	Static Pressure	Dynamic Pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal)	21 - 22 bar

Figure 209 F70 Compressor Control Diagram



Cold Head

General

Heat radiation plays the most important role when considering the heat impact to the cryo shield and insulation foils inside the magnet. Heat radiation must be reduced as much as possible to keep the helium boil-off rate to a minimum. This is achieved by the Cold Head maintaining the internal cryo shield at 50 Kelvin.

The magnet uses a SUMITOMO 4 K recondensing Cold Head which is able to liquefy the boiloff helium gas. A "sidesock" interface to the helium vessel allows easy replacement of the Cold Head at end of its lifetime.

In addition, the Cold Head recondenses any remaining boiloff by cooling the gas to 3.7 Kelvin (He liquification point = 4.2 Kelvin). Providing the heat load to the helium vessel is lower than the Cold Head cooling power, no helium will be lost.

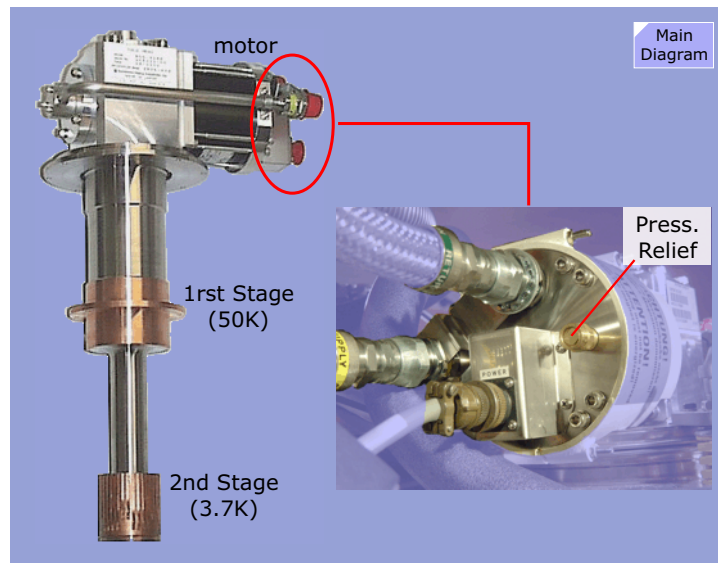
Function

The Cold Head expands the high pressure (approx. 21 bar) helium gas supplied by the compressor according to the Gifford-McMahon principle. Inside the Cold Head, a displacer piston is moved up and down in a metal cylinder, expanding the He gas in 2 chambers. Piston movement is controlled by a motor-driven Scotch Yoke drive, coupled to a rotary gas control valve. The Cold Head has two temperature stages:

- **50 Kelvin Stage (first stage)** - Connected to a single cryo shield.
- **3.7 Kelvin Stage (second stage)** - Located in the helium vessel above the liquid level. The boiloff He gas will condense on the second stage and return to the helium pool.

After expansion inside the Cold Head, the low pressure (approx. 7 bar) He-gas returns to the compressor (closed loop helium gas circuit). In case a compressor He gas refill is necessary, highly pure 5.0 grade helium gas must be used (purity > 99.9990 %).

Figure 210 Cold Head



Specifications

Cold Head	Specification
Refrigeration capacity 1st stage	50 W @ 30 K - 45 K
Refrigeration capacity 2nd stage	1 W @ 3.5 K - 4.2 K
Motor Rotation Speed @ 50 Hz / 60 Hz	60 rpm / 72 rpm
Weight	19.5 kg (42.9 lbs)

Service Routines

Disconnecting Cold Head Gas Lines

During normal operation the Cold Head contains small quantities of liquid helium inside. When the cold head is turned off this helium will expand resulting in high pressures that can damage the cold head (the PRV could stay open too long leading to contamination).

The gas lines will provide a sufficient buffer if they are left connected.

To avoid contamination when gas lines need to be removed:

- turn the compressor unit OFF and wait a minimum of 30 minutes to allow for the gas expansion
- disconnect the gas lines from the cold head and fit the Aeroquip connections with dust caps



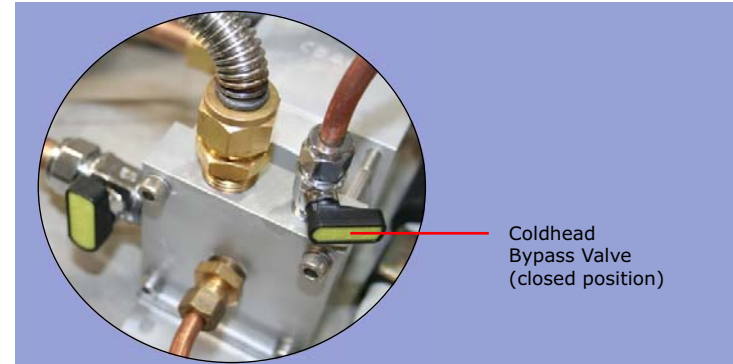
WARNING

Damage to the cold head occurs if sufficient time is not allowed for the gas to expand in the refrigeration circuit.

Wait a minimum time of 30 minutes before disconnecting the gas lines.

Cold Head Bypass Valve

Figure 211 Cold Head bypass valve *CLOSED*



The Cold Head is connected to the magnet turret via a permanent gas line.

To reduce static boil off during long periods when the Cold Head is turned off, the bypass line must be **OPEN**. During normal operation, the valve remains **CLOSED**.

Cooling System

Introduction

The Cooling System is needed to remove undesirable waste heat developed in several scanner components in order to assure and maintain system stability and reliability. Water is used as the primary coolant being applied to cooled components via **heat sinks** for direct cooling or through water-to-air **radiator heat exchangers** for indirect ambient cooling.

Components using heat sinks for direct cooling are the gradient amplifier, gradient coil, helium compressor, TX Box and the BB RFPA found in the MNO (multi-nuclei spectroscopy) option.

The RFCEL and ACS units are equipped with radiators for providing inter-cabinet air climatization primarily for cooling electronics boards.

Two cool system variants have been made available to meet customers needs:

SEP Variant

The SEP is a passive cooling aggregate using a heat exchanger requiring an external chill water source for heat transfer. The SEP variant is intended for customers having a central in-house chill water supply or an existing chiller unit.

Chiller/IFP Variant

The Chiller is an active cooling aggregate providing 20°C chill water to the scanner. There are two variants of this chiller

corresponding to the two gradient system strengths (see table below). The chiller is integrated into the MR system in that the scanner provides system on/off status to the chiller's controller for regulation purposes. The chiller in turn provides status feedback to the MR system for logged into the system's event log.

Free Cooler Option

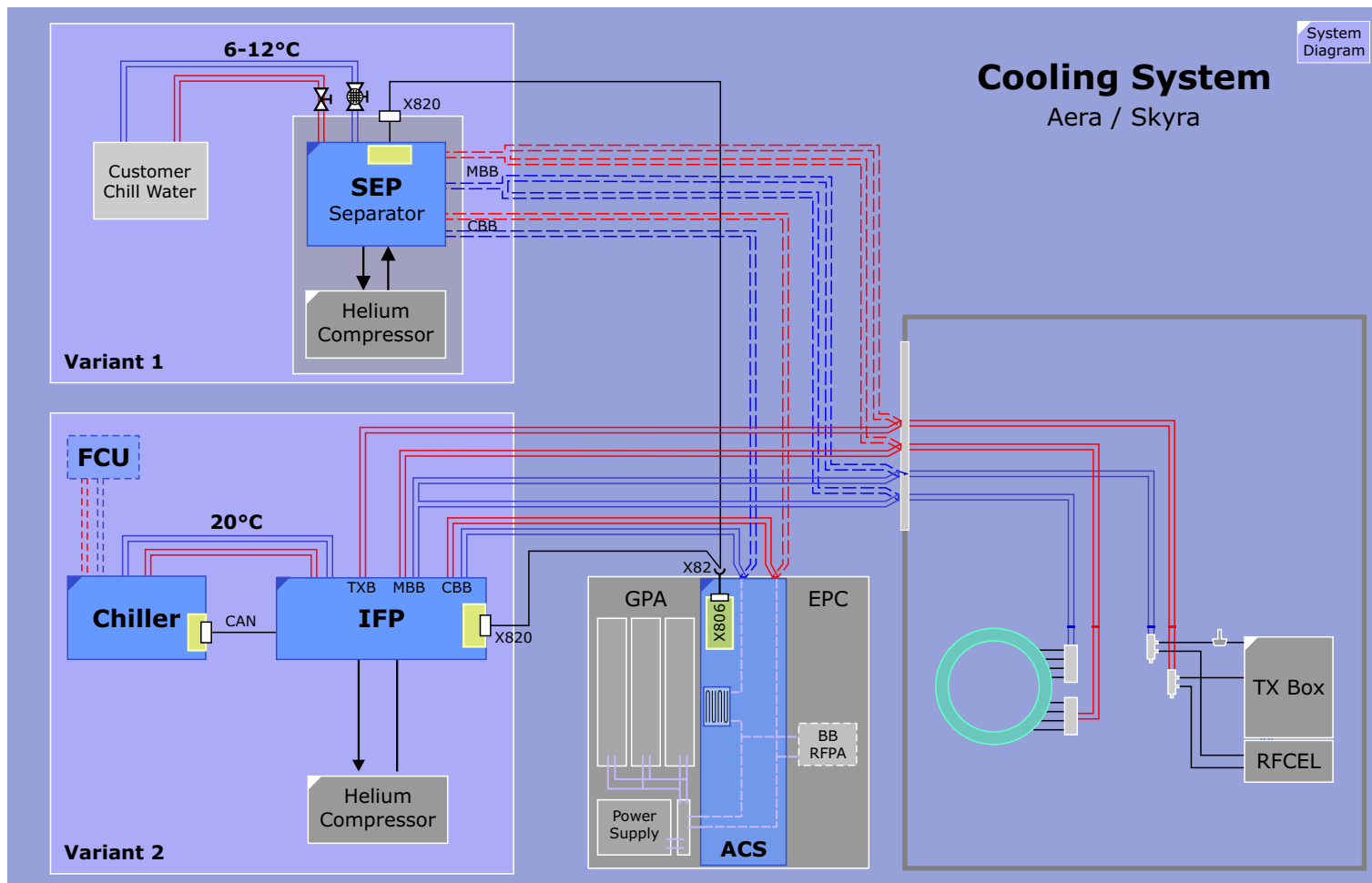
As option an additional passive cooling unit consisting of a radiator heat exchanger with a fan assembly can be installed to cool the chill water using outside ambient temperature. At 18°C and below the unit is automatically enabled. This feature has the potential to dramatically reduce running costs (electricity), up to 50%.

Cooling System Configurations

Hardware	SEP	Chiller
XJ-gradient	45 kW	
		45kW
		45 kw + optional FCU
XQ-gradient	60 kW	
		60 kW
		60 kW + optional FCU

Figure 212 provides an overview of both variants and their connections to the scanner components.

Figure 212 Cooling System Block Diagram



Air Conditioning System (ACS)

Overview

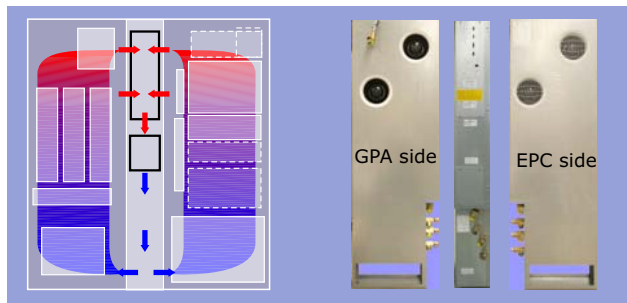
The ACS incorporates two functional blocks:

- EPC/GPA cabinet climatization
- Cooling water distribution to components in EPC/GPA
- Communications bridge for cooling system components

Cabinet Climatization

Blowers located on the Control-Blower Module draws in warm air from the GPA and EPC cabinets and forces it over a radiator supplied by 20°C coolant water from the cooling system. Blowers draw in air through cut-outs at the top and force it over the heat exchanger. through cut-outs at the bottom completing the air circulation path.

Figure 213 Cabinet Cooling Circulation



Air Temp Monitoring

Temperature sensors **R1** and **R2** measure the return air temperature and supply air temperature respectively. The thresholds for the air temperatures:

- air outlet temperature: 27°C = warning
- air inlet temperature: 38°C = warning
- air inlet temperature: 43°C = error - leads to measurement stop

The warning thresholds can be set in the **SeSo/ Magnet & Cooling/Cooling Setup** mask for test and service purposes.

Cooling Water Distribution

The 20°C coolant water supplied to the ACS is further distributed to components in the EPC or GPA cabinets requiring cool water:

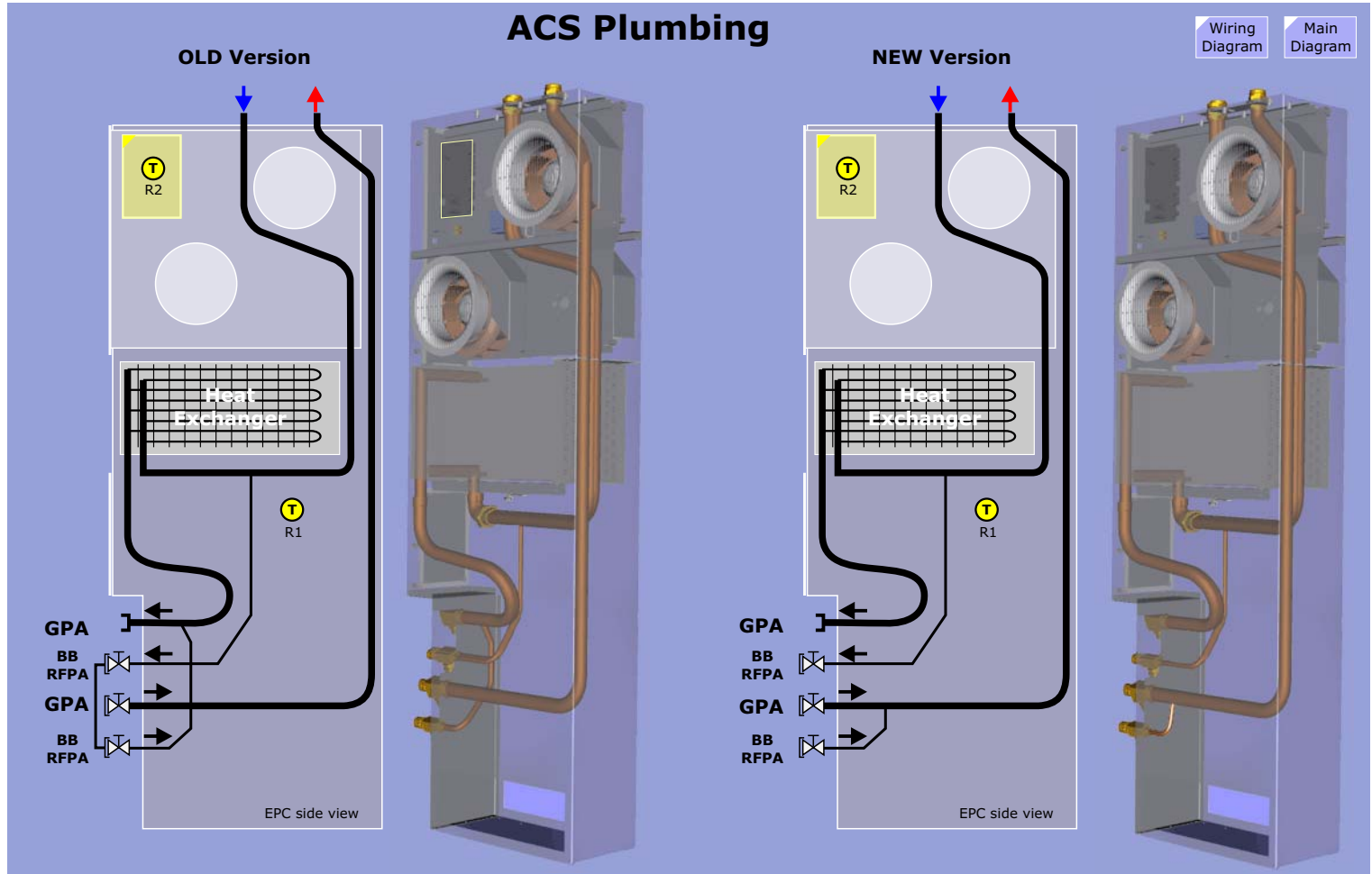
Sub-system	Cooling Capacity
GPA	
Final Stages	< 16 kW
Rectifiers	< 4 kW
ACS	
Flow / cool capacity	< 5 kW
Dressler BB RFPA (MNO option only)	
Flow / cool capacity	< 6 kW

There are no valves to adjust to attain these flows. Proper flow to the system components is assured by the diameter of the provided hosing and flow reducers inserted in the cooling water paths. As long as the pump is working properly, the flows will be ok. Flow sensors located in the SEP or on the IFP check only coolant flows to the EPC/GPA and RF room components. Flows through the individual components are not monitored by the cooling system, although some components do monitor coolant supply indirectly via internal temperature monitoring, i.e. the Dressler RFPA.

Communications Bridge

The ACS acts at connection point for the CAN bus communications from and to the SEP, Chiller and IFP components. The ACS however does not exercise control over these components.

Figure 214 ACS Plumbing Diagram



ACS Wiring

Power

The power is supplied from the LPD over two circuit breakers:

- **LPD.F35 - L**
fan power, is switched on/off with the system (after K2)
- **LPD.F32 - L'**
controller power, is NOT switched off with system (before K2)

WARNING Pin 4 is therefore ALWAYS hot!

Control Unit N1

The control unit N1 provides:

- air temperature monitoring of both return air (R2, found on the controller board) and air supply (R1, located beneath radiator)
- blower fan monitoring (rotation sensors)
- Gradient Filter blower fan monitoring (temperature sensors)
- Humidity sensor interface (option)

Service Comm Port

A RS232 port is provided (X808) for serial communication. The port is open and not password protected. Regulation and soft limits can be read out and changed, however, only some changed parameters will be over-written by the Host/MARS 2 at next boot. Other values will not. Bottom line: look, but don't touch.

Gradient Filter Temperature

The ACS is also responsible for evaluating the temperature switches monitoring the Gradient Filters. If a filter overheats one of the bi-metal temp switches will open and error will be reported to the system.

Humidity Sensors

Optional rH (relative humidity) sensors can be installed in the technical and examination rooms for monitoring purposes. Limits are set under SeSo/Magnet&Cooling. When limits are exceeded the error will be logged in the Event Log and can be read out over the CAN24 via SRS.

NOTE This option requires an ACS controller with version 3 or higher and will not be available with the MNO option.

SeSo / Magnet & Cooling / Cooling Status

Status

EPC	Air Return Temperature	28.1 °C
	Air Supply Temperature	21.3 °C
	GPA Fan 1	OK
	GPA Fan 2	OK
	EPC Fan 1	OK
	EPC Fan 2	OK

Equipment Room

GraFi Fans	OK
Relative Humidity	55.0 %
Room Temperature	21.9 °C
Absolute Humidity	9 g/kg
Atmospheric Pressure	1013 mbar

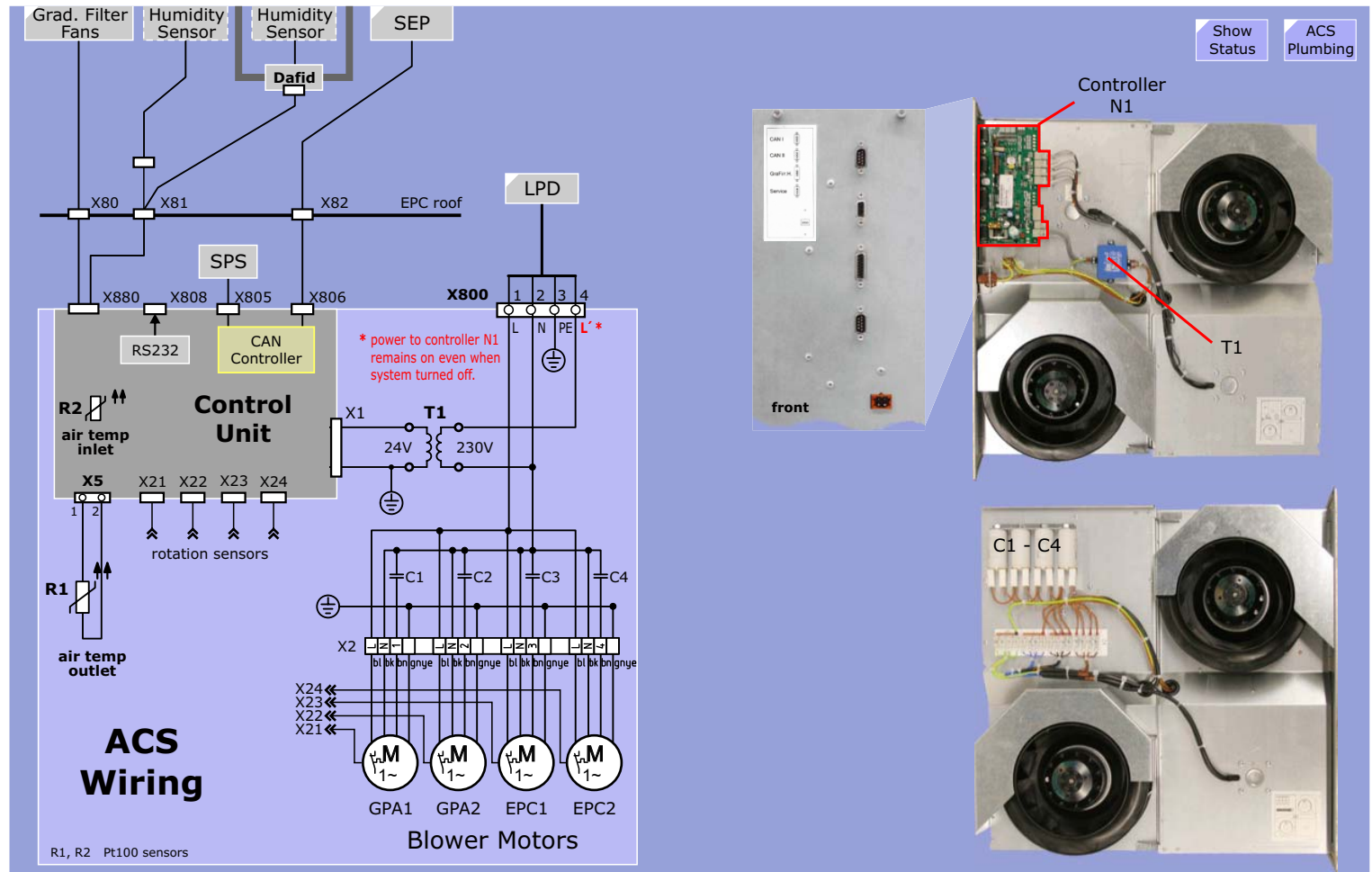
Setup

EPC	Air Return Temperature	Max.	38.0	°C
	Air Supply Temperature	Max.	27.0	°C

Equipment Room

Relative Humidity warning level	Min.	10.0	%
	Max.	80.0	%
Relative Humidity warning level	Max.	13.0	g/kg
	Min.	15.0	°C
Air Supply Temperature	Min.	15.0	°C
Air Supply Temperature	Max.	30.0	°C

Figure 215 ACS Wiring Diagram



SEP (Separator)

Overview

The Separator (SEP) cabinet houses both the SEP cooling unit as well as the MREF compressor.

The SEP cooling unit provides:

- regulation of 20°C secondary coolant water
- regulation of pressure
- shut-off valves and drain ports for service

Specifications

Specification	Value
Primary	
min. flow in primary circuit for XJ/XQ	80/100 l/min.
temperature bandwidth	6-12°C
maximal pressure difference	1 bar
Secondary	
nominal temp	20°C
temperature bandwidth (temp may leave this bandwidth for maximum 30 sec.)	19-22°C
alarm temp	≤15 or ≥26
nominal pressure difference	3.3 bar
max pressure difference	3.5 bar
Flow	regulated for demand

Water Circuit

Function

The SEP controller sensors the secondary water temperature and will regulate the secondary water flow through the heat exchanger using a 3-way valve in order to maintain a constant 20°C temperature. If the secondary water heats up, it will open the 3-

way valve to increase the flow through the heat exchanger (mix cooler) or close the 3-way valve to reduce the flow (mix warmer) as the heat in the water is dissipated to the primary water via the heat exchanger.

It is important that the primary water temperature stability of 2K/5min is maintained, otherwise the secondary temp cannot be kept constant. This may result in problems in image quality due to instabilities of the components being cooled, primarily the RF amplifier.

Temperature swings of the primary temp is not monitored, only the temp limits!

In case of a power failure the SEP must turn itself on again automatically.

The pressure differential (nominal 3.3 bar) between supply and return side of the SEP is regulated by the SEP pump using pressure sensors. This assures the proper flows to the components and the maximum allowed pressure (6 bar) is not exceeded.

Filling

A **pressure gauge** (18) displays static fill pressure and is provided for refilling. The **fill port** (5c) is located directly on the motor. A **safety valve** (15) with a water run-off hose located on the pump intake side is set for an over-pressure of 3 bar. This prevents the SEP from being over-filled.

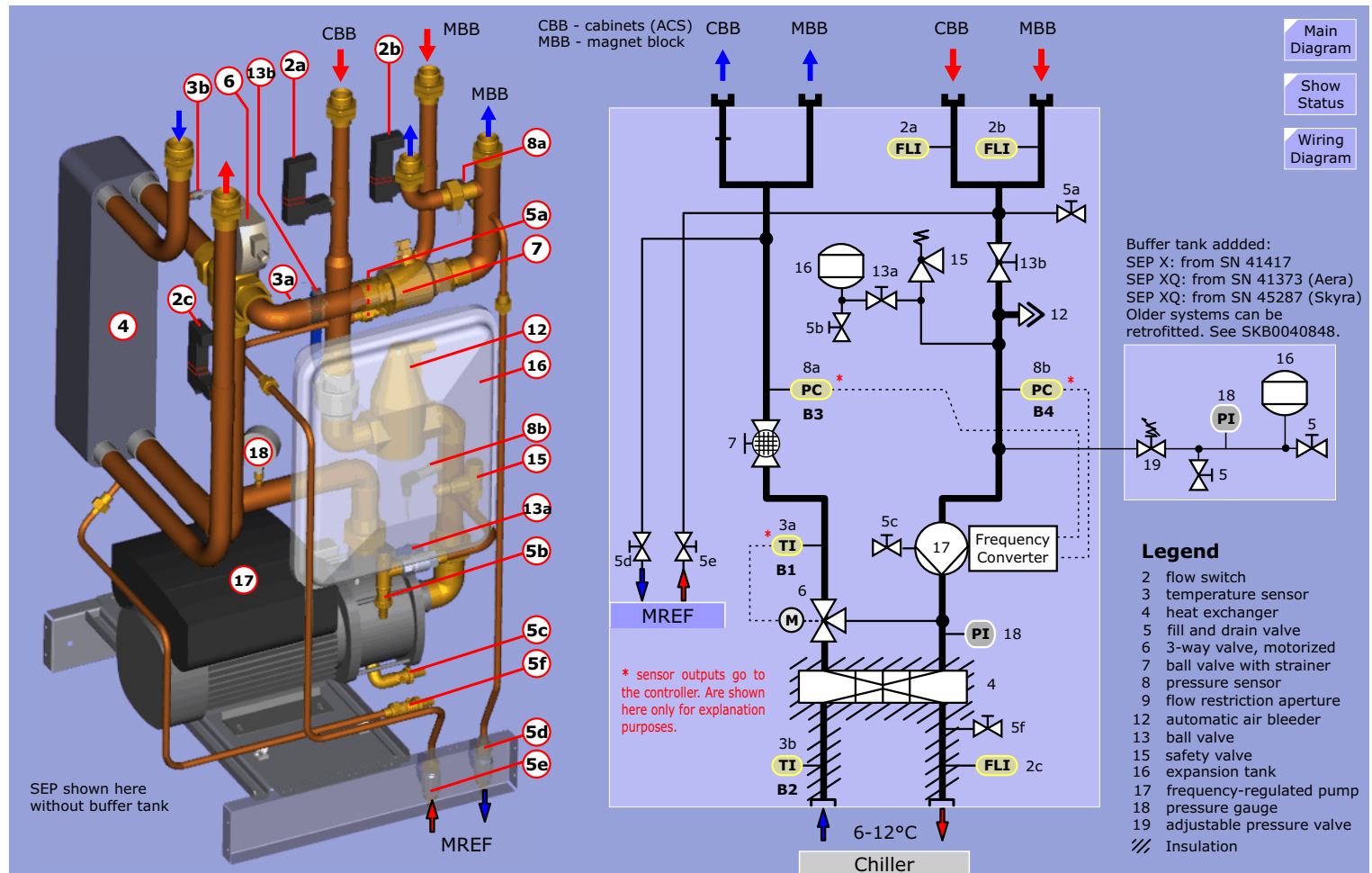
Maintenance

A unique **filter valve** (7) containing a built-in screen filter allows a very quick cleaning of the filter during maintenance. Close the valve and open the access screw cap to gain access to the filter. There is only about 1-2 cups of fluid trapped inside the valve.

Second Expansion Chamber

Beginning in May 2012 a second expansion chamber has been added to increase the water capacity, prolonging refill cycles. Retrofit of older systems is possible, check SKB0040848.

Figure 216 SEP Plumbing



Electrical Circuit

Overview

The **Control Unit N1** reads in the analog temperature and pressure sensor values and evaluates them against limits which are sent to it via the CAN bus. Soft limits can be entered into the **SeSo/Magnet and Cooling/Cooling Setup** mask.

Function

Circuit breaker **Q2** supplies the MREF (compressor) with AC power. If it trips, a signal is sent to the controller and LED **I1** goes out.

Relay **K1** is a phase rotation check device and will only enable **K3** (SEP pump supply) when the phase rotation is correct.

Circuit breaker **Q3** supplies the SEP water pump. If it trips it sends a signal back to the controller and LED **I5** goes out.

Flow sensors **FLI 1-3 (2a-2c)** monitor incoming primary and outgoing secondary water supplies. The limits are hard coded.

B1-B2 KTY temperature sensors measuring secondary and primary water temperatures respectively. B1 is used for the regulation.

B3-B4 pressure sensors indicating the dynamic (output pressure) and static (input pressure) of the secondary.

M5 drives the 3-way valve and is controlled by the Control Unit N1 via two on-board relays. The drive action is displayed by LEDs **O1** (make secondary colder) and **O2** (make secondary less colder).

Status is passed on to the MARS 2 via the CAN bus.

LEDs

LED	Description	normal condition
H1	CPU active	blink
H2	warning	OFF
H3	error	OFF

Inputs

I1	no overload MREF	ON
I2	flow switch MBB	ON
I3	flow switch CBB	ON
I4	flow switch primary	ON
I5	no overload pump	ON
I6	no error pump	ON
I7	--	--
I8	XQ ID	ON if XQ version of SEP

Outputs

O1	mix colder	ON indicates MORE flow through heat exchanger
O2	mix warmer	ON indicates LESS flow through heat exchanger
O3	--	--
O4	--	--
O5	pump enabled	ON

Figure 217 Separator Wiring Diagram

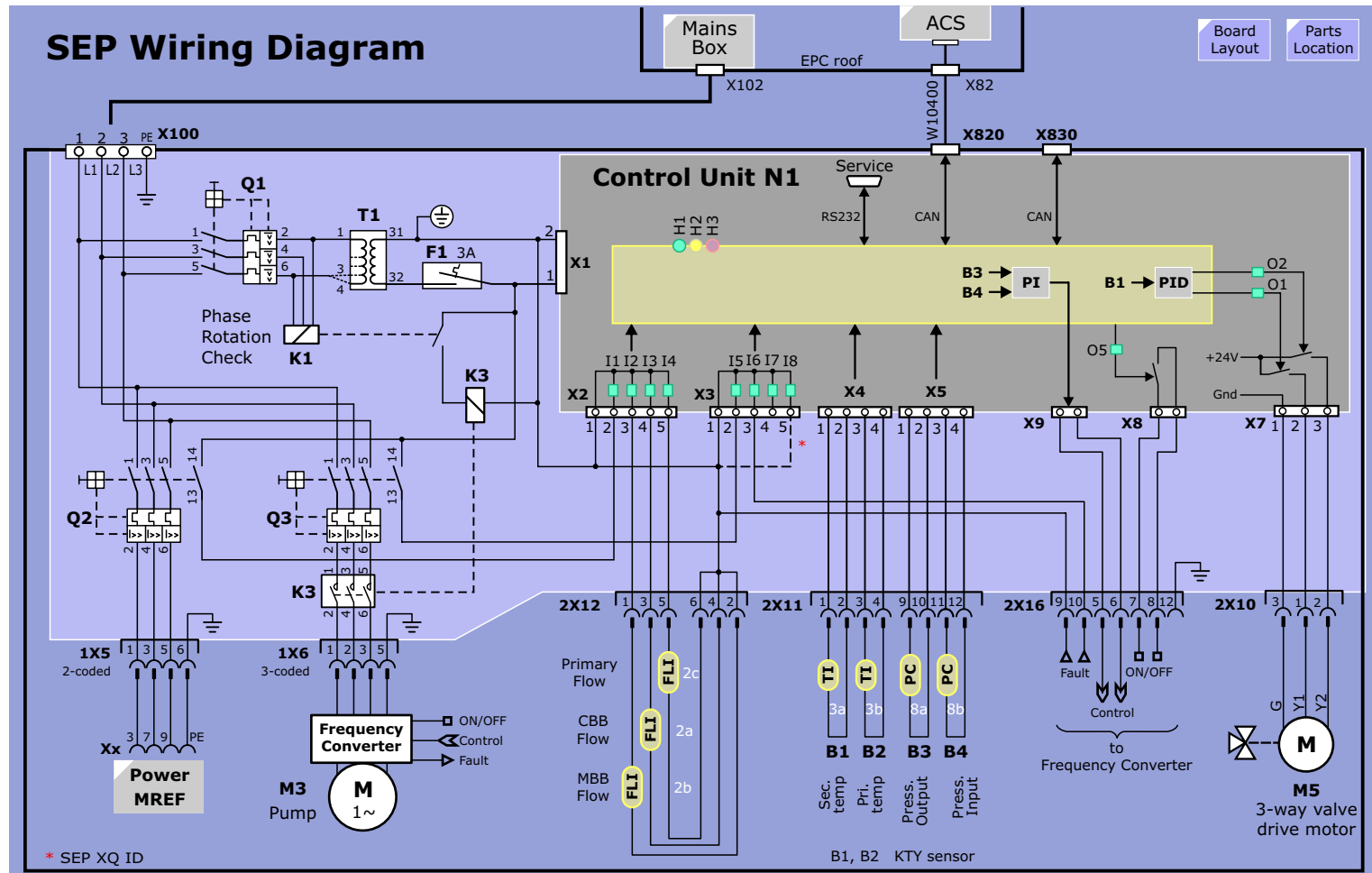
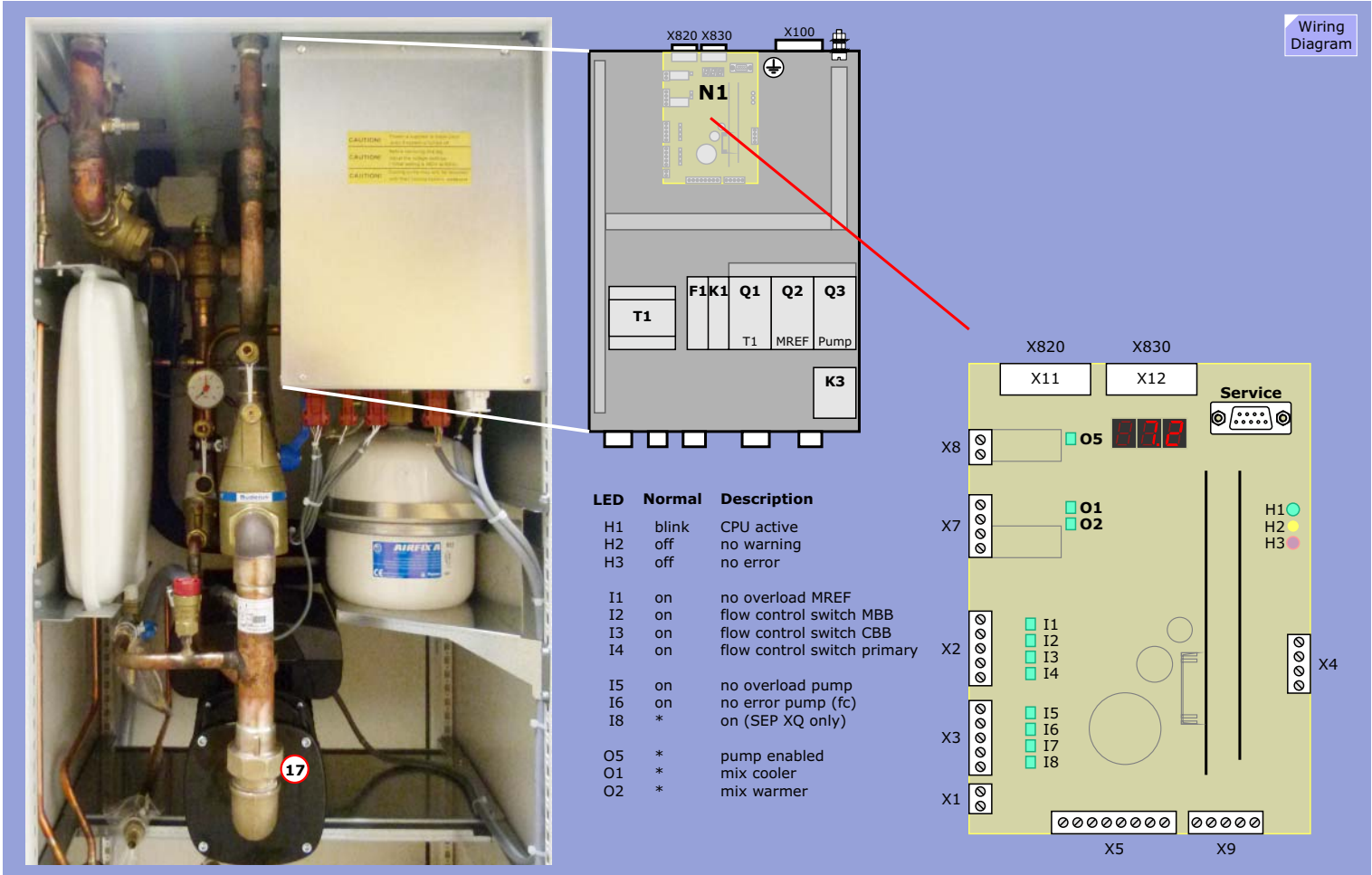


Figure 218 SEP Controller Layout



Display Codes

The Controller display can display the following information:

Value	Description
XX.X	The normal display is the primary water temperature in degrees Celsius
ECO	Ecological mode. When the system is put into standby or turned off, the SEP lowers the pump speed to reduce power.
CAL (Calibration)	If the SEP loses power and is turned back on again it needs to calibrate its 3-way valve (full open/full closed). The procedure takes about 170 seconds.
Fxx (error code)	Error codes. See Trouble Shooting Guide for the list of error codes and their meaning.

Monitoring Levels

Value	Limit
Primary	
Temperature WARNING - LOW	4°C
Temperature WARNING - HIGH	14°C
Temperature ERROR - LOW	1°C
Temperature ERROR - HIGH	25°C
Secondary	
Temperature WARNING - LOW	16°C
Temperature WARNING - HIGH	25°C
Temperature ERROR - LOW	10°C
Temperature ERROR - HIGH	30°C
Temperature STOP	35°C
Differential Pressure	
Set point - NORMAL Mode	3.3 bar
Set point - ECO Mode	1.4 bar
Pressure	
Supply WARNING	5.8 bar

Value	Limit
Supply ERROR	6.0 bar
Return WARNING - LOW	0,7bar
Return WARNING - HIGH	1,3 bar
Return ERROR - LOW	0,3 bar
Return ERROR - HIGH	1.8 bar

SeSo / Magnet & Cooling / Cooling Status

Figure 219 SEP Status and Setup Masks

Status

SEP	Secondary Water	Supply Pressure	4.28bar
		Return Pressure	0.97bar
		Temperature	20.0 °C
		Temperature Set Point	20.0 °C
	Primary Water	Flow MBB	OK
		Flow CBB	OK
		Pump	OK
		Pump Overload	OK
		Temperature	6.6 °C
		Flow	OK
		Pump Inverter	OK
		Pump Revolution	90.5%
		Mixer Position	21.2%

Setup

SEP	Secondary Water Supply Pressure	Max.	5.8	bar
		Max.	1.3	bar
	Secondary Water Return Pressure	Min.	0.7	bar
		Max.	25.0	°C
	Secondary Water Temperature	Max.	14.0	°C
		Min.	4.0	°C

Chiller

Overview

The Chiller and Interface Panel (IFP) supply the 20°C cooling water to the scanner components requiring coolant.

The Chiller can be blocked into three functional sections:

- Chill Water section
- Refrigerant section
- Control section

Chill Water Circuit

The chill water circuit consists of the components found in [Figure 221](#). The main component is the pump. Several sensors and indicators are provided for regulation of the water temperature and pressure. The input and output temperatures as well as the flow are monitored and evaluated. Values lying outside preset limits result in error messages which will be transferred to the MARS 2 via its CAN interface.

Refrigerant Circuit

The Refrigerant side of the Chiller can only be serviced by external service personnel. Refer to the Planning Guide for a list of service providers in your country.

Status Messages

The chiller monitoring issues error messages which differentiate between water and refrigerant circuit errors.

Error Indicators

That the chiller is outside and often inaccessible it was decided to put status LEDs relating to the Chiller's on the [IFP, which see](#).

SeSo / Magnet & Cooling / Cooling Status

Chiller	
Chiller	OK
Pump	OK
Fill/Static Pressure	1.29 bar
Water Circuit	OK
Refrigerant Circuit	OK
Pump Inverter	OK
Pump Revolution	58.0 Hz
Compressor Inverter	OK
Compressor Revolution	56.0 Hz

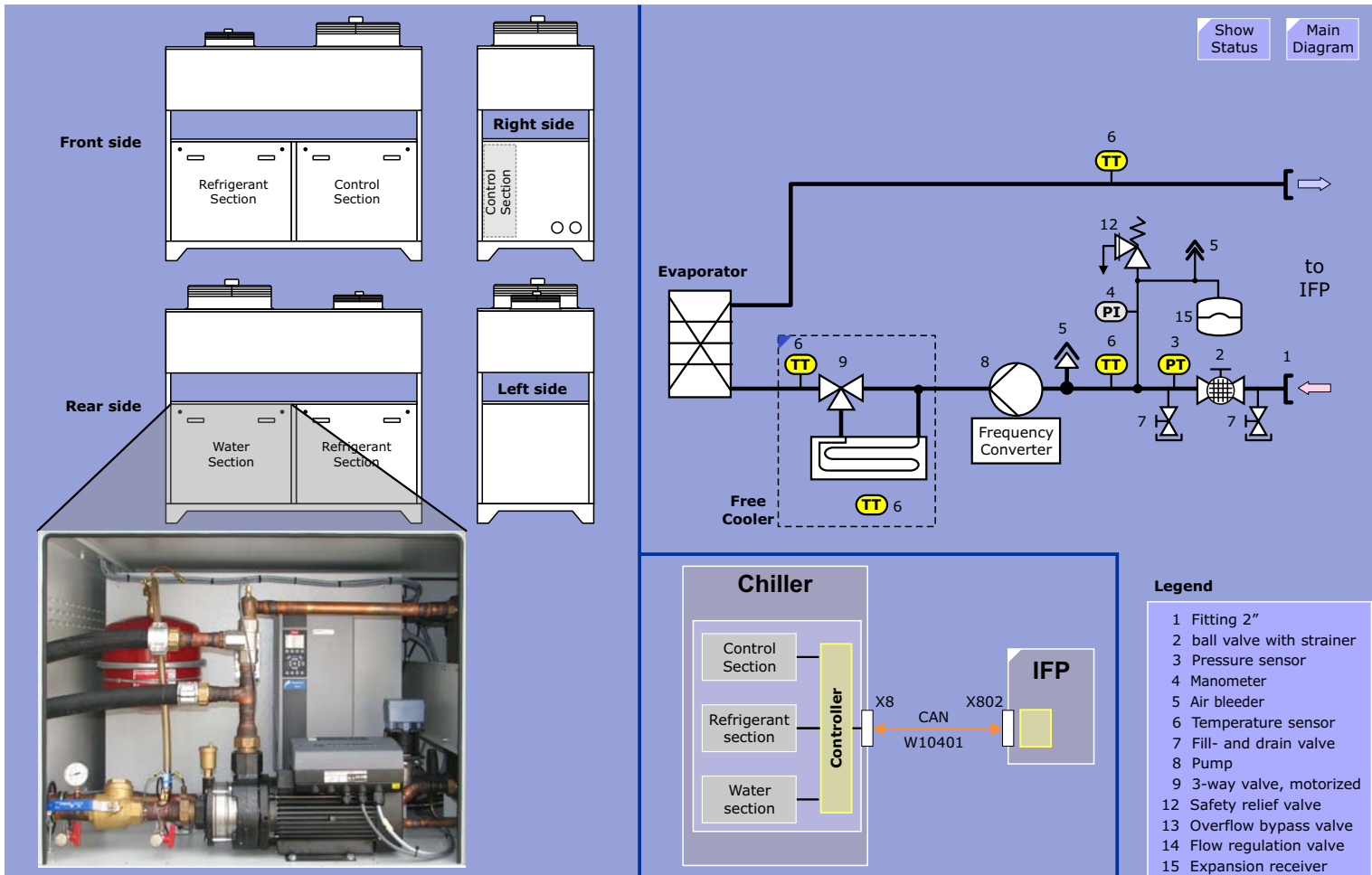
Free Cooler Unit (FCU)

As an option an external cooling fan is available. When installed it uses ambient air to cool the water. It is enabled when outside ambient air temperature is between 5-18°C.

Figure 220 Free Cooler



Figure 221 Chiller - Water Circuit Diagram



Interface Panel (IFP)

Overview

The IFP provides the following functions:

- cut-off valves for decoupling chiller from scanner
- Filter valve
- water distribution to scanner components
- monitoring of water temperature, flow and pressure
- water bypass (if incoming water temp falls below 16°C)
- booster pump for TX Box (creates under-pressure on return side of TX Box)
- automatic air bleeder
- power output for MREF compressor.

Functions

Monitoring

If the incoming cool-water from the chiller reaches 16°C, the 3-way valve will bypass the cool-water to prevent condensation. The bypass valve will be returned to normal when the water rises above 16°C.

The two pressure sensor outputs are used by the chiller to regulate the pressure. The chiller pump can regulate pressure differences when the chiller is placed up to 10m higher or up to 30m lower than the system.

SeSo / Magnet & Cooling / Cooling Status

IFP	Water Supply Pressure	5.18 bar
	Water Return Pressure	1.69 bar
	Water Temperature	18.9 °C
	Water Flow MBB	OK
	Water Flow CBB	OK
	Bypass	OK

LEDs

A set of 8 LEDs located on the IFP controller are used to display both the Chiller and IFP status. The first green LED is not part of the error codes and is on when performing a reset. The last 7 LEDs are used for error indication according to the following table:

LED Pattern	Description
	emergency stop (open circuit)
	refrigerant high pressure warning
	refrigerant low pressure warning
	water temp return high
	superheat <3K
	temp sensor suction gas
	superheat >15K
	water temp supply high
	low pressure chiller (refrigerant)
	sensor booster water
	sensor FCU water
	water return pressure sensor broken
	refrigerant high pressure error A17
	refrigerant high pressure error A18
	free cooling unit temp sensor malfunction
	water return press. <0.1bar, pump stopped
	phase error on AC power input
	flow error protection ~1bar press. difference
	thermal contact FAN2
	frequency drive pump
	thermal contact FCU Fan
	frequency drive compressor 1
	frequency drive Fan 1
	high pressure chiller refrigerant
	frequency drive booster fan
	motor protection compressor 2
	motor protection compressor booster

LED Pattern	Description
	motor protection fan 2
	motor protection FCU
	chiller comms down
	SD card removed
	IFP supply pressure AI2
	IFP return pressure AI3
	Ambient temp AI10
	refrigerant high pressure sensor booster AI9
	motor protection pump 1 TX Box
	flow switch pump 1 TX Box
	Service mode
	pipes connected wrong or sensors on IFP changed
	40°C safety cutoff for the pump
	water supply pressure high
	sensor temp return error chiller
	water temp supply low
	sensor temp supply error IFP
	return pressure low
	flow volume MBB (to magnet room)
	flow volume CBB (to cabinets)
	auxiliary switch MREF
	ambient high
	sensor temp supply error chiller
	supply pressure high

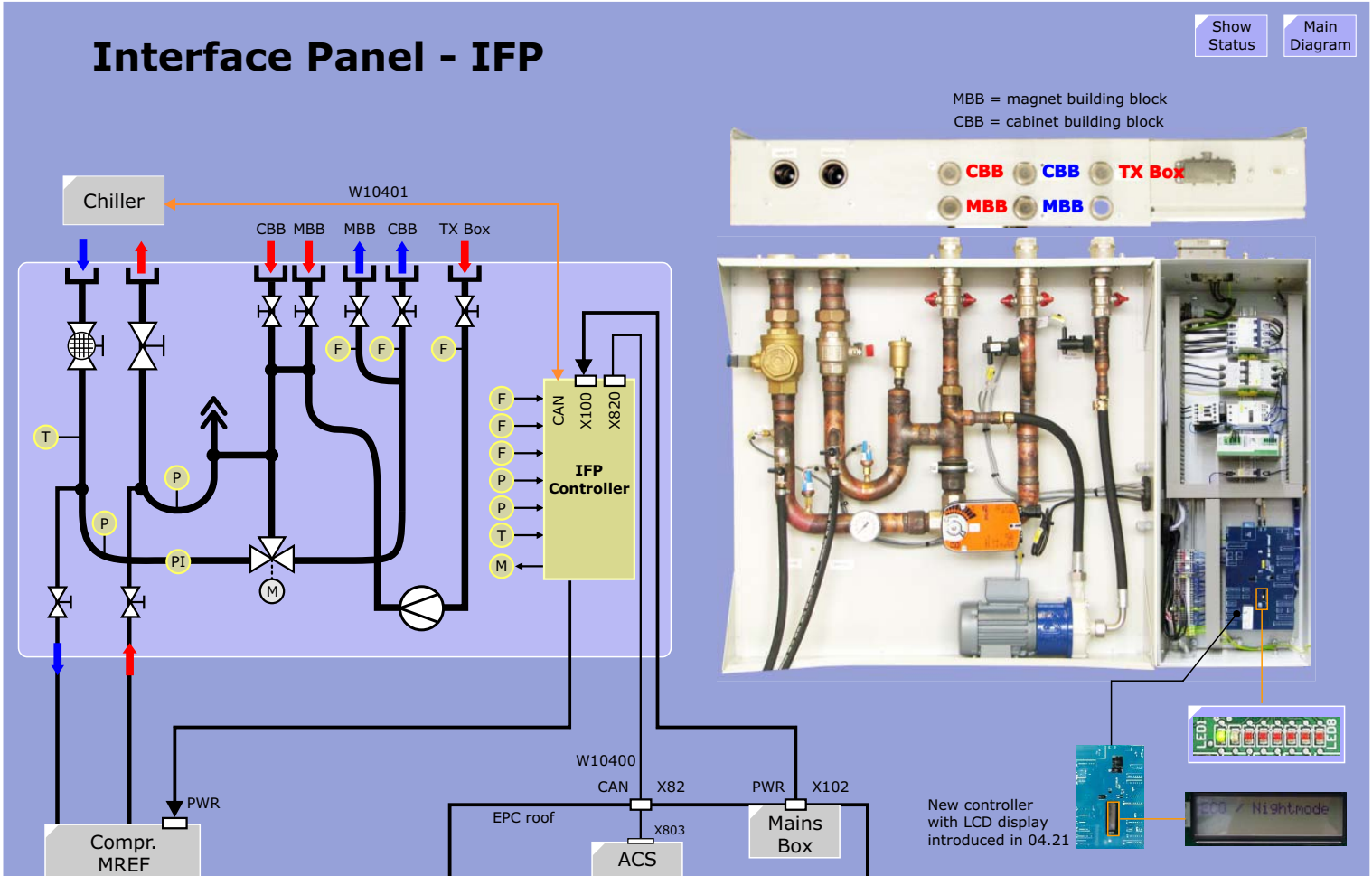
Controller with LCD Display

In April 2021 a new IFP Controller was introduced implementing an LCD display for the status and error messages.

Go back to [Chiller description](#)

Go back to [IFP Block Diagram](#)

Figure 222 Interface Panel



MISC Cooling Status

RFCEL

The RFCEL has a heat exchanger radiator for producing cooled air for both the RFCEL and TX Box. The air temperatures as well as the blower fan are monitored by the RFCEL_Com.

TX Box

The TX Box utilizes a heat sink for cooling the amplifier FETs. The supply and return water temperatures are monitored by the TX Box controller.

Gradient Coil

The gradient coil has 6 temp sensors attached to each of its primary and secondary (active shielding) coils. The monitoring is performed on the RFCEL_Satellite.

Magnet Room

A temperature sensor inserted in the tubing of the patient air measures the temperature of this air.

Cooling Status Mask

RFCEL	Air Temperature	21.5 °C
	Fan	OK

Coil Temperature Sensors

Sensor	Temp / °C	Status
GC1	22.8	OK
GC2	20.8	OK
GC3	22.8	OK
GC4	FAULT	FAULT
GC5	21.8	OK
GC6	21.8	OK

TX Box

Air Temperature	21.5 °C
Water Supply Temperature	22.5 °C

Magnet Room

Patient Air Temperature	24.3 °C
-------------------------	----------------

10 **Line Power Distribution**

Introduction

The LPD has the monumental task of distributing the lines power to the various components within the system. The LPD includes a transformer for isolation to the customers mains and to adapt the customer mains voltage to the power requirements of the system. Circuit breakers of input (primary) and transformer outputs (secondaries) are used extensively to allow removal of power to the various system components for safety and servicing purposes.

Diagrams

Simplified LPD diagrams (where cable assembly and/or manufacturing remarks have been removed) are used on the following pages for explanatory purposes. The version used in this description may deviate from the official diagram version found in the Diagrams binder. Please refer to these documents for trouble shooting.

Figure 223 Line Power Distributor Diagram, page 1

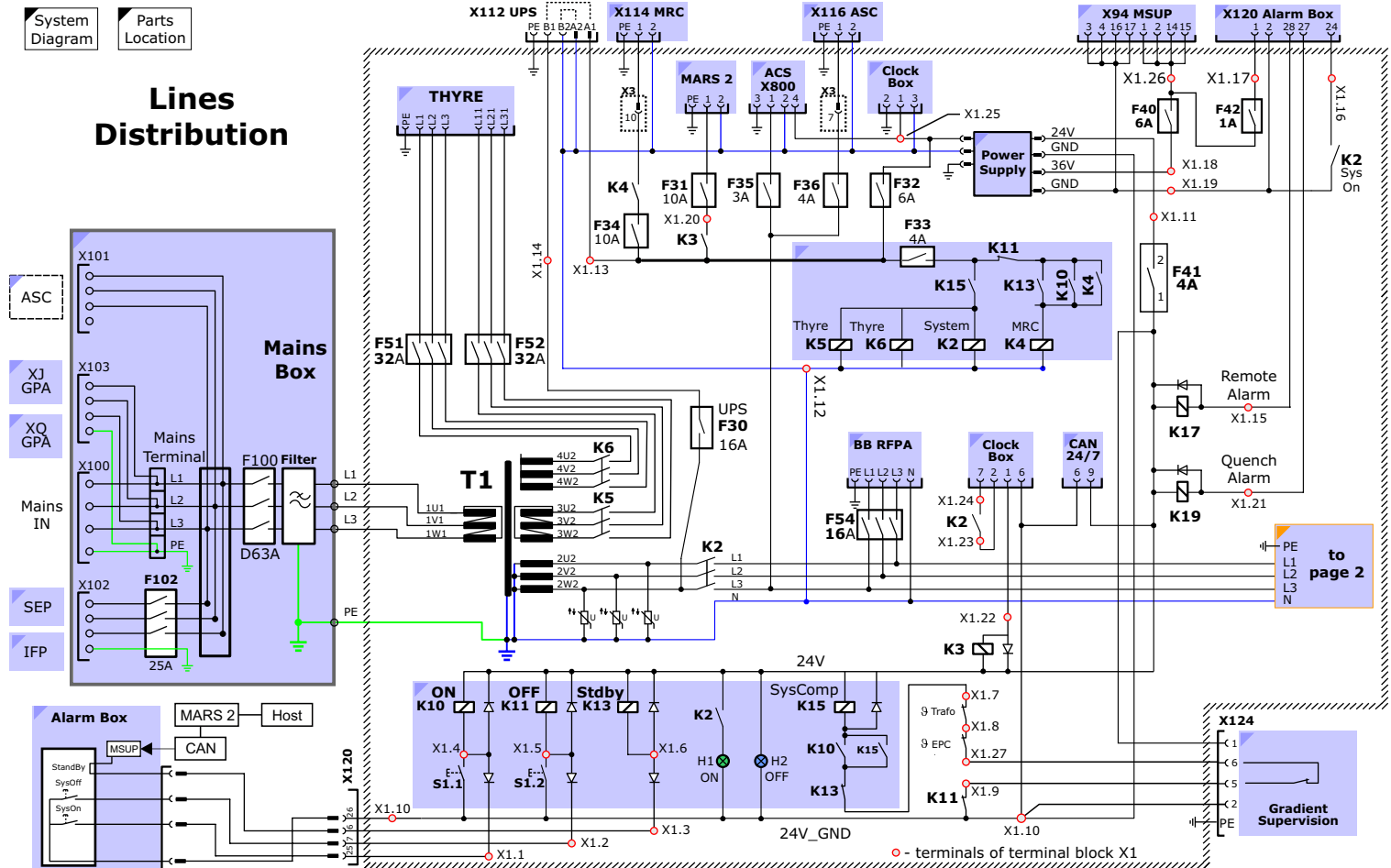
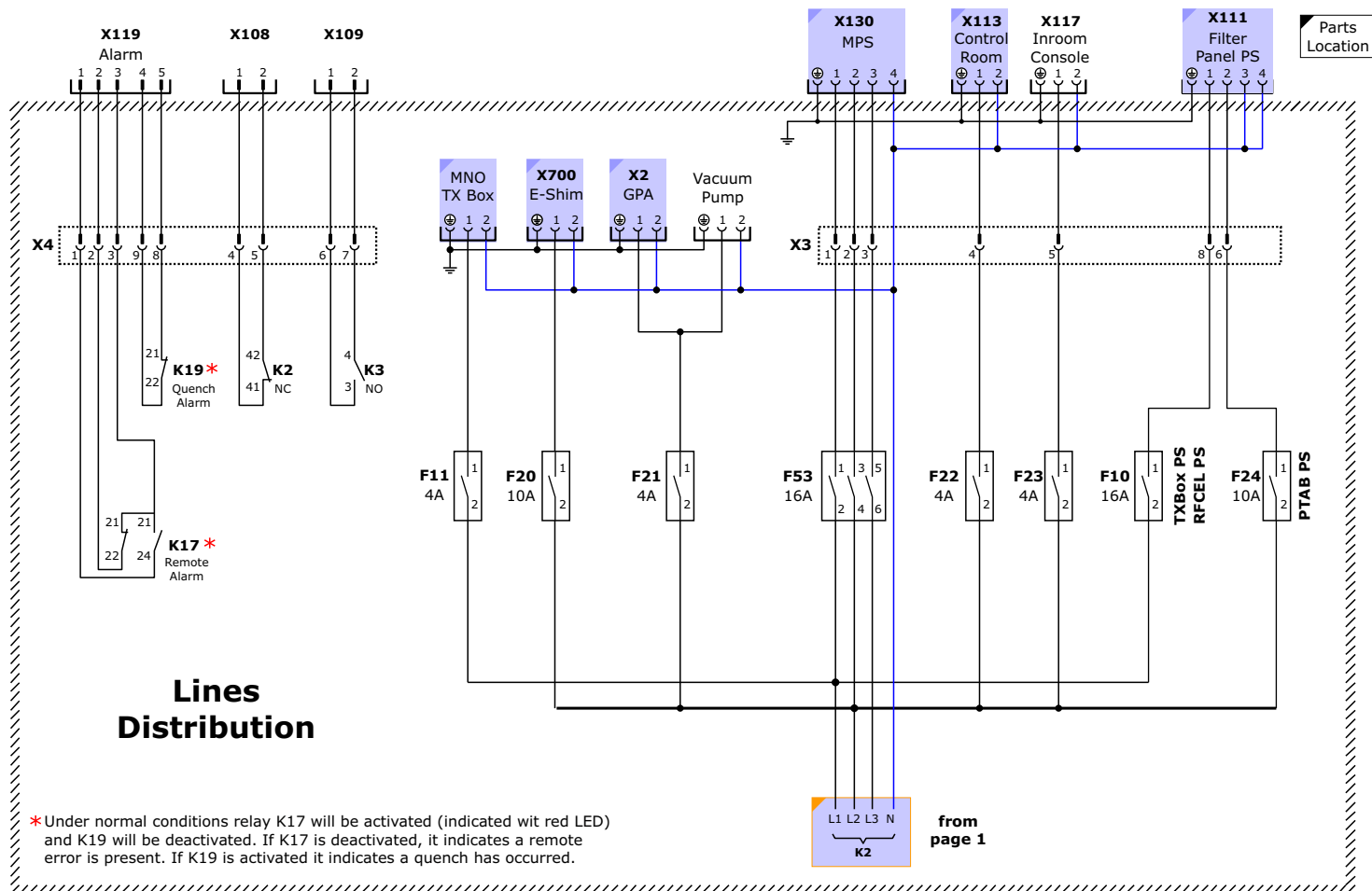


Figure 224 Line Power Distributor Diagram, page 2



Power ON/OFF Relay Logic

The following refers to the diagram in [Figure 225](#).

System On

When either the **System On** button on the [Alarm Box](#) or the System On button in the LPD is pressed, **K10** is temporarily activated which activates relays **K15** and **K4**.

- **K15**
Relay K15 has one help-contact for self-hold to remain on and a main output-contact supplying power to relays **K2**, **K5** and **K6**. When K2 is activated it supplies power to most of the scanner components and relays K5 and K6 supply power to the THYRE.
- **K4**
Relay K4 has one contact for self-hold and one to supplies power to the MRC Host and keeps itself on via a self-hold contact.

System Off

By pressing the **System Off** buttons on the Alarm Box or LPD, **K11** is temporarily activated. K11 has two contacts:

- one deactivates **K15**. With **K15** deactivated, relays **K2**, **K5**, **K6** are deactivated removing power to these components.
- the other deactivates **K4** which removes power from the MRC.

System Standby

There is no System Standby button on the [Alarm Box](#). It is a software button on the *System / Control / Meas&Recon* tab card. When activated

- the Host sends a command to the MSUP via the MARS 2-CAN24-CAN bus.

- The MSUP applies a hardware signal to the LPD through the Alarm Box. To get through the Alarm Box the 36V supply voltage from the 24/36V power supply in the LPD must be present.

Assuming all of the above components are functioning, the signal gets to relay **K13** in the LPD.

- **K13**
When **K13** activates it opens a contact supplying **K15** which in turn deactivates and removes power to relays **K2**, **K5** and **K6** which remove power from all system components except the MRC Host which is supplied over **K4**. A second contact of **K13** closes to assure **K4** remains activated.

Figure 225 System Power On/Off Switch Logic

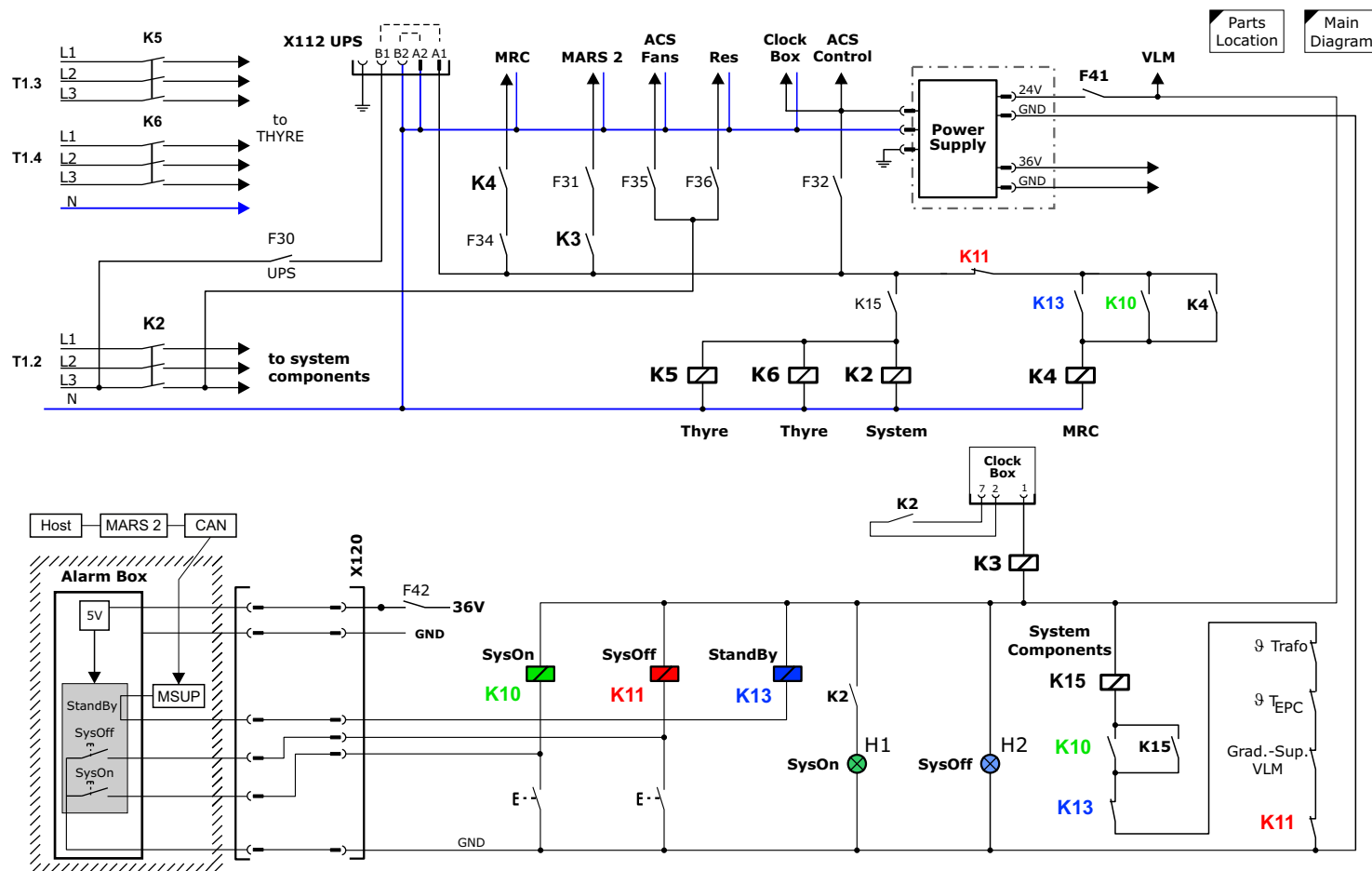


Figure 226 Line Power Distributor Parts Location

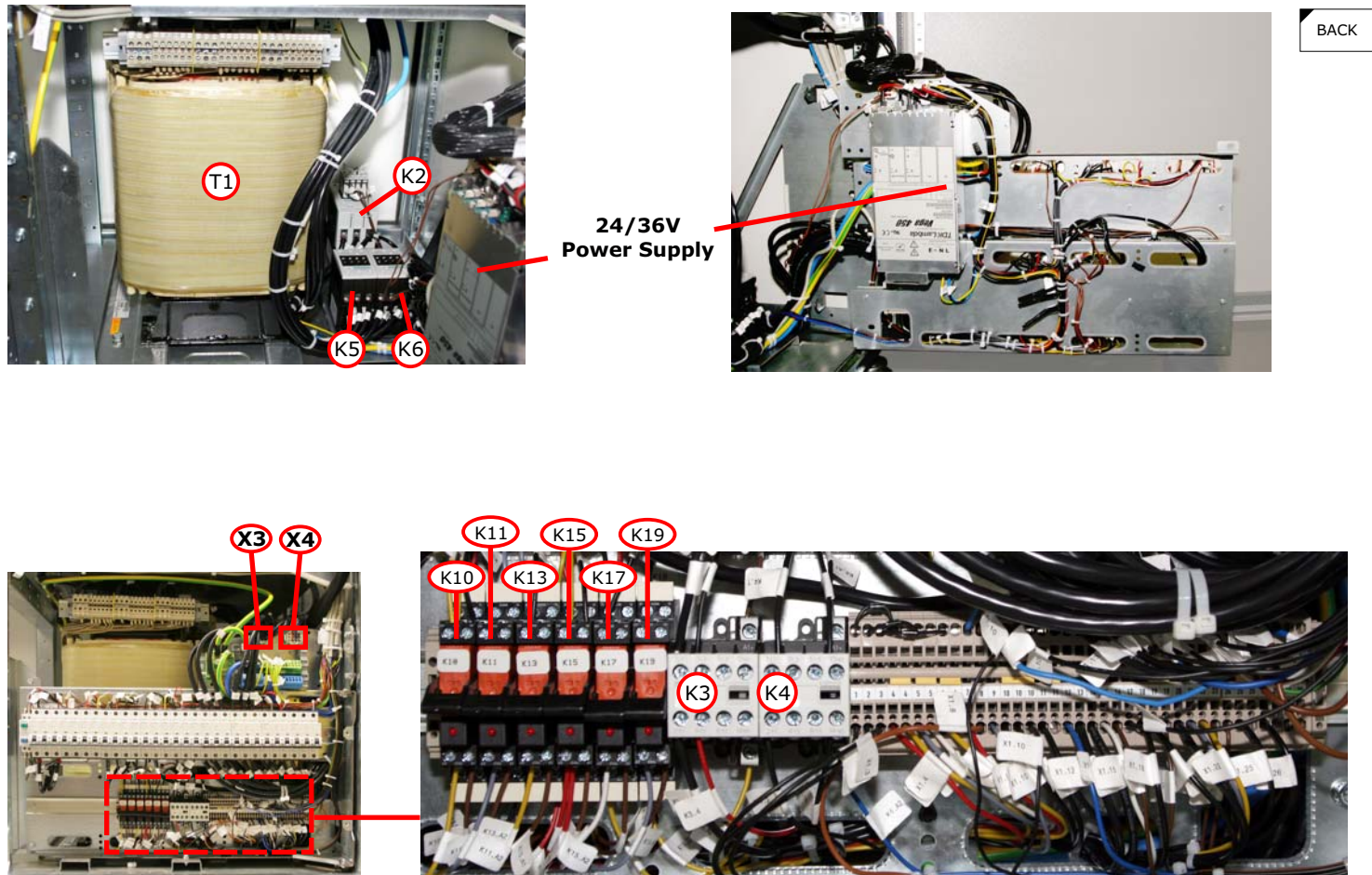


Figure 227 Circuit Break Layout

Circuit Breaker Layout

BACK

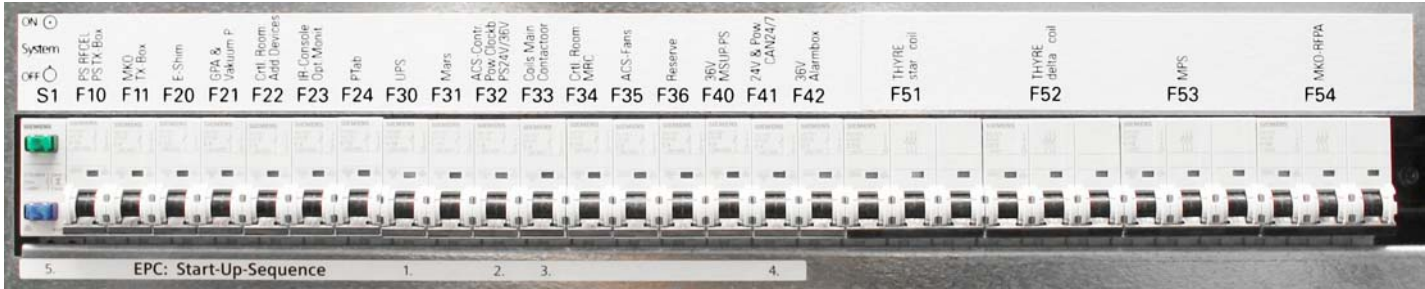
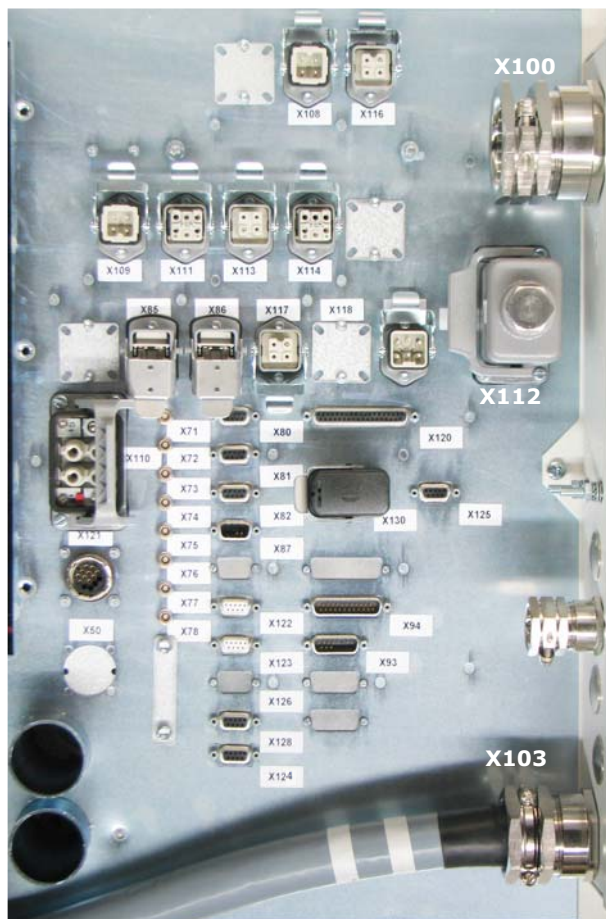


Figure 228 Mains Box and EPC Roof Connections



EPC Roof connections

X50	Transmit cable (MNO TX Box)	X108	K2 active (for customer)
X71	10 MHz from RFCEL to Clock Box	X109	K3 active (for customer)
X72	320 MHz from RFCEL to Clock Box	X110	130VDC power to TX Box
X73	100 MHz to RFCEL from Clock Box	X111	1p power to FP power supplys
X74	320 MHz to RFCEL from Clock Box	X112	1p power to/from UPS
X80	Gradient Filter Fan sensor	X113	1p power to Control Room
X81	rH/temp sensor (option)	X114	1p power to MRC
X82	CAN SEP/IFP	X116	1p power (reserve)
X85	Service port ethernet	X117	1p power to In-room console
X86	MARS 2 to Host ethernet	X118	reserve
X87	Door Contact	X119	remote alarm (for customer)
X93	Compressor Control	X120	to/from Alarm Box
X94	MSUP Control	X121	E Shim output (to Grad coil)
X100	Mains input	X122	CAN (reserved)
X102	3p power to SEP	X123	CAN to Injector
X103	3p power to GPA	X124	CAN/PWR to Gradient Supervision
		X125	CAN to MPS
		X128	Serial to Host
		X130	3p power to MPS

Main
Diagram



11 Tune Up / QA

Introduction

The Tune Up is a collection of procedures used to correct or compensate for hardware and environment variations in order to optimize the scanners performance. The Tune Up (TU) must be performed when the MR unit is initially installed and subsequently when certain hardware components have been replaced requiring one or more Tune Up procedures to be performed. This information is found in the Replacement of Parts procedures.

The QA offers a set of procedures intended for one to verify the scanner is operating within specified parameters and secondly to help identify and determine causes of poor system performance.

This description will include physical background information of the individual Tune Up and QA procedures. The more simplified and automatized the service-software has become, the more important it is to have a good understanding of these procedures and what they are doing.

Tune Up Procedures

The procedures are performed in the following sequence:

- Feedback Loop Calibration
- Rx Gain Calibration
- BC Tuning
- Gradient Regulator
- Table Height
- Light Marker
- Phantom Shim
- Body Coil Power Losses (CPL)
- Cross Term Compensation (CTC)
- Eddy Current Compensation (ECC)
- Gradient Delay
- Gradient Sensitivity
- BC Image Brightness
- Dynamic Field Map

Go back to [SYSTEM DIAGRAM](#)

Procedure Status

If a procedure is successful, it will be set to **Done**.

A status of "To Do" will be set at the initial installation or when a procedure having interdependencies was performed retrospectively. As example, performing a regulator adjustment

sets the status of ALL following gradient procedures to "To Do".

If a Tune Up procedure has lost its "Done" status then the corresponding Quality Assurance procedure(s) will also be affected and changed to "To Do".

The ERROR status is set if a procedure has stopped or aborted due to an error or time-out.

Reports

A report is generated for every procedure or set of procedures as they have been selected in the menu and performed. They are found under "Reports", "Session History".

Active Coil Menu

The Active Coil menu shows the name of the currently selected coil that will be used by the Tune Up procedures. Since the body coil is always connected, it will always be the default selection when the Tune Up platform is opened.

Dependencies

All selected procedures are performed from top to bottom. The individual steps follow a specific order and should not be changed due to the dependencies between those prior and those that follow.

If a Tune Up procedure has changed its status then the procedure(s) in the Quality Assurance menu for the corresponding coil are changed to To Do.

NOTE

The Tune Up procedures measure and **SAVE** system parameters. Therefore the Tune Up procedures should NOT be used for trouble shooting or when the system is not in proper working order. The **Quality Assurance** procedures only measure and **VERIFY** system specifications without saving any system parameter.

Expert Mode

This mode should only to be used by experienced CSEs that know and understand the consequences of performing system-altering procedures. If improperly used the system may be improperly adjusted and adversely affect the image quality.

Quality Assurance Procedures

The description for the QA procedures begins on [page 322](#).

Tune Up / Quality Assurance

System Diagram

HOME

Workflow

➔ Tune Up

- RX Gain Calibration
- BC Tuning
- Gradient Regulator
- Lightmarker
- Phantom Shim
- Cross Term Compensation
- Eddy Current Compensation^
- BC Power Losses
- Gradient Delay
- Gradient Sensitivity
- Dynamic Field Map
- Image Brightness

📁 Tune Up Expert

- Extended Frequency Adjustment
- Gradient Regulator
- Phantom Shim
- Cross Term Compensation
- Eddy Current Compensation
- Gradient Delay
- Gradient Sensitivity

☐ Feedback Loop Calibration
☐ RX Gain Calibration
☐ BC Tuning
☐ Gradient Regulator
☐ Table Height
☐ Lightmarker
☐ Phantom Shim
☐ Cross Term Compensation
☐ Eddy Current Compensation
☐ Coil Power Losses
☐ Gradient Delay 1.5T - LP 3T - SP
☐ Gradient Sensitivity
☐ BC Image Brightness
☐ Dynamic Field Map

SP = small phantom
 LP = large phantom

Estimated Time: 0:50:00

HOME

Workflow

➔ Quality Assurance

📁 Quality Assurance - General

- Feedback Loop Check
- RX Gain Calibration Check
- BC Tuning Check
- Gradient Regulator Check
- Phantom Shim Check
- BC Power Losses Check
- Cross Term Compensation Check
- Eddy Current Compensation Check
- Gradient Delay Check
- Gradient Sensitivity Check
- BC Image Check
- Image Orientation Check
- Gradient Rise Time Check
- Spike Check
- RF Noise Check
- Stability Check
- Synthesizer Check
- Long Term Stability Check

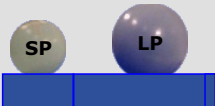
📁 Quality Assurance - Coil

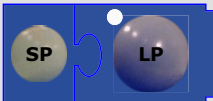
📁 Quality Assurance Expert

- Feedback Loop Calibration Check
- Coil Check
- Image Orientation Check
- Phantom Shim Check
- Cross Term Compensation Check
- Eddy Current Compensation Check
- Gradient Regulator Check
- Gradient Delay Check
- Gradient Sensitivity Check
- Gradient Rise Time Check
- Spike Check
- RF Noise Check
- Stability Check
- Synthesizer Check
- Field Stability Check
- Long Term Stability Check
- Dynamic Field Map Check
- Position Check

☐ Feedback Loop Calibration Check
☐ RX Gain Calibration Check
☐ BC Tuning Check
☐ Gradient Regulator Check
☐ Phantom Shim Check
☐ BC Power Losses Check
☐ Cross Term Compensation Check
☐ Eddy Current Compensation Check
☐ Gradient Delay Check
☐ Gradient Sensitivity Check
☐ Gradient Rise Time Check
☐ BC Image Check
☐ Image Orientation Check
☐ Spike Check
☐ Stability Check
☐ Synthesizer (NCO) Check
☐ Field Stability Check
☐ Long Term Stability Check
☐ RF Noise Check
☐ Gradient Rise Time Check

Estimated Time: 1:27:00





TuneUp checks

Quality Assurance checks

Tune Up

Feedback Loop Calibration

The TX-Box has an internal feedback loop for regulating the B1 field at the Body Coil. This TU procedure determines crucial parameters (feedback delay and phase) that are needed for an optimized operation of this regulation loop.

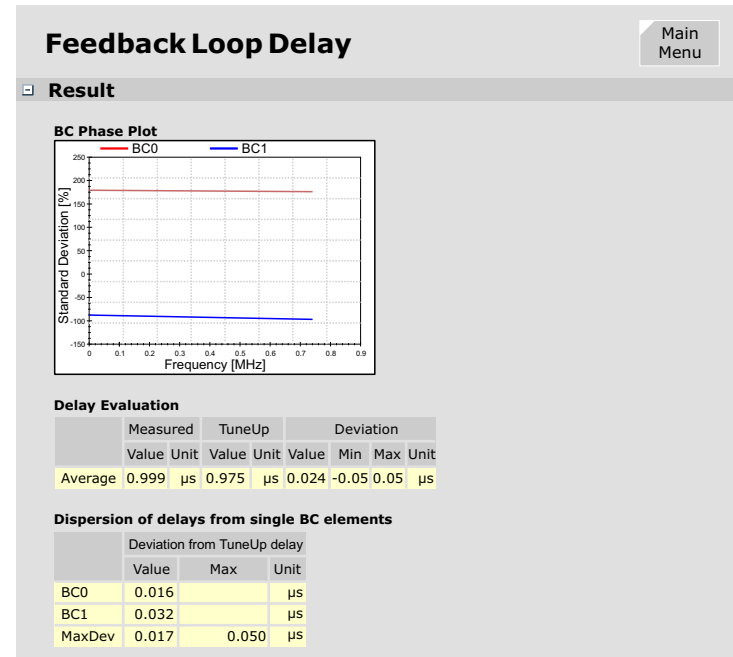
NOTE An improper feedback phase results in suboptimal dynamic behavior and can in the worst case even cause oscillation.

Measurement

The tuncal sequence is invoked which feeds an RF signal over a wide frequency range to both BC systems (0° and 90°) individually. The frequency dependence of the signal phase received at the DICOs is then analyzed and the feedback loop delay is determined. The measurement is then also performed for the LC path to calibrate phase differences between LC and BC channels.

From the measured phase values the new delay value and a new file "nb_hybrid_compen.dat" is calculated and flashed into the TX Box.

Figure 230 Feedback Loop Calibration



Receive Gain Calibration

The coil combination and selection flexibility of the Tim technology presents a **challenge** in that any of the connected coil elements can be routed to any available receive channel. The problem lies in the fact that each coil element and receive channel will have slightly different electrical/frequency characteristics causing deviations of the received signal amplitudes which, if not corrected, would result in deviations to the image brightness. It is necessary, therefore, to normalize the signal intensity of each coil and receive channel so that the image intensity of the final image is the same regardless which coil/receiver combination is used.

The coil-related deviations have been established at the factory and assigned a correction factor and stored - along with other pertinent coil-specific details - in files for each coil (coil files) on the Host.

The receiver-related deviations are determined with the Receive Gain Calibration. There are two measurements made:

- Channel correction - determine deviations caused by the switch fabric
- High/Low gain correction - determine amplitude and phase deviations between high and low gain settings of the SGA amplifiers. This is especially important for 3D sequences where the gain is switched dynamically during the sequence.

Measurement

The test signal (BC_RX) is generated in the TX Box and transferred to the IF_RCCS over coax cable. This signal is then routed through LC Input Mux 1 to the switch fabric where it is sequentially routed through to each of the SGAs and ADCs on the IF_RX32 boards.

There are two measurements made per channel at high and low gain settings.

Evaluation

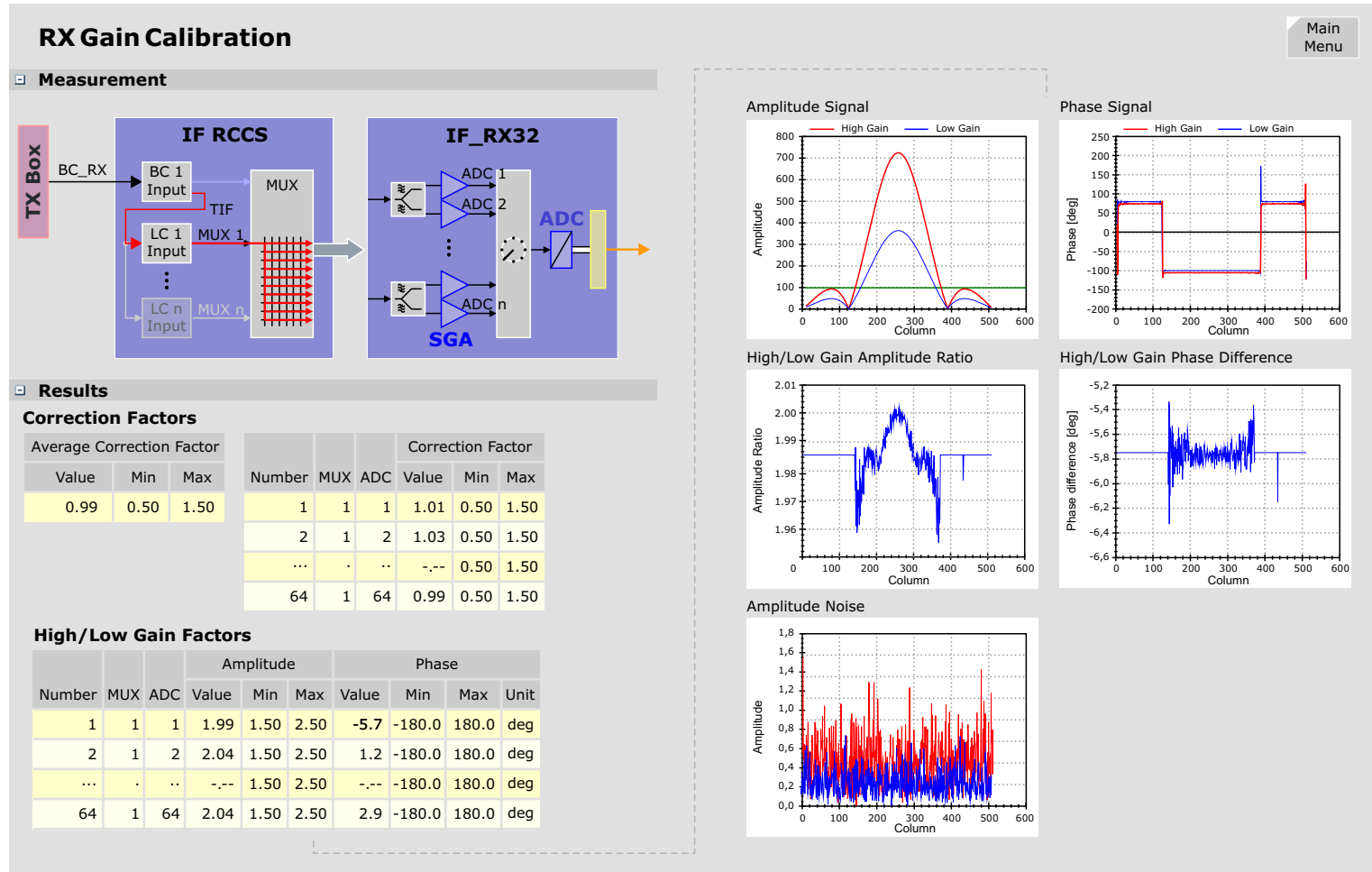
Channel Correction

The signal values of the high-gain measurements of all channels are determined and an average taken. The individual channel correction factor values are the measured channel deviations in relation to this average. This has the advantage that component tolerances are better taken into account.

High/Low Gain Factors

These differences in amplitude and phase of the low and high gain settings of the SGAs on the IF_RX32 are determined and listed as Amplitude value and Phase value.

Figure 231 Receive Gain Calibration



Body Coil Tuning

This procedure is used to measure and, if necessary, adjust the central *resonant frequency* of each of the two Body Coil systems as well as minimize the *coupling* between them.

Measurement

The measurement is performed by taking 80 samples of the forward and reflected values at 10 kHz intervals covering the transmitters full operating frequency range. The result of each measurement is the r-factor = U_r/U_f .

Evaluation

BC Reflection

The measured r-factors are plotted along the frequency axis and display the Body COIL's frequency response.

The evaluation determines:

- the coil's center (tuned) frequency is within a specified range, i.e. is not too far from the center of the operating range
- that the center frequencies of both coil systems are tuned to within specified limits of each other
- that the reflection factors are within expected limits

The resonant frequency is found at that point where the reflection of the BC has its minimum.

Transmission (coupling)

The transmission factor describes the amount of coupling between the two Body Coil systems, corresponding to a mutual inductance of the Body Coil systems primarily due to coil construction. The transmission is determined by sending a signal into the 0°-system and measuring what comes out of the 90°-system.

The transmission factor is determined at that frequency where the signal strength (coupling) is at its maximum.

Adjust Procedure

This is a manual adjustment requiring access to the Body Coil's three 50-turn trim capacitors (refer to [Figure 232](#) below), two for adjusting the frequency of each Body Coil system (T1 and T2 - not to be confused with the relaxation times of soft tissues) and one for minimizing the coupling between the two (TD).

Tuning

The trim capacitors **T1** and **T2** vary the resonance frequency of coil systems 1 and 2 respectively.

Decoupling

TD is also a 50-turn capacitor affecting the de-coupling between the two BC systems. It is not possible to know which direction to take at first, but there is only one minima.

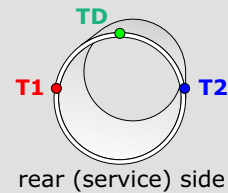
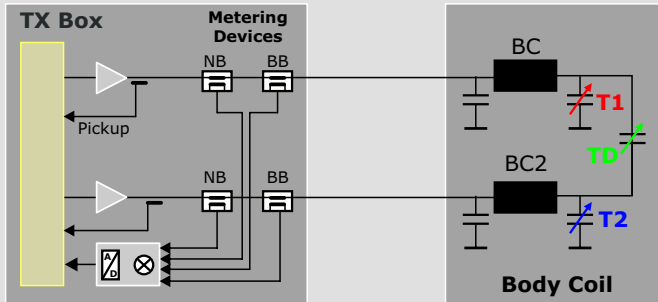
NOTE There is a dependency between all three adjustments! The BC Tune adjustment procedure is found in the Replacement of Parts document.

Figure 232 Body Coil Resonant Frequency and De-coupling Adjustments

BCTuning

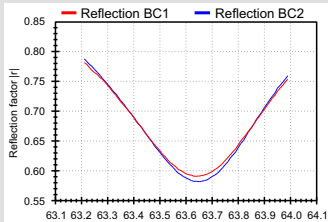
Main
Menu

Measurement

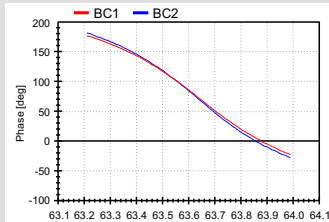


Results Reflection

Reflection Amplitude



Reflection Phase

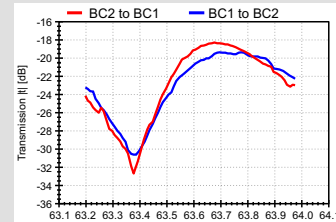


Evaluation results reflection

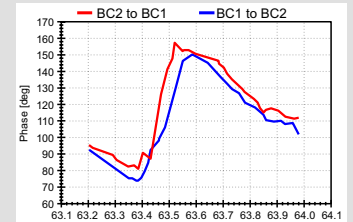
BC element	Frequency (at min)				factor (at min)			Phase		Q-factor
	Value	Min	Max	Unit	Value	Min	Max	Value	Unit	
BC	63.638	63.5	63.8	MHz	0.59	0.40	0.80	84.19	deg	327.88
BC2	63.645	63.5	63.8	MHz	0.58	0.40	0.80	86.39	deg	339.57
Delta	7.0	0.0	100.0	kHz						

Results Transmission

Transmission Amplitude



Transmission Phase



Evaluation results transmission

Transmission path	Frequency (at max)				decoupling (at max)			
	Value	Min	Max	Unit	Value	Min	Max	Unit
BC1 to BC2	63.696	63.2	64.0	MHz	-22.13	-40.00	-18.00	dB

Gradient Related Tune Up

Regulator Adjust

The regulators in the newest very-high-power Gradient Amplifiers (500A and above at 2000V) have a **proportional**, P, (= gain), **integral**, I, (= time) and **differential**, D, regulation characteristics, thus each regulator consists of a P, I and D regulator circuit in series. The purpose of this procedure is to find the optimal P, I and D regulator adjustment values for an optimized regulation characteristic.

Measurement

The regulators are tuned one after the other in the order P, I, and D. Beginning with a configured setting each regulator is consecutively measured and incriminated until an optimized setting for each regulator is found. Finally, a configured *safety factor* is added to the regulator.

The GradReg program then performs the following:

- checks whether the data is in the real or imaginary part
- displays the measured pulse: the first graphic shows the complete gradient pulse, the second graphic displays the overshoot of the gradient pulse which is used for the evaluations (described below)

Evaluation

The results are then displayed: the first graphic shows the complete gradient pulse, the second graphic displays the overshoot of the gradient pulse which is used for the evaluations

The pulse overshoot is evaluated upon three criteria:

- **Overshoot** amplitude to the nominal pulse amplitude (in %)
- **Decay time** for overshoot to decay to 15% of its amplitude
- **Smoothness** is the maximum positive deviation of the gradient pulse top portion.

The smoothness factor is the maximum difference determined by the above process divided by the nominal pulse amplitude, expressed in %

Pulse Check

A check if the regulator is being over-driven or the pulse signal oscillates is also made, indicated by 2 or more sign changes in the data points of the overshoot.

If there are points with an amplitude below the nominal amplitude an undershoot after an existing overshoot is detected. That means over-driven regulator or oscillating signal.

The final regulator adjustment values and achieved values are displayed in tabular form.

After the regulators have been optimized it is checked if the signal is in specification. If in spec the regulator values are saved permanently. If an error occurs or the program is aborted and the old regulator values will be restored.

Figure 233 Regulator Signal Evaluation

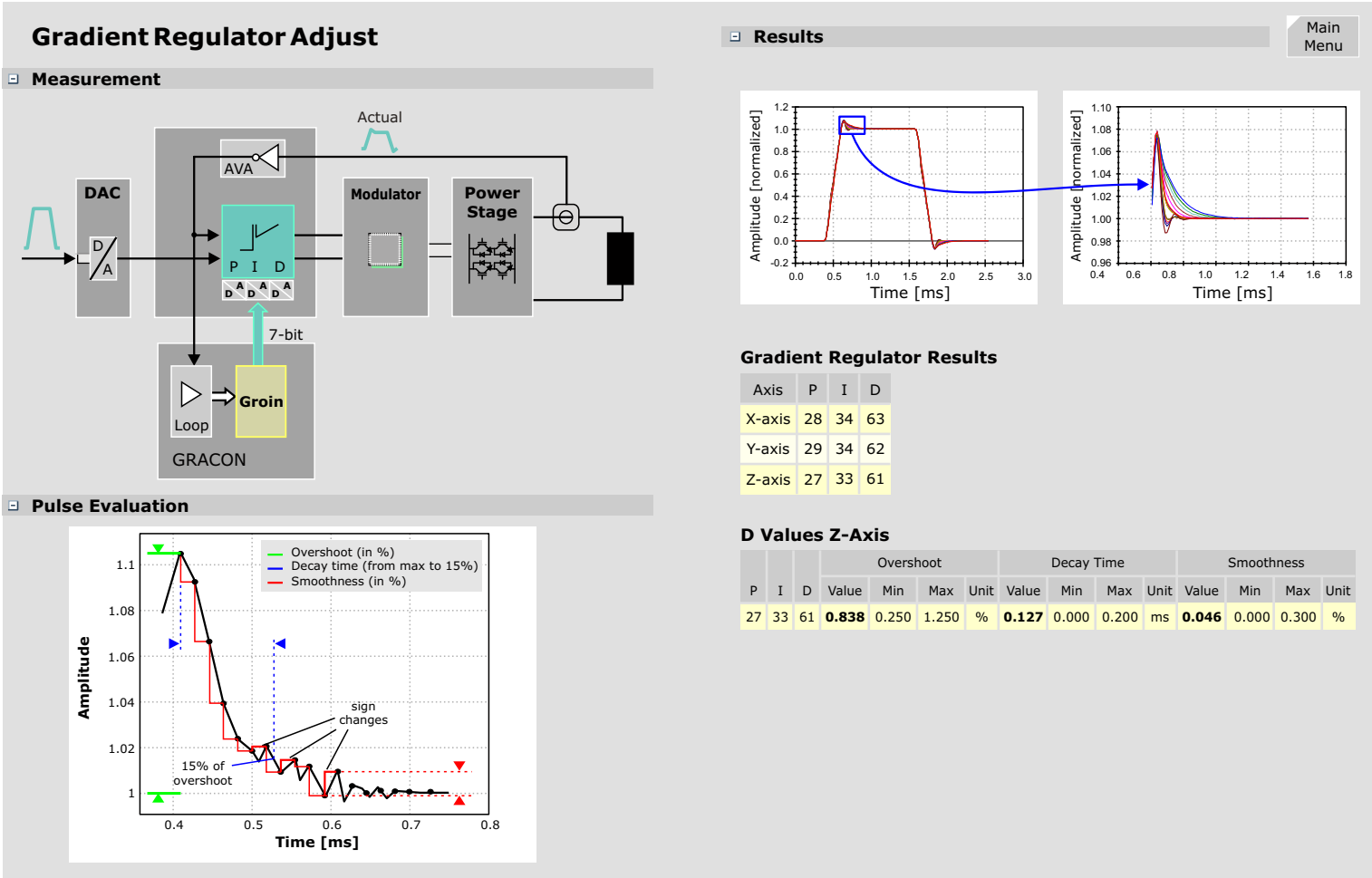


Table Height

The PSY unit mounted on the scissor assembly of the table contains an inclination sensor. The value of this sensor is read out and saved by this Tune-up procedure with the table being in the vertical home position. The vertical home position is set with a reflector located on the face of the tunnel cover which is adjustment during the system installation.

Due to the nature of a mobile table, its height can vary depending on use. For this reason it is not known at the time of docking whether the table is above or below the vertical home position.

For faster detection of the vertical home position the Table Height value is requested by the PTAB after docking. It will then be used to determine whether the vertical home position is above or below its current position and drive to the home position.

[BACK TO MENU](#)

Light Marker

The goal of the Table Adjustment is to determine:

- center positioning of the phantom in the magnet iso-center
- distance between light marker and magnet iso-center

The Tune Up procedures which use MR-signals require a centered phantom for accurate results. The adjustment procedure determines the phantom position with respect to the center of the gradient coils (iso-center). From this, the distance from the light marker to the magnetic iso-center is determined.

Measurement

The measurement uses a double echo sequence. The first echo is acquired without applying a readout gradient resulting in an FID the length of which is a rough check of the fundamental imaging volume homogeneity.

During the second echo a readout gradient is applied resulting in a

signal projection of the phantom. The center of the phantom's signal spectrum is compared to the center of the scan volume frequency window to calculate the phantom position in readout direction. The sequence measures the three axis in this way. [Figure 234](#) shows as example only the X axis.

Evaluation

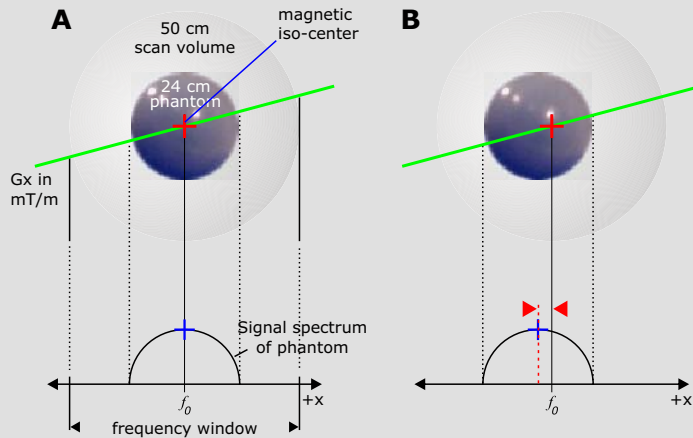
The results of the Z axis measurement is used to determine the table position and will be corrected automatically and stored as the new light marker distance. All three phantom position values are displayed in a table, shown in the table in the graphic. Corrections necessary in the X direction are made by re-positioning the phantom.

Corrections in the Y axis may be made by correcting an improperly adjusted Body Coil mechanical height (see RF-REP) and/or readjusting the vertical table height (described in the PTAB-TSG).

Figure 234 Light Marker

Light Marker

Measurement



With a gradient applied, a linear frequency deviation over distance results providing a Hz/mm relationship.

A If the phantom is centered within the scan volume, the center of its signal spectrum will correlate with the center of the frequency window defined by the applied gradient. The magnetic center of the gradient coil defines the iso-center of the scan volume.

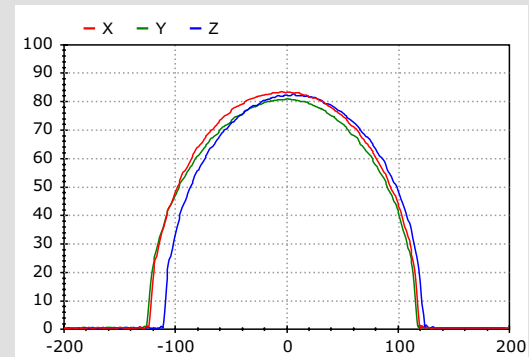
B If the phantom is off-center, its signal spectrum will not be centered to the frequency window. The deviation from center in mm can be derived from the correlation of frequency per mm value known from the gradient amplitude in mT/m.

Results

[Main Menu](#)

Phantom Position

Axis	Position				Diameter				SN	
	Value	Min	Max	Unit	Value	Min	Max	Unit	Value	Min
X	-1.17	-10.00	10.00	mm	238.3	210.00	245.00	mm	714.24	100.00
Y	-3.13	-10.00	10.00	mm	239.1	210.00	245.00	mm	772.93	100.00
Z	0.78	-2.00	2.00	mm	229.7	210.00	245.00	mm	736.51	100.00



Phantom Shim

Phantom Shim is a procedure which measures and generates a "field map", i.e., a 3D spatial plot of magnet field inhomogeneities for the purpose of determining the gradient offset and shim coil currents (option) necessary for optimization. It is performed at this point in the Tune Up set as preparation for the following procedures using the large spherical phantom for measurement. The correction currents determined here also become the default values for the pre-sequence adjust shim procedure used for patient imaging.

Measurement

FID Check

Prior to the measurement an FID measurement is performed and the FID amplitude and decay time determined. This will be used as an indicator that the solutions for the corrections are indeed giving an improvement.

Field Map

The sequence is a 3D-DESS-sequence. The echo times are so chosen, that fat and water are in phase, hence, the sequence is prepared for patient use as well.

The measured volume is 35 x 35 x 35 cm with a resolution of 32 (Aera) or 64 (Skyra) isotropic voxels. A major advantage gained using this method is in having actual field values for each of the voxels and thus allowing the user to define any specific volume within the measurement volume that is to be optimized. This is called local volume shimming. Also, the evaluation algorithm detects voxels with little or no signal and will not use the data for correction further improving the optimization.

As seen in [Figure 235](#) the Phase_{GR} line represents the phase rotation due to the readout gradient and the Phase_{static} is the phase rotation due to the static field inhomogeneities which is the factor we are looking to determine.

Evaluation

The field difference between neighboring voxels is obtained by comparison of the phases of the magnetization vectors. With an echo time difference of 9.4ms a shift of 10 degrees from voxel to voxel in one direction results in a linear homogeneity of about 9μT/m in that direction.

The equation minimizes the difference between:

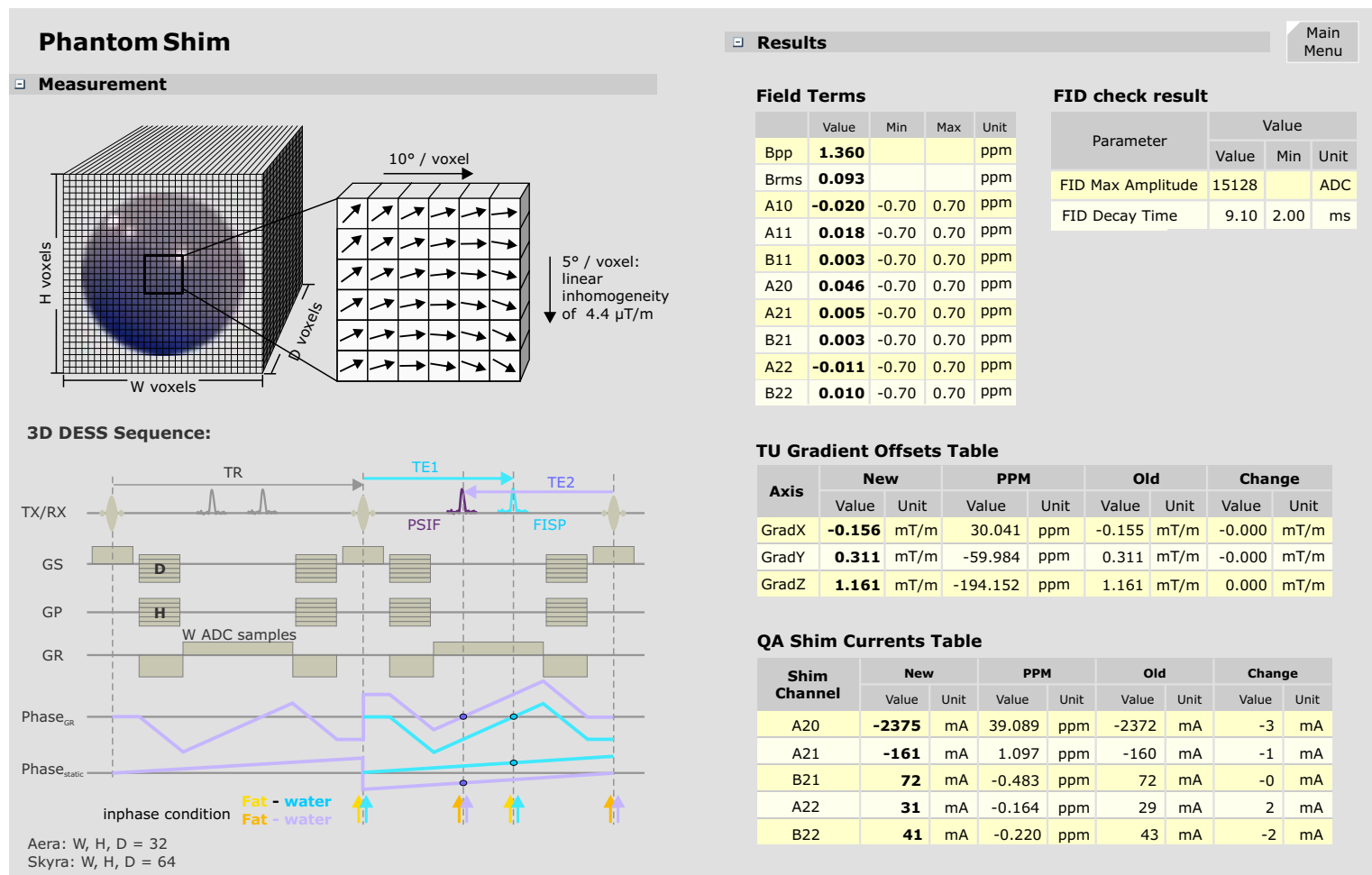
- the field generated by the three gradient coils and the 5 shim coils
- the measured magnetic field inhomogeneities.

The first table displays the measured field terms for each of the 8 terms that can be corrected. The 3 first order field terms and the 5 second order field terms that can be optimized are:

Spherical Harmonic	Field	Correction Device
First Order Terms		
A(1,0)	Gz	Z gradient coil
A(1,1)	Gx	X gradient coil
B(1,1)	Gy	Y gradient coil
Second Order Terms		
A(2,0)	$z^2 - (x^2 + y^2)/2$	Shim Coil
A(2,1)	xz	Shim Coil
B(2,1)	yz	Shim Coil
A(2,2)	$(x^2 - y^2)/2$	Shim Coil
B(2,2)	xy	Shim Coil

The tables in the graphic below give the correction currents for the gradient and shim coils. The gradient offset current are expressed in mT/m and the shim coil currents in mA.

Figure 235 Phantom Shim



Cross Term Compensation (CTC)

An applied gradient field in any axis produces not only eddy currents along that axis, but also small amounts of eddy currents are coupled over the conductive magnet components onto the other two axis', the so-called cross terms. The CTC (cross term compensation) procedure measures the eddy currents created by the cross terms. Compensation is made by pre-distortion of the gradient waveform. Cross term amplitudes typically arrive at values of up to 0.2% of the main applied gradient amplitude. Unlike the main eddy currents which are complicated and need to be modeled with 5 time constants, the Cross Term currents decay exponentially in time in general with only one time constant.

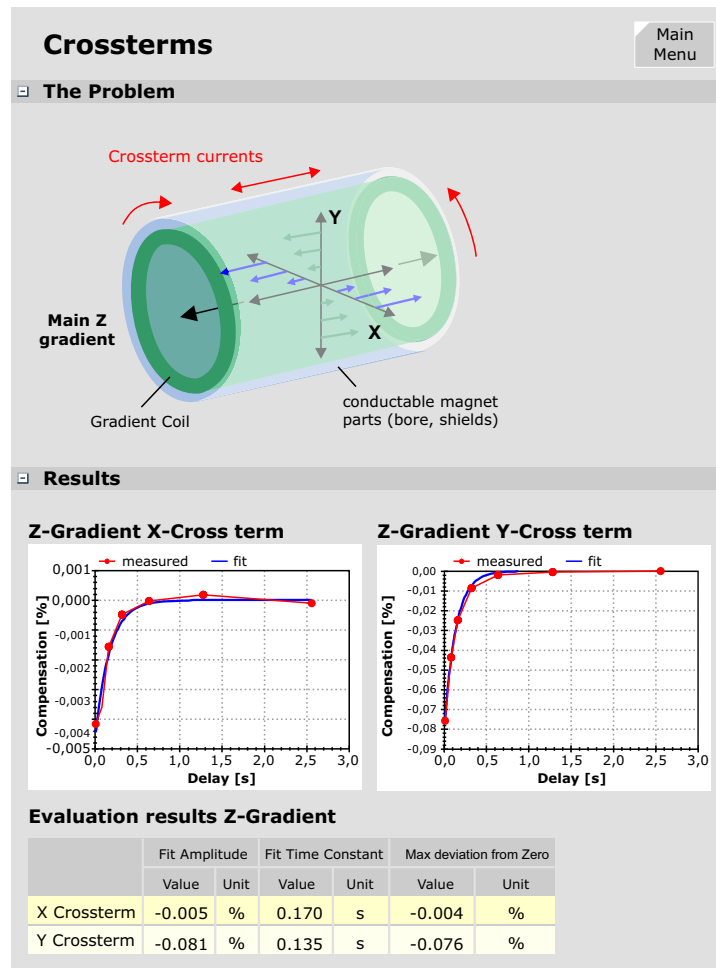
Measurement

The procedure measures the decay of the eddy currents caused by the cross terms. Amplitudes and time constants of the eddy current contributions are then calculated. The compensation is straightforward: gradient fields of equal and opposite amplitude and time constants to the measured cross terms are applied perpendicular to the main applied gradient field. The result is a much cleaner main gradient with very small residual cross terms improving the final image quality.

Evaluation

The amplitude and time constants of the compensations are displayed graphically and in tabular form. Only the Z axis is shown here.

Figure 236 Cross term Currents



Eddy Current Compensation (ECC)

The dynamic gradient fields produce eddy currents in all surrounding conductive structures (mainly the Body Coil, magnet bore and the cryo-shields) resulting in magnetic fields which oppose and distort the applied gradient fields. Eddy currents in the warm components (Body Coil and Magnet bore) have relatively short decay times, whereby eddy currents developed in the cold cryo-shields can have relatively long time constants.

The eddy currents have to be accurately quantified and characterized by their time constants and amplitudes to achieve a good compensation. The compensation is made by adding the reciprocal of the measured eddy currents to the gradient pulse thus neutralizing their effects. For accurate compensation 5 time constants are used.

Spatial Dependency of Eddy Currents

Experience has shown, that for smaller slice shifts it is sufficient to express the eddy fields into a 0th and a 1st order term.

Term	Description
0th	This term arises from an asymmetry of the gradient coil with respect to the magnet bore and cryo shields. This term is space independent and present in the complete imaging volume and adds to the nominal B_0 -Field. The amplitude (x) is given in the unit $\mu\text{Tm/mT}$. The time constant of the most significant 0th term component is about 500ms, although there are shorter time constant components as well.
1st	This is the most important term as it has the same symmetry as the gradient field itself. The amplitude is given in % of the applied gradient pulse.
2nd or above	There are also high ordered eddy currents present, they are usually small and negligible as long as the slice shifts and off center zooms are not too large. No compensation for the high order eddy currents can be made.

Measurement

First, the phantom center position is checked. The same preparation measurement is used as for Phantom shim, however, the small spherical phantom is used. If the phantom is not correctly positioned the measurement stops with an error.

The sequence generates several spin echoes with differing delay times of up to 9 seconds after the applied gradient.

The measurement is made at the moderate slice shifts of ± 5 cm to prevent contributions from higher order terms.

Evaluation

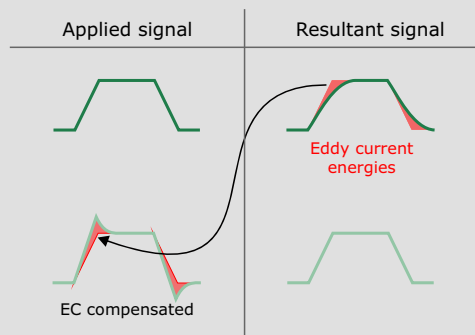
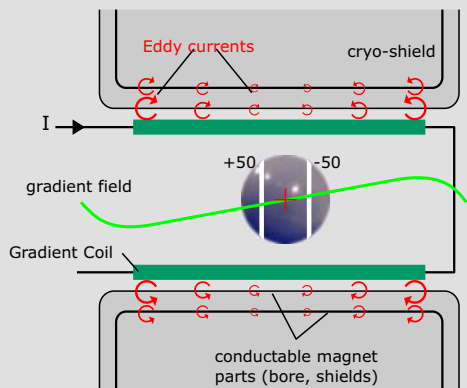
Signal Nr. 1 is measured for frequency offset. Signal 2 is measured to correct for the eddy currents generated by the slice-select gradient (measurement of long delays).

After the sequence was run for the two slice positions + 50 mm and - 50 mm the data first will be added up and then subtracted in order to separate the 0th order term and 1st order term.

Figure 237 Eddy Current Compensation - The Measurement

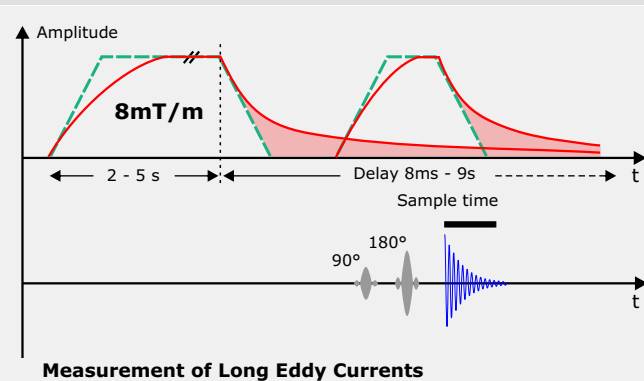
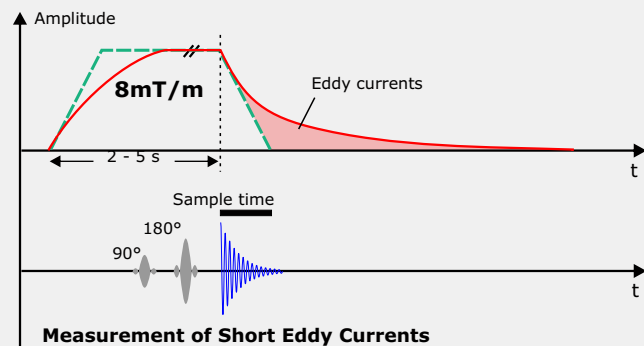
Eddy Currents

The Problem



Measurement

Results



The program has calculated:

Gradient Compensation (1st order term)

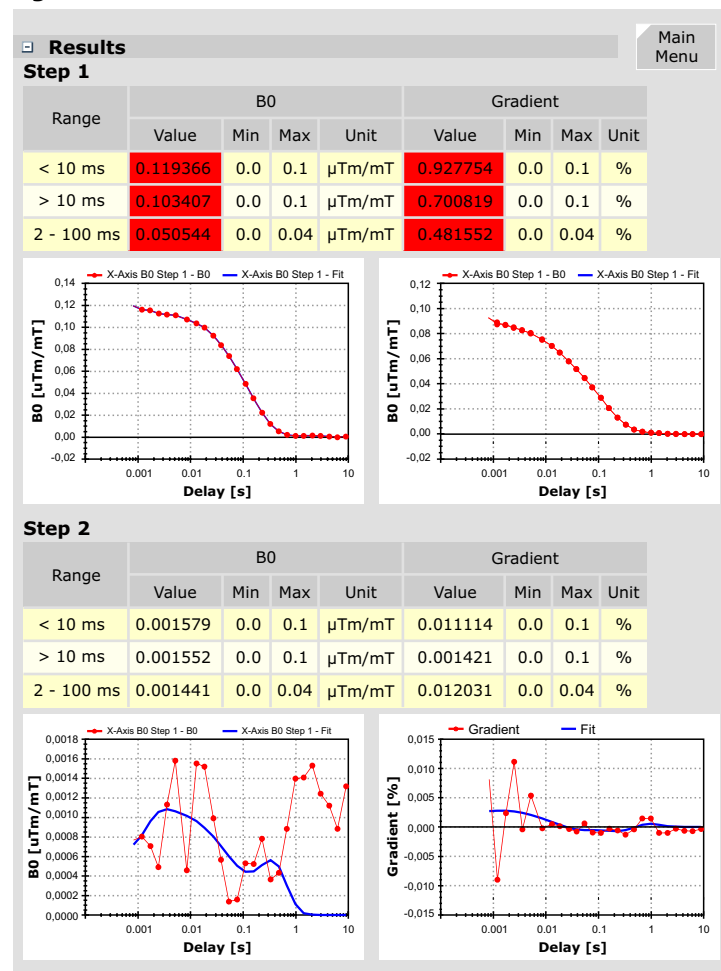
In this example the overshoot is 0.29% higher than the normal value without eddy current compensation.

B₀ Compensation (asymmetry)

The first plot of the uncompensated eddy fields shows a smooth curve. The next measurement will establish already a large improvement.

The final iteration, usually after 2 or 3 measurements, shows an oscillating pattern indicating the remaining field perturbations which are too small to be compensated.

Figure 238 ECC Results



Gradient Delay

The gradient amplifiers have a delayed response time (t_{nom} to t_{act}), due mainly to the coil inductance and is slightly different for each of the three gradient axis. The Gradient Delay measurement determines the delay time for each gradient axis. A precise timing of gradient pulses and RF is very important for good image quality.

Measurement

The measurement uses a Spin Echo sequence without phase encoding, hence only one line is measured. The gradient to be tested acts as the readout gradient and is switched on once between the 90°- and the 180°-pulse and then again during the readout time of the spin echo.

The first gradient pulse causes a dephase of the magnetization vector in the direction of the gradient axis. The amount of dephase is proportional to the time-amplitude integral of the gradient pulse. Subsequently, a 180° refocusing pulse is applied normally resulting in a spin echo at TE/2 time later. However, due to the dephasing of the vector by the first gradient pulse, the echo will only land in the center of the ADC window (which is also centered exactly around the TE time point) when an equal rephasing gradient has been applied at the proper time so that the rephasing is finished at the center of the ADC cycle.

For a non-corrected gradient delay the two gradient pulses are time-delayed. This does not influence the dephasing effect of the first gradient pulse, however the delayed second gradient pulse causes a rephasing delay and thus a time-delayed echo signal. From the echo signal delay the required correction time can be.

Difference Deviation Check

Each gradient is measured at three different gradient strengths - 1.5, 4 and 8 mT/m - giving three gradient delays for each gradient. For an ideal gradient system, all delays should be the same.

The **Max. Delay Deviation** is the maximum difference of the delay times with different gradient strengths, i.e. high gradient

pulses may be different from low amplitude pulses. If the tolerance is exceeded, this indicates some kind of nonlinearity in the gradient system.

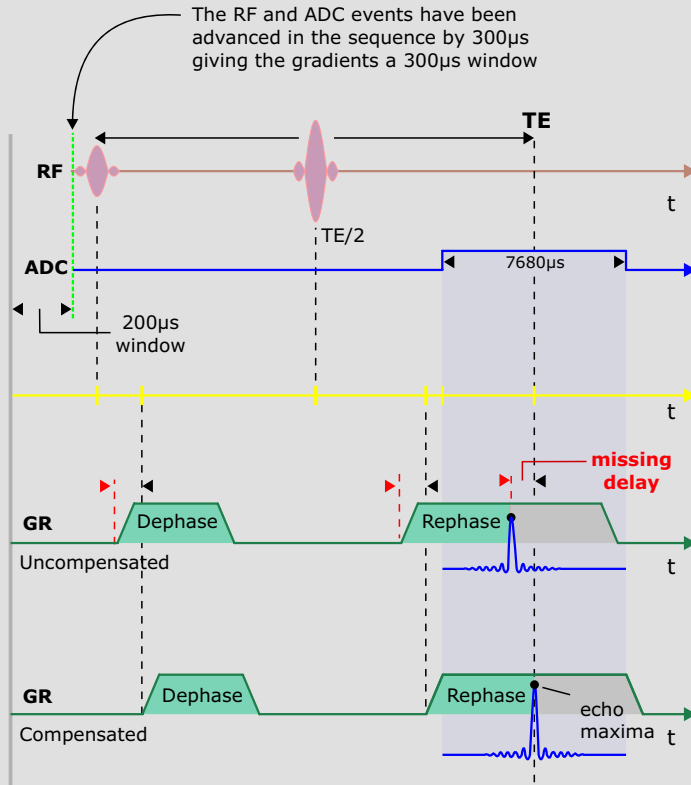
Evaluation

The new delays and maximum deviation values are displayed in two separate tables. If the procedure was successful, the values will be saved.

Figure 239 Gradient Delay Measurement and Results

Gradient Delay

Measurement



Results

Gradient Delay Values

	Old Delay				New Delay			
	Value	Min	Max	Unit	Value	Min	Max	Unit
X Gradient	35.0	0.0	200.0	us	34.9	0.0	200.0	us
Y Gradient	36.0	0.0	200.0	us	36.9	0.0	200.0	us
Z Gradient	32.0	0.0	200.0	us	31.5	0.0	200.0	us

Max deviation between different gradient strengths

	Max Delay Deviation			
	Value	Min	Max	Unit
X Gradient	2.3	0.0	5.0	us
Y Gradient	0.3	0.0	5.0	us
Z Gradient	0.2	0.0	5.0	us

Main
Menu

Gradient Sensitivity

The purpose of this measurement is to establish the actual LSB weighting (sensitivity) of the gradient amplifier so that accurate gradient field strengths can be calculated assuring a correct determination of image size and position in all three spatial directions.

Nominal Sensitivity

The nominal sensitivity (gradient field / bit) is the nominal gradient strength in mT/m divided by the digital resolution of the DAC. In the example below, the gradient system has a specified field strength of 40 mT/m and a 20-bit DAC, whereby 1 bit is used for polarity ($\pm 10V$ on the DAC) leaving 19 bits for the dynamic range. This gives a nominal sensitivity of $40\text{mT/m}/2^{19} = 0.07629 \mu\text{T/m}$. The first Gradient Sensitivity measurement will use this value.

Actual Sensitivity

The actual sensitivity, however, will depend on the characteristics of the DAC, the amplifier and the coil. In order to determine the actual gradient sensitivity an **object of known diameter** (the large 24 cm spherical phantom) is measured using a 35cm FoV. The resultant image is evaluated for the actual phantom diameter. The actual sensitivity is the old value times the correction factor which is determined by dividing the nominal phantom diameter with the actual measured image diameter.

Table Distance

During this procedure the patient table distance - the distance from the light localizer to the iso-center - is also measured (again, it was first measured some steps ago) and, if necessary, automatically corrected. If a correction is necessary the measurement is repeated.

Measurement

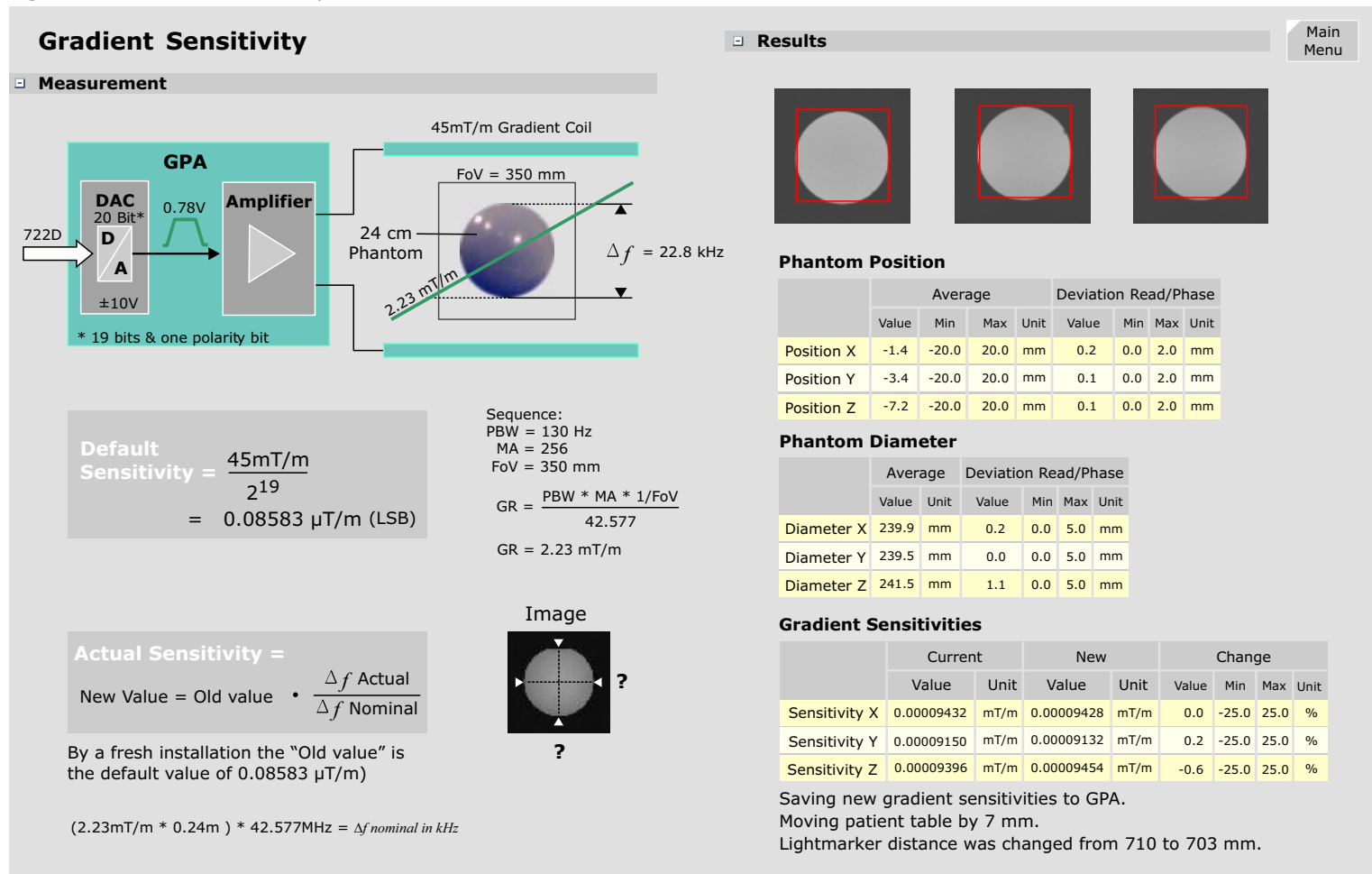
The measurement uses the gradsens sequence from which three images are generated, one for each orientation. For the sagittal orientation the usual orientation of the readout and phase encoding gradients are swapped.

Evaluation

The 24cm phantom is measured and checked initially for size and position. Deviations from the expected are corrected by assigning new correction factors.

The phantom position is checked again and if necessary the light distance will be corrected for the last time.

Figure 240 Gradient Sensitivity Measurement and Results



Coil Power Losses (CPL)

The purpose of this procedure is to measure the amount of power loss of the Body Coil. This value is used by the RF-Safety Watchdog (RFSWD) monitor to achieve a more accurate SAR measurement.

Measurement

A complete in-line adjustment with a frequency and transmitter adjustment performed. The voltage from the Adjust/Transmitter result is used to calculate the reference power.

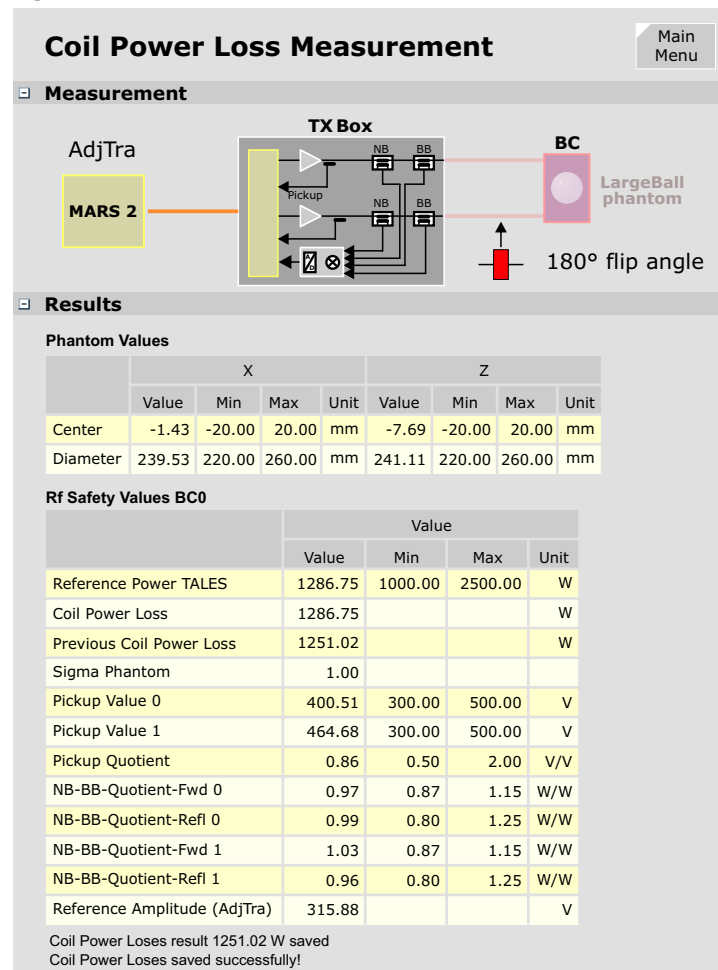
The large spherical oil phantom is used to reduce the dielectric resonance effects. A subsequent measurement is made to verify that the large (24cm diameter) spherical phantom is used and properly centered.

Evaluation

The measured reference power from the metering circuit in the TX Box is checked against expected specification parameters.

The value for the Coil Power Losses is calculated.

Figure 241 Coil Power Loss Measurement



BC Image Brightness

As mentioned in the Receive Gain calibration description, coil-specific characteristics influence the signal intensities and must be corrected for to avoid deviations in image brightness. The correction factors for local coils are established by the factory but for the Body Coil will be measured empirically using this procedure.

Measurement

The measurement uses the SN sequence from which three images are generated, one for each orientation.

Evaluation

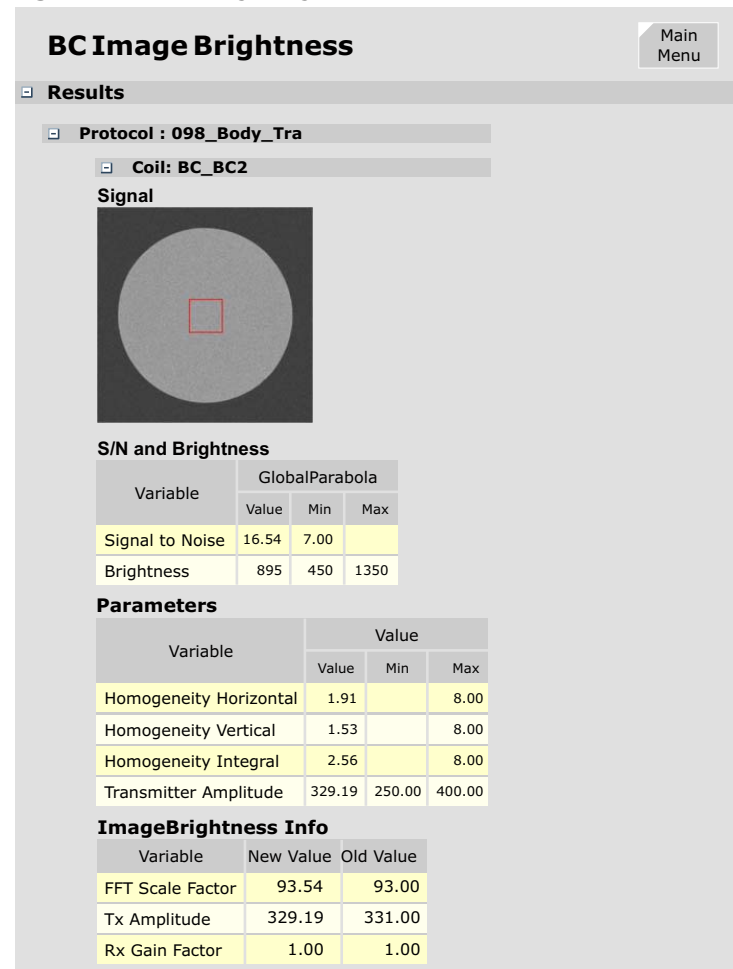
From each image the following values are determined:

- S/N value
- Brightness
- Homogeneity in horizontal and vertical directions

The brightness value is the signal intensity. The FFT scale factor is adapted to adjust the brightness to the middle of the specified value range (min and max). The transversal image is used for determining the FFT scale factor.

The transmitter amplitude (for 180°) for the sequence is also evaluated and is expected to lie with the middle of the value range (min and max). This value is specific to the phantom.

Figure 242 BC Image Brightness



Dynamic Field Map

Gradient pulses are always accompanied by unwanted time-dynamic field perturbations caused by e.g. eddy current or hysteresis effects. These field perturbations comprise multiple terms with different spatial dependencies. The 0th and 1st order terms are analyzed and compensated by the ECC and CTC Tune-up procedures and checked versus specification by the respective QA procedures.

The purpose of the dynamic field mapping function is to measure and analyze the residual field perturbations with sufficient spatial resolution in order to identify in particular the higher order terms.

It supplements the existing tools such that all spatial order terms are detected and checked versus specification.

Measurement

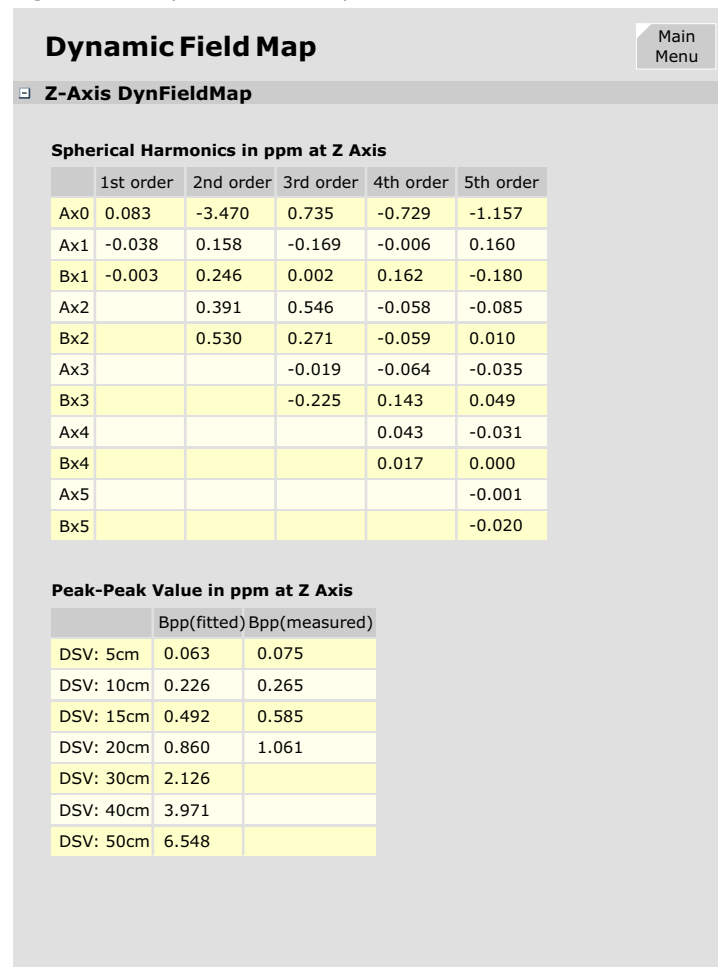
The dynamic field map service sequence is executed to measure the dynamic field perturbations using the large spherical phantom.

Results

- Peak-peak values in ppm
- Spherical harmonics up to the 5th order

After the measurement, data for image based correction of dynamic field perturbations in diffusion imaging will be saved.

Figure 243 Dynamic Field Map



Quality Assurance

The QA procedures are a set of procedures which are used to verify system performance and include all the procedures found in the Tune Up as well as additional procedures for testing overall system performance. Only these will be described since those procedures found in Tune Up are identical, with the exception that no values are saved:

- If you are out of specification after a Quality Assurance step, you must perform the corresponding Tune Up step.
- Every Tune Up and Quality Assurance procedure can be run in either Normal or Expert mode with one exception: the "field stability" procedure under Quality Assurance (very useful during shim procedure or trouble shooting stability problems) can only be run under Expert mode.

Go back to [MAIN MENU](#)

Coil Check

This procedure determines:

- Signal to noise ratio S/N
- Image brightness
- Image uniformity (inhomogeneity)

One or more protocols have been defined for each coil. After each protocol is measured, the routine will analyze the resulting image or images for S/N and in some cases the image intensity profile and image size. For transmit-capable coils the specific absorption ratio (SAR) is checked as well.

For coils where only one phantom position is necessary, the procedure performs all required measurements without user intervention. If more than one protocol is required, the user will be guided on screen through the various phantom positions.

QA Procedures

The procedures can be put into two major groups:

- **Tune-up Checks**
These procedures are identical to those found under Tune Up with the exception that the QA procedures only measure and evaluate without making changes and as such will not be described further.
- **System Performance**
 - Image Orientation
 - Spike Check
 - Stability Check
 - Field Stability Check
 - Long Term Stability Check
 - Synthesizer (NCO) Check

NOTE Some QA checks are safety relevant and if they can not be brought into specification the system can not be used for clinical imaging.
Refer to SKB entry SKB0102928.

Go back to [MAIN MENU](#)

Image Orientation

The 6 electrical connections between gradient amplifier and coil are checked for swaps between either axis or polarity which would cause an incorrect image orientation.

These checks are performed:

- swaps between the three axis (x to y, y to z, etc.)
- correct polarity of each axis

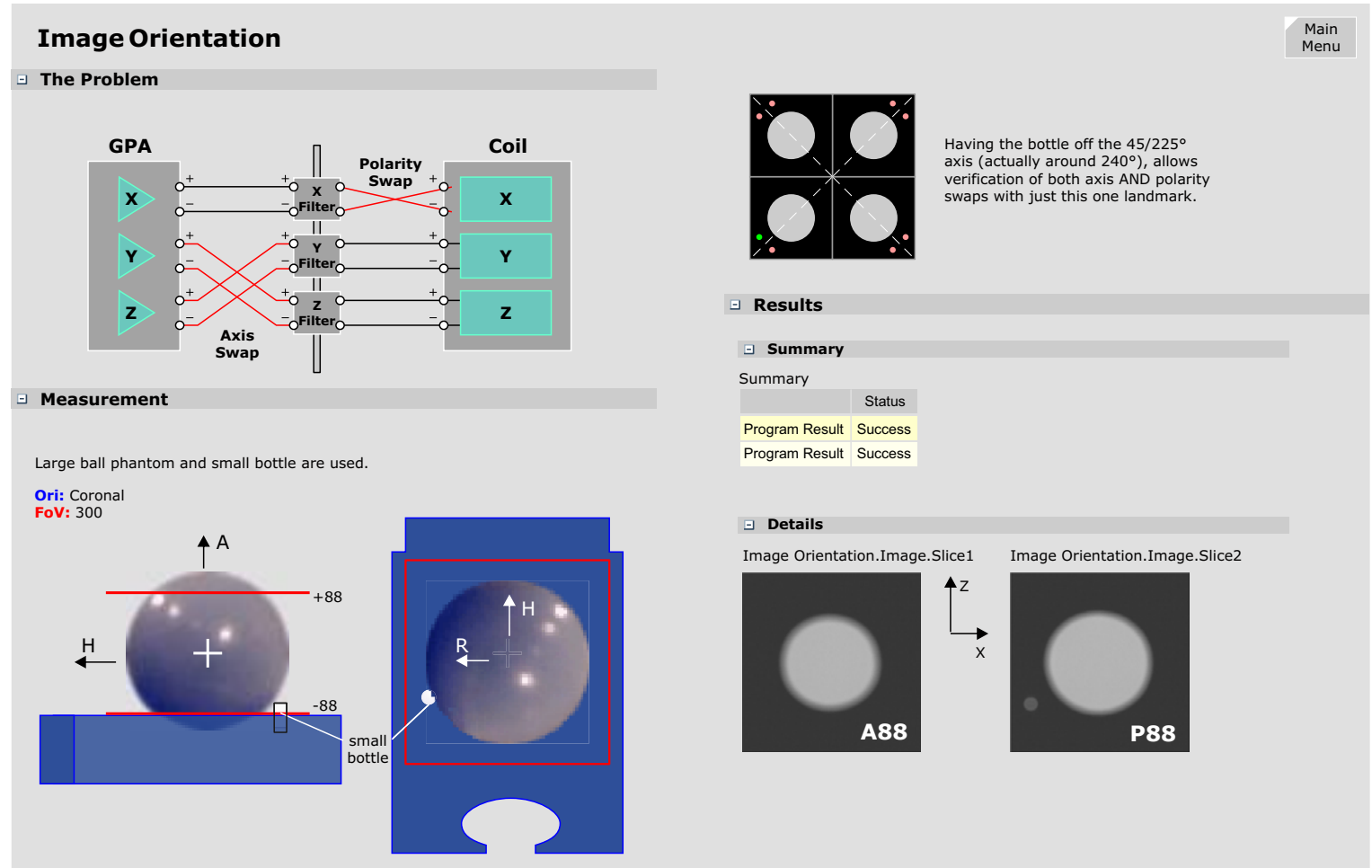
Measurement

Two coronal images are generated, the first at position A88 and the second at P88. The little bottle is expected in the second image in the bottom left corner.

Evaluation

The two measured images are automatically evaluated following the measurement. If the bottle is not in the expected position the program evaluates the image and automatically makes further measurements to locate the bottle and identify what has been swapped.

Figure 244 Image Orientation



Spike Check

The Spike Check looks for occurrences of "spikes" or RF interference. Moving components (usually caused by the pulsed gradients) in the RF room can cause electrostatic buildup and discharges which, if picked up by the RF coils during the measurement, create "spikes" within the raw data. Spikes are very short in time and fill only one or a couple points in the raw data set and will have amplitudes above the typical noise level. How the spikes affect the image depend on where the spikes are in the raw data set and how big their amplitudes are. Typical patterns in the image caused by spikes can be seen as a periodic intensity modulation or structure and can occur diagonally or even straight and in any direction.

Measurement

Normal

The spike sequence starts with an initial series of strong gradient pulses of both positive and negative polarities on all three gradients intended to cause a buildup of static electrical energy caused by any moving components (cables, covers, etc.). The ADC is then enabled and again a pattern of gradient pulses with amplitudes ranging from max positive to max negative are applied. Any static electrical discharges caused by moving or loose parts will be picked up by the receiving coil (usually the body coil) and ADC and stored in the raw data. The evaluation looks for intensities in the raw data set exceeding a threshold (usually noise) and counts both the number and intensities of the spikes and displays them numerically and graphically. The electrical discharges will be seen as "spikes" in the raw data. The position of the spikes in the raw data set relates to the gradient that was on when the spike occurred which may give an indication of the direction in which the cause of the spike may be found.

Advanced Spike Check

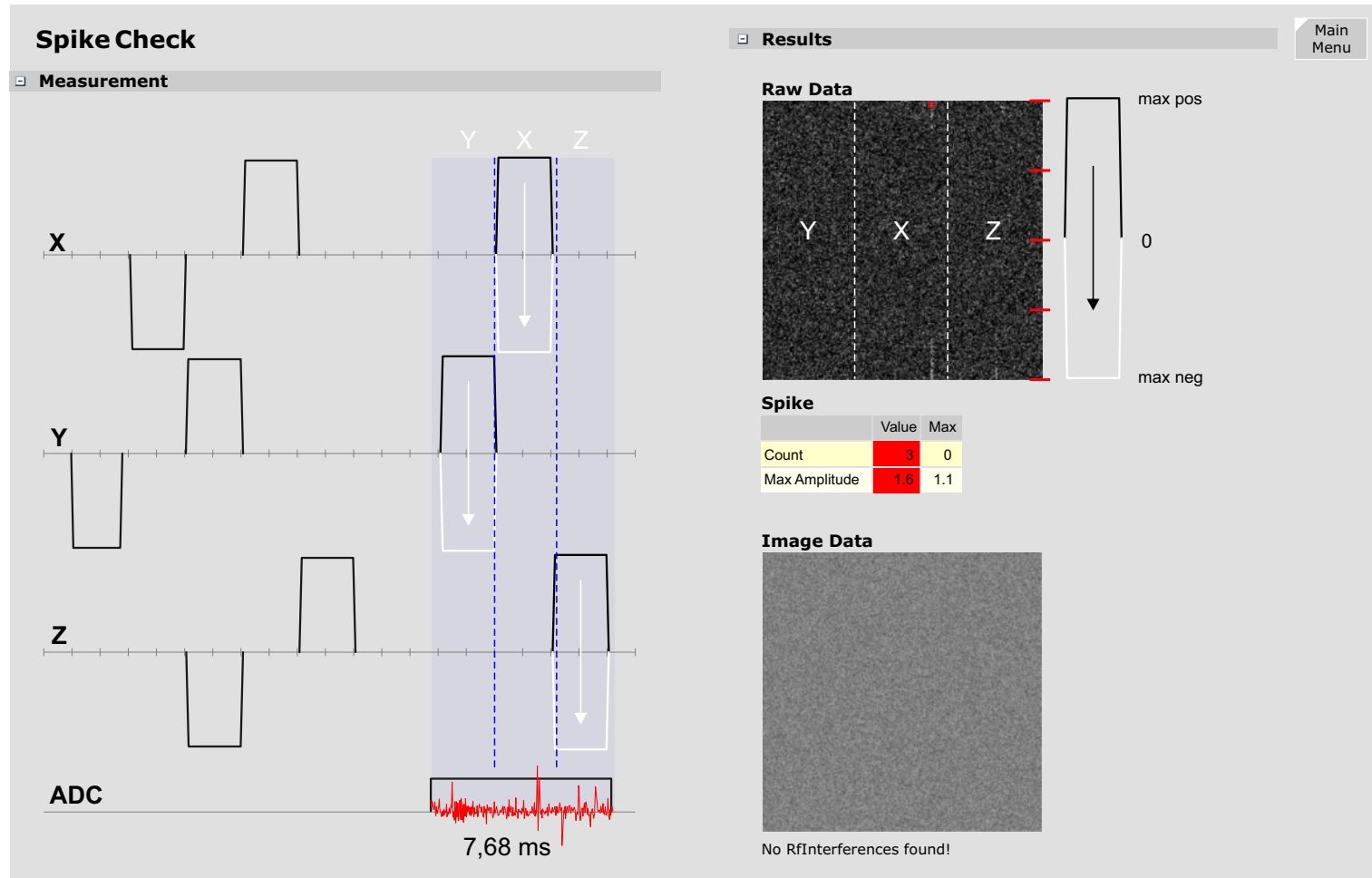
TRUFI and HASTE sequences and sequences used for EPI or BOLD-

Imaging may generate spikes which cannot be verified by the normal "Spike Check". In these cases the "Advanced Spike" can be used for spike trouble shooting. The "Advanced Spike" should be repeated at least three times to be assured the system is free of spikes. But be aware, if there weren't any spike-causing problems present before using this measurement, there may be afterwards!

Evaluation

The measured raw data and RF-interference image will be loaded and displayed.

The same algorithm for spike searching is applied to the image data of the RF-interference measurement. "Spikes" in the RF interference image are actually RF-interference patterns that have been picked up during the noise measurement and then fourier transformed resulting in a spike in the image data, i.e. the RF-interferences will result in pixels of increased intensity in the image data.

Figure 245 Spike Check

Stability Check

The process of acquiring the large number of MR signals necessary for an MR image over a relatively long acquisition time, usually in the order of several minutes, puts a high demand on the field generating systems (RF, gradients, magnet and shim) to have good **stability** and **reproducibility**. Instabilities in any of these systems result in smearing (aka ghost) artefacts. The Stability test is a useful tool which, when used in tandem with the Field Stability and Synthesizer checks, can help localize the section responsible for such artefacts.

Measurement

The stability test consists of a series of measurements using both spin echo and gradient echo sequences. These sequences are **without phase encoding** so all echoes should be identical. All three orientations are measured so that each gradient is performed as a read-out gradient. In addition, measurements are made at center and off-center (+50 mm) positions.

☒ Measurement

Coil Body

Sequence

☒ Spin echo sequence

☐ Gradient echo sequence

Repetition time 301

Echo time 20

Orientation

☒ Sagittal slice

☐ Coronal slice

☐ Transversal slice

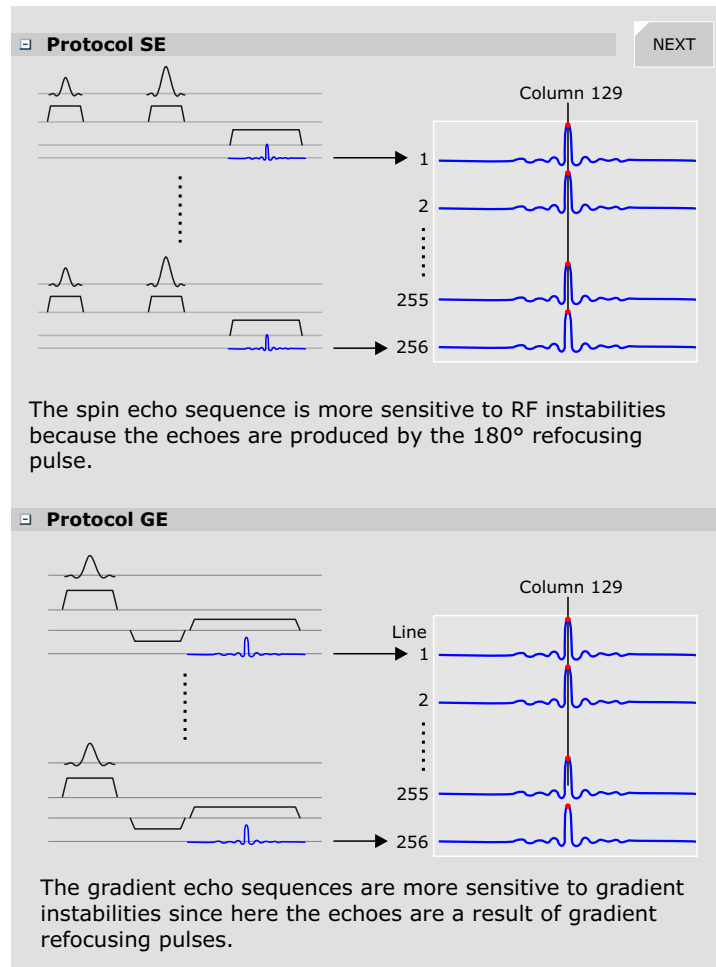
Slice Shift

☒ center (no shift)

☐ off-center

The spin echo sequence is more sensitive to RF instabilities because the echoes are produced by the 180° refocusing pulse. The gradient echo sequences are more sensitive to gradient instabilities since here the echoes are a result of gradient refocusing pulses.

Figure 246 Stability Sequences



measurements are also more sensitive to gradient instabilities at the center slices due to the enhanced effects of the further away from center the measurement

center (no shift) ☒ off-center (+50mm) NEXT



The field deviations caused by an instable gradient are greater off-center than at center. Therefore gradient instabilities will effect off-center slices more than center slices. The center slice is sensitive to B_0 field changes as well.

determines the role of each gradient axis. For some there is an instability in the X gradient. The initial orientation (X = slice selection) may be in c for the transversal orientation (X = read out) c.

gittal slice RO = Y (Y swapped for RO)
 ronal slice RO = Z
 ansversal slice RO = X



An instable gradient axis will cause greater deviations when used as a read out gradient.

Evaluation

Spin Echo

The spin echo sequence is most sensitive to 50/60 Hz when the echo time is set to 20ms (default).

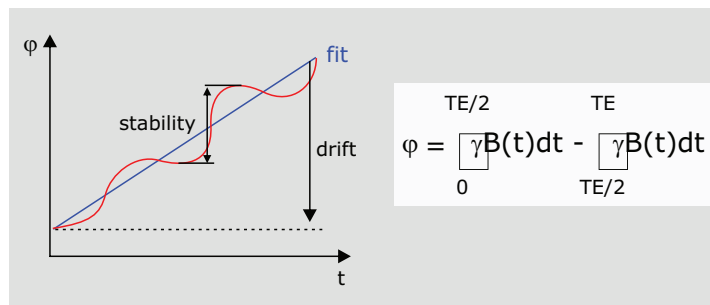
Gradient Echo

The gradient echo/flash sequence is sensitive from DC up to a few Hz which can be caused by mechanical vibrations.

Amplitude Stability

The amplitude value in the column of maximum signal is displayed in the graphical output.

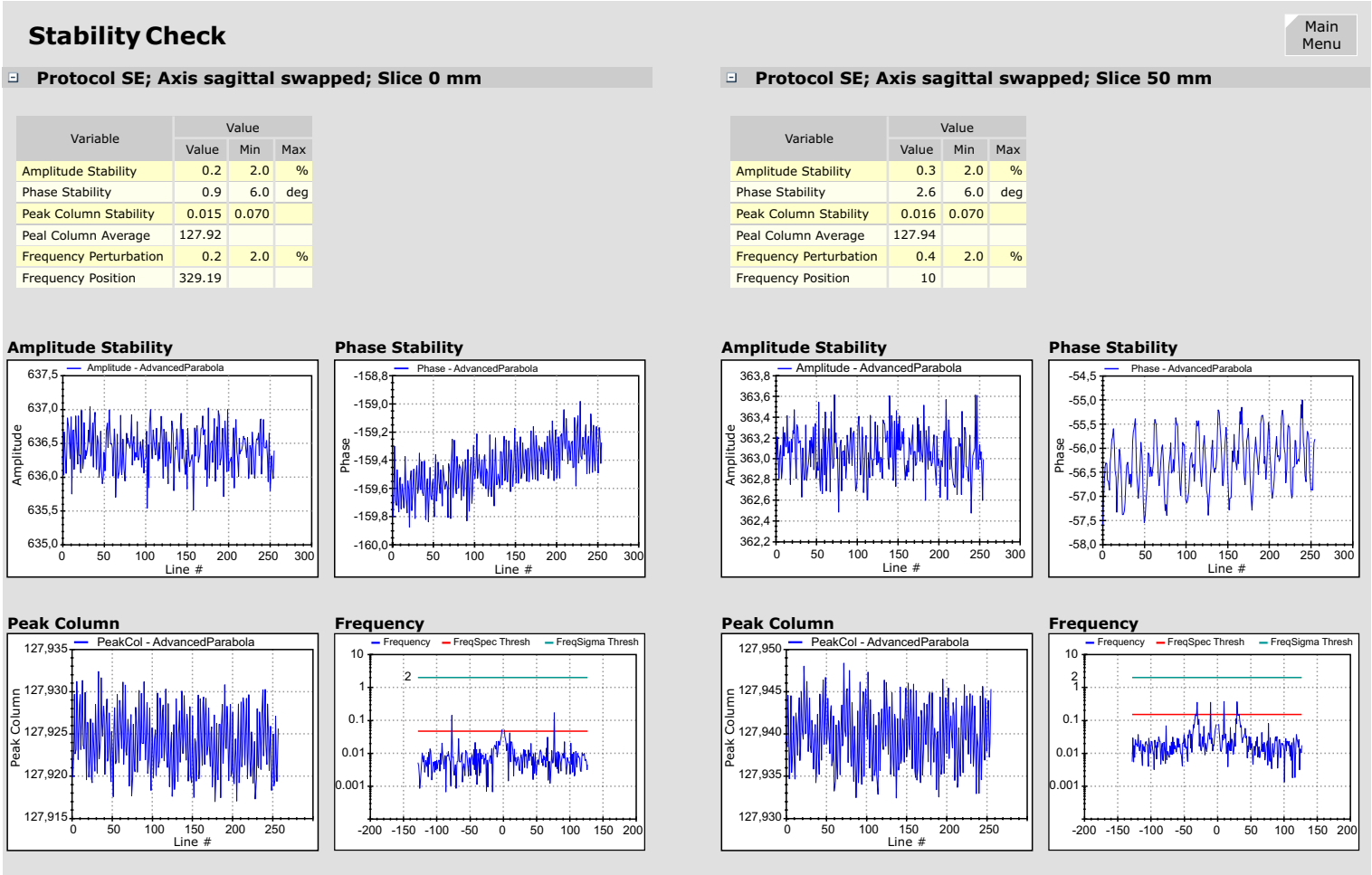
The phase shift of a spin echo is determined by the integral over the magnetic field from 90° to 180° pulse minus the integral from 180° pulse to echo:



For a constant field or a slowly varying field, both integrals cancel each other. However, for an oscillating field with a period of TE (20 ms = 1/50 Hz) this expression becomes maximum.

For the gradient echo sequence, the phase stability is evaluated by fitting a straight line to the phase data. Subtracting the value of the linear fit from the phase data results in the phase stability changes. The slope of the straight line indicates the linear phase drift.

Figure 247 Stability Check Evaluation



Synthesizer (NCO) Check

The NCO Check procedure is an important test to verify the proper operation of the synthesizer. It is used to validate the frequency and phase quality of the synthesizer with respect to the accuracy of the slice position.

Measurement

In the first step the reference signal is varied in frequency over the whole range of the MR system.

In the second step the phase is incremented in fixed intervals. In this way, the entire operating range of the oscillator can be tested.

The test is performed for one RX-NCO from group 1 and one RX-NCO from group 2. The two RX-NCOs are validated and can be used for testing the other NCOs.

Loop-measurements between the tested RX-NCOs and the TX-NCOs are performed. In this way, the correct functionality of the TX-NCOs can be approved.

In additional steps, all other RX-NCOs are tested against the approved TX-NCOs by testing group 1 RX-NCOs and group 2 RX-NCOs separately.

Evaluation

The measured values are evaluated according the expected frequency and phase changes and the deviations are displayed in a table.

NCO Check

[Main Menu](#)

MR, TXGroup 1, RXGroup 1, Testing RXNCO

Frequency and Phase Deviations

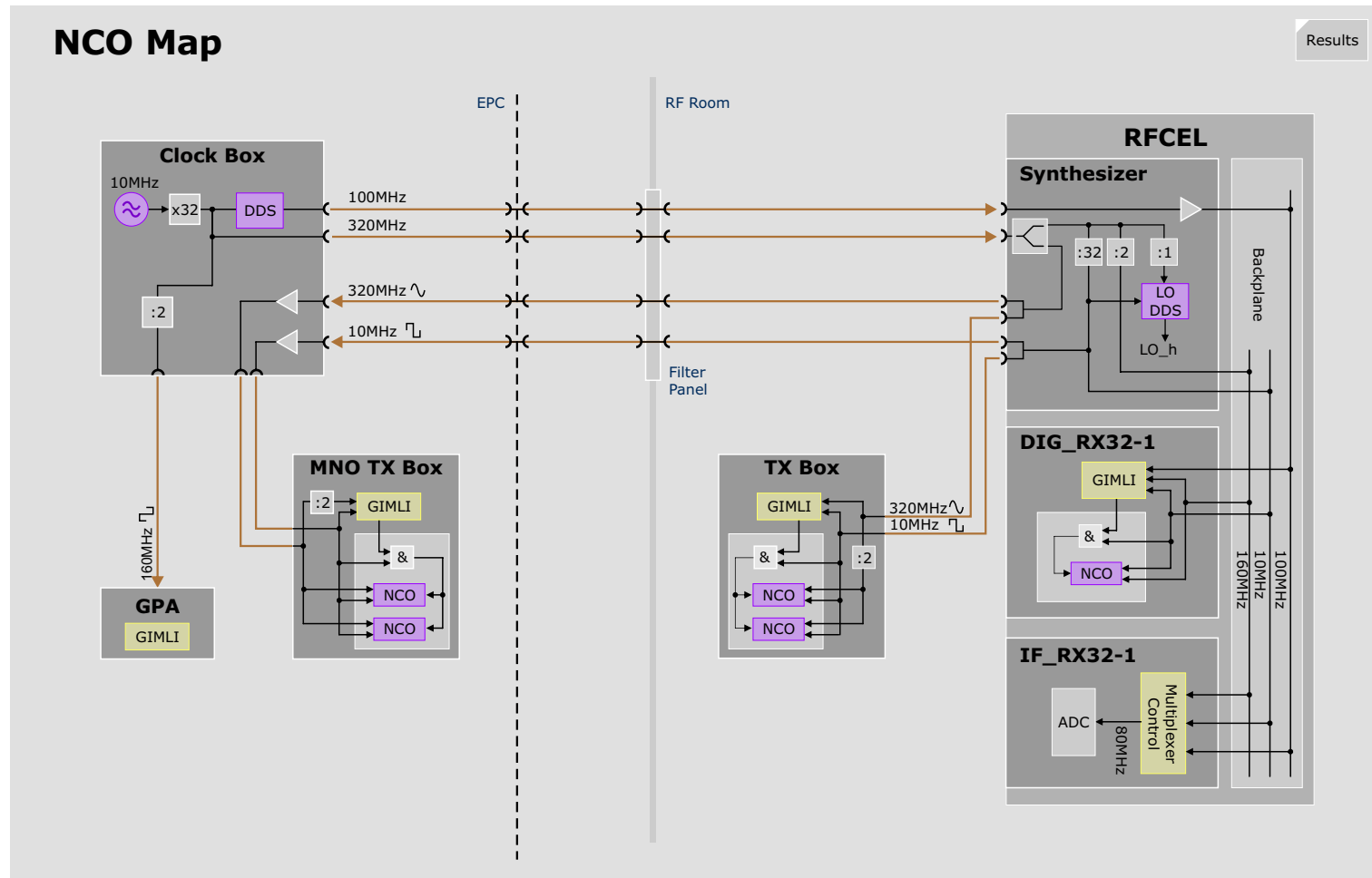
NCO	Frequency Deviation				Phase Deviation			
	Value	Min	Max	Unit	Value	Min	Max	Unit
1	0.34	0.00	10.0	Hz	2.27	0.00	10.0	Hz

Loop, TXGroup 1, RXGroup 1, Testing RXNCO

Frequency and Phase Deviations

NCO	Frequency Deviation				Phase Deviation			
	Value	Min	Max	Unit	Value	Min	Max	Unit
1	0.21	0.00	10.0	Hz	0.33	0.00	10.0	Hz
2	0.23	0.00	10.0	Hz	0.30	0.00	10.0	Hz
3	0.22	0.00	10.0	Hz	0.32	0.00	10.0	Hz
4	0.22	0.00	10.0	Hz	0.32	0.00	10.0	Hz
5	0.27	0.00	10.0	Hz	0.31	0.00	10.0	Hz
6	0.29	0.00	10.0	Hz	0.33	0.00	10.0	Hz
7	0.21	0.00	10.0	Hz	0.27	0.00	10.0	Hz
8	0.21	0.00	10.0	Hz	0.33	0.00	10.0	Hz

Figure 248 NCO Map



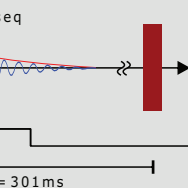
ity

useful in detecting magnet field changes. To
es by the gradients and RF a very simple FID
ed.

s used and is shown below.

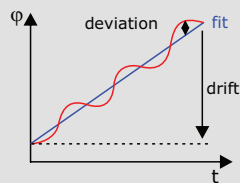
Stability Sequence

y



$$\frac{d\phi(t)}{dt} \propto \gamma B \cdot t$$

c ratio of the proton
nt magnetic field
every 301 ms.



Os are measured over a 154 second period
301 ms). From every FID the phase over time is
linear fit is performed, the slope giving the

he fit should be long to get a high accuracy but
y the length of the FID. A good compromise is
ges over 50Hz line hum which is usually of no
surement.

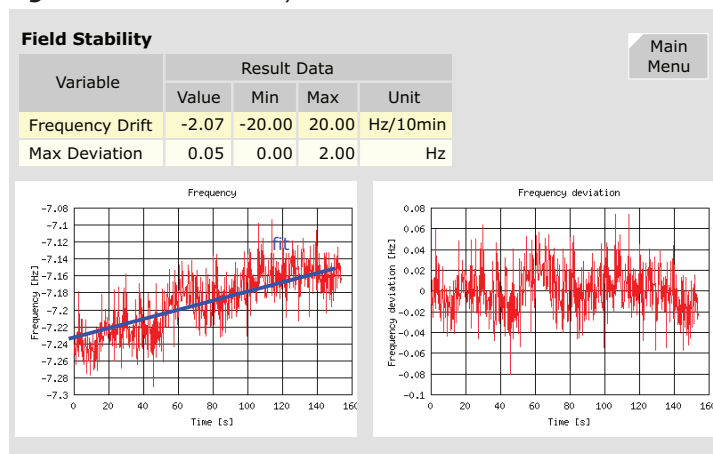
Evaluation

To replace the integral, it is assumed that the field $B(t)$ does not change much during sampling of the FID. So the field can be calculated directly from the phase at time t . However, because of hardware imperfections it is better to take the slope of the phase over time which also gives the field.

The result of the field measurement is the field (=frequency) over the measurement time. As a measure of quality the linear drift over 10 min (typical measurement time of an imaging sequence) and the maximum deviation from this linear drift are taken. When the measurement time is different from 10min, the drift is corrected for 10min by multiplying with the ratio (10min / actual measurement time).

The drift over 10min should not exceed the pixel bandwidth of low bandwidth sequence (e.g. 20Hz). The max deviation value is important for gradient echo sequences to avoid smearing. To be able to use long echo times it should be typically less than 2Hz.

Figure 250 Field Stability Check Evaluation



Stability_LongTerm Check

A new application of MR-Imaging is to make activities in the brain visible. For that purpose the patient will be exposed to a periodical optical, acoustical or other stimulation. During the exposition, a series of MR-images will be measured. Afterwards the images will be evaluated for changes of the signal intensity. The regions of the brain activated by the stimulation can be localized by a very small variation of signal intensity. These variations can easily be hidden by various instabilities of the system and/or the environment. Therefore the instability of the MR-system and interferences from environment must be below a specified level.

The Stability_LongTerm Check determines the sum of all instabilities while doing a phantom measurement. Typical causes of instabilities are:

- Local B0 field variation caused by temperature variation in the shim-unit generate frequency instabilities. The result is a virtual movement of the measured object during the scans.
- B1 field variations by instable RF produce image intensity variations.
- Gradient fields generate frequency and phase variations, causing also virtual movement of the object.
- Mechanical vibrations, e.g. by cold head, produce signal intensity variation by the movement of the measured object relatively to the magnetic iso-center.
- Movement of phantom fluid cause phase errors.

Measurement

NOTE Between phantom positioning and measurement start wait at least 10 min, since even small movements of phantom liquid will affect measurement results.

The measurement is done with the 8 Channel Head Matrix coil and the large bottle phantom (7.3 l).

A series of 512 measurements (EPI_FID sequence) with 11 slices will be performed and repeated once again.

Prior to the measurements, the system performs a warm-up scan to achieve thermal equilibrium of the hardware components. The first measurements are performed with a flip angle of 30° (evaluation of B1 field stability), the second with a flip angle of 90° for 1.5T systems and 70° for 3T systems (minimization of unstable RF, detection of B0 and gradient-like interferences).

Evaluation

The following graphical and numerical evaluations will be performed:

- Mean values of a 15x15 ROI related to the mean value of all measurements as function of image number
- Standard Deviation of a 15x15 ROI mean values (linear and logarithmic plot)
- Center position in x and y direction of phantom, dependent of image number (Expert mode only)
- Measurement protocol values
- Statistics of Mean values (for ROI size 15x15 and 25x25):
 - Mean Image Intensity
 - Variation peak-peak
 - relative Variation peak-peak
 - Standard Deviation
 - total relative linear Drift
 - Standard Deviation from linear Fit
- Image quality parameters such as SNR and Ghosting
- Statistics of Phantom Movement (Expert mode only)

Normally these evaluations will not be displayed.

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Figure 251 Longterm Stability

Longterm Stability Check

Measurement

Flip Angle = 30 deg

General Information

Name	Value	
	Value	Unit
Multi Slice	17 slices, 5x5 mosaic	
Slice Orientation	transversal	
Slice Thickness	5	mm
TR	1000	ms
TE	30	ms
FOV	1200	mm
Flip Angle	30	deg
Phase Correction	3-Line B0	
No of Measurements	512	
No of Prep. Scans	5	

First Image of Series

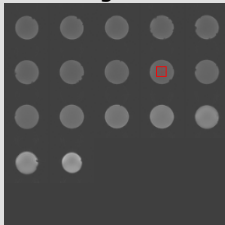


Image Quality

Name	SNR	Ghosting	
		Value	Unit
Mean Value	102.308	43.471	%
Standard Deviation	4.192	0.304	%

Flip Angle = 90 deg

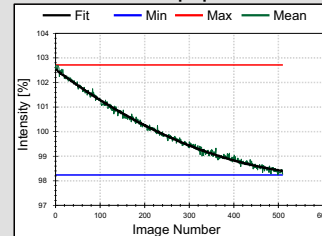
Results

Main
Menu

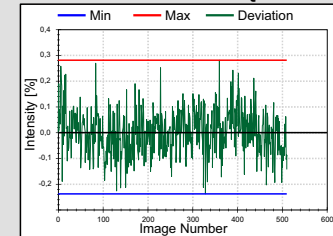
Mean Values

Name	15x15 ROI				25x25 ROI			
	Value	Min	Max	Unit	Value	Min	Max	Unit
Mean Image Intensity	532.801				543.616			
Peak to Peak Variation	23.813				24.095			
Relative Peak to Peak Variation	4.469			%	4.432			%
Standard Deviation	1.218			%	1.211			%
Total Relative Linear Drift	-4.134			%	-4.114			%
Standard Deviation from Linear Fit	0.224			%	0.218			%
Peak to Peak Deviation from Linear Fit	1.092			%	1.065			%
Total Relative Quadratic Drift	4.134	-8.000	8.000	%	4.114	-8.000	8.000	%
Standard Deviation from Quadratic Fit	0.087	0.000	0.220	%	0.073	0.000	0.200	%
Peak to Peak Deviation from Quadratic Fit	0.518	0.000	1.100	%	0.471	0.000	1.000	%

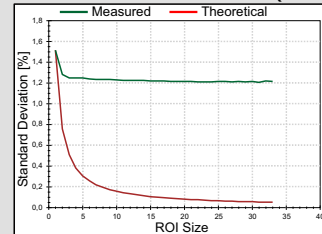
Mean Values of ROI in prop. to Overall Mean



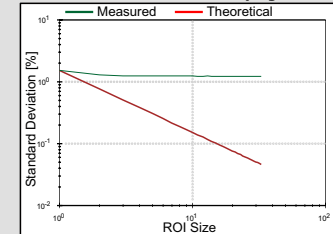
Deviation of Mean Values from Quadratic Fit



Std. Deviation of ROI Mean Values (Linear)



Std. Dev. of ROI Mean Values (Logarithmic)



RF Noise Check

The RF Noise Check is used to measure RF interferences.

Measurement

The noise measurement is performed with the service sequence rf_noise. The sequence is without RF or gradient pulses, just an open receiver.

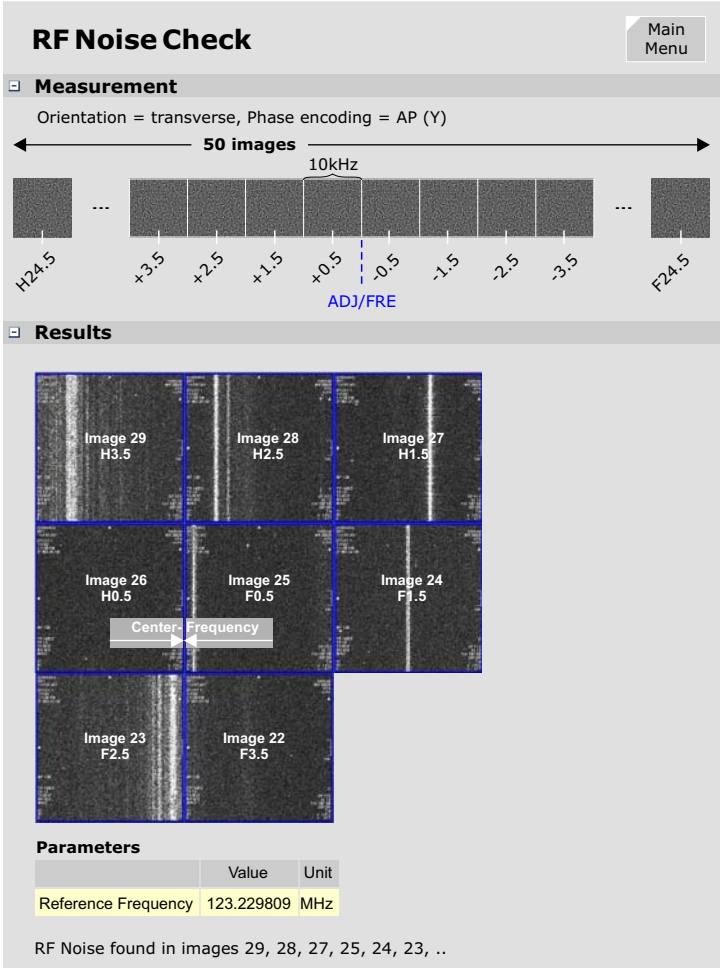
The sequence produces 50 images (i.e., slices). The difference between the images is the variation of the center frequency. Each image covers 10 kHz (256 times 39 Hz) so that the complete receiver bandwidth of +-250 kHz is covered. The border between the center slices 25, 26 is the current center MR-system frequency.

Evaluation

The evaluation part of the procedure is comparable to the Spike Check but the measurement is performed with limited bandwidth.

The evaluation algorithm looks for RF interference in the acquired image data i.e., dots and streaks in phase-encoding direction and noise level.

Figure 252 RF Noise Check



Gradient Rise Time Check

The Gradient Rise Time Check measures the minimal gradient rise time or the so-called "Slew Rate".

Measurement

The measurement uses a spin-echo EPI sequence; the tested gradient channel has phase-encoding as well as readout functionality. In the readout interval, develop five echoes because of the bipolar triangle gradient pulses. These echoes should appear according to the pre-dephasing exactly at the pulse spikes. During the measurement, the pulse amplitudes run the range from -Gmax to +Gmax.

Simultaneously with the pulse amplitudes, the ascent of the pulse ramp (with constant pulse ramp time) changes and determines the limits of the gradient system.

If there is a deviation of the echo position from the nominal position, the required slew rate could not be reached.

The limits can be determined and calculated from the raw data (representation of magnitude with corresponding windowing).

Sequence of procedures:

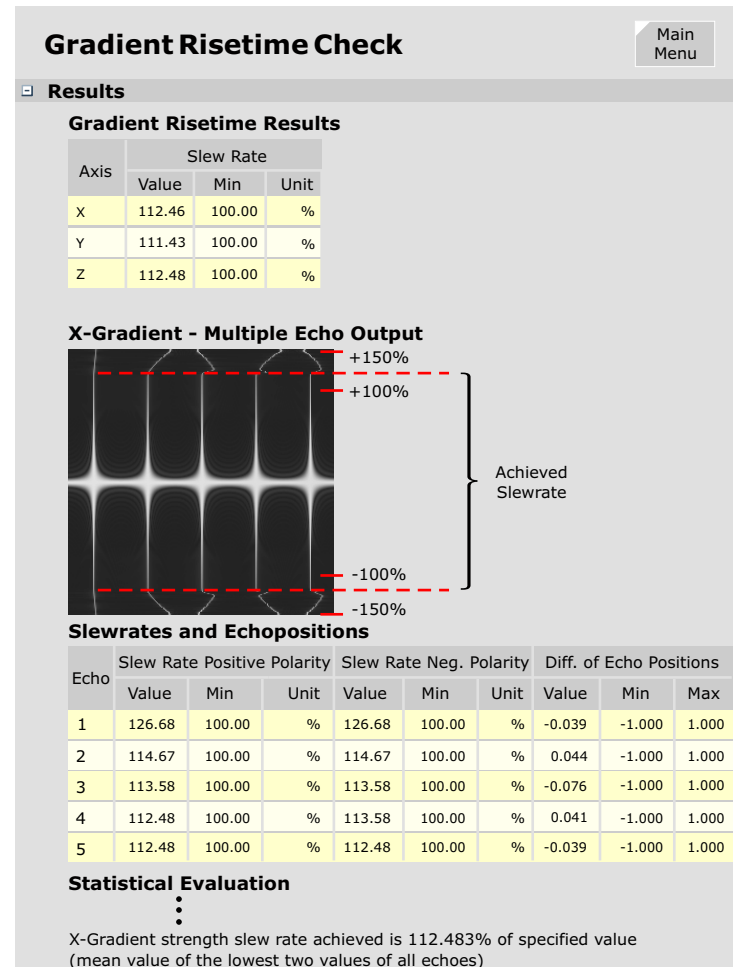
- Verification check of the used phantom
- Gradient Risetime measurement for all three gradient axes
- Evaluation of the raw data image and calculation of gradient rise time
- Numerical and graphical output

Evaluation

The following output data will be shown in the report:

- Graphics of echo maximum vs. gradient slew rate, echo 1 -5
- Statistics of measured values against specifications
- Y/Z/X- Gradient Rise Time Results
- Y/Z/X Gradient Rise Time Multiple Echo Output (raw data image)

Figure 253 Gradient Rise Time Check



Section

12 Service Tools

Introduction

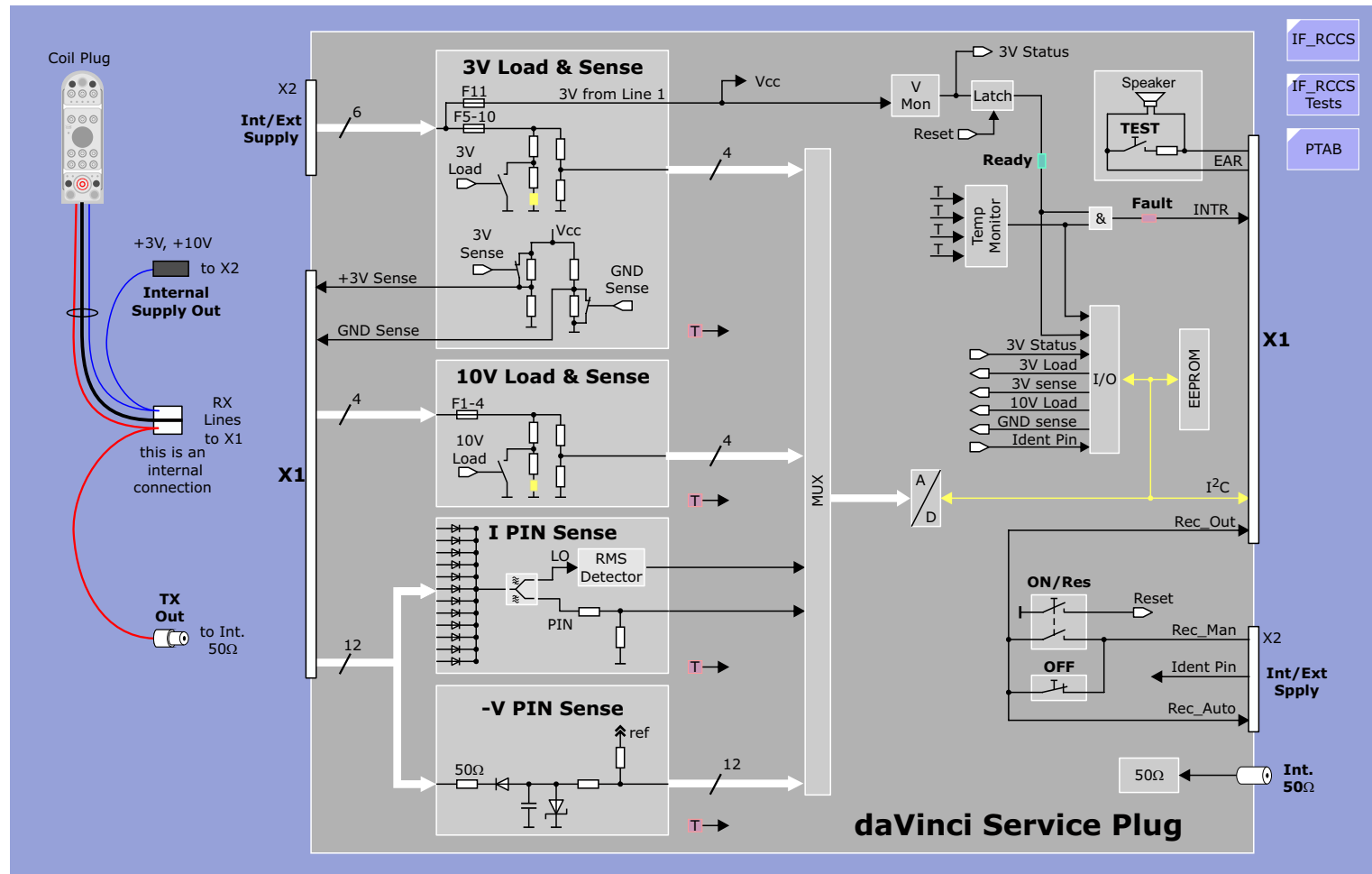
This section provides a functional description for any service tools having electronic or mechanical functions in and of themselves.

These include:

- [Service Plug](#)
- [Helium Service Syphon Kit](#)
- [Magnet Power Supply](#)

Go back to [System Diagram](#)

Figure 254 Service Plug Block Diagram



Helium Service Syphon Kit

Due to the no boil-off concept, a Helium Fill Syphon is not standard (except for mobile systems). If a refilling is necessary (e.g. after Cold Head service), an adaptor - the service syphon kit - provides access into the magnet turret.

The Helium Service Syphon Kit is used for helium filling on all 4 K magnets and is compatible to the existing dewar syphon hardware.

Table 2 Service Syphon Use for OR98/OR99/OR103 Magnets

	130*mm leg	190*mm leg	End Pieces
Top Fill	2	N/A	parts 6 and 7
Bottom fill	2	1	part 8

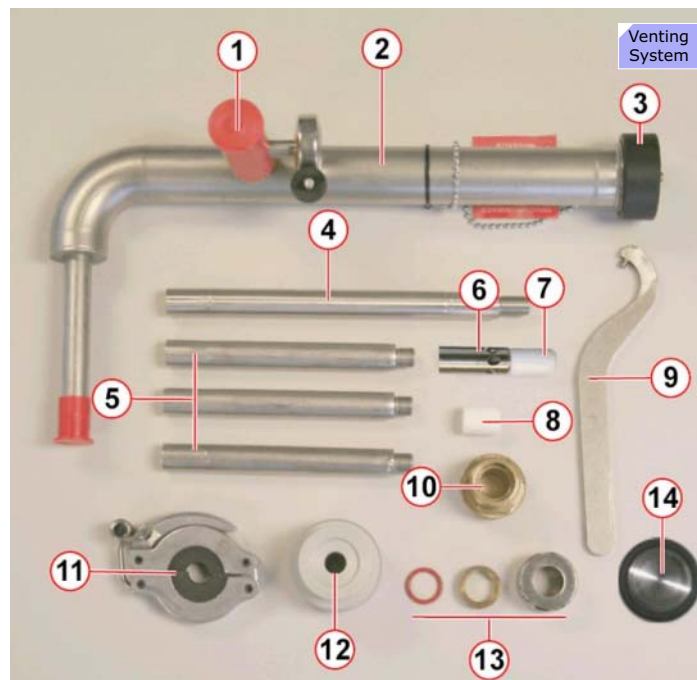
The syphon diffuser should optimally be 50-80 mm above the syphon cone for top filling.

Two fill configurations are possible:

- **Top-Fill** - This is the normal helium fill procedure configuration when the magnet is LHe-filled under field, during standard service or routine maintenance.
- **Bottom-Fill** - This fill procedure must be used when the magnet level is reading zero percent, or the magnet temperatures have risen above helium temperature (check Ceramic Carbon Resistor (CCR) values. It will avoid unnecessary helium loss and fill time. Bottom filling is required only in certain circumstances: after a quench or long transportation times.

NOTE The service syphon kit, part no: 10126651 is a service tool. To ensure that the LHe-filling procedure is performed correctly, refer to Installation>Start-up.

Figure 255 Helium Service Syphon Kit



- 1) Vacuum valve
- 2) Service Syphon assembly
- 3) Syphon end cap
- 4) 190 mm extension leg (1 x)
- 5) 130 mm extension legs (3 x)
- 6) 6 hole de-ice and helium fill diffuser tip (top fill)
- 7) PTFE de-ice extension tip (top fill)
- 8) 22 mm PTFE seal (bottom fill)
- 9) Syphon "C" spanner/wrench
- 10) Syphon adaptor bush
- 11) Syphon leg support clamp
- 12) Nitrogen fill adaptor and O-ring seal
- 13) Syphon ring nut, brass syphon flange, syphon O-ring seal
- 14) NW25 flow restrictor

Magnet Power Supply Unit 3600

Overview

The Magnet Power Supply Unit (MPSU 3600) is an air-cooled, portable service tool used to ramp up (energize) and ramp down (de-energize) the magnet for start-up and service purposes.

The MPSU is software controlled requiring a laptop or computer (the Host) which will be connected to the MPSU via a RS232 serial cable, included with the supply. For ramping the MPSU needs to communicate with the Magnet Supervision Unit requiring an additional cable to connect to the CAN bus of the MR system. for line power supply, and for connecting the magnet coil via ramp cables.

The MPSU consists of:

- **MPSU 3600**
- **Line power connection cable**
- **Ramp cables** - Delivered with the system (or extension kit where required)
- **RS 232 connection cable**
- **CAN bus connection cable**
- **Safety Manual**

Physical Description

The MPSU is enclosed in a robust blue shipping case, mounted on a trolley. The trolley has wheels to enable movement when required, but can be easily lowered on to skids when in position.

Removal of the lid of the shipping case exposes the working face of the MPSU, enabling all the connections to be made and monitoring of local indicators to take place.

The MPSU may be lifted clear of the trolley, if required, but care must be exercised as it weighs 80 kg (176 lbs).

The long sides of the MPSU have open slot grills to allow the flow of cooling air.

Servicing

The MPSU is a field replaceable unit (FRU), there are no user serviceable parts inside the 3600 MPSU. **The cream metal covers must not be removed.**

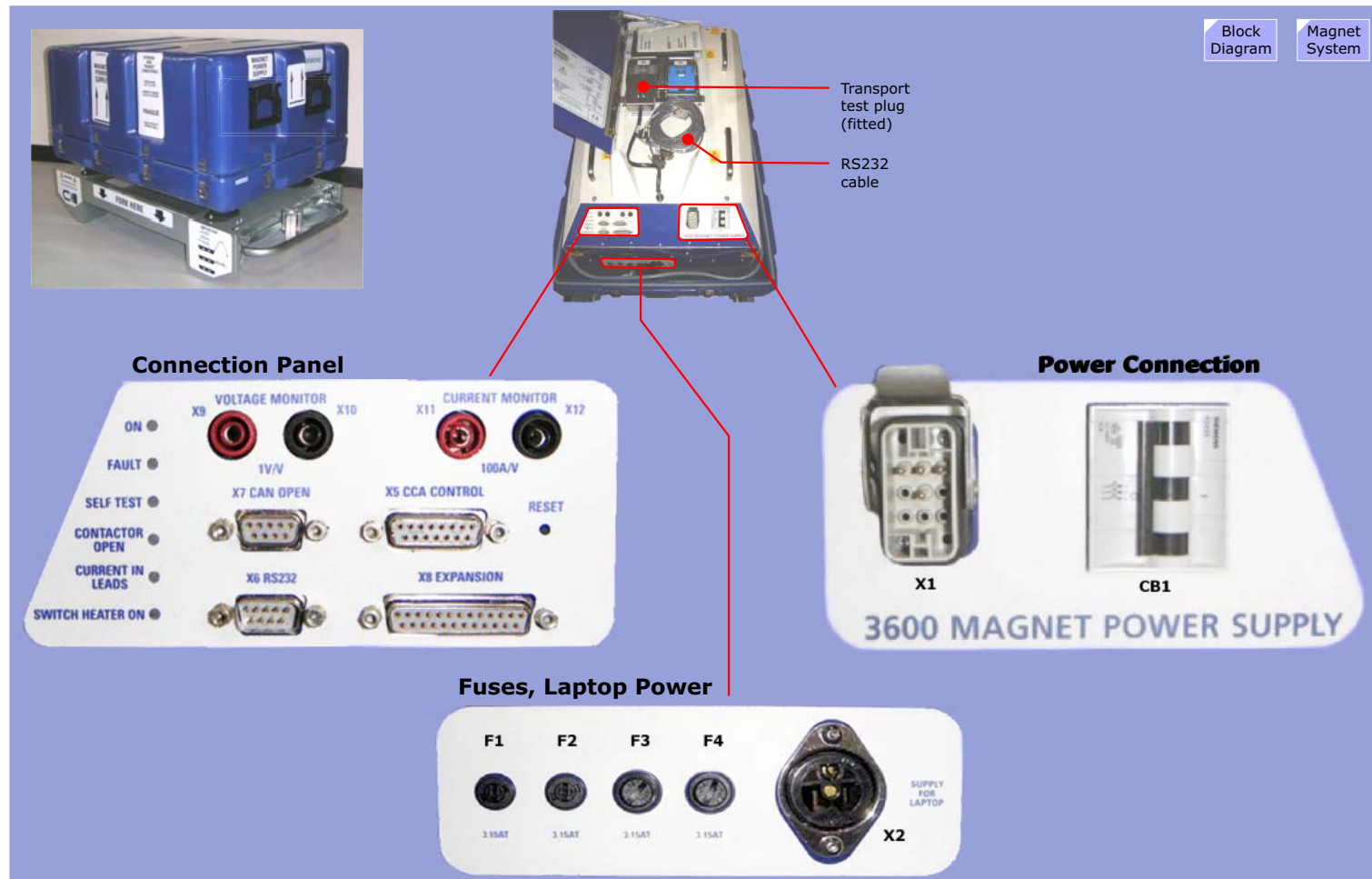
Maintenance is restricted to checking and replacing fuses **F1** to **F4** on the front panel, and external cleaning, using a slightly damp cloth. If the 3600 MPSU fails to operate, it must be returned to a local tooling centre for repair, and a replacement obtained.

Handling and Storage

The module may be stored in an environment as follows:

Parameter	Specification
Ambient temperature	-20 °C to 55 °C (-4° to 131° F)
Ambient pressure	700 to 1060 mbar (10 to 15 psi)
Relative humidity	40 % to 80 % non condensing

Figure 256 Magnet Power Supply Connections



Function

Communications

Two types of communication links are provided:

- **CAN** - for communication to the magnet supervision unit
- **RS232** - for menu-driven software control

Power

The 400 VAC 50/60 Hz 3-phase line voltage is fed via **CB1** to a line filter. Two phases are supplied to transformer **T1** primary over fuses **F3** and **F4**. T1 secondary output, fused with **F1** and **F2**, is used for feeding an internal logical power supply, fans, output contactor and a power outlet for a laptop.

Control Circuit

The control circuit monitors and controls the operation of the MPSU. Ramping commands issued by the operator are performed fully automatic by the MPS. Magnet-specific data required by the controller is obtained directly from the Magnet Supervision (MSUP) via the CAN-bus connection **X7**. If the CAN-bus is disconnected, the MPSU is inhibited and the fault is displayed at the host or Service laptop PC.

Ramping Power Supply

The 3-phase input is rectified and regulated via a MOSFET switch and available at output connectors **X3** and **X4** (max. 725 A @ 12 V).

Run Down Load, MOSFET switch

To de-energize the magnet the energy stored in the magnet coil is applied to and dissipated by diodes in the Run Down Load (RDL) which are embedded in heat sinks and cooled by fans.

Shorting Contactor

The MPSU is equipped with a shorting contactor that is normally closed. During different ramp phases, the control circuit will open the contactor when:

- the MPSU ramps up the magnet
- the MPSU ramps down the magnet

NOTE

Should the MPS shut off during a ramp (up or down) the magnet current will safely run down via the ramp cables and closed shorting contactor.



WARNING

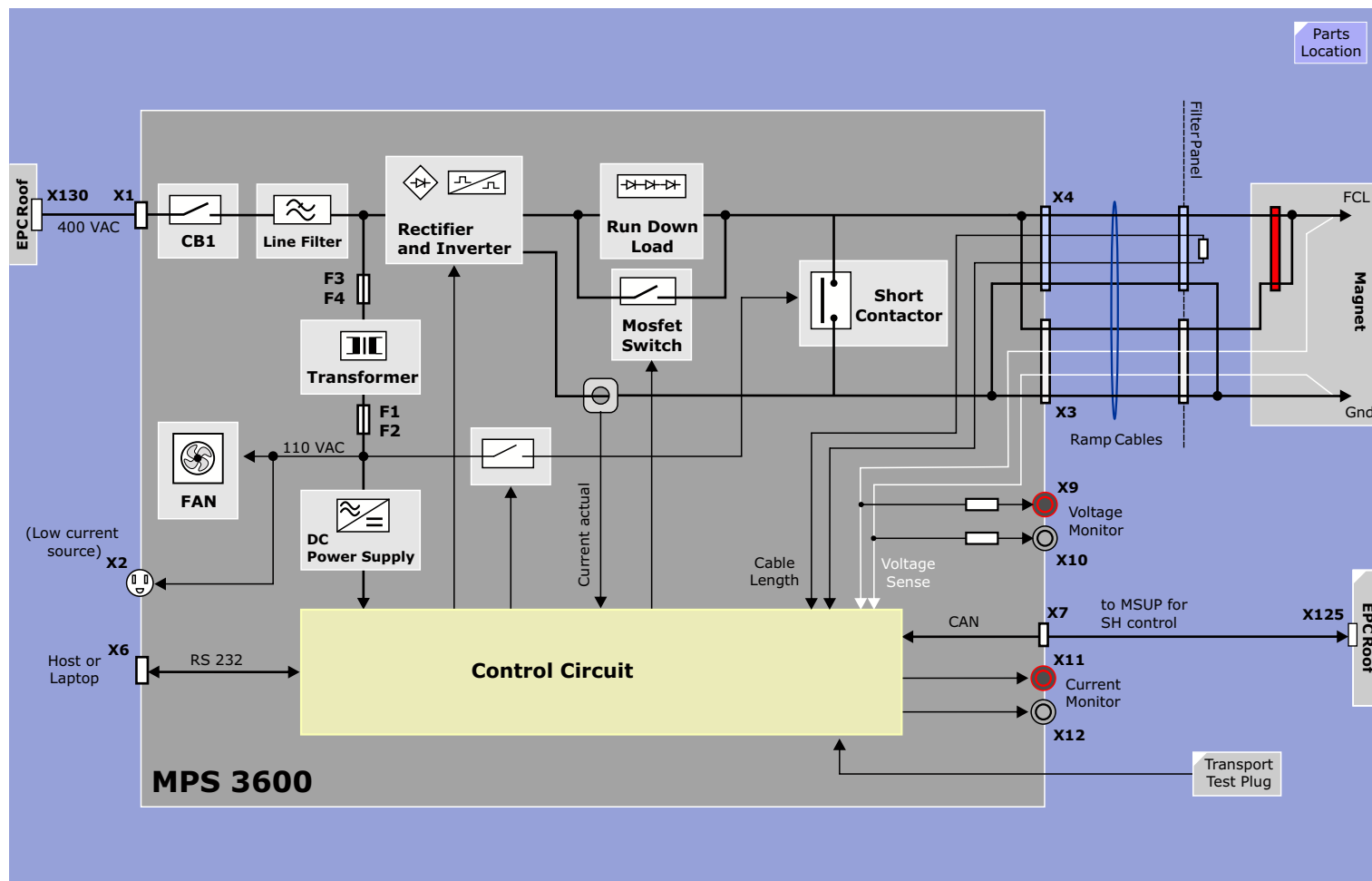
Risk of electrocution. Failure to observe the following safety warning may result in physical injury and equipment damage. If you suspect power is flowing in the ramp leads, DO NOT DISCONNECT THE RAMP LEADS from the MPSU or the magnet.

Actual output current is measured by a current transducer and can be measured at connectors **X11** and **X12**.

Current/Voltage Monitoring

The MPSU output current and voltage are monitored by the Control Circuit. It is reported via RS 232 to the host or Service laptop PC. Additionally, a DVM can be connected to the voltage monitor terminals (**X9**, **X10**) or current monitor terminals (**X11**, **X12**) on the MPSU connector panel.

Figure 257 Magnet Power Supply 3600 Block Diagram



Initialization

When the MPSU is switched on (at **CB1**), a 'boot up' screen is displayed. Operator initials are entered.

The MPSU retrieves the **magnet type, magnet serial number, last set current, He level, date** and **time** from the MSUP.

Transport Test

The transport test checks the MPSU for internal damage that may have occurred during transportation. The test is only required the **first time** the MPSU used on site. The test is performed **prior** to connecting the ramp leads.

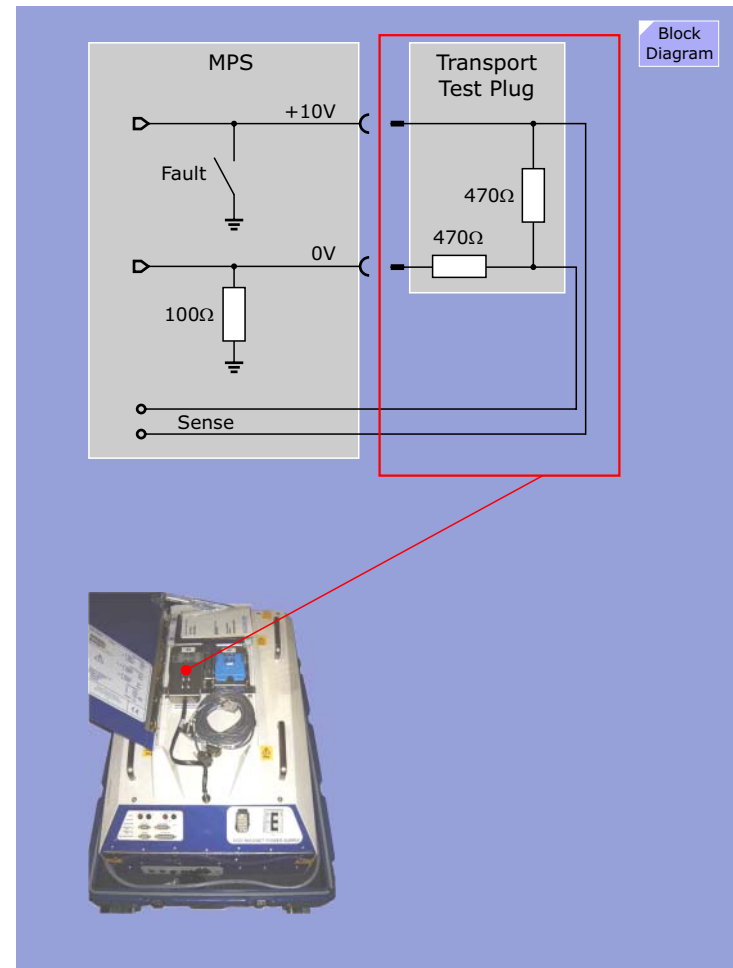
The test uses a test plug consisting of two 470 ohm terminating resistors with a center tap which the MPS can use to check its electrical and operational integrity.

If there is a fault, it can be determined by measuring at the voltage monitor connectors X9 and X10 and using the information provided in the table below.

Table 3 Transport Test fault condition

Condition	Magnet voltage detected
-ve bus bar fault.	+ 9.5 V
Test passed	+ 8.5 V
Test plug ground lead not connected.	+ 4.7 V
+ve bus bar fault.	0 V
Test plug not connected.	0 V
Rundown load diode shorted to chassis	0 - 4.5 V

Figure 258 Transport Test Plug



Self Test

The MPSU is run up to 700 A with the shorting contactor closed. The voltage drop across the MOSFET switch is monitored. When the current reaches its peak, the voltage is recorded (V1). Next, the MOSFET switch is opened and the current runs through the Run Down Load. Before the current starts to decay, the voltage drop across the RDL is recorded at Peak Current (V2).

Errors

If a fault occurs, an error screen is displayed automatically. To access it manually, the command "G" can be entered (see table above). The Magnet Peak Current is recorded on this screen, so if the magnet should quench, the level current at that time is stored.

MPSU log file

The last ramp data is stored in a log file in non-volatile memory within the 3600 MPSU, even if the MPSU is switched off. Data is overwritten when the next ramp is started. It is recommended that after each ramp, the log file is downloaded and stored for future analysis.

The Magnet Peak Current is recorded. If the magnet should quench, the recording will tell you at what current level the magnet quenched.

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13 Tim4G Upgrades

Introduction

This description covers the Tim 4G upgrades of the MAGNETOM Avanto and MAGNETOM Verio systems.

The upgraded Avanto will be named **Avanto^{fit}**.

The upgraded Verio will be named **Skyra^{fit}**.

The Tim4G upgrade brings Tim4G+Dot to the installed base by providing daVinci hardware to the MAGNETOM Avanto and MAGNETOM Verio systems. The upgrade package shall provide the new Tim4G technologies, Dot and the new cover design.

Product Profiles

The MAGNETOM Avanto and Verio systems have an outstanding reputation and acceptance in the market, a result of the highly successful Tim technology, first introduced with Avanto, as well as their magnet and gradient performance. The upgraded systems shall keep these components and combine them with the new Tim4G hardware and Dot software technologies.

Software Versions

System	Version
Aera, Skyra	VD13A
Tim4G fit upgrades	VD13B
TimTX TrueShape	VD13C
Prisma	VD13D

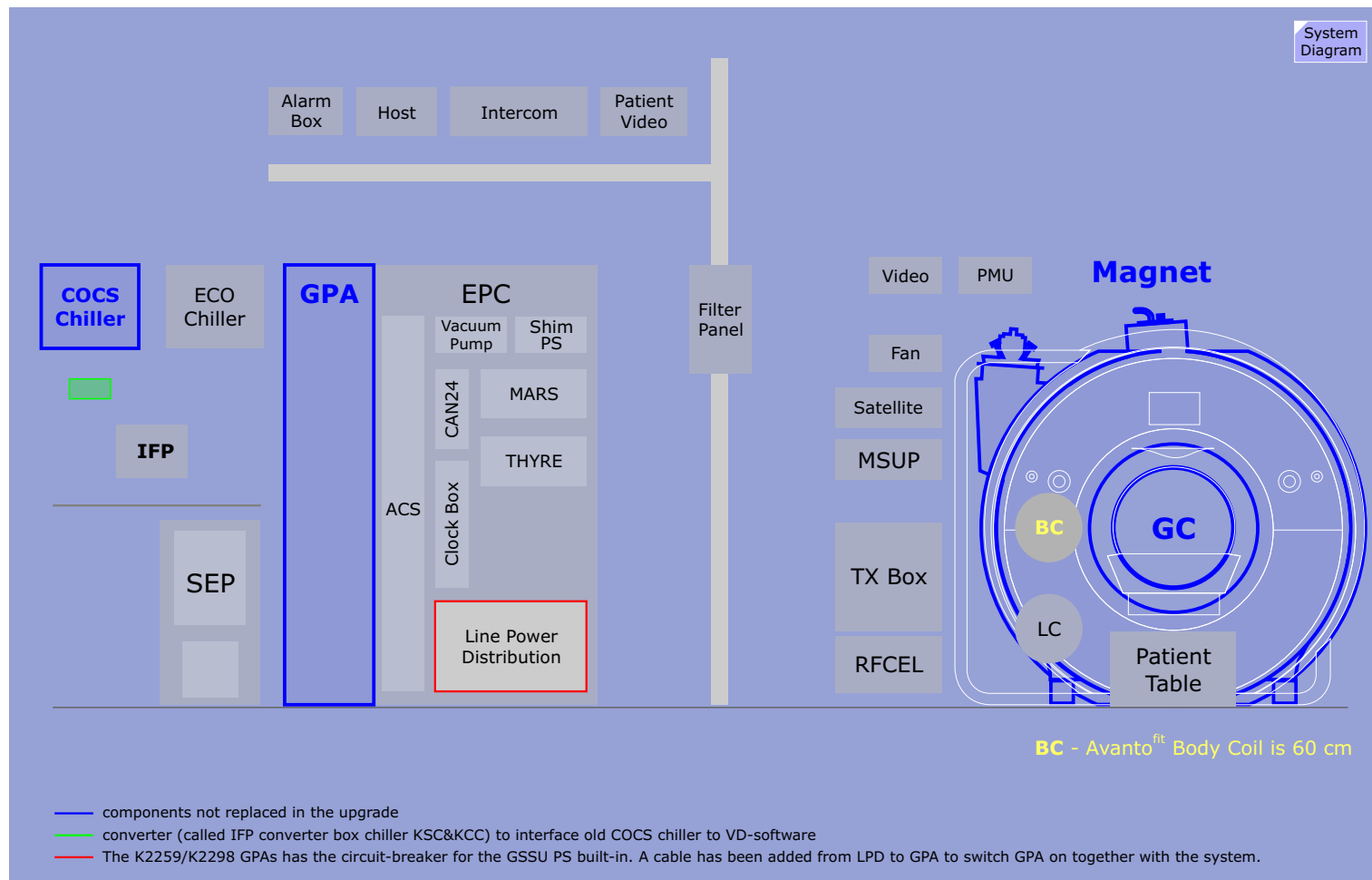
Avanto - Avanto^{fit}

Component	Avanto	Avanto ^{fit}
Magnet	OR105	same
Gradient	Q or SQ gradient	SQ Gradient
Tim	[32x8] [76x18] [76x32]	[204x48]
Dot	not available	Dot engines
Table	fixed	fixed or mobile
Software	VB line	VD line
MNO	available	NOT available

Verio - Skyra^{fit}

Component	Verio	Skyra ^{fit}
Magnet	OR97	same
Gradient	VQ gradient	same
Tim	[102x8, 8] [102x8, 18] [102x8, 32]	[204x48] or [204x64]
Dot	not available	Dot engines
Table	fixed	fixed or mobile
Software	VB line	VD line
MNO	available	available

Figure 259 Overview of Differences Avanto/Verio to Avanto^{fit}/Skyra^{fit}



Common Differences

- **General** - With the exception of the magnet, GPA and gradient coil all components will be replaced with daVinci components.
- **Software** - Software version is VD13. SeSo functions are nearly identical. Any differences are due to the newer hardware.
- **Receive Coils**
 - Head and Neck Matrix replaced with Head/Neck 20
 - Spine Matrix with Spine 32
 - Body Matrix is replaced with a Body 18
 - A Flex Coil Interface with one Flex Large 4 and one Flex Small 4 are included
 - One Tim Adapter is included to use existing Matrix coils
- **Service Tools**
 - The Avanto^{fit} requires a 60 cm array shim device.
 - The Skyra^{fit} requires a 70 cm array shim device.

Avanto^{fit} Differences

- **Body Coil** - The Body Coil diameter for Avantofit will remain 60 cm due to the fact that the Avanto 005 gradient coil (not replaced in the upgrade) can only accept a 60 cm Body Coil.
- **OLM and Patient Light** - The OLM cover patient light components have been adapted to the 60cm Body Coil
- **Patient Tables**
 - The Avanto^{fit} tables will only be rated for patient weights up to 200kg.
 - The whole-body option requires no additional rear support, the upgrade only requires a license.
- **Phantoms** - Along with the upgrade come new phantoms. The phantom holder for Avanto^{fit} is designed for a 60cm bore and has the colors Blue/Red.

Differences field-upgrade and ex-Factory

- **Covers**
 - The Avantofit ex-factory will be delivered with covers unmounted and must be mounted on site.
 - There will be differences in the cover hangers mounted on the magnet. Field-upgrades will have these holders glued. Ex-factory versions will have addition studs welded to the magnet to allow to have these holders bolted.
- **Chiller**
 - If the upgraded system has an older COCS chiller it may be retained. It will get the new IFP panel and an adapter box to interface the VD software to the older hardware.
 - An ex-Factory Avanto^{fit} will get the new ECO chiller.

Skyra^{fit} Differences

- **Phantoms** - The phantoms and phantom holders are identical to Skyra (70cm bore) and have the colors Blue/Yellow.
- **Service Tools** - The Skyra^{fit} requires a 70 cm array shim device.
- **Patient Tables** - The whole-body option requires no additional rear support, the upgrade only requires a license.

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14 Prisma / Prisma^{fit}

Introduction

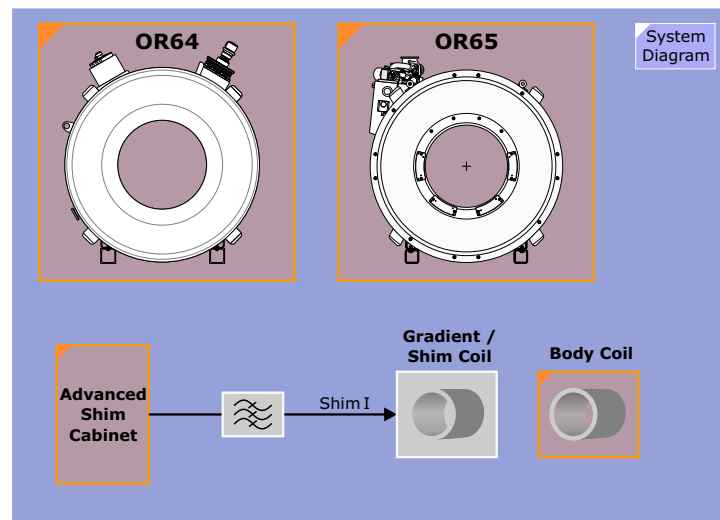
This description covers those components of the MAGNETOM **Prisma/Prisma^{fit}** which in any way deviate from the Skyra platform.

Product Profiles

The table below summarizes the differences between the two Prisma variants. The Prisma^{fit} is a forklift upgrade or any Trio.

Component	Prisma	Prisma ^{fit}
Magnet	OR65 (new)	OR64 (Trio)
Gradient	XR 80/200 gradient	same
RX	Tim4G [204x64] Tim4G [204x128]	Tim4G [204x48] Tim4G [204x64] Tim4G [204x128]
TX	TimTX TrueShape 2-Channel Transmitter	same
Tim Table	fixed, mobile	fixed, mobile
Software	VDline	same
MNO	available	available
Bore size	60 cm	same

Figure 260 Differences between Skyra to Prisma/Prisma^{fit}



OR64 Magnet

Introduction

The OR64 is an actively shielded superconductive magnet providing a nominal field strength of 2.98 Tesla. The magnet is comprised of a core assembly consisting of the magnet windings: 6 primary and 2 active shielding coils. The core is sealed within a welded stainless-steel vessel (cryostat) to be filled with liquid helium. The magnet core (cryostat) is completely encased in two aluminium tube/bore assemblies (cyro-shields) and suspended in an outer steel shell by rods made of a fiberglass composition. The magnet is put under a vacuum and the two inner aluminum shields will be actively cooled by a refrigeration system. All of these measures are taken in order to reduce the boil-off rate of the liquid helium.

The magnet is fitted with several thermal sensors, helium fill level sensors and heaters required for monitoring and servicing.

The connections to these sensors are mounted on a service turret which also houses the neck tube assembly, providing the only opening to the magnet core. The neck tube assembly provides a two access ports: the ECL probe port (for a current probe use to energize and de-energize the magnet) and a liquid helium fill port. The Service Turret also provides a venting system consisting of various valves and tubing, provide paths for exiting helium gasses.

Helium level after a quench

To generate a field strength of 3T there is a larger magnetic field energy stored in the magnet compared to 1.5 T magnets. After a quench, as a rule, the magnet will be empty, regardless of the helium level before the quench. It is therefore recommended to have at least 500l liquid helium available when ramping the magnet. Helium refilling has to occur as quickly as possible after a quench, otherwise the magnet warms up to a point (anywhere above the temperature of liquid nitrogen) where it becomes

necessary to perform re-cooling of the magnet by a process called thermo-cycling. This is indeed a very uncomfortable situation because this procedure takes 2 weeks!

Specifications

Specifications	Value
Nominal Field	2.895T
0.5mT field	z - 6m x,y - 3.9m
weight (full, no coils)	9970 kg
length (no covers)	2.13 m
height	2.53 m
diameter	2.15 m
ramp time	120 m
current (typical)	484 A
Helium volume	1200 l
boil off (Cold Head on and for typical clinical use)	0.15 l/h
boil off (Cold Head off)	3% / day

The Magnet can be broken down to these major components:

- Magnet Core
- Service Turret
- Cold Head

Magnet Core Construction

The OR64 magnet core consists of 6 inner coil journals and 2 outer coils that provide the active shield (i.e. reducing the magnetic fringe field).

Main Coil

The superconductive magnet coil is designed to generate a homogeneous, stable main magnetic field B_0 at a minimum fringe field.

Switch Heater

A switch made of special superconductive wire is electrically heated by a resistive heater to enable energized and de-energized of the magnet.

Quench Heaters

In case of emergency (accidents, fire, etc.), the magnetic field has to be de-energized immediately to avoid danger to personnel. Two quench heaters are placed close to the main coil. When activated, they cause the magnet wire to loose superconductivity. The current is brought to zero within 20 seconds (quench). The thermal energy released will boil off ALL of the liquid helium.

Quench Protection System

During a quench, up to 6 kV can be induced across the coil. To avoid damage, diodes clamp the voltage to a maximum of approximately 12 V.

External Interference Screen (EIS)

The EIS is a separate set of superconductive windings placed over some of the primary and both active shielding windings intended to inductively absorb external magnetic transients to prevent B_0

fluctuations. The currents induced in the EIS by the transients, as well as the current induced in the EIS by the steady field decay of the magnet (caused by the rest resistance of the coil joints) produces a counter-field that deteriorates B_0 field homogeneity. Therefore, the EIS has to be quenched once a day. This is performed by the Magnet Supervision Unit (MSUP). To avoid image quality problems, the EIS must be reset when the system is not being used (default time of the reset is 2:00 am but can be programmed to any time).

Figure 261 Magnet Schematic

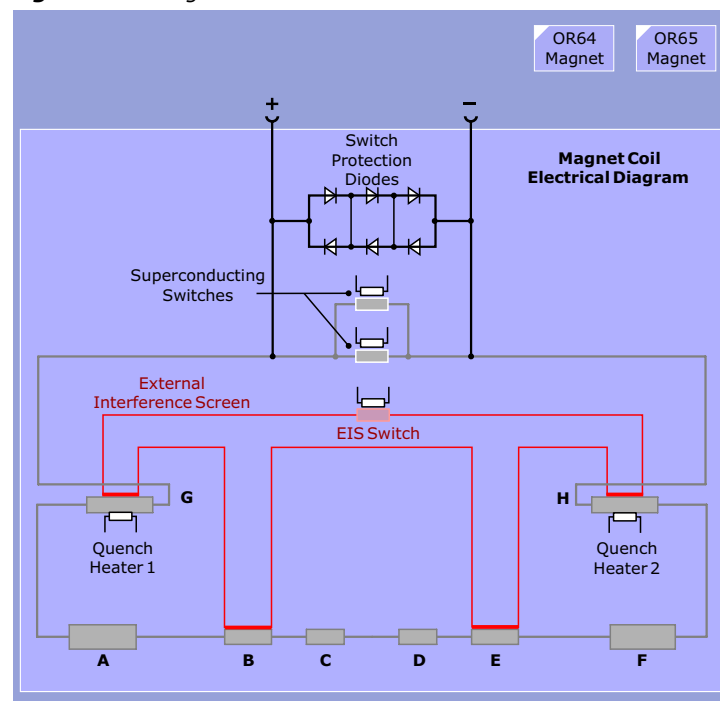
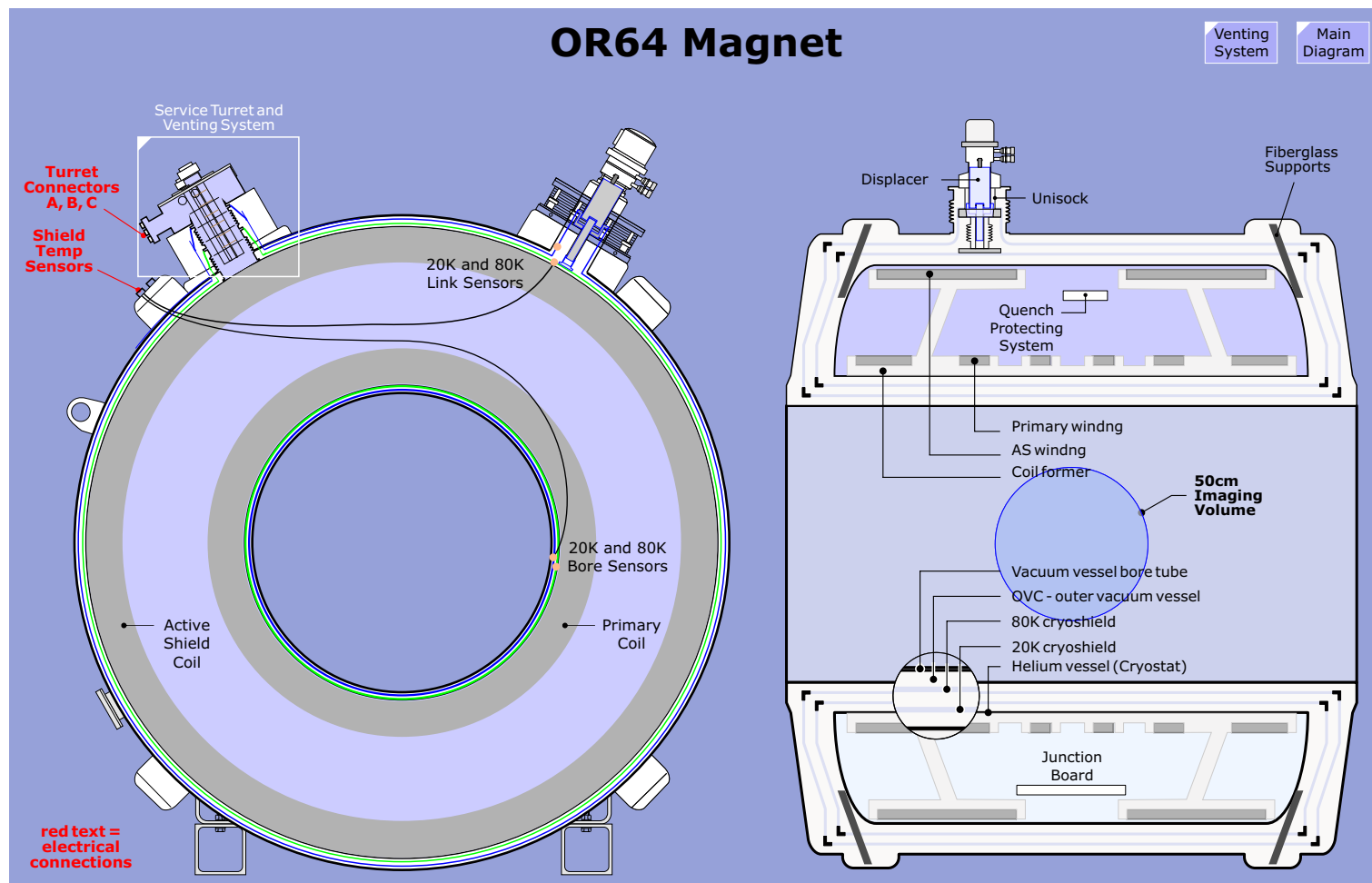


Figure 262 OR64 Magnet

Service Turret

The Service Turret is placed at the top of the magnet. It contains:

- Venting system
- helium fill port
- Enhanced Current Leads (ECL) port
- Electrical connections of internal circuitry

Venting System

Vent Valve

A vent valve is fitted in parallel with the Quench valve. It vents boil-off gasses automatically when the internal magnet pressure exceeds rating of the valve (0.33 psi). This purpose of this is to establish a positive pressure in the magnet to prevent air ingress.

Depressurization (Bypass) Valve

A hand-operated valve in parallel to the vent valve and is used to de-pressurize the helium vessel. This is required before opening the magnet which is necessary for some service procedures.

Quench Valve

The quench valve is a spring-loaded clap valve which opens at 2.2 psig. The clap valve incorporates a graphite disk which will burst at 10 psig should the valve not open. The valve also includes a silicon o-ring for sealing. Normally the burst disk and o-ring survive a quench, nevertheless, the complete valve assembly must be checked for integrity after a quench.

Pressure Gauge

The pressure gauge indicates the internal magnet pressure. Pressure during normal operation should be always positive (i.e., above atmospheric pressure), close to 1/3 psi if the vent valve is working properly. It is important to know the internal magnet pressure before making any opening to the magnet. If it is under atmospheric pressure (i.e., vacuum, the red zone on the gauge)

opening up to the magnet will cause air to ingress and cause ice formations within the magnet, a very serious situation. If the magnet is under a positive pressure (indicated by the yellow and green zones), it has to be depressurized before opening to prevent the extreme cold temperature of the exiting gas from eventually causing cold burns to those working on the magnet.

NOTE After a quench, the valves of the venting system must be checked for integrity. Please refer to the Troubleshooting Guide "Post Quench" procedure.

Helium Fill and Vent Lines

The helium fill line is permanently installed. It is accompanied by the fill vent line which is used to purge the helium fill line before use.

ECL Port

The magnet is energized via demountable Enhanced Current Leads (ECL). It is inserted only for magnet ramping.

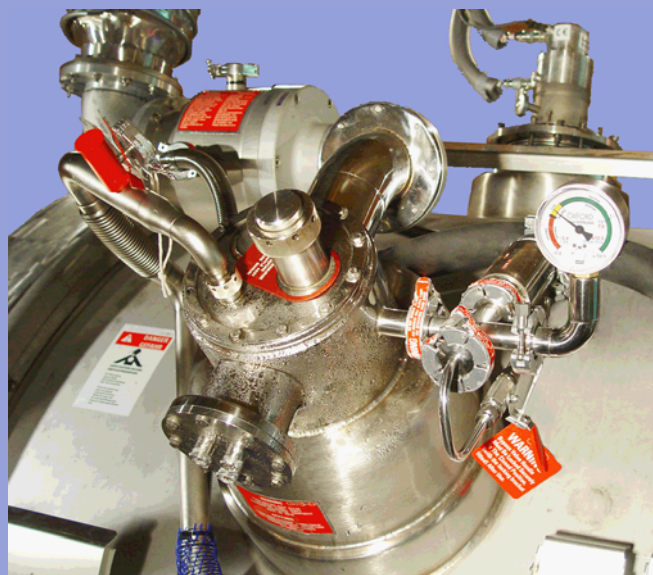
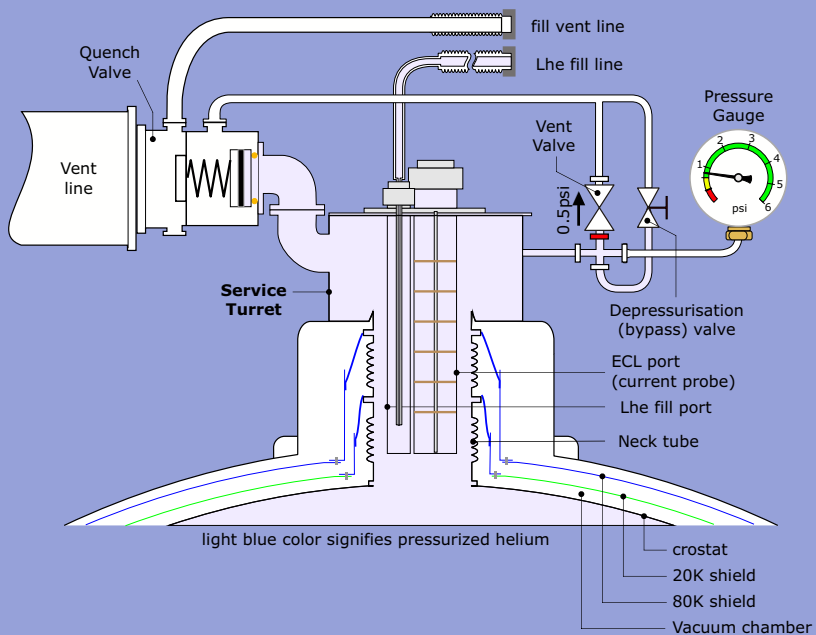
Enhanced Current Leads (ECL)

The ECL is inserted into the magnet where it makes connections between the Magnet Power Supply and the magnet coil for energizing the magnet (ramping). This procedure is described in the "Start-up" section under "Ramping the magnet".

Figure 263 OR64 Venting System

OR64 Venting System

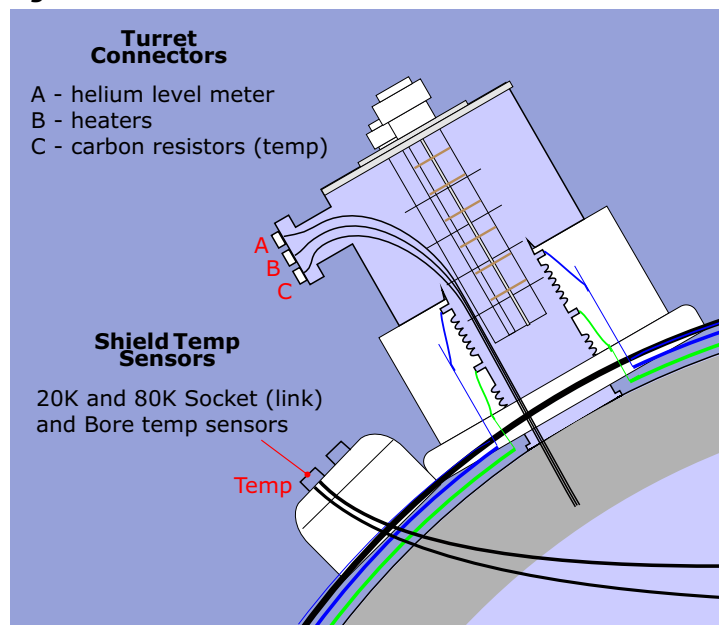
Magnet
Overview



Electrical Connections

Connections to internal magnet sensors and heaters are provided by three connectors (labelled **A**, **B** and **C**) on the Service Turret and an additional connector (labelled **Temp**) located to the right of the service turret. These are connected to the Magnets Supervisory (MSUP) for monitoring purposes.

Figure 264 *Electrical Connections*



Cold Head

Heat radiation plays the most important role when considering the heat impact to the cryo-shield and insulation foils inside the magnet. This heat impact has to be reduced as much as possible to keep the helium boil-off rate as low as possible. This is obtained by cooling two cryo-shields to low temperatures around 20 Kelvin and 80 Kelvin, using a cold head.

The OR64 magnet system employs a 2 stage cold head manufactured by APD. It is fitted into a closed interface (Unisock).

Magnet Supervision

The Magnet Supervisory model 2306 has been modified to accept the sensors of the OR64 magnet.

Differences are:

OR65 Magnet

The OR65 is a hybrid magnet comprised of the exceptional OR64 magnet core and the 4K technology of the OR99/OR103 shell (side socket design with integrated service turret and 4K cold head). The OR64 magnet core provides the extremely homogeneous magnetic field and the OR99 shell provides the advantage of 4K technology for exceptionally low life-cycle costs.

Service wise the OR65 is, for all practical sakes and purposes, identical to the other daVinci family magnets. Their venting and electrical interface are identical.

The [OR99/103 description](#) in the Magnet System section can be used for the OR65.

Shim Options

- Standard Configuration
one Shim PS providing 5 shim channels with 5A/channel (same as previous systems)
- Autonomous Shim Cabinet 2
two Shim PS providing 5 shim channels with 10A/channel

ASC 2

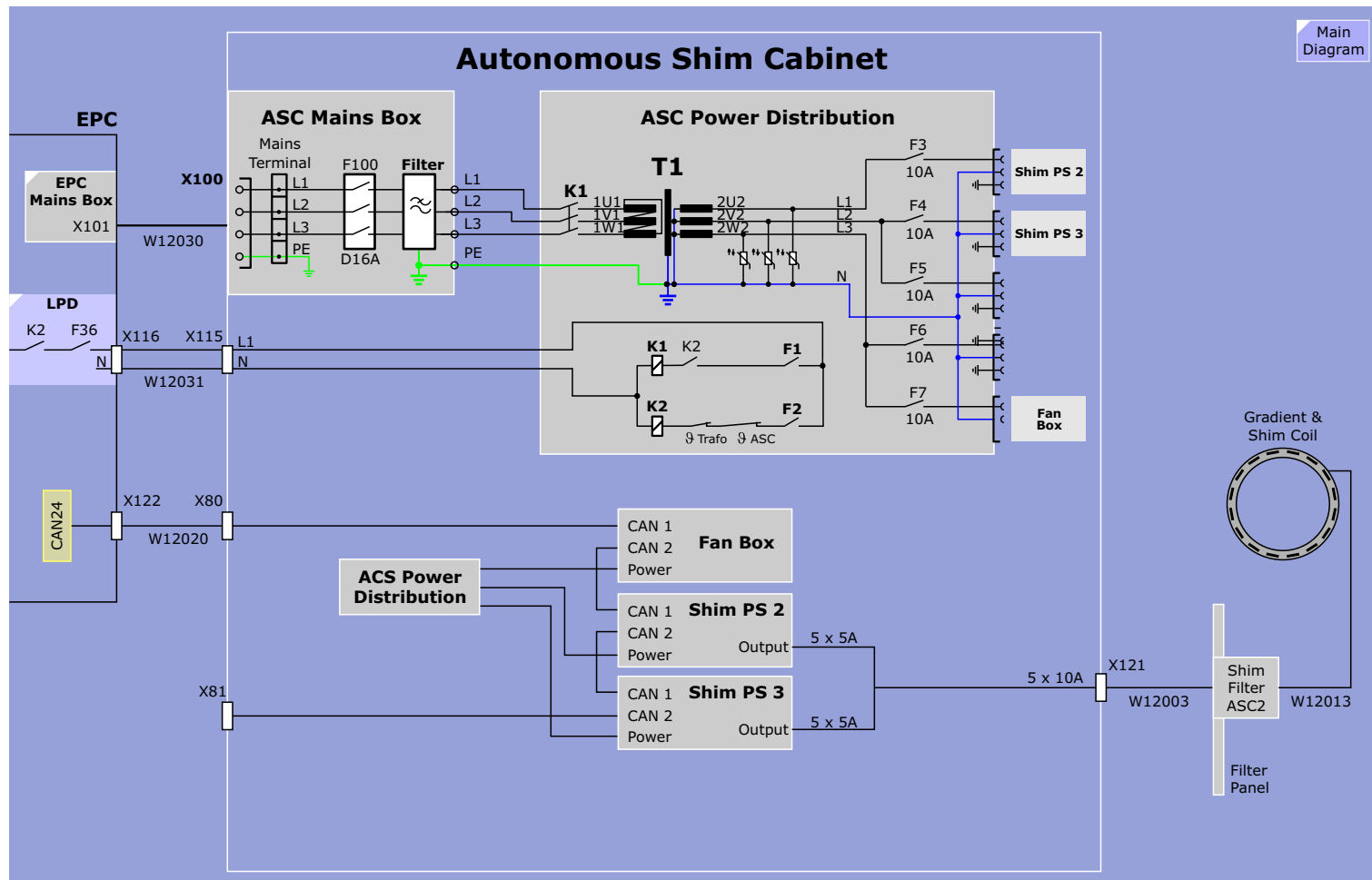
The Autonomous Shim Cabinet is comprised of 2 Shim Power Supply modules providing 5 Shim channels of 10A each.

When the ASC-2 Option is installed the Shim PS in the EPC will not be used.

The ASC-2 uses an external Filter box (Filter box ASC-2) located at the standard Filter Panel. Mechanically, the external filter box is compatible to the standard Filter box and uses the same cutout on the filter panel.

Term	SPS	Channel	Pin connectors	Max I
A(2,0)	2, 3	1	1, 2	10A
A(2,1)	2, 3	2	3, 4	10A
B(2,1)	2, 3	3	5, 6	10A
A(2,2)	2, 3	4	7, 8	10A
B(2,2)	2, 3	5	9, 10	10A

Figure 265 Autonomous Shim Cabinet Block Diagram



Prisma Body Coil

The Body Coil for the Prisma Prisma-fit systems works on the same principles as all other birdcage designs but has been given two additional tuning capacitors, T1a and T2a. These were necessary to achieve the tuning range for this particular coil. Referring to the Tune Up instructions, the tuning capacitor pairs T1/T1a and T2/T2a will be adjusted together as if comprising one single adjustment.

Function

Figure 266 Prisma BC Parts Location

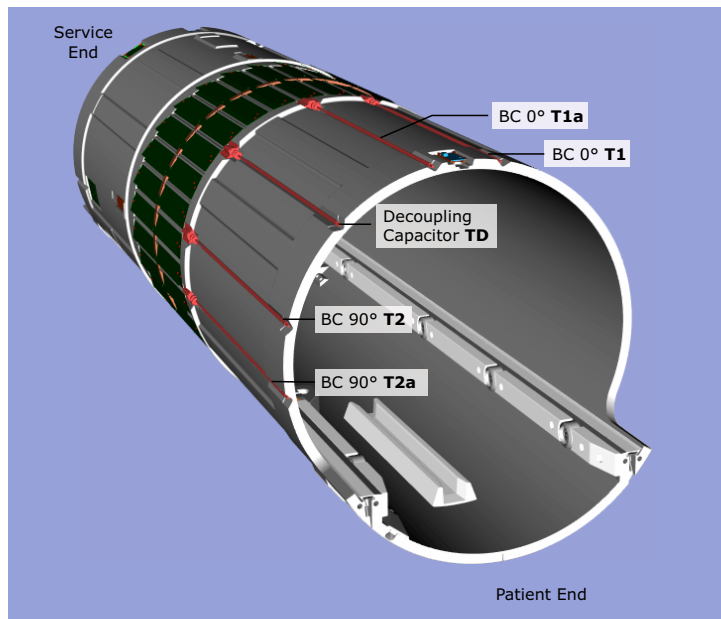
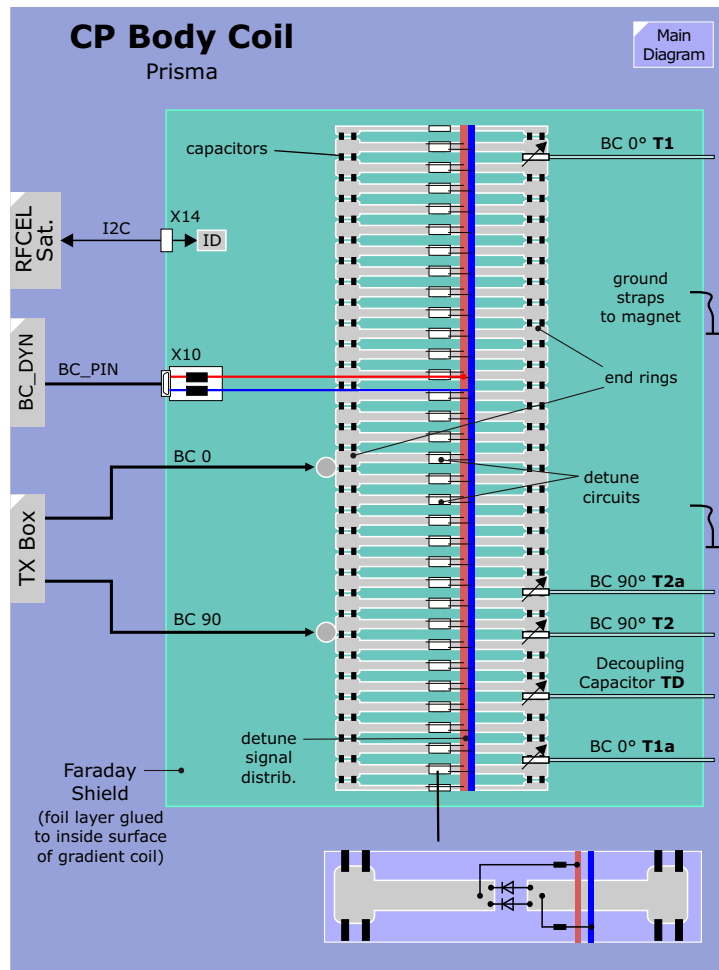


Figure 267 Prisma Body Coil Block Diagram



15 Changes

Changes in this Version

Section 1 - System

[Page 4](#)

- added MAGNETOM Skyra Tim [204x24] XQ Gradient and Aera Tim [204x24] XQ Gradient configurations

- added new wording for Tim

- corrected last sentence in RoHS paragraph to: "Components having an RoHS label are free from these materials or substances."

[Page 8](#) - Corrected filter panel power connections from filters Z15-21 (exam room side), and added connector labels to PTab power supply.

Section 3 - Host

[Page 39](#) - Revised text of Remote Monitor section for better clarity and removed TCP port numbers since these may vary on make and model of KVM devices.

All graphics have been adapted for longevity.

Section 5 - RF System

[Page 62](#) - reformulated Tim 4G Coils section to match marketing

[Page 64](#) - shortened RF System specification list

[Page 68](#) - Corrections to TX Box 2 LED descriptions (V8, V28, V51, V60)

[Page 71](#) - Edited text on Thyre Function. Corrected graphic: Filter Plate X40A changed to X22A.

[Figure 86](#) - Corrected error: 300MHz clock from Clock Box is 320MHz

Section 6 - Patient Handling

[Page 143](#) - updated texts on the obsolete OLM and OSC.

[Page 126](#) - corrected maximum patient table load from 250kg to 200kg for Avanto fit.

[Page 158](#) - Updated descriptions for Docking Unit and Emergency Release.

[Page 167](#) - power cables to displays (W15815, W15816) were swapped

[Page 180](#) - added text in Battery Charger section referring to an external power supply for the charging of the PERU and PPU

Section 7 - Gradient System

[Page 181](#), 184, 186, 230 - corrected tables, the E-Shim is **not available** for the Aera XJ Gradient Coil and standard for the Aera XQ Gradient Coil.

[Page 209](#) - added X16.2 jumper

[Page 224](#) - added X16.2 jumper on D70

[Page 225](#) - added Avanto fit specific informations

[Page 226](#) - revised CCU Overview, replacing sensor company names (LEM, Danfysik) to their functions.

[Page 228](#) - Removed texts on the number of possible NTCs and updated the text on the Gradient Connections.

[Page 232](#) - Power Supply section: corrected connector reference (X128 to X124)#

Section 8 - Magnet System

[Page 238](#) - removed helium levels and added note to refer to SKB for current recommendations

[Page 248](#) - replaced Hyperterminal with Putty

Section 9 - Cooling System

[Page 277](#) - added nominal pressure difference, added SKB reference to expansion chamber.

[Page 278](#) - corrected arrows indicating primary water flow, they were reversed

[Page 282](#) - corrected Monitoring Level values and updated Cooling Mask values.

[Page 286](#) - amended text on error LEDs

[Page 287](#) - added new IFP controller with LCD display introduced in 04.2021.

Change History

Same page references may no longer correspond those in this version due to the additions/changes of diagrams and/or texts in the older versions.

Changes in Version 9

Section 2 - Software

Page 32 - corrected MARS IP address in the text.

Section 4 - MARS and Control

Page 60 - Completed last sentence of first paragraph to enhance clarity.

Section 8 - Magnet System

Pages 242-245 - Added note stating OVC burst disk and vent will no longer be fitted.

Page 272 - Updated text on CH gas line removal.

Page 249 - added 5A fuse to Li battery output

Section 9 - Cooling System

Page 280 - added note on buffer tank retrofit

Page 288 - added description of IFP controller LEDs

Changes in Version 8

Section 1 - System

Page 4 and Page 5 - Added 24 channel option

Section 3 - Host

Page 38 - Added new HP Z420 model

Section 4 - MARS and Control

Pages 46, 48, 50 - Description adapted for 24-channel configurations

Page 57 - Added Gradient Cable Supervision

Pages 58, 59 - Updated for latest CAN24 version (LEDS modified)

Section 5 - RF System

Pages (many) - added or modified description to include the new 24 channel configuration (new IF_RX24 board).

Section 7 - Gradient System

Page 230 - manometers are no longer used for the gradient coil transport air cushions

Page 233 - improved cabling to CAN24

Section 8 - Magnet System

Page 268 - Modified text on compressor lines voltage and second adsorber-change timer.

Section 9 - Cooling System

Page 285 - Added tables of Display Codes and Monitoring Levels.

Changes in Version 7

Section 2 - Software

Page 21 - update System Manager - Periphery graphic

Section 5 - RF System

Page 70 - updated TX Box 2 LED description

Page 99 - added external PMU trigger input

Page 116 - really updated LED description

Section 6 - Patient Handling

Page 152 - improved diagram readability

Page 168 - corrected text under "Power Switching"

Section 7 - Gradients

Page 205 - Added cable from GRACON to X89 on EPC roof (CHARM MR_00388775)

Page 278 - Updated note in Humidity Sensor section

Section 10 - LPD

Page 292 - added ASC option

Section 14 - Prisma

new section

Changes in Version 6

Section 4 - MARS and Control

Page 50 - added note on 100MHz clock

Page 51 - added note on 160MHz clock

Page 54 - added DPD to map

Page 57 - corrected connector number of CAN24 - SRS Router from X76 to X85

Section 5 - RF System

Page 67 - Updated TX Box variant table to include latest 2 channel TX Box and corrected TX Box variants table.

Page 70 - Updated LED description for TX Box version 2

Page 71 - Added 2-Channel TX Box

Page 73 - Added list of input voltage values and changed signal name THYRE_ENABLE to THYRE_On in the diagram.

Page 116 - Added Tim Adapter LED description

Section 6 - Patient Handling

Page 131 - corrected cable labels of W14306 and W14380, were labelled wrong at the DPD.

Pages 131, 139 and 147 - connectors X2 and X3 on PAM were swapped.

Page 163 - Texts describing V1/V2, V3/V4 were swapped

Section 9 - Cooling System

Page 275 - Water flow in ACS corrected

Page 277 - Added new ACS version

Page 278 - Added humidity sensor option.

Page 285 - status mask changed in VD13 software

Page 286 - added description of Free Cooler

Section 11 - Tune Up / QA

Page 302 - Added new Feedback Loop Calibration procedure

Section 13 - Tim4G Upgrades

Page 349 - New section.

Changes in Version 5

Section 3 - Host

Page 38 - Section MRAWP - new hosts added.

Section 4 - MARS and Control

Page 46 - Updated MARS configurations for VD13 software

Page 50 - error correction in table: "Data transferred over PCIE bus"

Page 51 - error correction in table: "Data transferred over ISO-Ring bus"

Page 53 - Scanner Software section updated

Page 54 - Added MNO TX-Box (option)

Page 56 - CAN24 description updated

Section 5 - RF System

Page 64 - Removed bogus statement about the Body Coil having a temp sensor.

Page 67 - Updated TX Box description to reflect changes to THYRE PS output voltage of +110V DC for TX Box version 2. The TX Box block diagram updated to include the names of internal components which are also used in many TX Box related error messages.

Page 71 - Correction of THYRE LEDs

Page 103 - Correction of LED description for MSM

Page 116 - Pinout of SlideConnect corrected

Page 117 - MNO (multi nuclei option) description added

Section 6 - Patient Handling

Complete section reworked. Links added.

Section 7 - Gradient System

Page 192 - internal interrupt signal N_INT_INT removed

Page 200 - D41 "common fault 1" changed

Page 201 - error correction of FOCs U13 and U14

Page 209 - D40 replaced by D41, power stage supply voltage corrected to 277 VAC

Page 216 - AC line voltage monitoring from D110 removed

Page 228 - New Section "Gradient Supervision" added

Section 8 - Magnet System

Added new OR103 Magnet

Page 237 - Correction of EIS windings

Page 245 - Correction of fuses

Page 251 - Section Magnet Pressure Regulation completely reworked

Section 9 - Cooling System

Page 277 - added marker 3a and line 12 in the Legend.

Page 279 - corrected pinout of power connector to MREF.

Page 280 - added K1 in controller layout.

Page 285 - added link to Chiller.

Page 285 - Cooling Status Mask, removed the following line: RFCEL > Water Temperature. There is no water temp sensor.

Section 10 - Line Power Distribution

Page 288 - Gradient Supervision SIEMENS_VLM added.

Changes in Version 4

Section 7 - Gradient System

Page 247 - Modulator D40 corrected to Modulator D41

Page 249 - Modulator D40 corrected to Modulator D41

Changes in Version 3

Section 1 - System

Page 4 - error correction available system variants

Pages 7,8 - corrected labelling of Filter Panel - TXBOX PS and RFCEL PS labels were reversed.

Page 14- editorial changes to RoHS

Section 4 - MARS - Control

Page 86 - corrections to connector designation in FPGA Topology diagram Figure 51

Section 5 - RF System

Pages 140-152 have been added to the Tim 4G Coil section

Section 6 - Patient Handling

General structure changed - component description grouped under Tim Table, Tim Dockable Table and Common Table Functionalities.

Graphics and navigation built-in.

Page 152 - chapter Introduction - description of Whole Body Option added

Page 153 - chapter Introduction - description of Information Flow added

Page 160 - chapter Tim Table > Vertical lift > Powerstage Y (PSY) D4110 - description of Service Procedure Table Height Adjustment added.

Page 161 - Figure 109. - chapter Tim Table > Vertical lift > Powerstage Y (PSY) D4110 - hydraulic overview diagrams added

Page 170 - chapter Tim Dockable Table > Vertical lift - description of Hydraulic pump assy + Y-Drive D4312 added

Page 183 - chapter Common Table Functionalities > Horizontal Drive - description of DPD D4320 programming added

Page 192 - chapter Cabin User Interface (CUI) - error correction and functional details added

Page 205 - chapter PMU - error correction and functional details added

Section 8 - Magnet

Page 274 - chapter Service Turret > Fixed Current Lead - description of Voltage Sense Interface added.

Page 277 - chapter Magnet Supervision (MSUP) > ERDU Battery - error correction MSUP connector X1 for +36 V Service power supply. CHARM MR_00394754.

Page 289 - chapter Alarm Box > Function > Magnet Status Indicator > Alarm Buzzer - note added. CHARM MR_00398356.

Section 9 - Cooling

Page 309 - X80 and X81 swapped in the ACS Wiring diagram

Section 10 - LPD

Page 322 - Block Diagram "Line Power Distributor" and "System Power On/Off Switch Logic" - error correction standby signal. CHARM MR_00395945.

Section 11 - Tune Up / QA

Page 329 - complete section newly added.

Changes in Version 2**Section 1 - System**

Corrected Tim configuration list

Section 4 - MARS & Control

Added boot flowchart

Section 7 - Gradients

Added links

Section 8 - Magnet

Editorial changes and added links.

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